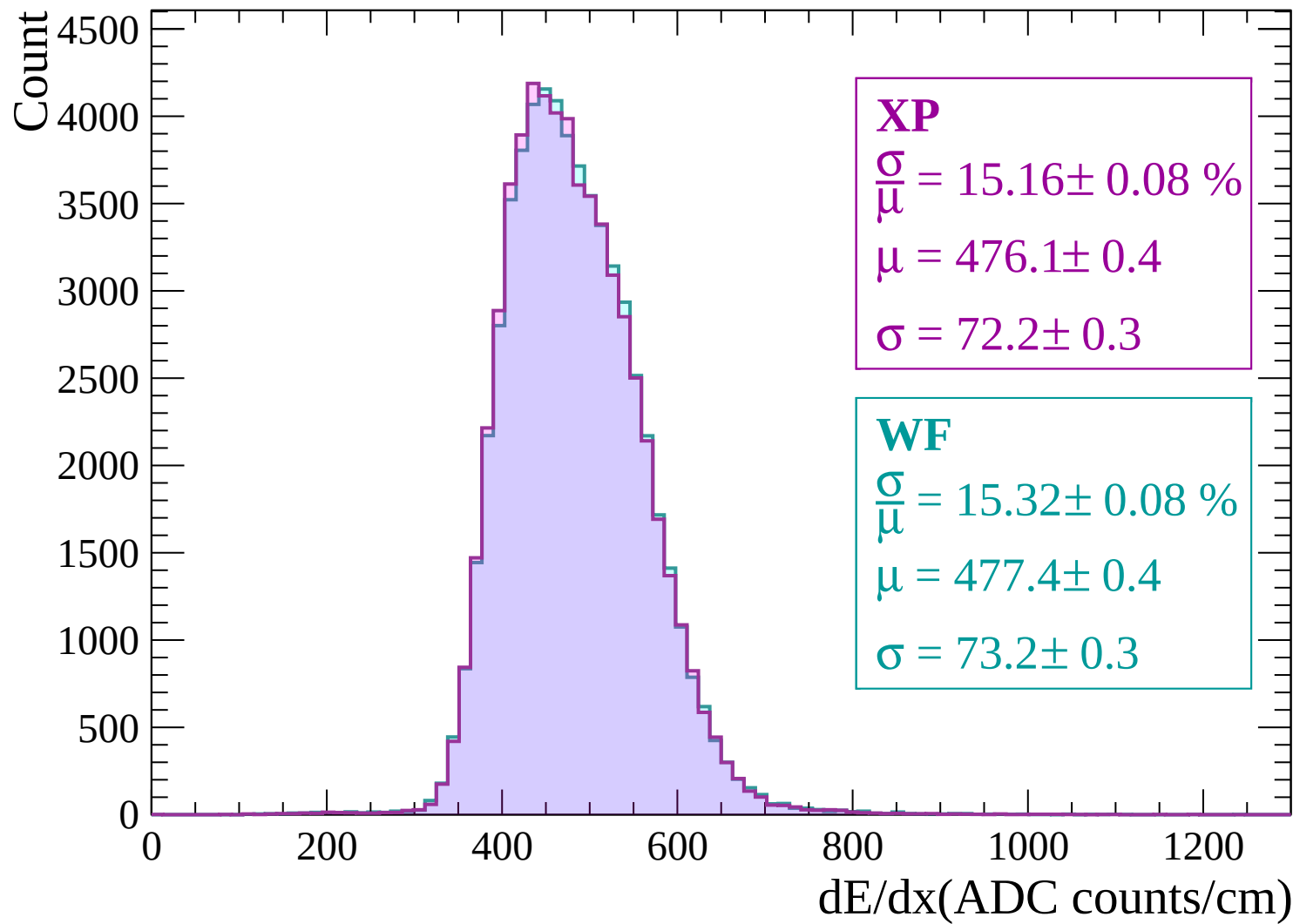
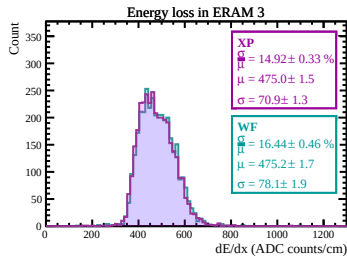
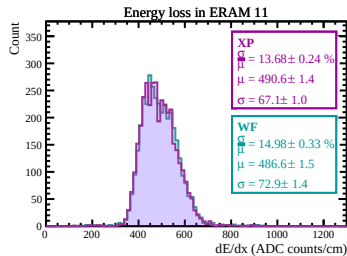
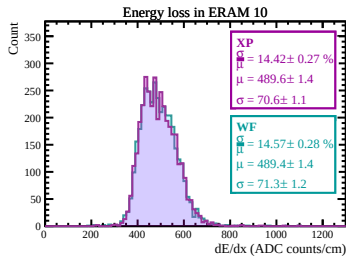
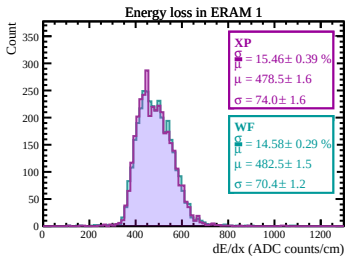
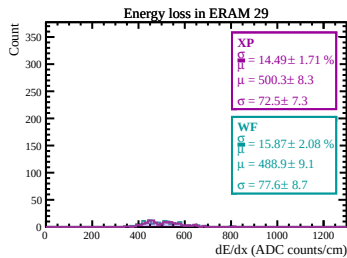
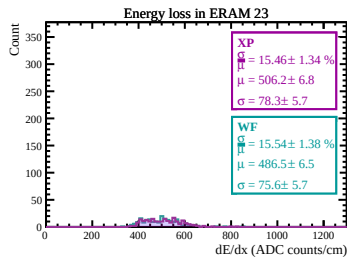
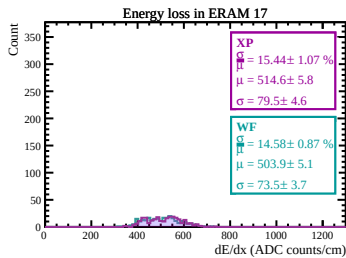
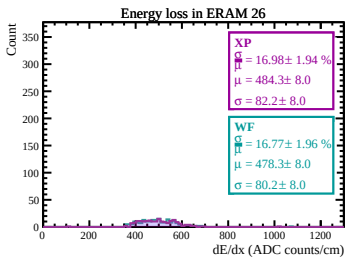
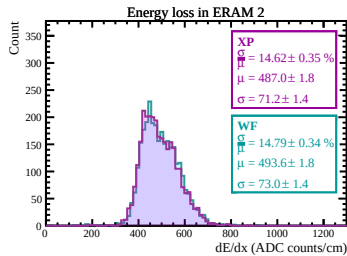
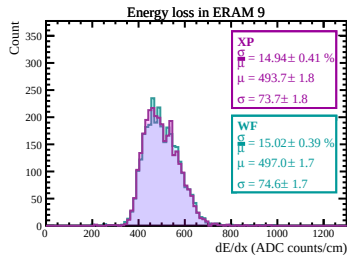
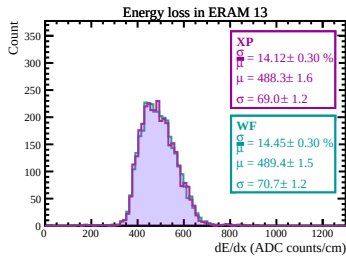
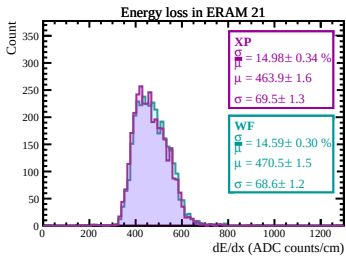
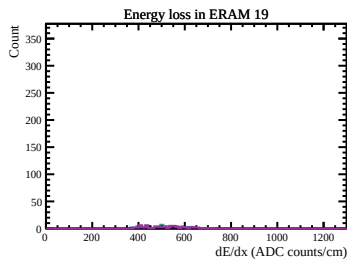
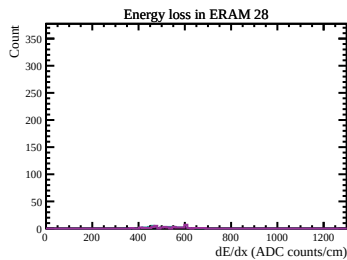
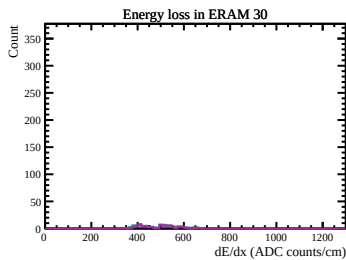
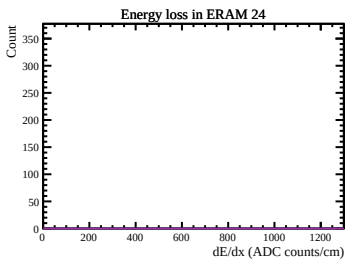
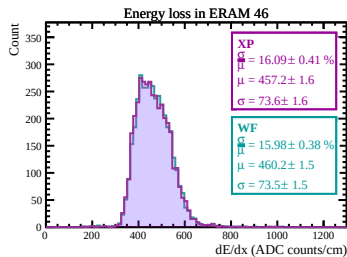
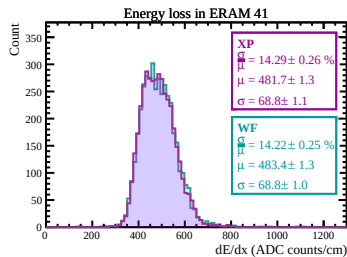
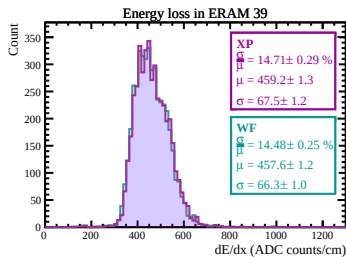
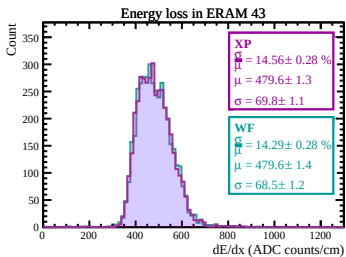
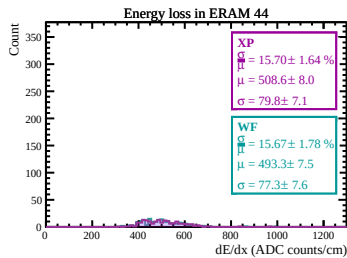
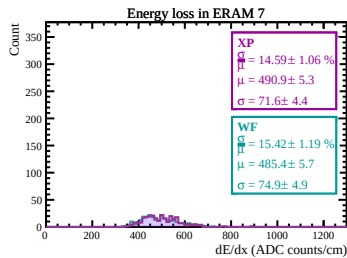
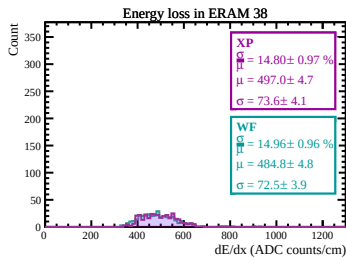
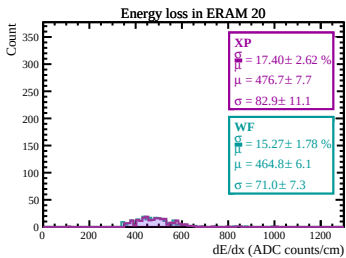
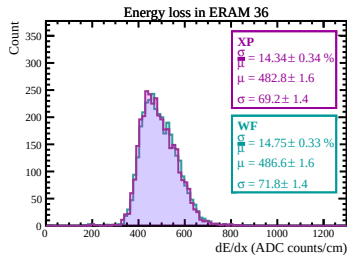
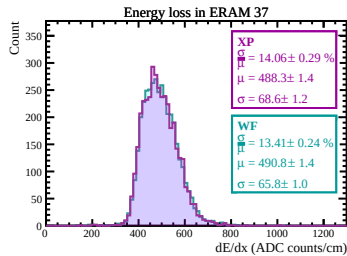
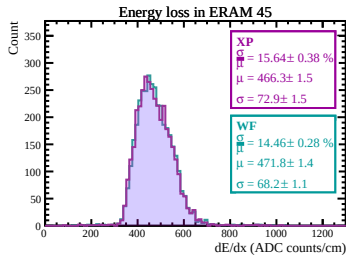
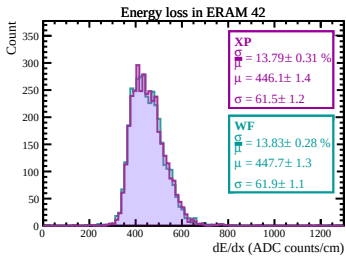
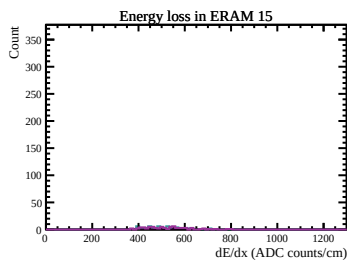
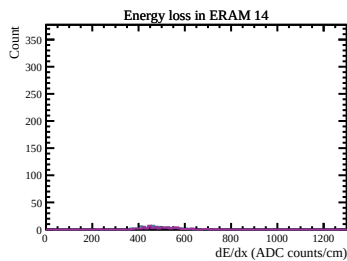
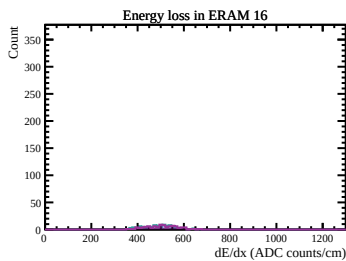
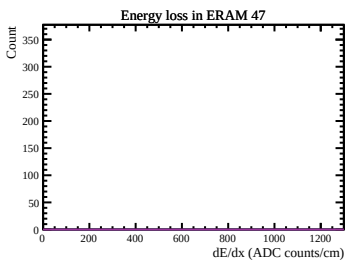
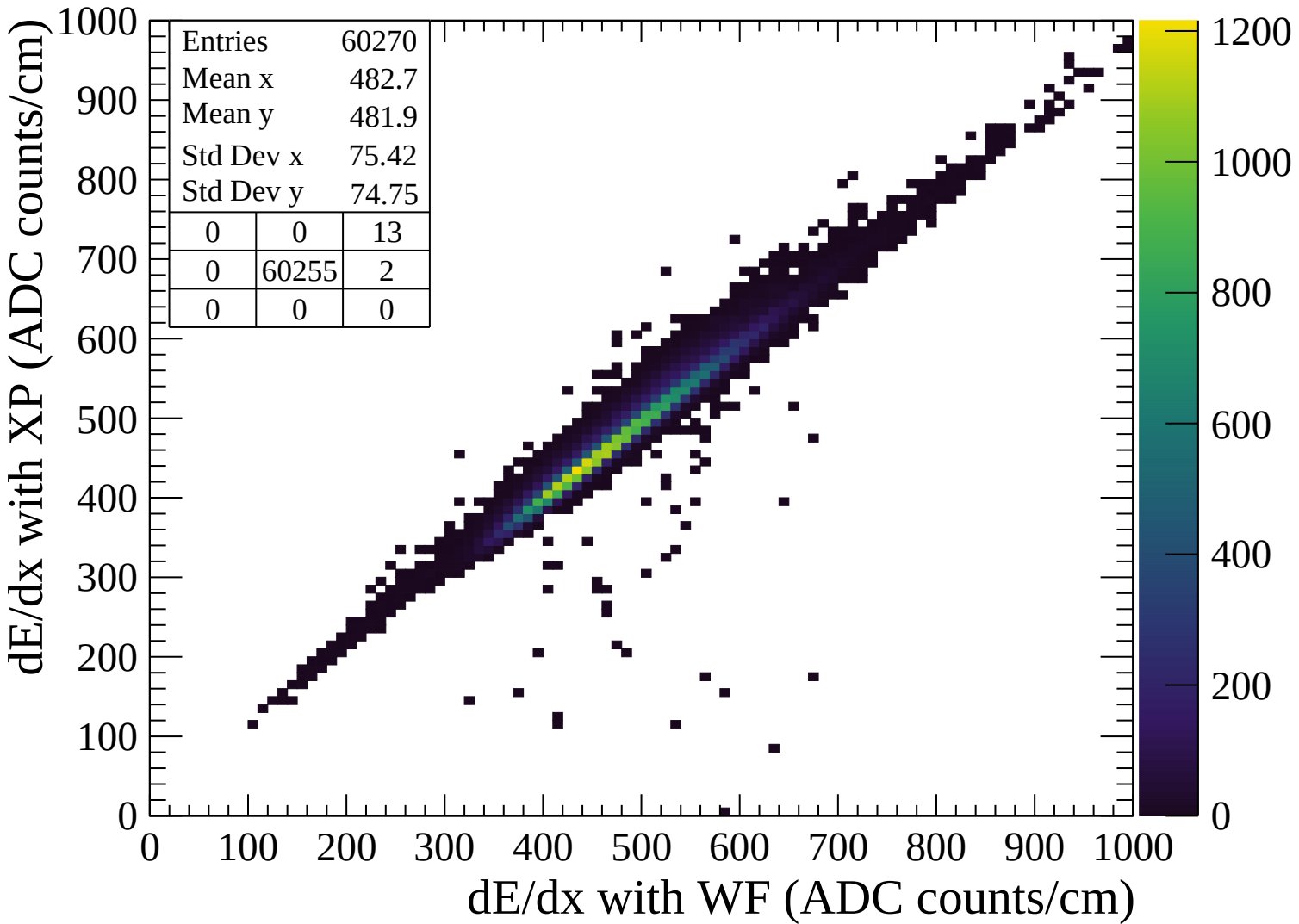
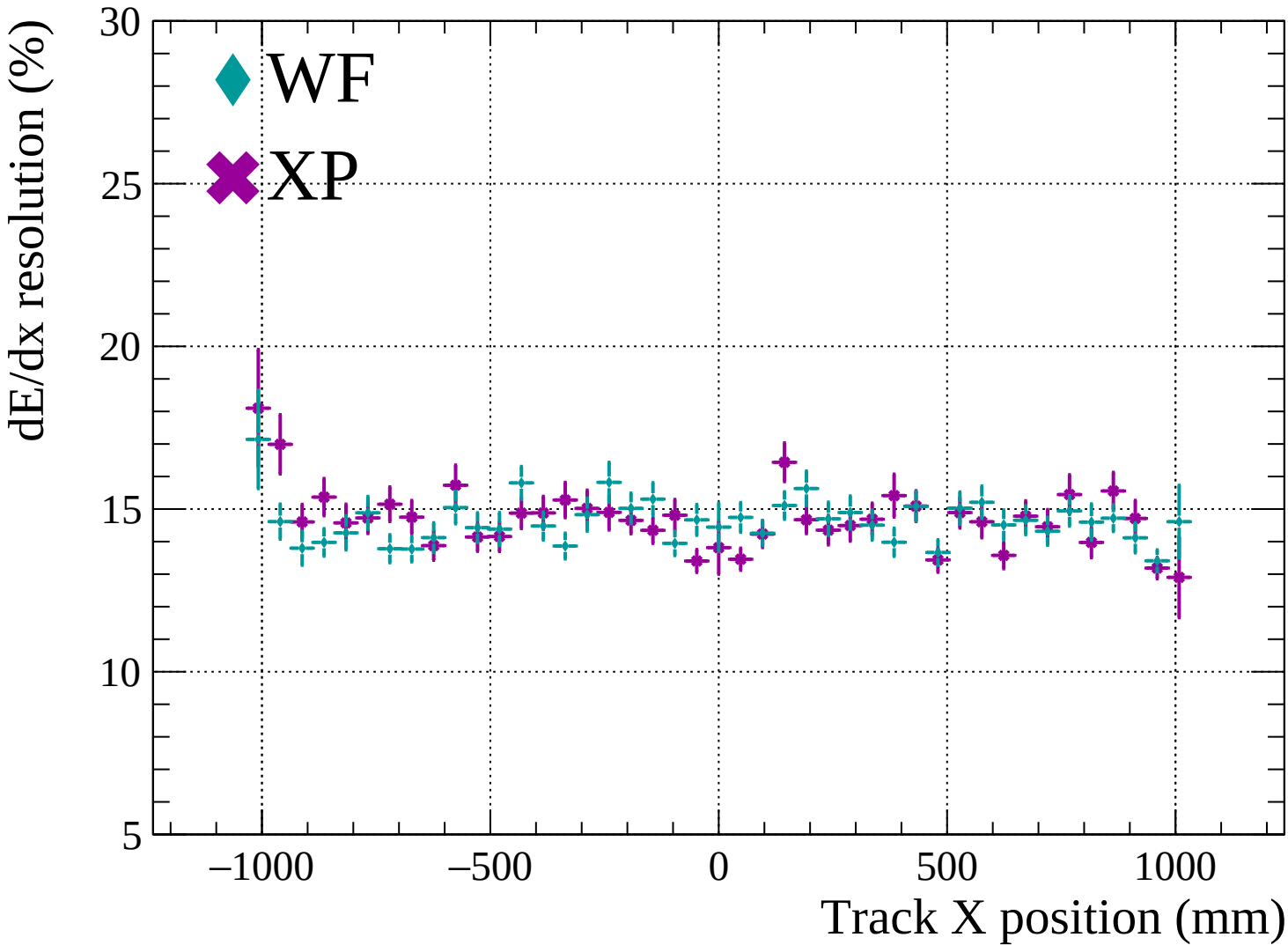


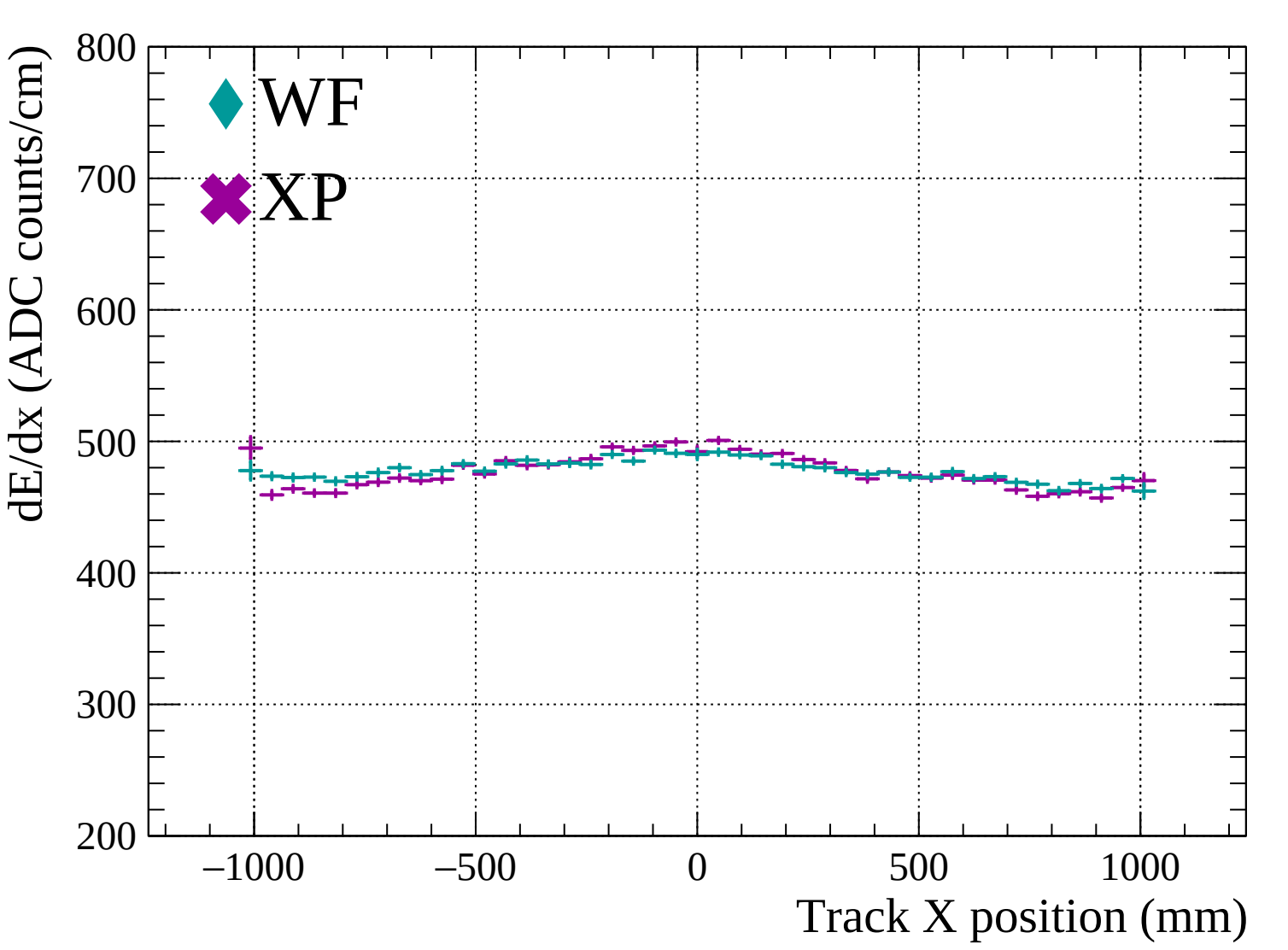


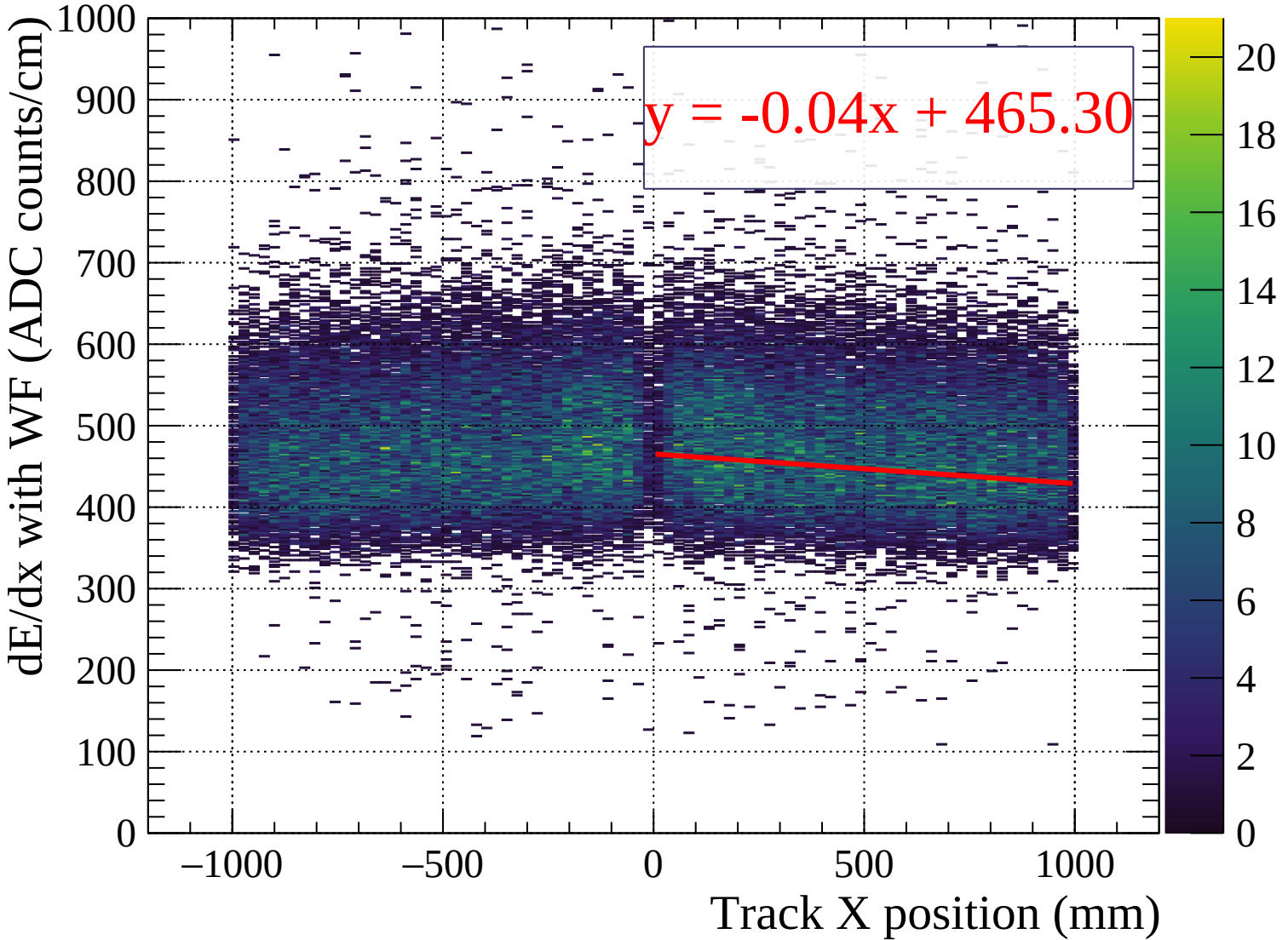


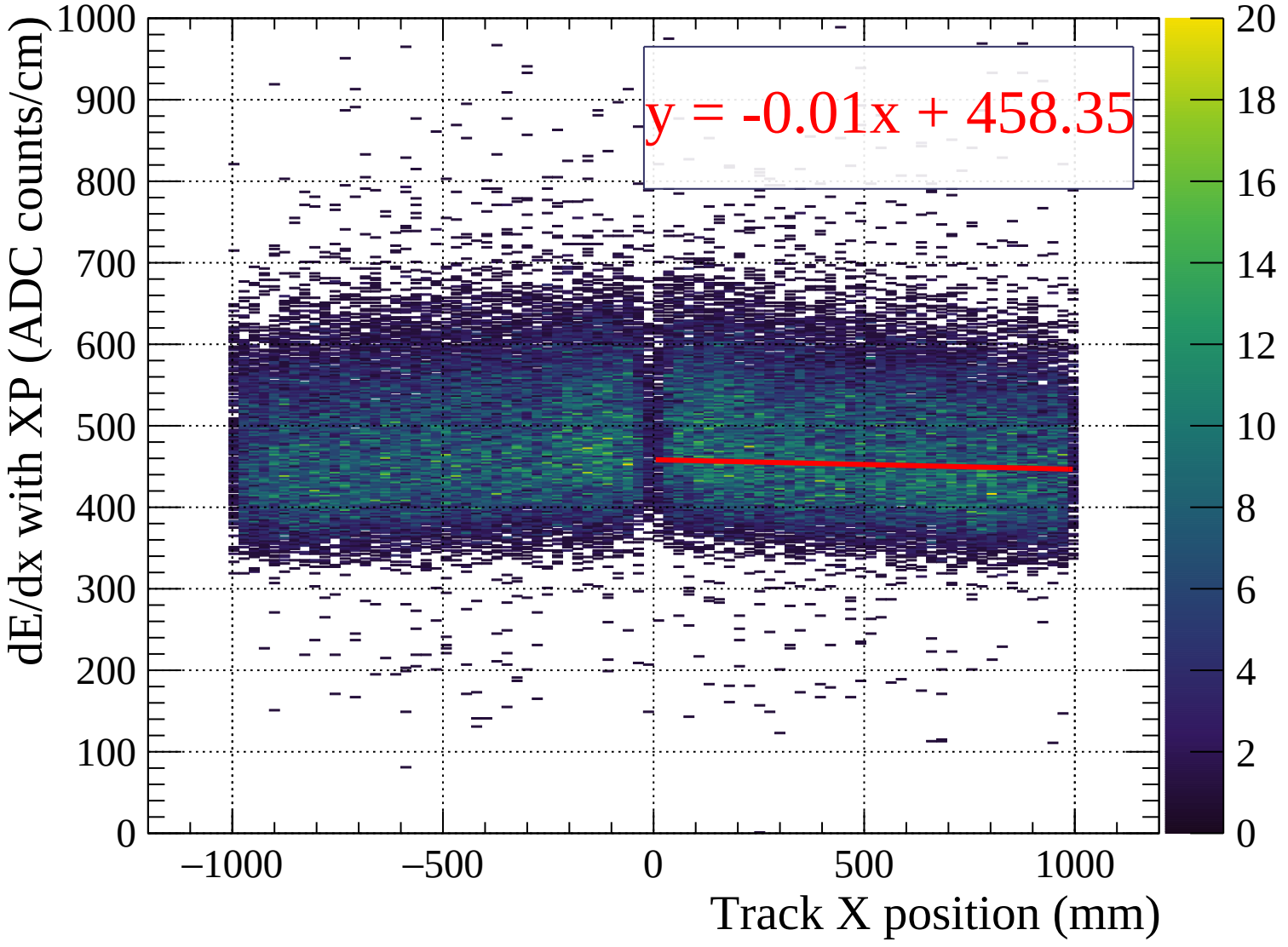


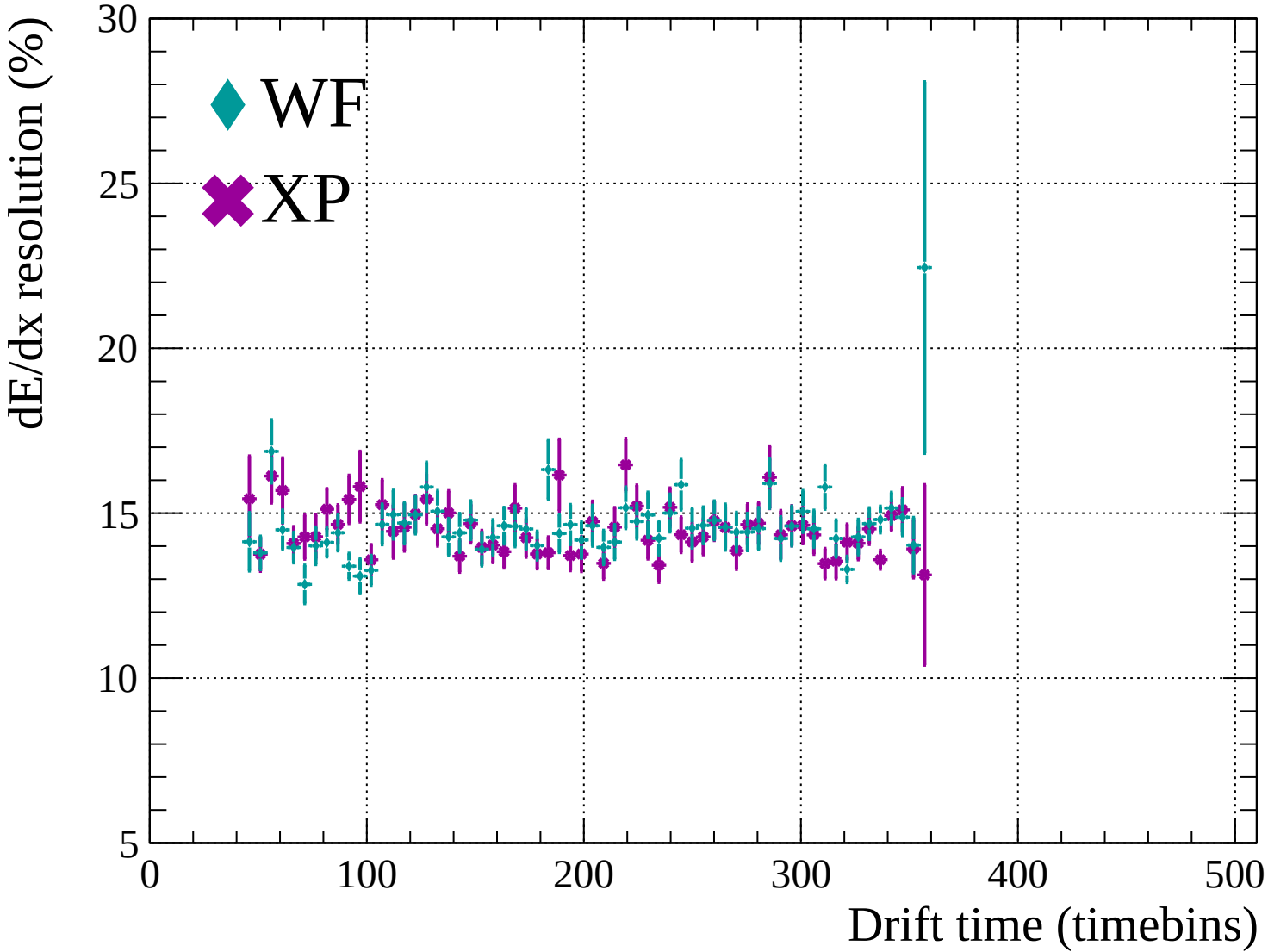


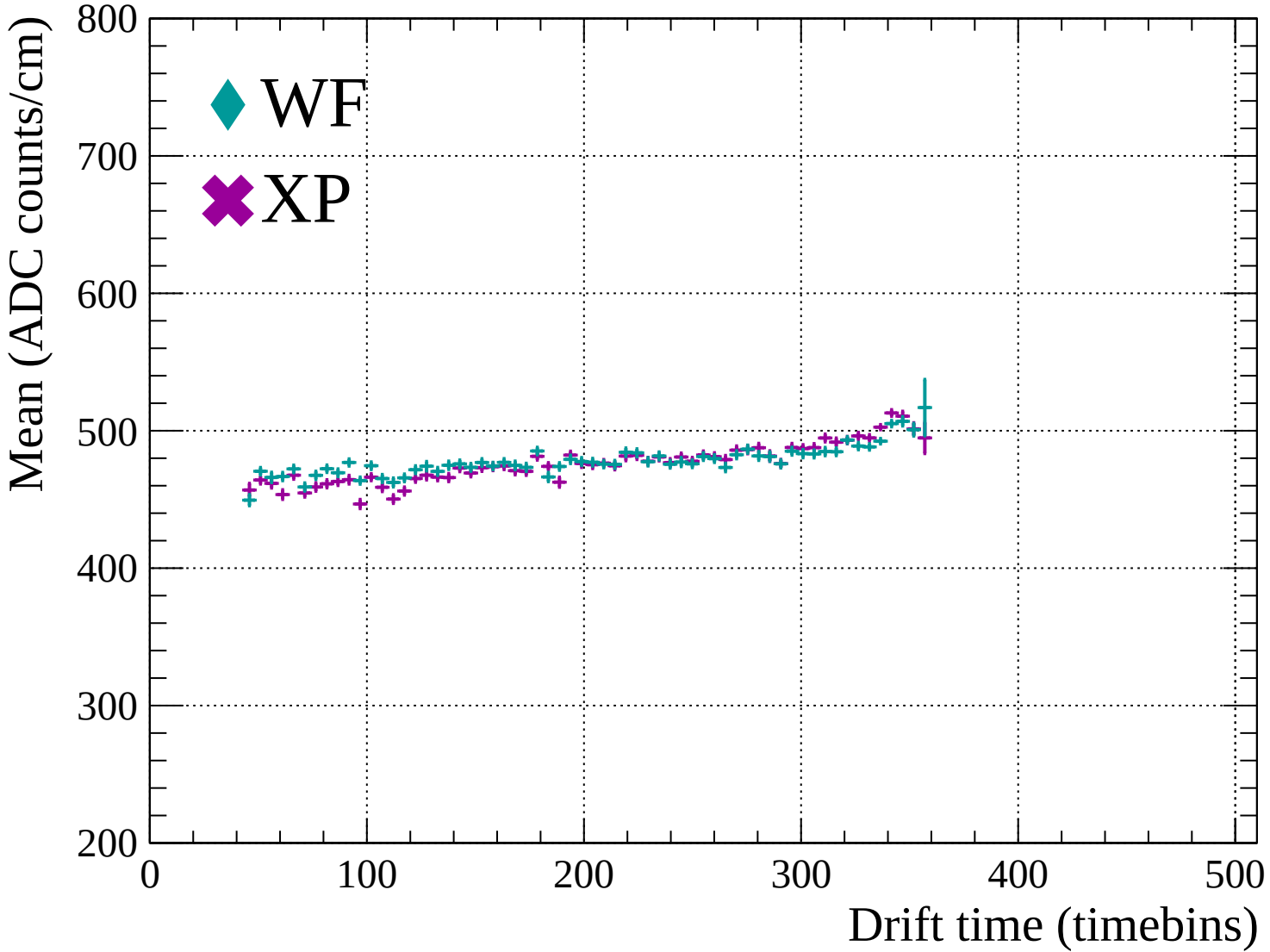


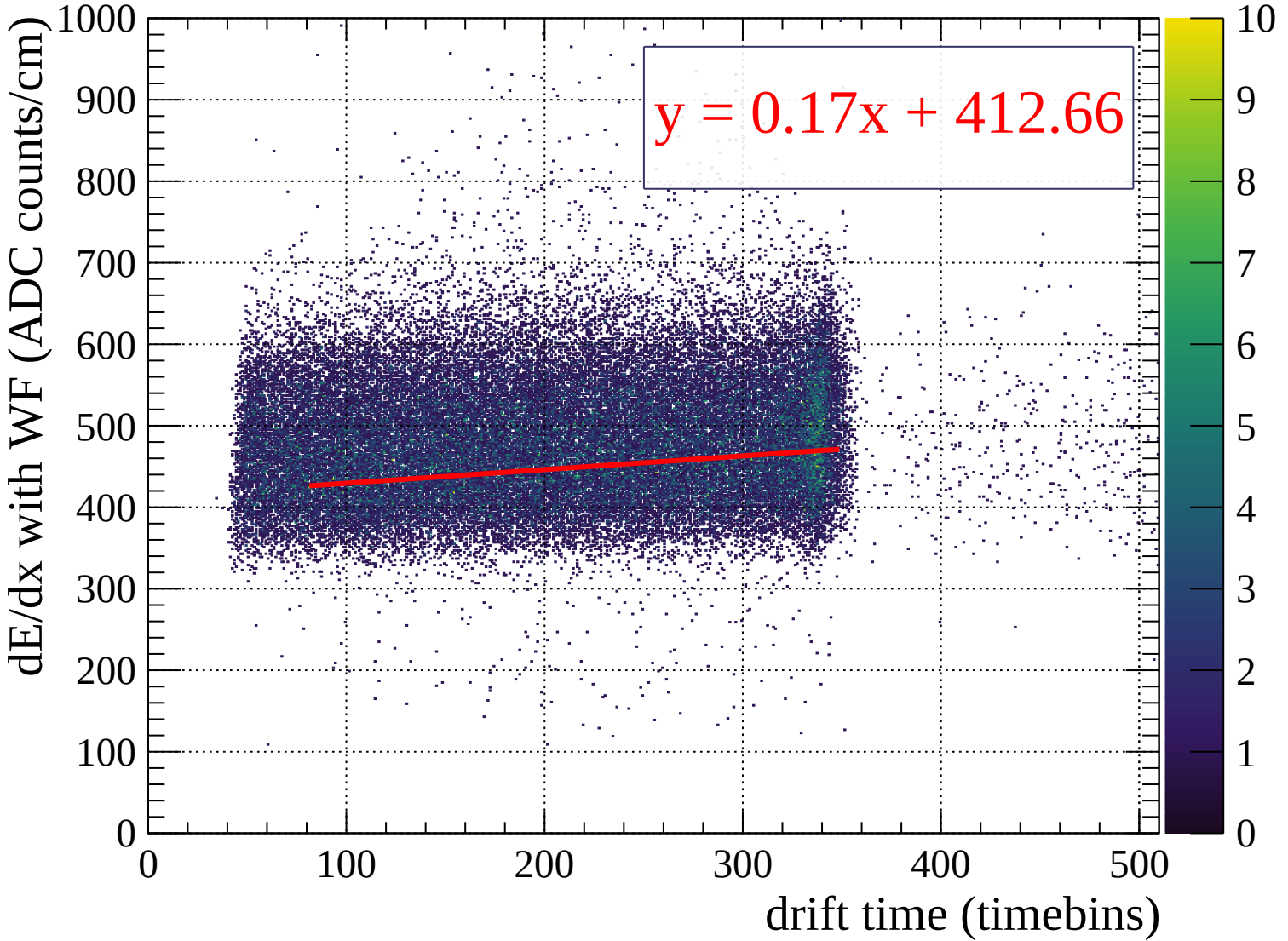


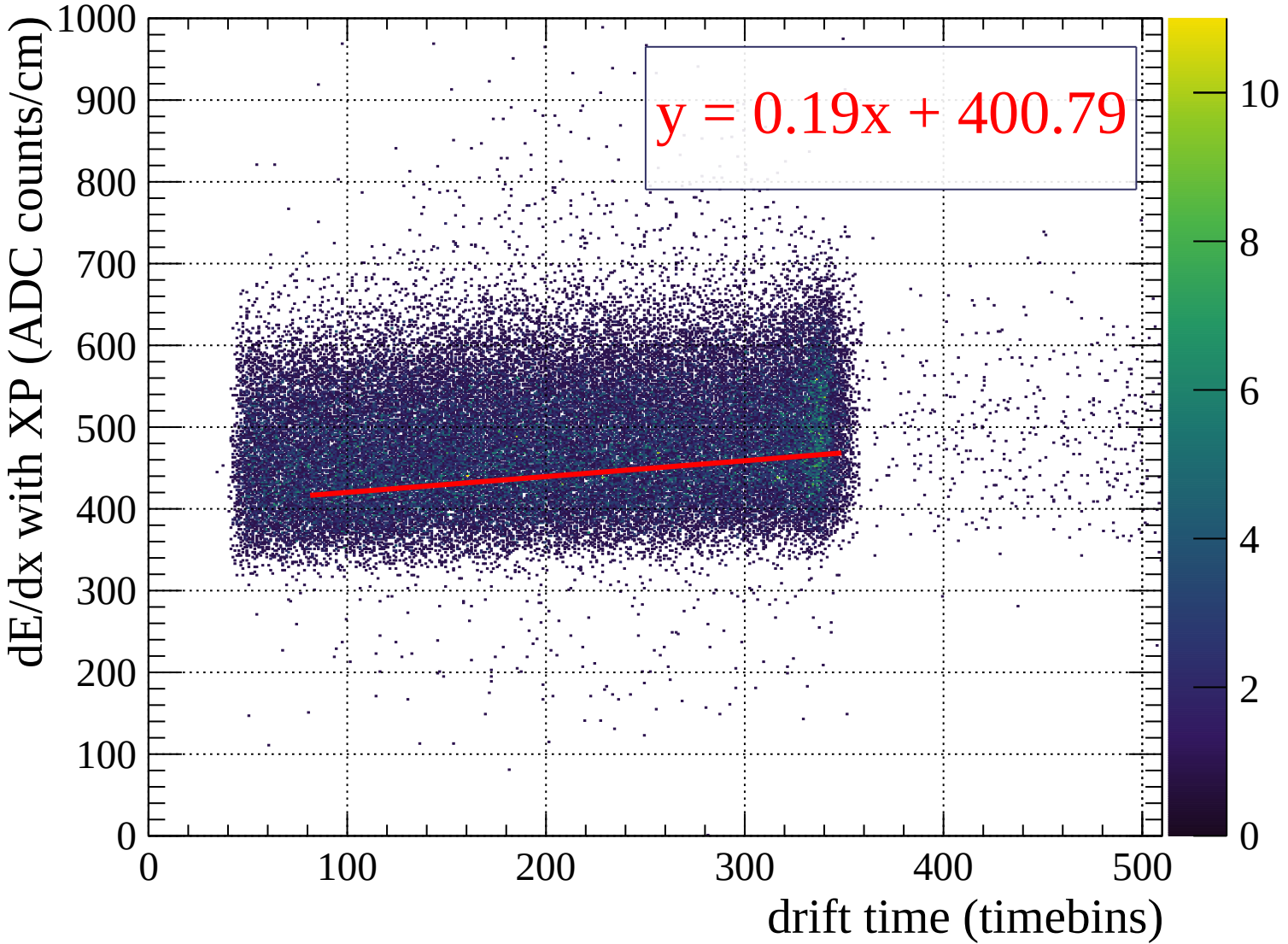


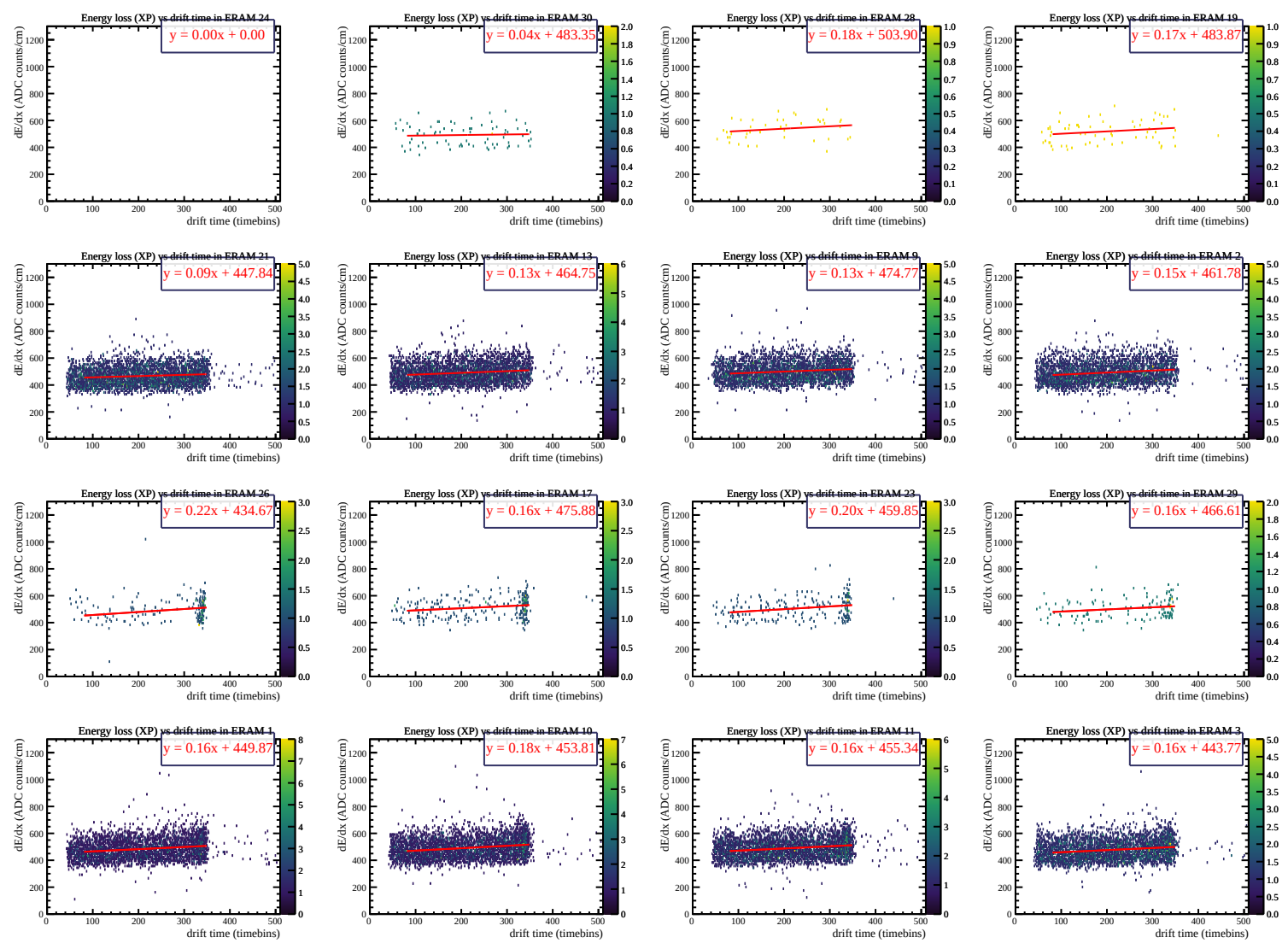


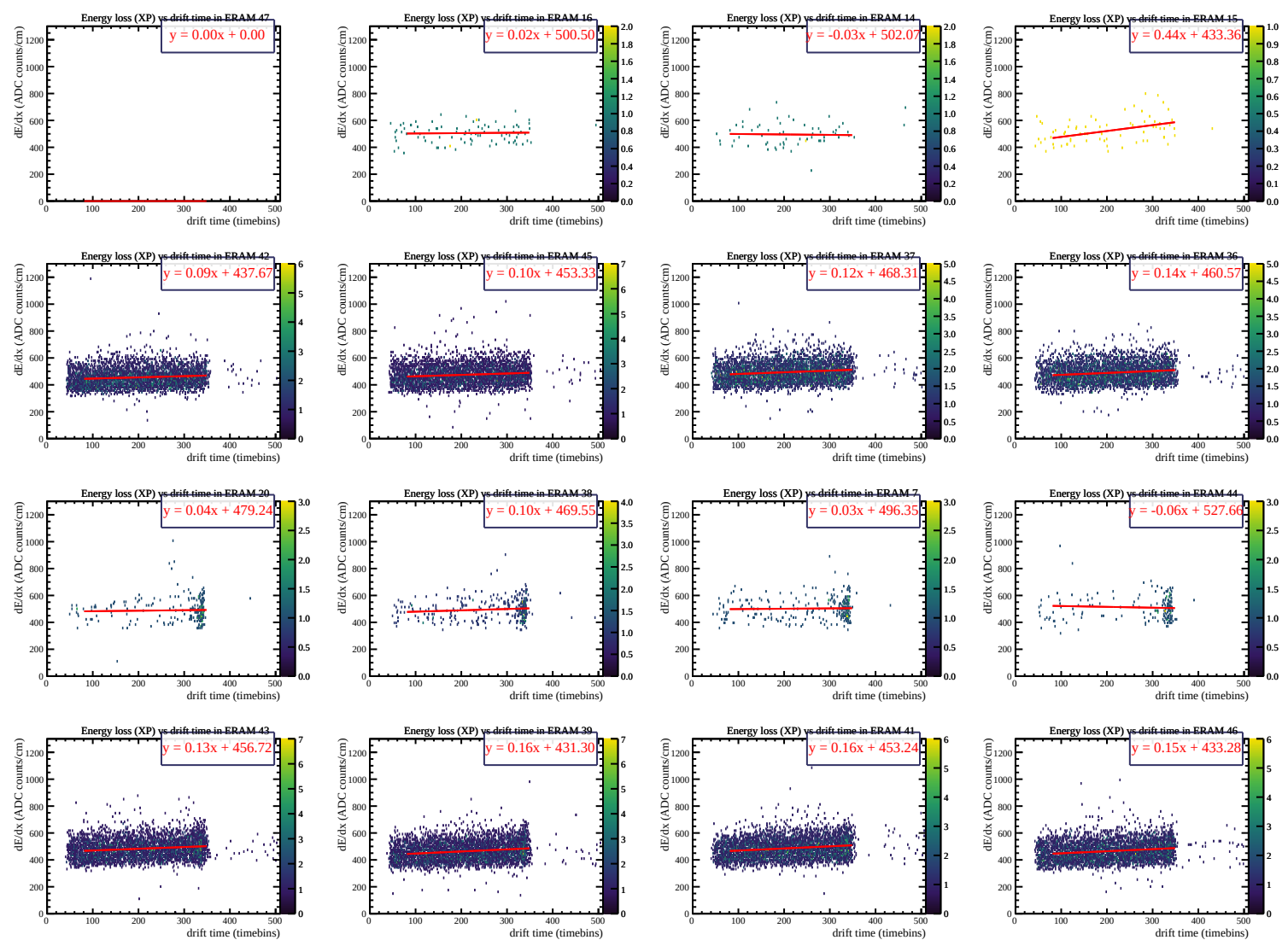


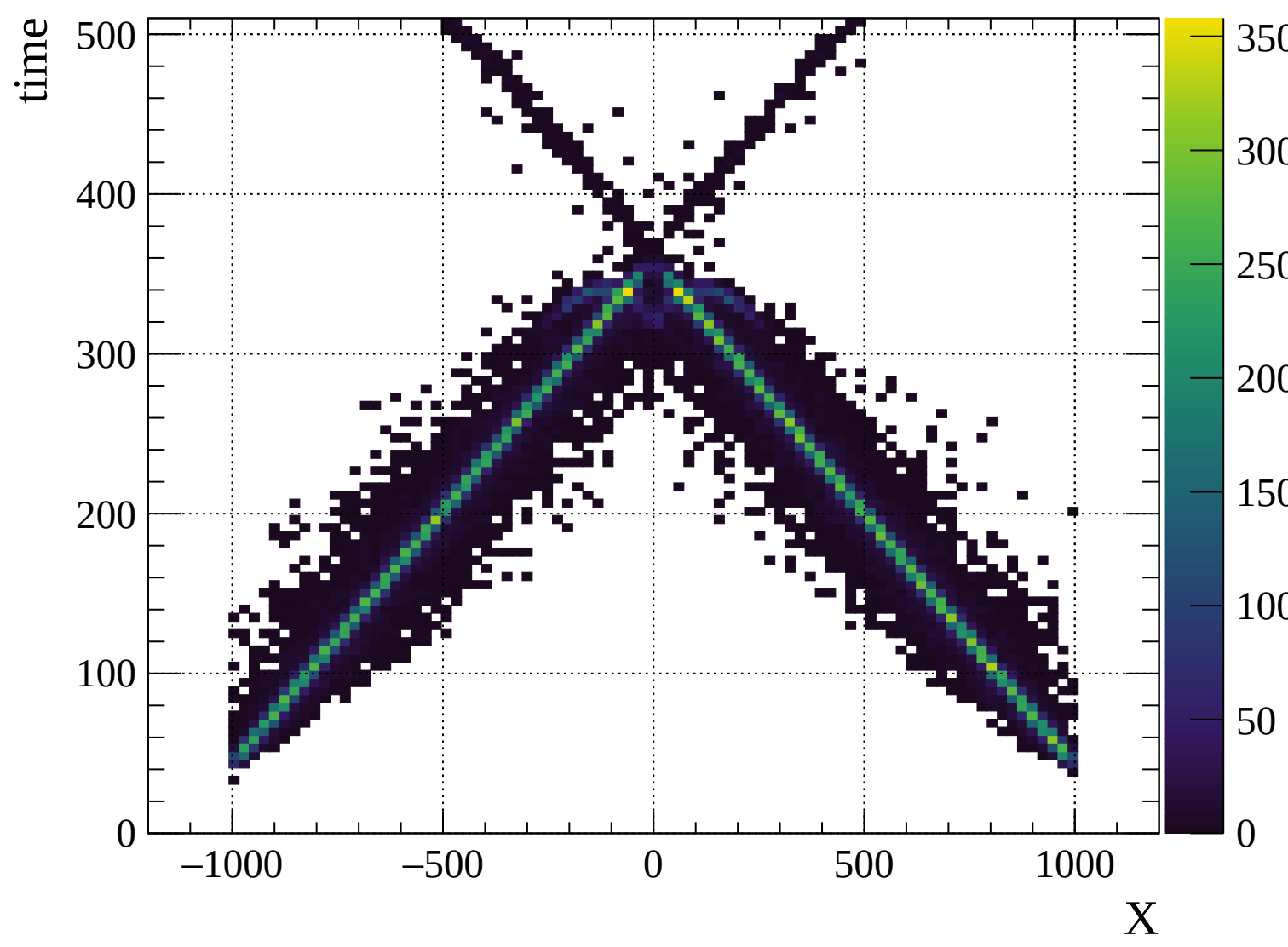


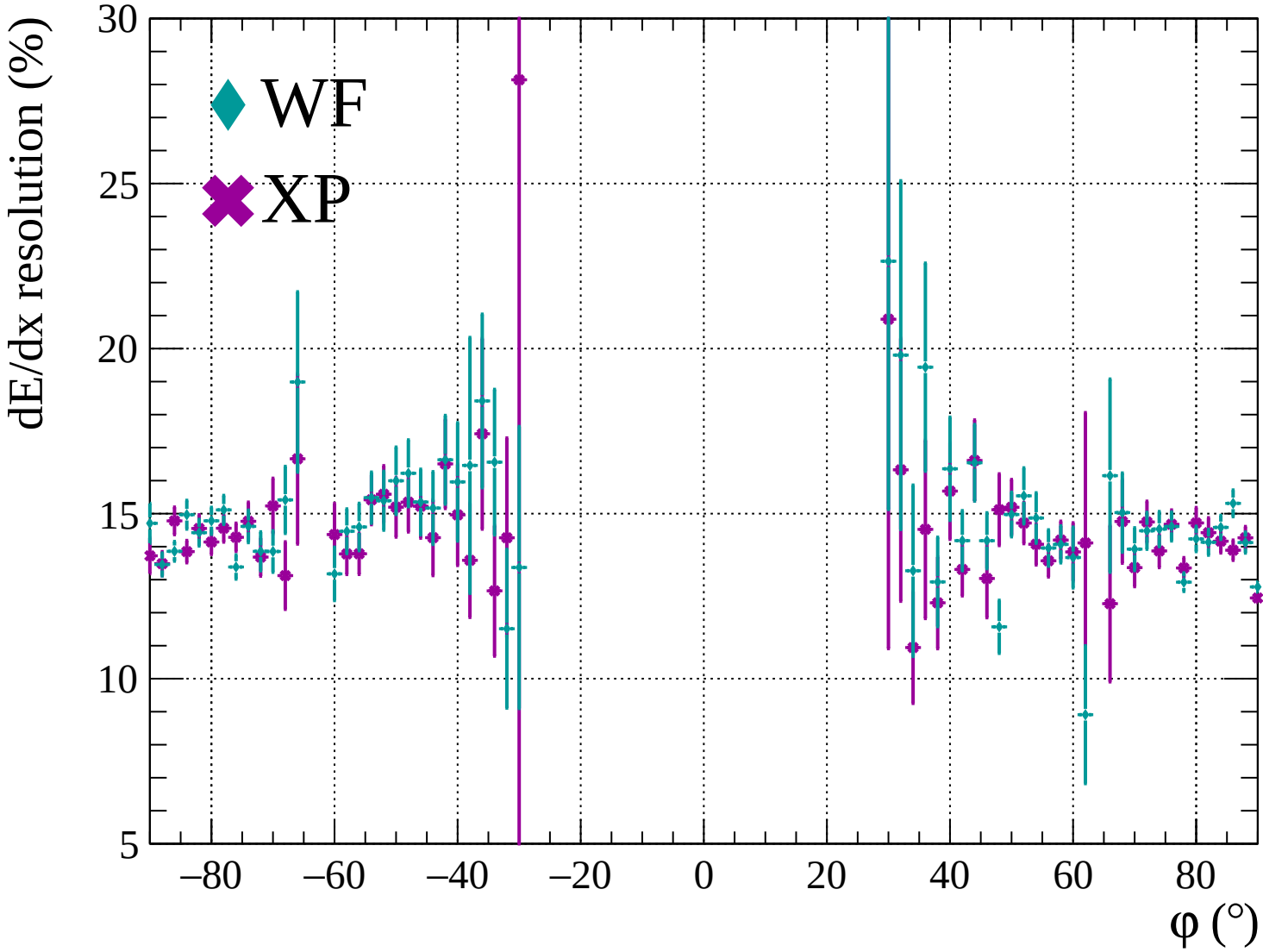


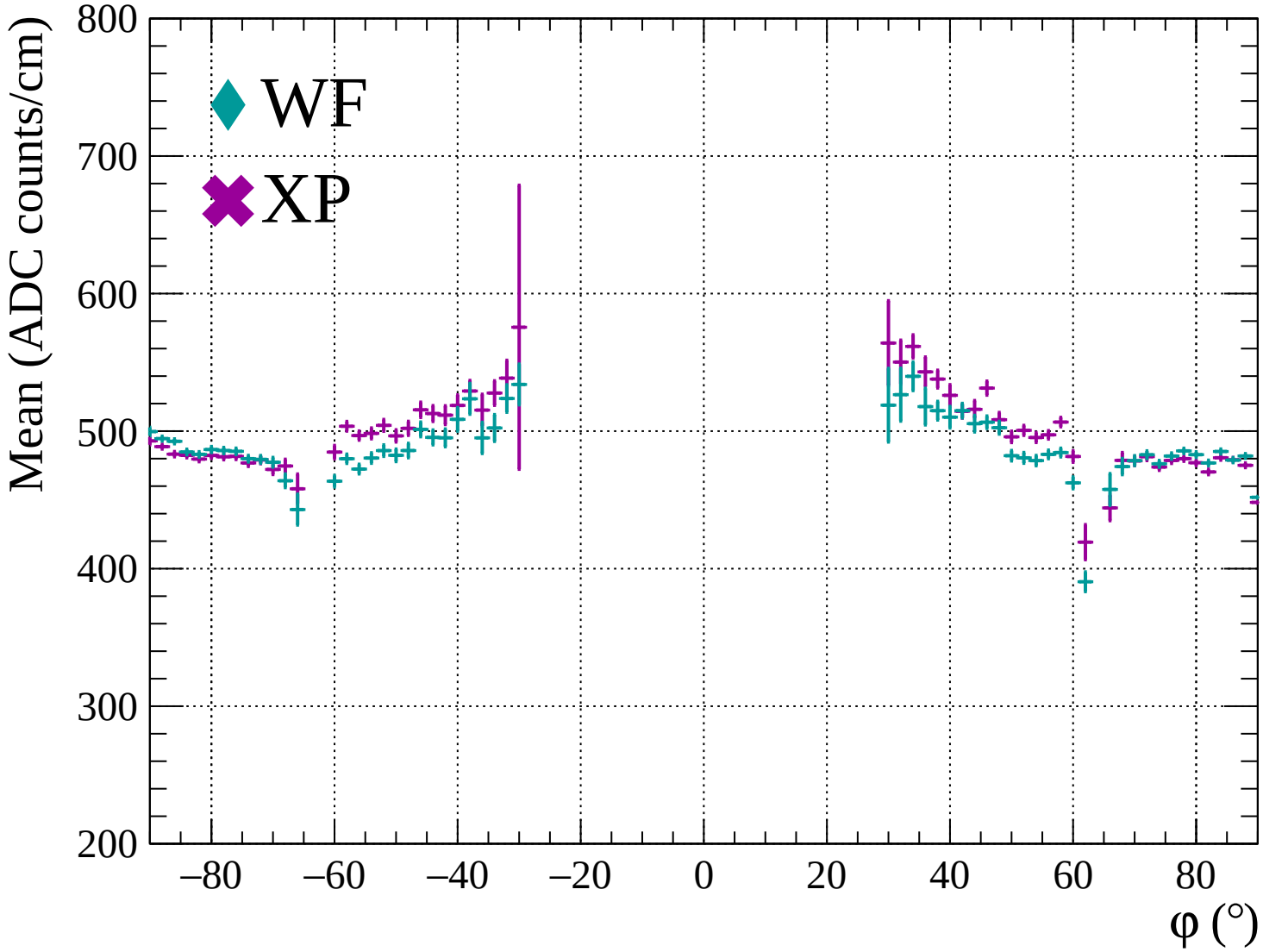


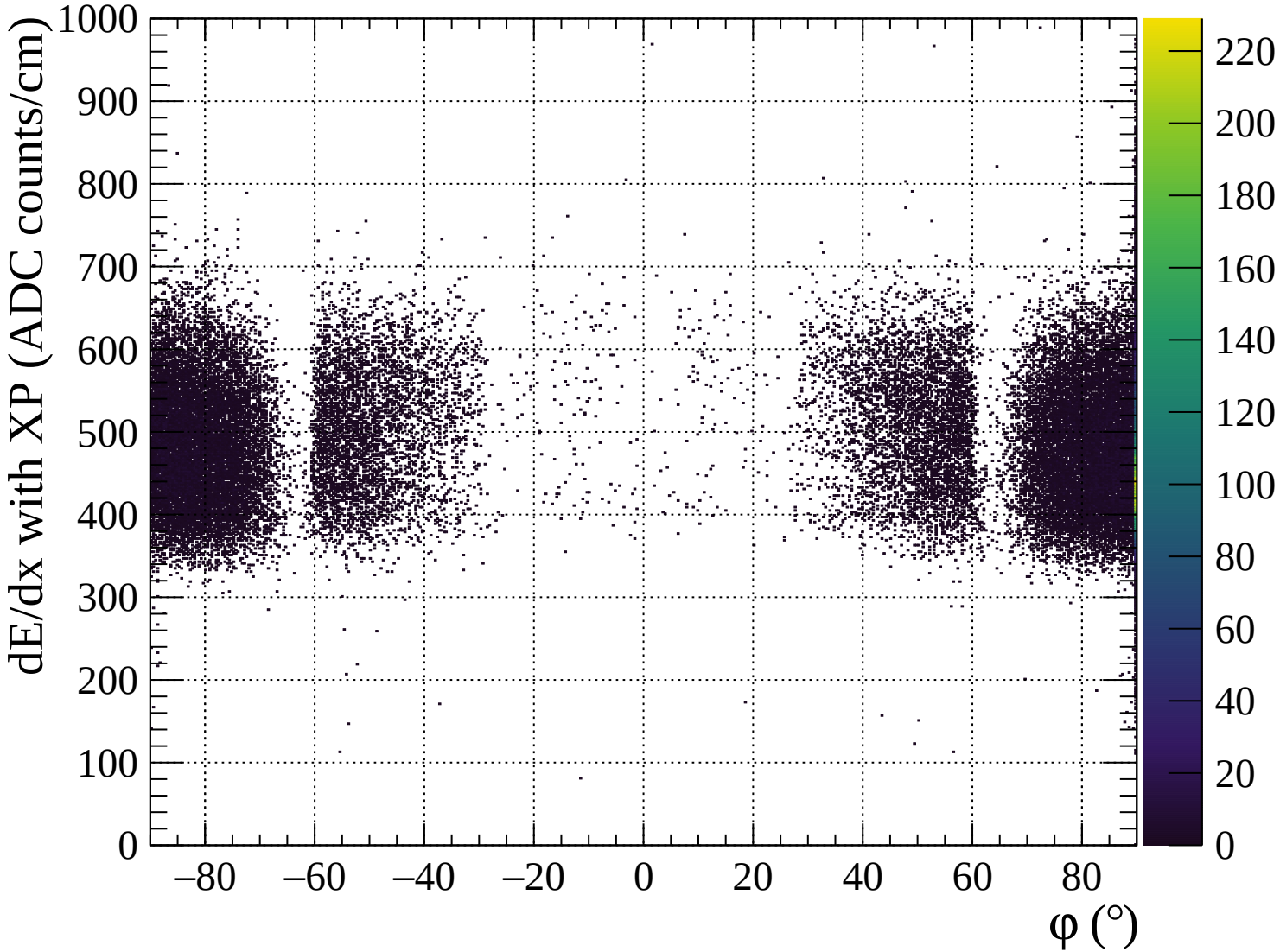


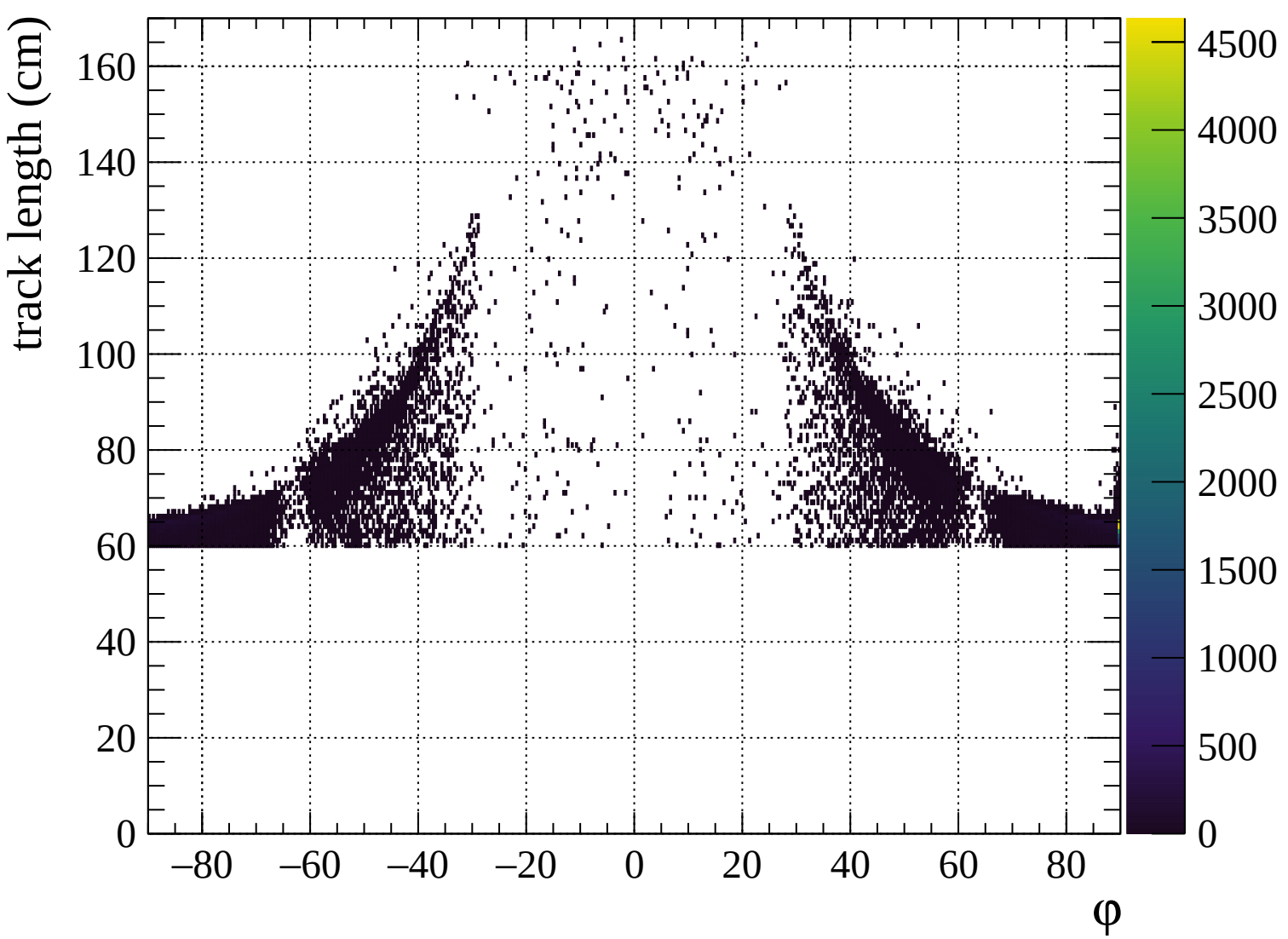


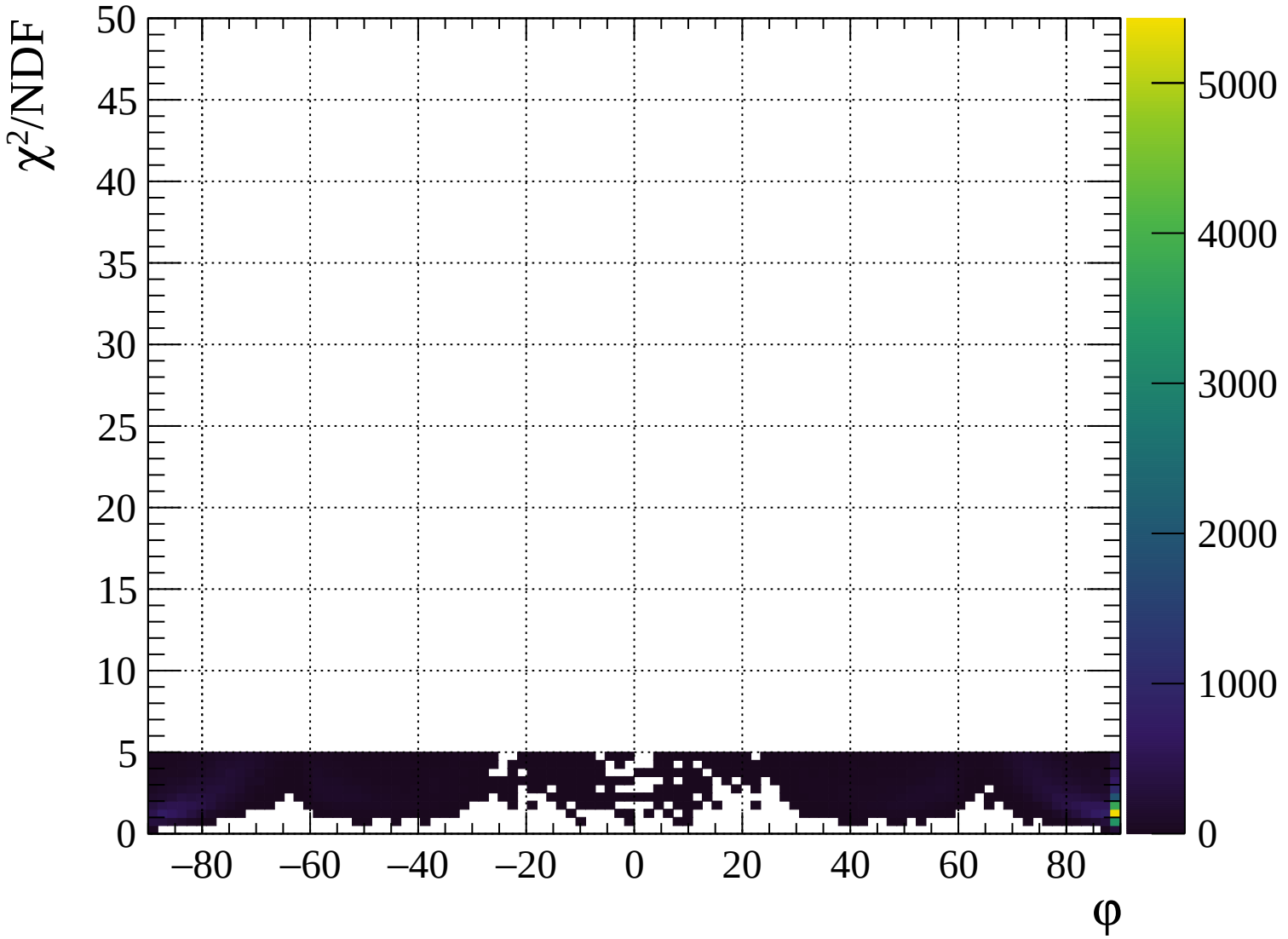


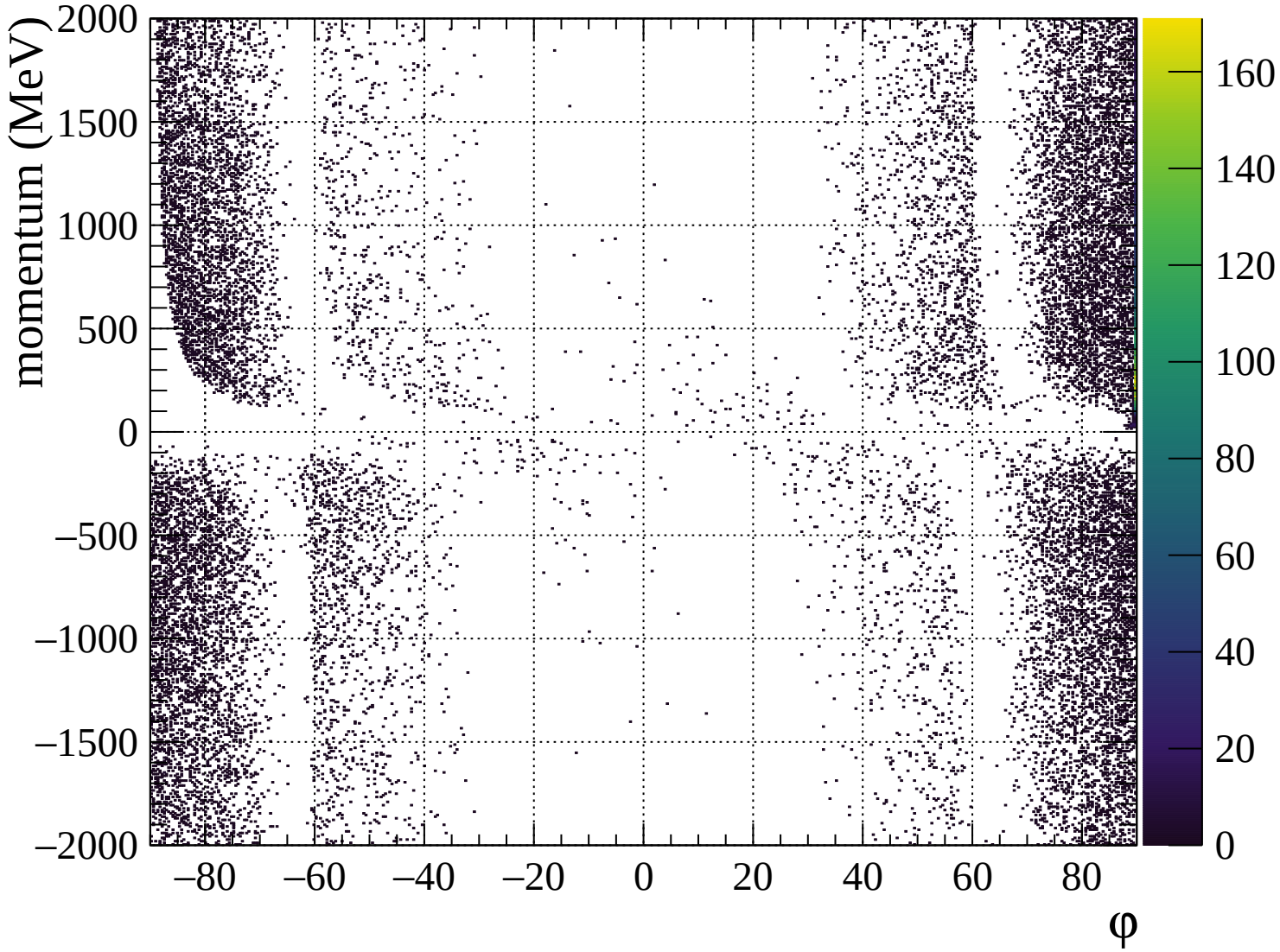


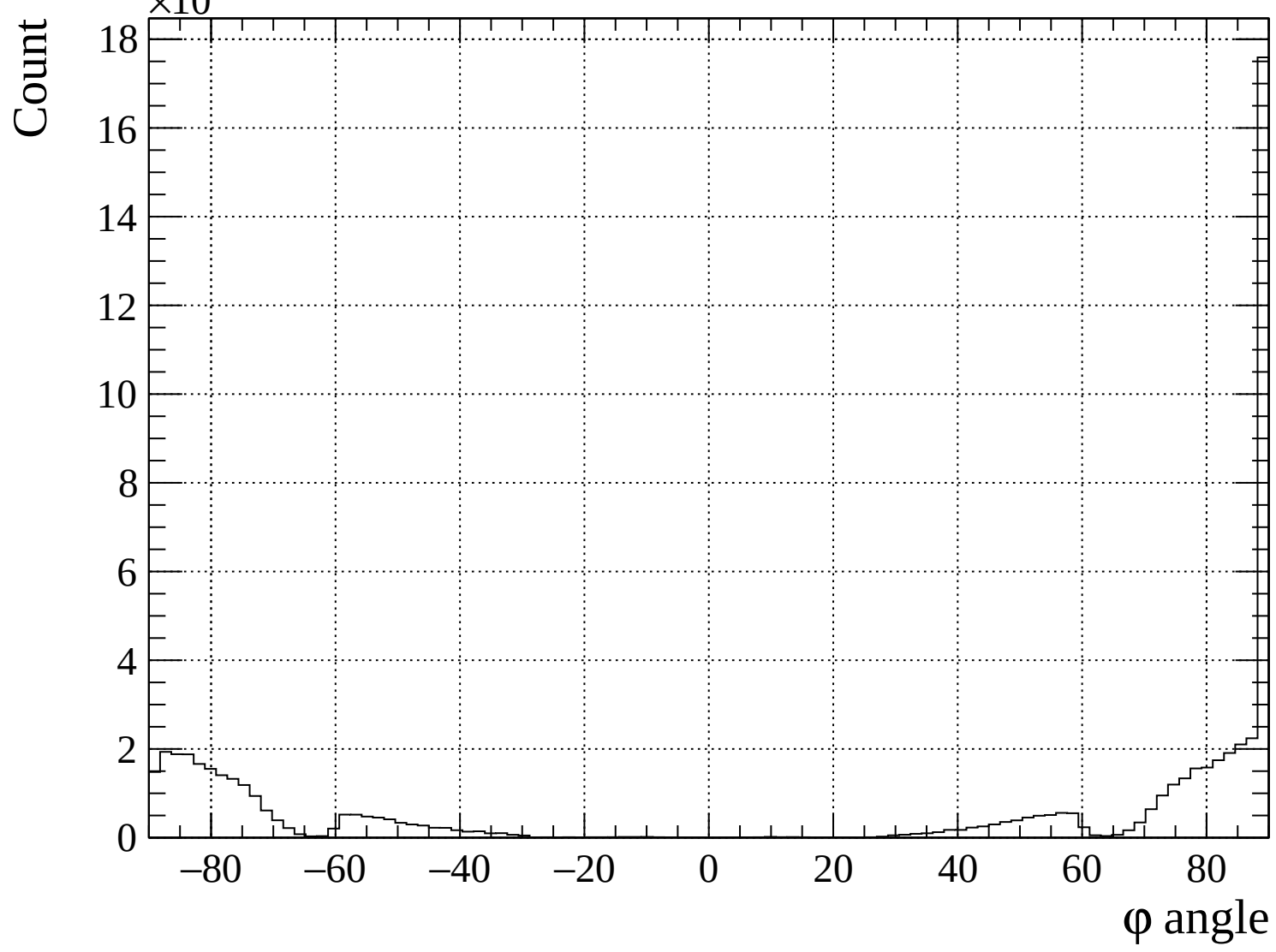


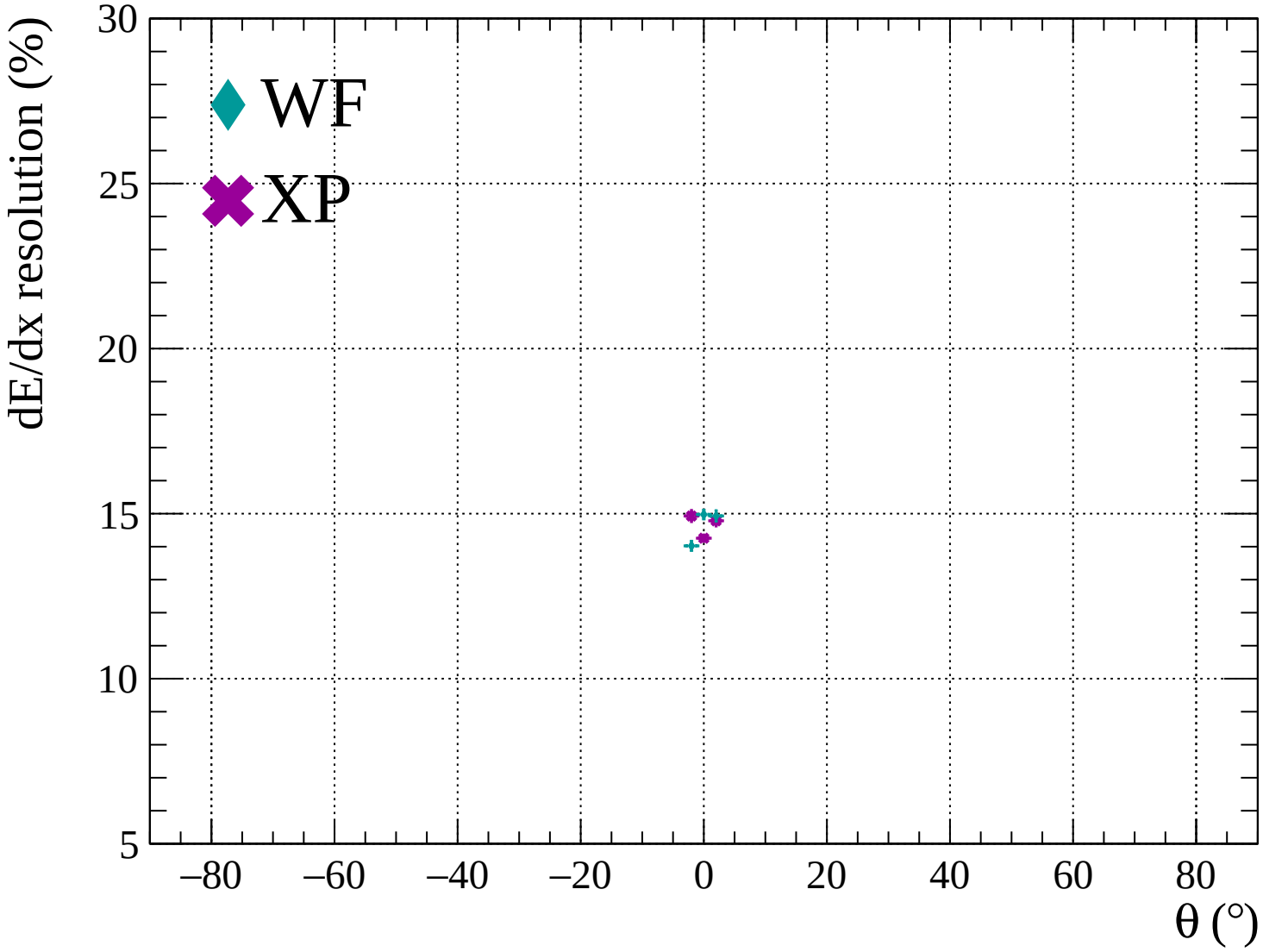


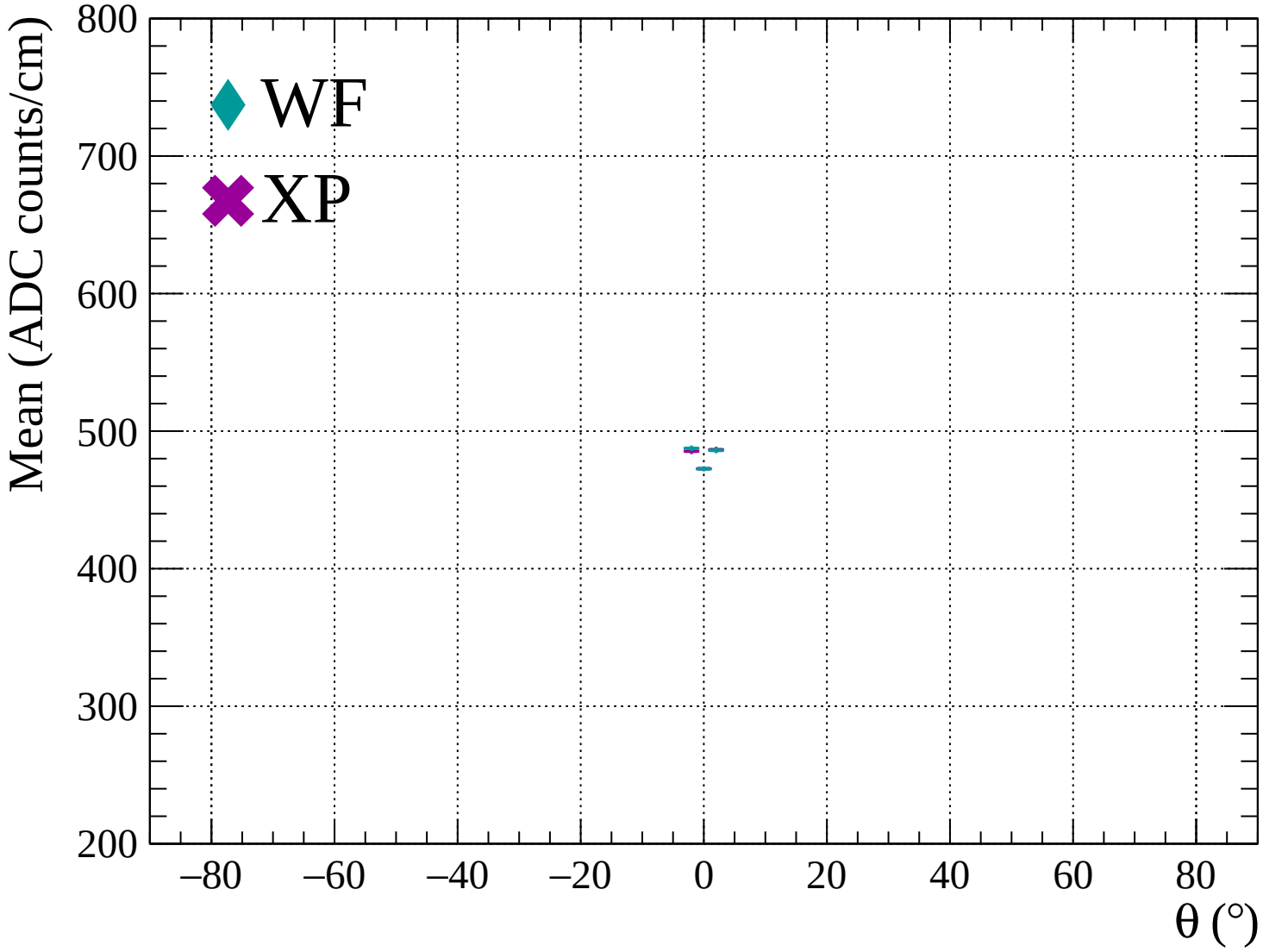


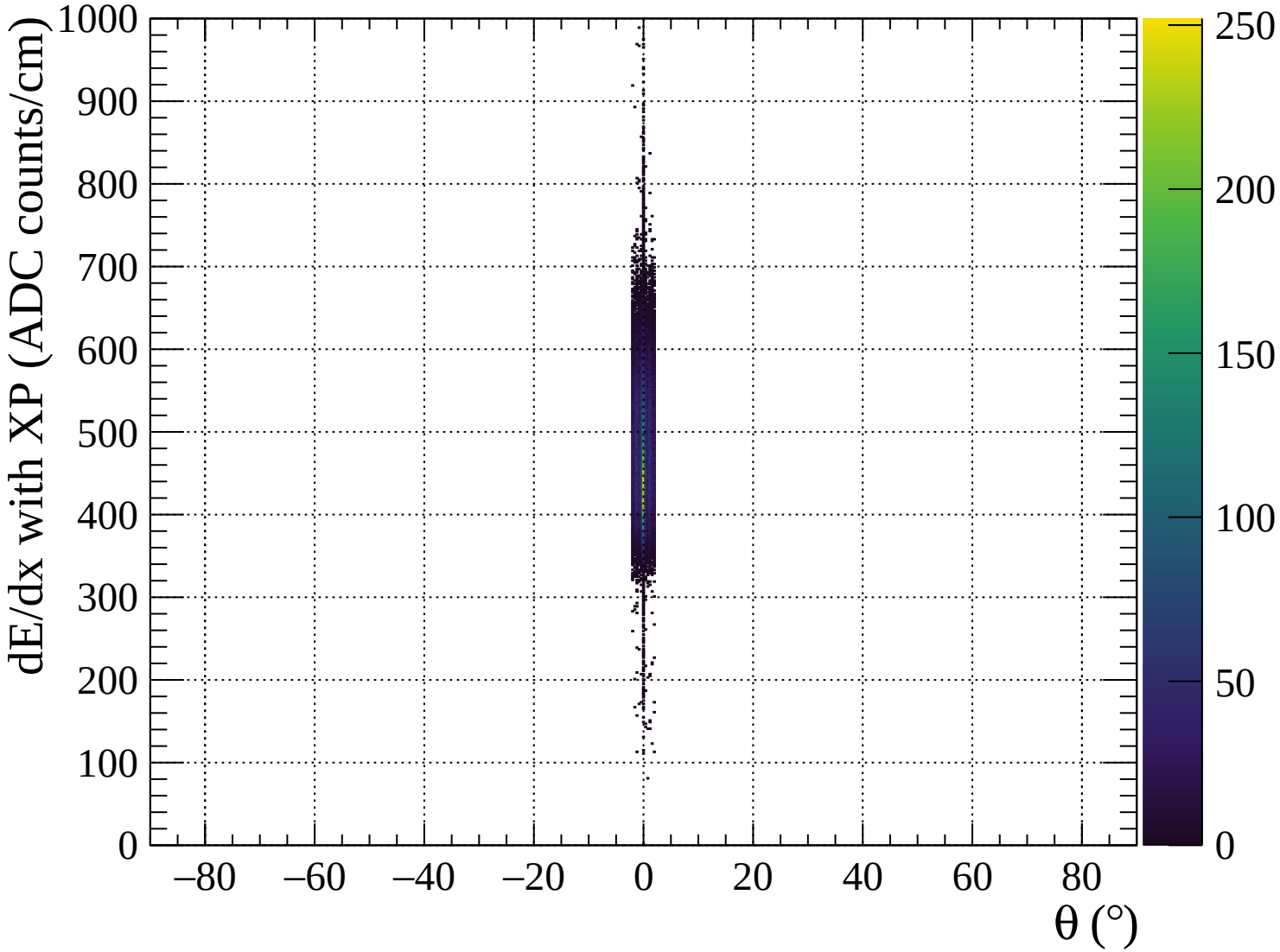


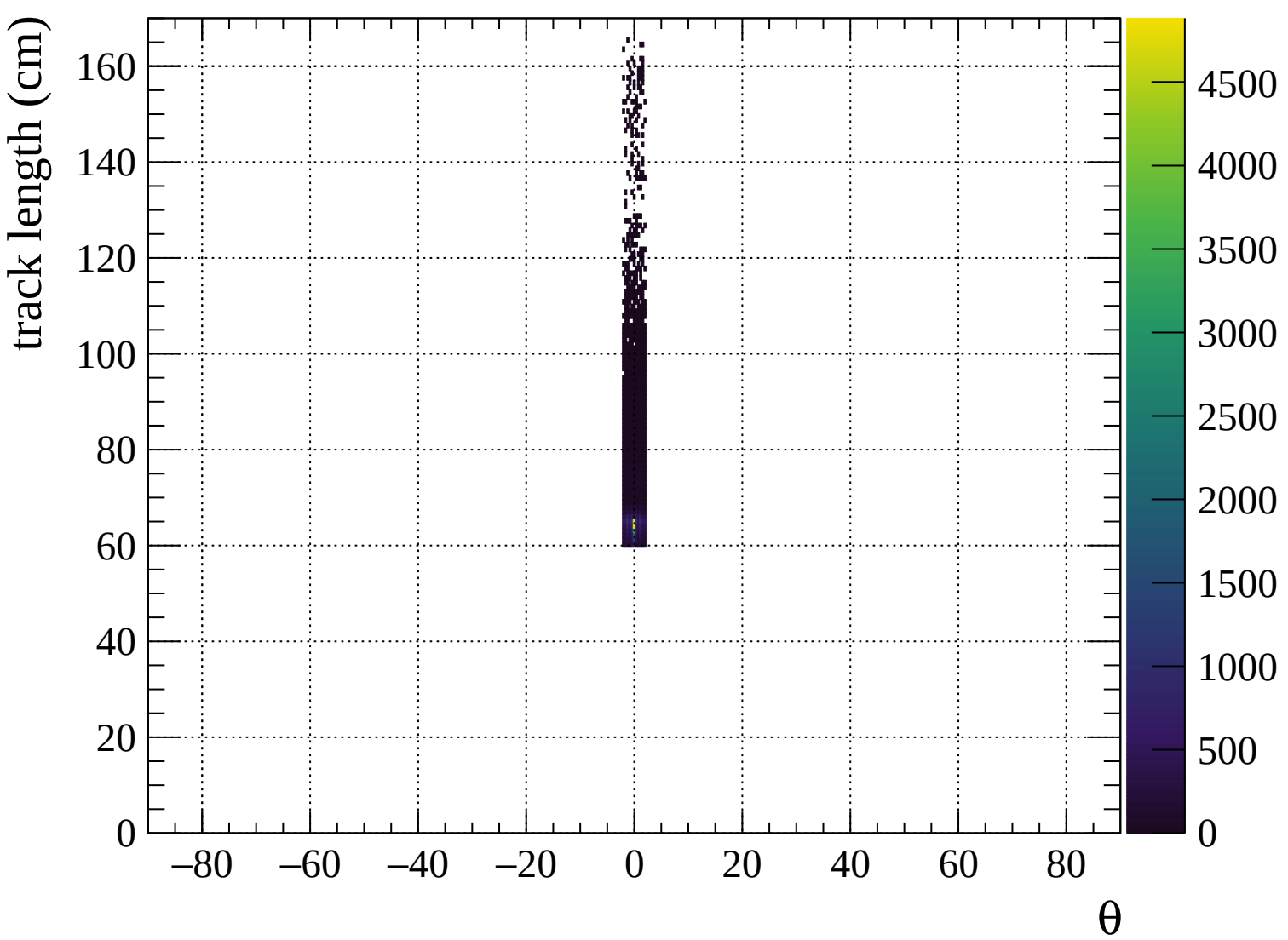


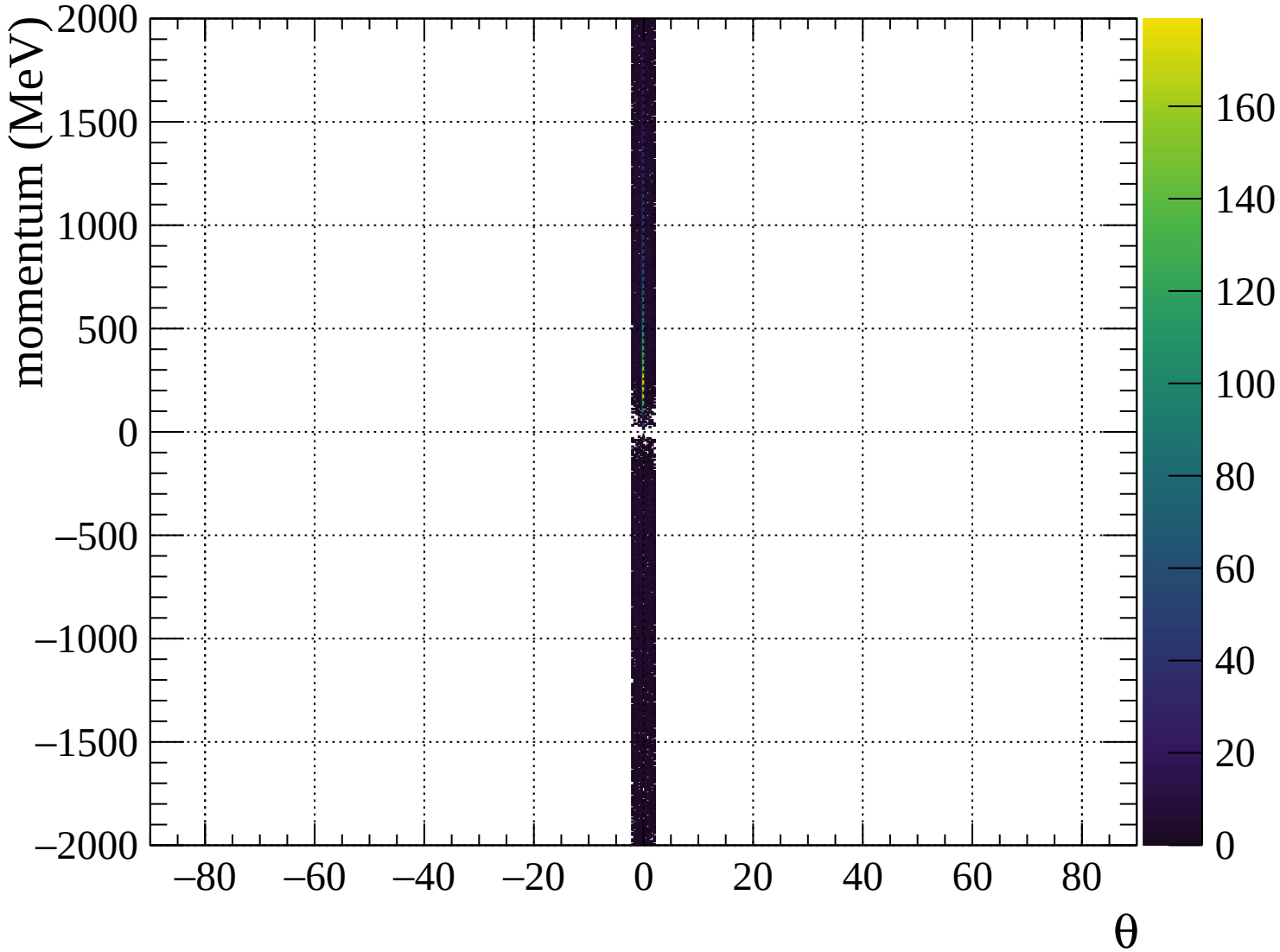


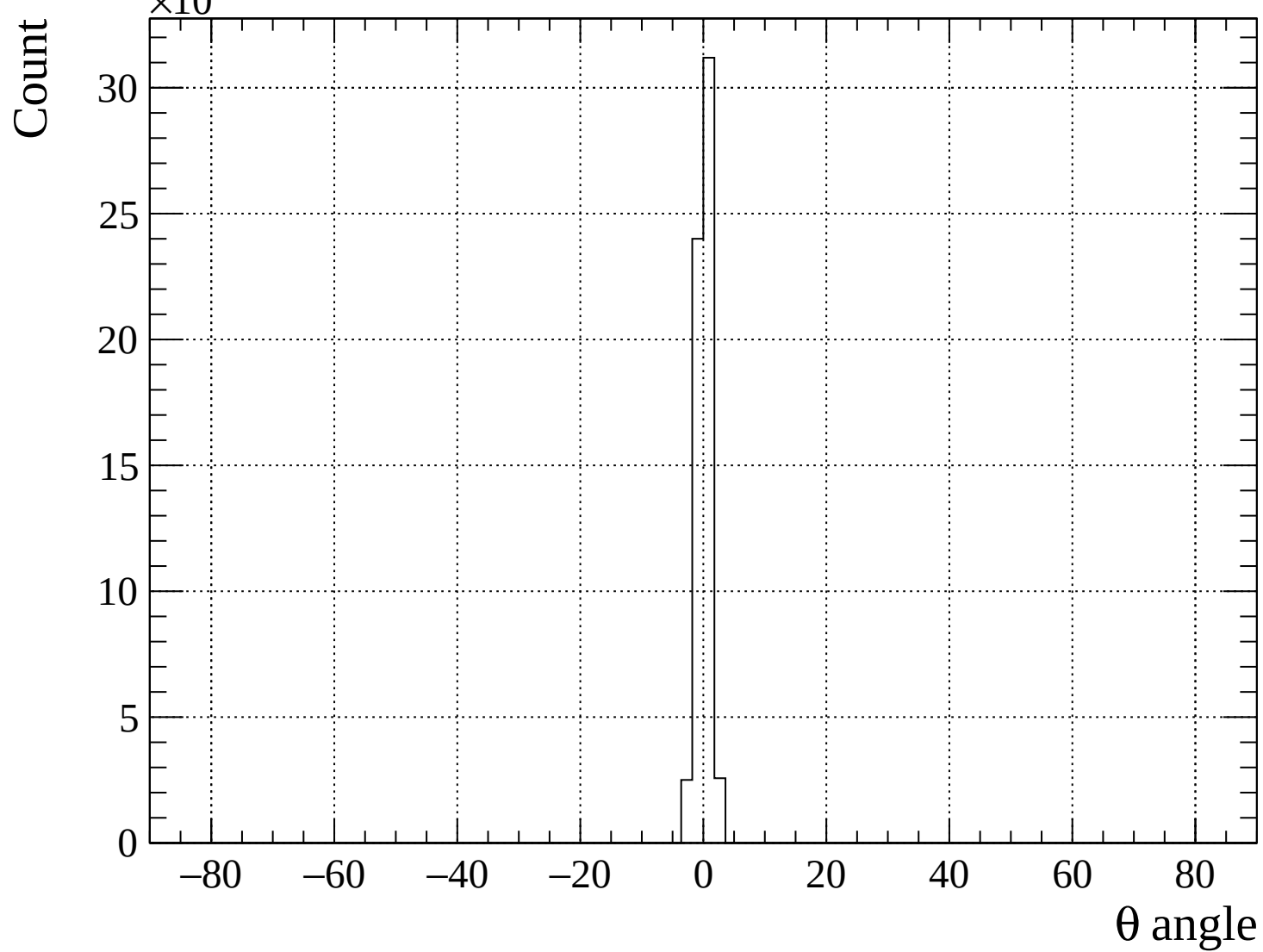


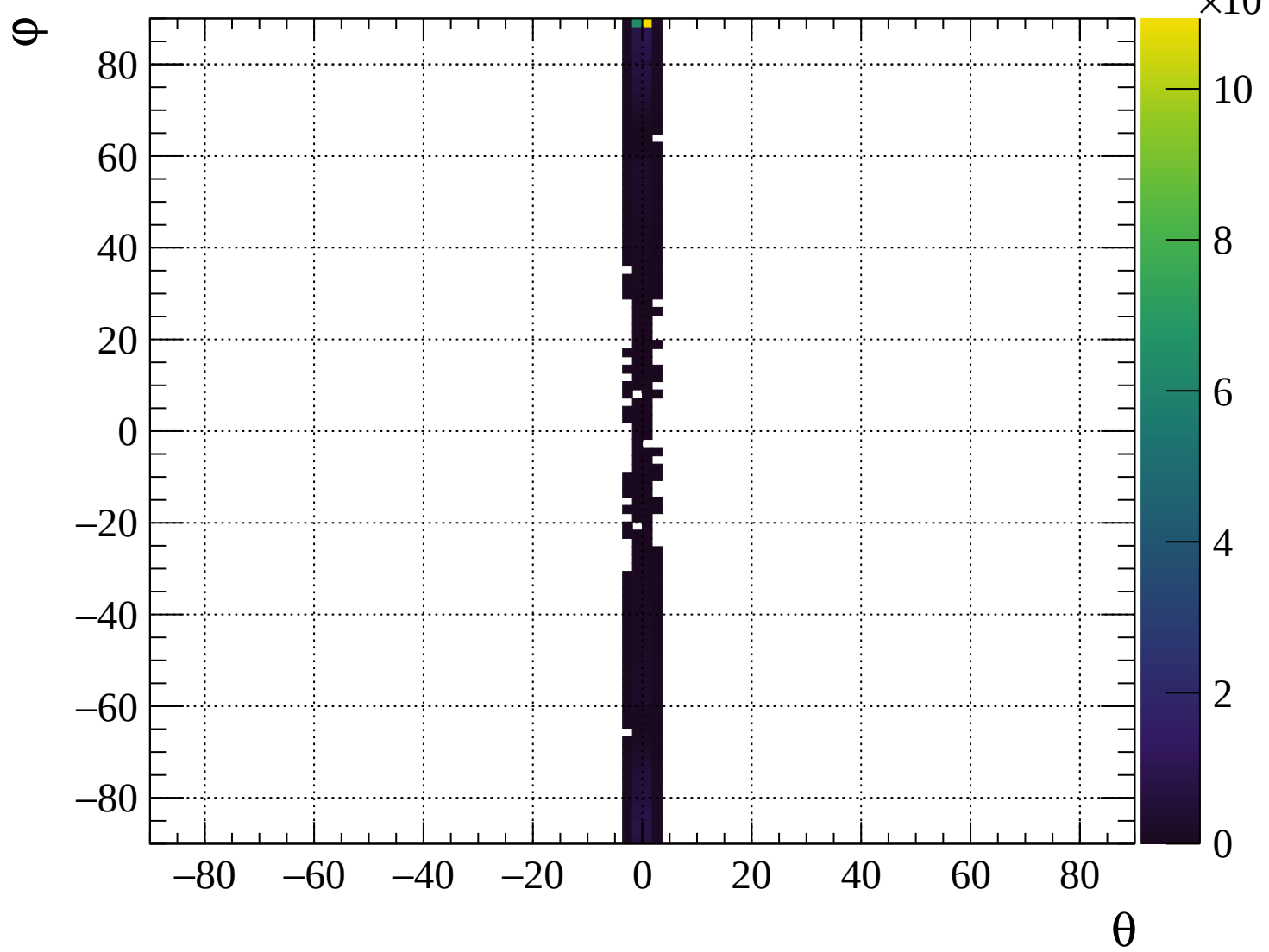


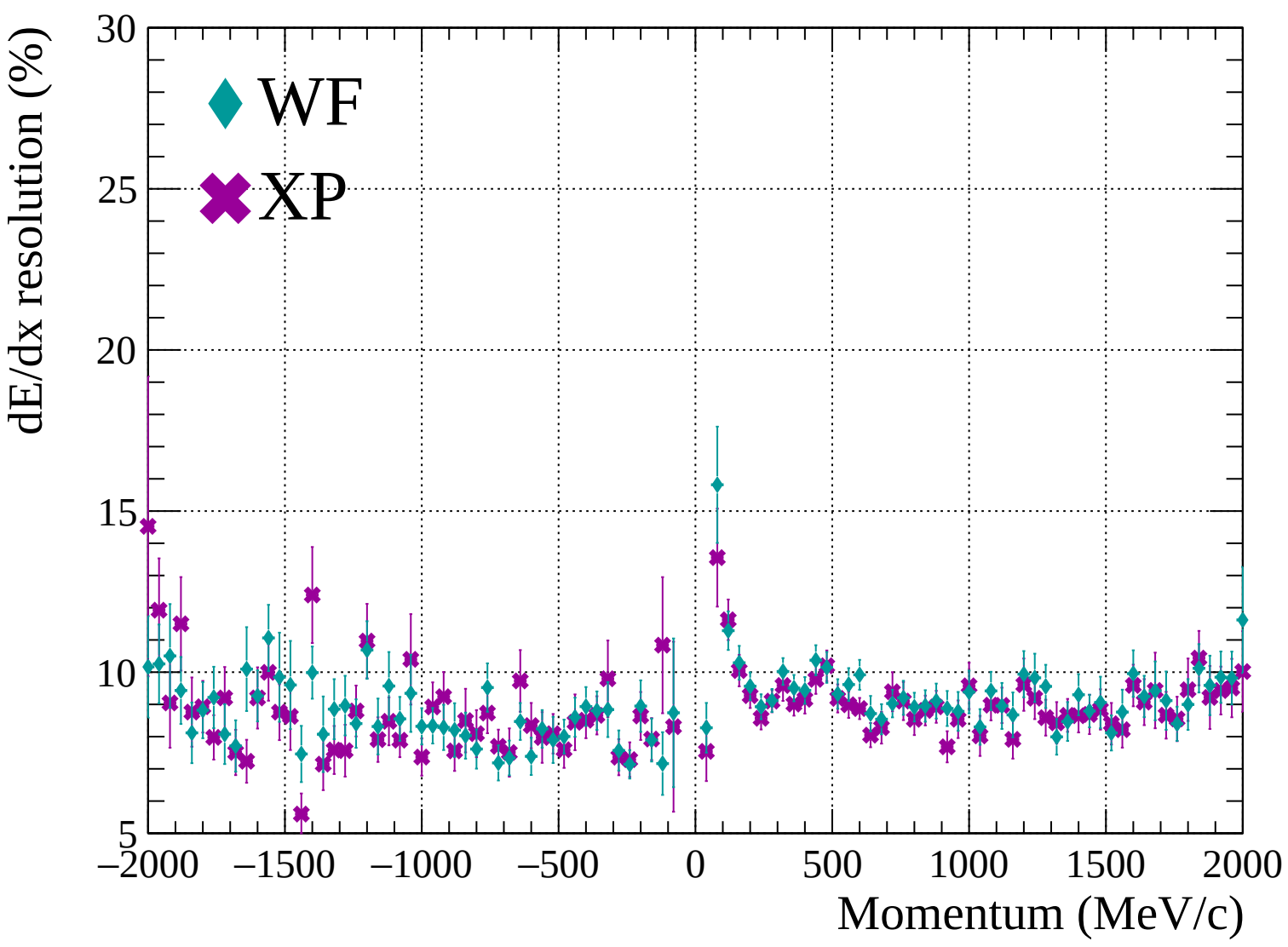


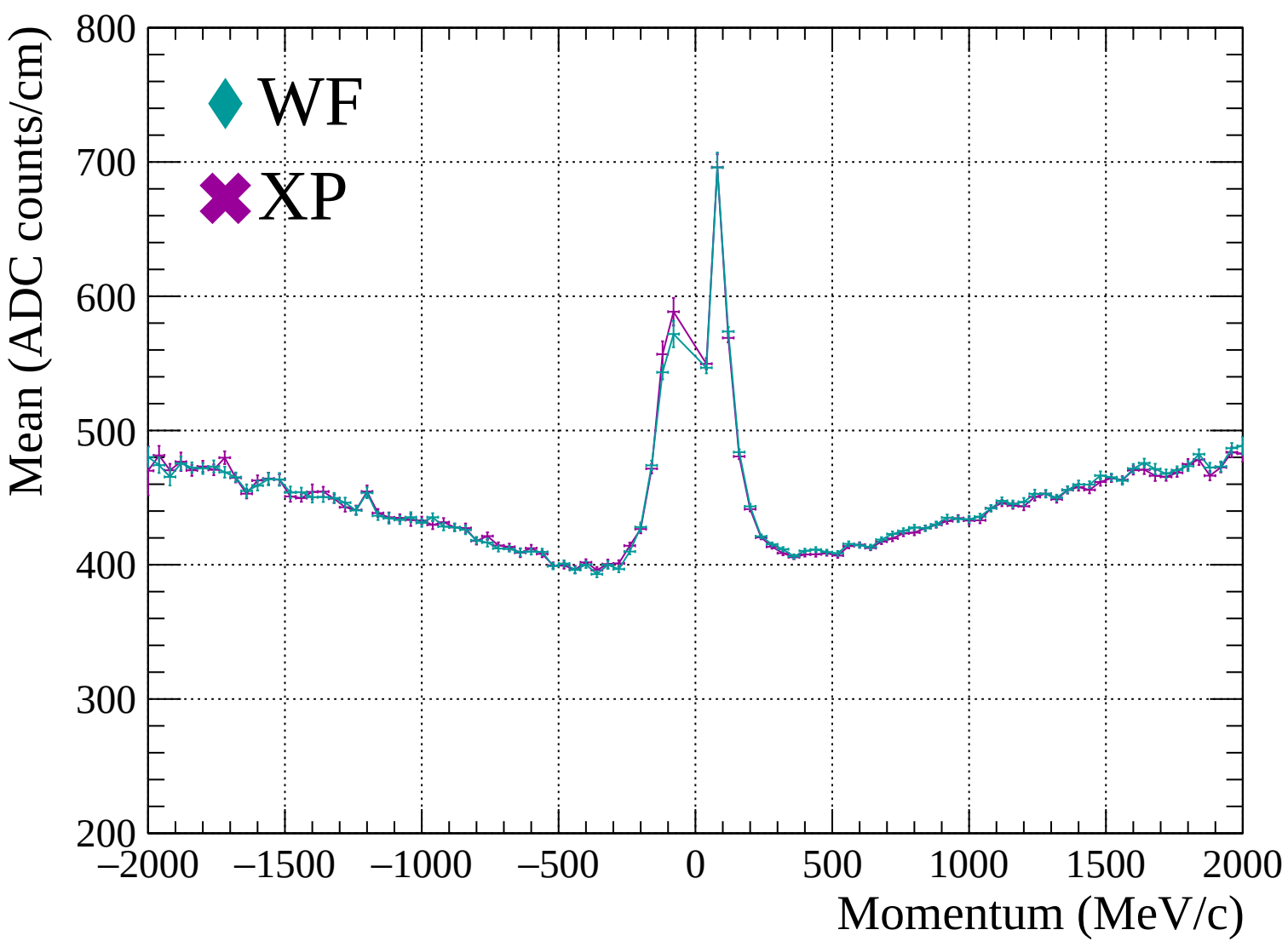


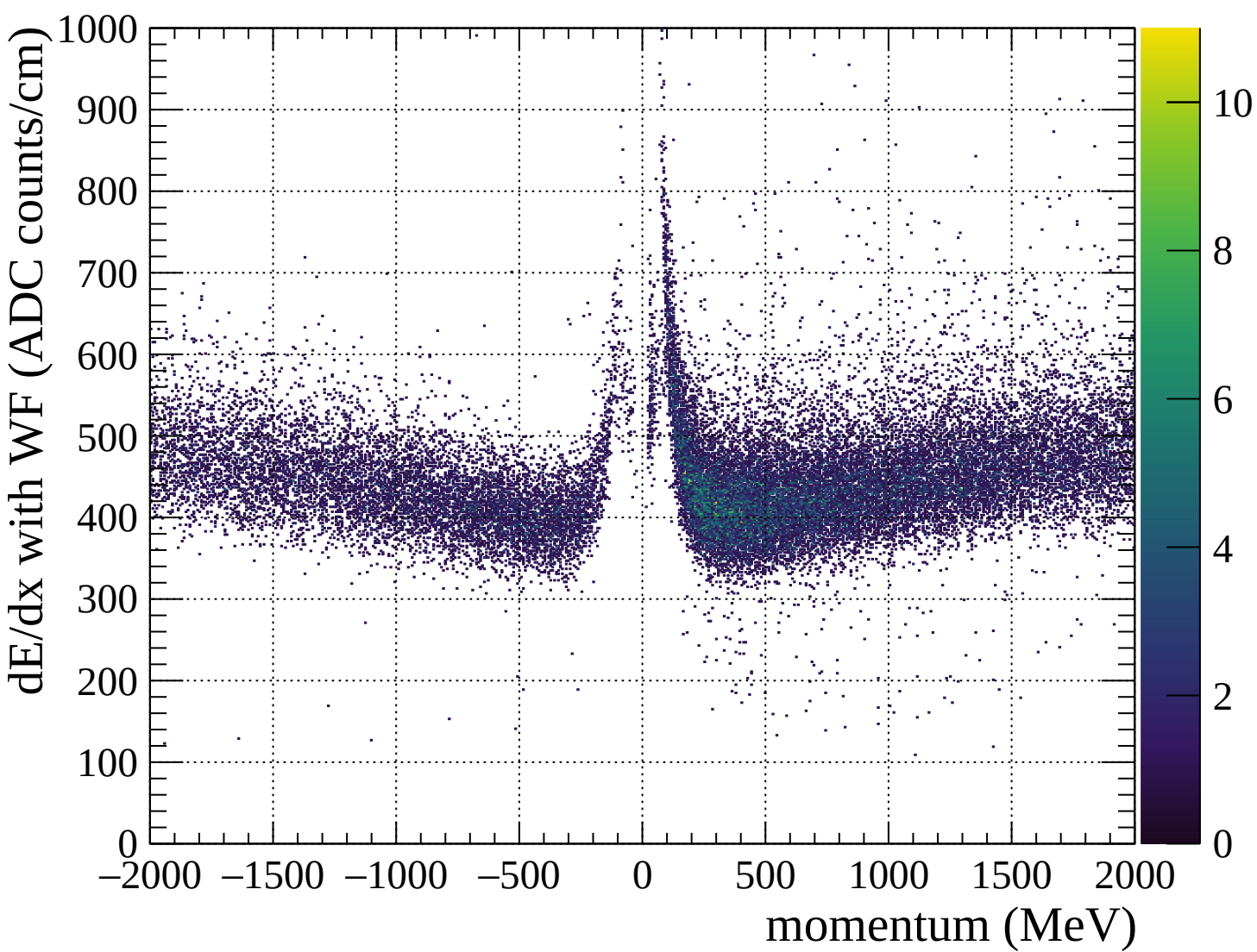


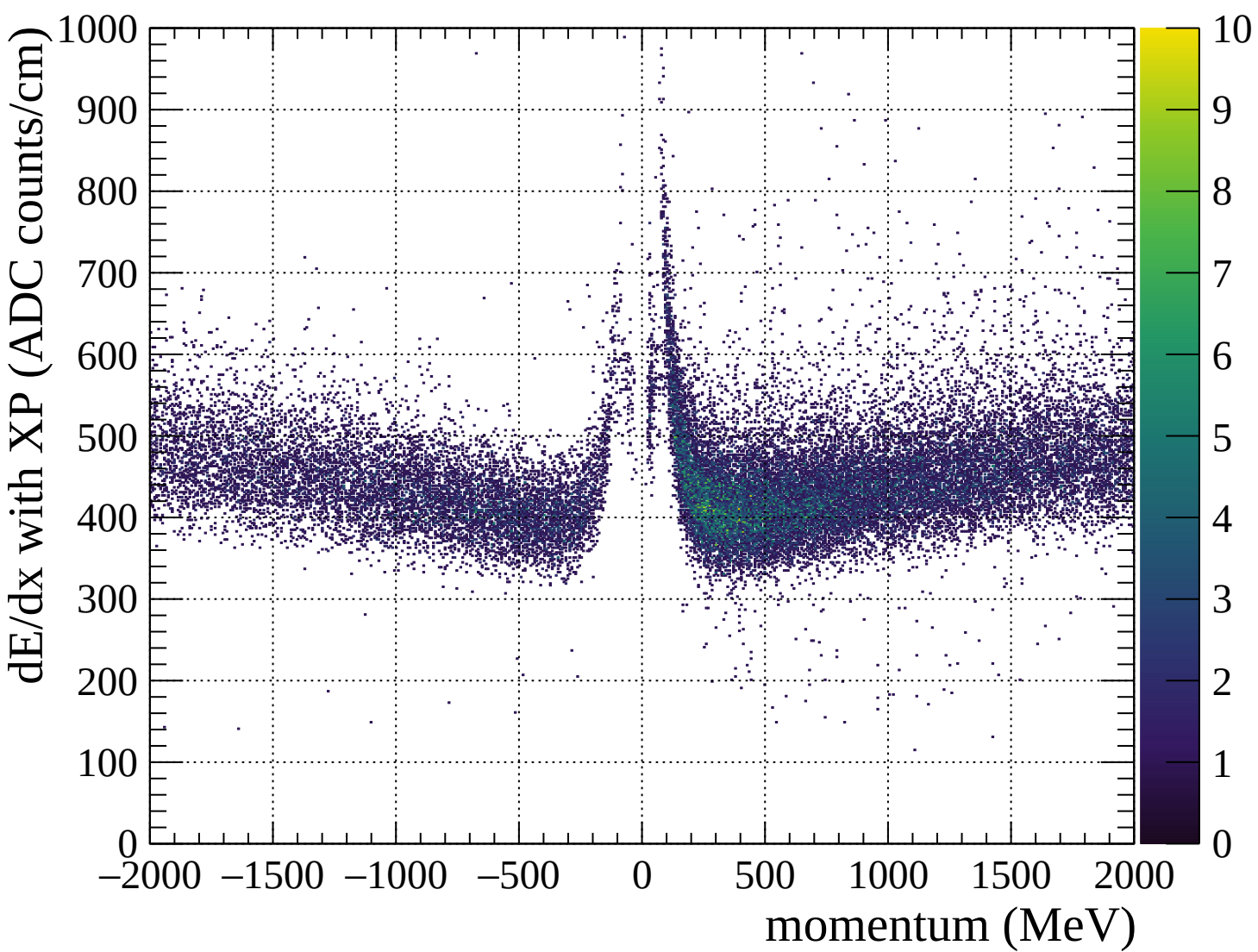


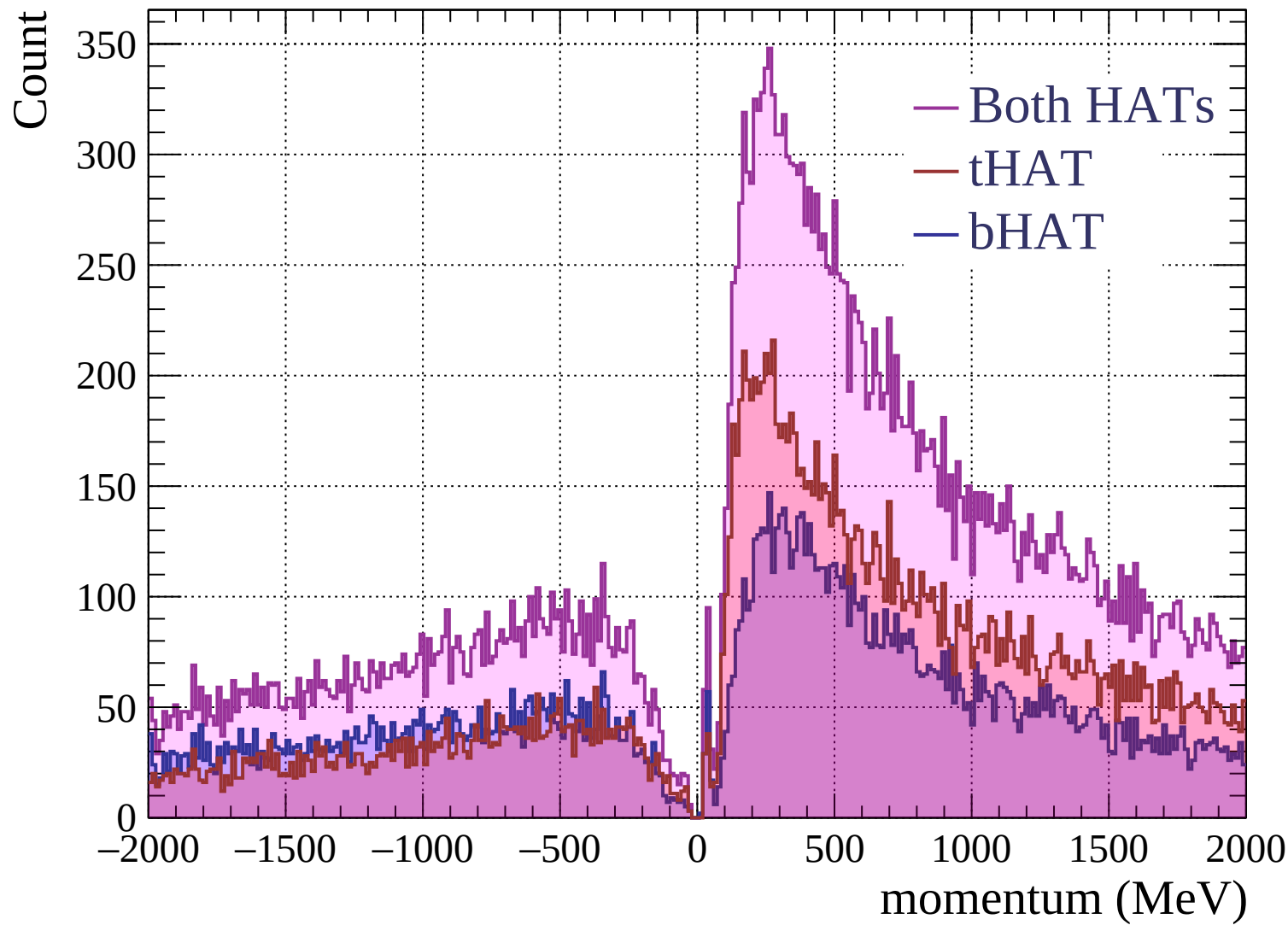


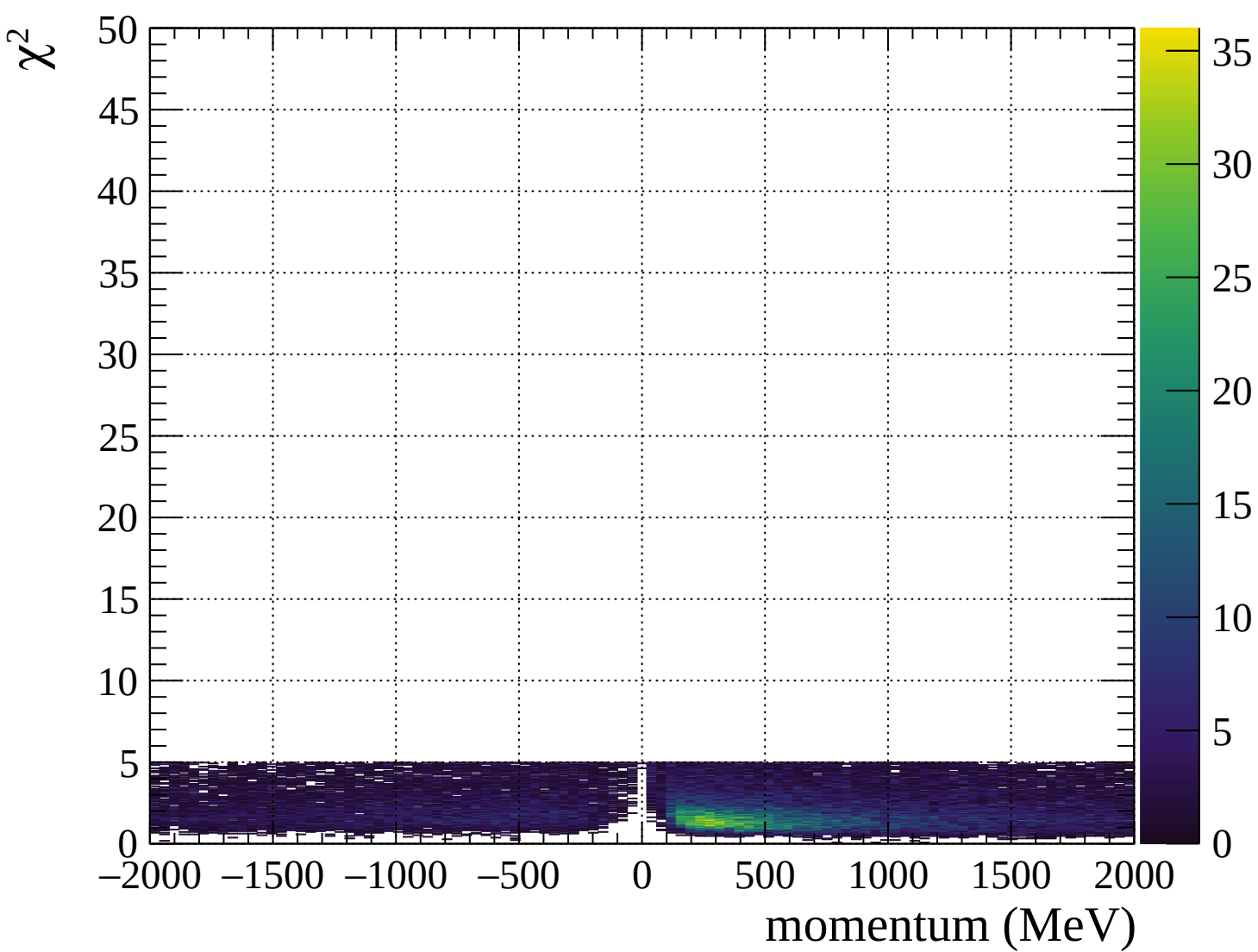


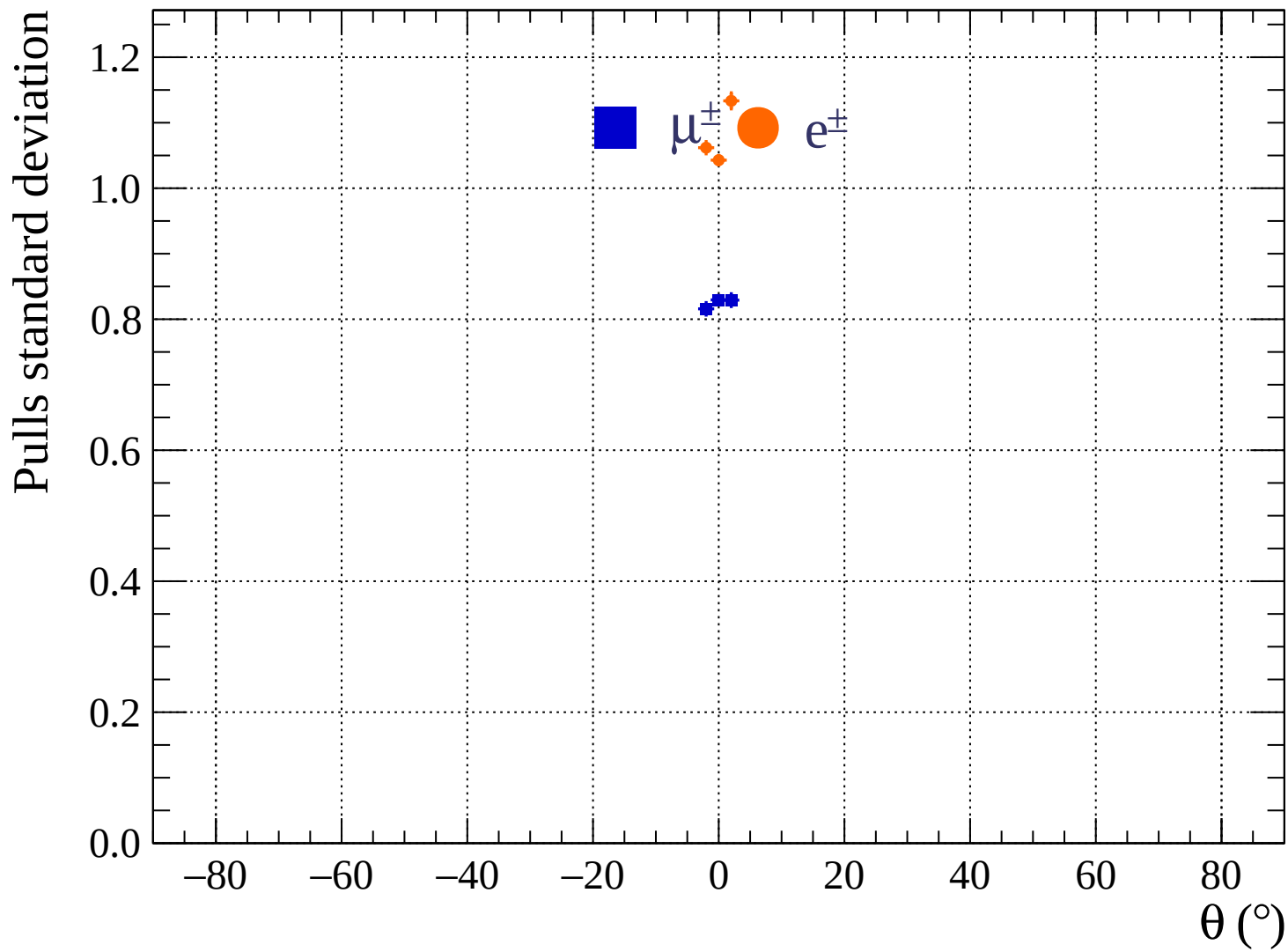


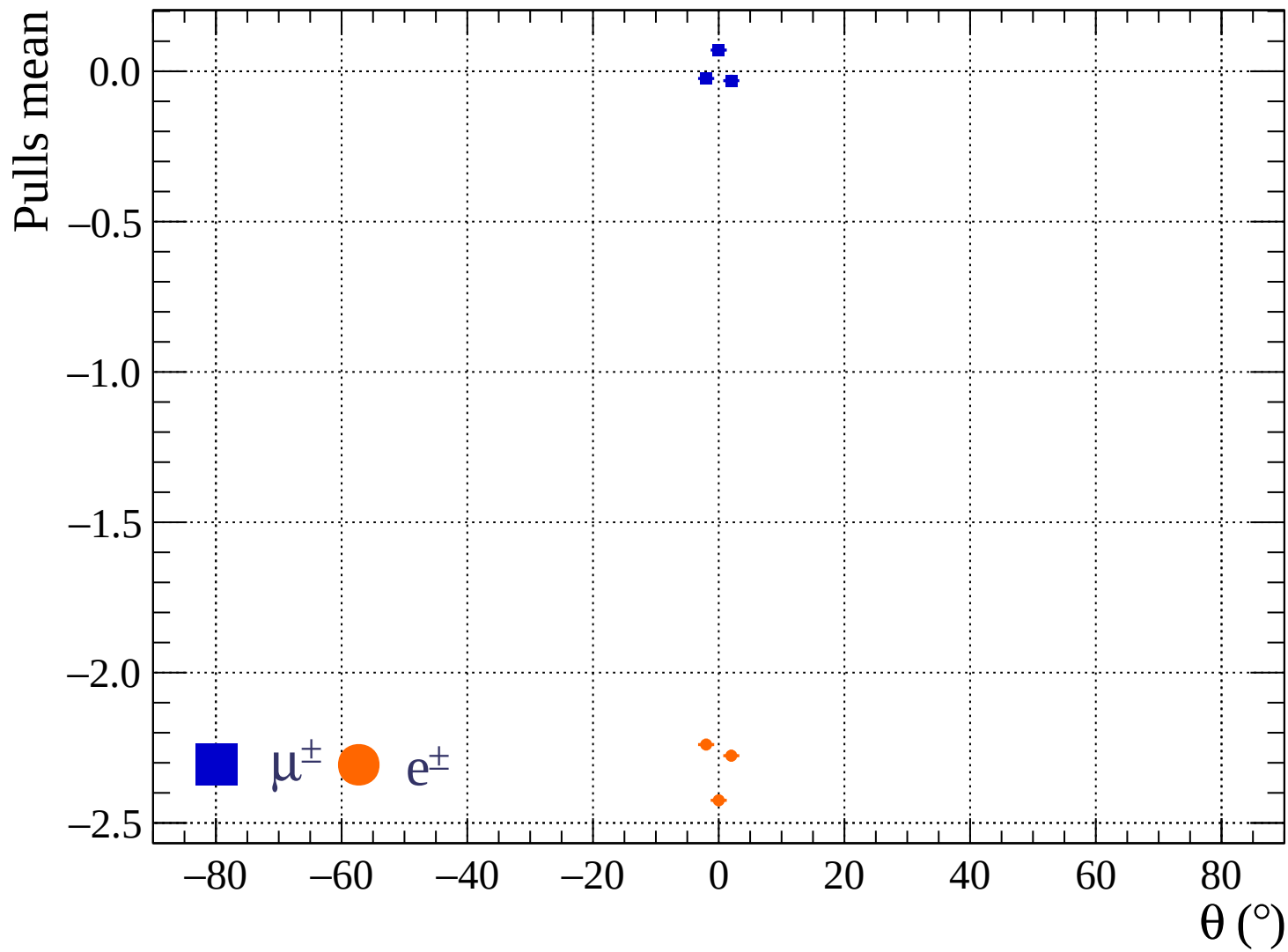




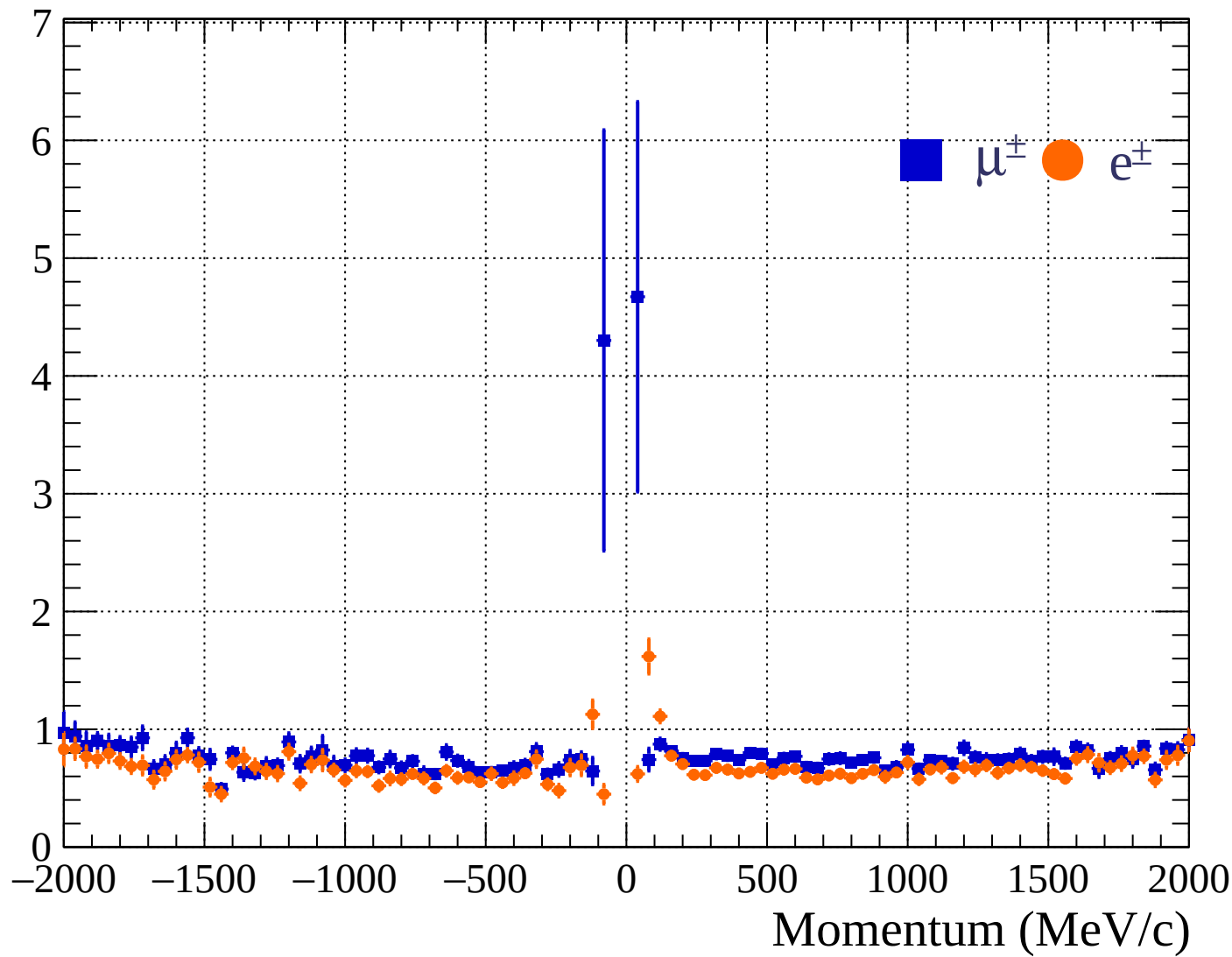




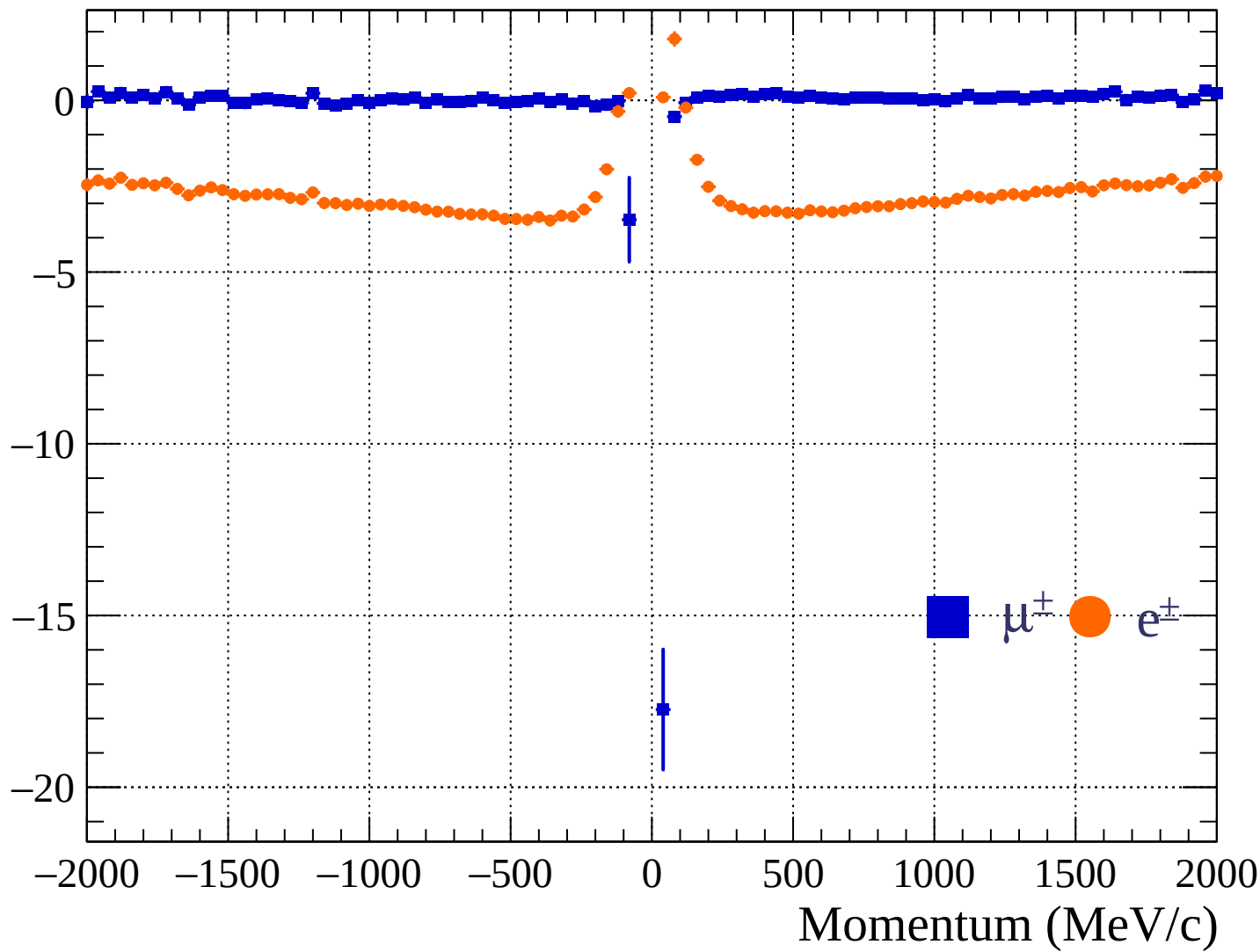


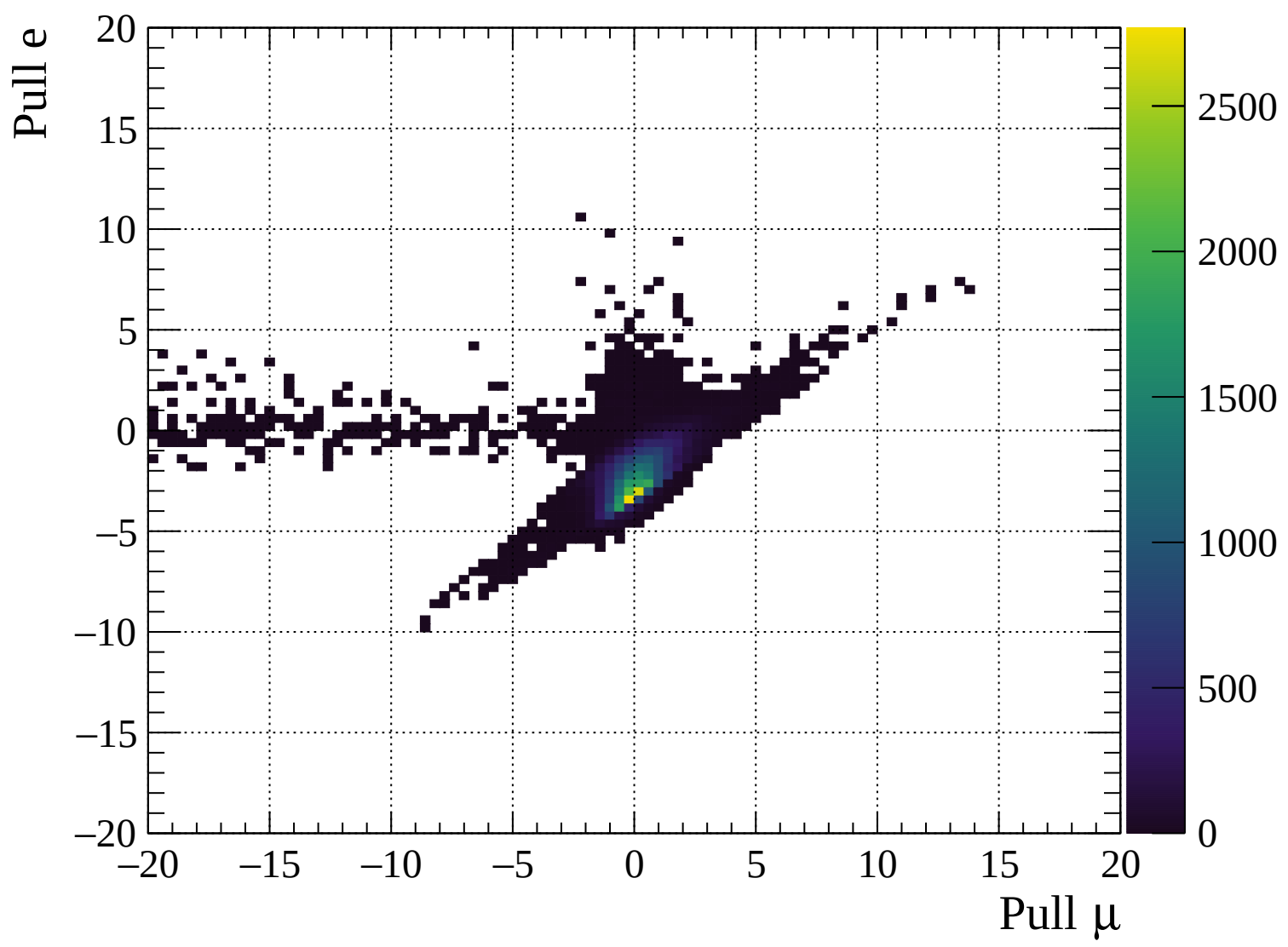


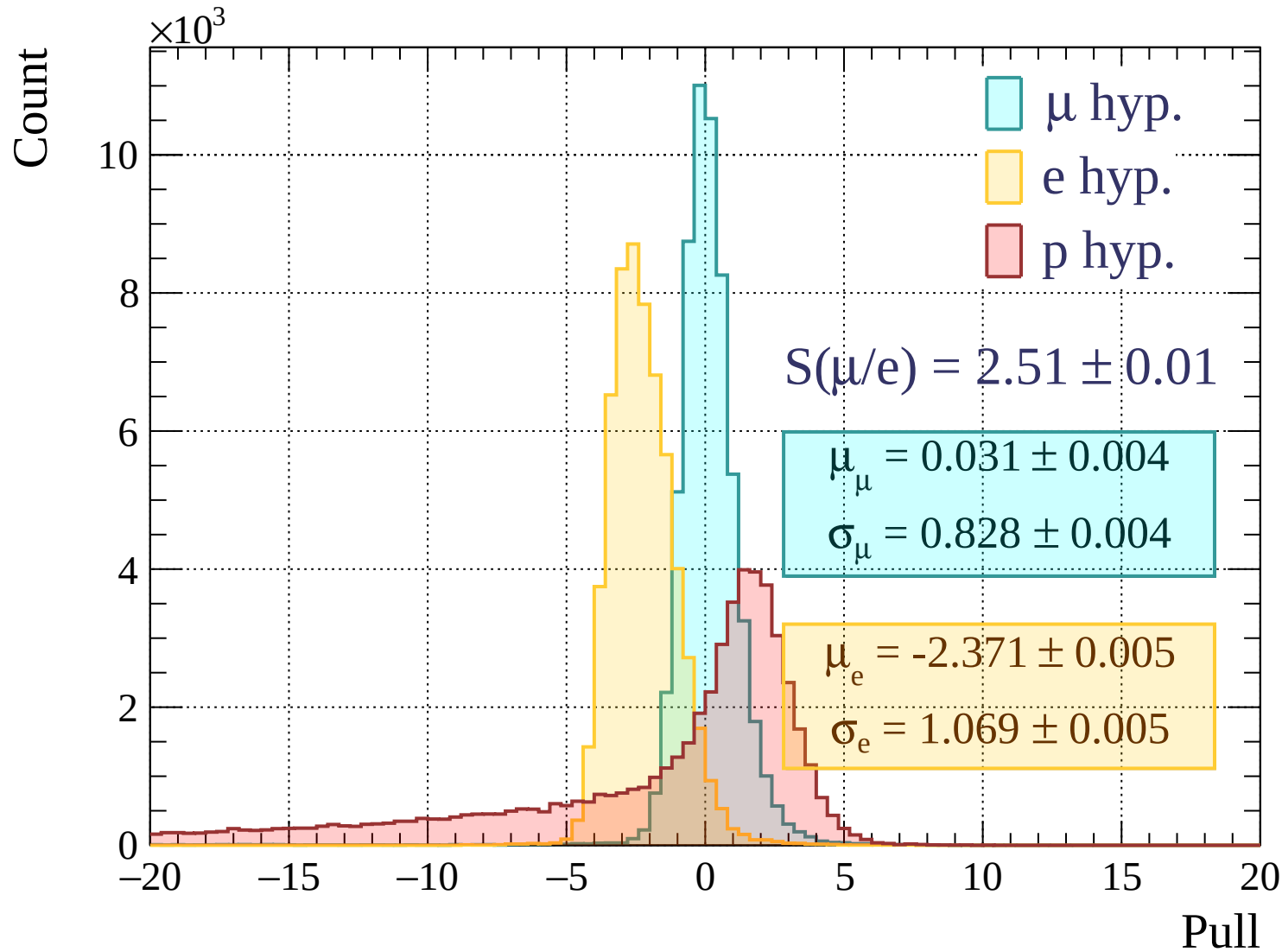
Pulls standard deviation

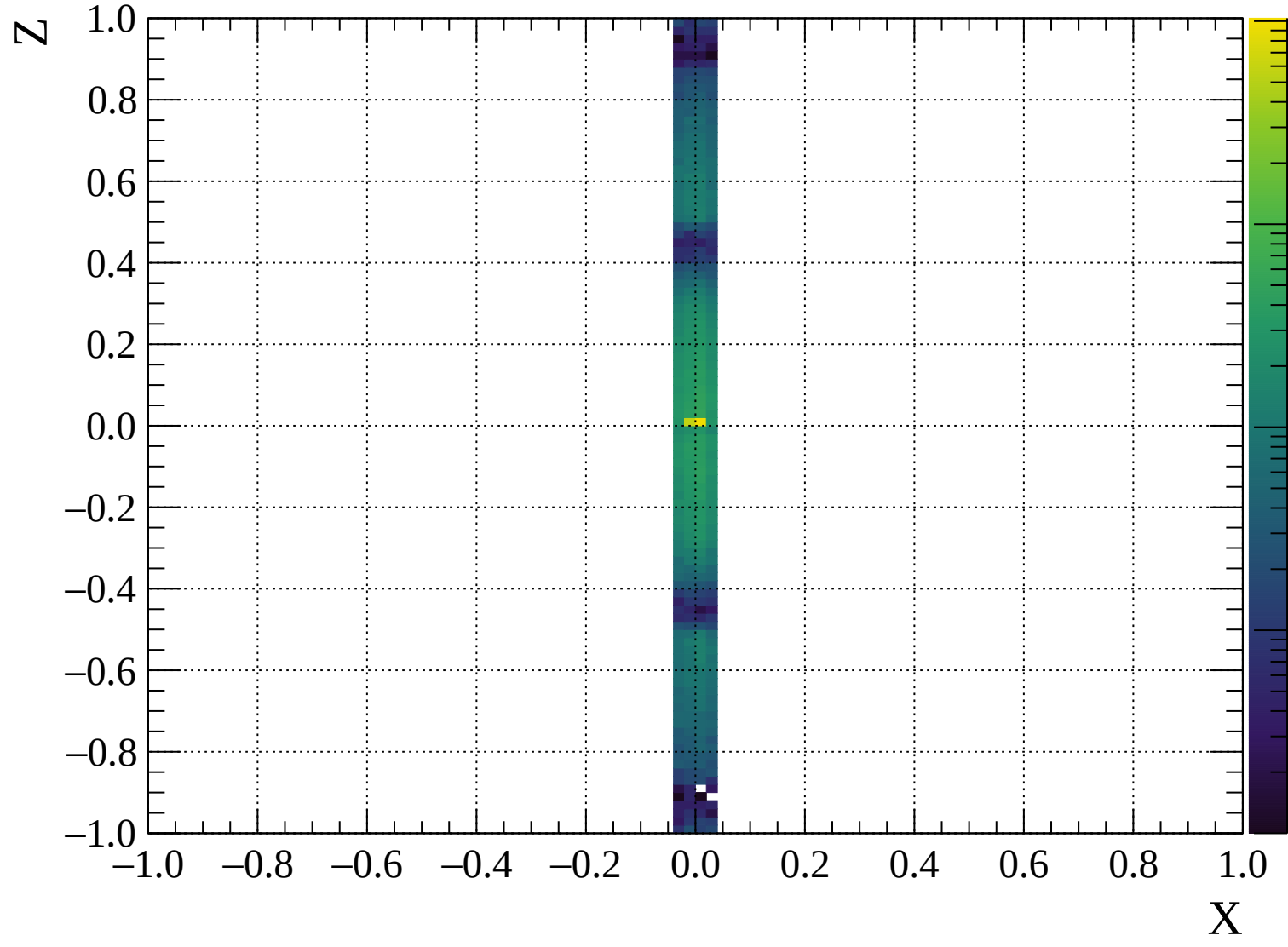


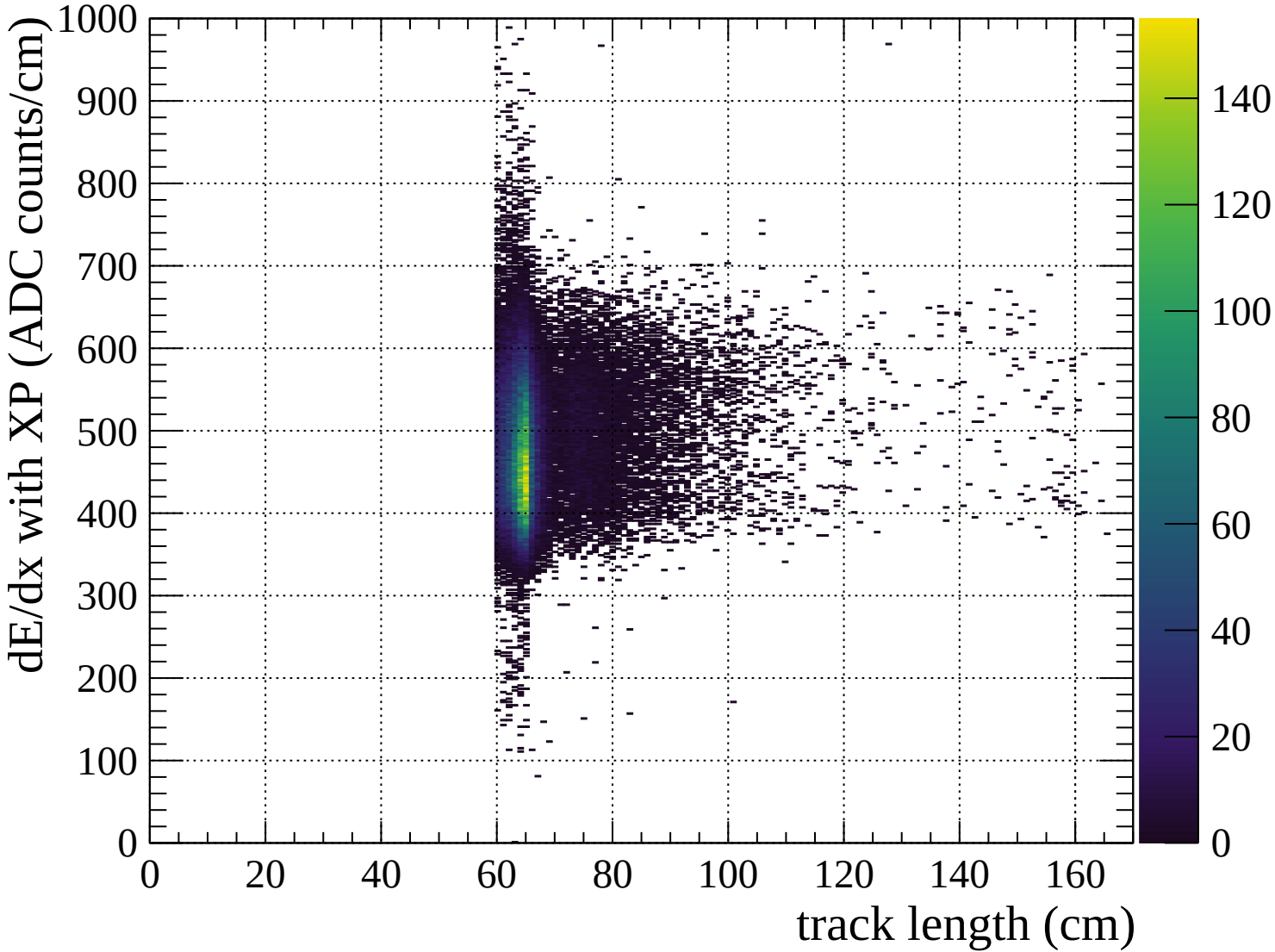
Pulls mean

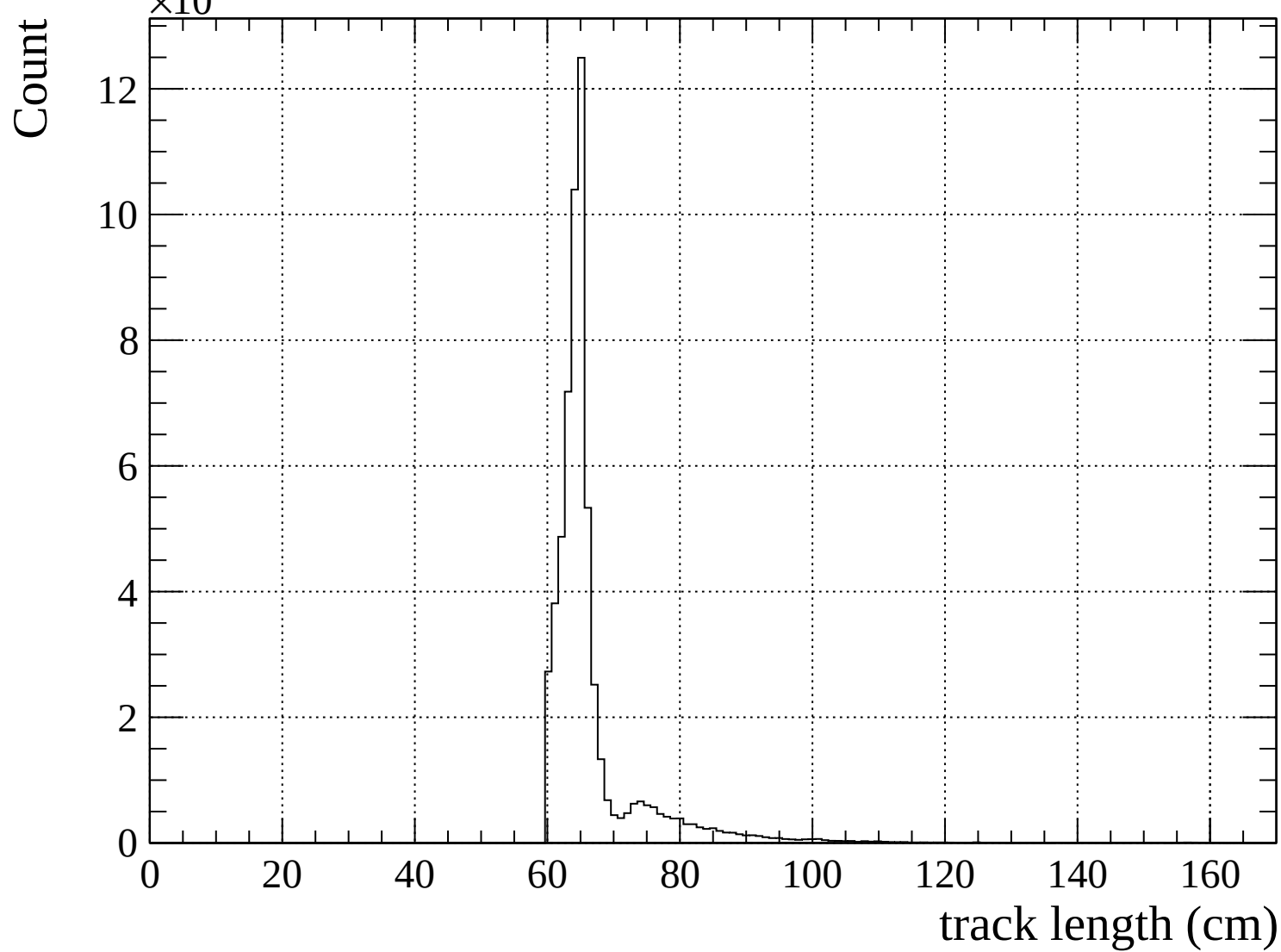


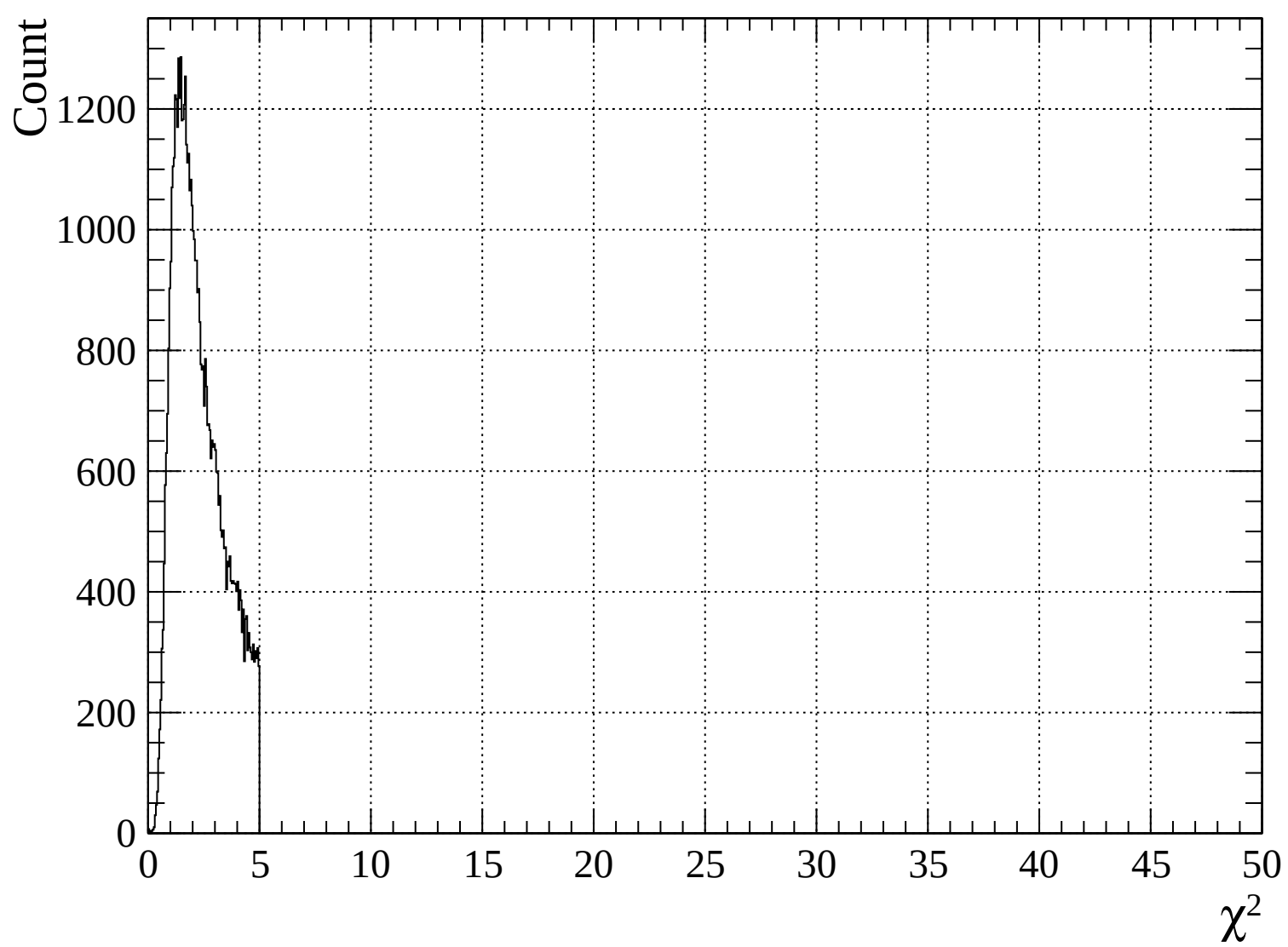


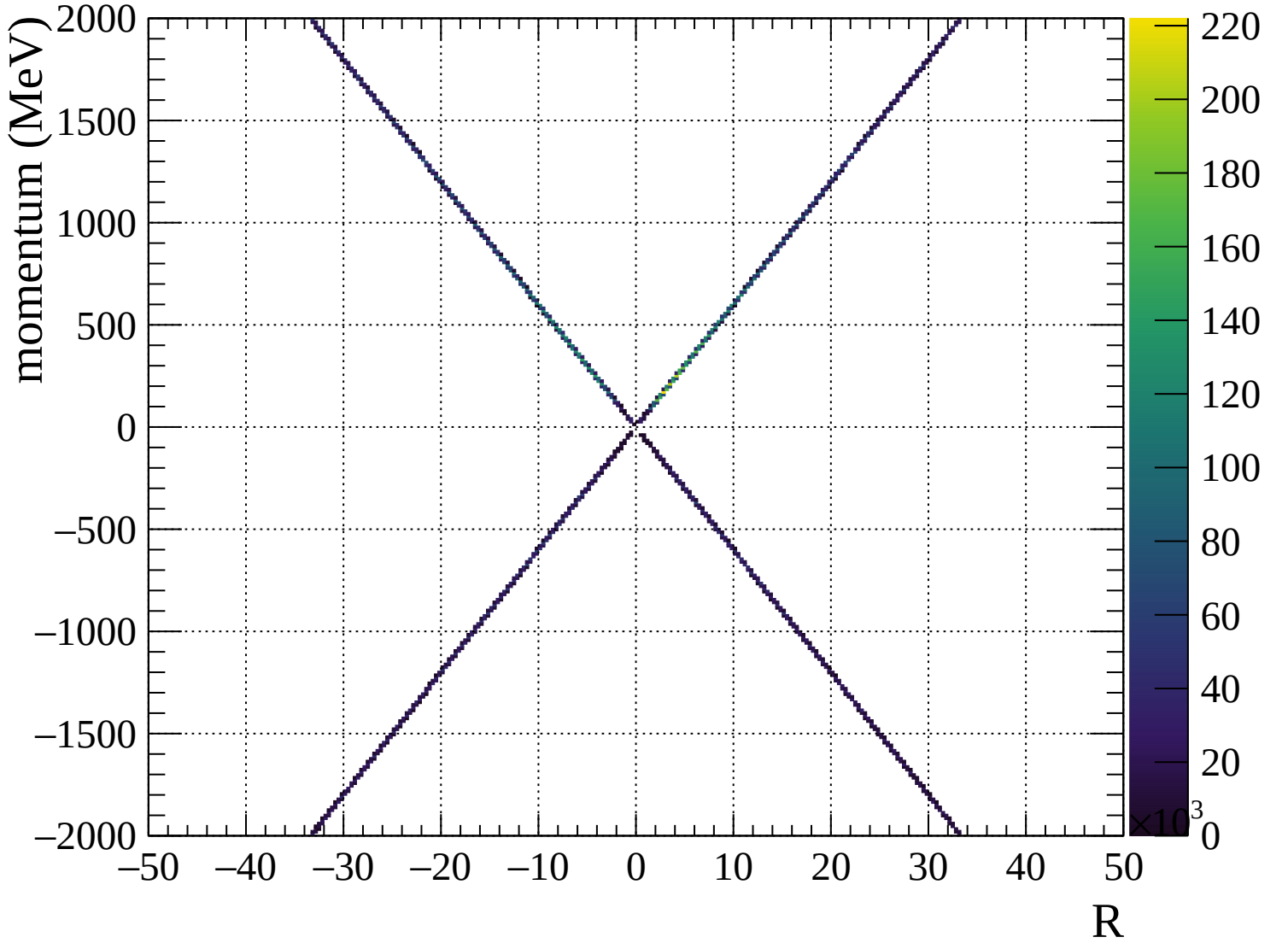


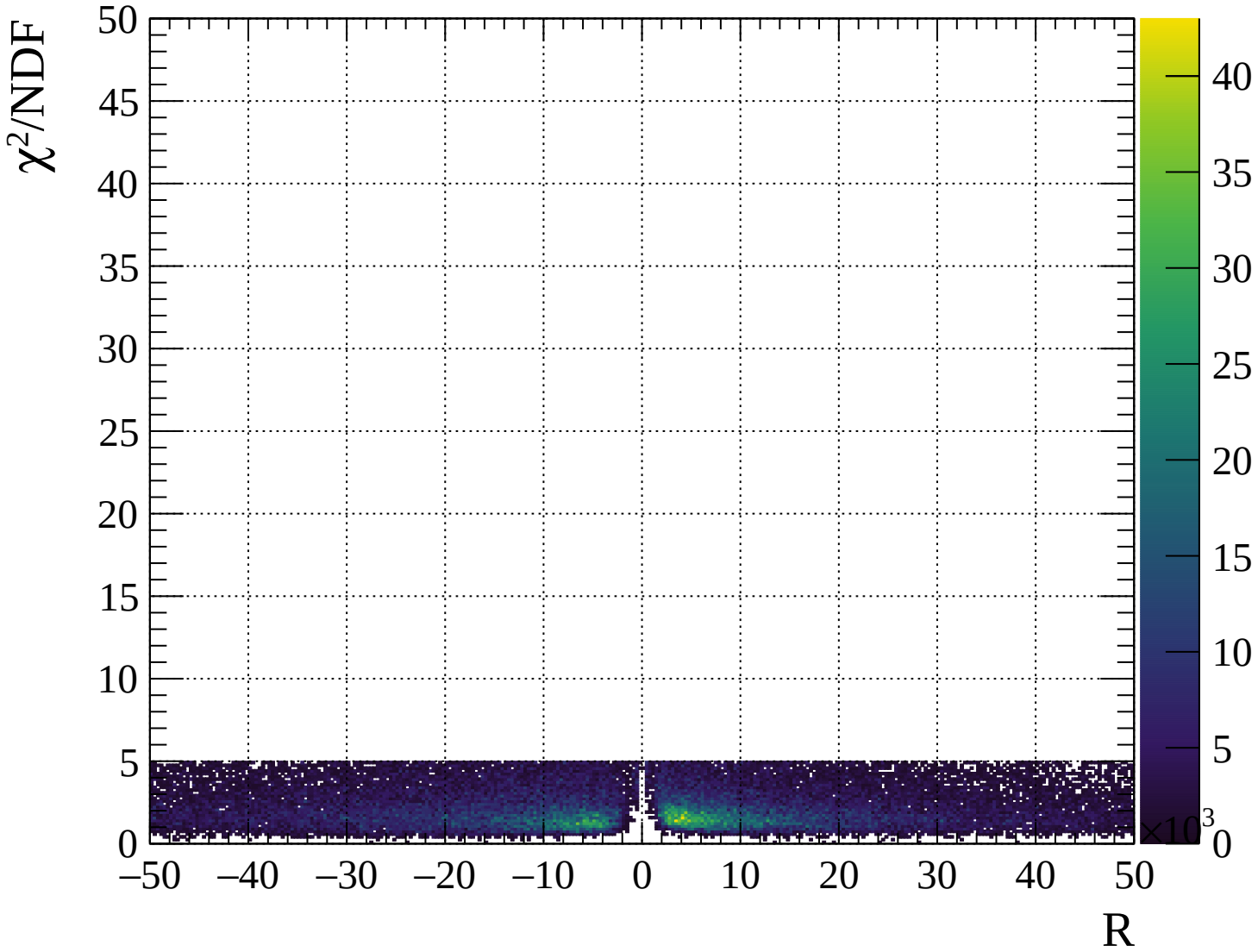






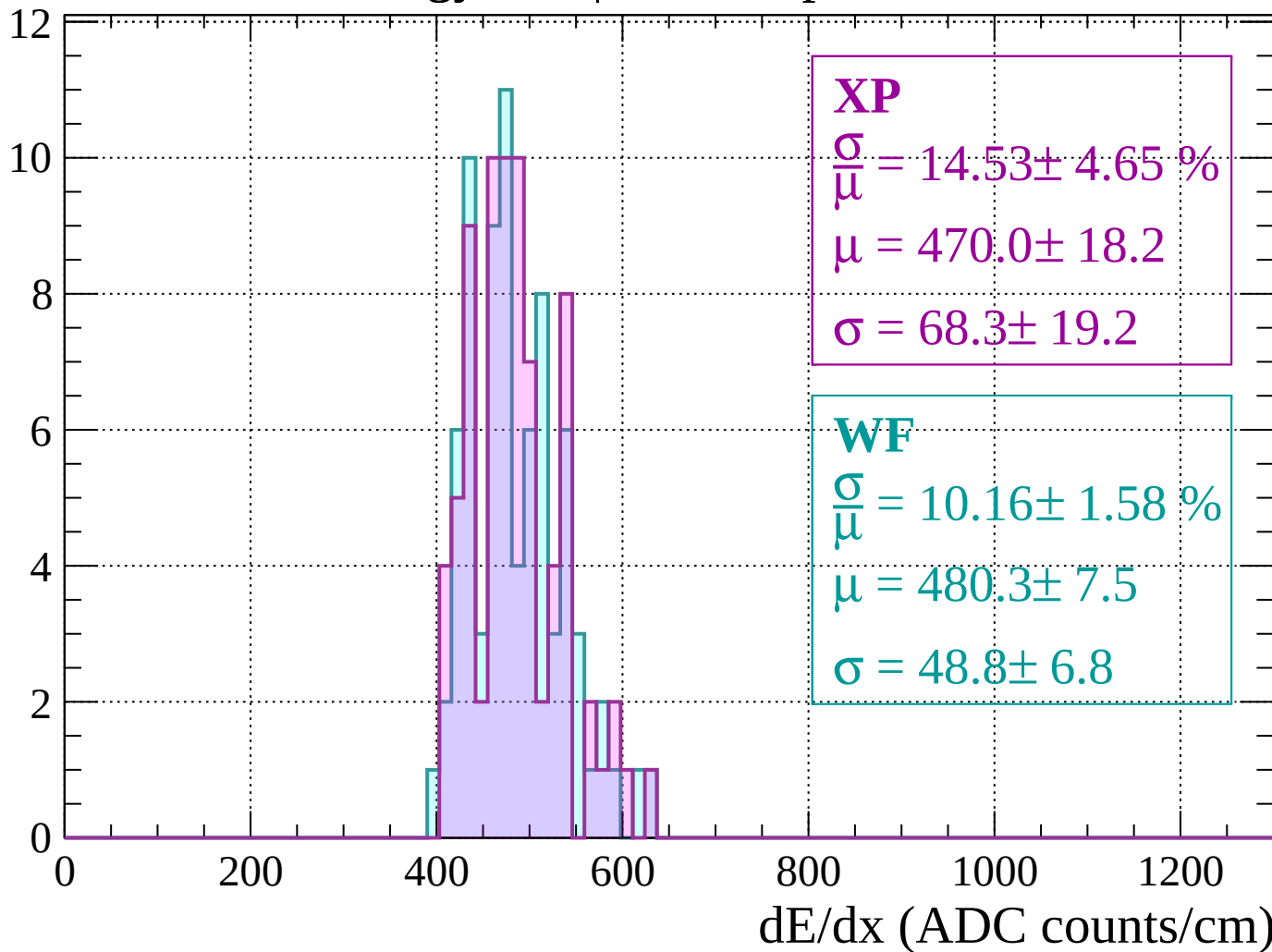






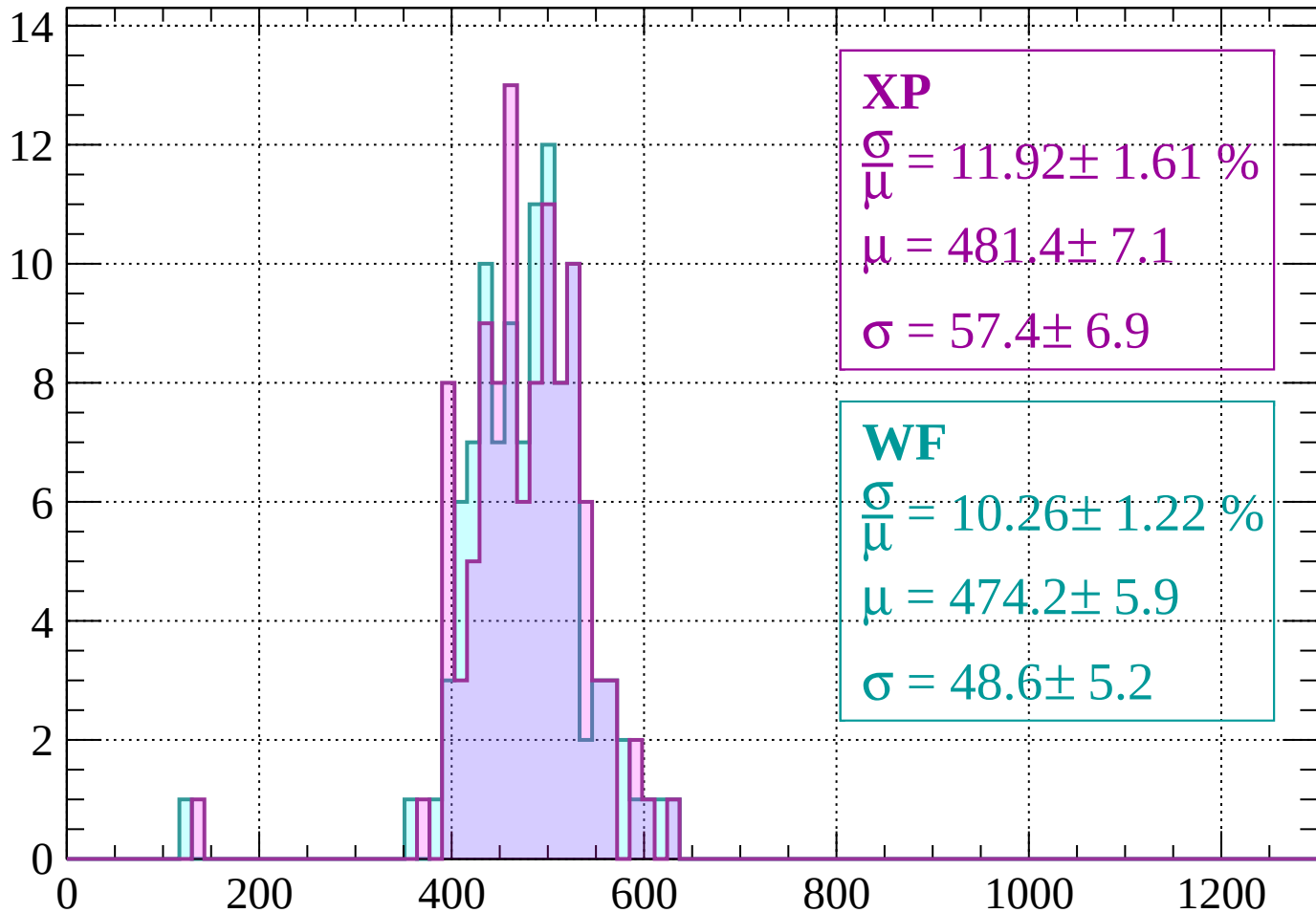
Energy loss | -2000 < p < -1960

Count



Energy loss | -1960 < p < -1920

Count



XP

$$\frac{\sigma}{\mu} = 11.92 \pm 1.61 \%$$

$$\mu = 481.4 \pm 7.1$$

$$\sigma = 57.4 \pm 6.9$$

WF

$$\frac{\sigma}{\mu} = 10.26 \pm 1.22 \%$$

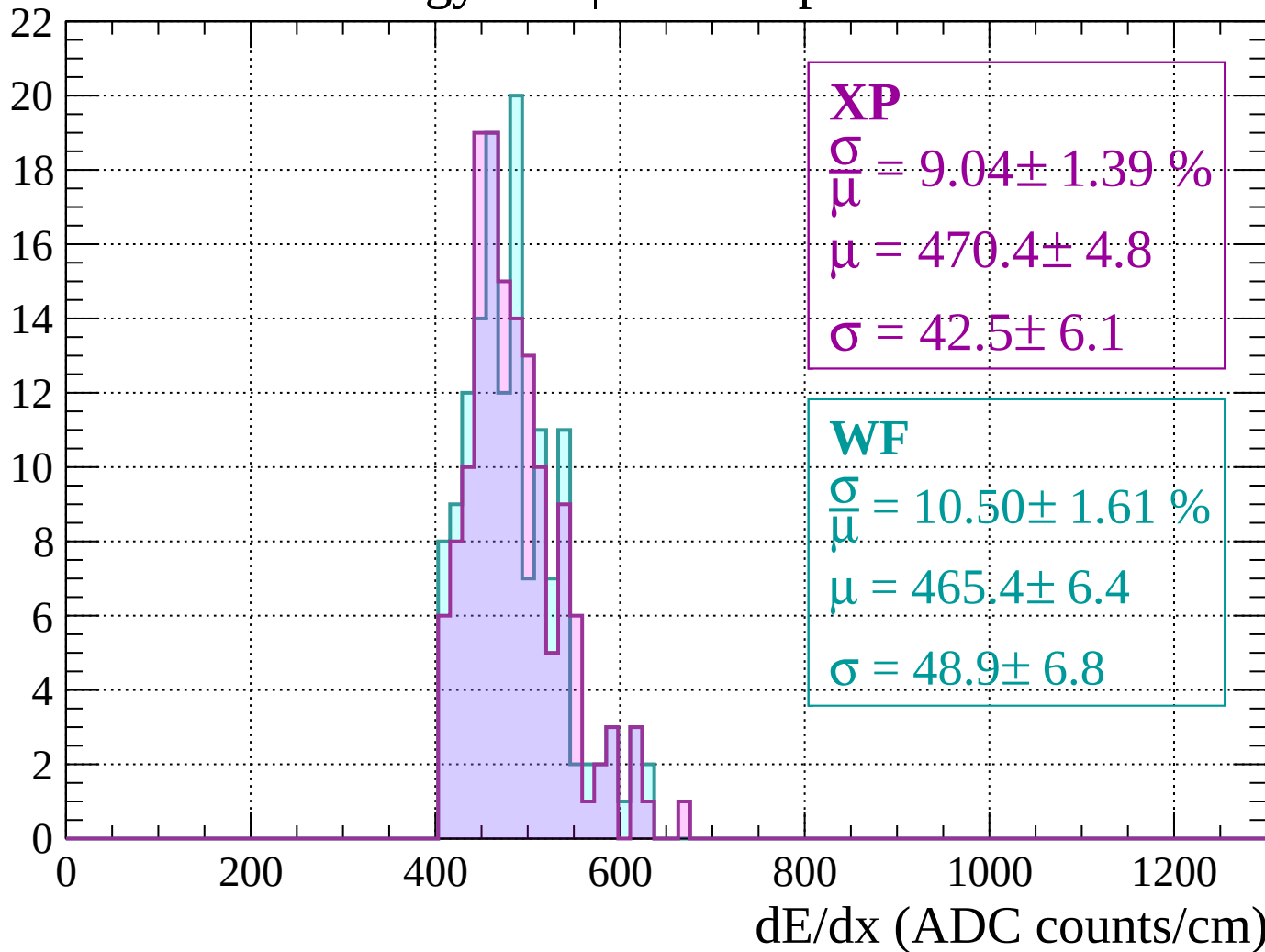
$$\mu = 474.2 \pm 5.9$$

$$\sigma = 48.6 \pm 5.2$$

dE/dx (ADC counts/cm)

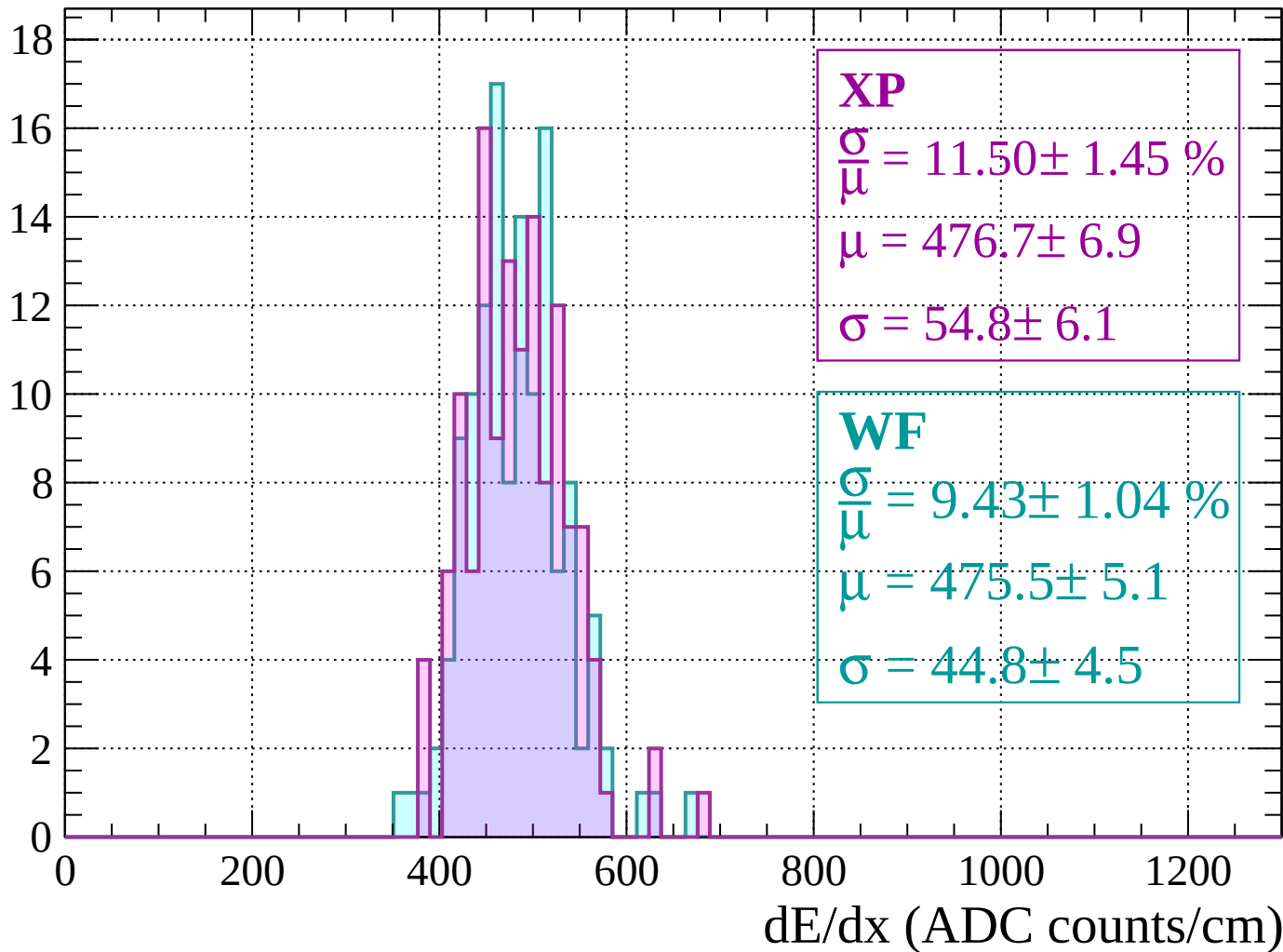
Energy loss | -1920 < p < -1880

Count



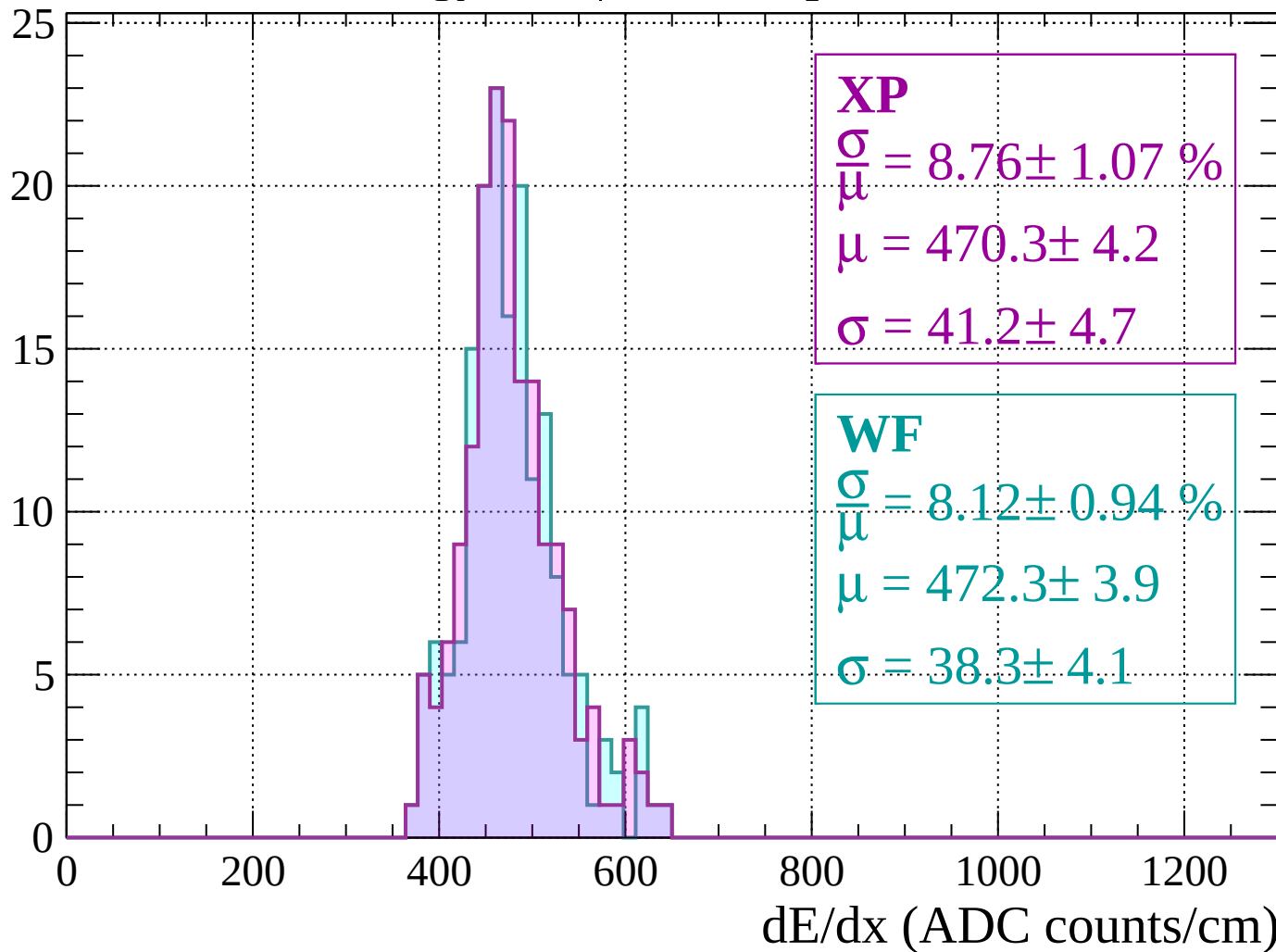
Energy loss | -1880 < p < -1840

Count



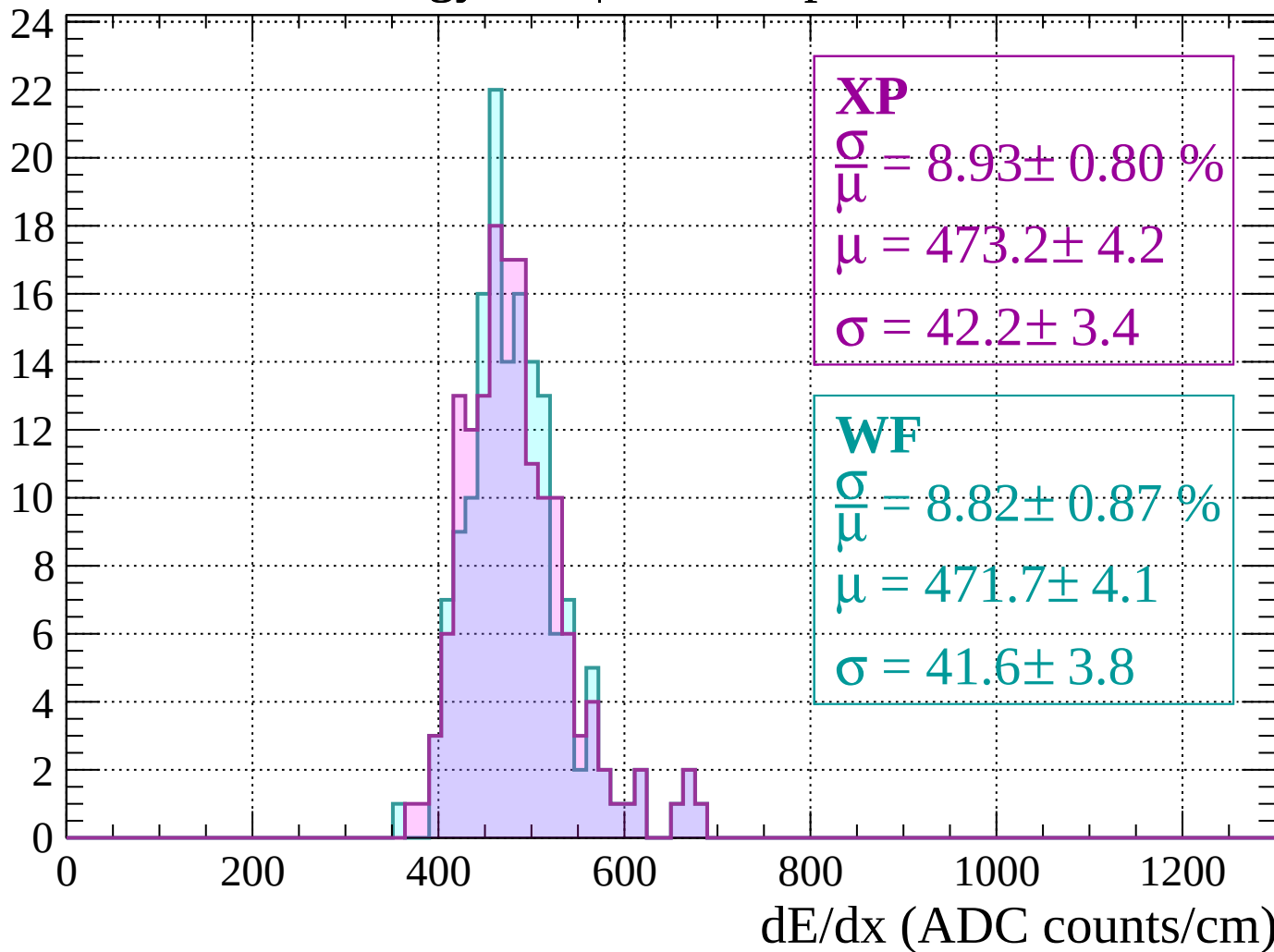
Energy loss | -1840 < p < -1800

Count



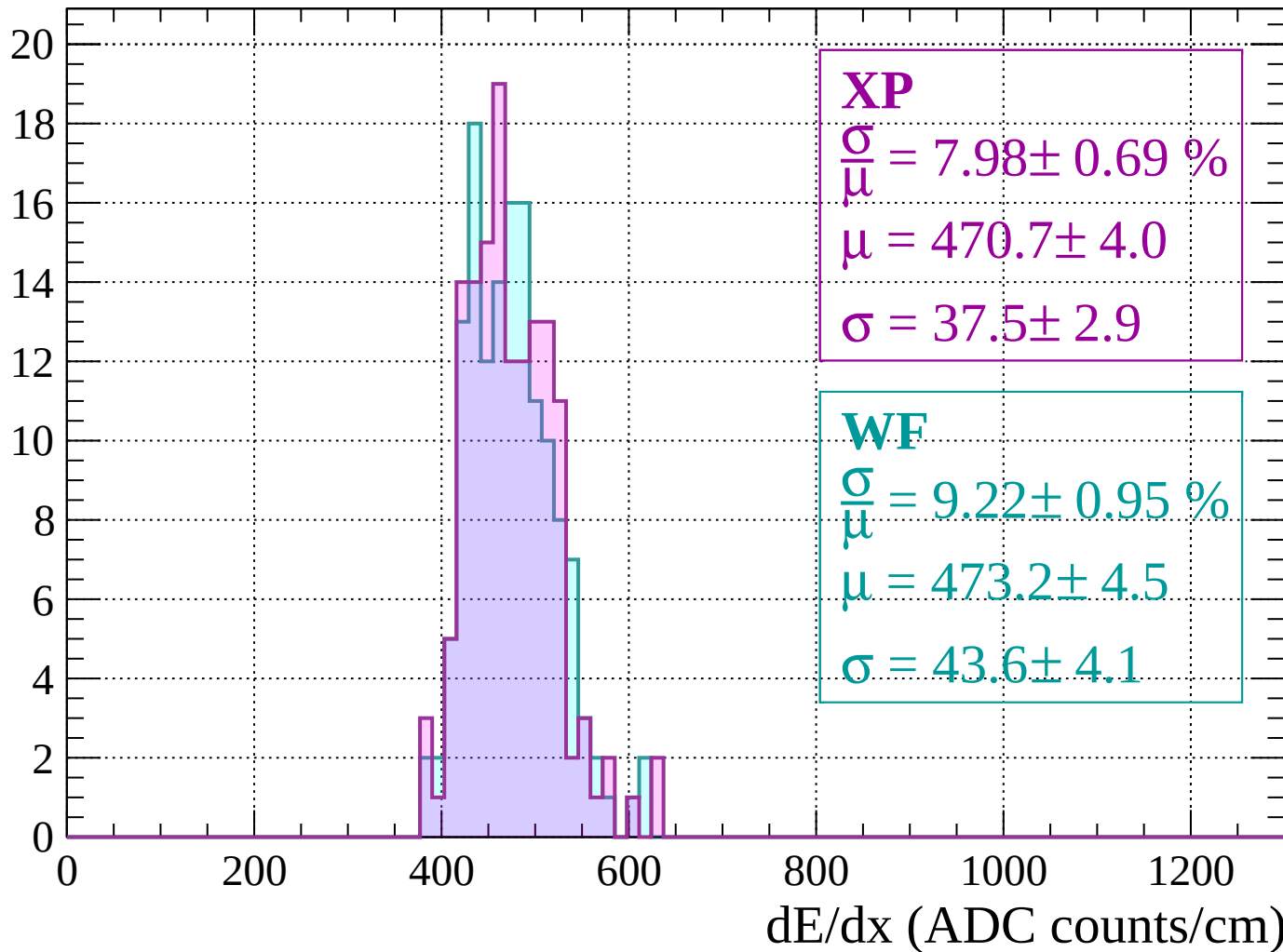
Energy loss | -1800 < p < -1760

Count



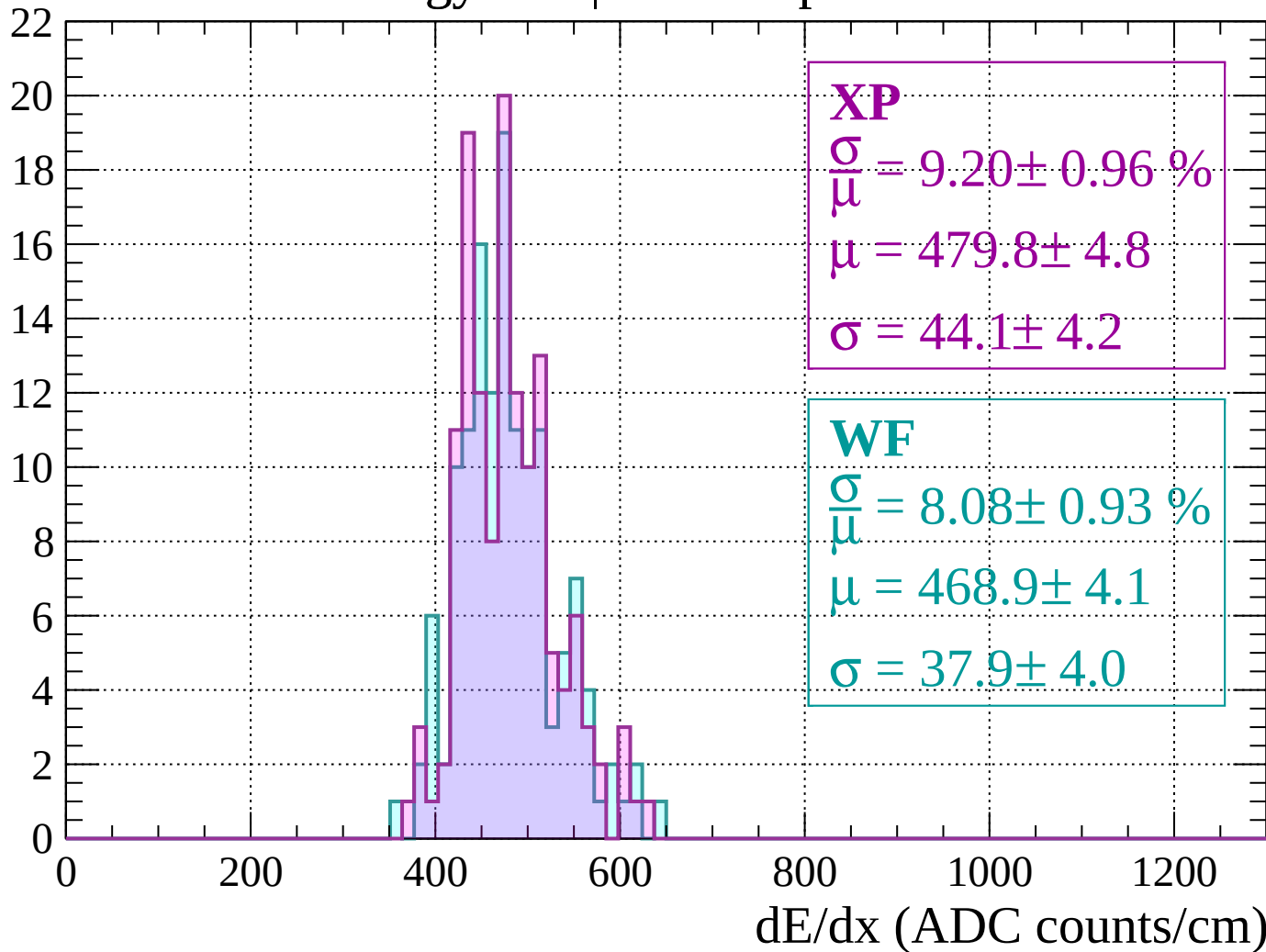
Energy loss | -1760 < p < -1720

Count



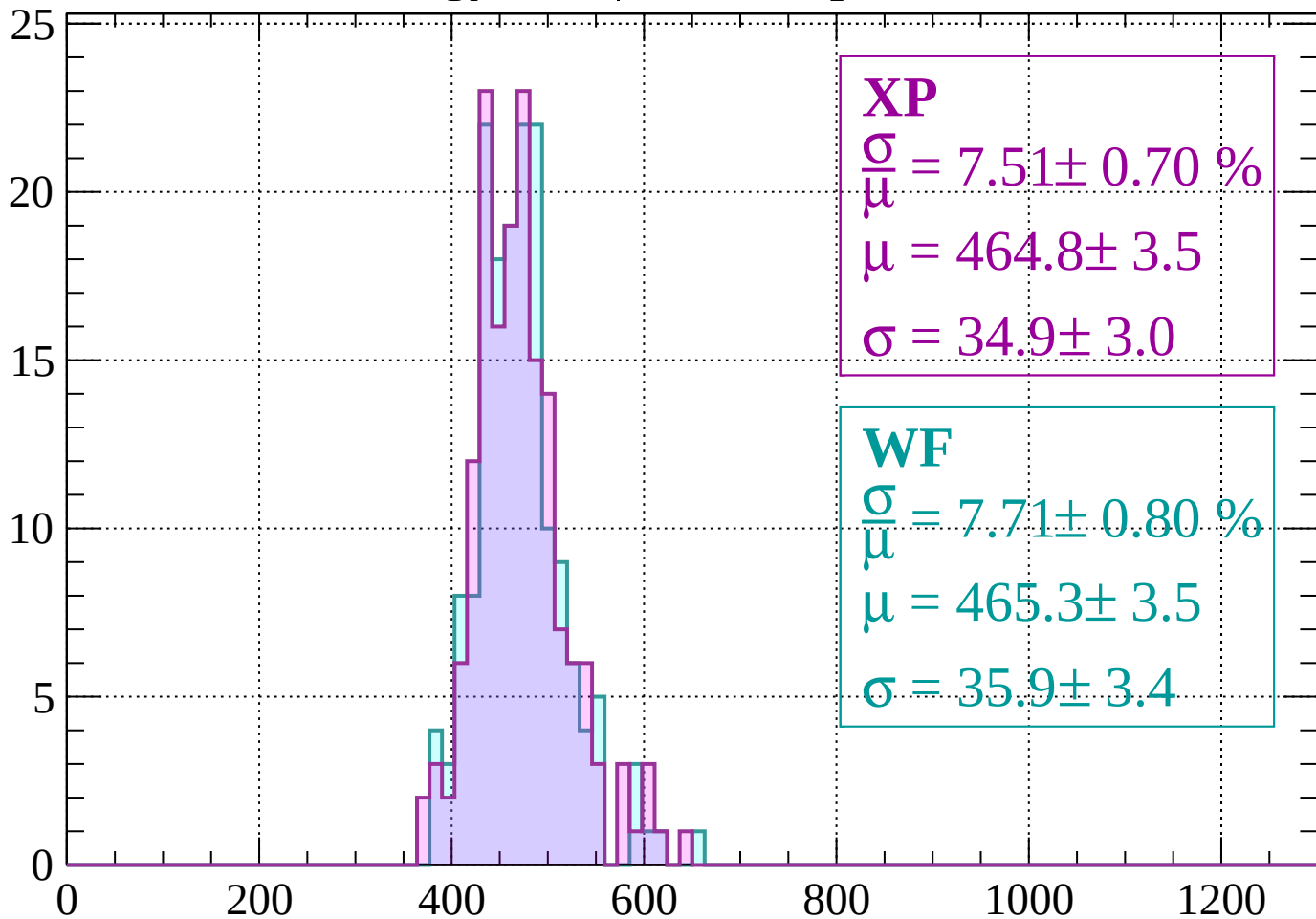
Energy loss | $-1720 < p < -1680$

Count



Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 7.51 \pm 0.70 \%$$

$$\mu = 464.8 \pm 3.5$$

$$\sigma = 34.9 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 7.71 \pm 0.80 \%$$

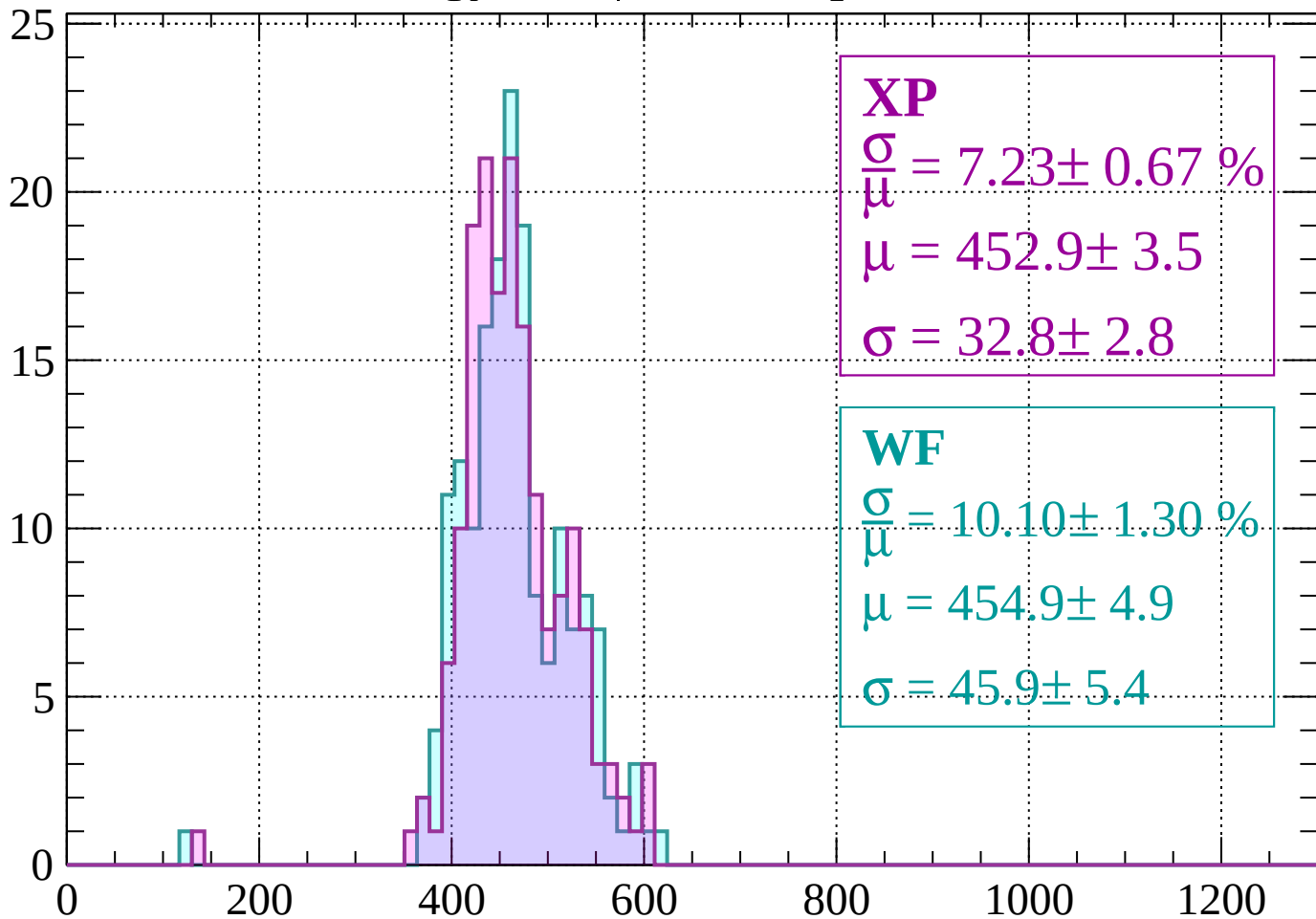
$$\mu = 465.3 \pm 3.5$$

$$\sigma = 35.9 \pm 3.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.23 \pm 0.67 \%$$

$$\mu = 452.9 \pm 3.5$$

$$\sigma = 32.8 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 10.10 \pm 1.30 \%$$

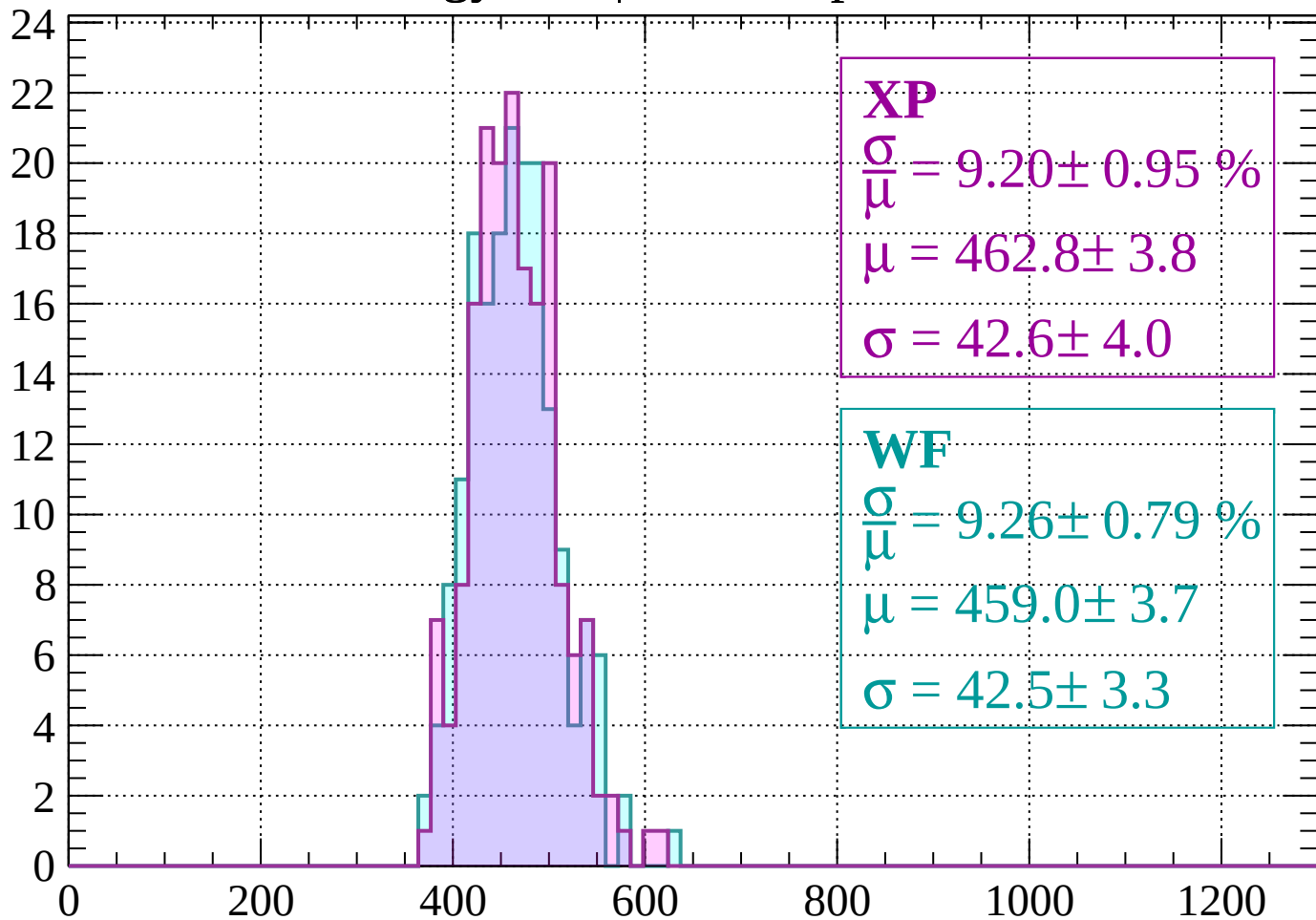
$$\mu = 454.9 \pm 4.9$$

$$\sigma = 45.9 \pm 5.4$$

dE/dx (ADC counts/cm)

Energy loss | -1600 < p < -1560

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.20 \pm 0.95 \%$$

$$\mu = 462.8 \pm 3.8$$

$$\sigma = 42.6 \pm 4.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.26 \pm 0.79 \%$$

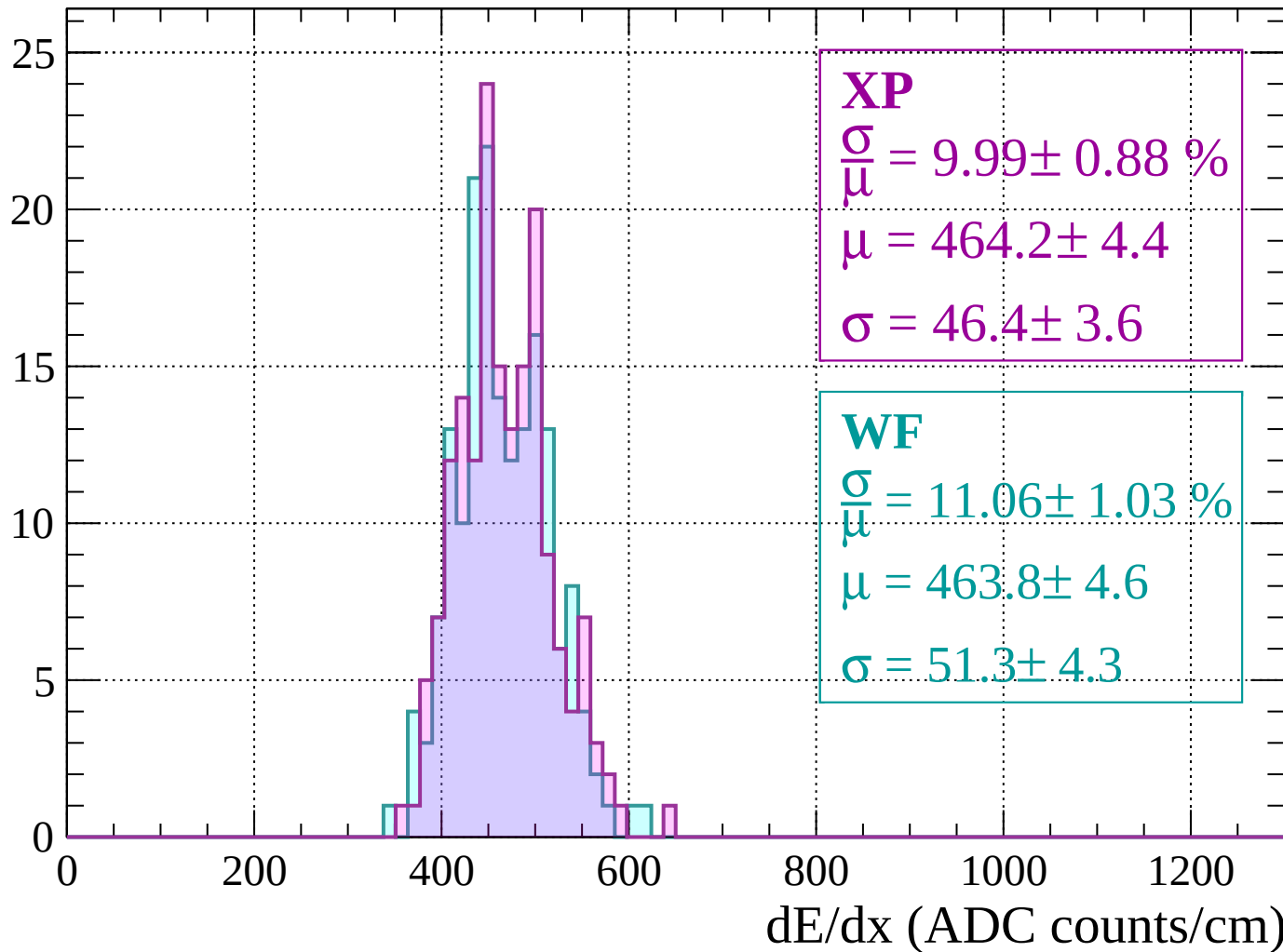
$$\mu = 459.0 \pm 3.7$$

$$\sigma = 42.5 \pm 3.3$$

dE/dx (ADC counts/cm)

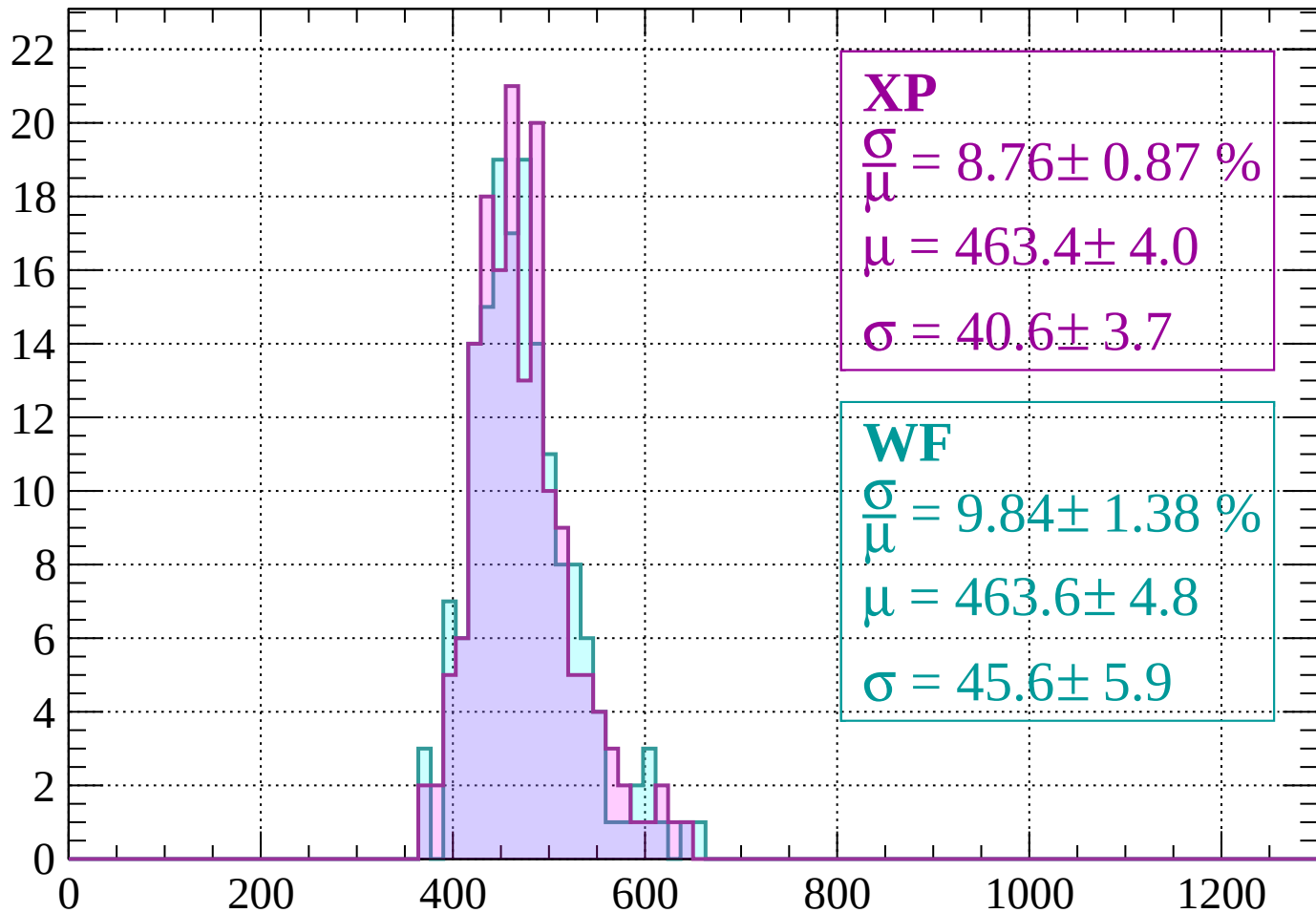
Energy loss | $-1560 < p < -1520$

Count



Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 8.76 \pm 0.87 \%$$

$$\mu = 463.4 \pm 4.0$$

$$\sigma = 40.6 \pm 3.7$$

WF

$$\frac{\sigma}{\mu} = 9.84 \pm 1.38 \%$$

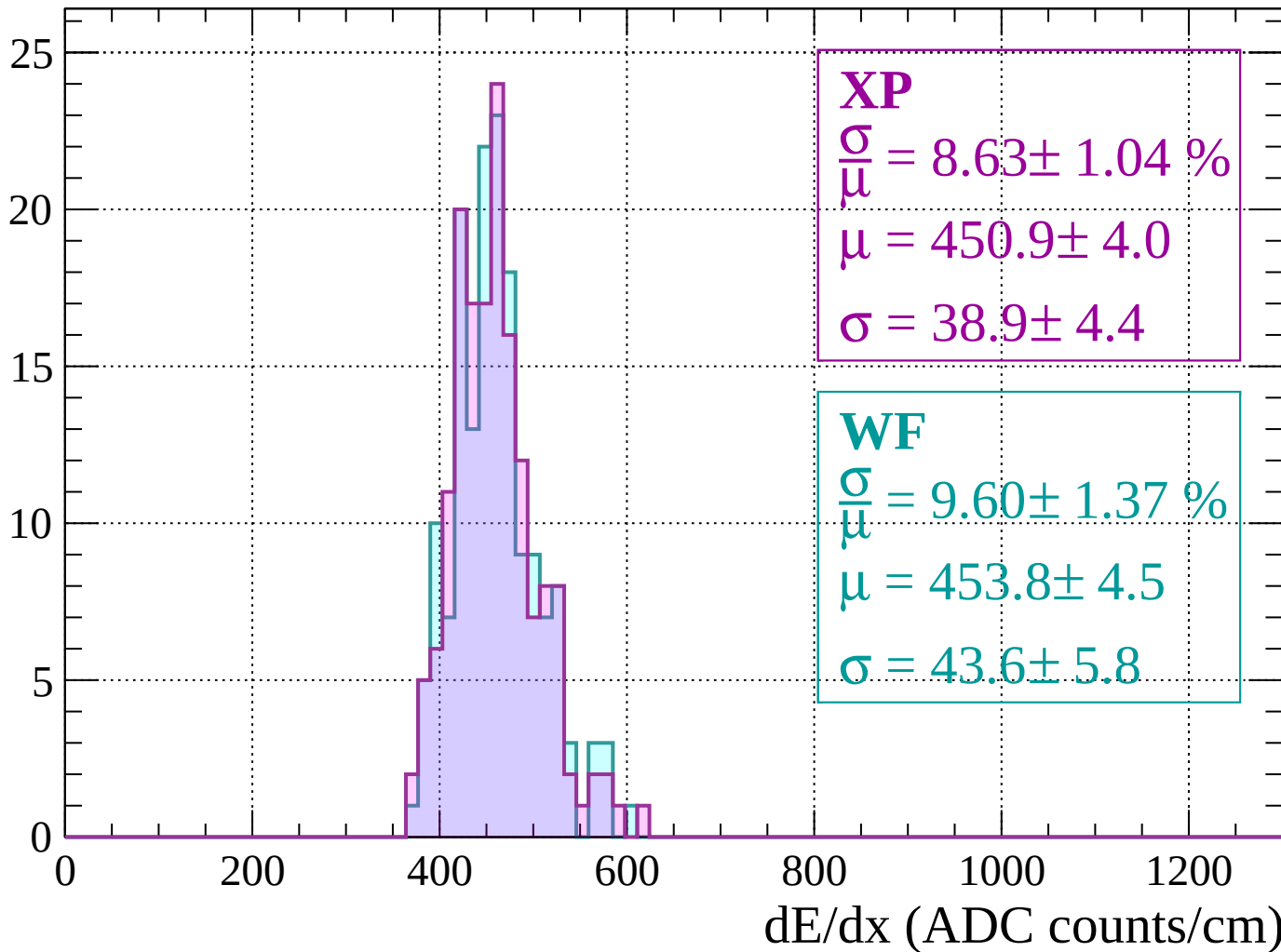
$$\mu = 463.6 \pm 4.8$$

$$\sigma = 45.6 \pm 5.9$$

dE/dx (ADC counts/cm)

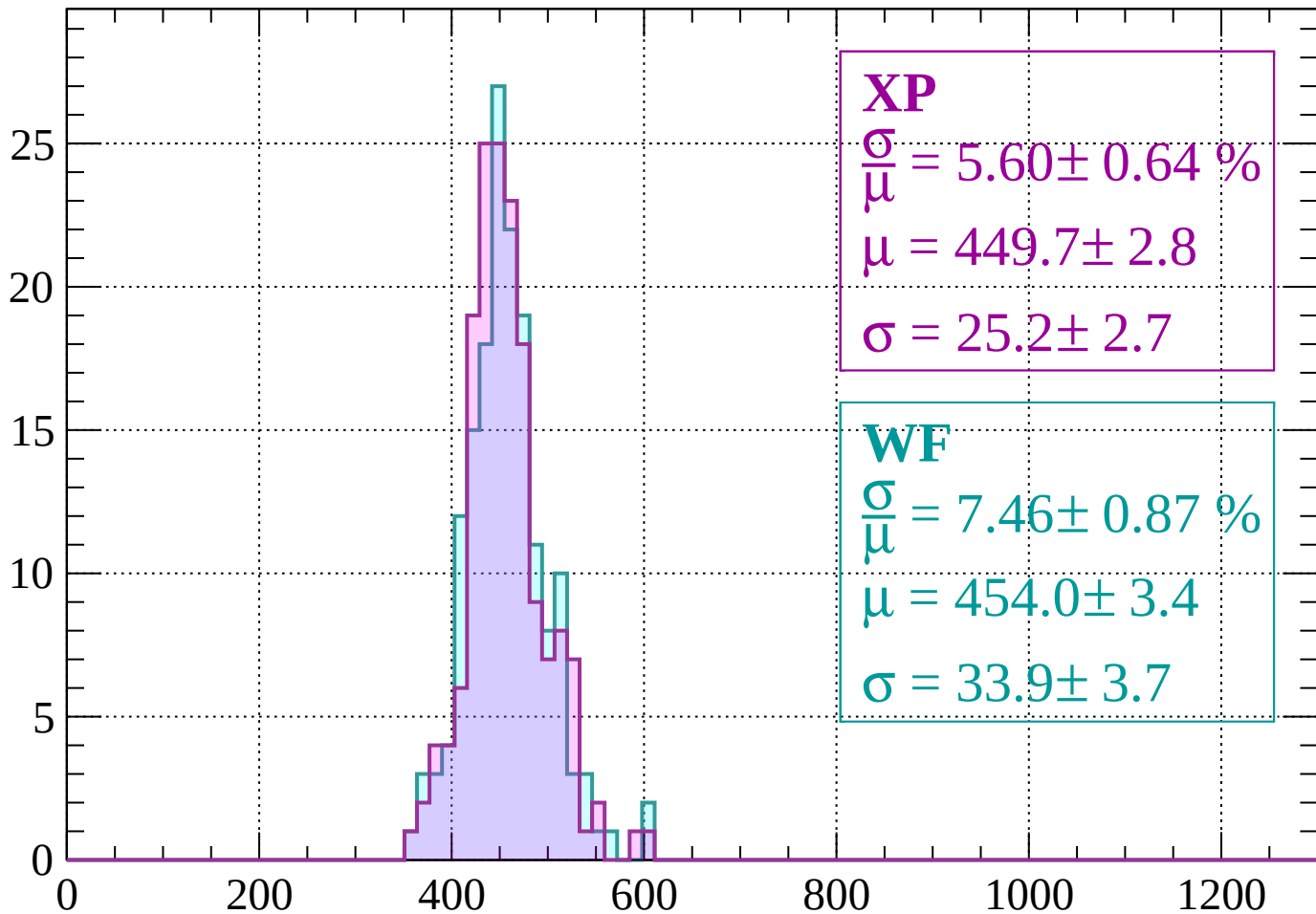
Energy loss | $-1480 < p < -1440$

Count



Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 5.60 \pm 0.64 \%$$

$$\mu = 449.7 \pm 2.8$$

$$\sigma = 25.2 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 7.46 \pm 0.87 \%$$

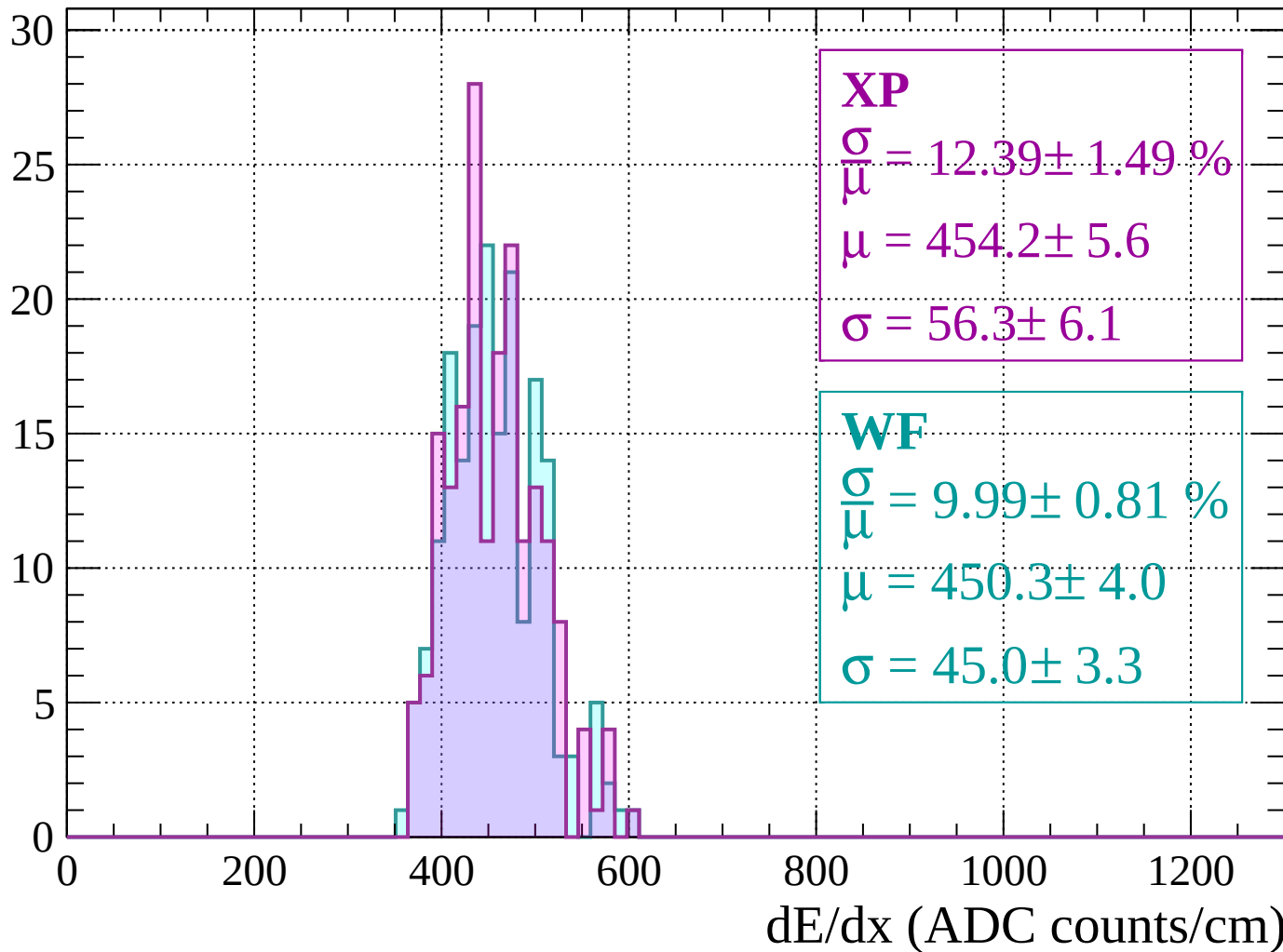
$$\mu = 454.0 \pm 3.4$$

$$\sigma = 33.9 \pm 3.7$$

dE/dx (ADC counts/cm)

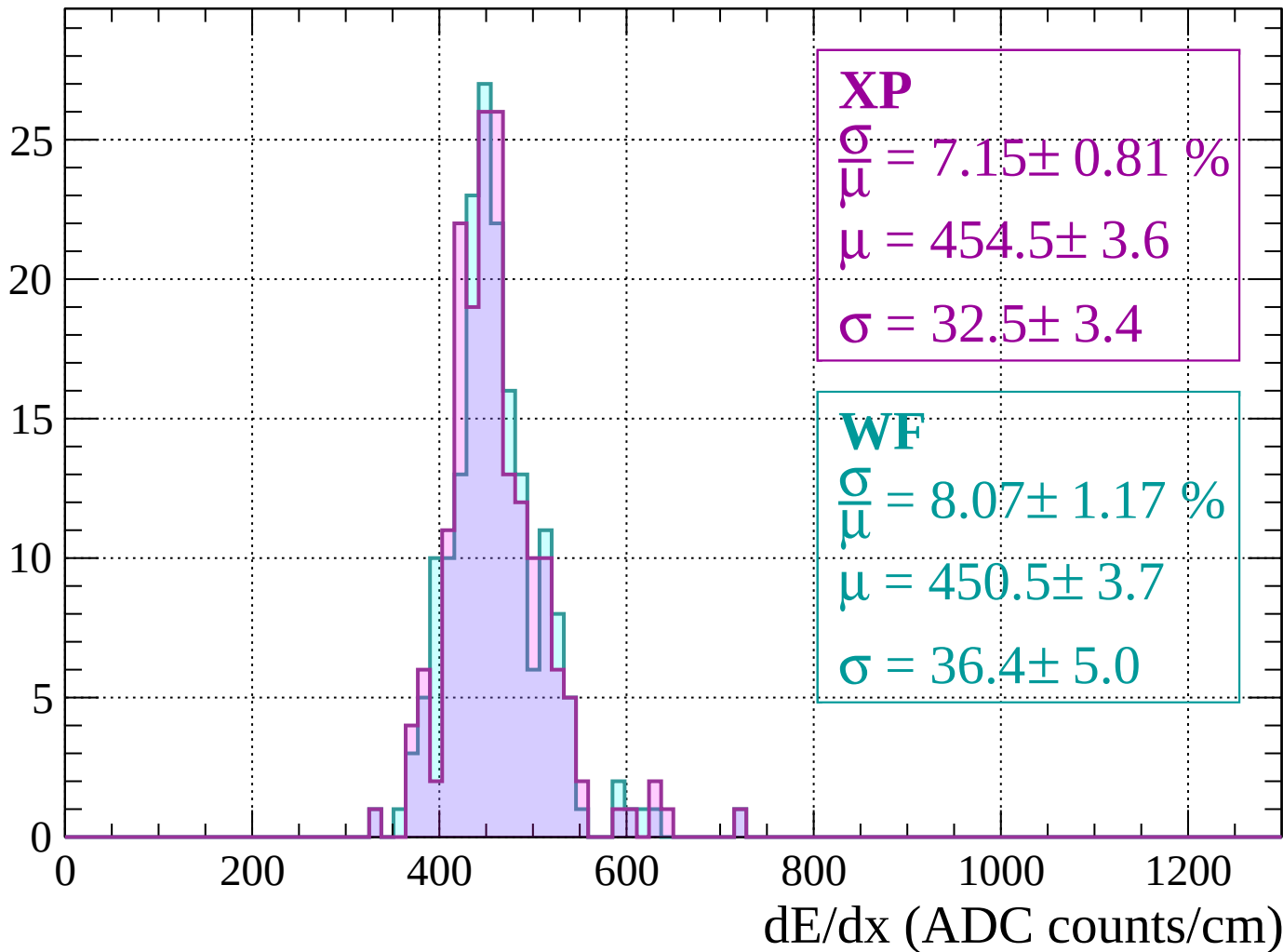
Energy loss | $-1400 < p < -1360$

Count



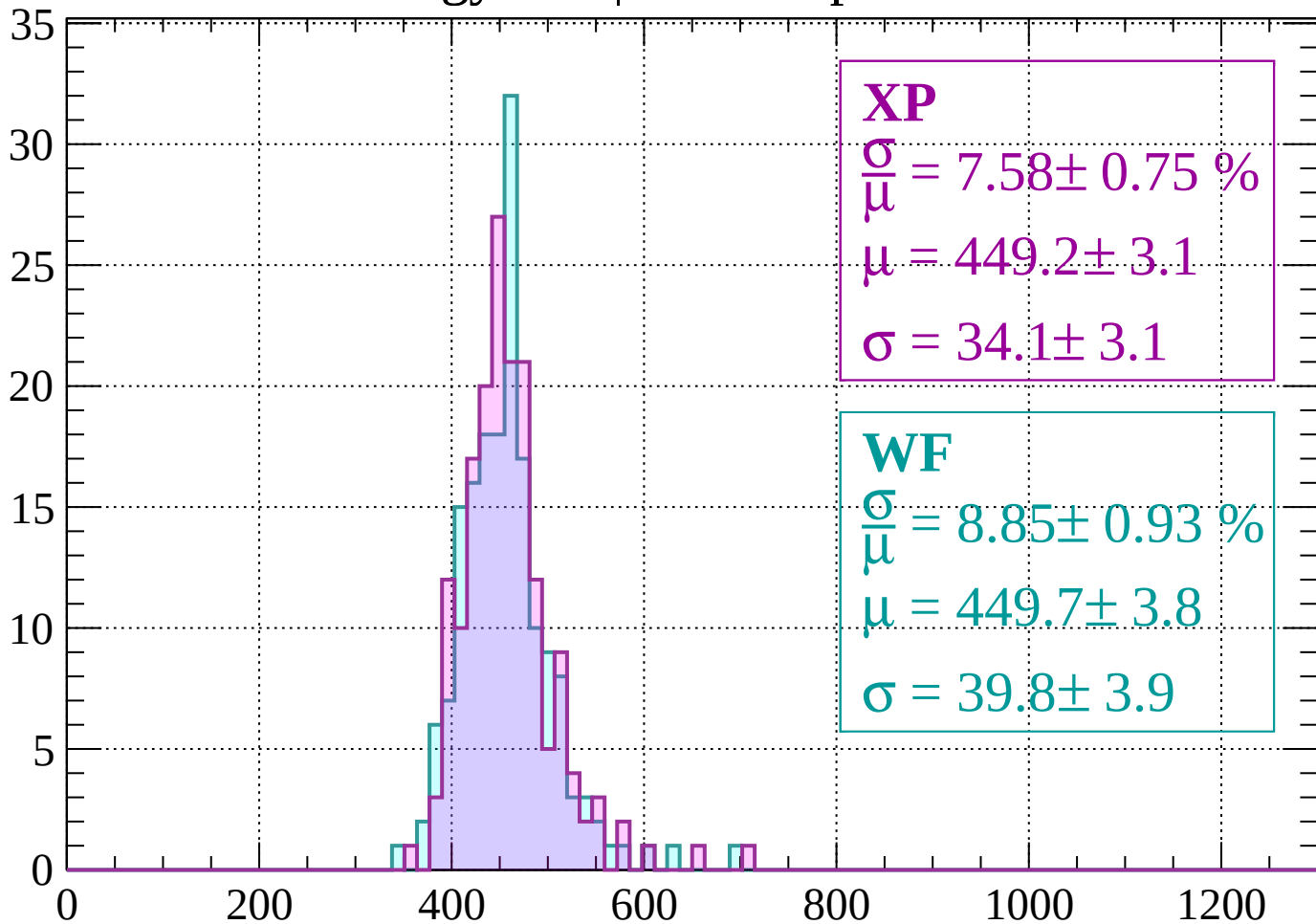
Energy loss | $-1360 < p < -1320$

Count



Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.58 \pm 0.75 \%$$

$$\mu = 449.2 \pm 3.1$$

$$\sigma = 34.1 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 8.85 \pm 0.93 \%$$

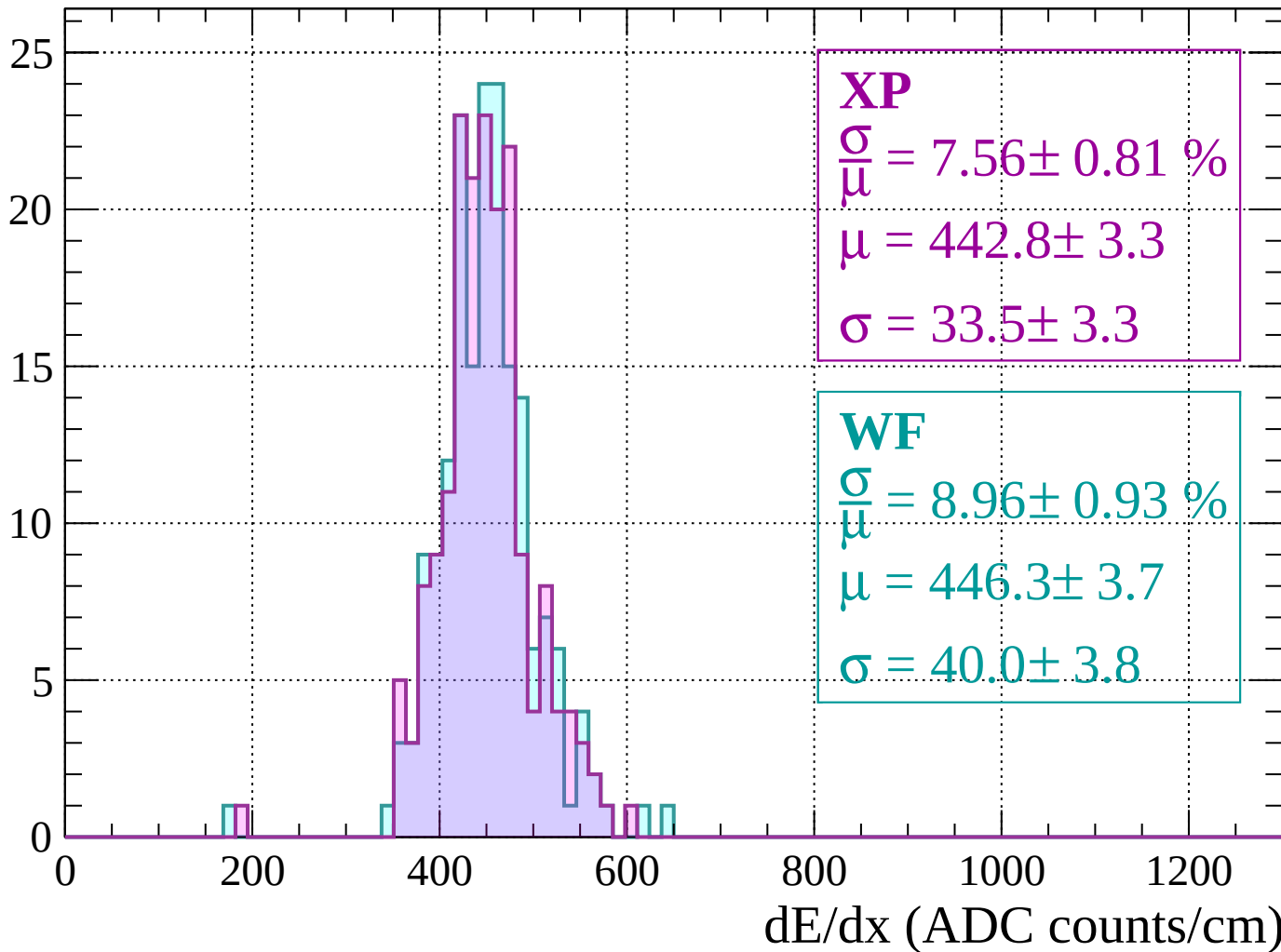
$$\mu = 449.7 \pm 3.8$$

$$\sigma = 39.8 \pm 3.9$$

dE/dx (ADC counts/cm)

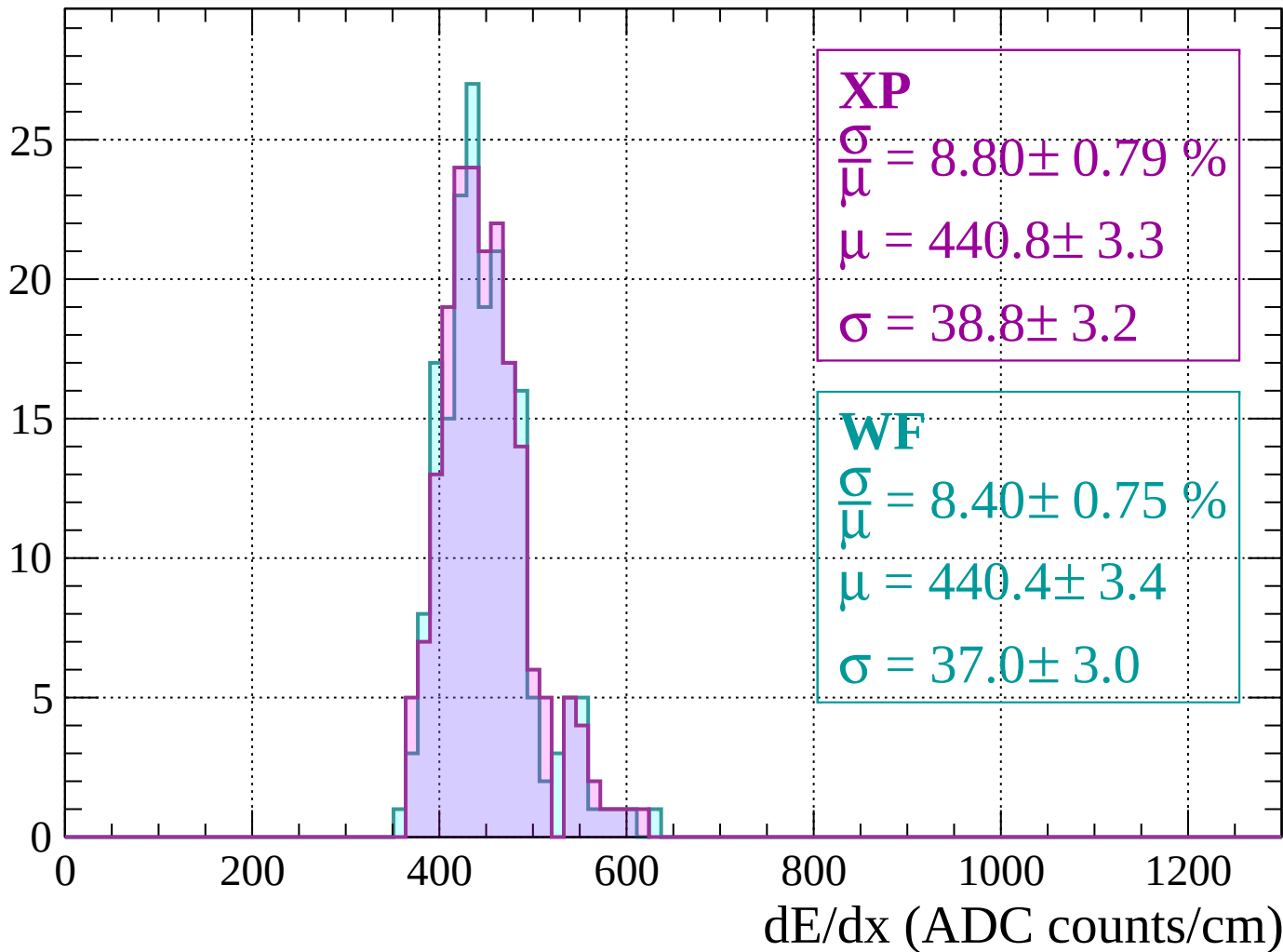
Energy loss | $-1280 < p < -1240$

Count



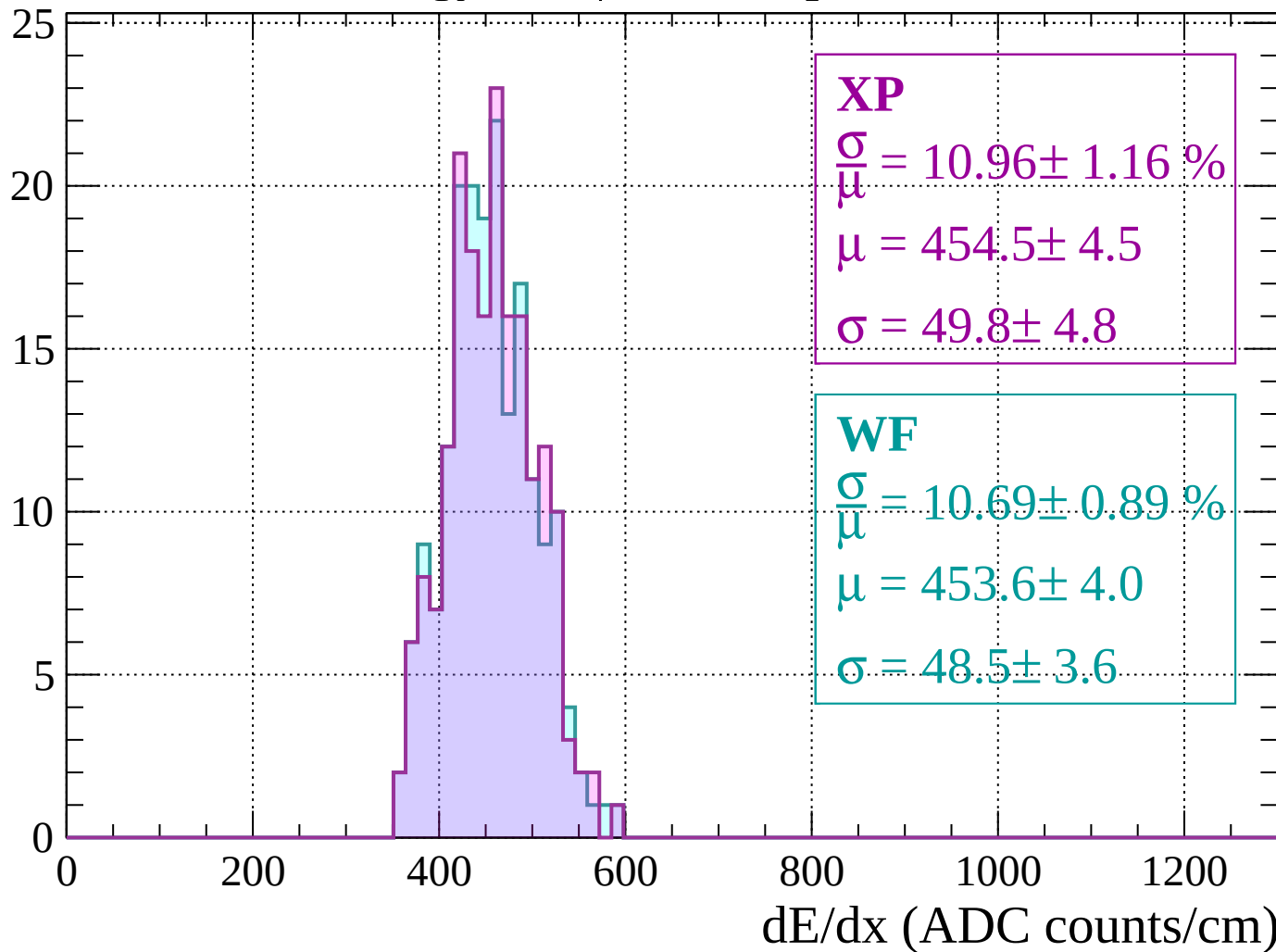
Energy loss | -1240 < p < -1200

Count



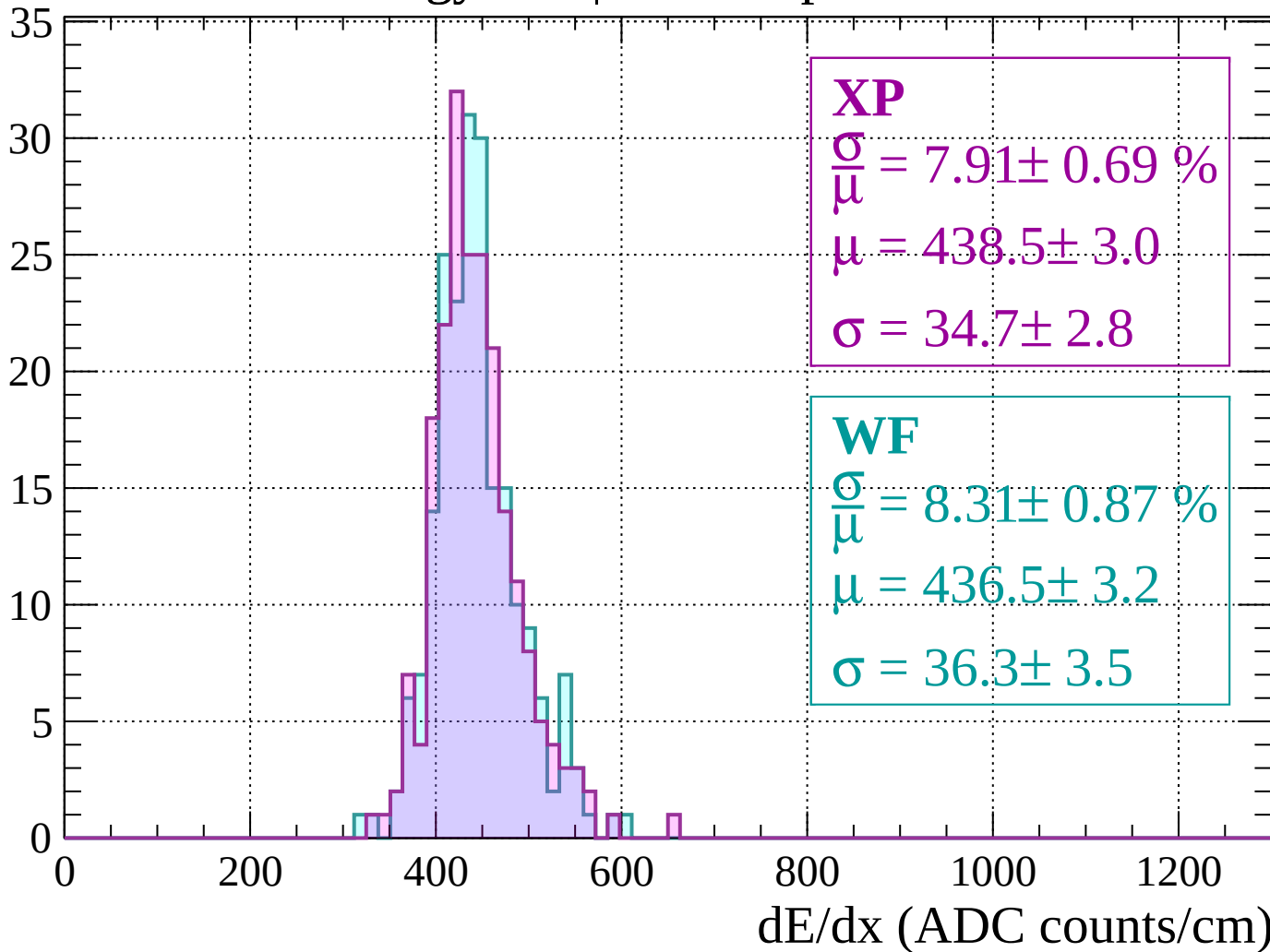
Energy loss | -1200 < p < -1160

Count



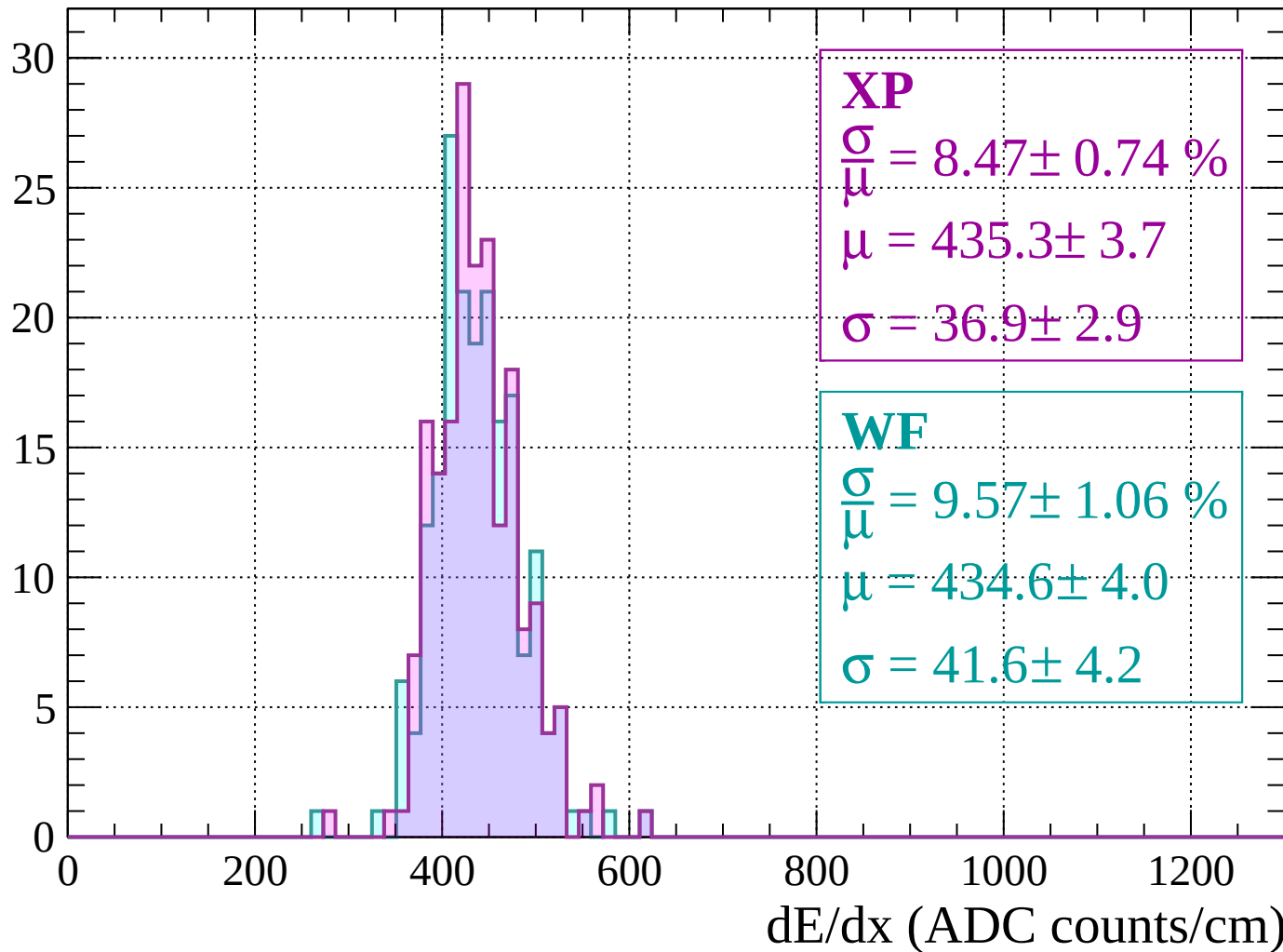
Energy loss | $-1160 < p < -1120$

Count



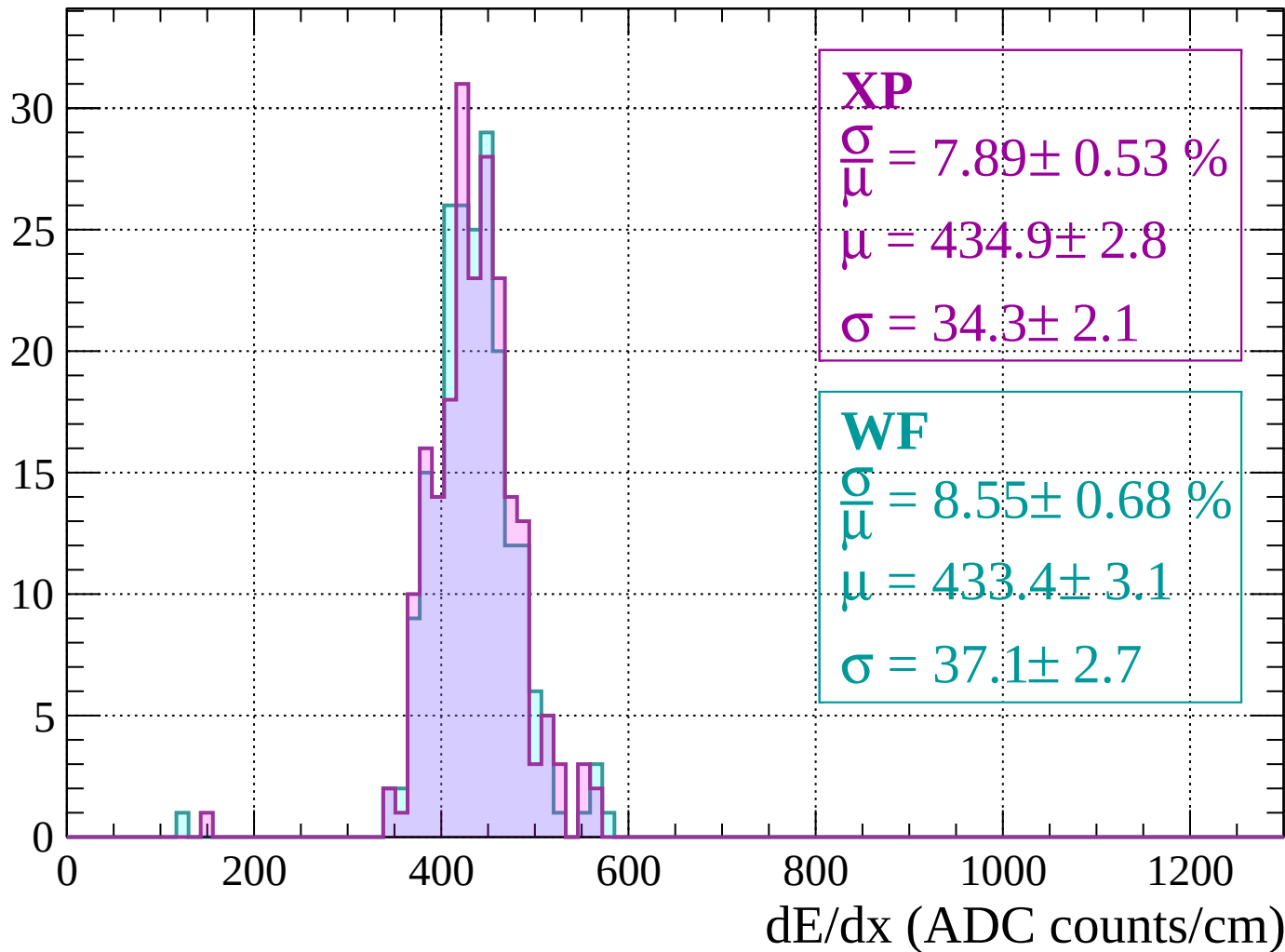
Energy loss | -1120 < p < -1080

Count



Energy loss | $-1080 < p < -1040$

Count



Energy loss | $-1040 < p < -1000$

Count

XP

$$\frac{\sigma}{\mu} = 10.40 \pm 1.40 \%$$

$$\mu = 433.4 \pm 4.5$$

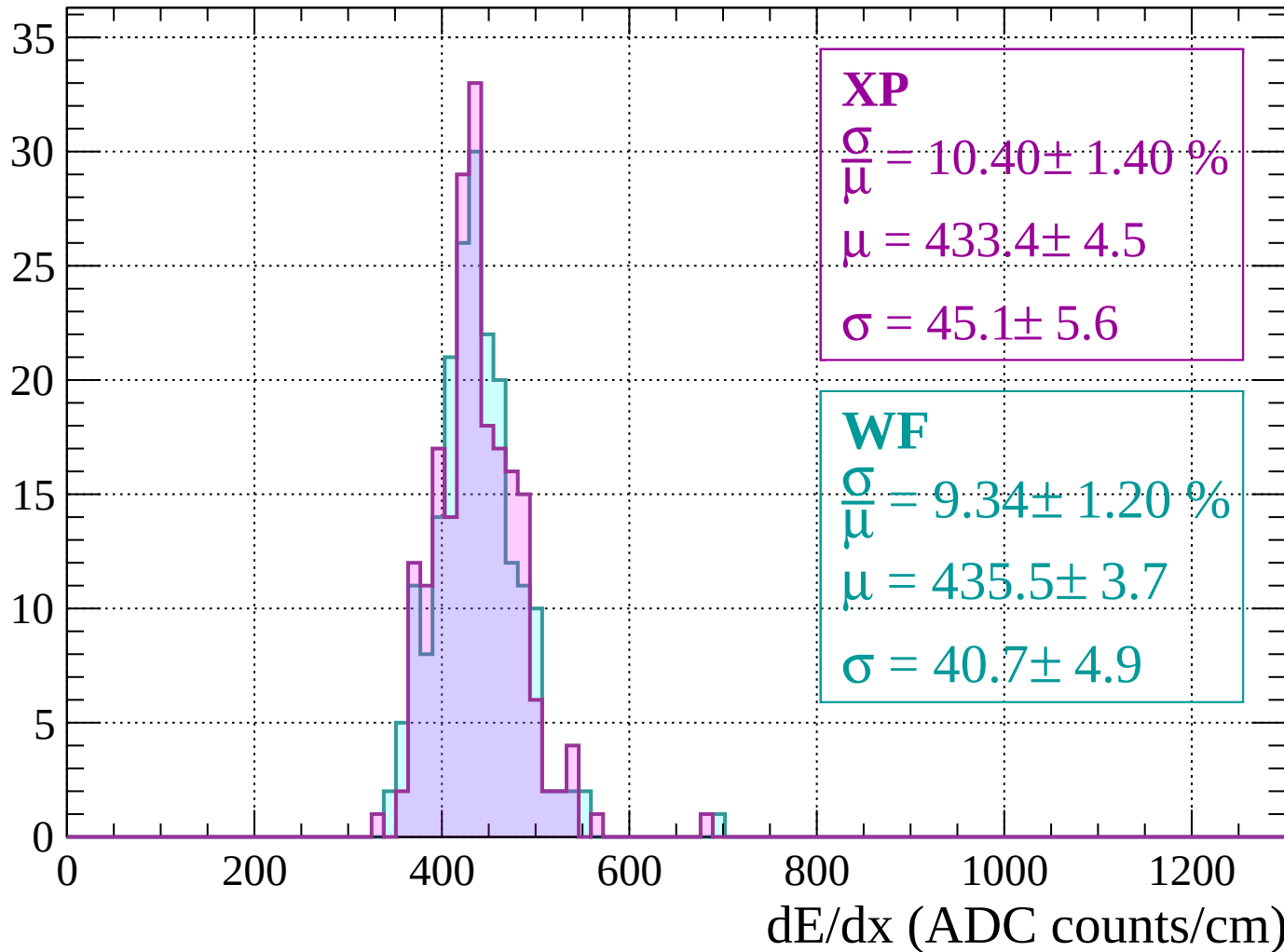
$$\sigma = 45.1 \pm 5.6$$

WF

$$\frac{\sigma}{\mu} = 9.34 \pm 1.20 \%$$

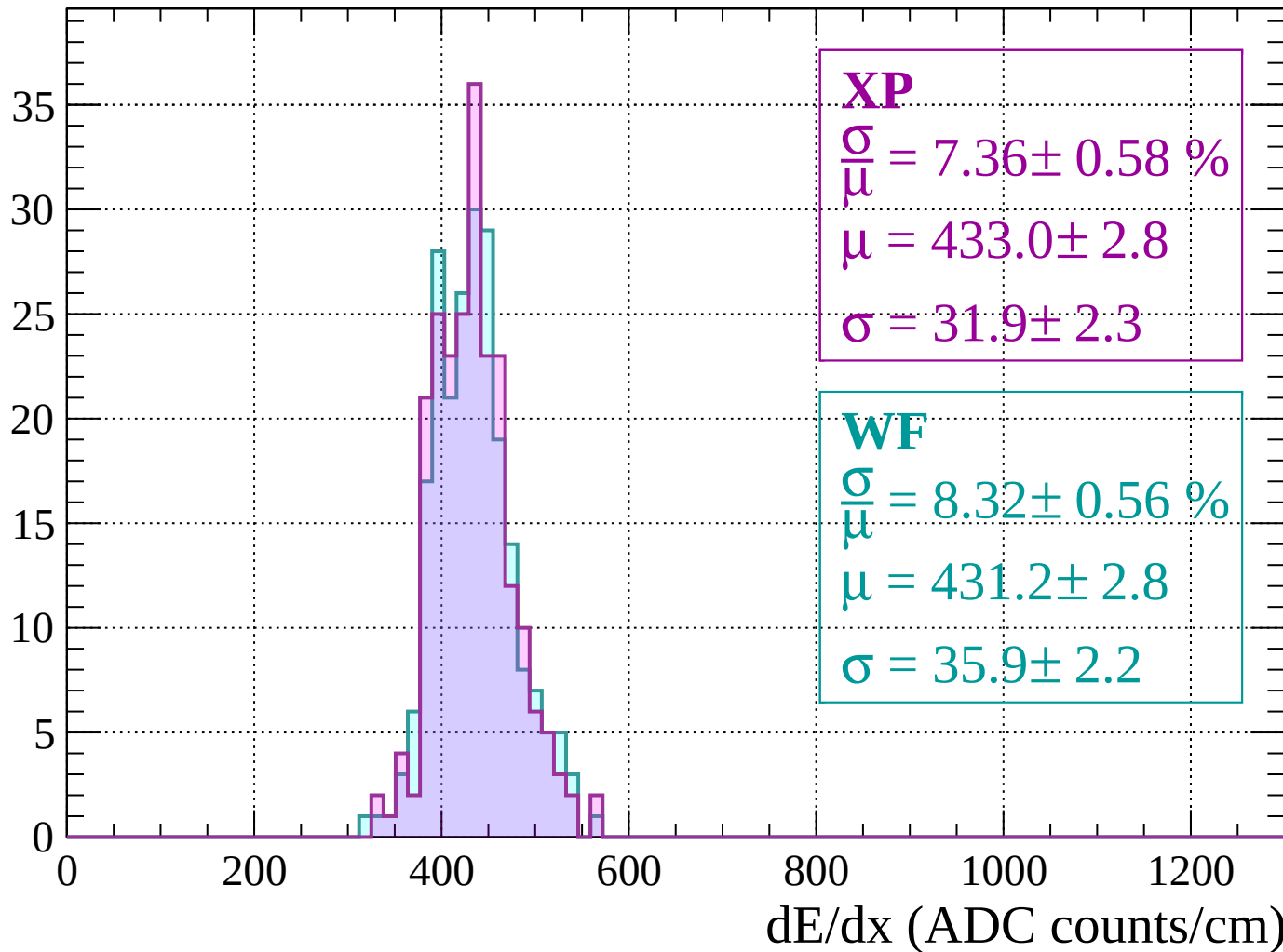
$$\mu = 435.5 \pm 3.7$$

$$\sigma = 40.7 \pm 4.9$$



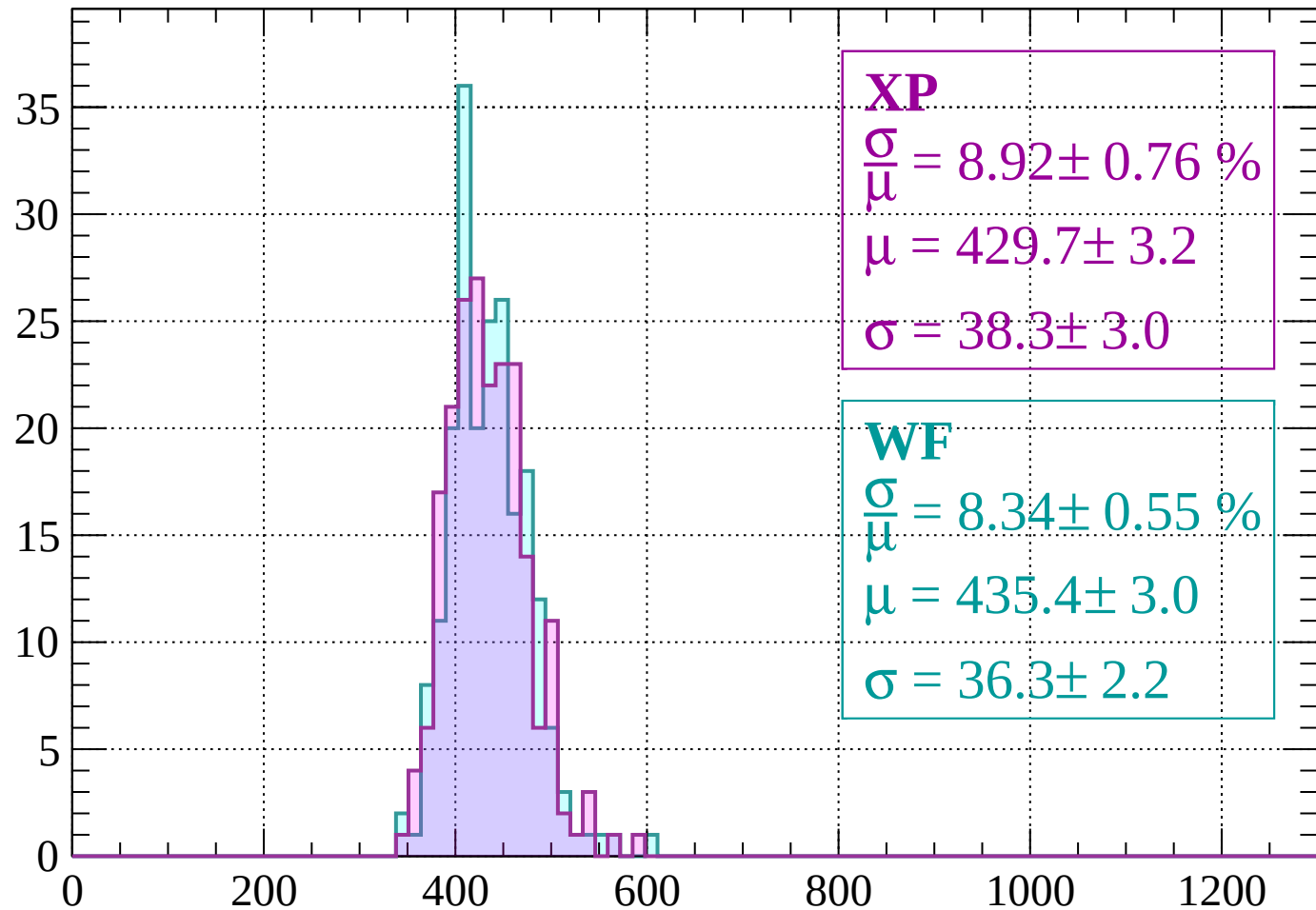
Energy loss | $-1000 < p < -960$

Count



Energy loss | -960 < p < -920

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.92 \pm 0.76 \%$$

$$\mu = 429.7 \pm 3.2$$

$$\sigma = 38.3 \pm 3.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.34 \pm 0.55 \%$$

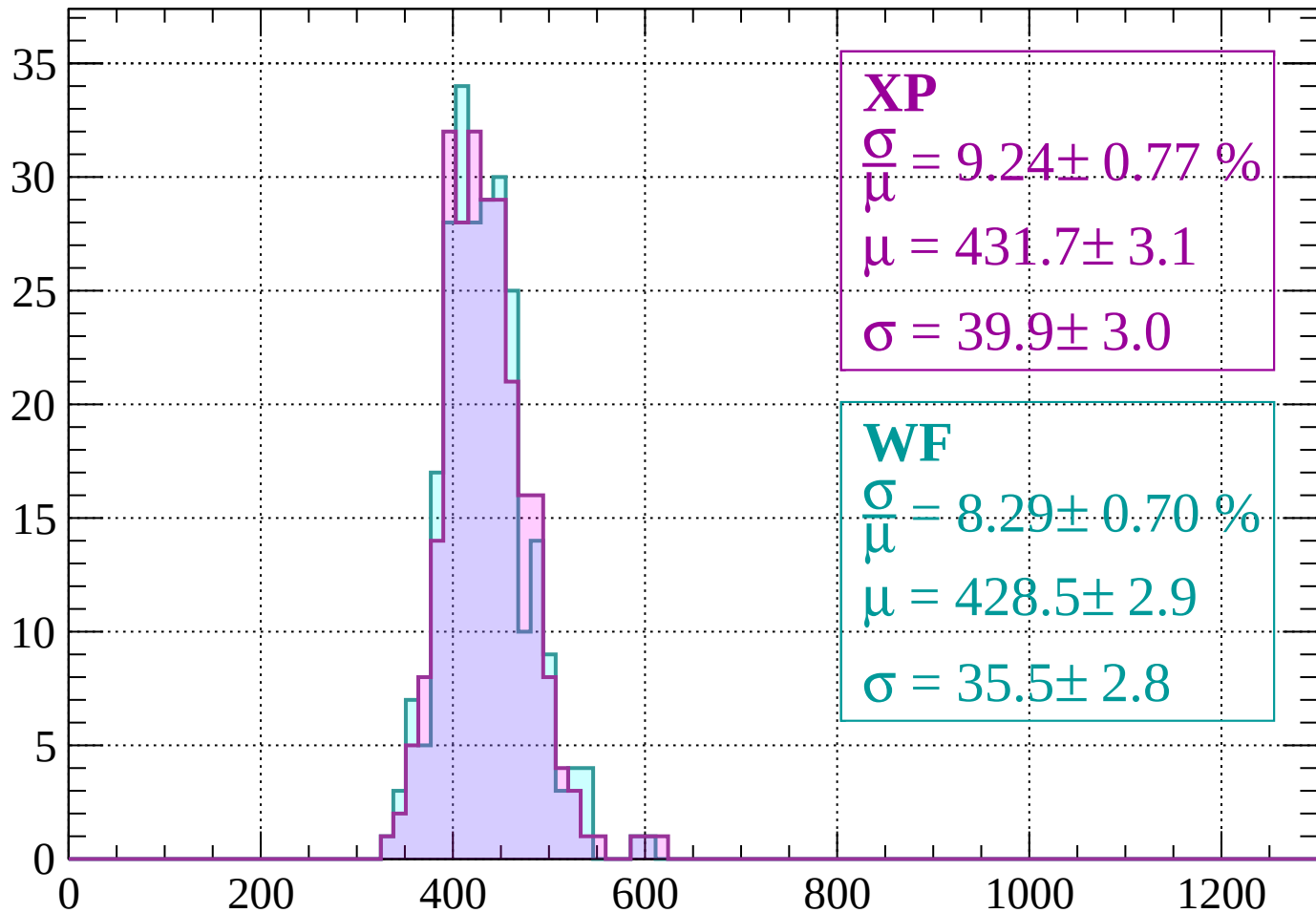
$$\mu = 435.4 \pm 3.0$$

$$\sigma = 36.3 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 9.24 \pm 0.77 \%$$

$$\mu = 431.7 \pm 3.1$$

$$\sigma = 39.9 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 8.29 \pm 0.70 \%$$

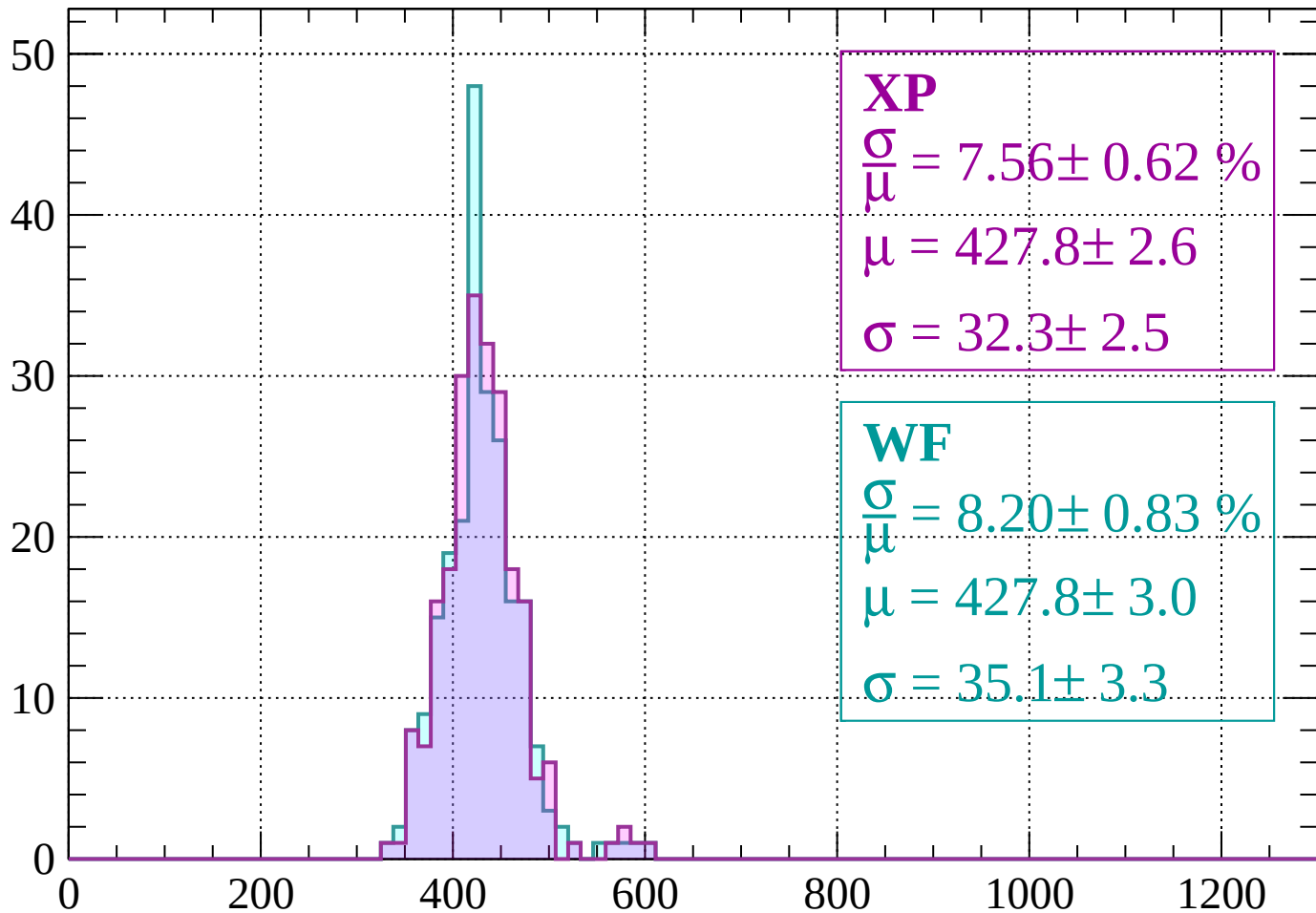
$$\mu = 428.5 \pm 2.9$$

$$\sigma = 35.5 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 7.56 \pm 0.62 \%$$

$$\mu = 427.8 \pm 2.6$$

$$\sigma = 32.3 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.20 \pm 0.83 \%$$

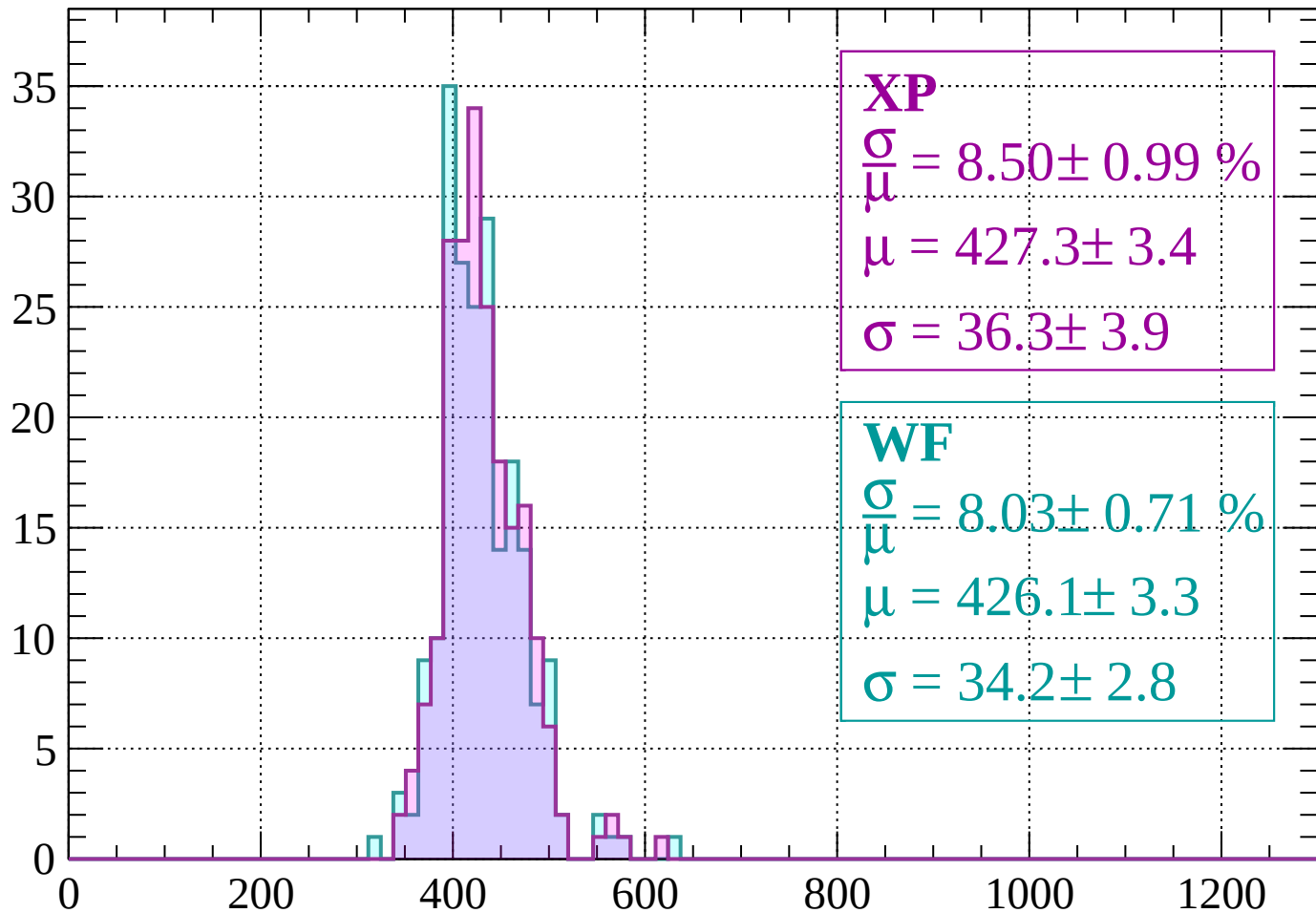
$$\mu = 427.8 \pm 3.0$$

$$\sigma = 35.1 \pm 3.3$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 8.50 \pm 0.99 \%$$

$$\mu = 427.3 \pm 3.4$$

$$\sigma = 36.3 \pm 3.9$$

WF

$$\frac{\sigma}{\mu} = 8.03 \pm 0.71 \%$$

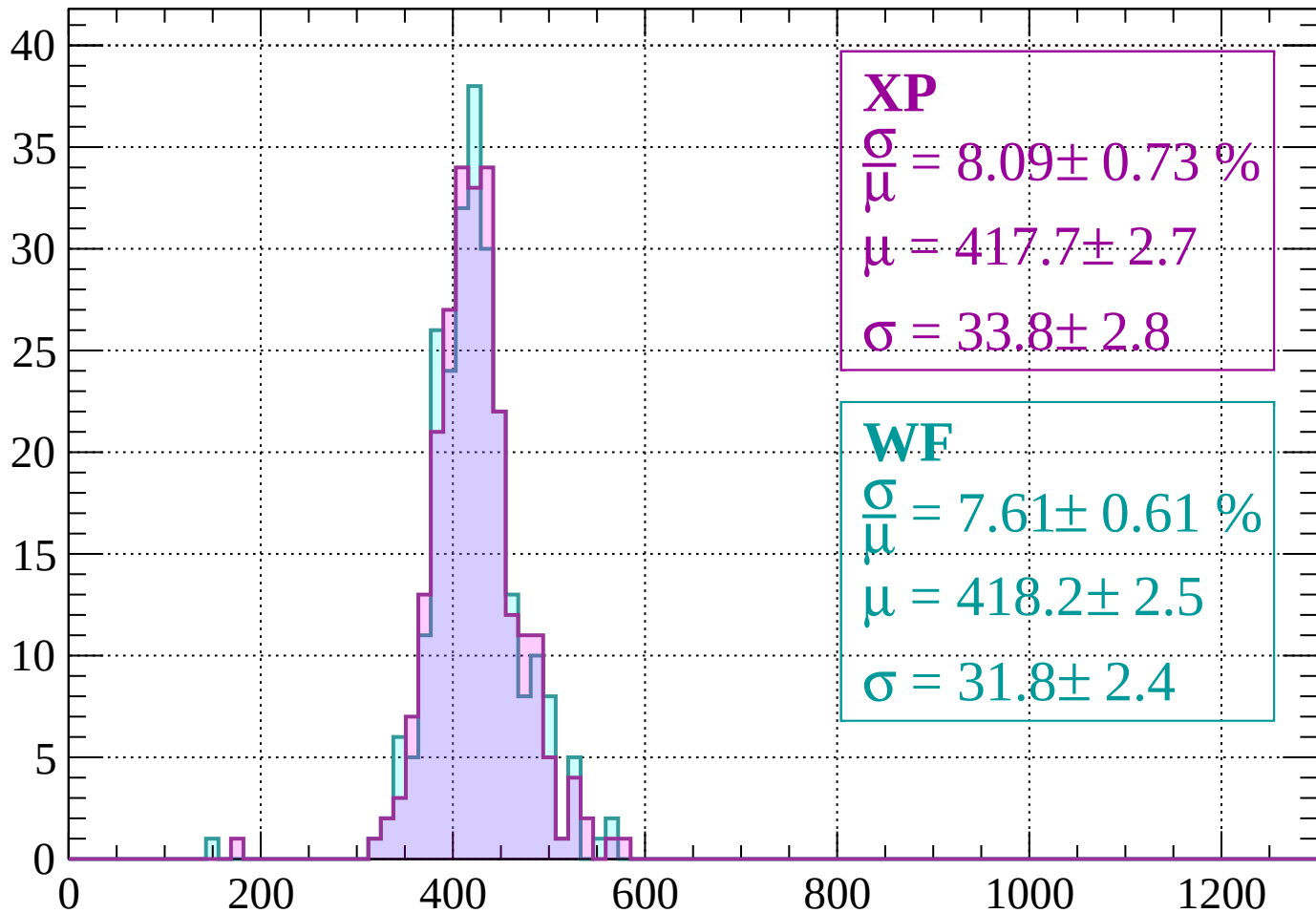
$$\mu = 426.1 \pm 3.3$$

$$\sigma = 34.2 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 8.09 \pm 0.73 \%$$

$$\mu = 417.7 \pm 2.7$$

$$\sigma = 33.8 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.61 \%$$

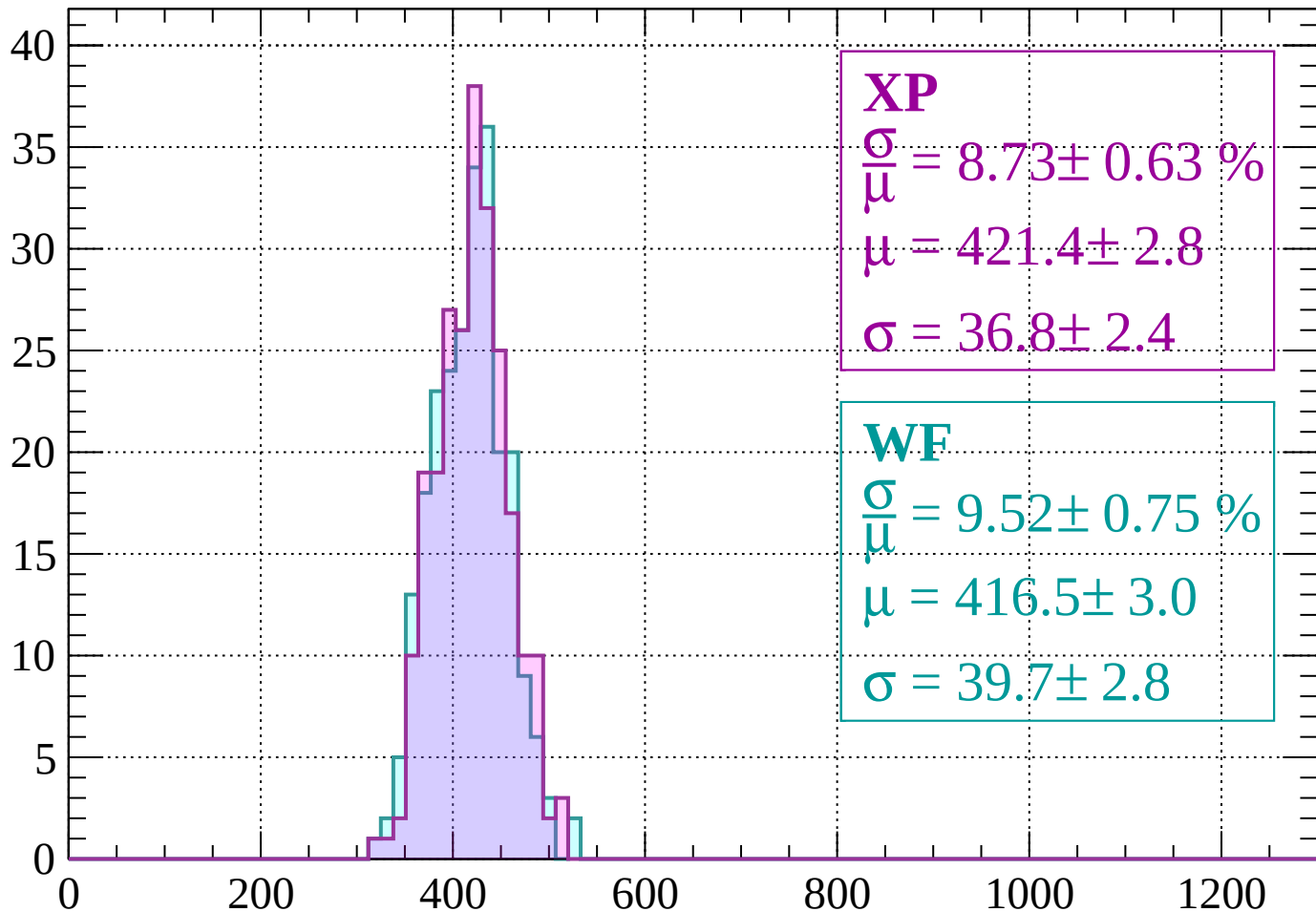
$$\mu = 418.2 \pm 2.5$$

$$\sigma = 31.8 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 8.73 \pm 0.63 \%$$

$$\mu = 421.4 \pm 2.8$$

$$\sigma = 36.8 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 9.52 \pm 0.75 \%$$

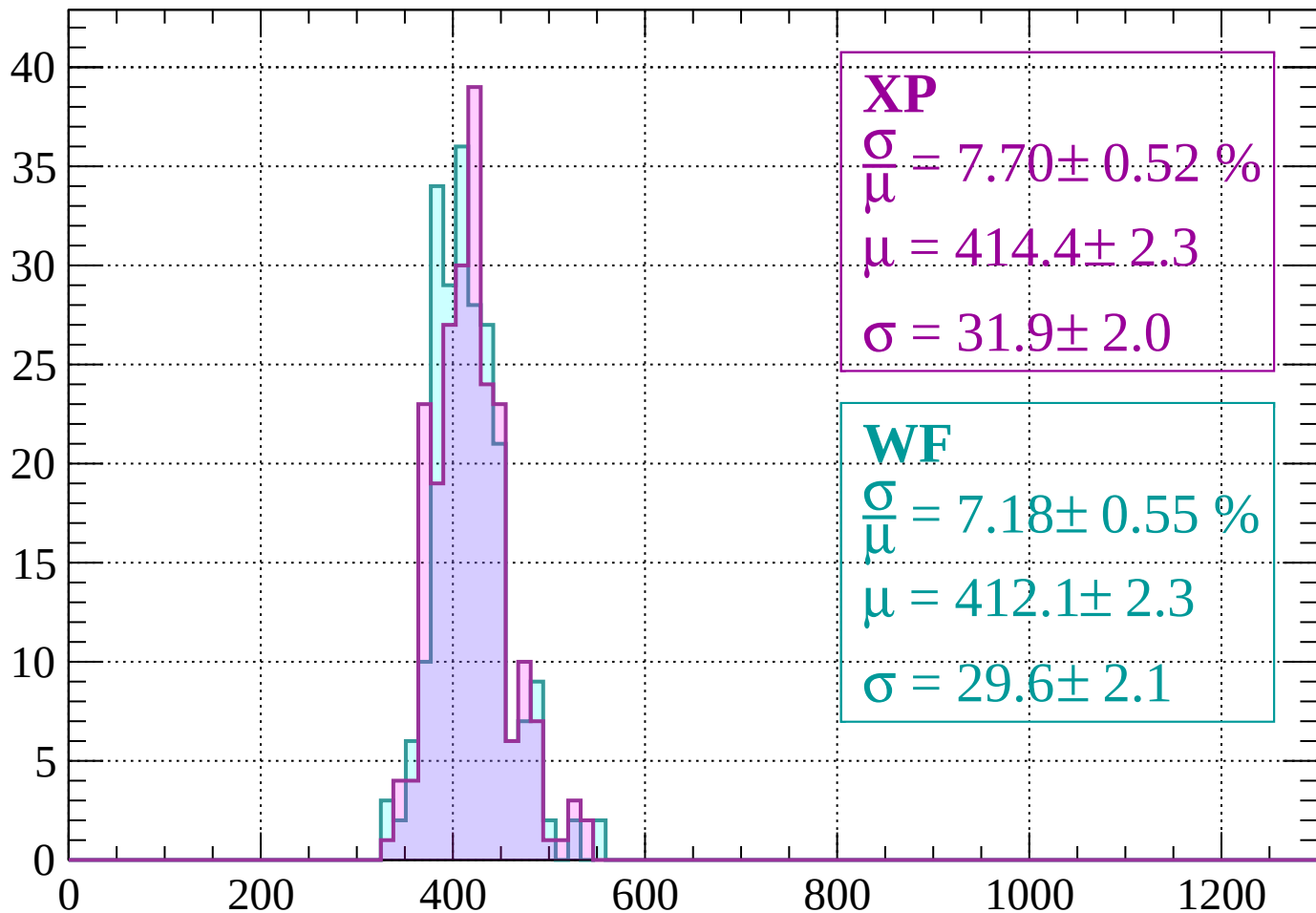
$$\mu = 416.5 \pm 3.0$$

$$\sigma = 39.7 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 7.70 \pm 0.52 \%$$

$$\mu = 414.4 \pm 2.3$$

$$\sigma = 31.9 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 7.18 \pm 0.55 \%$$

$$\mu = 412.1 \pm 2.3$$

$$\sigma = 29.6 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count

XP

$$\frac{\sigma}{\mu} = 7.51 \pm 0.75 \%$$

$$\mu = 413.3 \pm 2.4$$

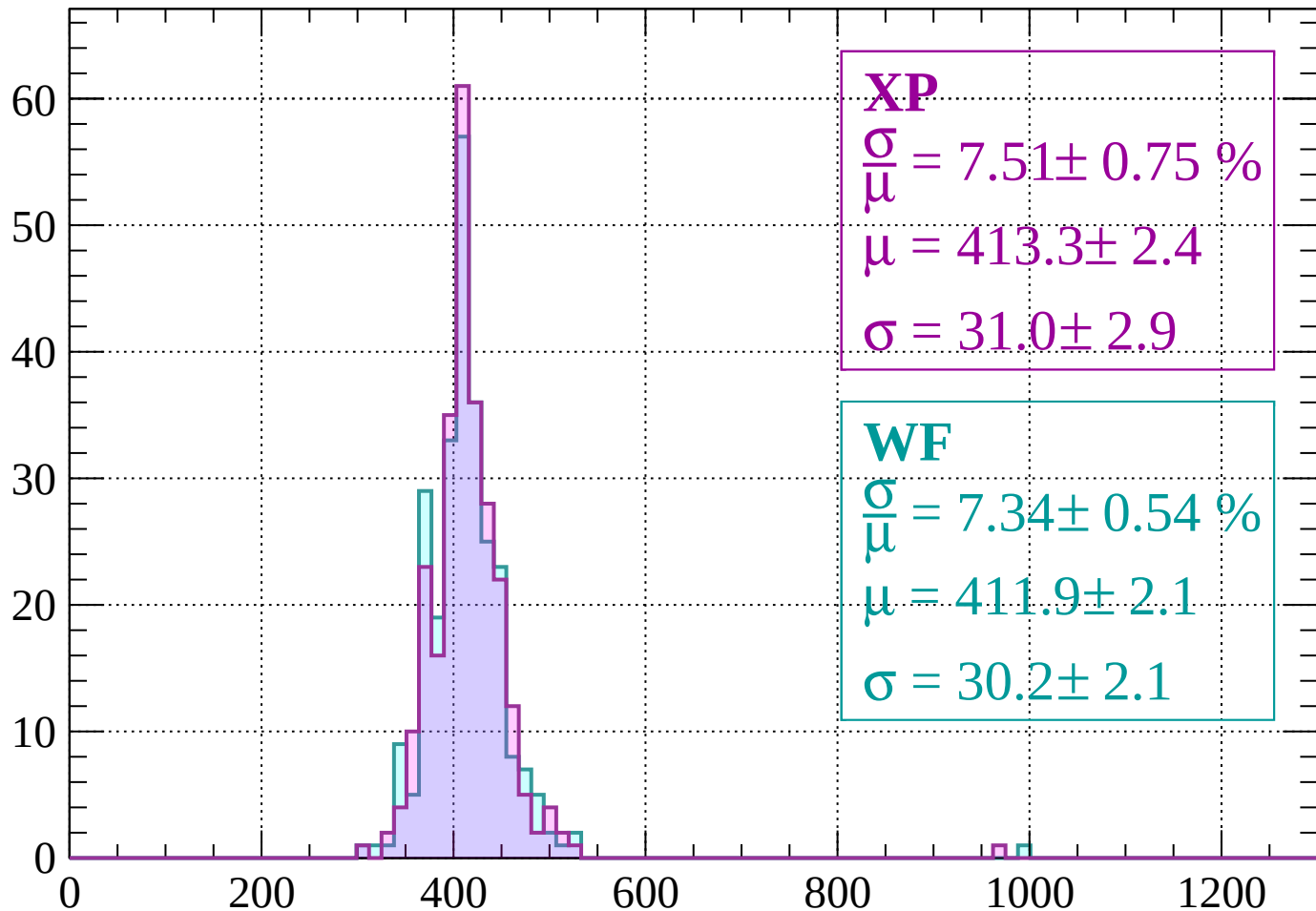
$$\sigma = 31.0 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 7.34 \pm 0.54 \%$$

$$\mu = 411.9 \pm 2.1$$

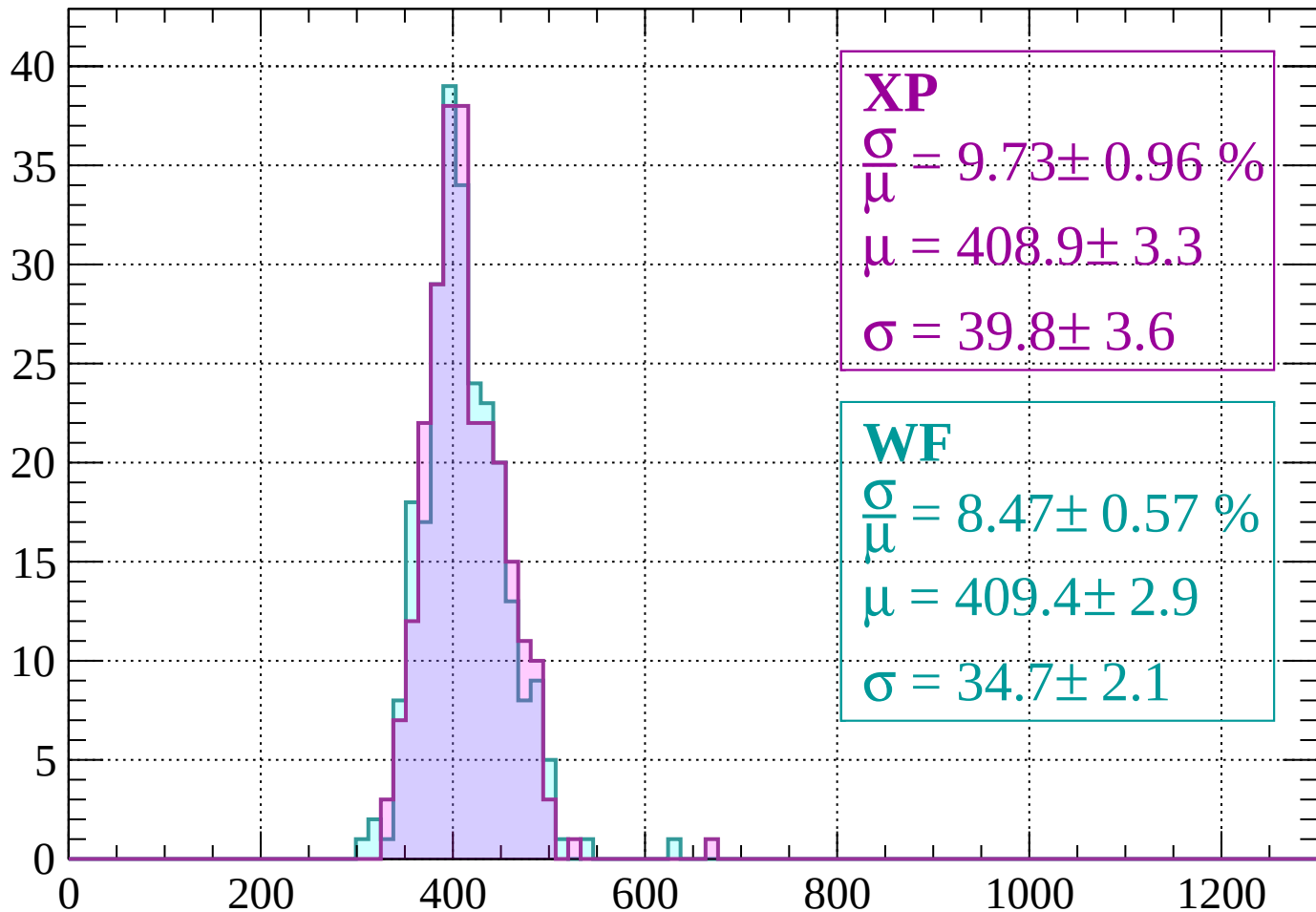
$$\sigma = 30.2 \pm 2.1$$



dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.73 \pm 0.96 \%$$

$$\mu = 408.9 \pm 3.3$$

$$\sigma = 39.8 \pm 3.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.47 \pm 0.57 \%$$

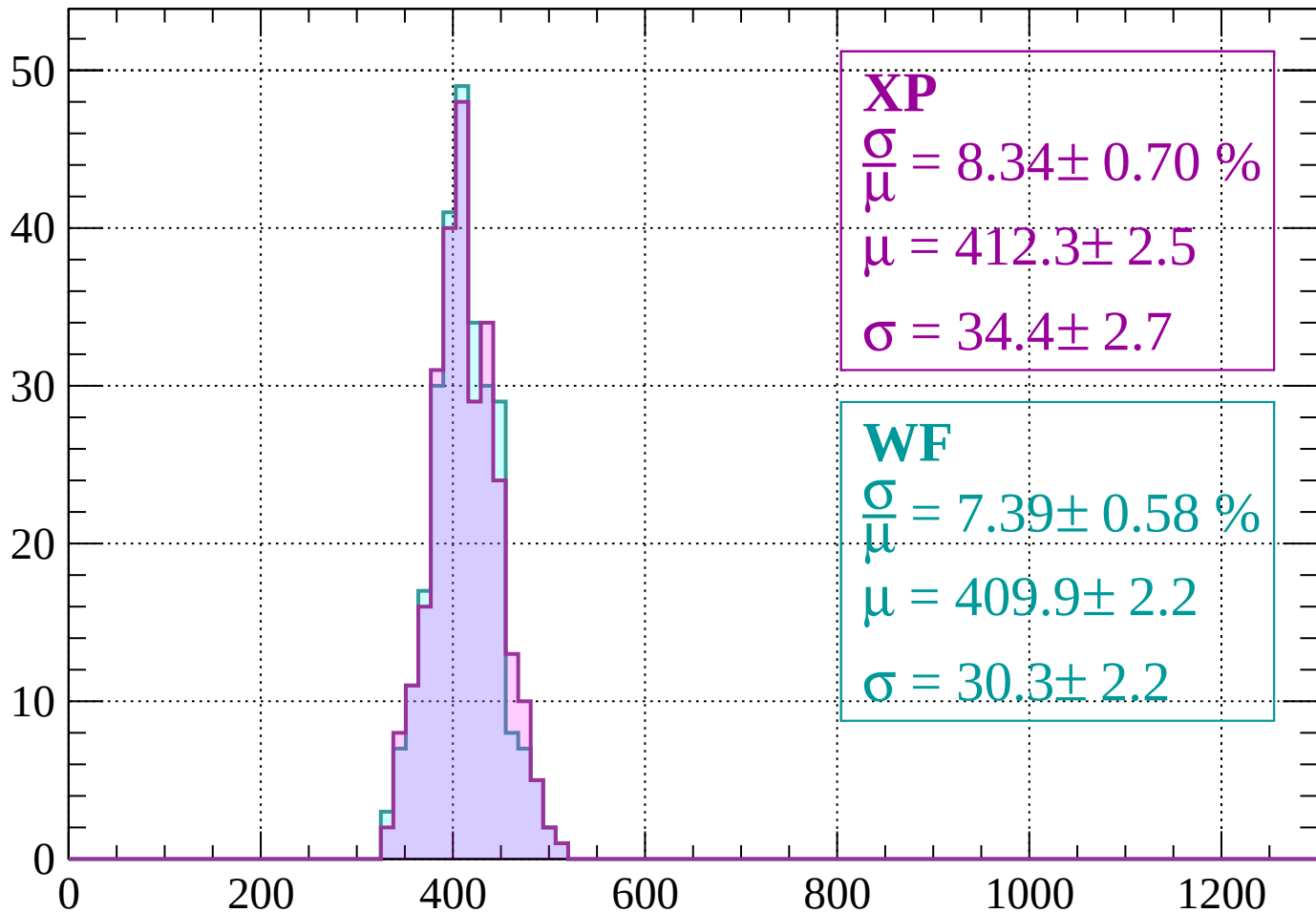
$$\mu = 409.4 \pm 2.9$$

$$\sigma = 34.7 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.34 \pm 0.70 \%$$

$$\mu = 412.3 \pm 2.5$$

$$\sigma = 34.4 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.58 \%$$

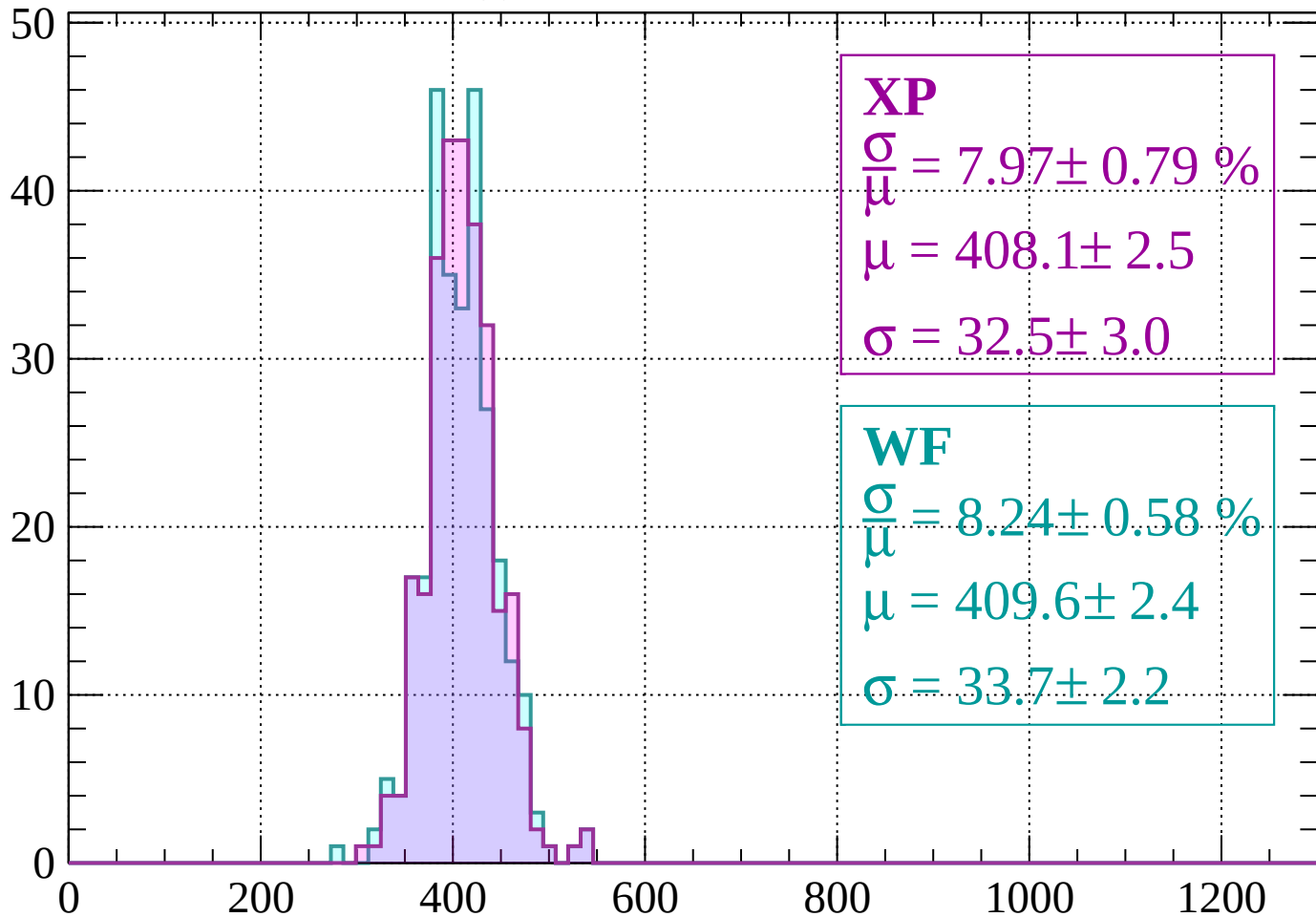
$$\mu = 409.9 \pm 2.2$$

$$\sigma = 30.3 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 7.97 \pm 0.79 \%$$

$$\mu = 408.1 \pm 2.5$$

$$\sigma = 32.5 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 8.24 \pm 0.58 \%$$

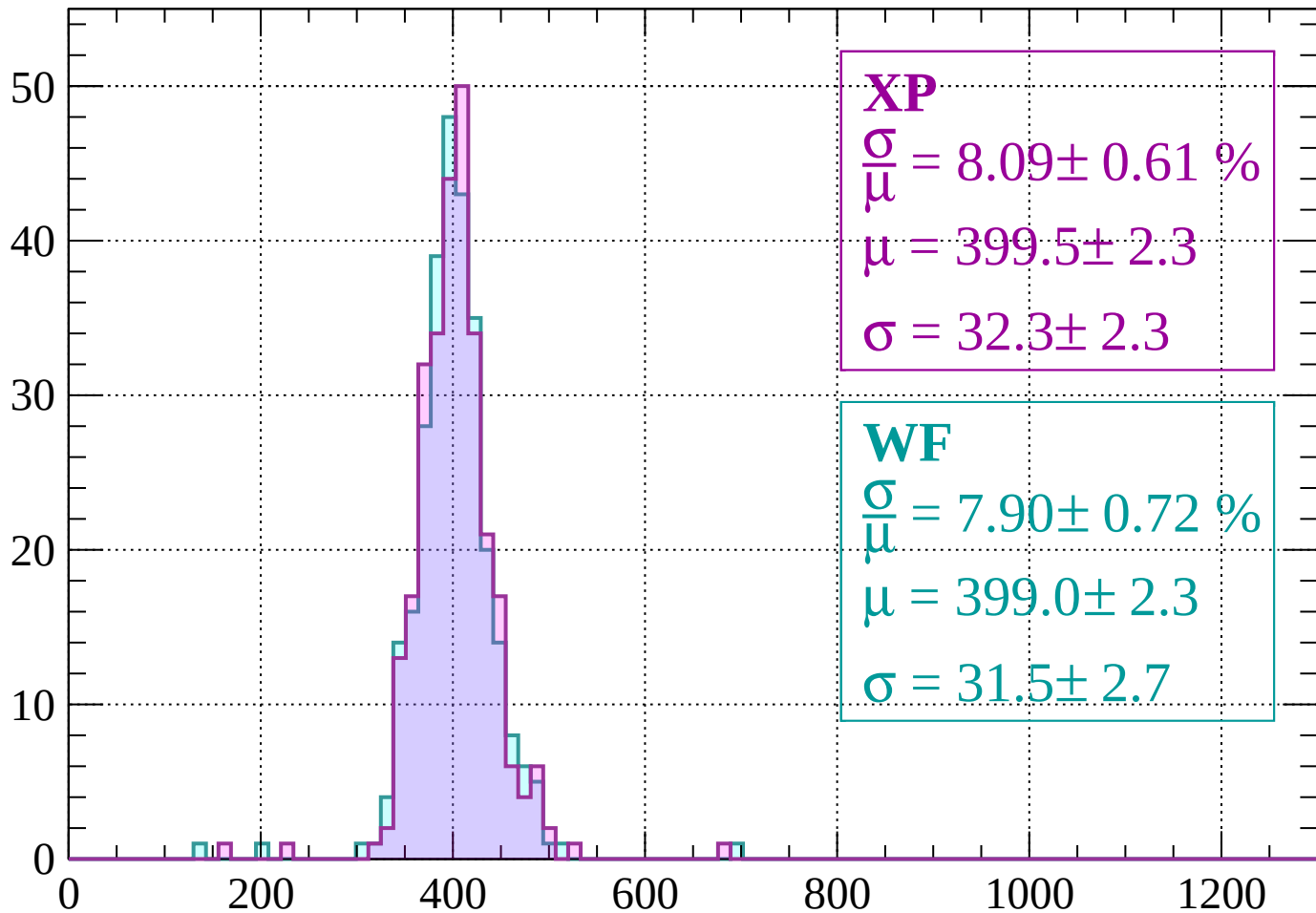
$$\mu = 409.6 \pm 2.4$$

$$\sigma = 33.7 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.09 \pm 0.61 \%$$

$$\mu = 399.5 \pm 2.3$$

$$\sigma = 32.3 \pm 2.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.90 \pm 0.72 \%$$

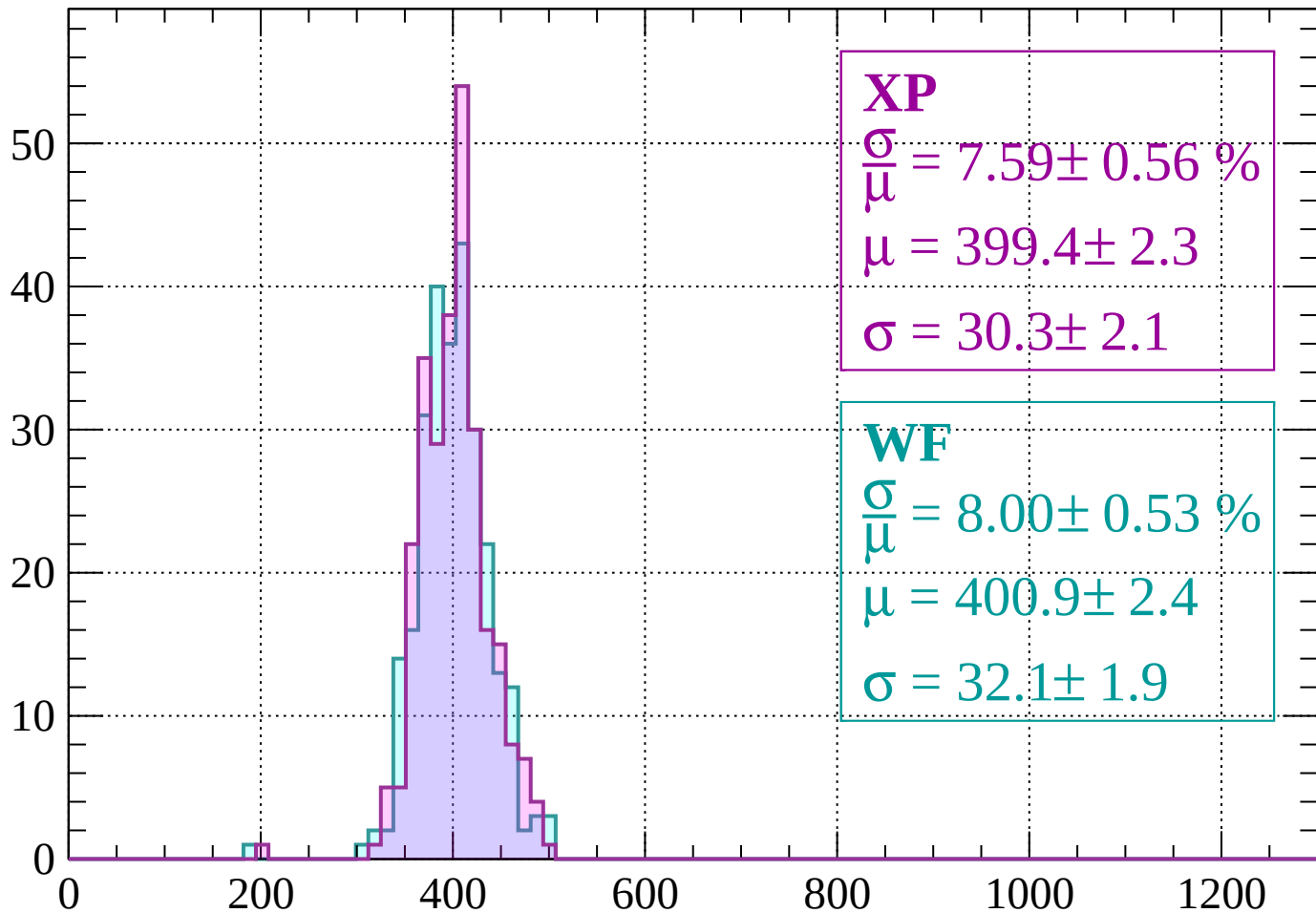
$$\mu = 399.0 \pm 2.3$$

$$\sigma = 31.5 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 7.59 \pm 0.56 \%$$

$$\mu = 399.4 \pm 2.3$$

$$\sigma = 30.3 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 8.00 \pm 0.53 \%$$

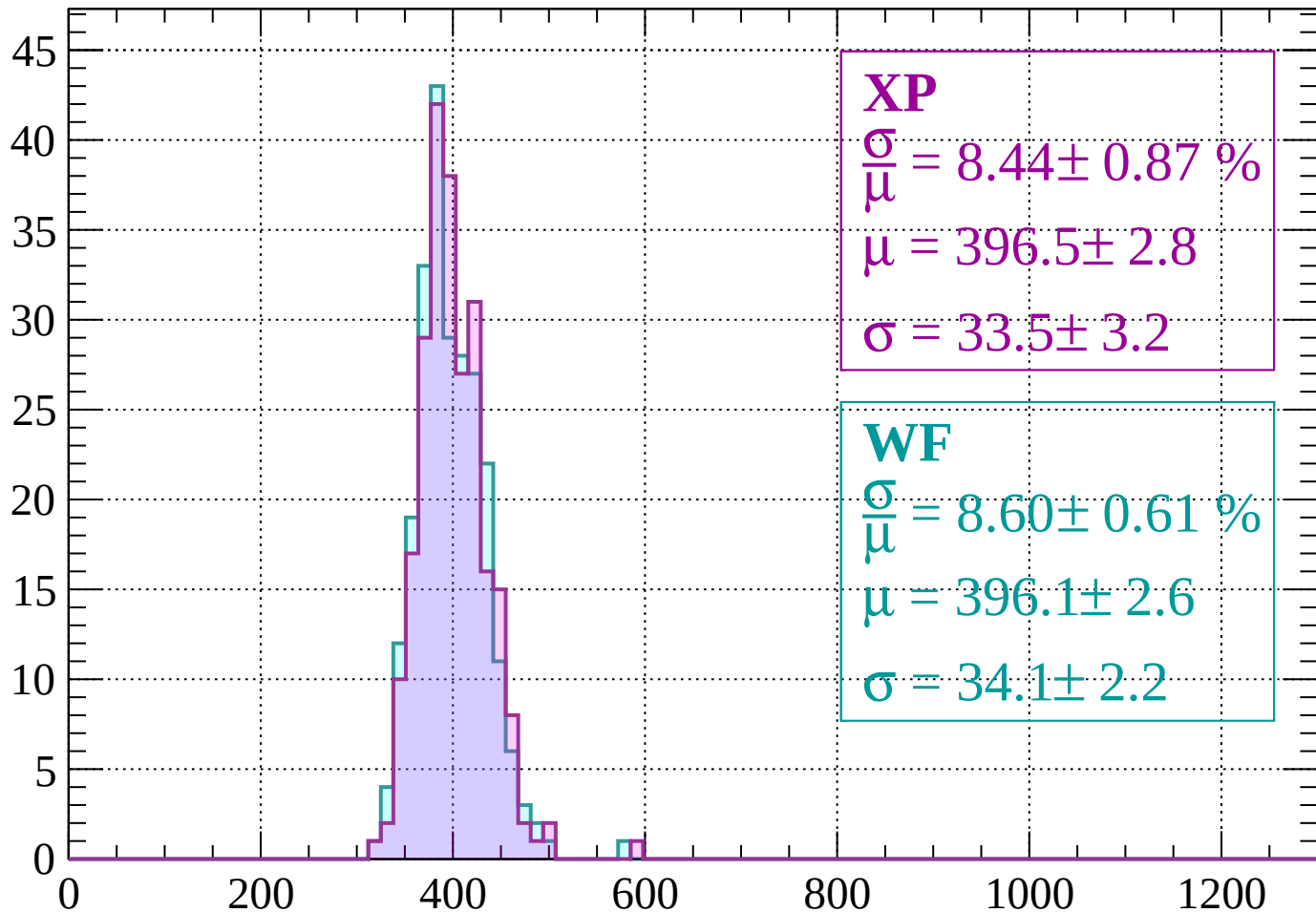
$$\mu = 400.9 \pm 2.4$$

$$\sigma = 32.1 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 8.44 \pm 0.87 \%$$

$$\mu = 396.5 \pm 2.8$$

$$\sigma = 33.5 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 8.60 \pm 0.61 \%$$

$$\mu = 396.1 \pm 2.6$$

$$\sigma = 34.1 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count

XP

$$\frac{\sigma}{\mu} = 8.51 \pm 0.56 \%$$

$$\mu = 401.8 \pm 2.4$$

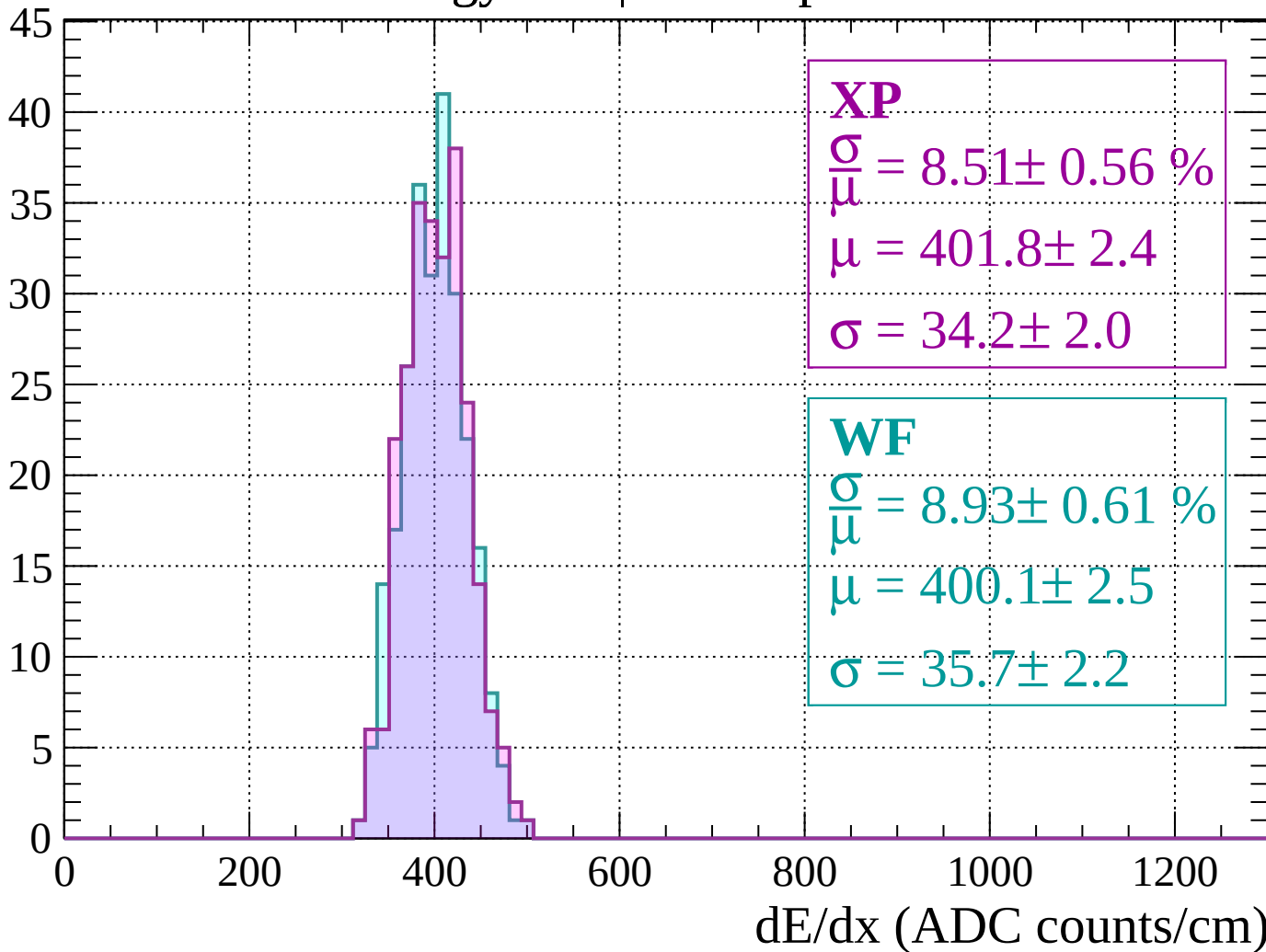
$$\sigma = 34.2 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 8.93 \pm 0.61 \%$$

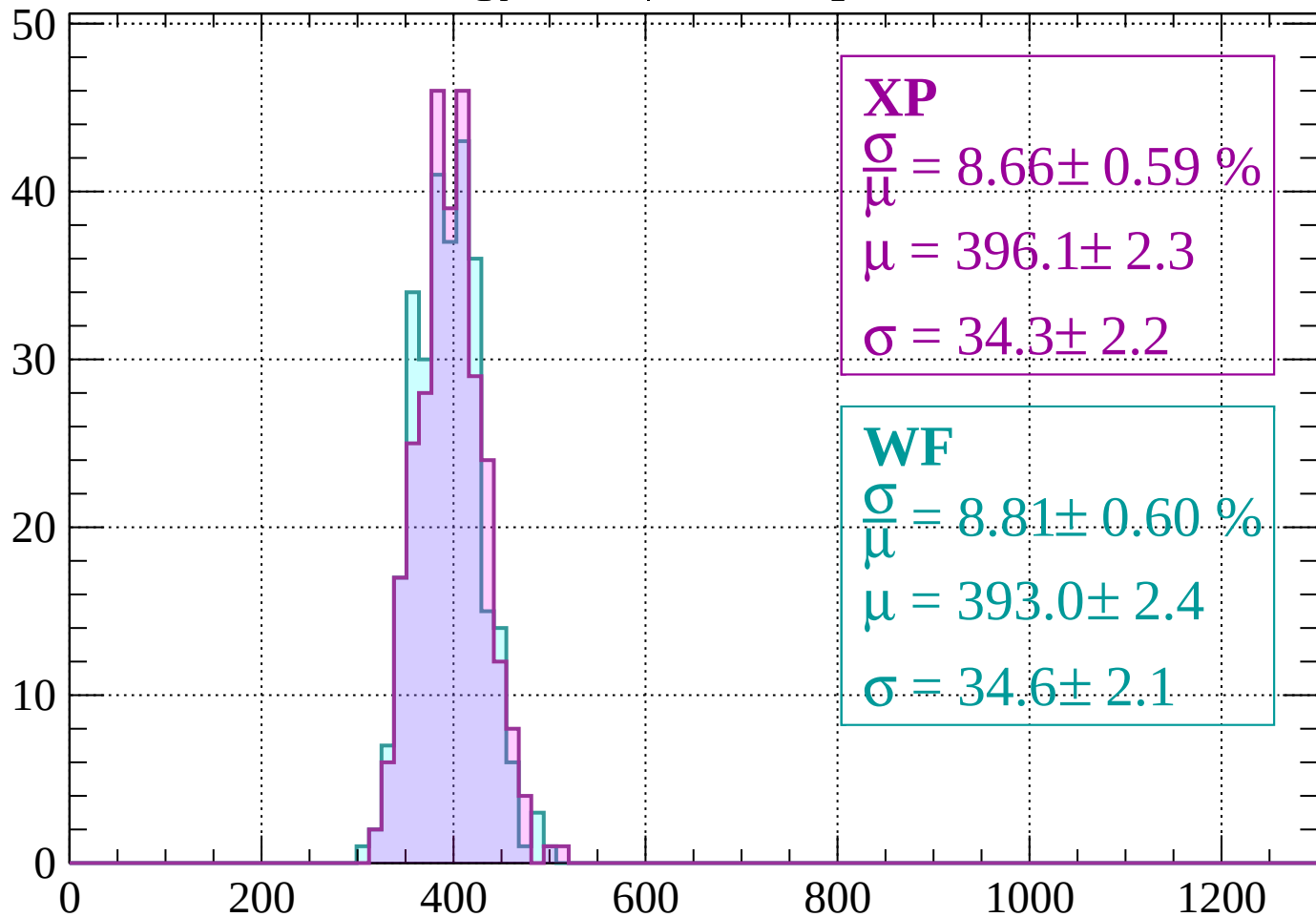
$$\mu = 400.1 \pm 2.5$$

$$\sigma = 35.7 \pm 2.2$$



Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.59 \%$$

$$\mu = 396.1 \pm 2.3$$

$$\sigma = 34.3 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.81 \pm 0.60 \%$$

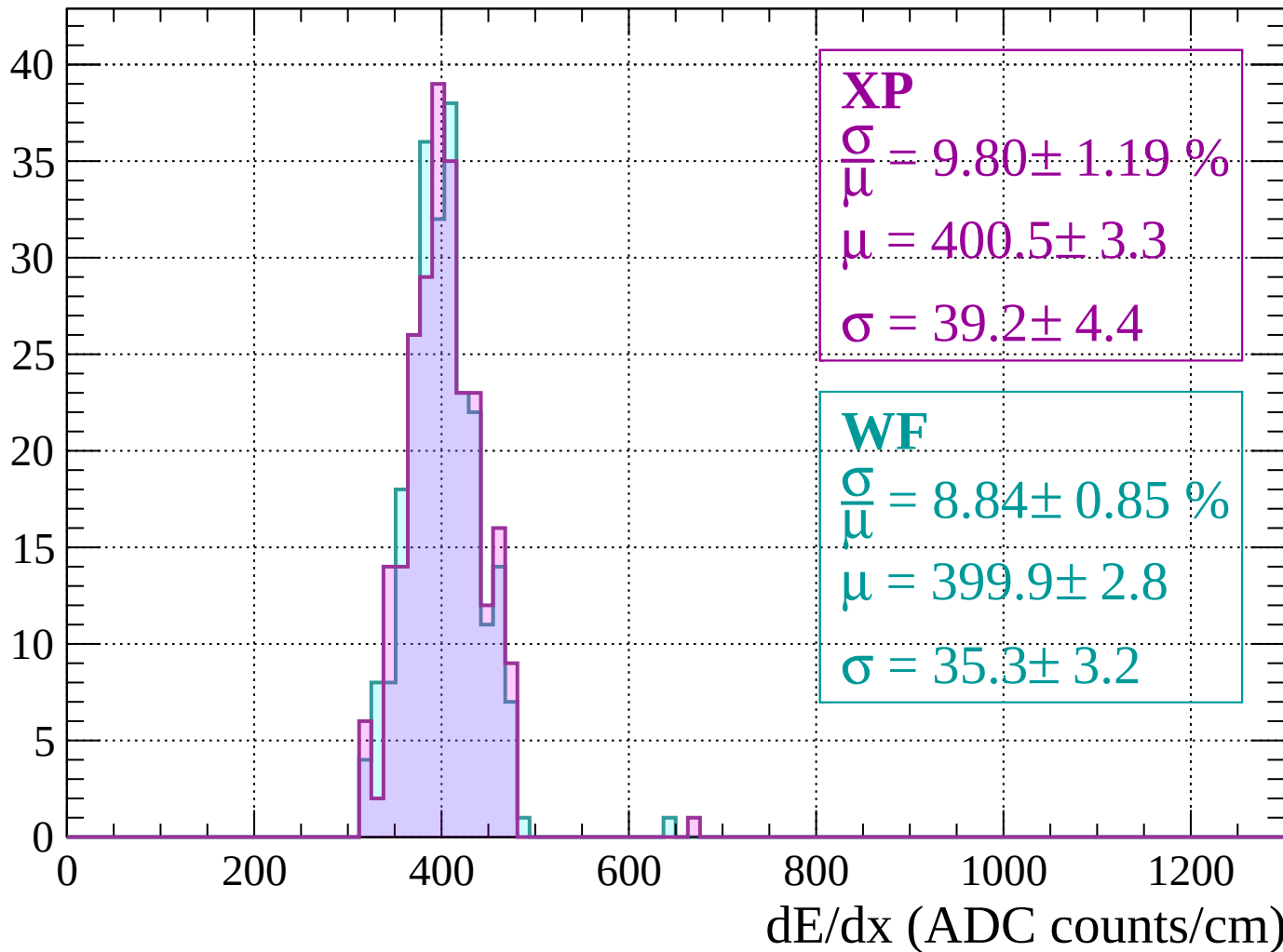
$$\mu = 393.0 \pm 2.4$$

$$\sigma = 34.6 \pm 2.1$$

dE/dx (ADC counts/cm)

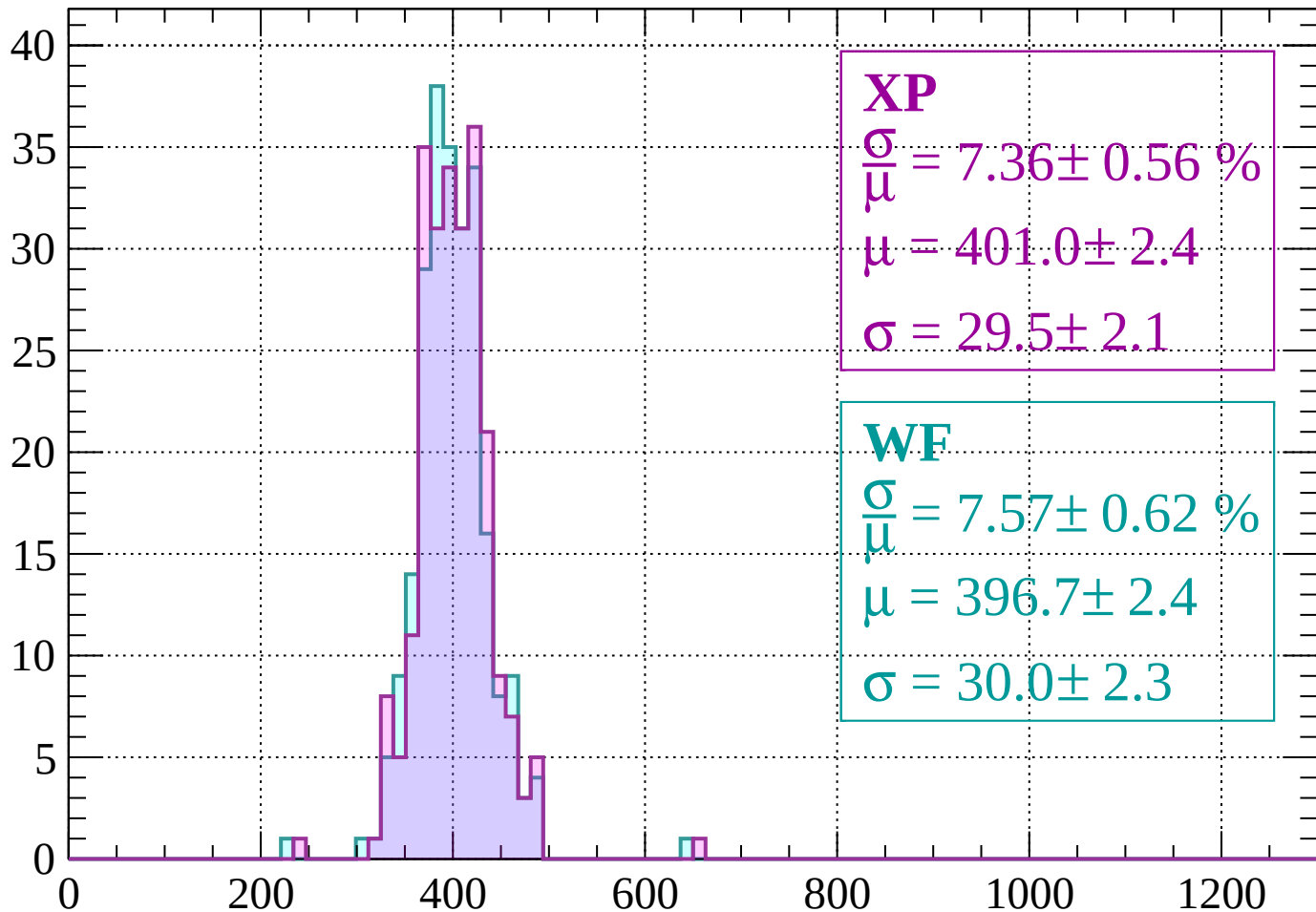
Energy loss | $-320 < p < -280$

Count



Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 7.36 \pm 0.56 \%$$

$$\mu = 401.0 \pm 2.4$$

$$\sigma = 29.5 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.57 \pm 0.62 \%$$

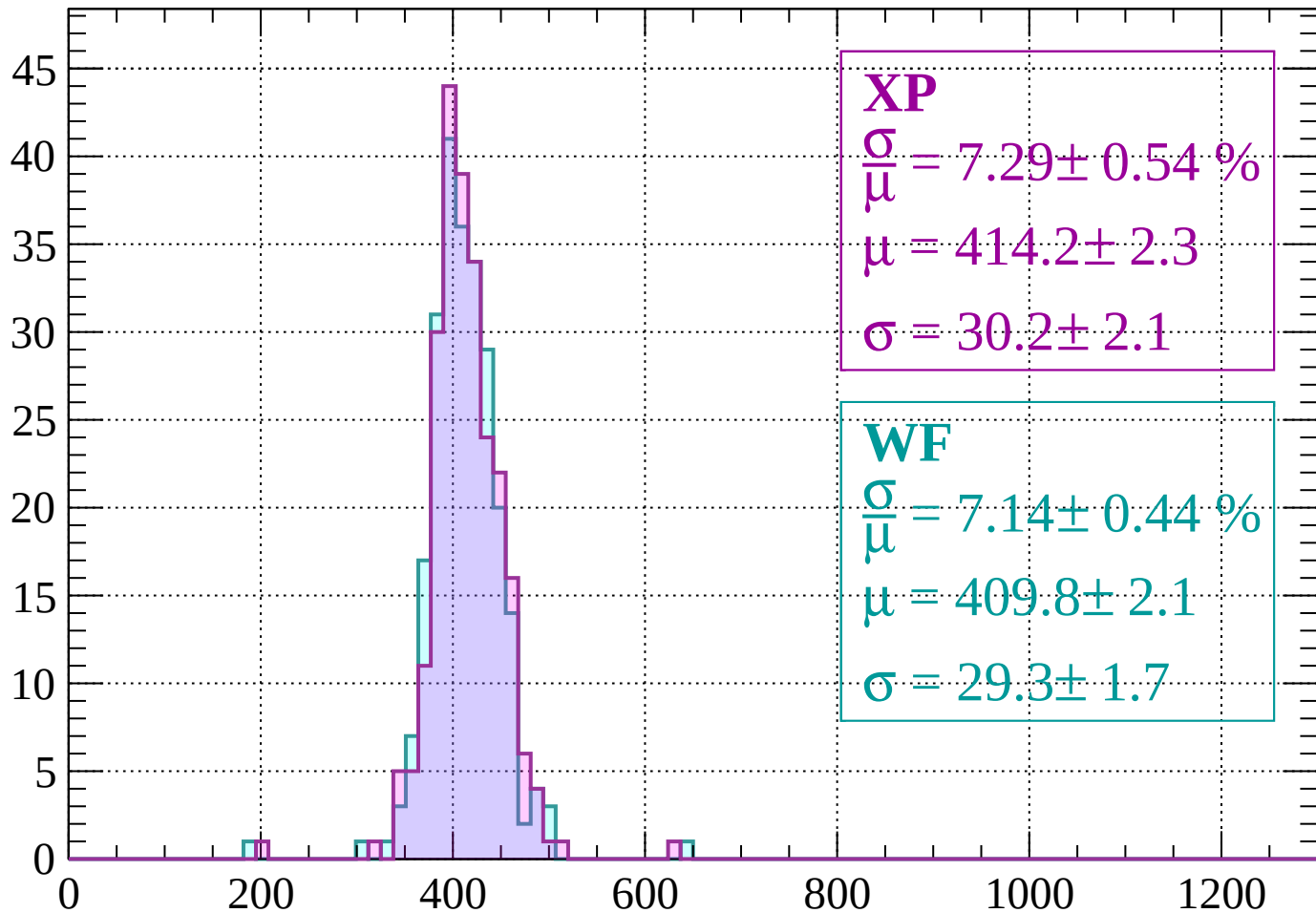
$$\mu = 396.7 \pm 2.4$$

$$\sigma = 30.0 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

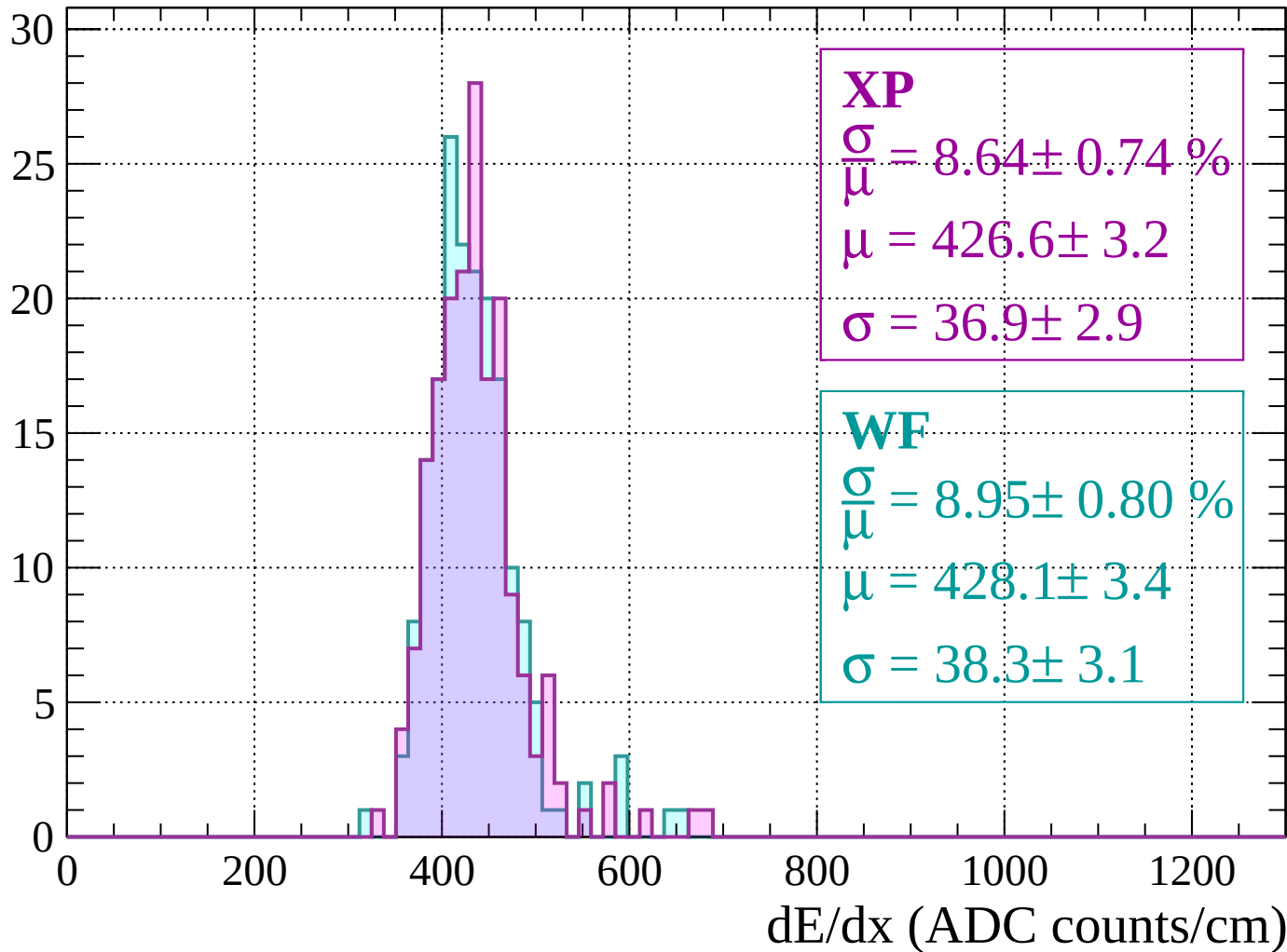
Count



dE/dx (ADC counts/cm)

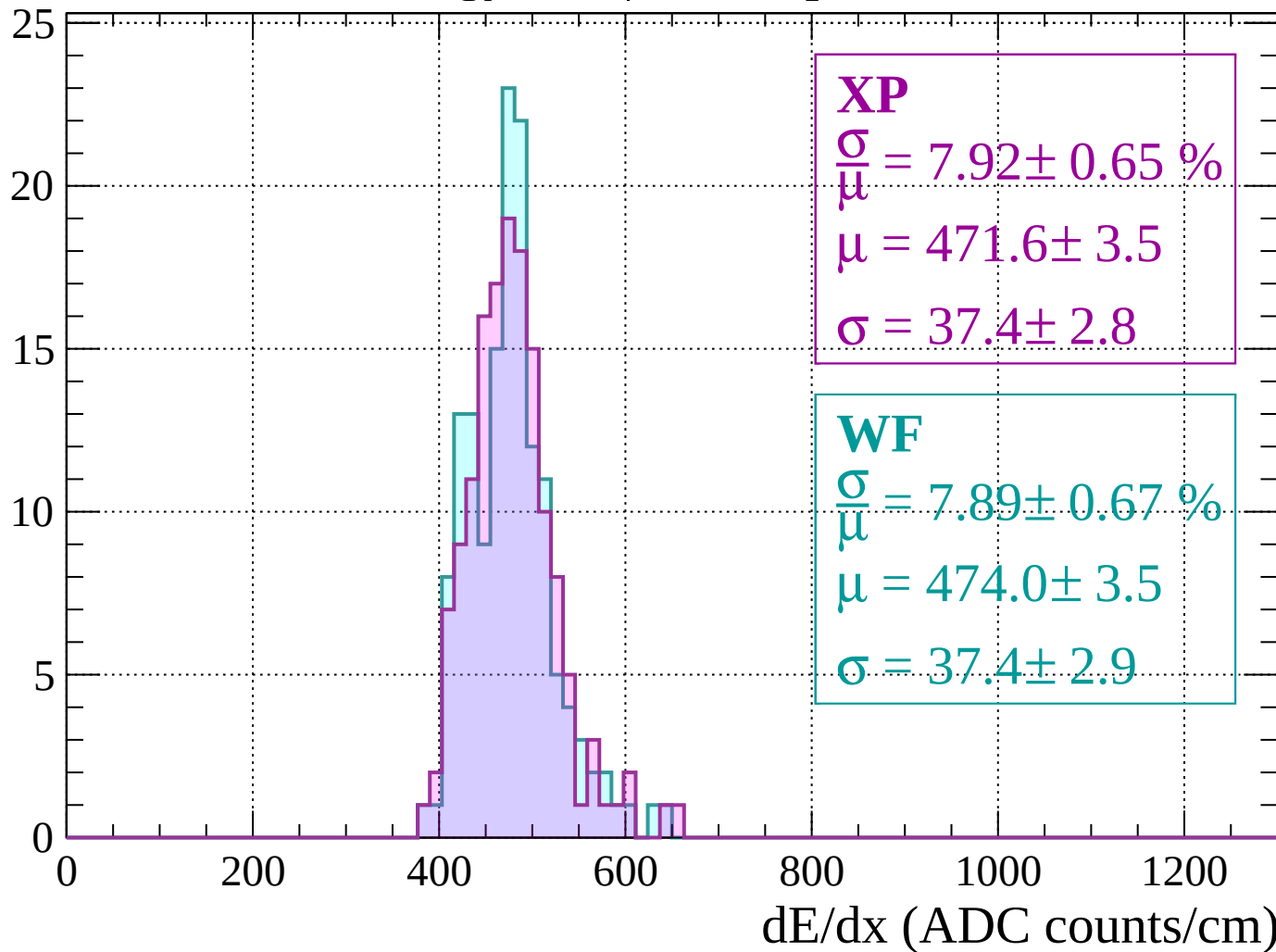
Energy loss | $-200 < p < -160$

Count



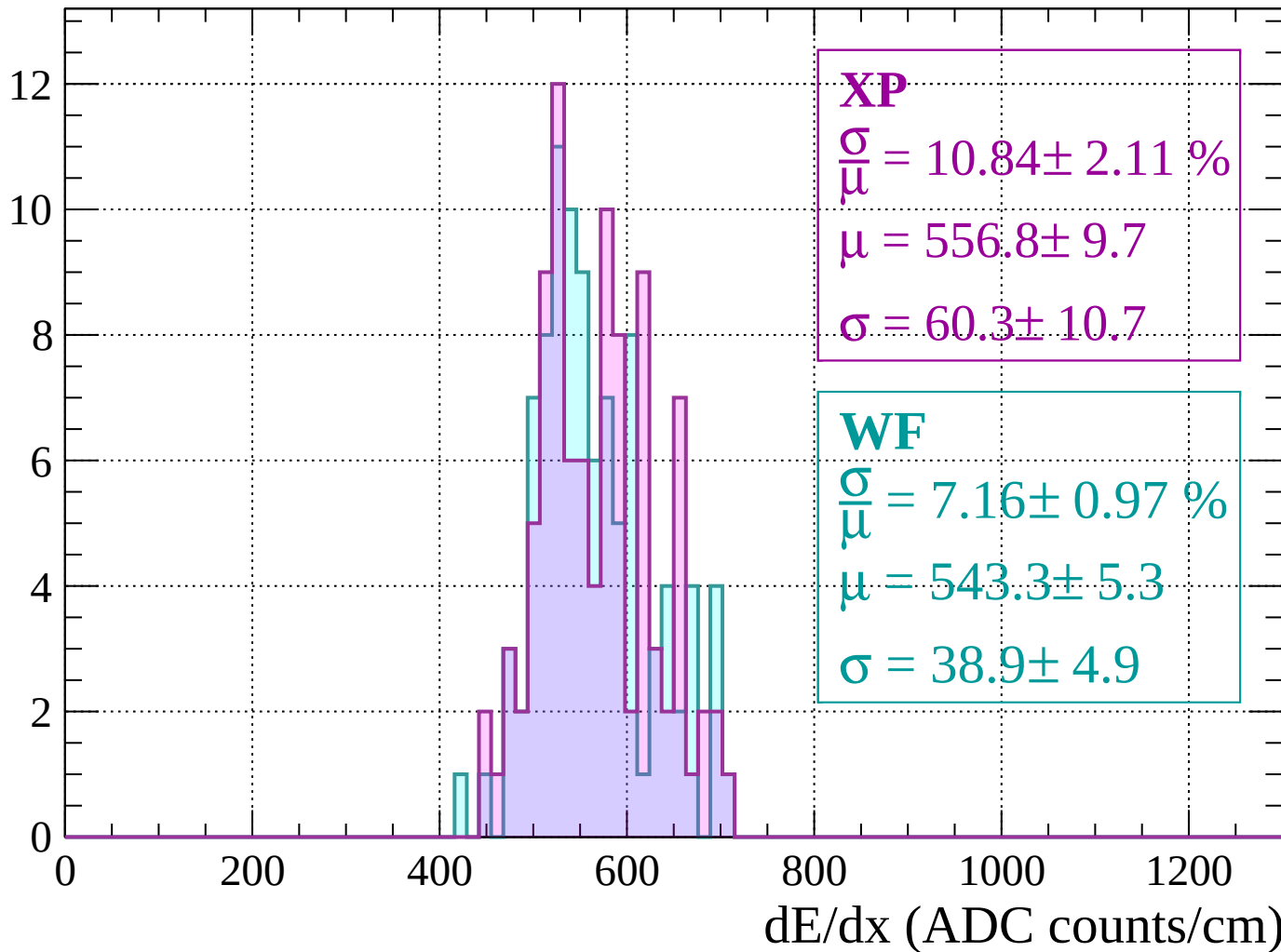
Energy loss | $-160 < p < -120$

Count



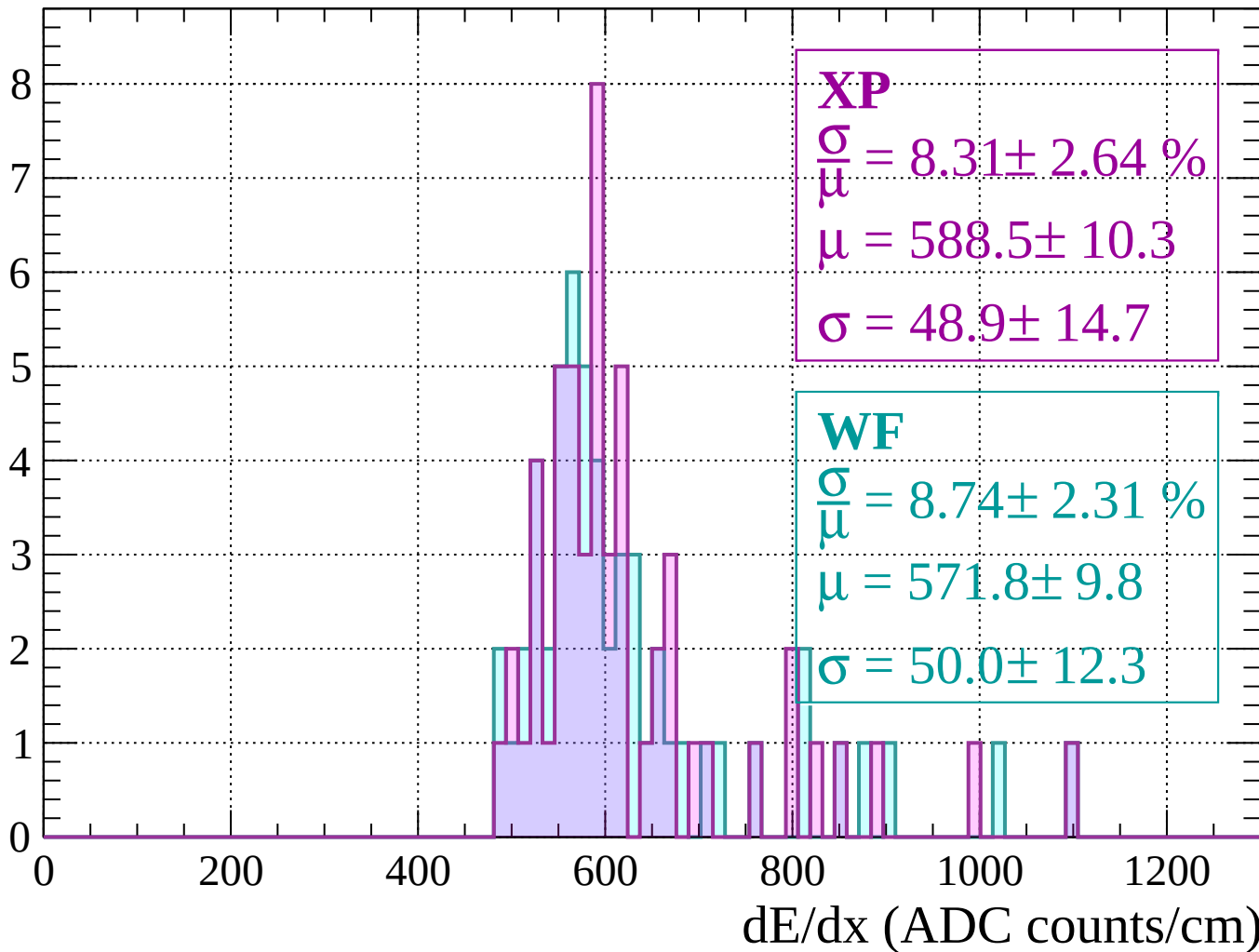
Energy loss | $-120 < p < -80$

Count



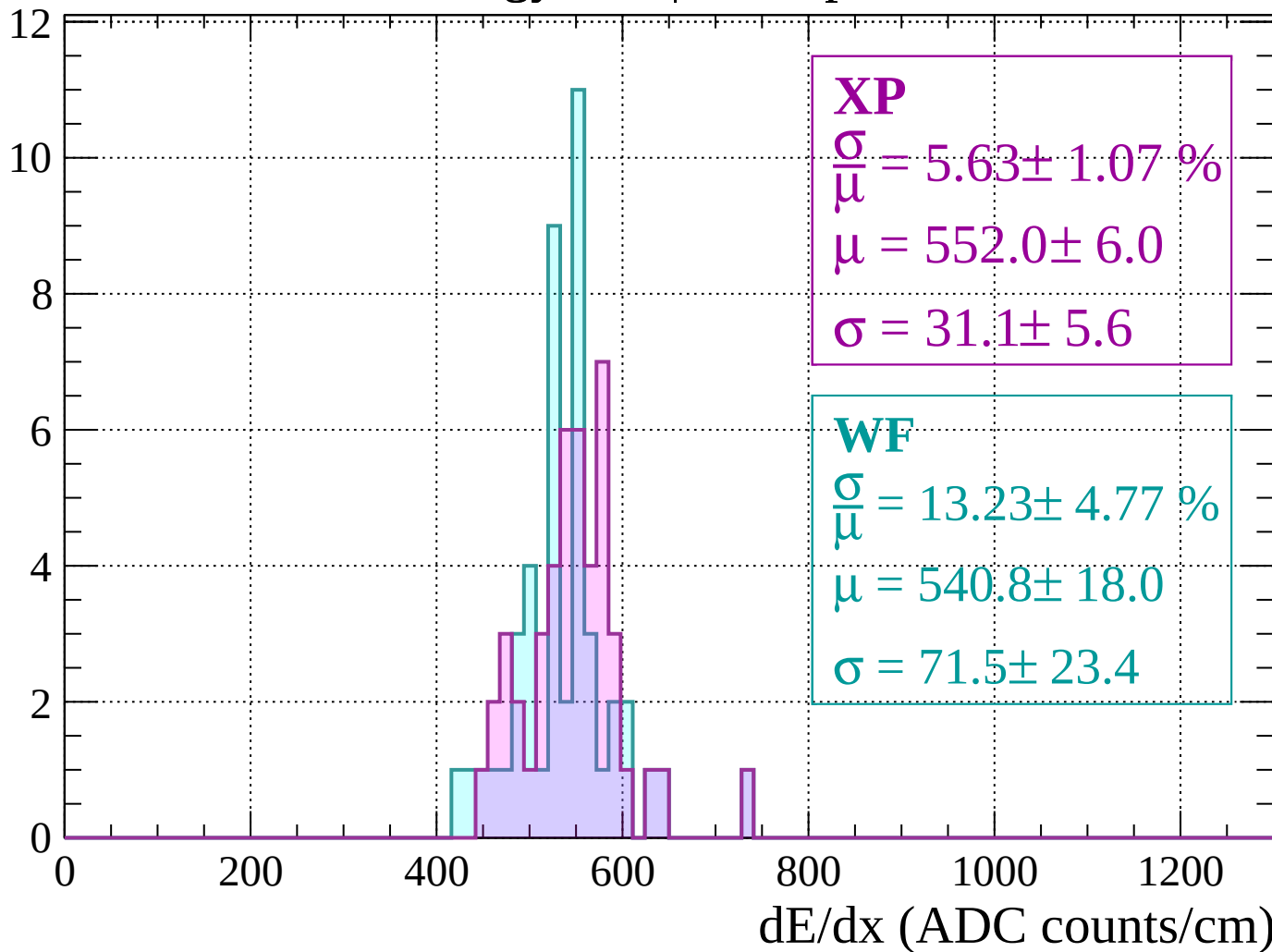
Energy loss | $-80 < p < -40$

Count



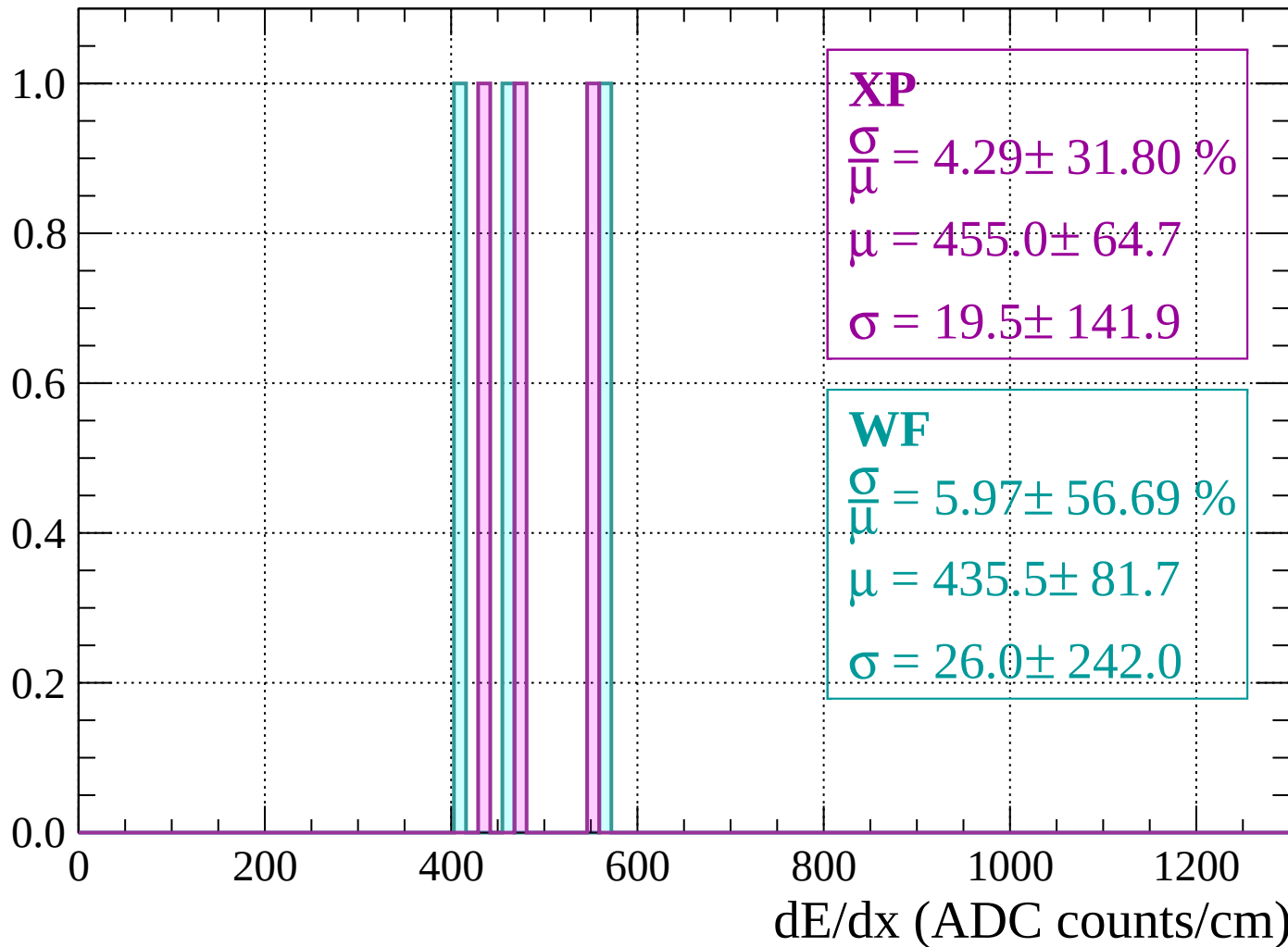
Energy loss | $-40 < p < 0$

Count



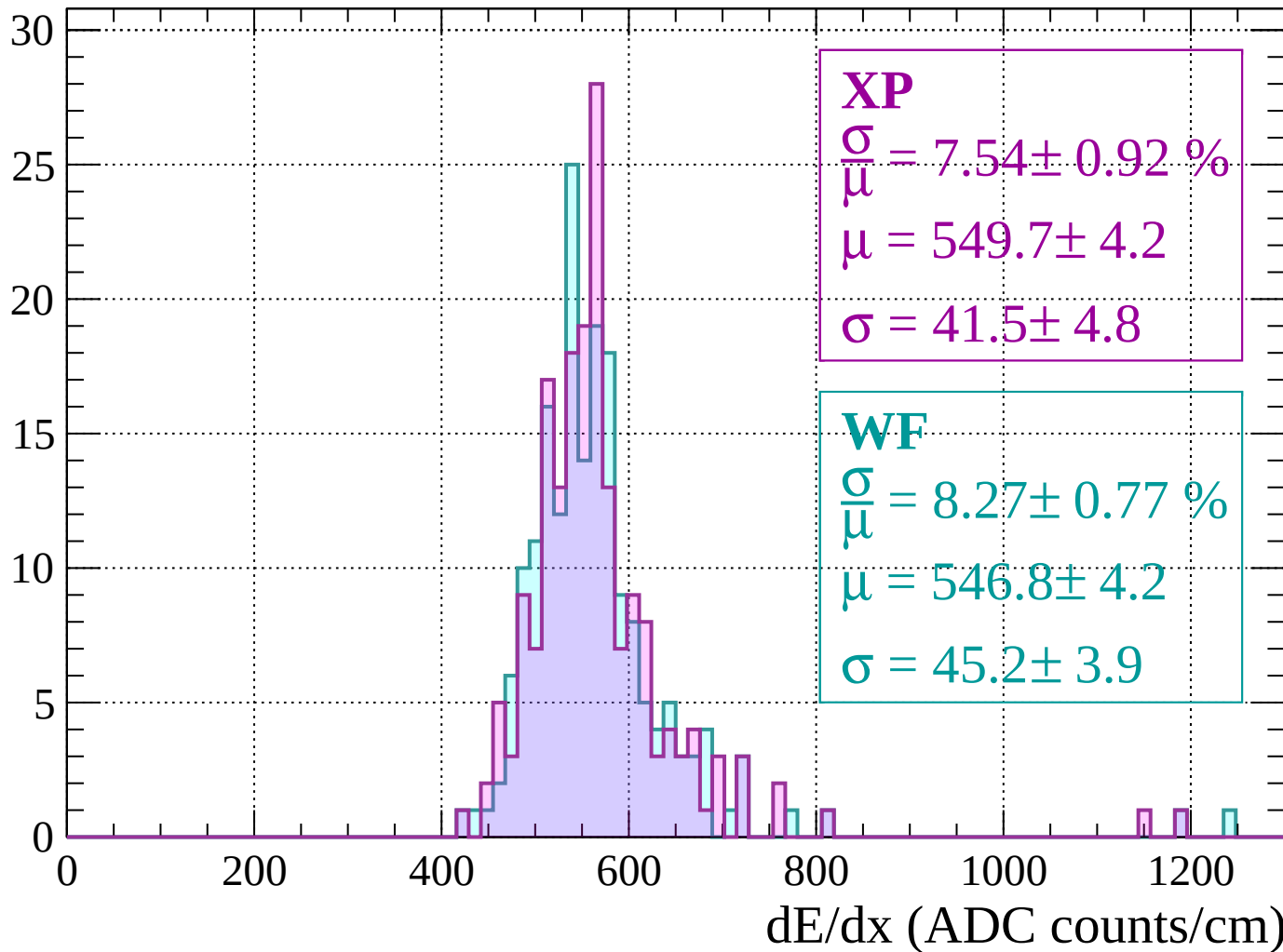
Energy loss | $0 < p < 40$

Count



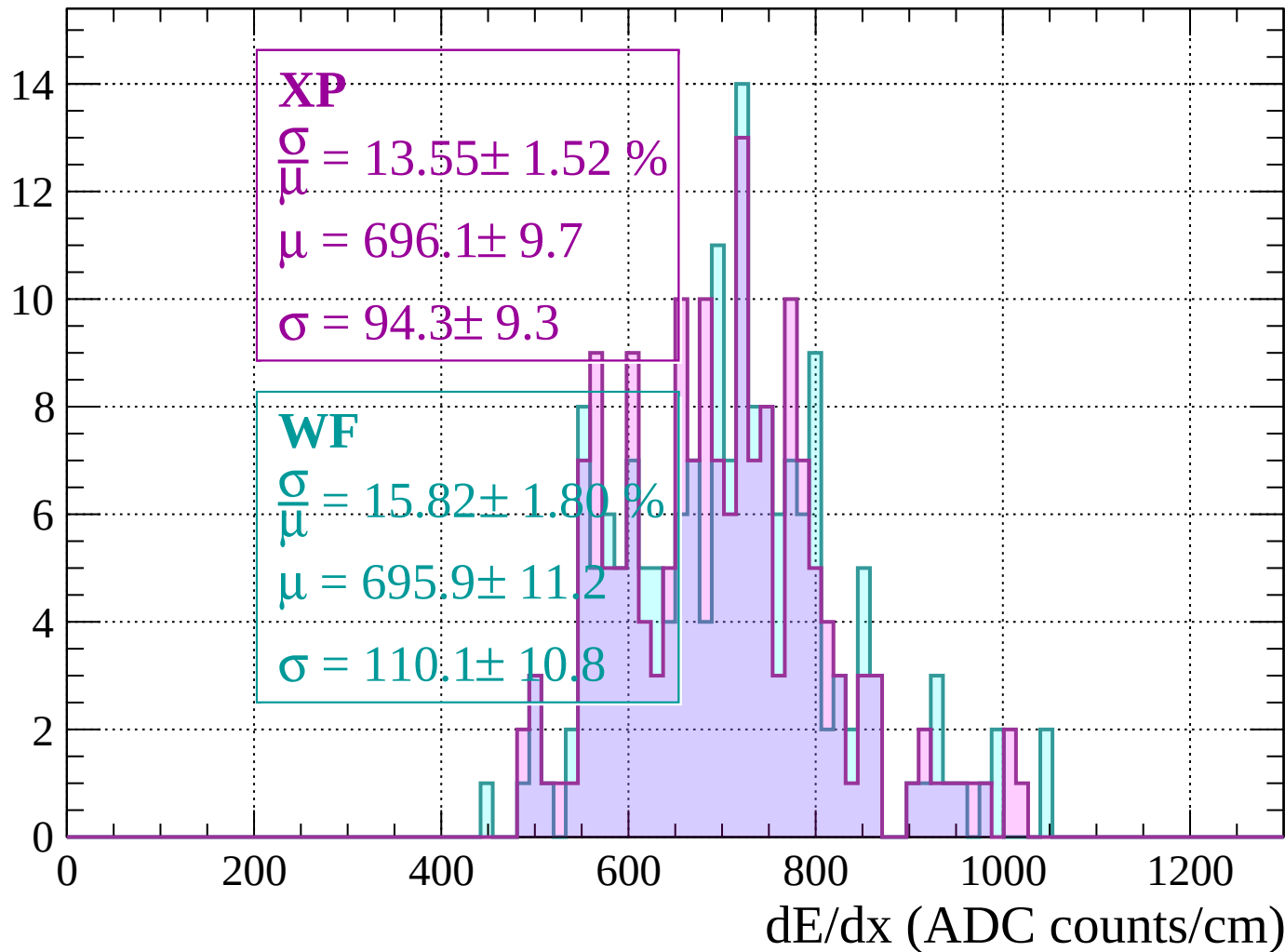
Energy loss | $40 < p < 80$

Count



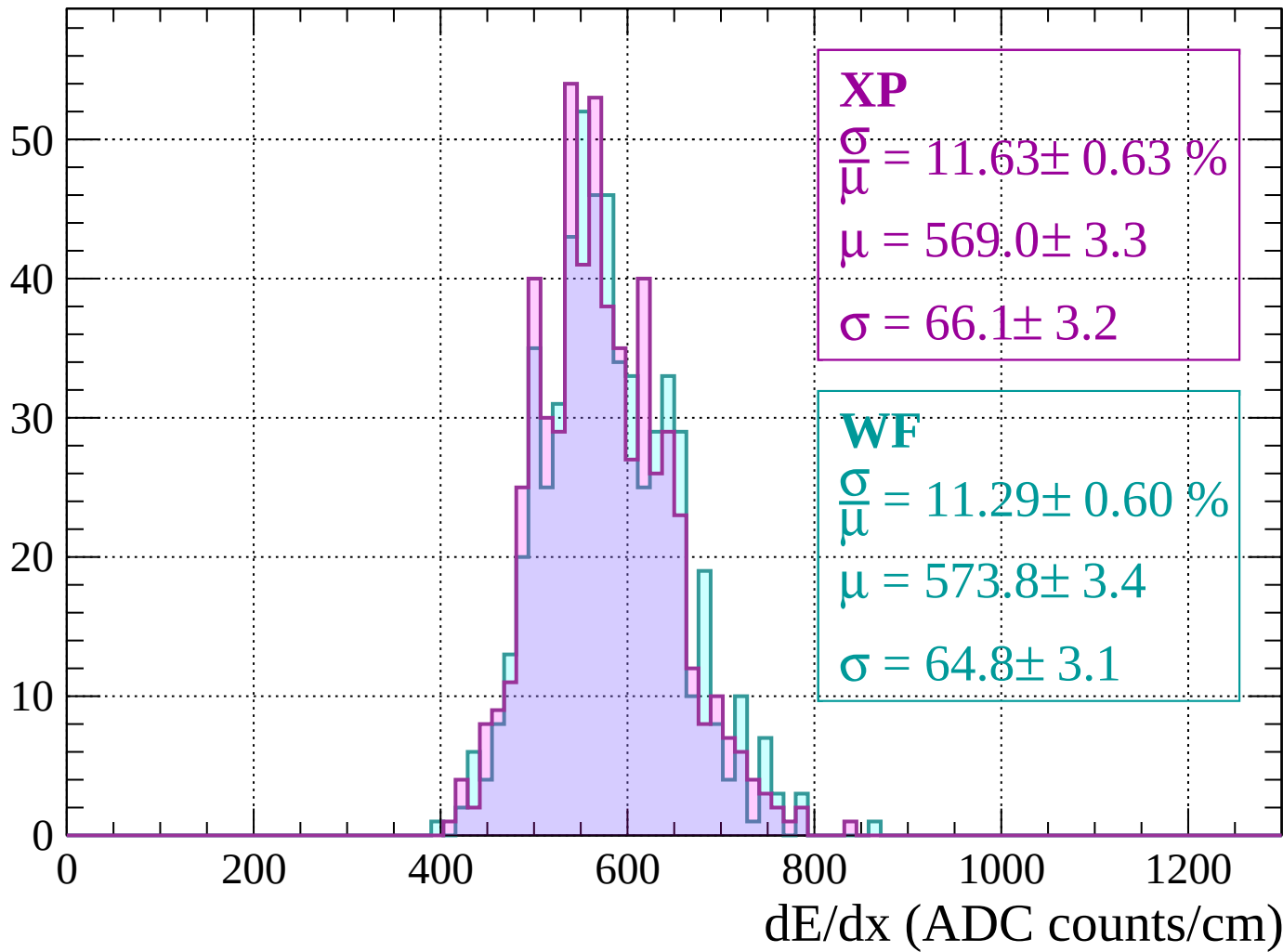
Energy loss | $80 < p < 120$

Count



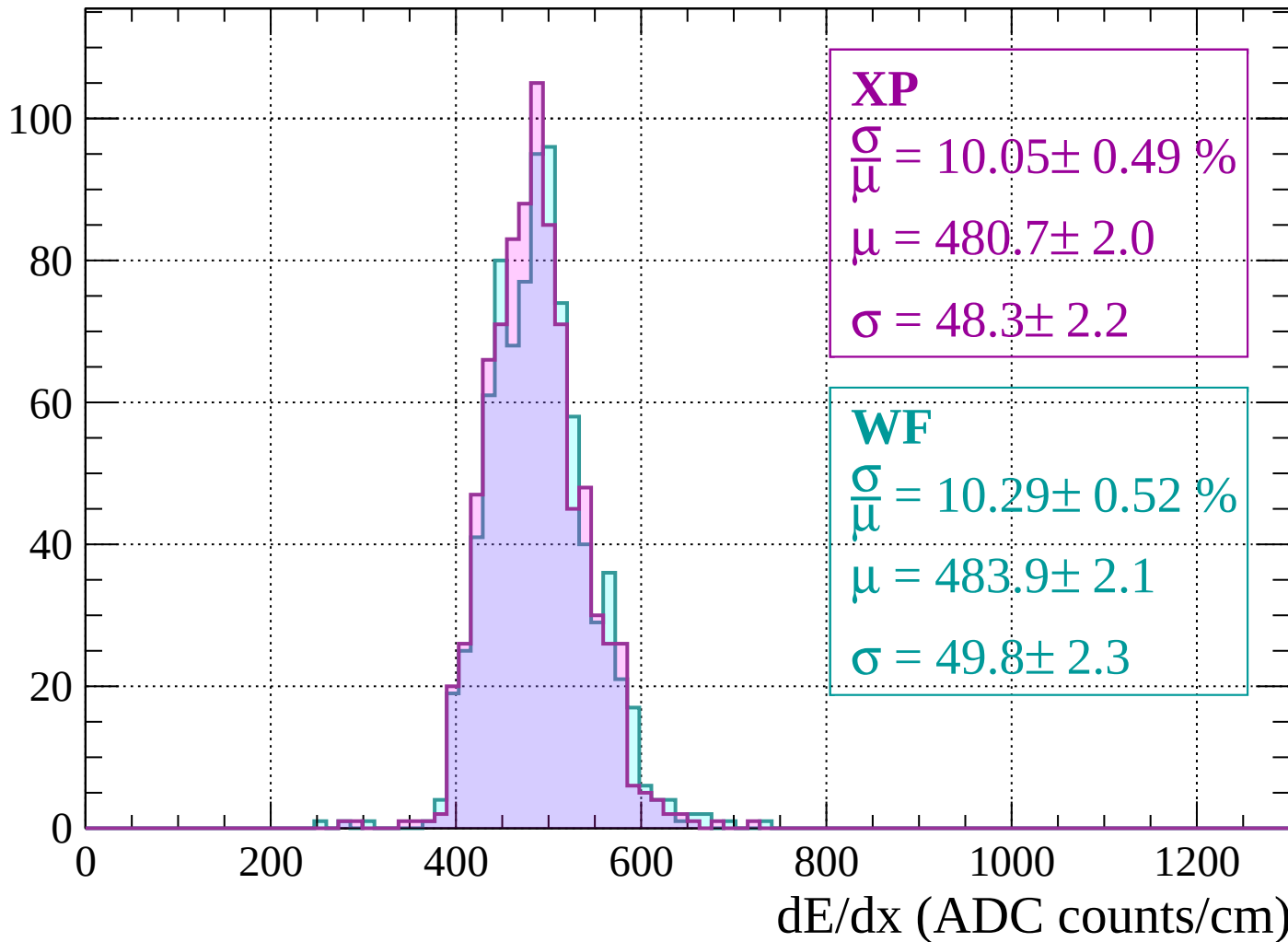
Energy loss | $120 < p < 160$

Count



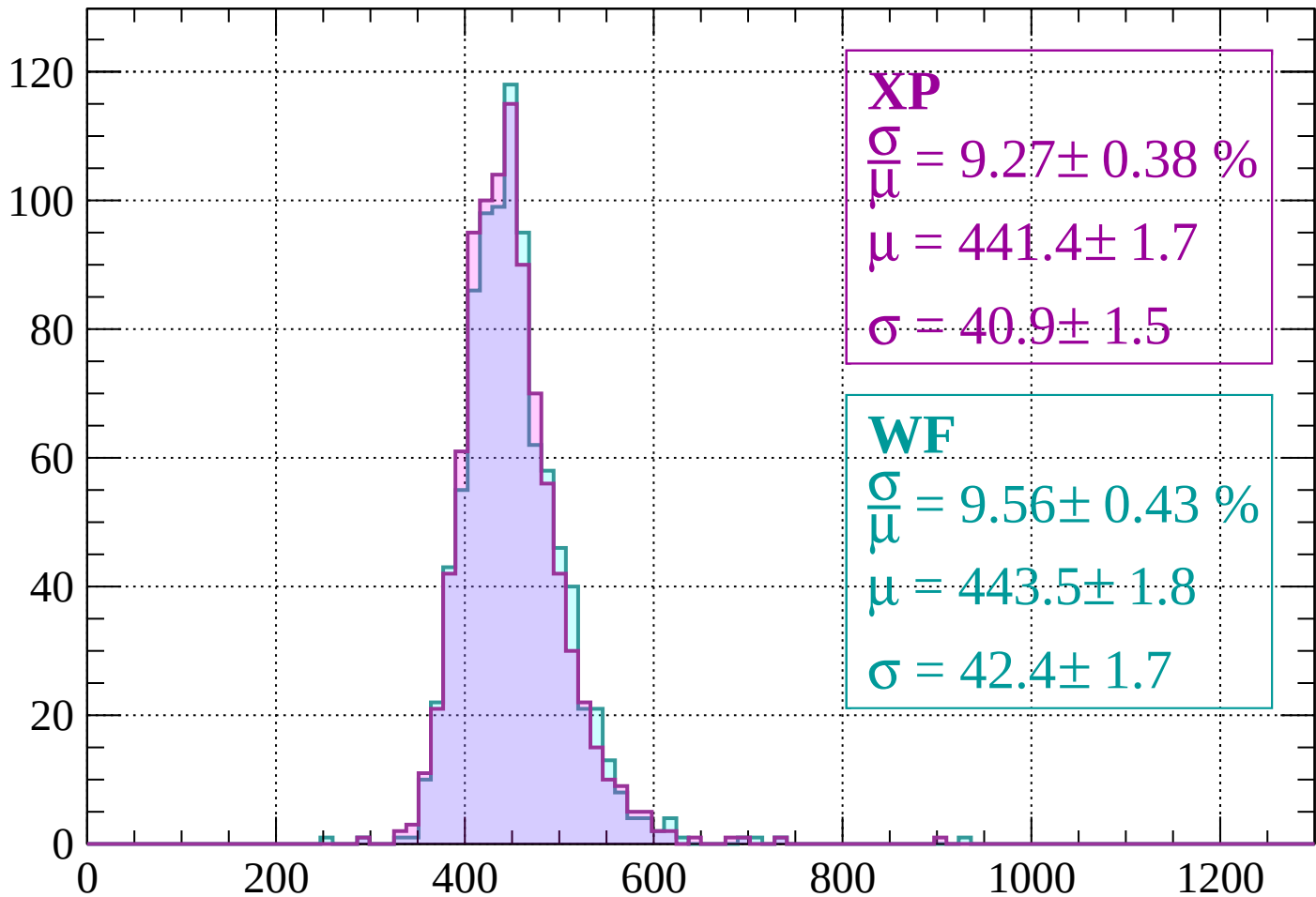
Energy loss | $160 < p < 200$

Count



Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 9.27 \pm 0.38 \%$$

$$\mu = 441.4 \pm 1.7$$

$$\sigma = 40.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.43 \%$$

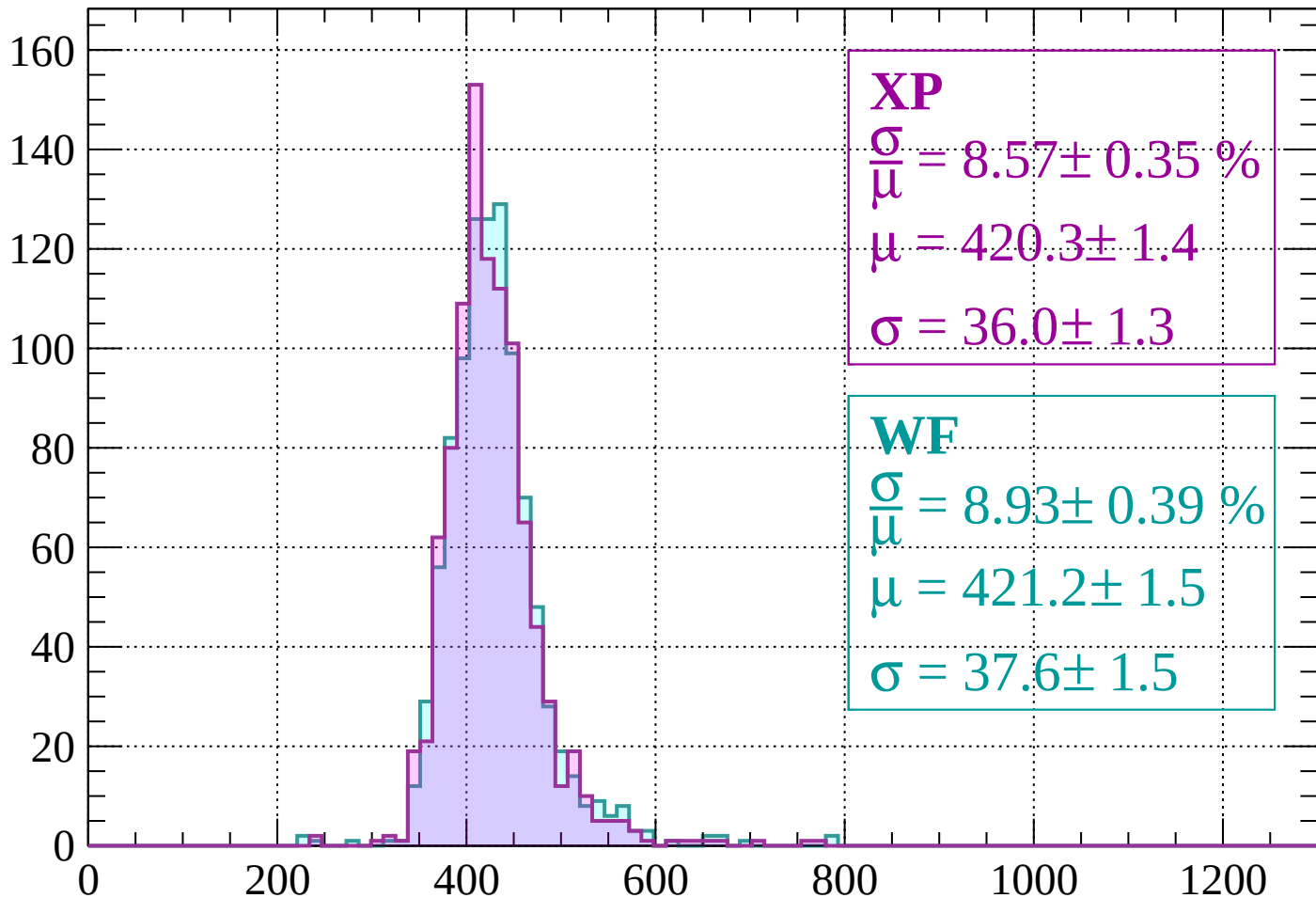
$$\mu = 443.5 \pm 1.8$$

$$\sigma = 42.4 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 8.57 \pm 0.35 \%$$

$$\mu = 420.3 \pm 1.4$$

$$\sigma = 36.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.93 \pm 0.39 \%$$

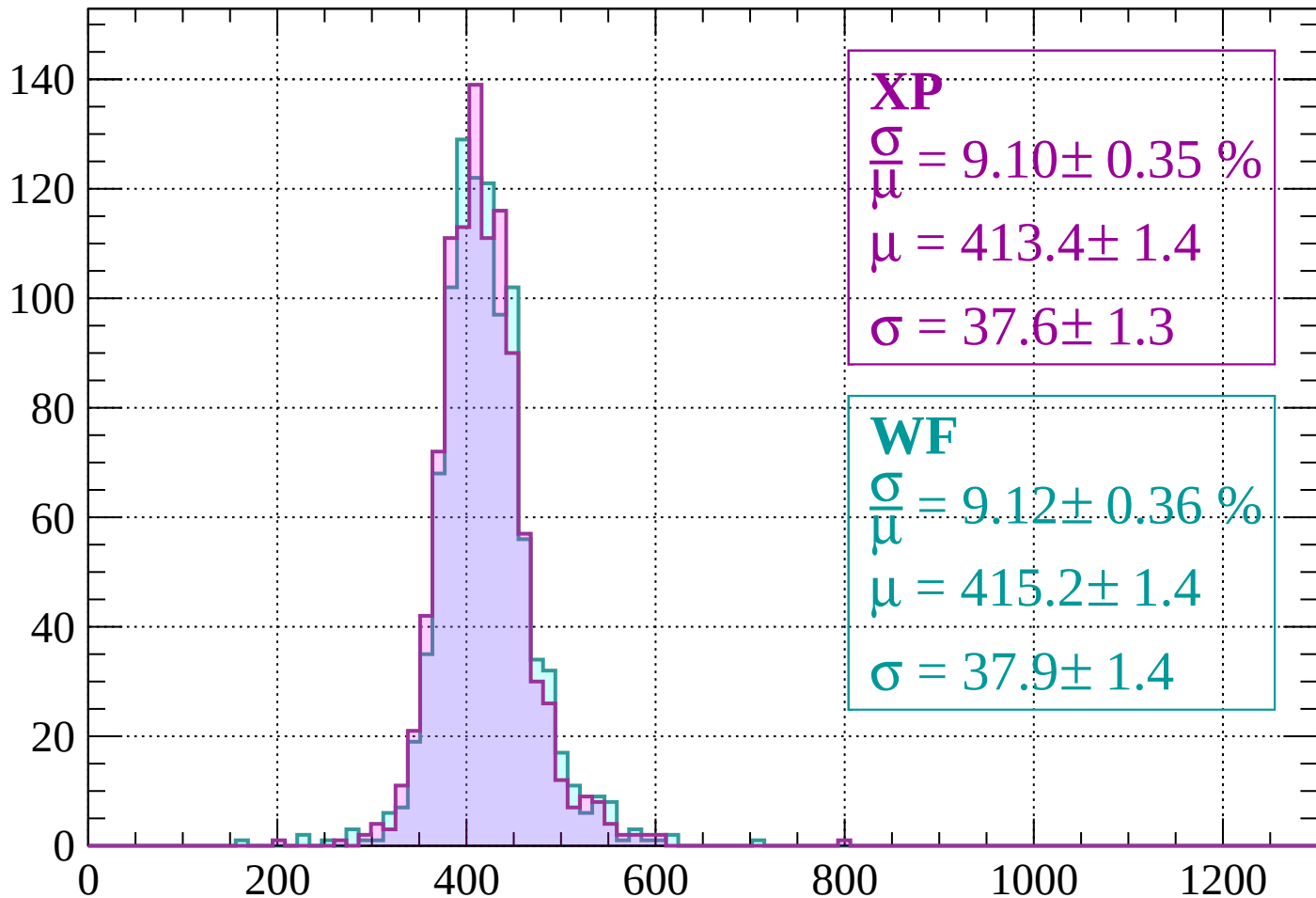
$$\mu = 421.2 \pm 1.5$$

$$\sigma = 37.6 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.35 \%$$

$$\mu = 413.4 \pm 1.4$$

$$\sigma = 37.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.12 \pm 0.36 \%$$

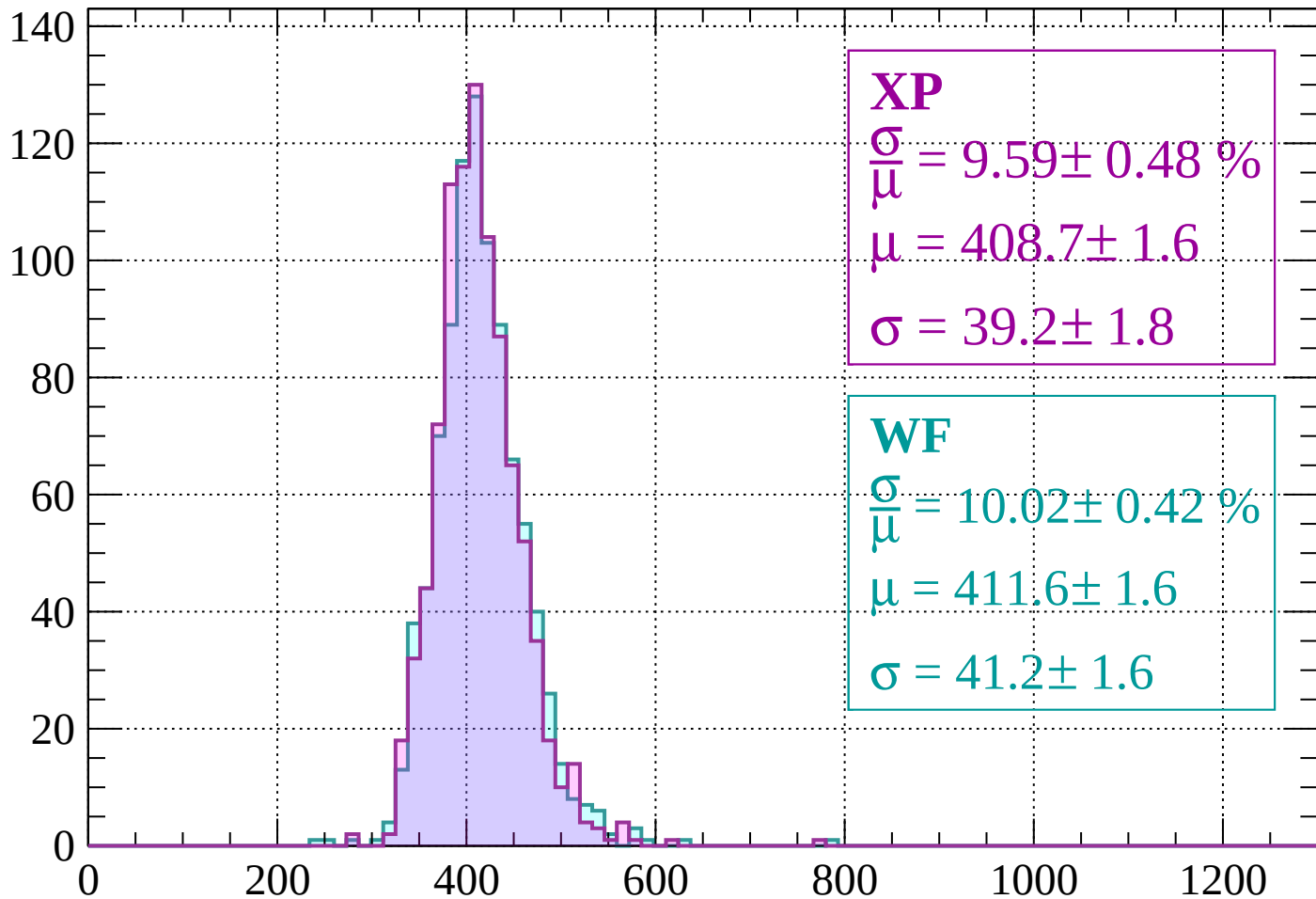
$$\mu = 415.2 \pm 1.4$$

$$\sigma = 37.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 9.59 \pm 0.48 \%$$

$$\mu = 408.7 \pm 1.6$$

$$\sigma = 39.2 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 10.02 \pm 0.42 \%$$

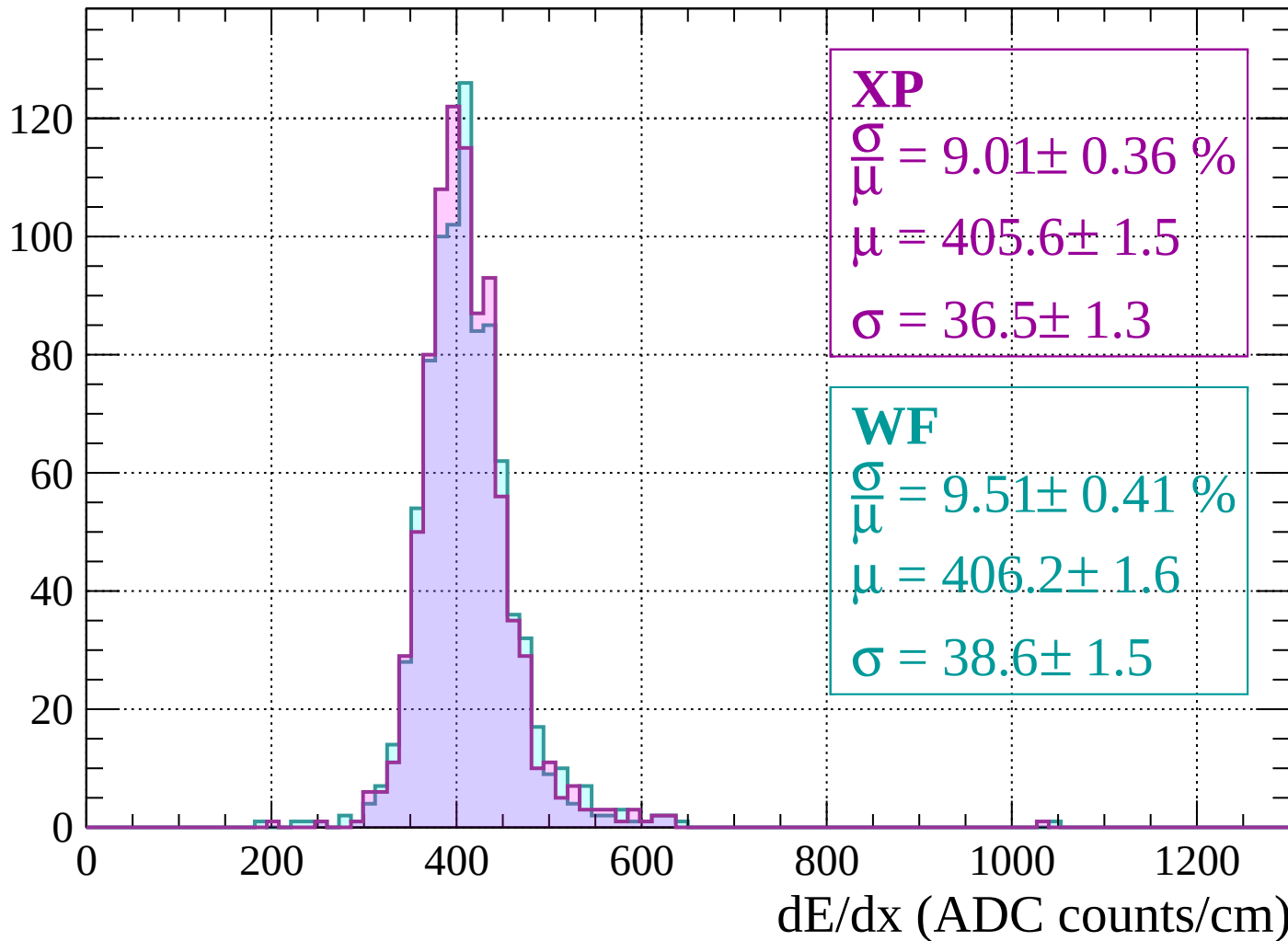
$$\mu = 411.6 \pm 1.6$$

$$\sigma = 41.2 \pm 1.6$$

dE/dx (ADC counts/cm)

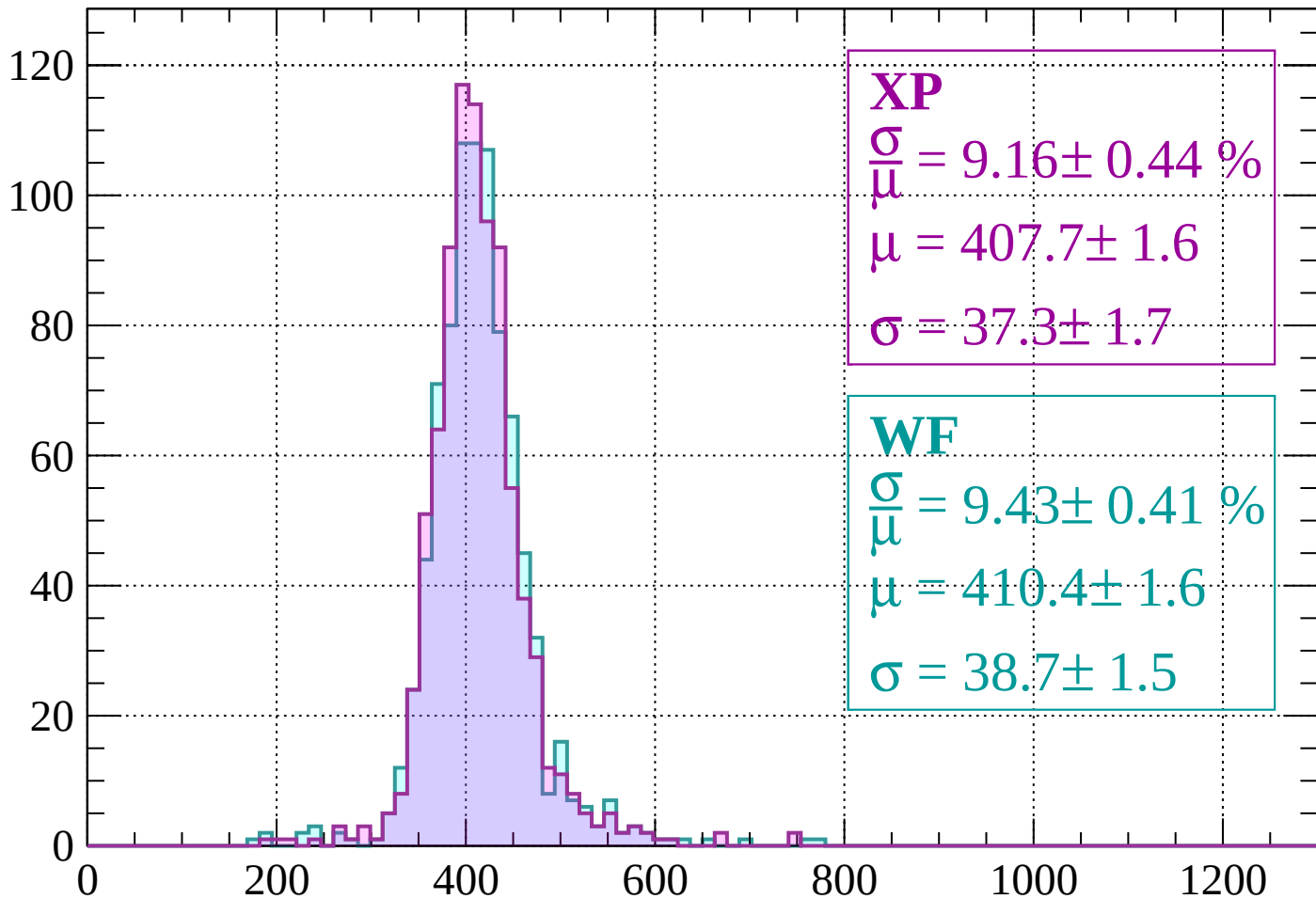
Energy loss | $360 < p < 400$

Count



Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 9.16 \pm 0.44 \%$$

$$\mu = 407.7 \pm 1.6$$

$$\sigma = 37.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.43 \pm 0.41 \%$$

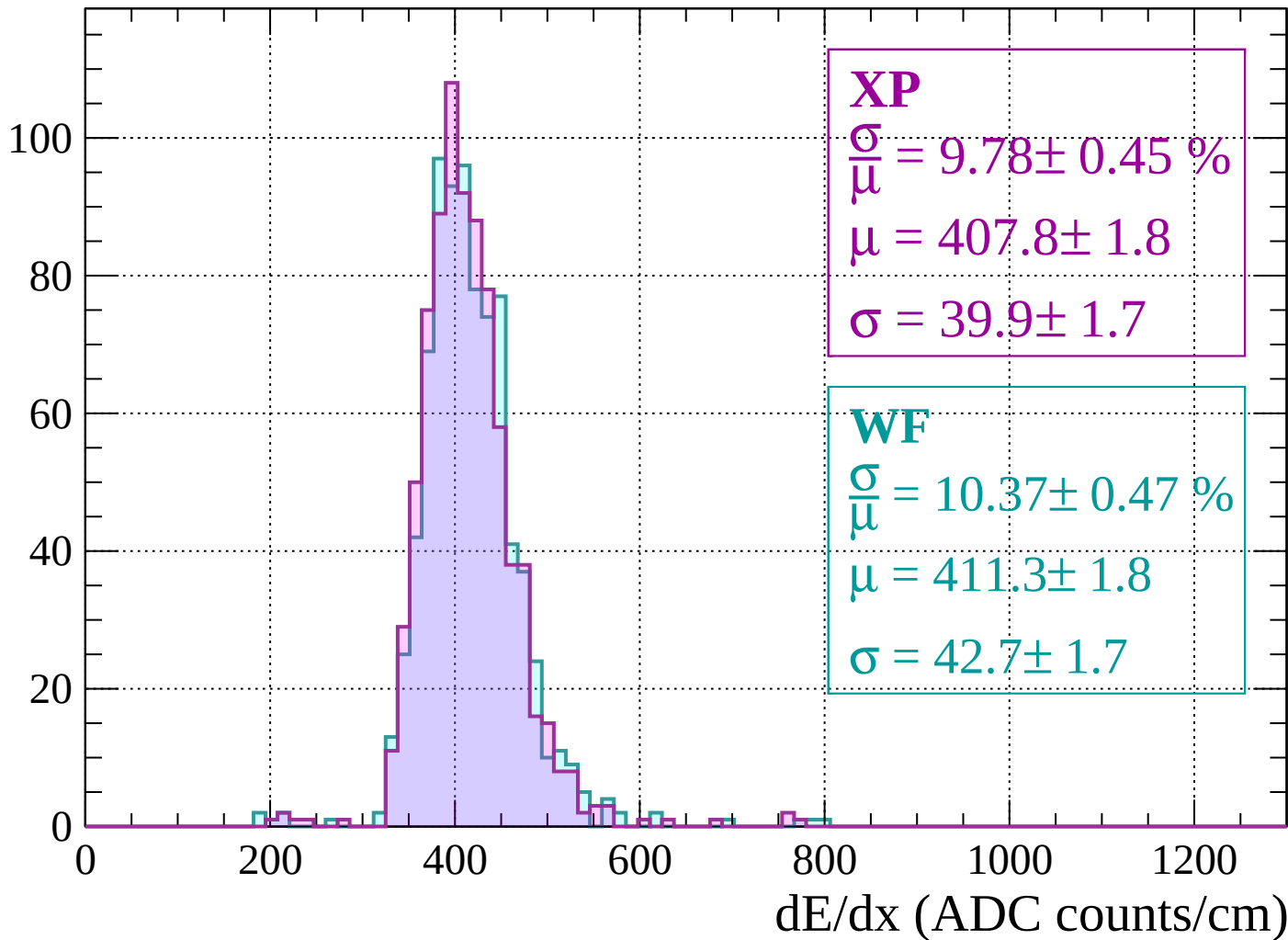
$$\mu = 410.4 \pm 1.6$$

$$\sigma = 38.7 \pm 1.5$$

dE/dx (ADC counts/cm)

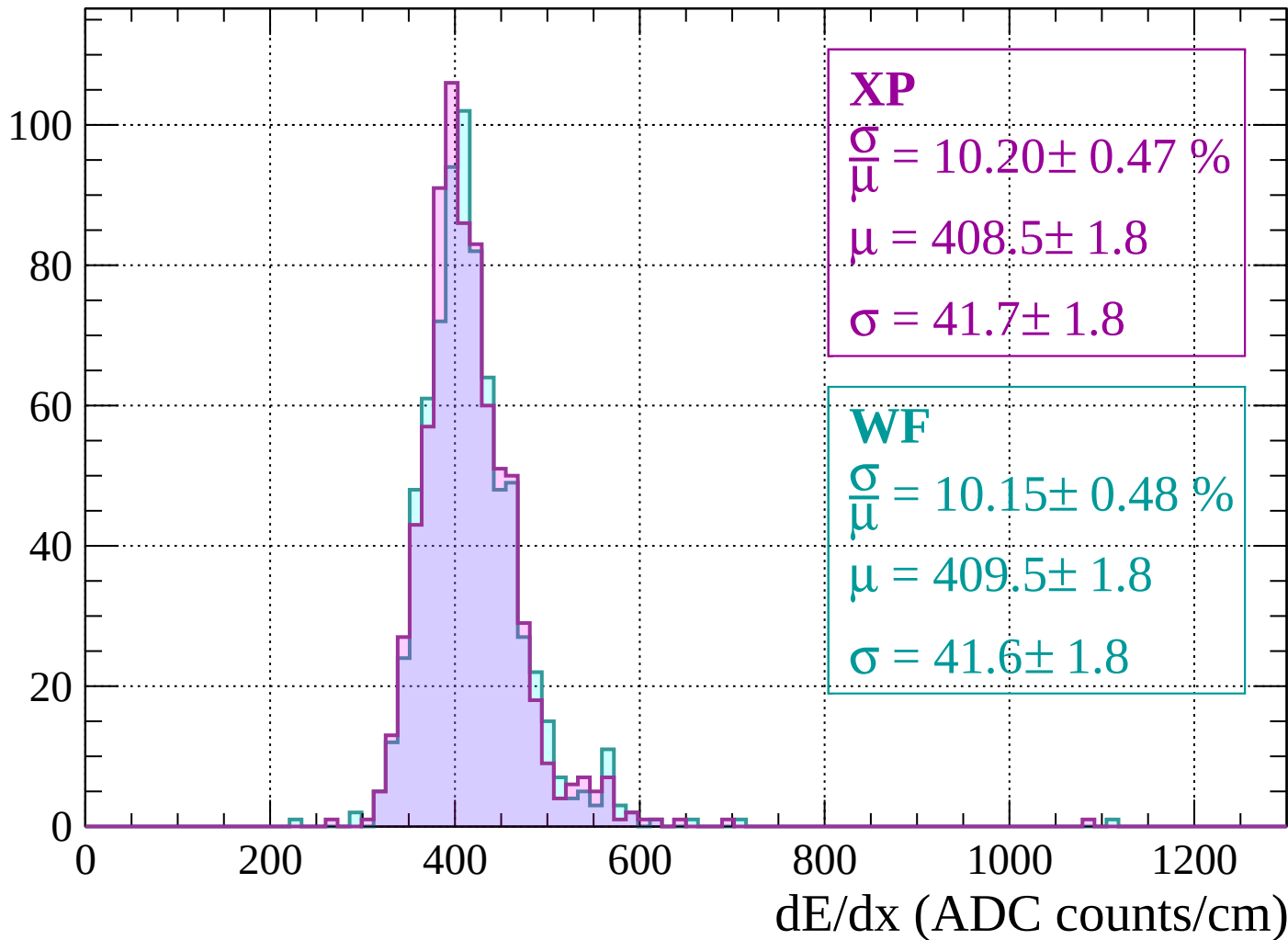
Energy loss | $440 < p < 480$

Count



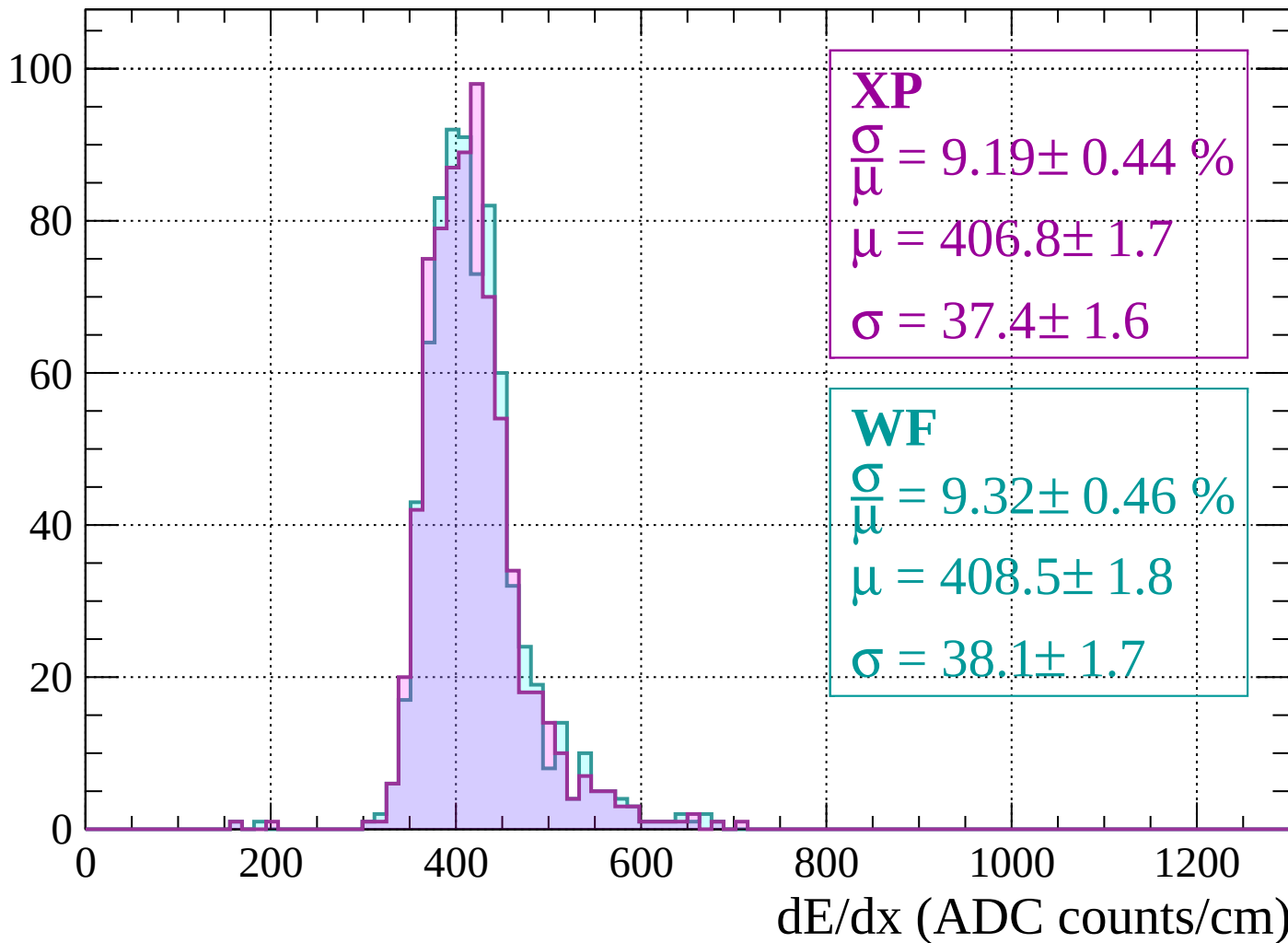
Energy loss | $480 < p < 520$

Count



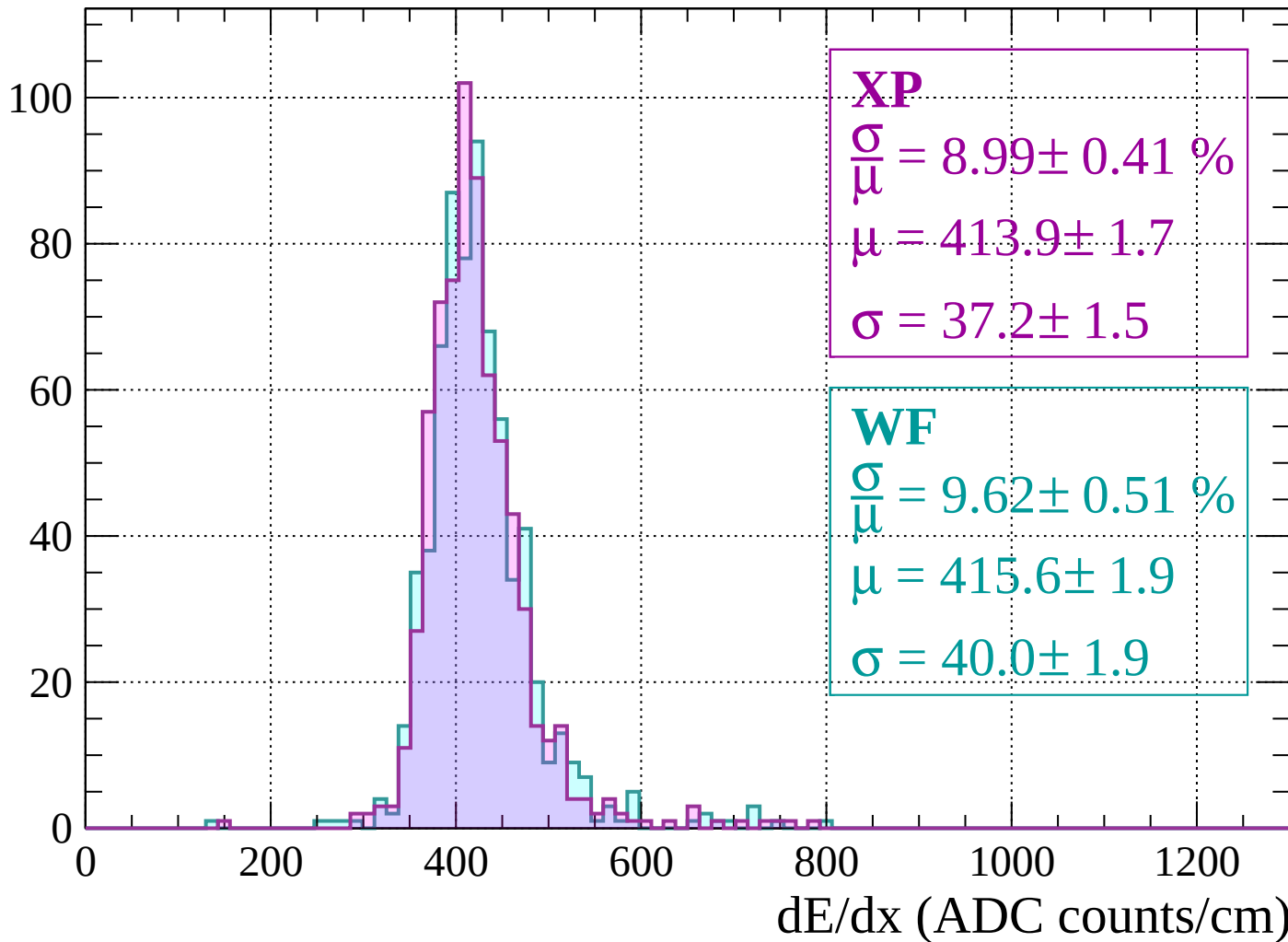
Energy loss | $520 < p < 560$

Count



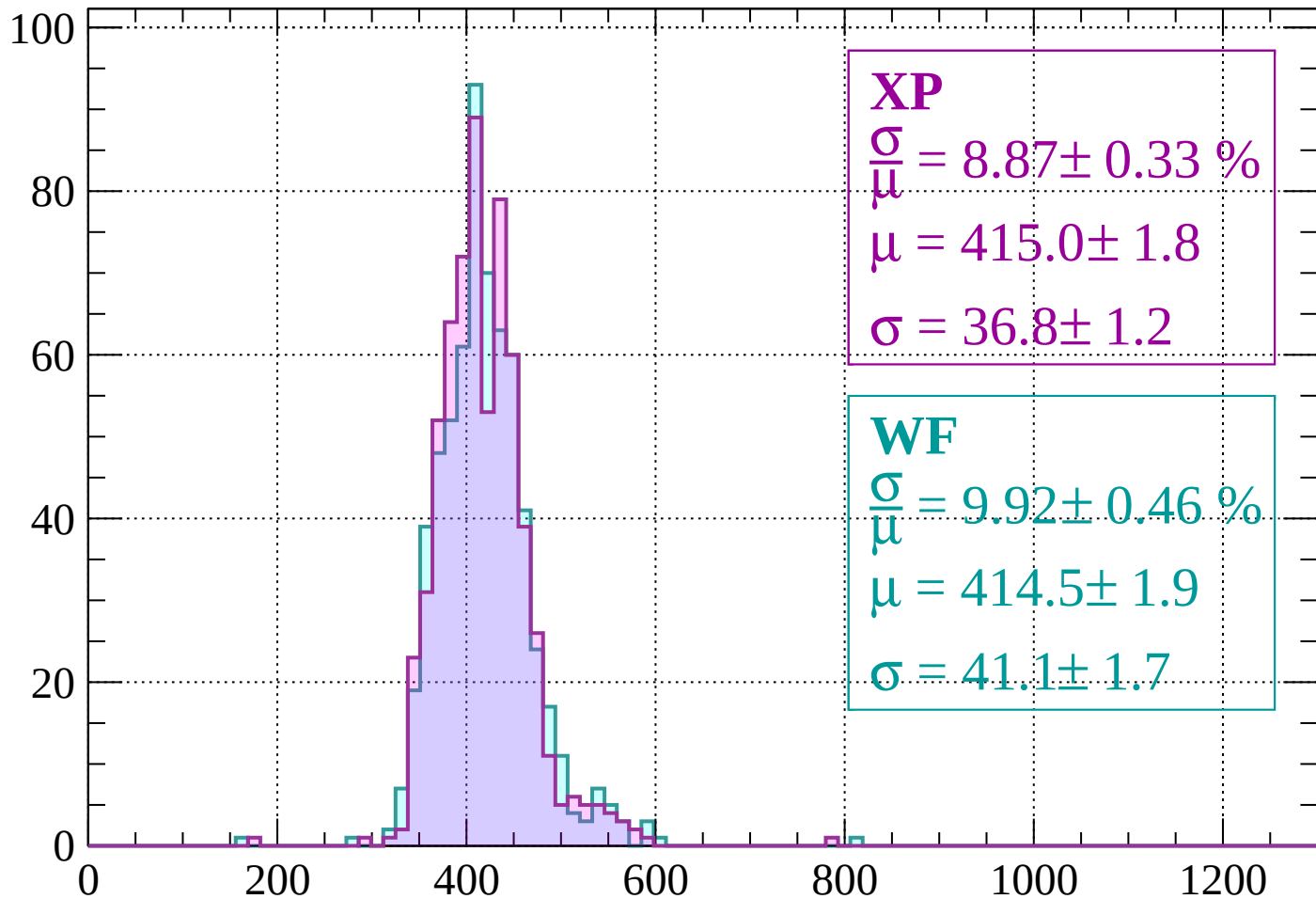
Energy loss | $560 < p < 600$

Count



Energy loss | $600 < p < 640$

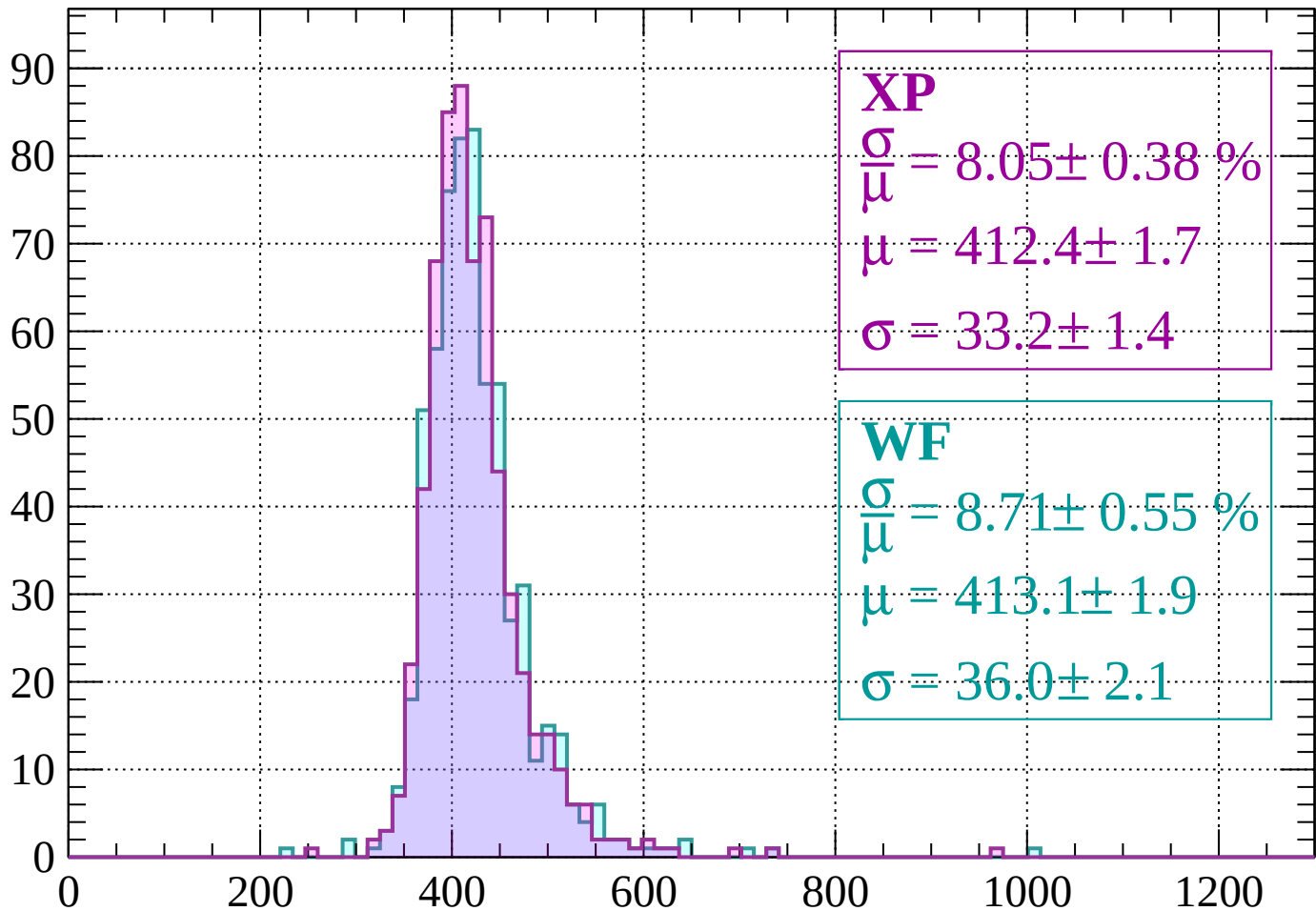
Count



dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.05 \pm 0.38 \%$$

$$\mu = 412.4 \pm 1.7$$

$$\sigma = 33.2 \pm 1.4$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.71 \pm 0.55 \%$$

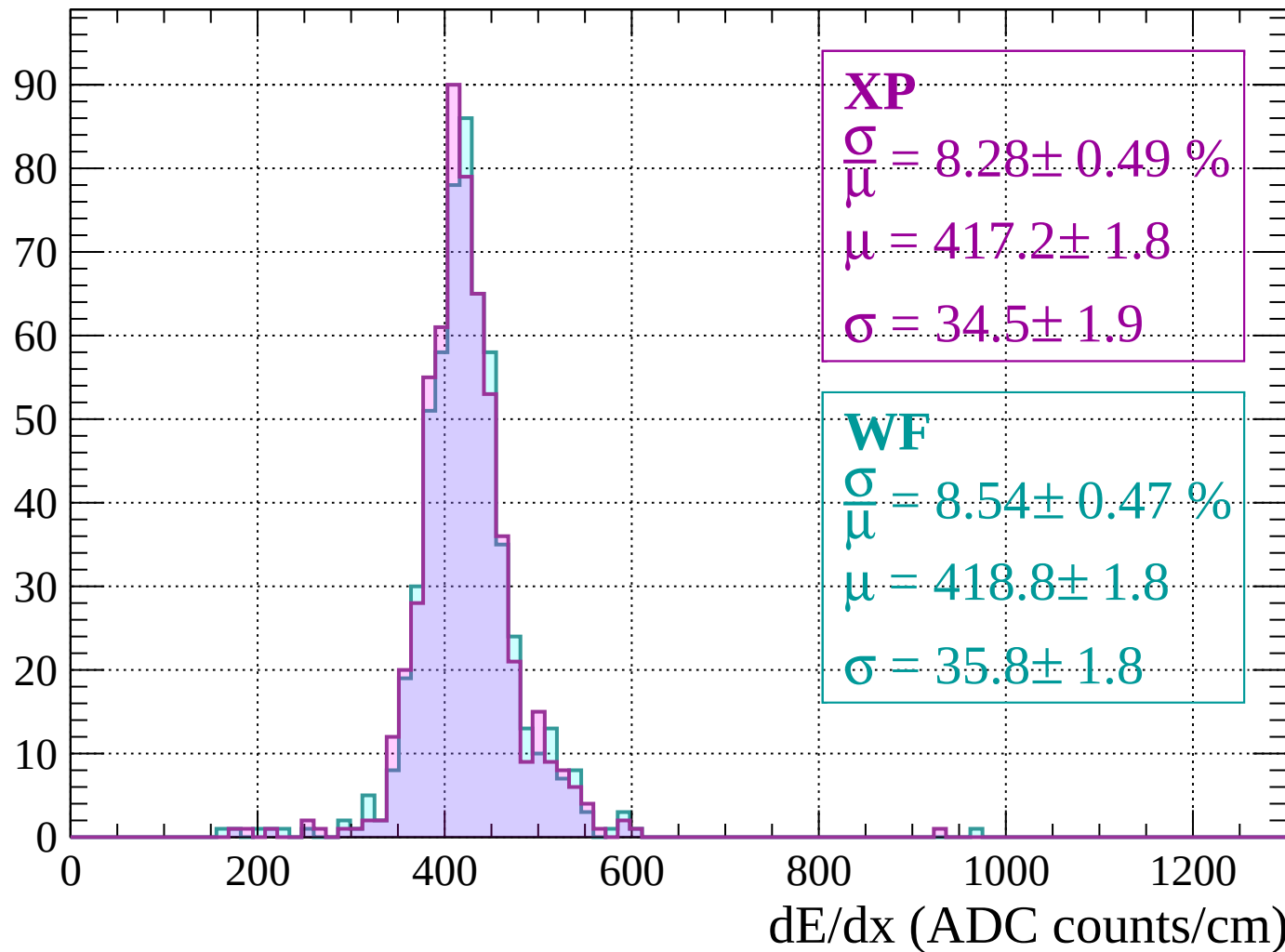
$$\mu = 413.1 \pm 1.9$$

$$\sigma = 36.0 \pm 2.1$$

dE/dx (ADC counts/cm)

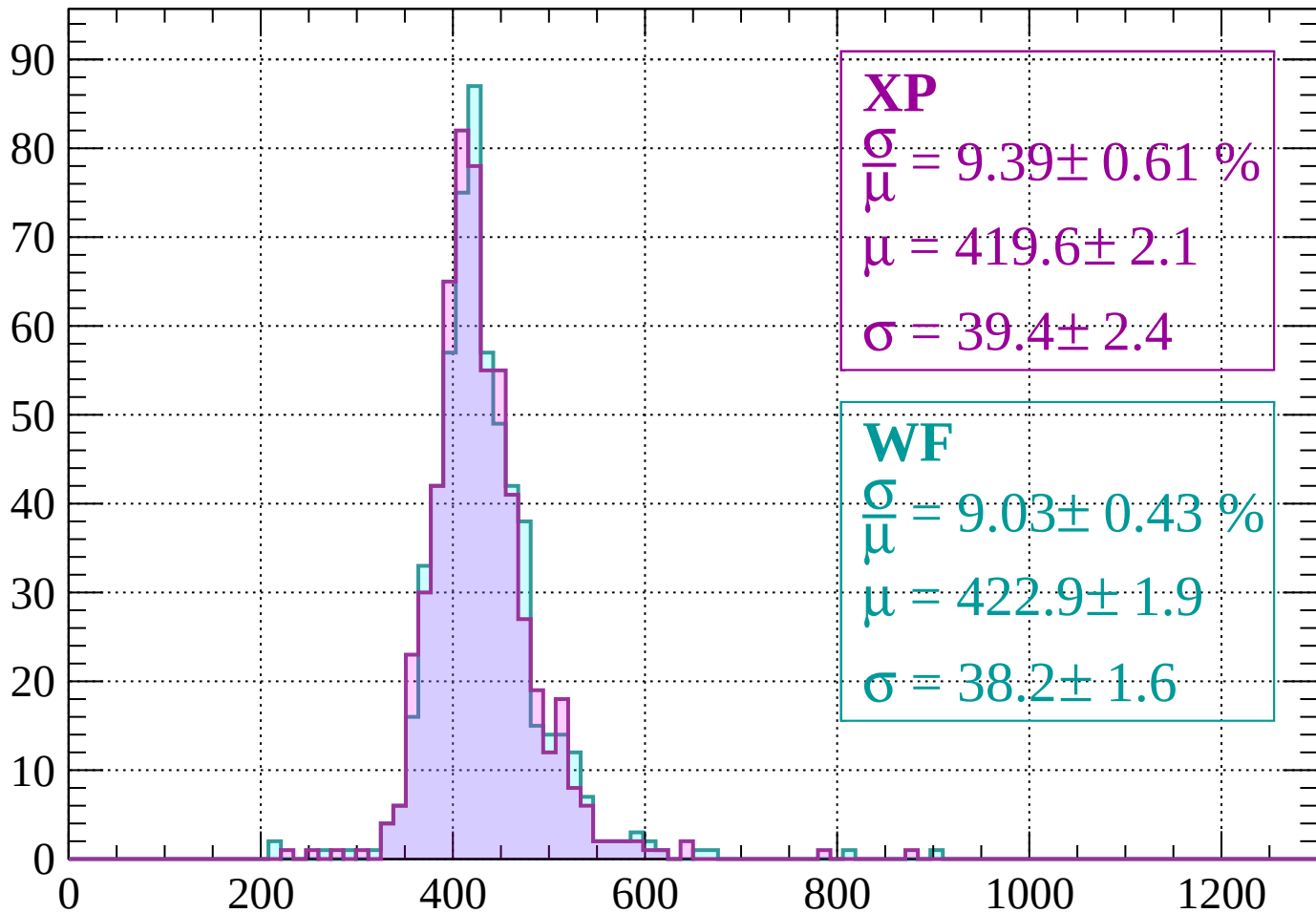
Energy loss | $680 < p < 720$

Count



Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.61 \%$$

$$\mu = 419.6 \pm 2.1$$

$$\sigma = 39.4 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 9.03 \pm 0.43 \%$$

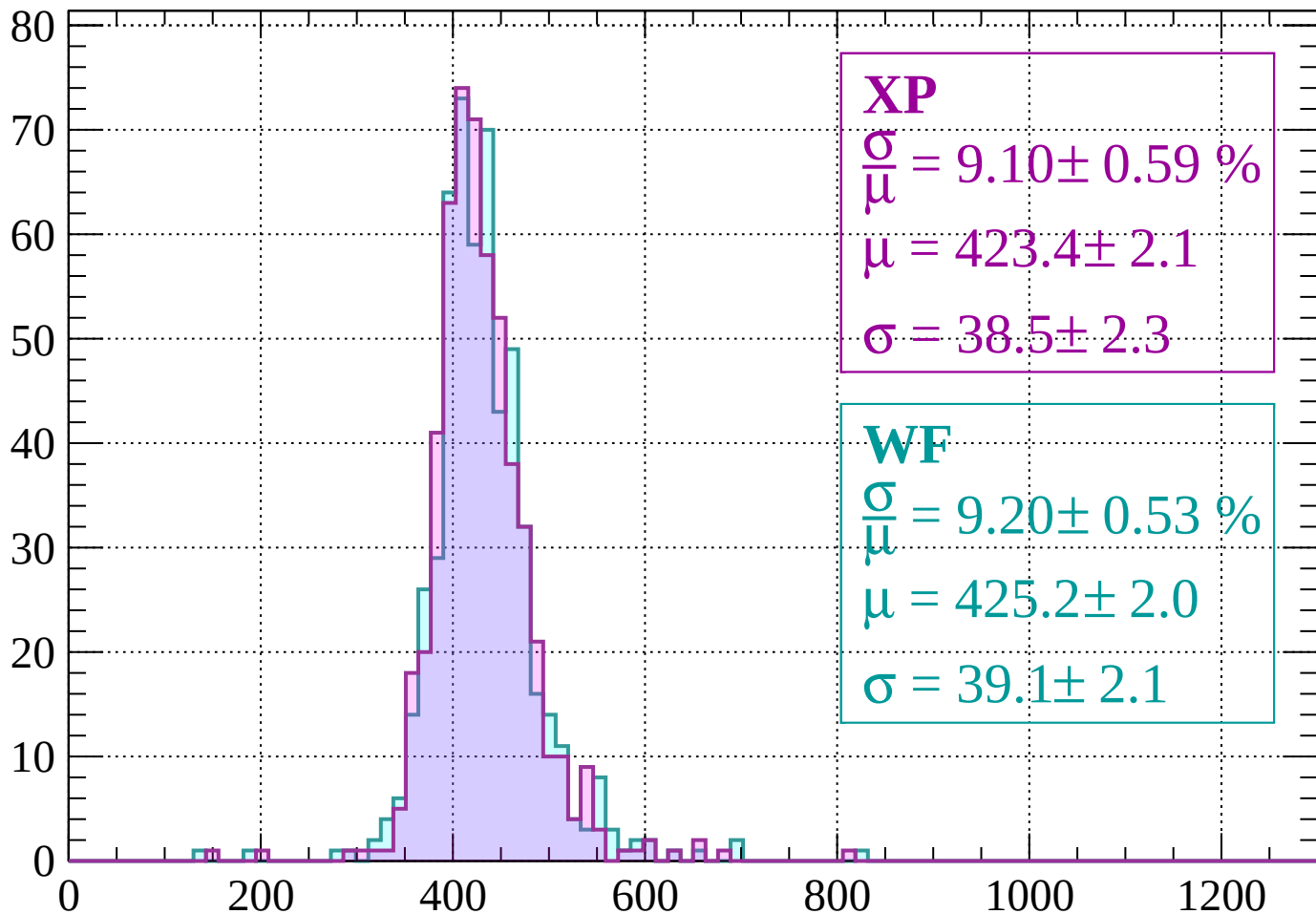
$$\mu = 422.9 \pm 1.9$$

$$\sigma = 38.2 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.59 \%$$

$$\mu = 423.4 \pm 2.1$$

$$\sigma = 38.5 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 9.20 \pm 0.53 \%$$

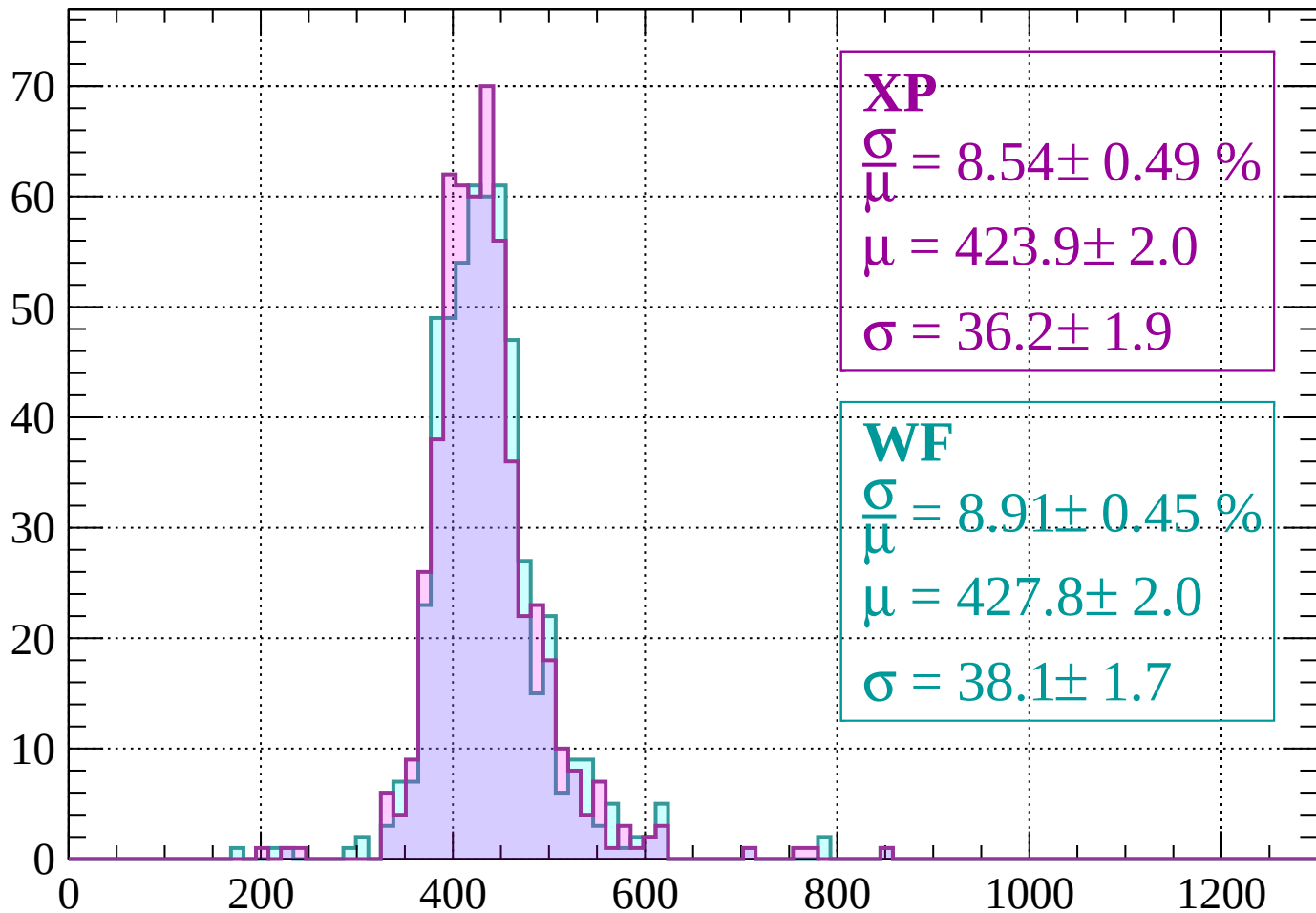
$$\mu = 425.2 \pm 2.0$$

$$\sigma = 39.1 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 8.54 \pm 0.49 \%$$

$$\mu = 423.9 \pm 2.0$$

$$\sigma = 36.2 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 8.91 \pm 0.45 \%$$

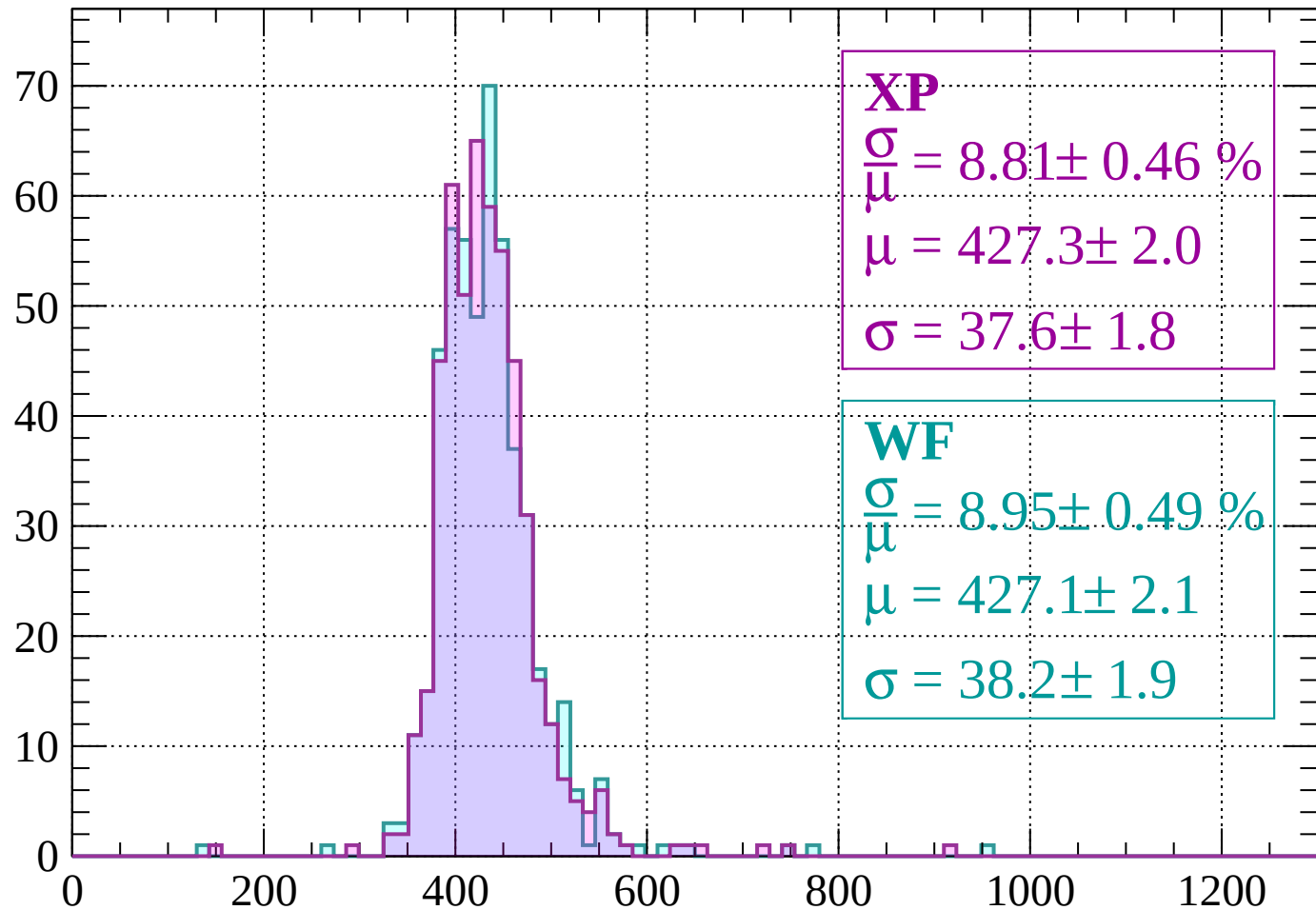
$$\mu = 427.8 \pm 2.0$$

$$\sigma = 38.1 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.81 \pm 0.46 \%$$

$$\mu = 427.3 \pm 2.0$$

$$\sigma = 37.6 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.95 \pm 0.49 \%$$

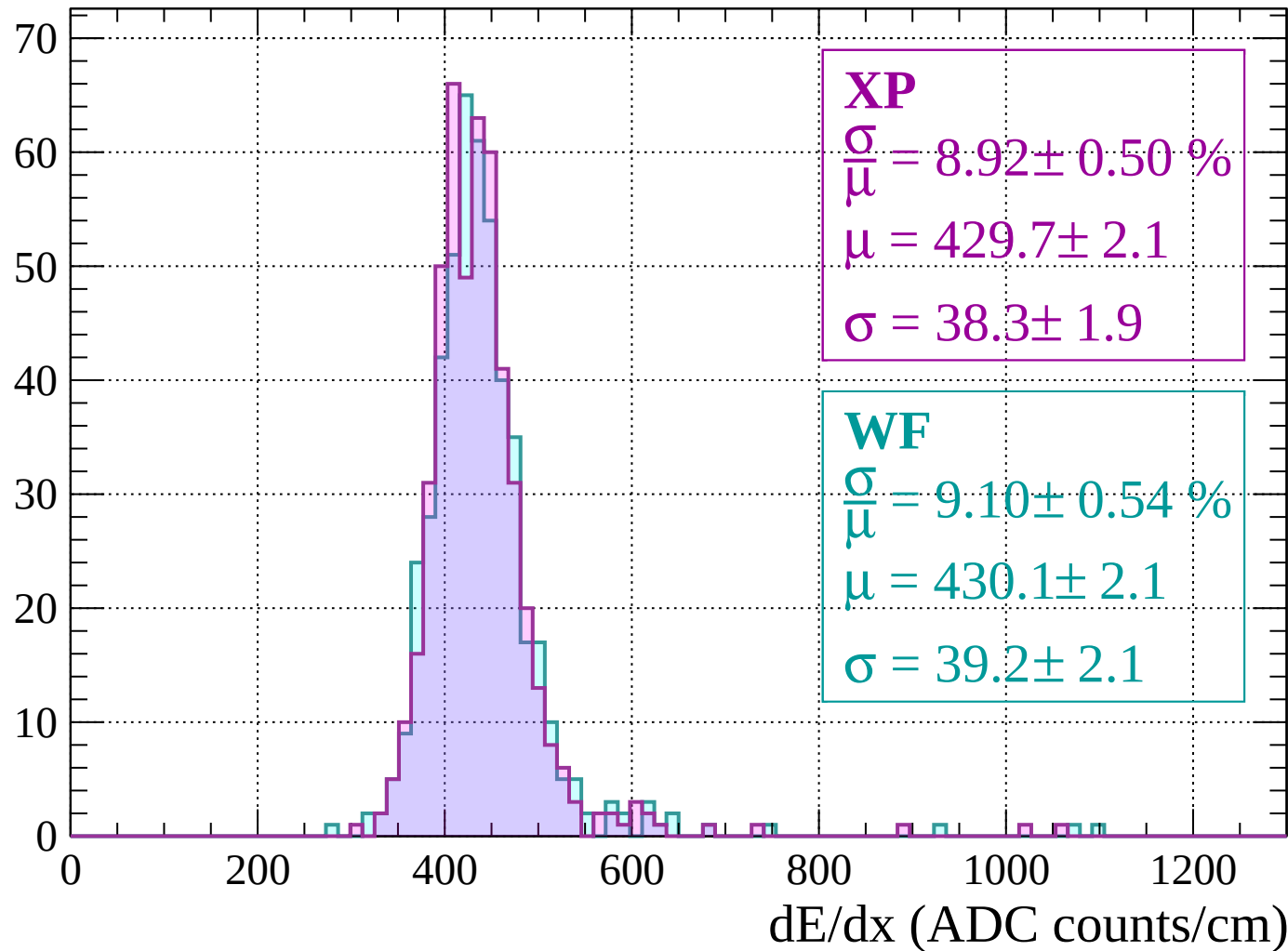
$$\mu = 427.1 \pm 2.1$$

$$\sigma = 38.2 \pm 1.9$$

dE/dx (ADC counts/cm)

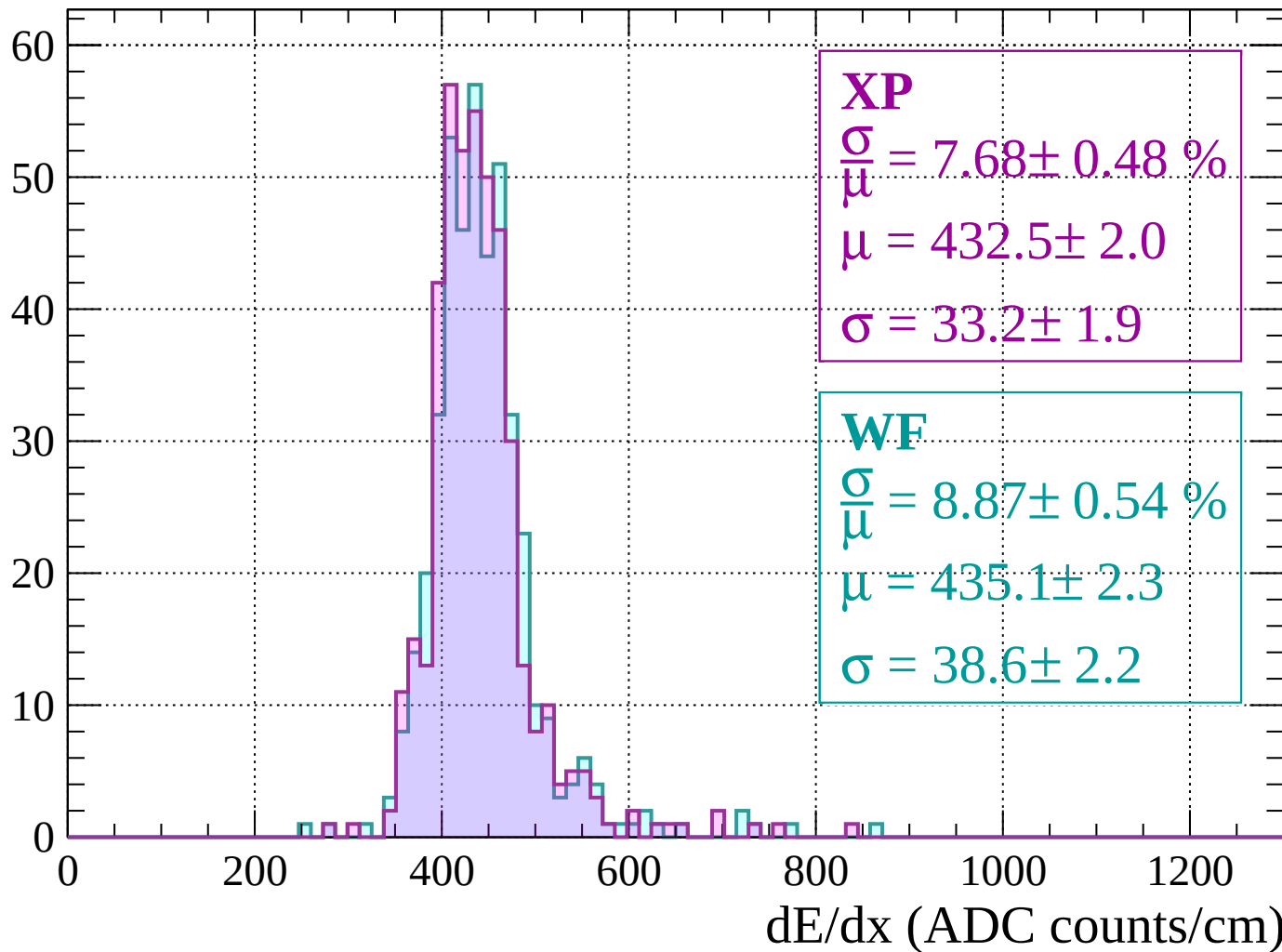
Energy loss | $880 < p < 920$

Count



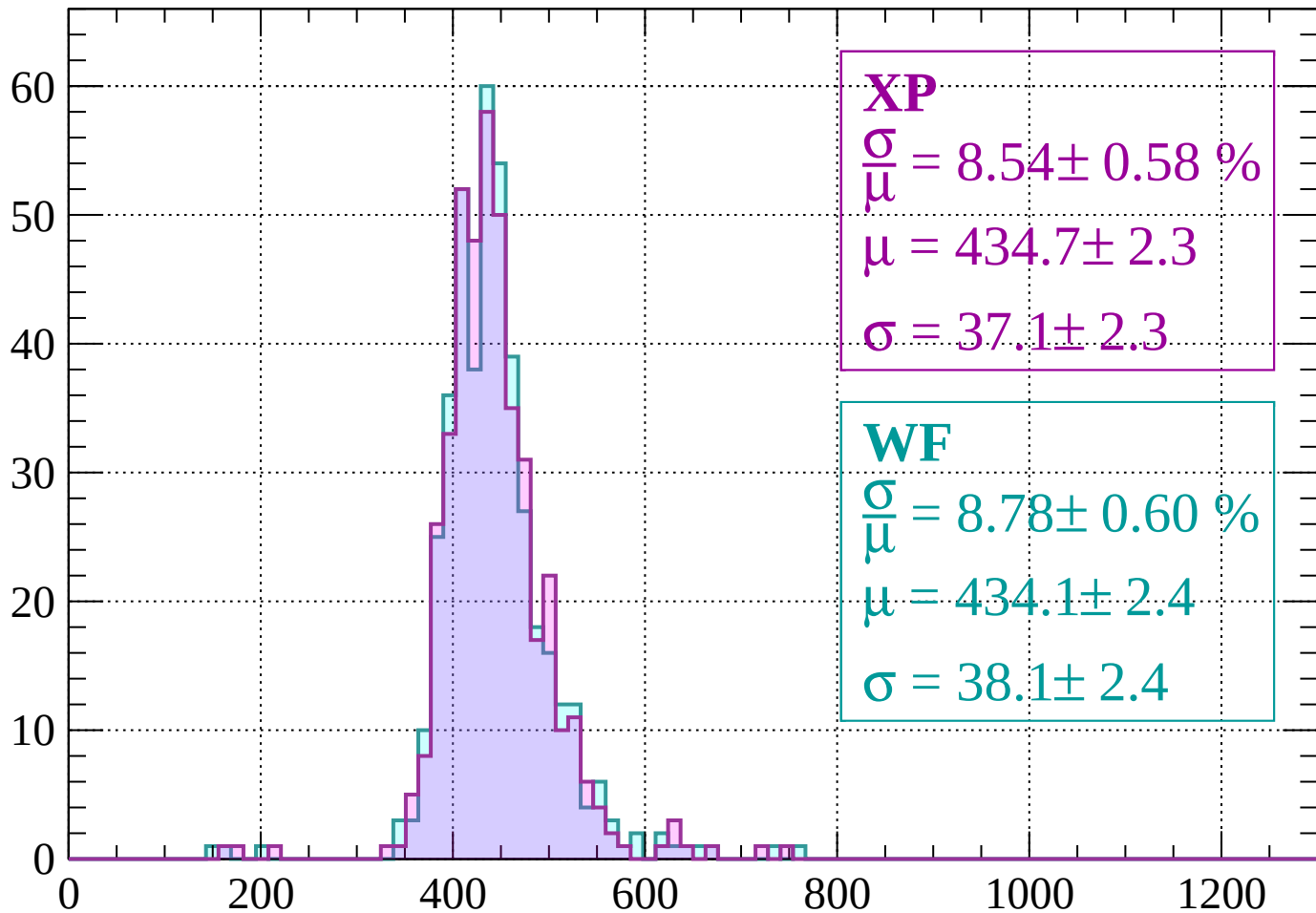
Energy loss | $920 < p < 960$

Count



Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\mu} = 8.54 \pm 0.58 \%$$

$$\mu = 434.7 \pm 2.3$$

$$\sigma = 37.1 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 8.78 \pm 0.60 \%$$

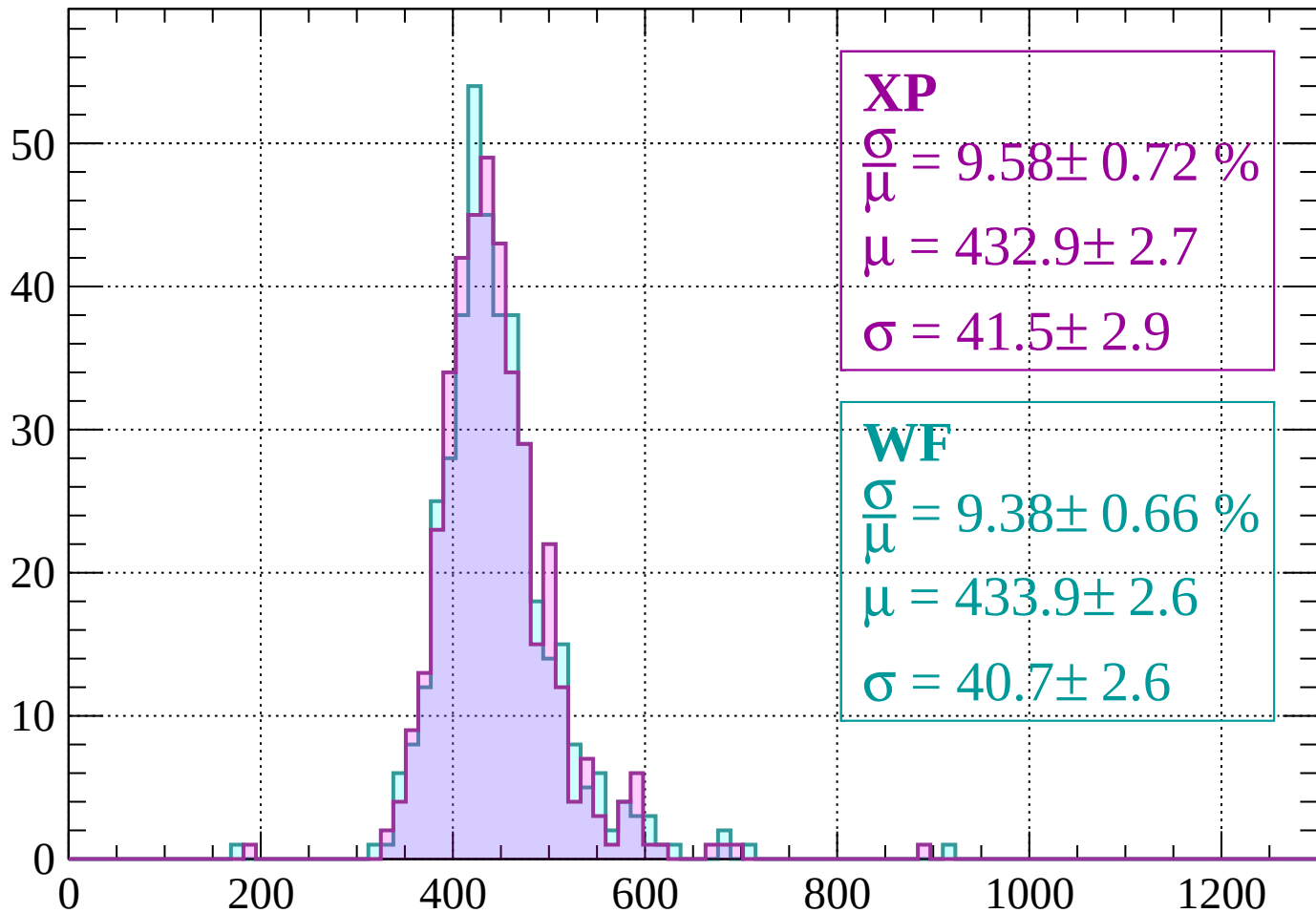
$$\mu = 434.1 \pm 2.4$$

$$\sigma = 38.1 \pm 2.4$$

dE/dx (ADC counts/cm)

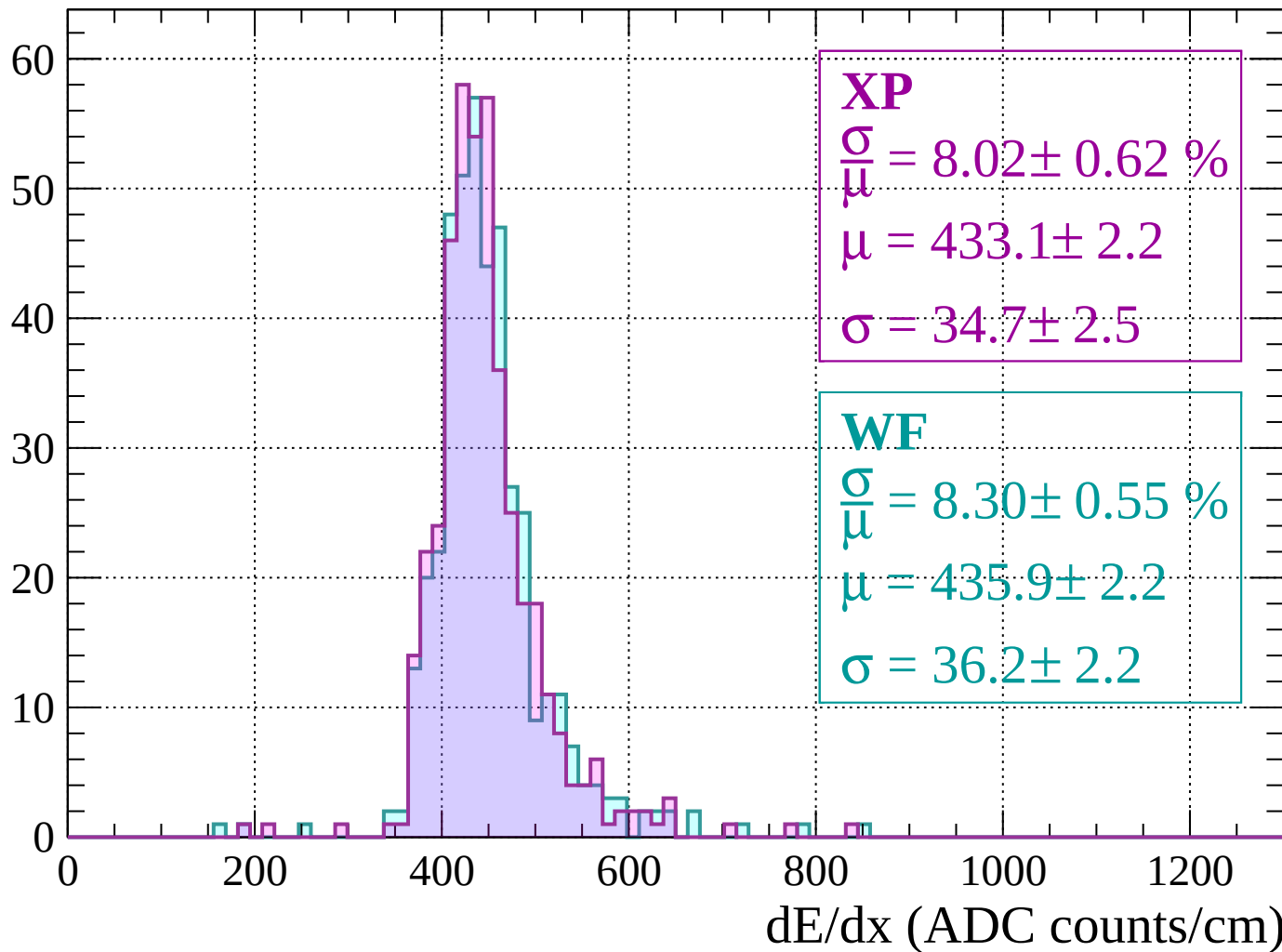
Energy loss | $1000 < p < 1040$

Count



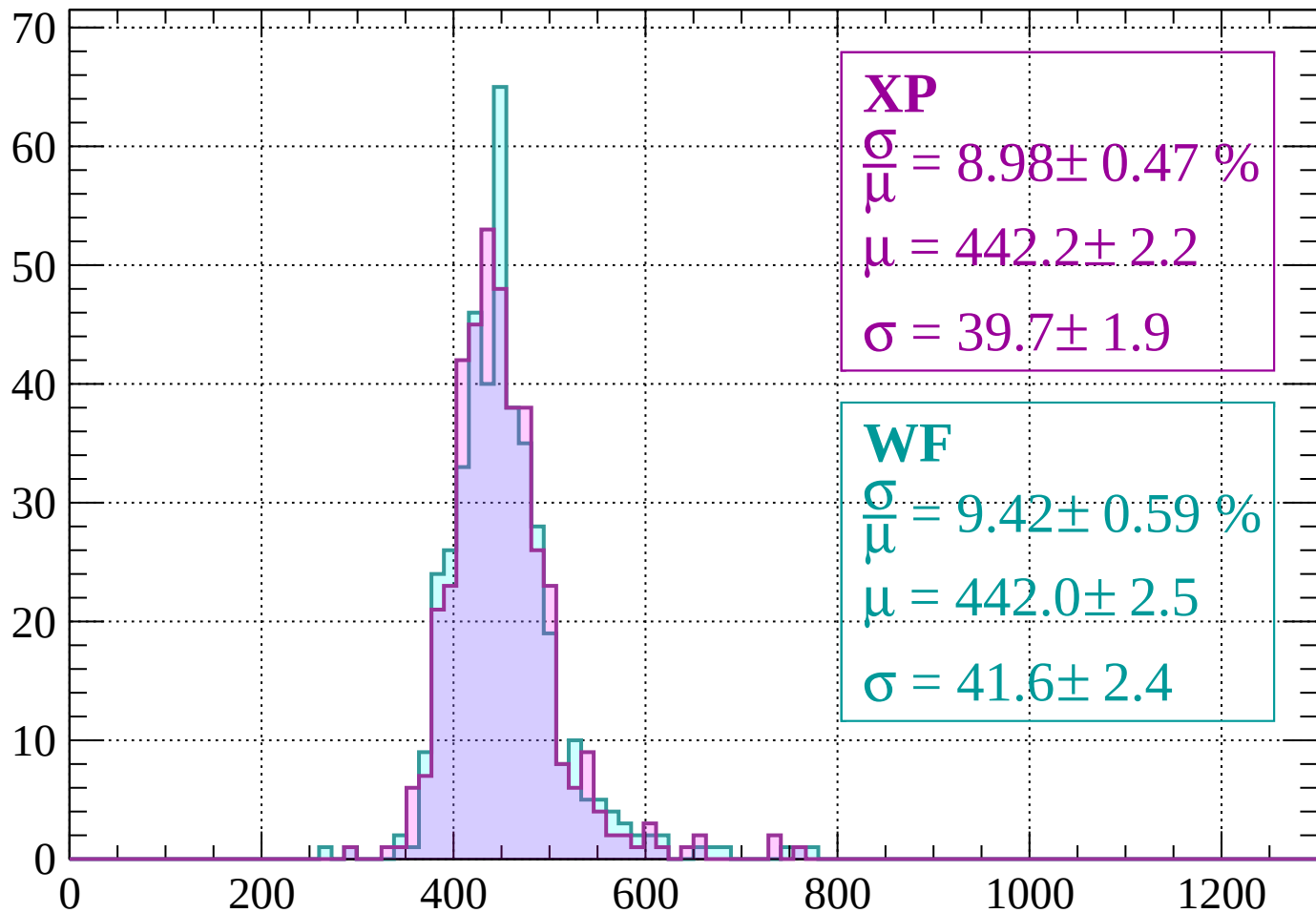
Energy loss | $1040 < p < 1080$

Count



Energy loss | $1080 < p < 1120$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.98 \pm 0.47 \%$$

$$\mu = 442.2 \pm 2.2$$

$$\sigma = 39.7 \pm 1.9$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.42 \pm 0.59 \%$$

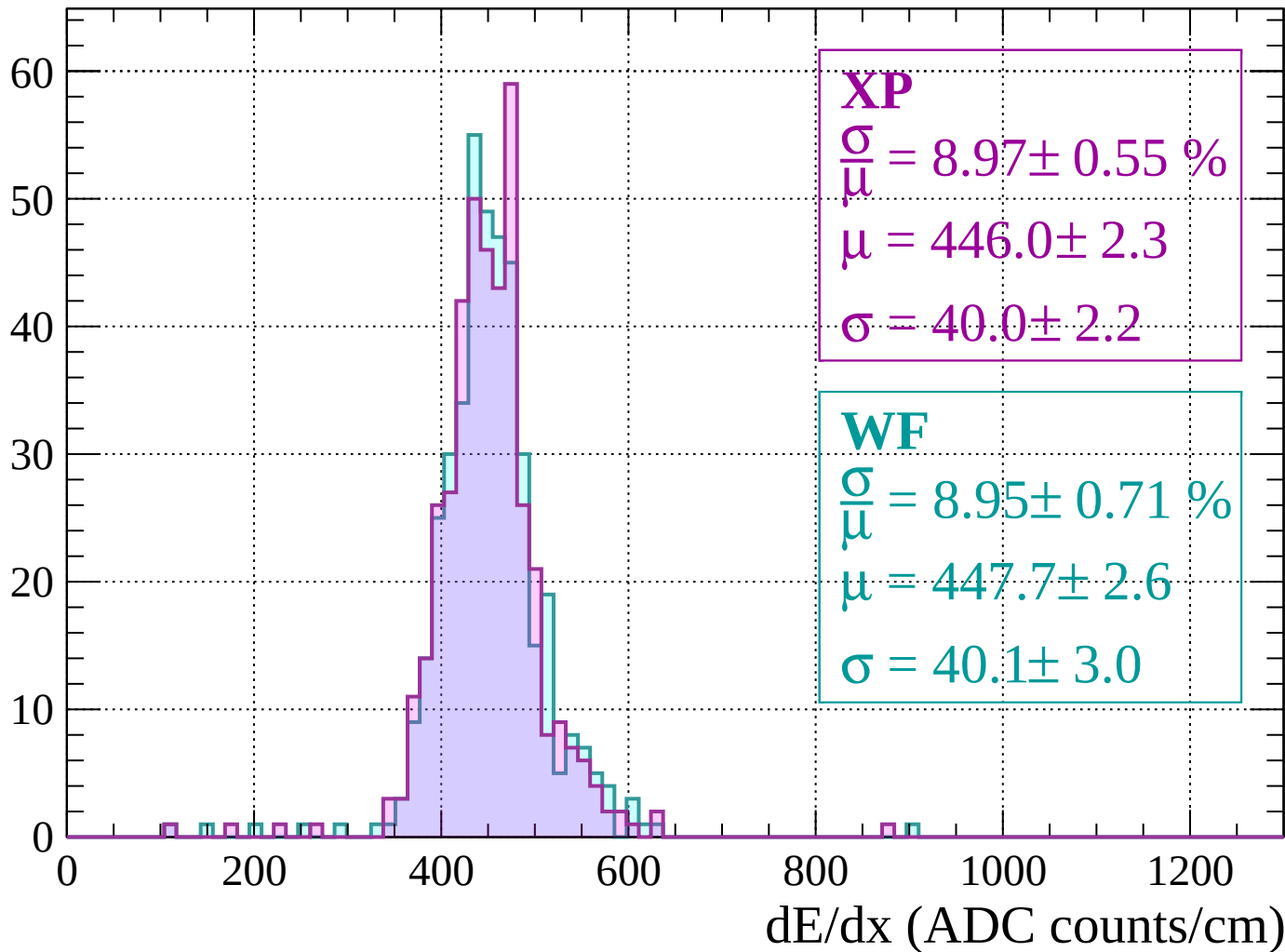
$$\mu = 442.0 \pm 2.5$$

$$\sigma = 41.6 \pm 2.4$$

dE/dx (ADC counts/cm)

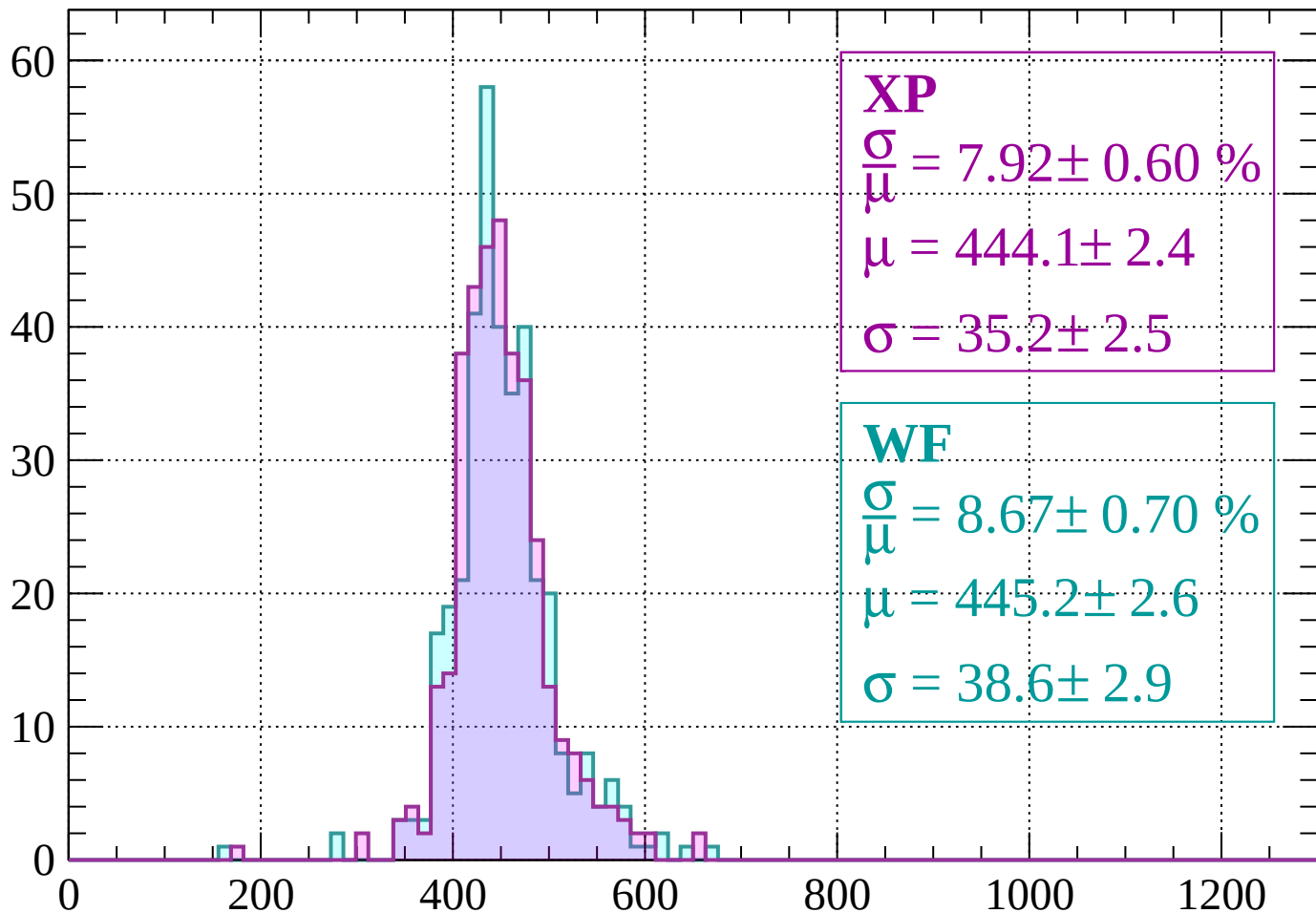
Energy loss | $1120 < p < 1160$

Count



Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 7.92 \pm 0.60 \%$$

$$\mu = 444.1 \pm 2.4$$

$$\sigma = 35.2 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.67 \pm 0.70 \%$$

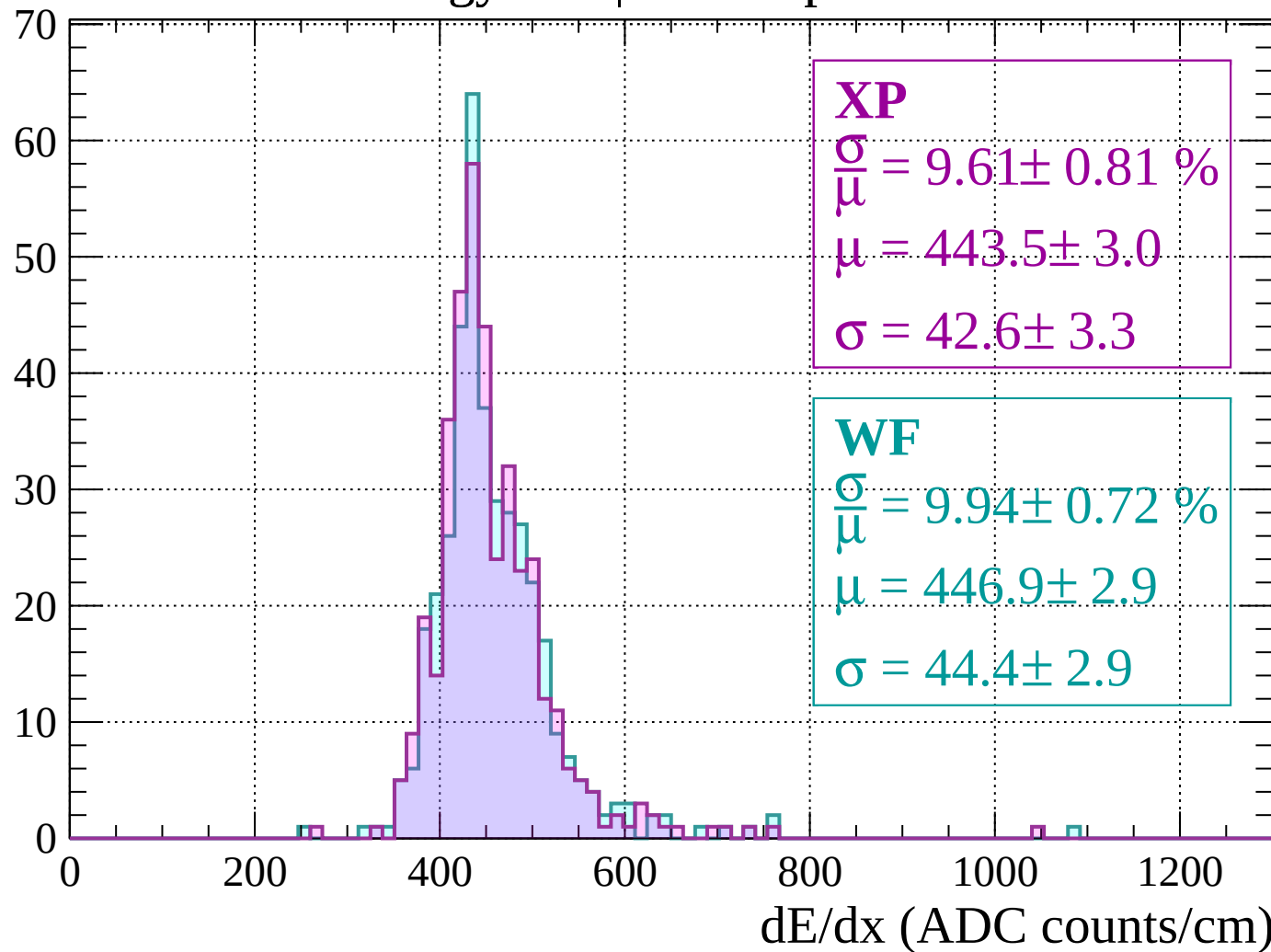
$$\mu = 445.2 \pm 2.6$$

$$\sigma = 38.6 \pm 2.9$$

dE/dx (ADC counts/cm)

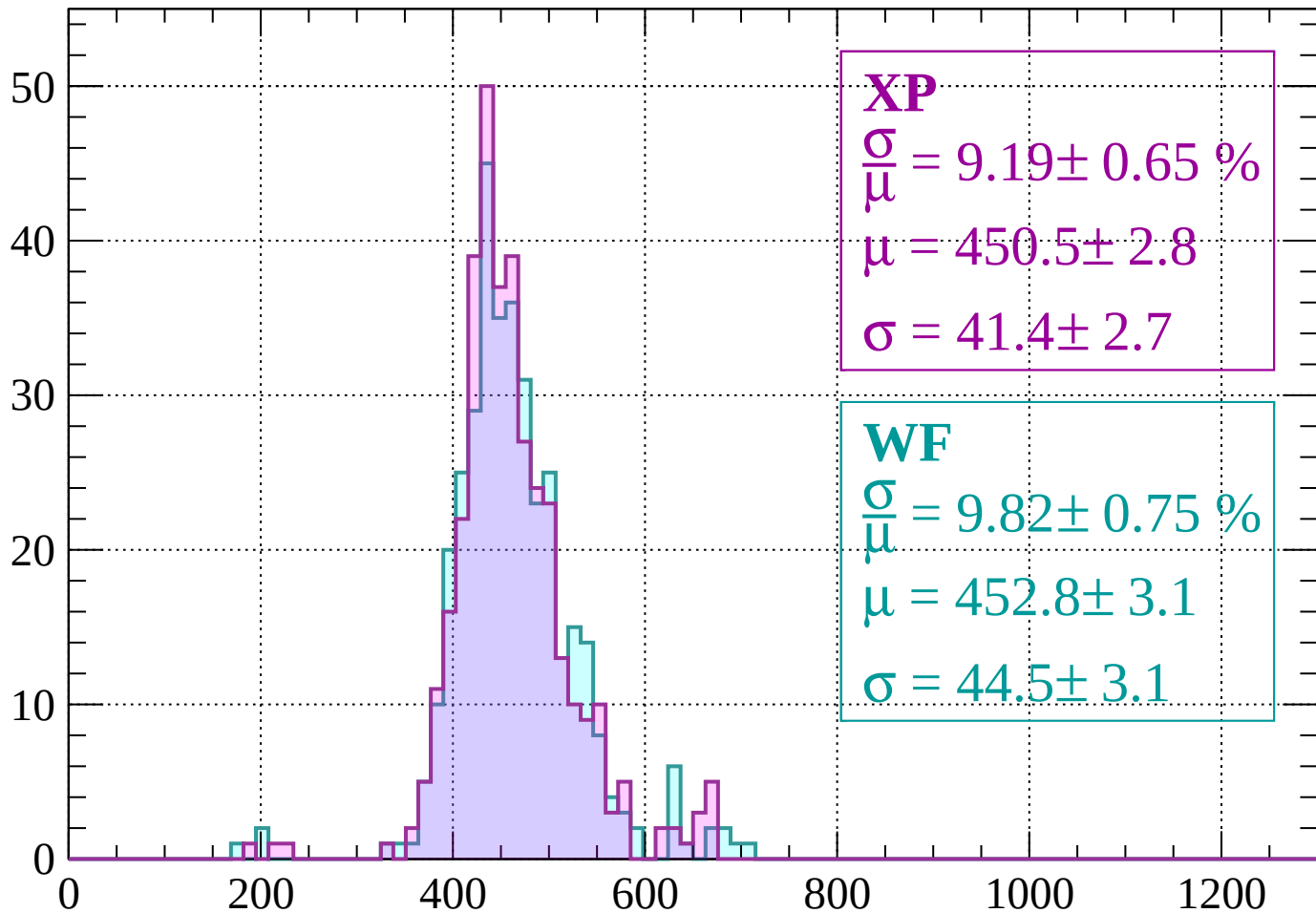
Energy loss | $1200 < p < 1240$

Count



Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 9.19 \pm 0.65 \%$$

$$\mu = 450.5 \pm 2.8$$

$$\sigma = 41.4 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 9.82 \pm 0.75 \%$$

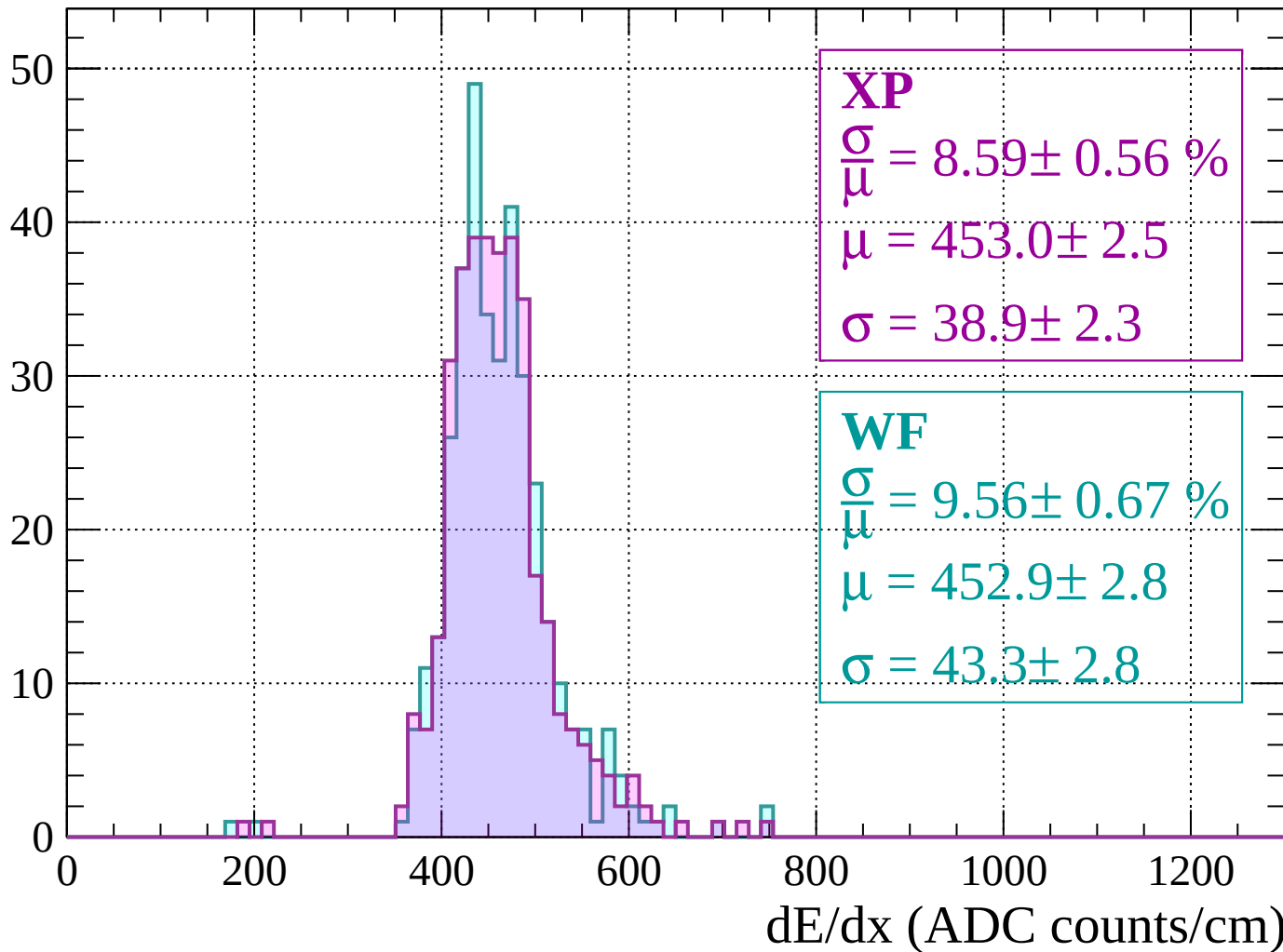
$$\mu = 452.8 \pm 3.1$$

$$\sigma = 44.5 \pm 3.1$$

dE/dx (ADC counts/cm)

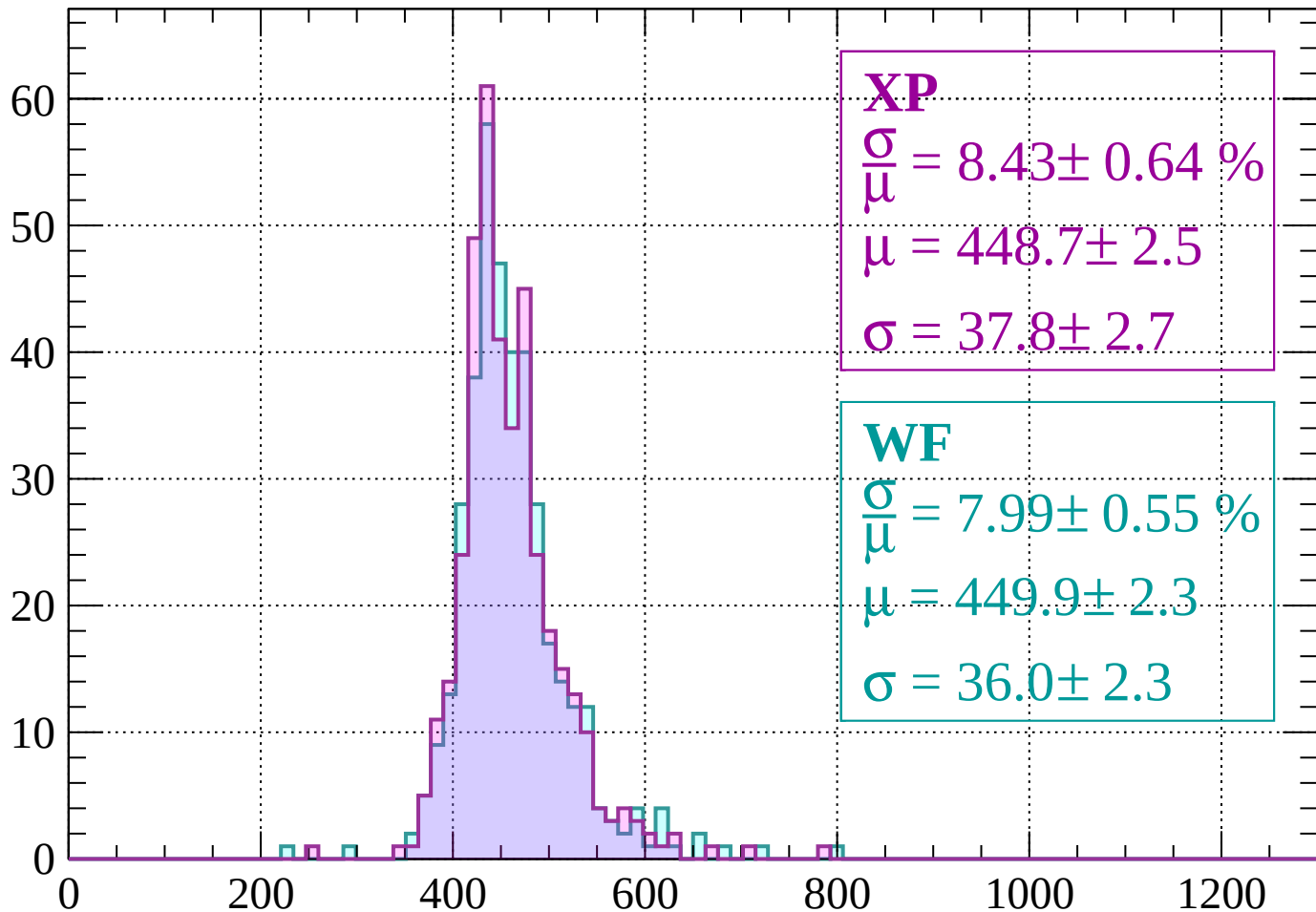
Energy loss | $1280 < p < 1320$

Count



Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.43 \pm 0.64 \%$$

$$\mu = 448.7 \pm 2.5$$

$$\sigma = 37.8 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 7.99 \pm 0.55 \%$$

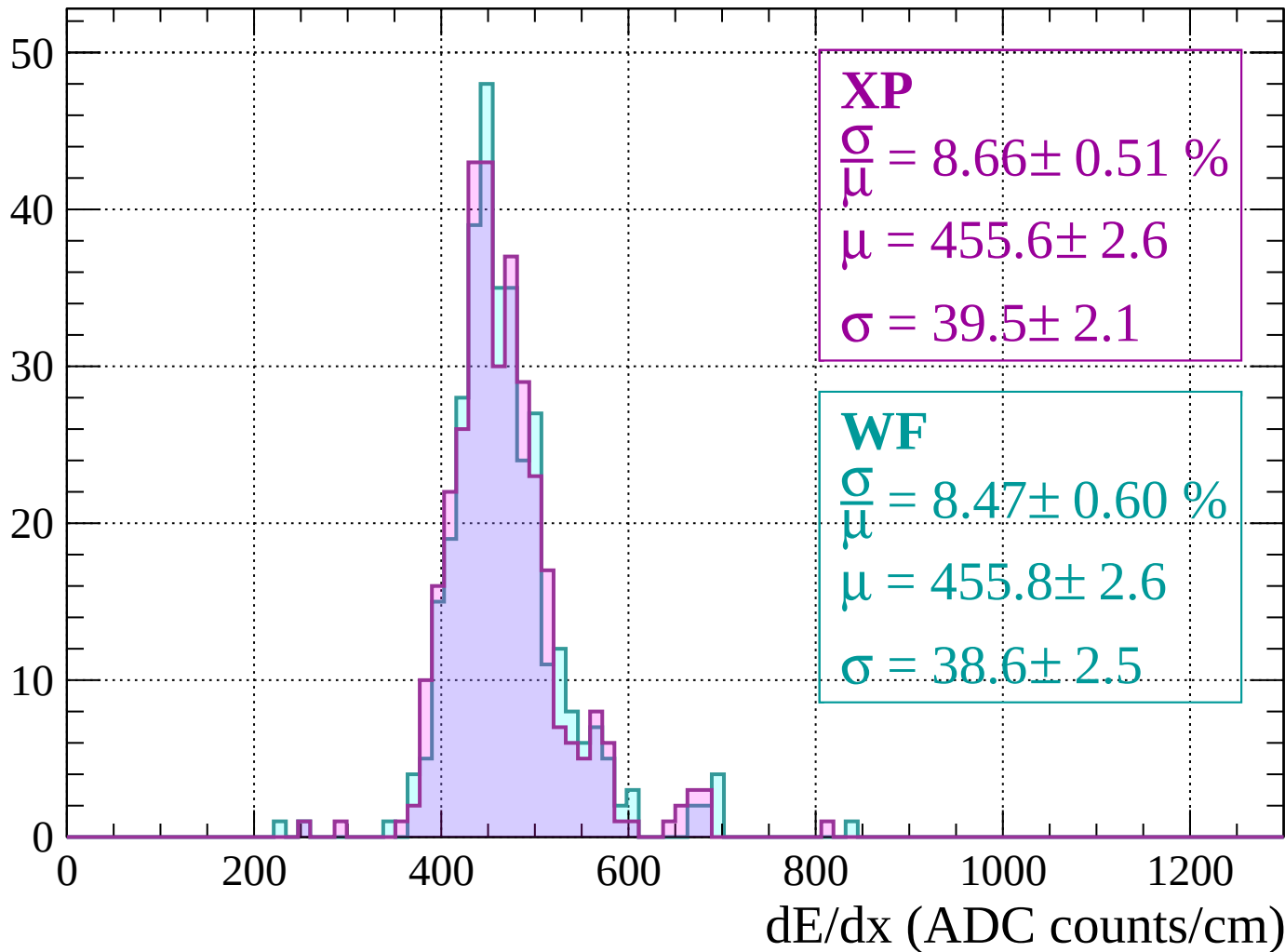
$$\mu = 449.9 \pm 2.3$$

$$\sigma = 36.0 \pm 2.3$$

dE/dx (ADC counts/cm)

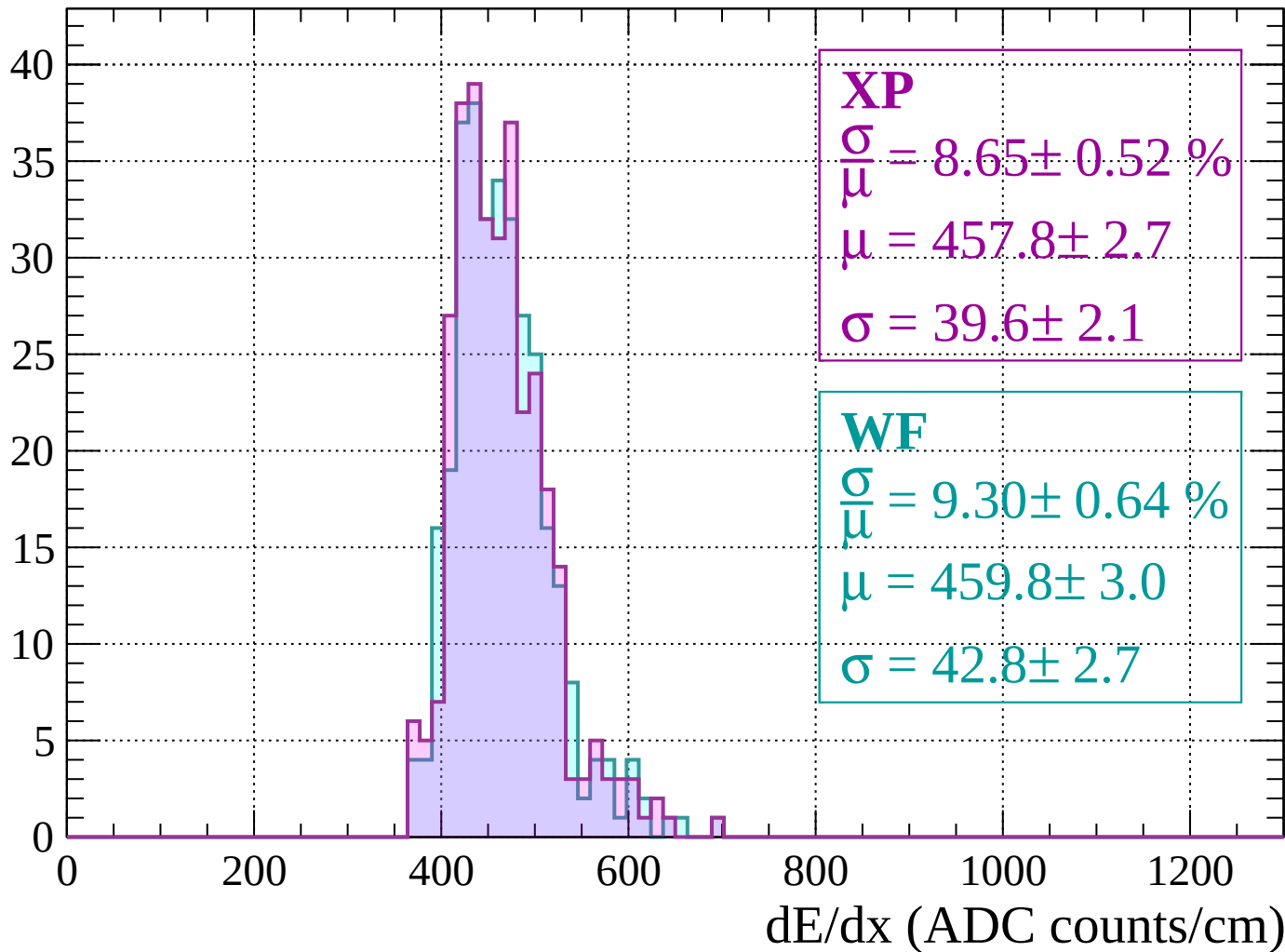
Energy loss | $1360 < p < 1400$

Count



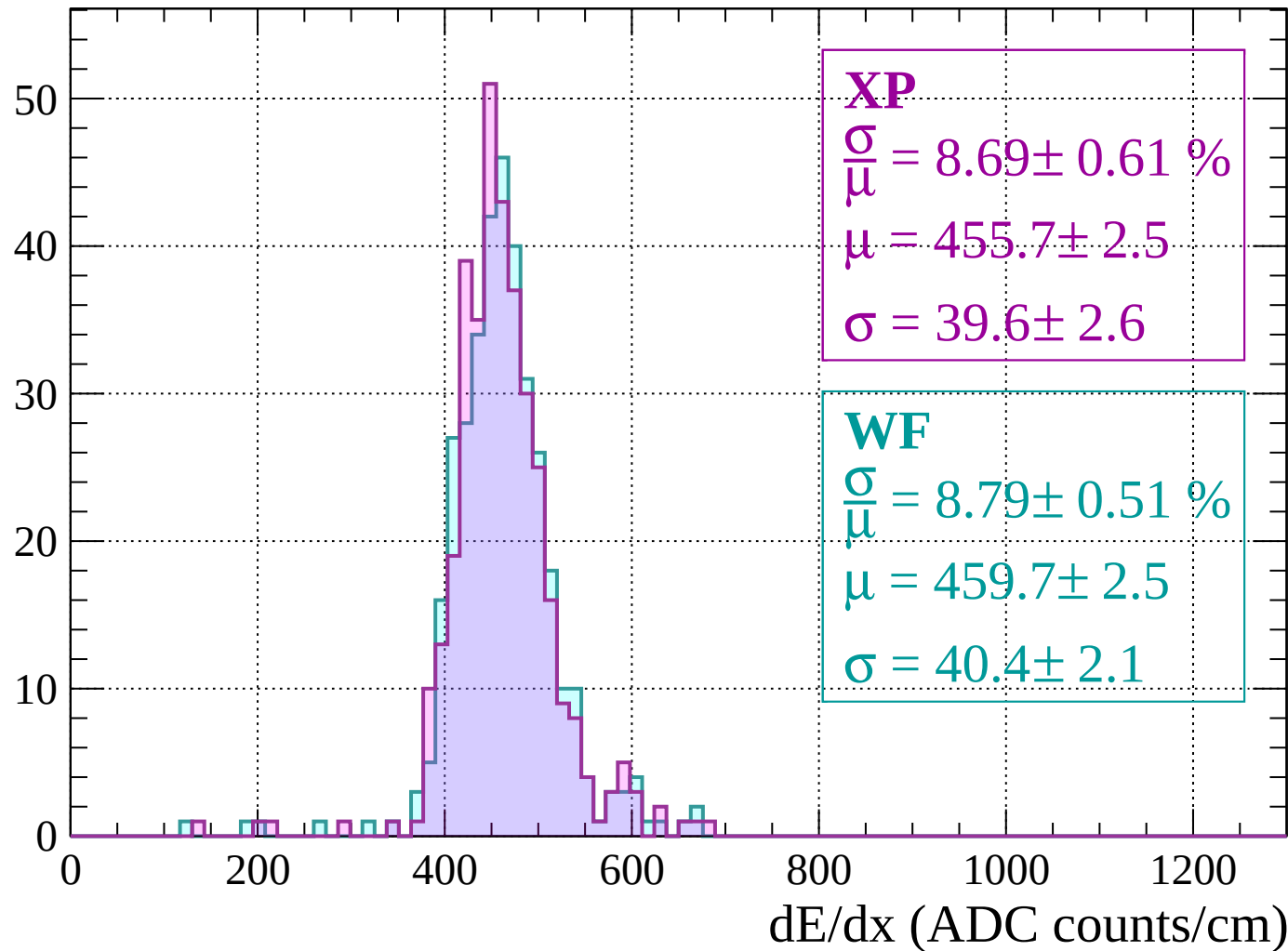
Energy loss | $1400 < p < 1440$

Count



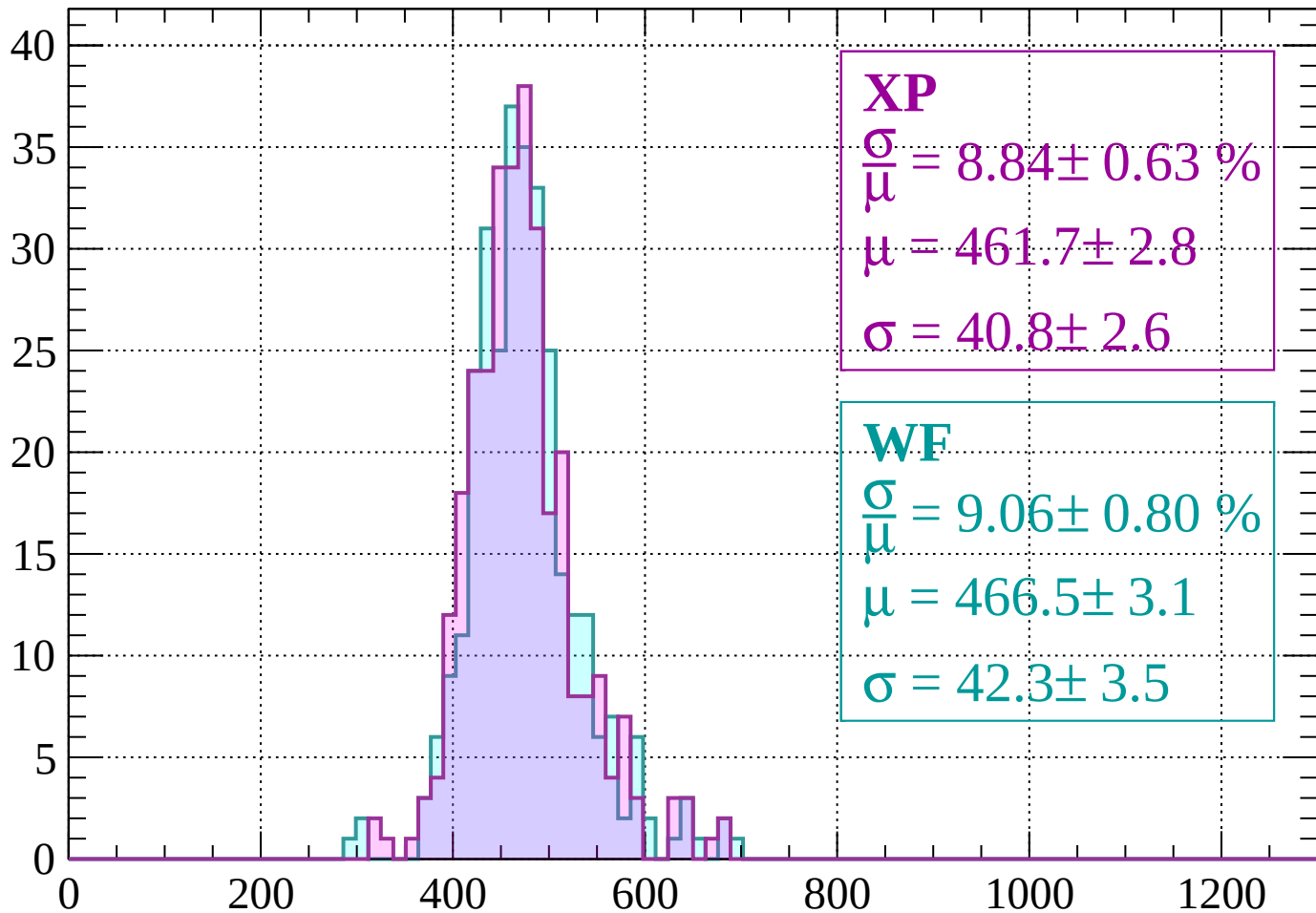
Energy loss | $1440 < p < 1480$

Count



Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.84 \pm 0.63 \%$$

$$\mu = 461.7 \pm 2.8$$

$$\sigma = 40.8 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 9.06 \pm 0.80 \%$$

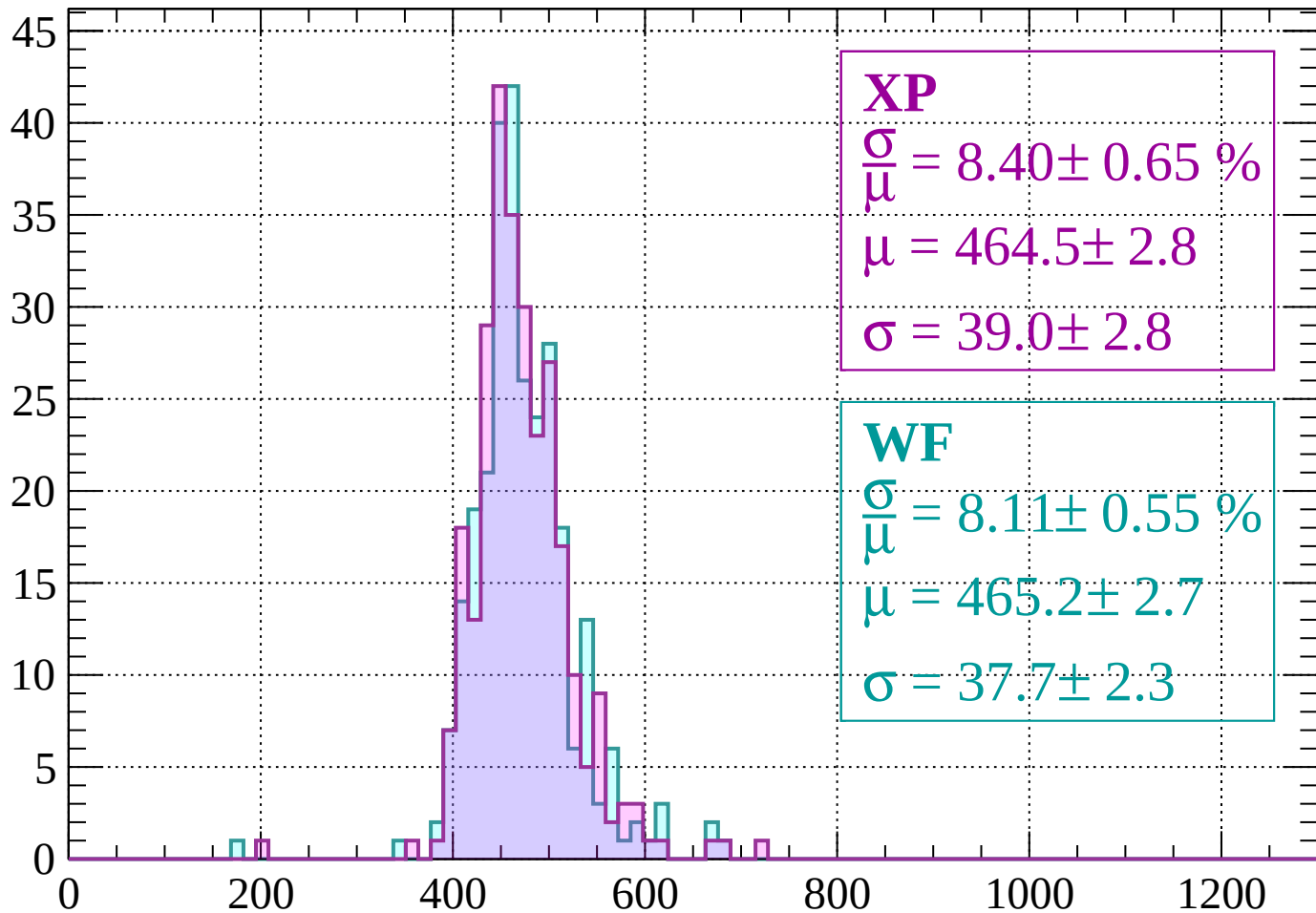
$$\mu = 466.5 \pm 3.1$$

$$\sigma = 42.3 \pm 3.5$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 8.40 \pm 0.65 \%$$

$$\mu = 464.5 \pm 2.8$$

$$\sigma = 39.0 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 8.11 \pm 0.55 \%$$

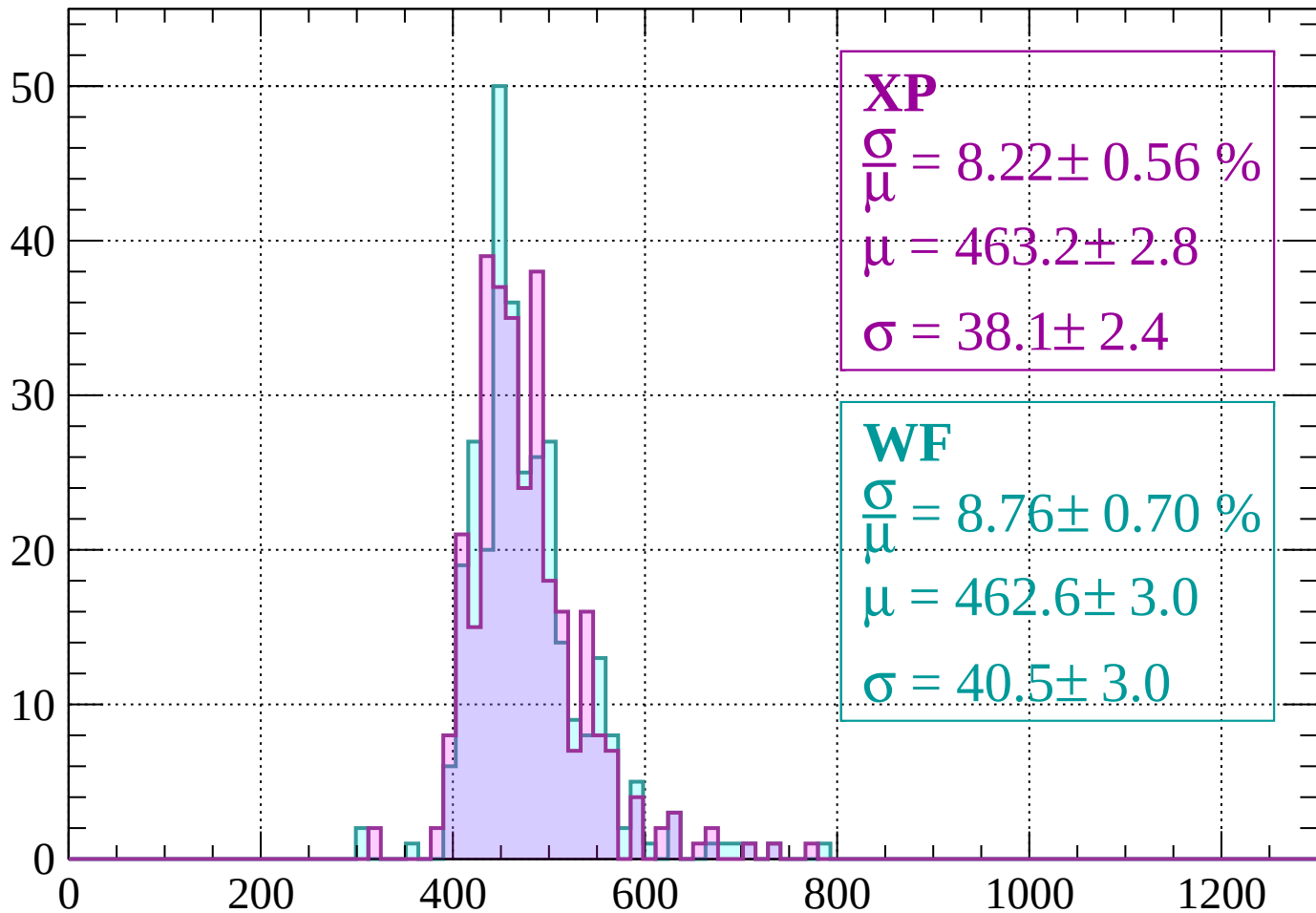
$$\mu = 465.2 \pm 2.7$$

$$\sigma = 37.7 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.56 \%$$

$$\mu = 463.2 \pm 2.8$$

$$\sigma = 38.1 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 8.76 \pm 0.70 \%$$

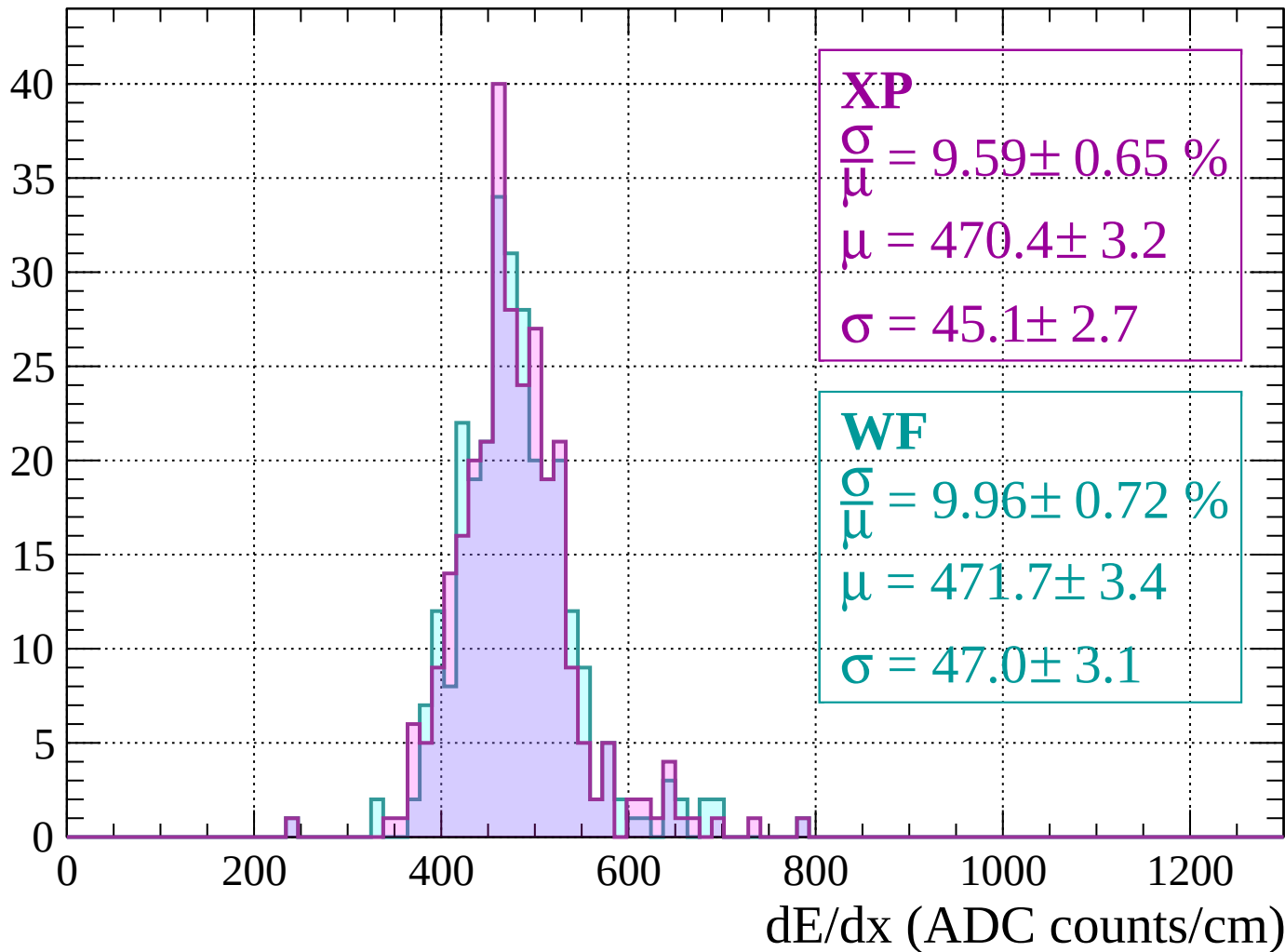
$$\mu = 462.6 \pm 3.0$$

$$\sigma = 40.5 \pm 3.0$$

dE/dx (ADC counts/cm)

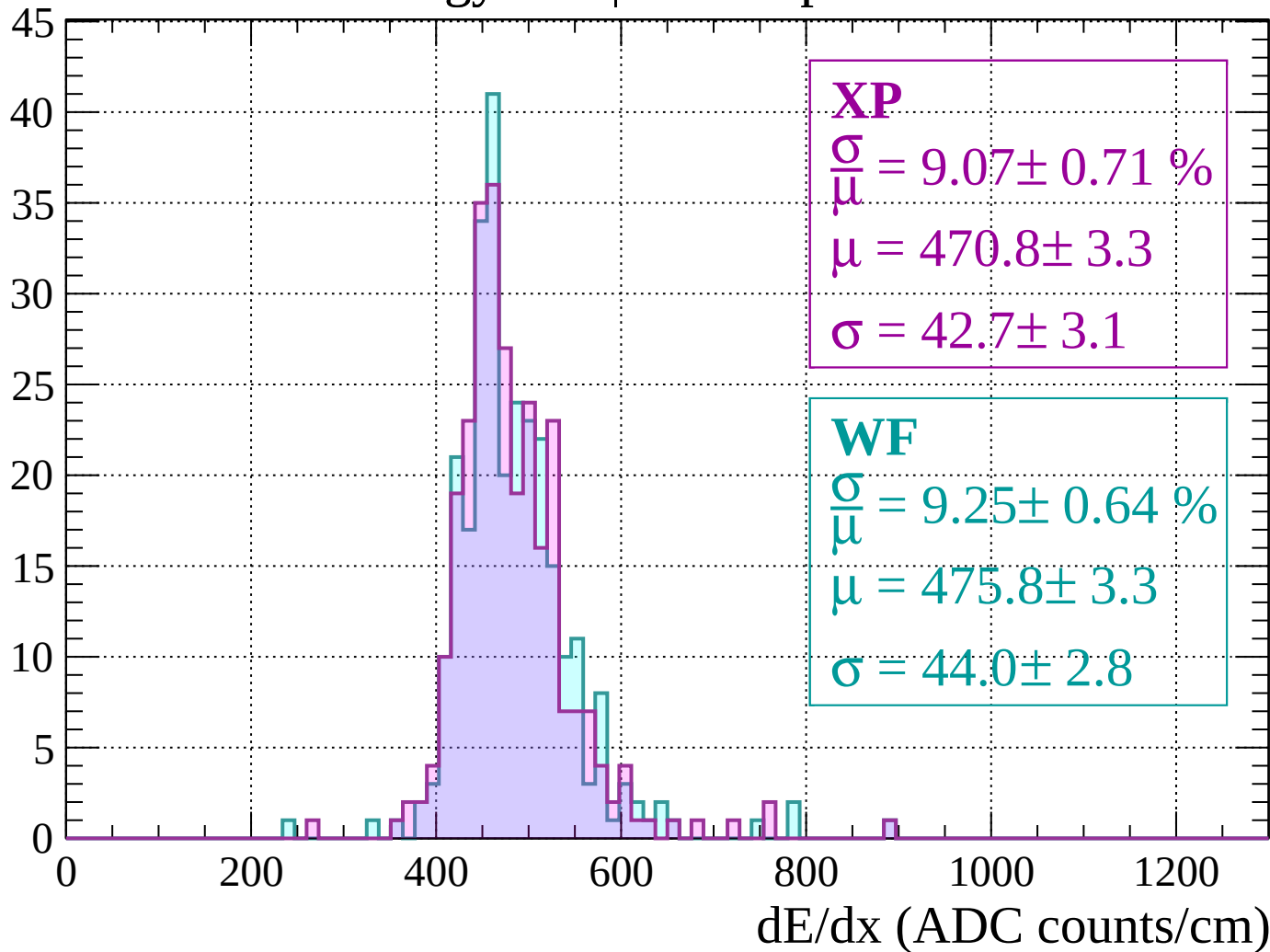
Energy loss | $1600 < p < 1640$

Count



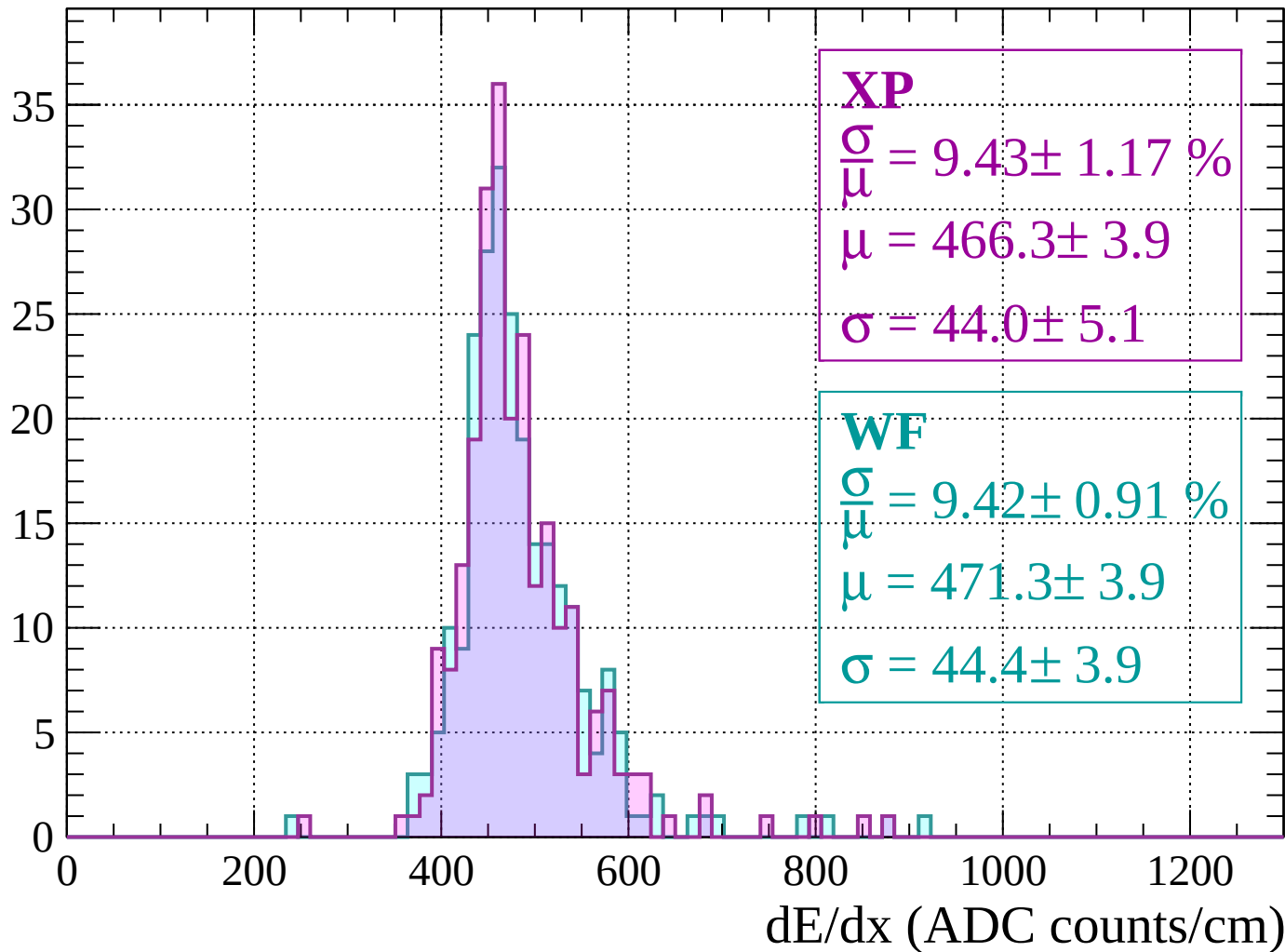
Energy loss | $1640 < p < 1680$

Count



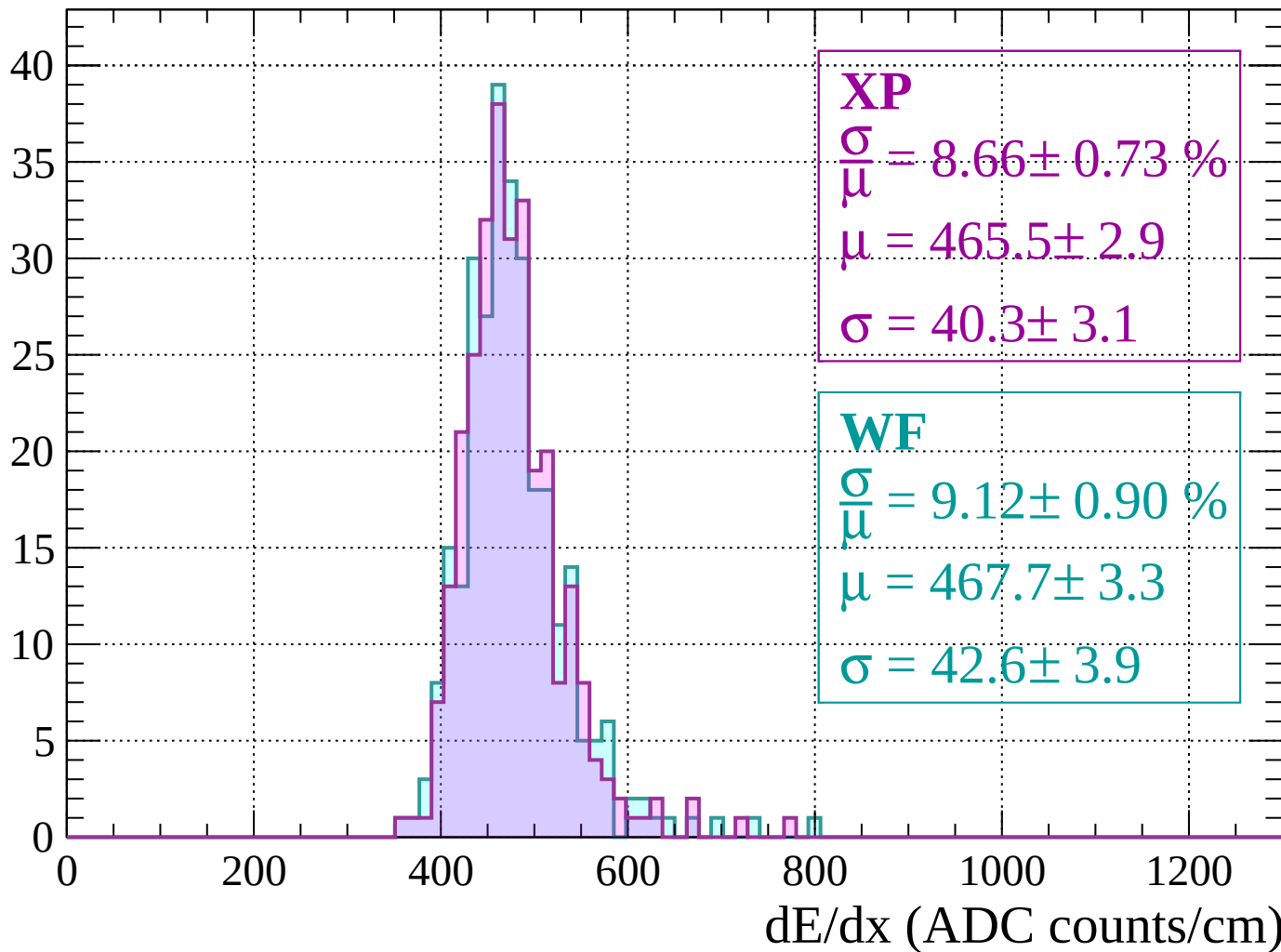
Energy loss | $1680 < p < 1720$

Count



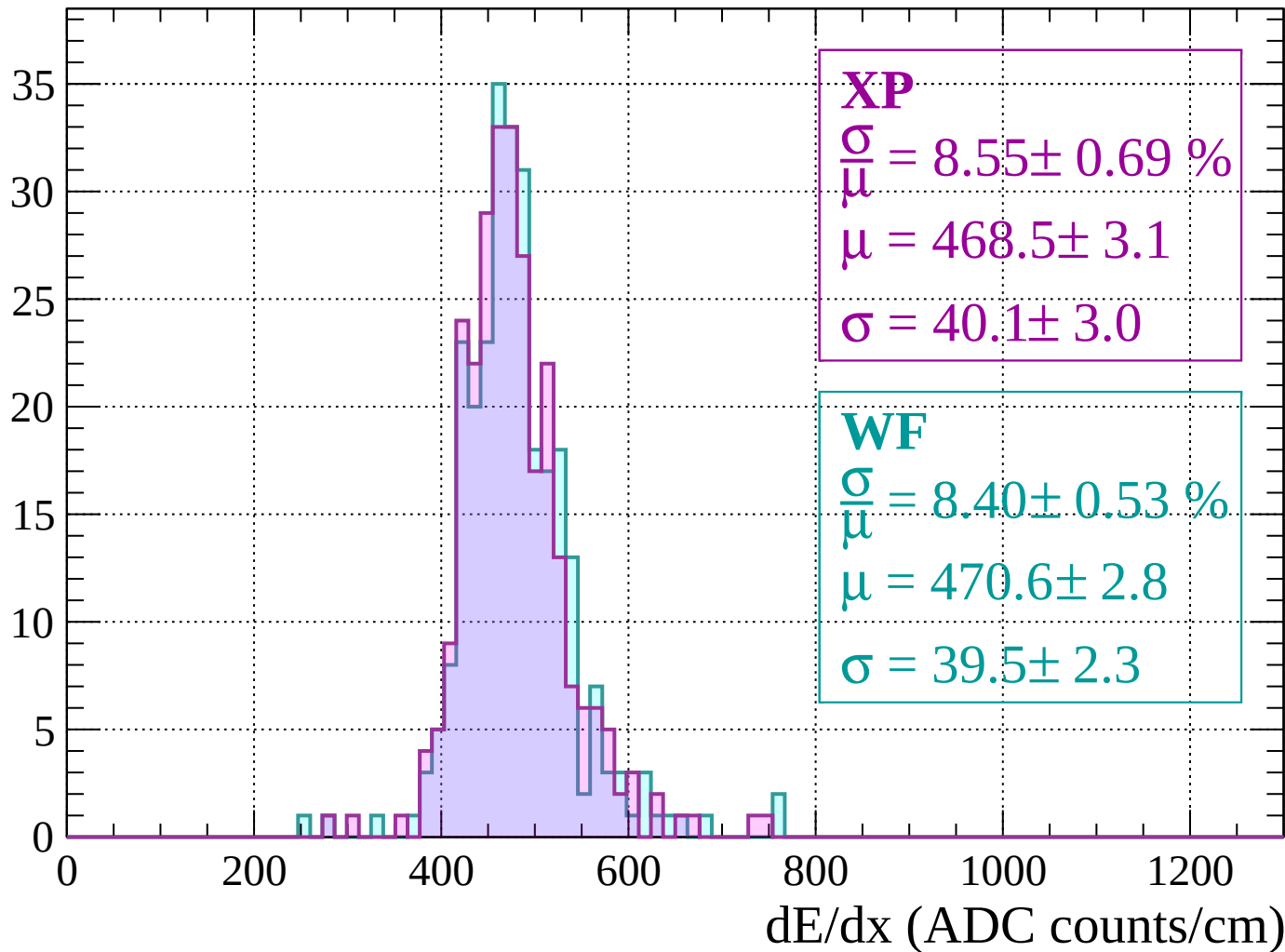
Energy loss | $1720 < p < 1760$

Count



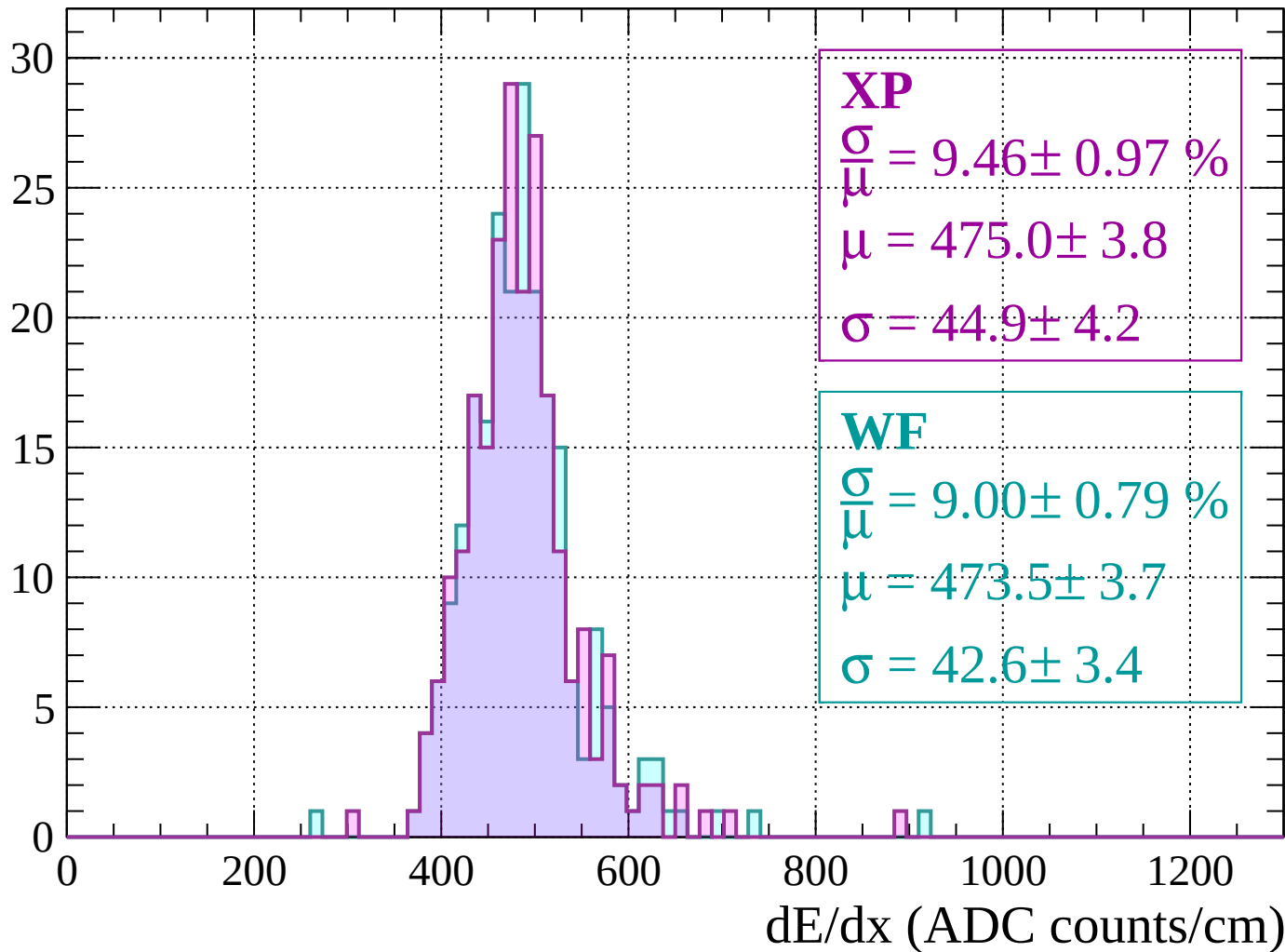
Energy loss | $1760 < p < 1800$

Count



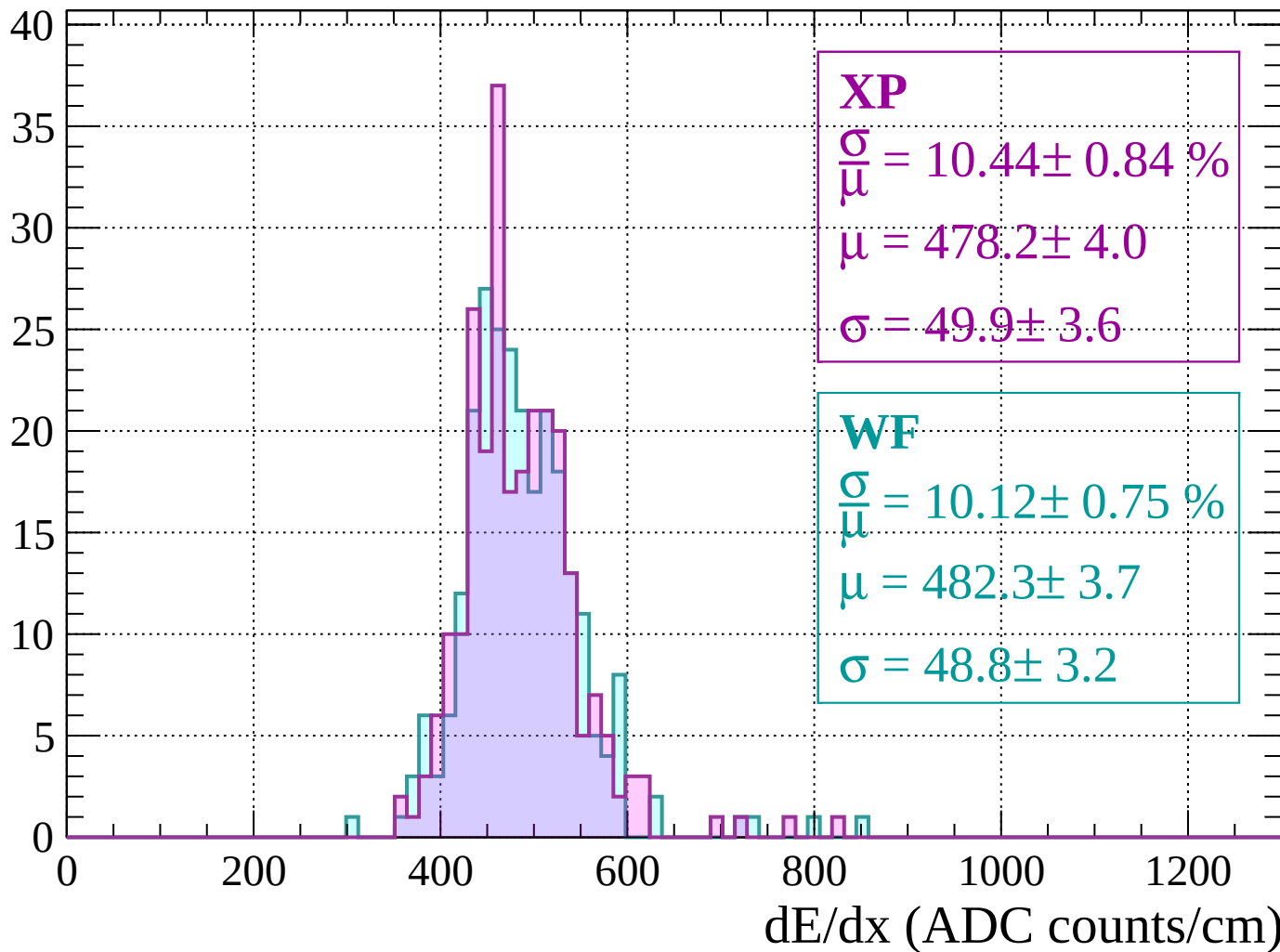
Energy loss | $1800 < p < 1840$

Count



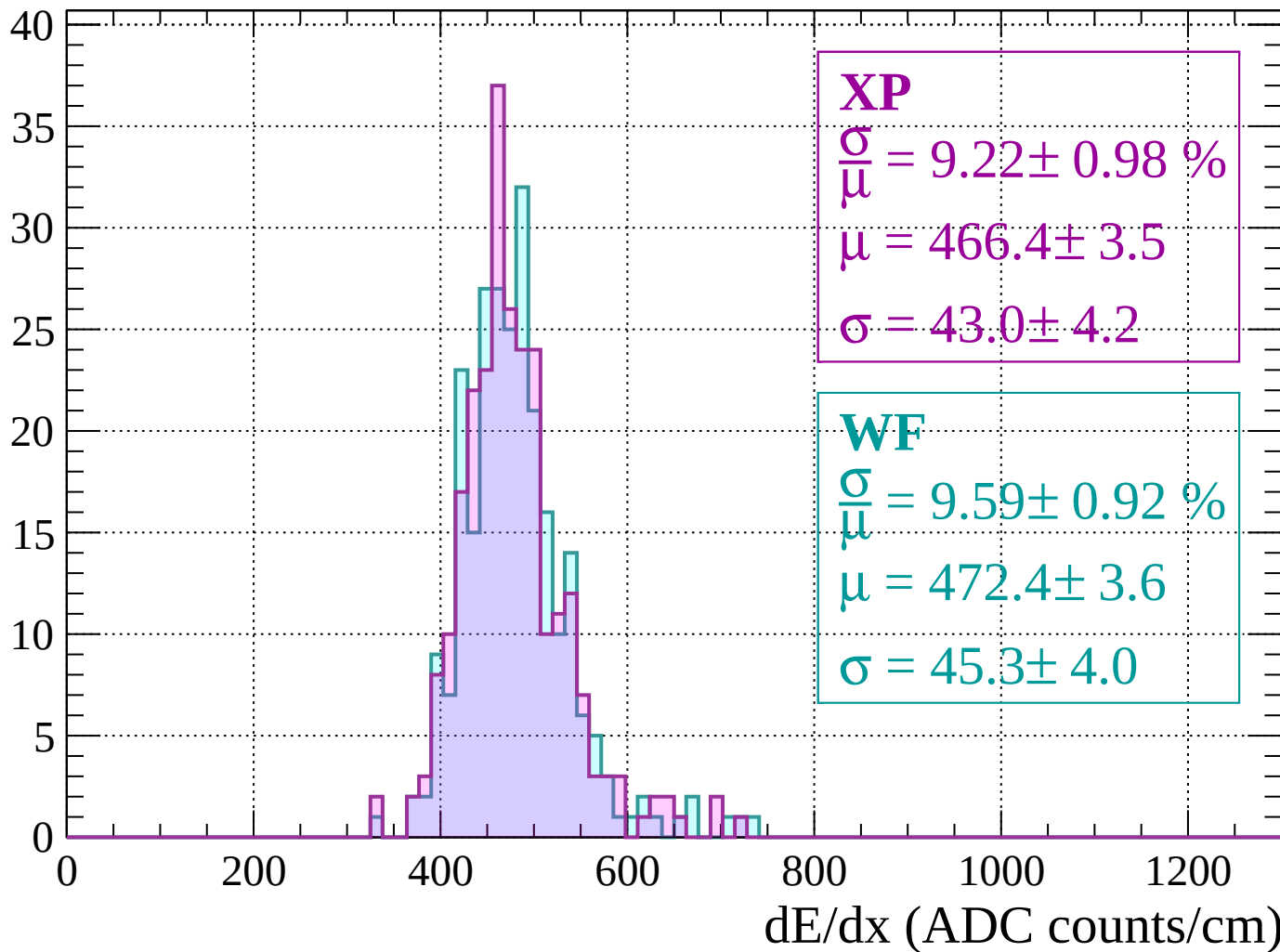
Energy loss | $1840 < p < 1880$

Count



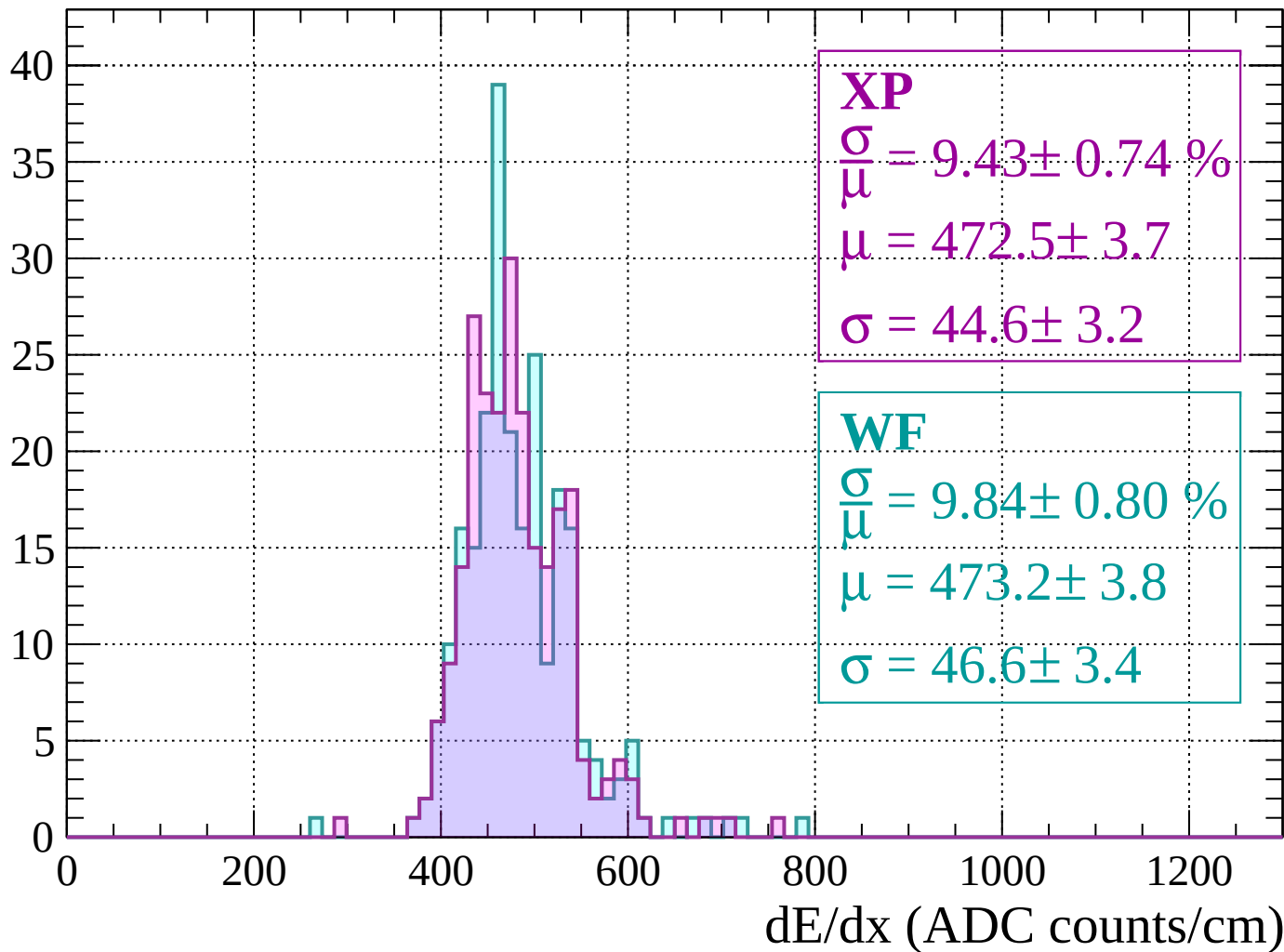
Energy loss | $1880 < p < 1920$

Count



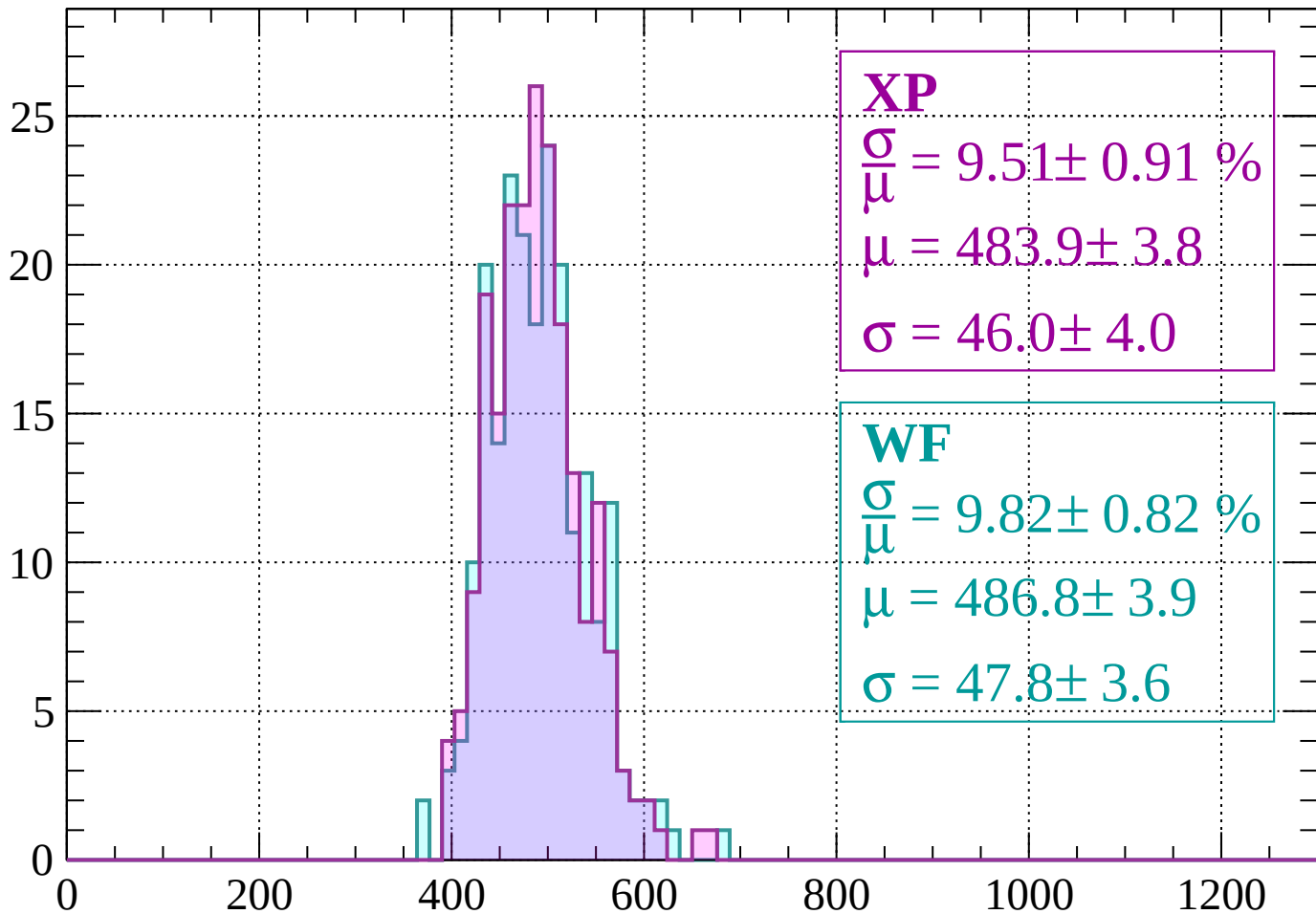
Energy loss | $1920 < p < 1960$

Count



Energy loss | 1960 < p < 2000

Count



XP

$$\frac{\sigma}{\mu} = 9.51 \pm 0.91 \%$$

$$\mu = 483.9 \pm 3.8$$

$$\sigma = 46.0 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 9.82 \pm 0.82 \%$$

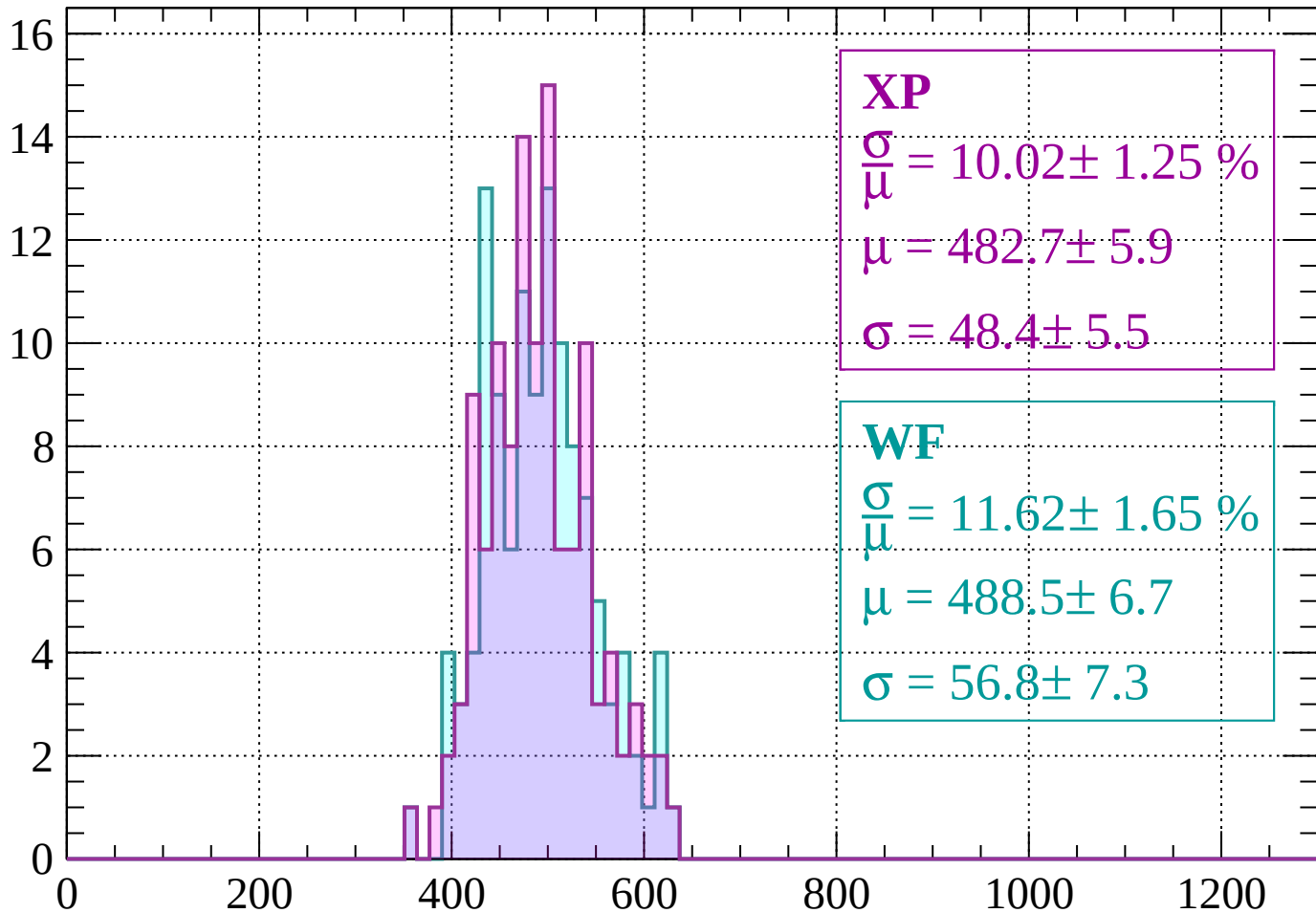
$$\mu = 486.8 \pm 3.9$$

$$\sigma = 47.8 \pm 3.6$$

dE/dx (ADC counts/cm)

Energy loss | $2000 < p < 2040$

Count



XP

$$\frac{\sigma}{\mu} = 10.02 \pm 1.25 \%$$

$$\mu = 482.7 \pm 5.9$$

$$\sigma = 48.4 \pm 5.5$$

WF

$$\frac{\sigma}{\mu} = 11.62 \pm 1.65 \%$$

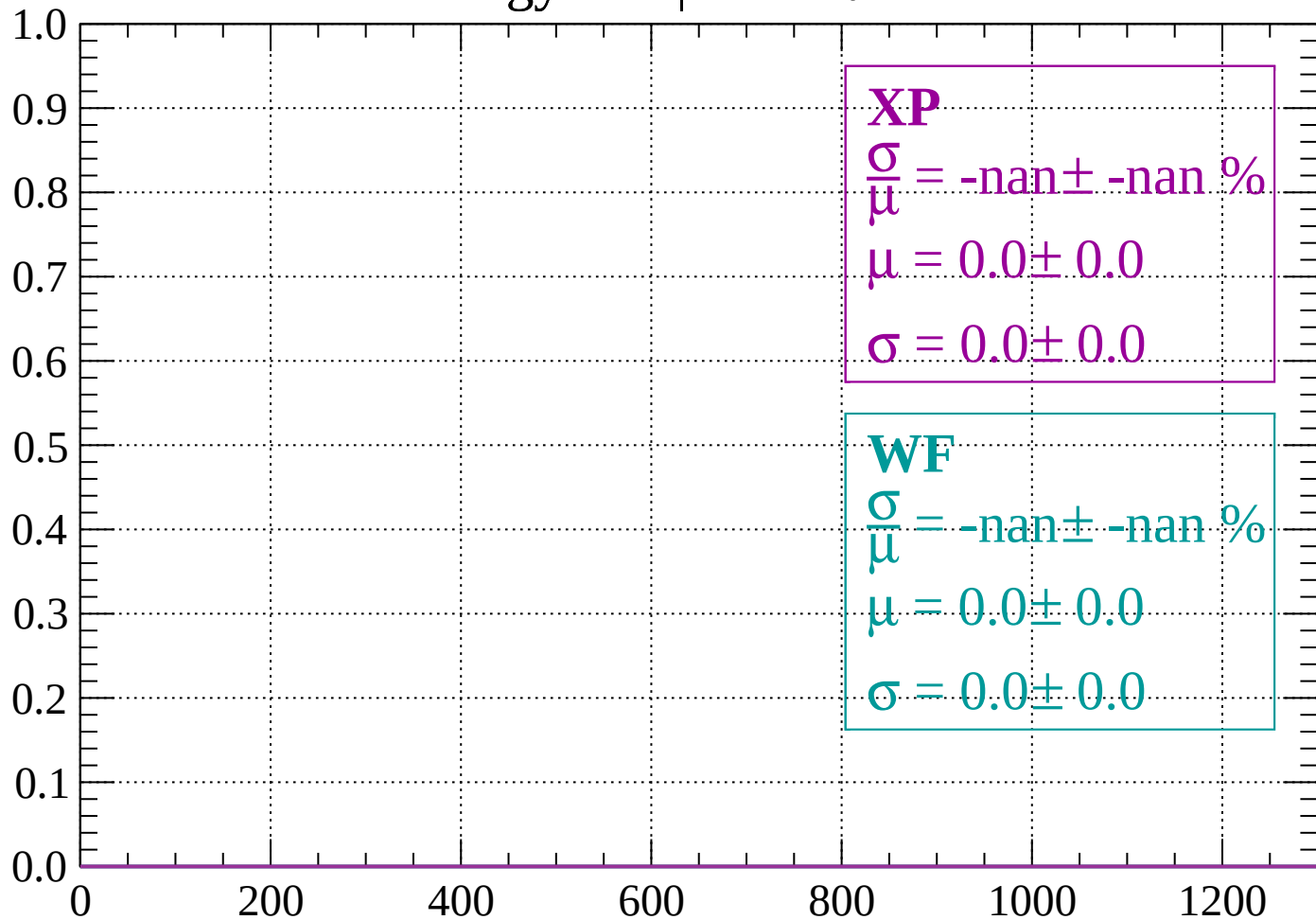
$$\mu = 488.5 \pm 6.7$$

$$\sigma = 56.8 \pm 7.3$$

dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

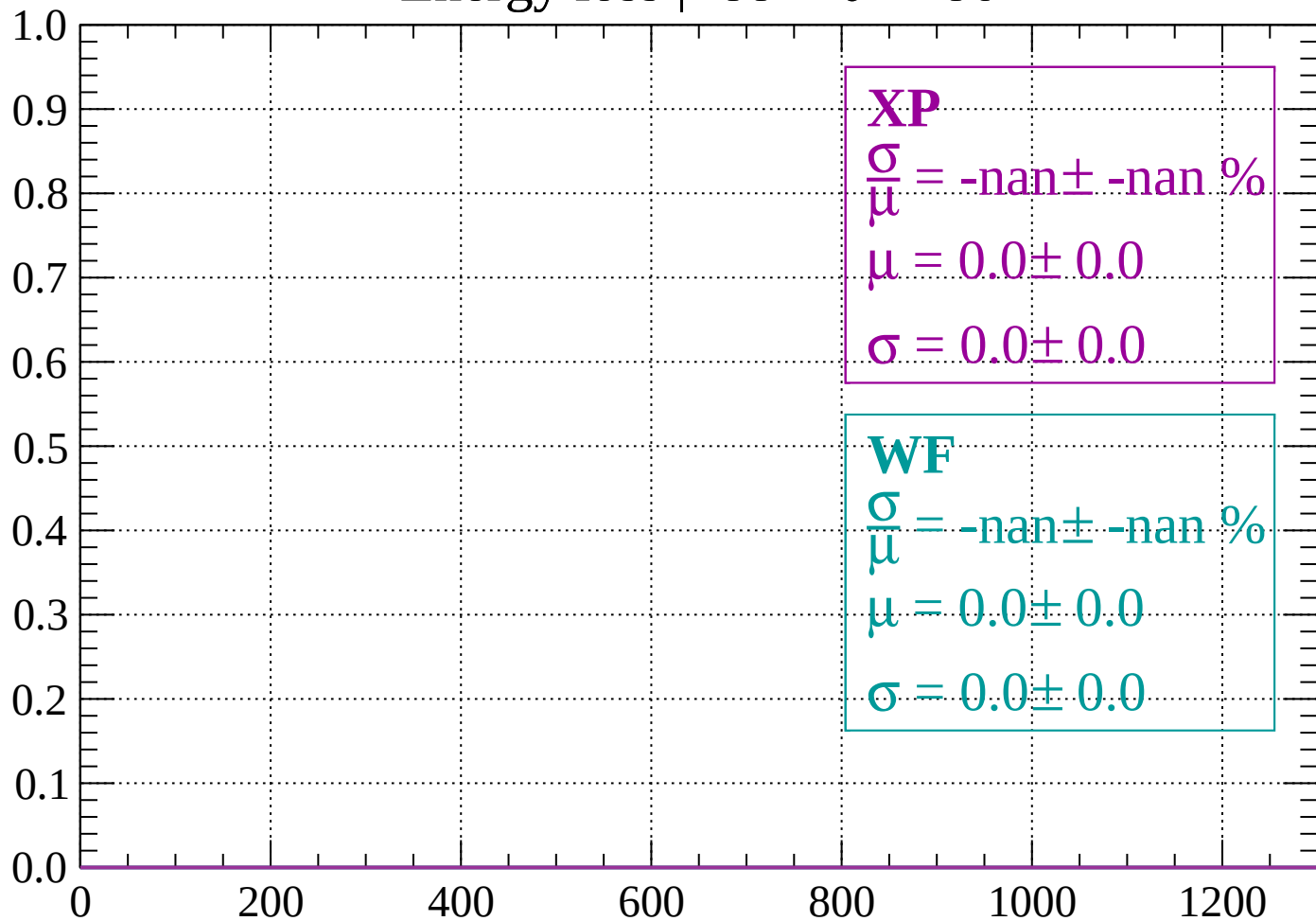
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

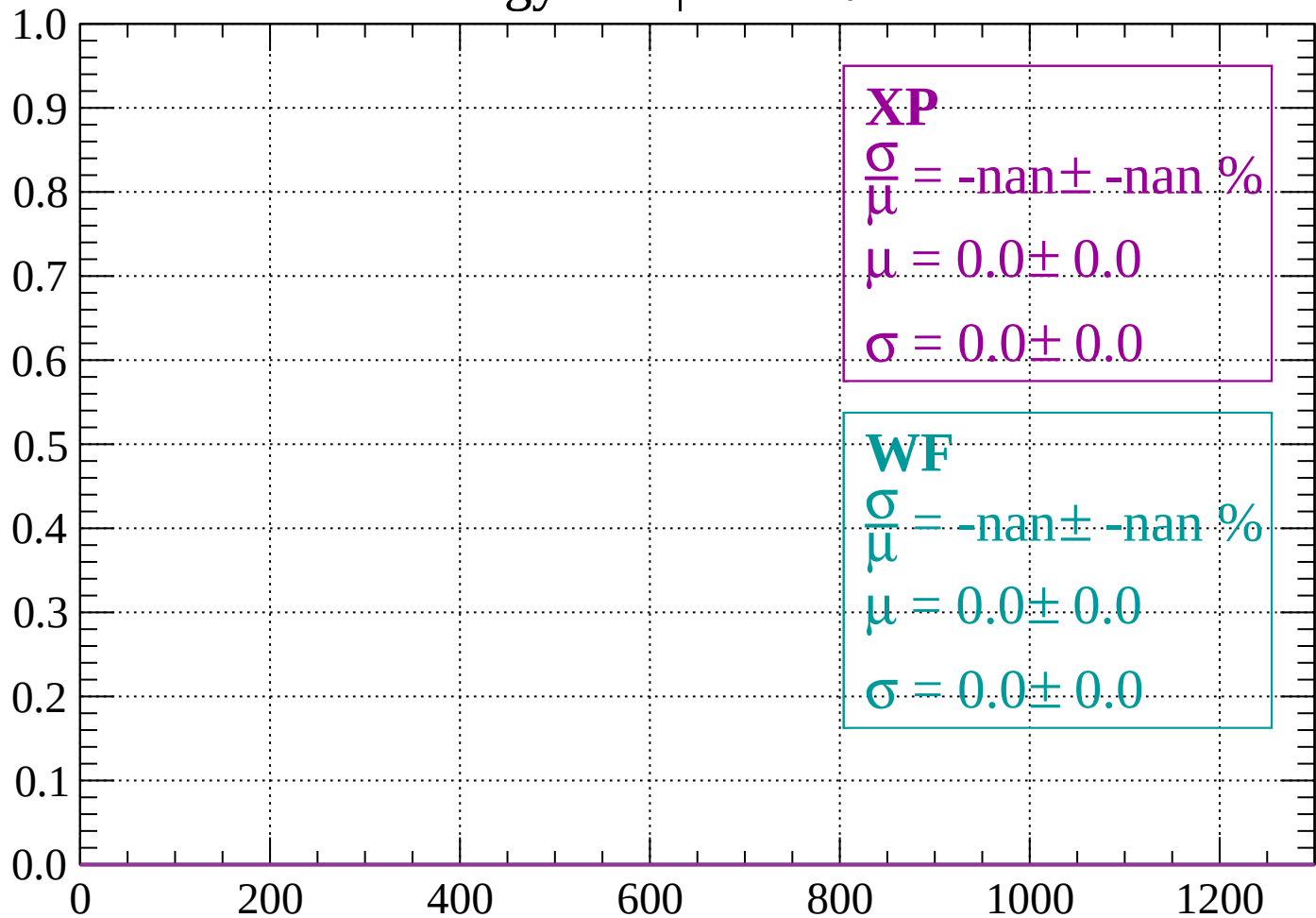
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

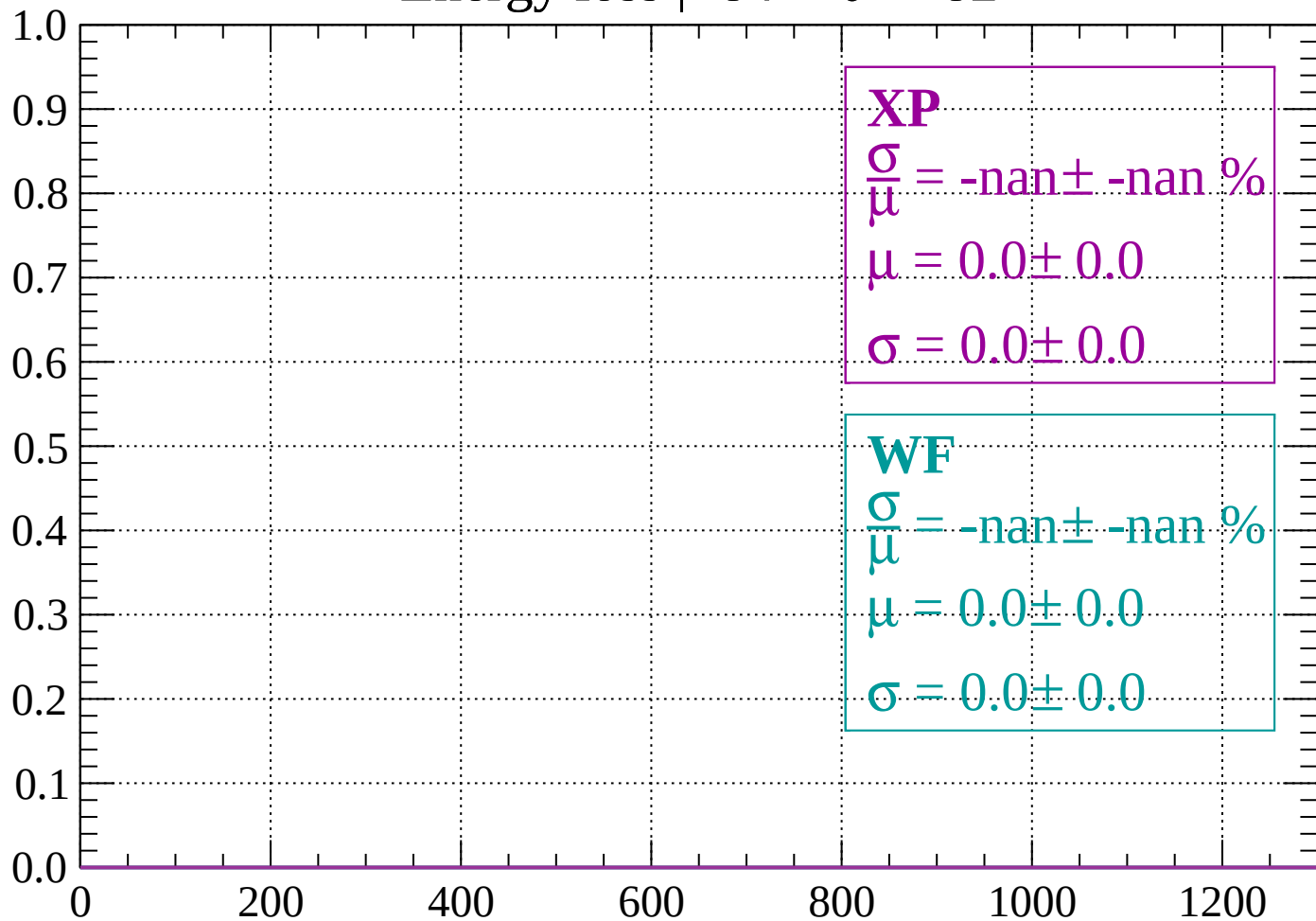
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

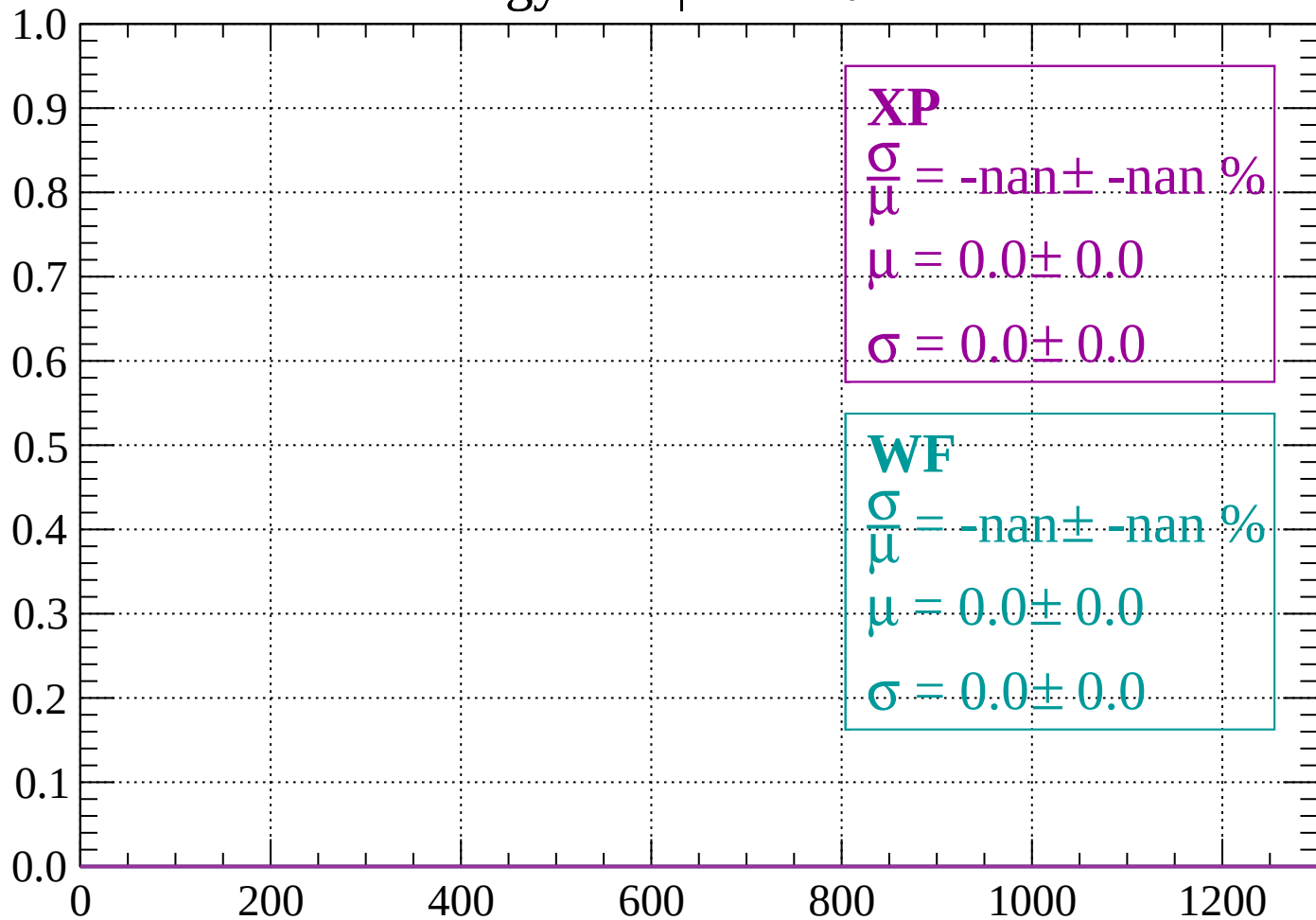
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

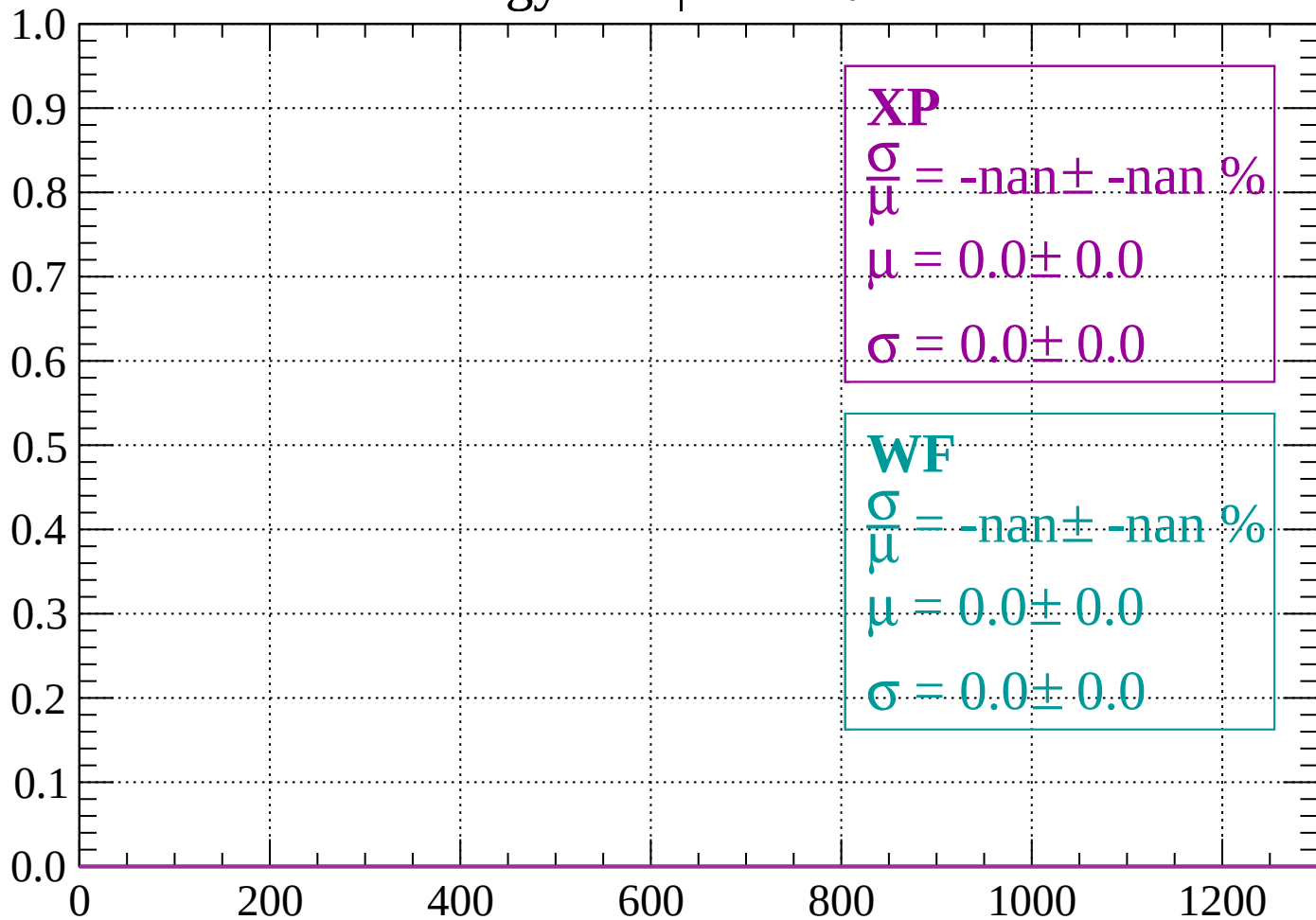
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

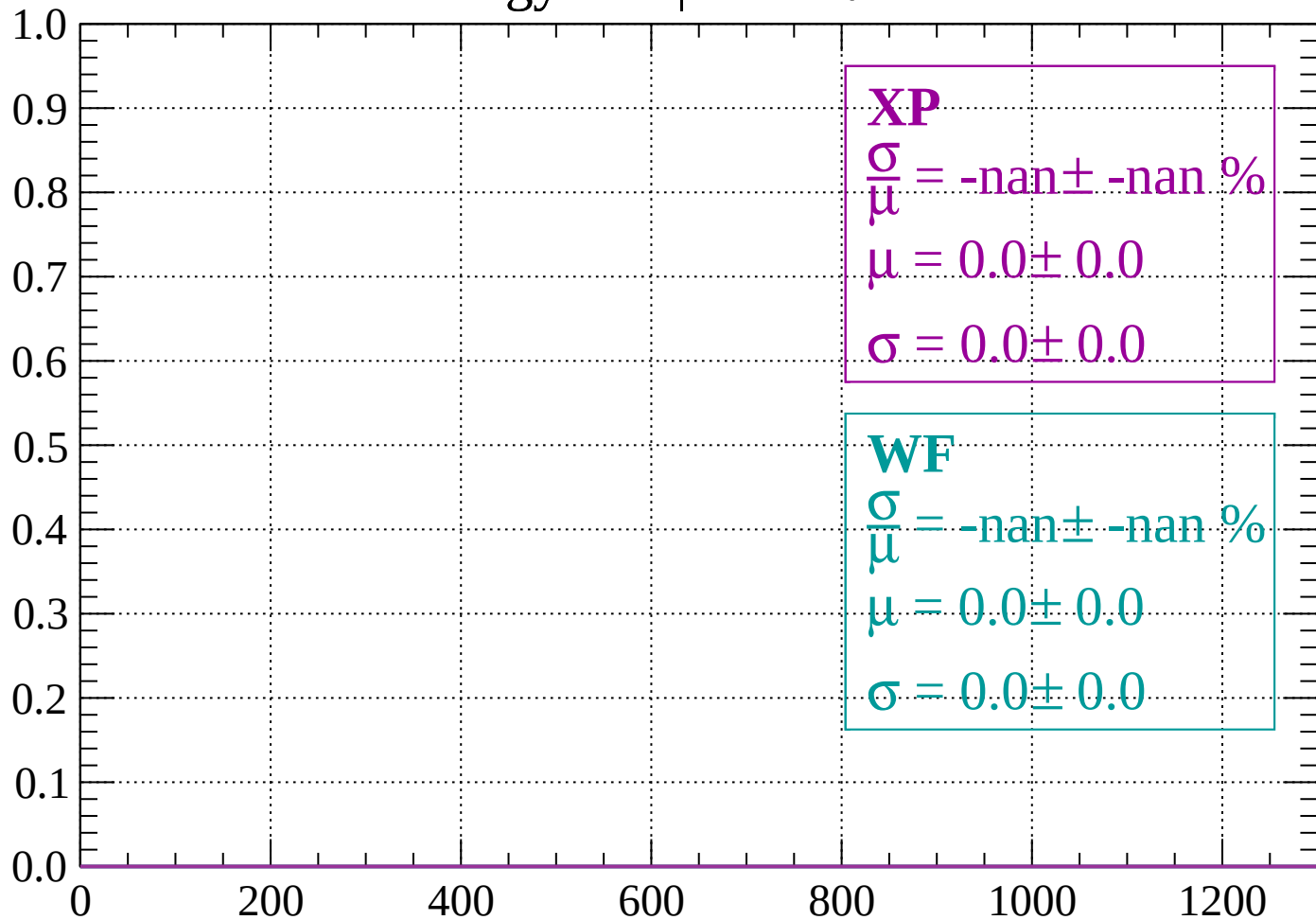
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

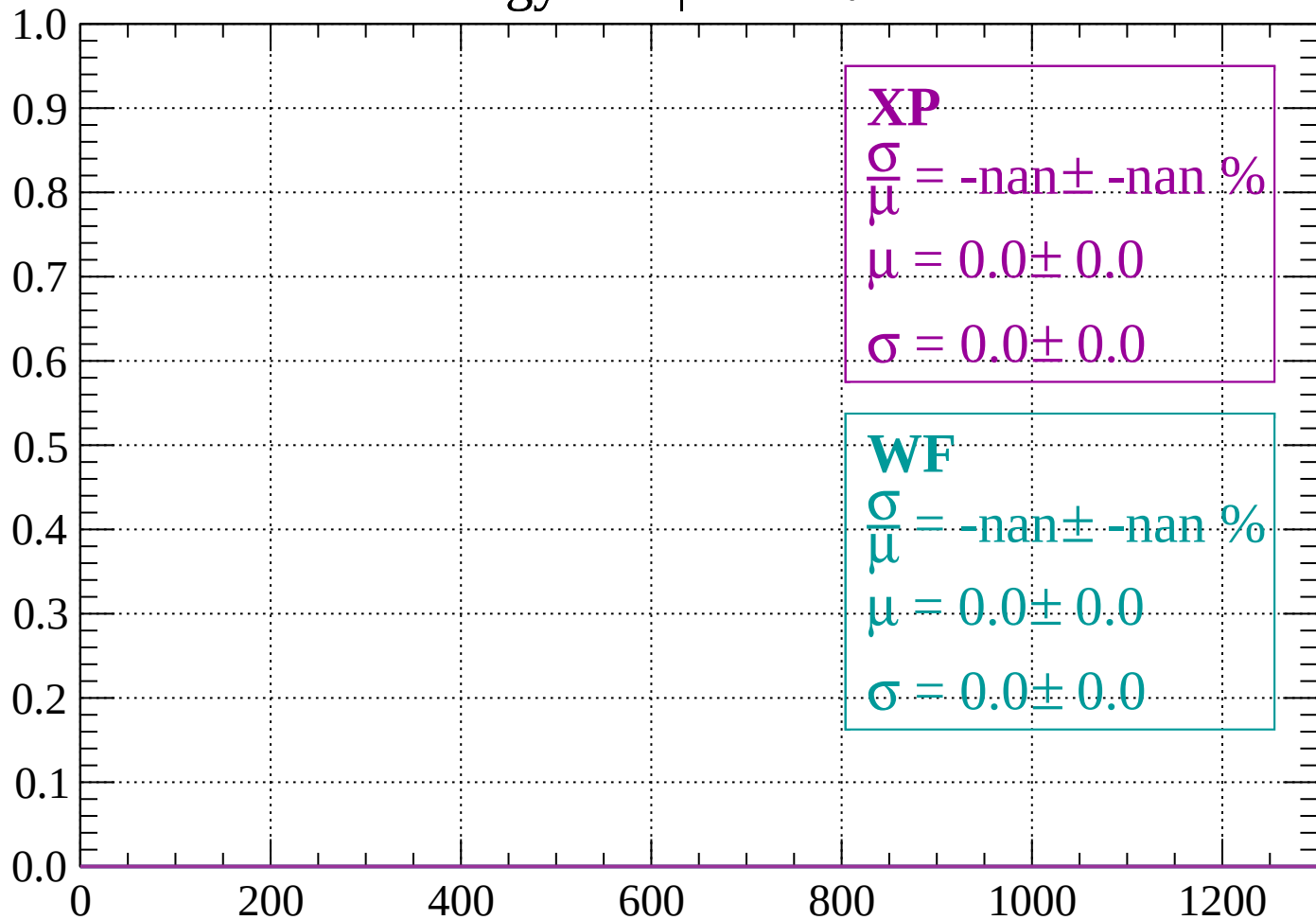
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

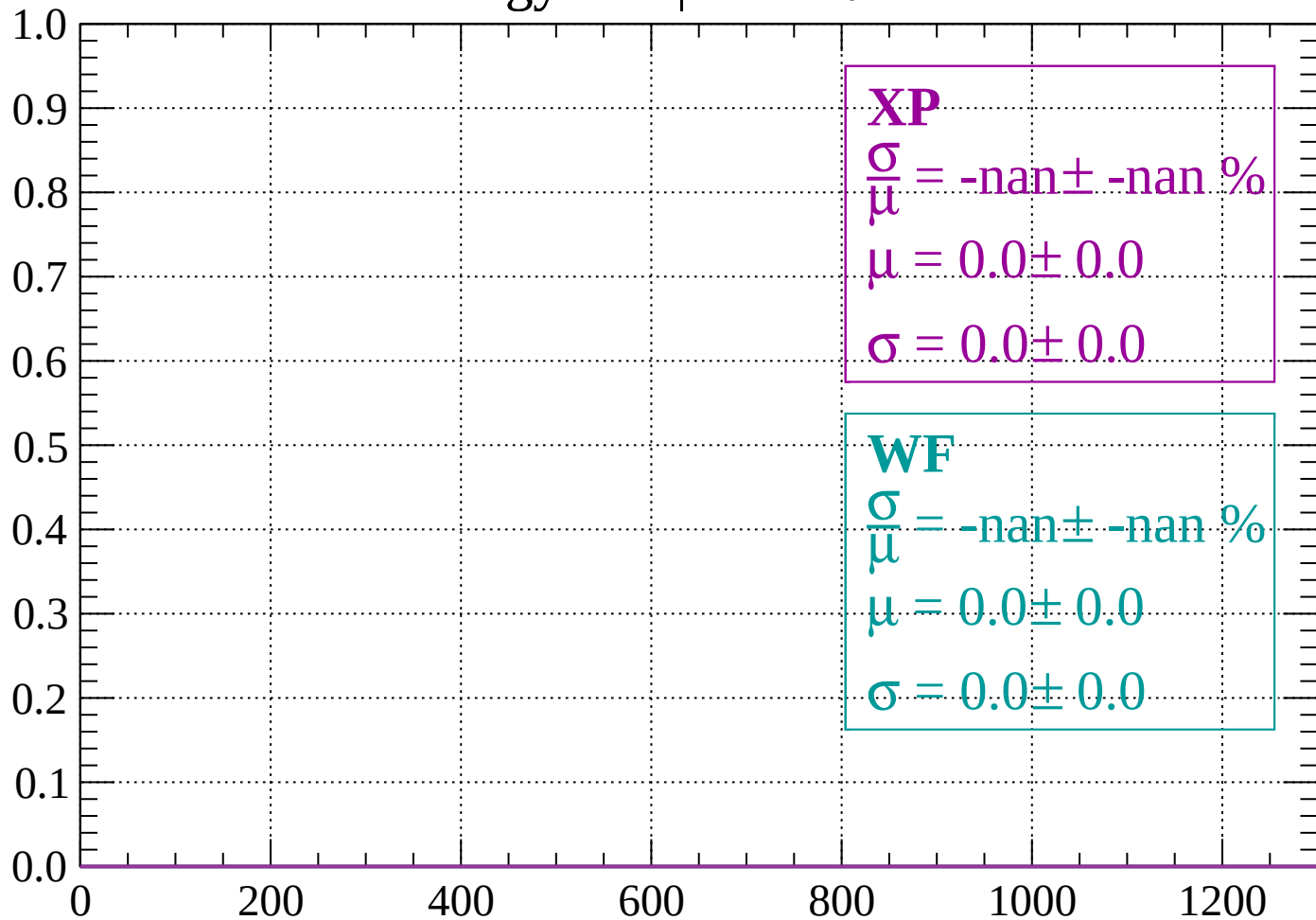
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

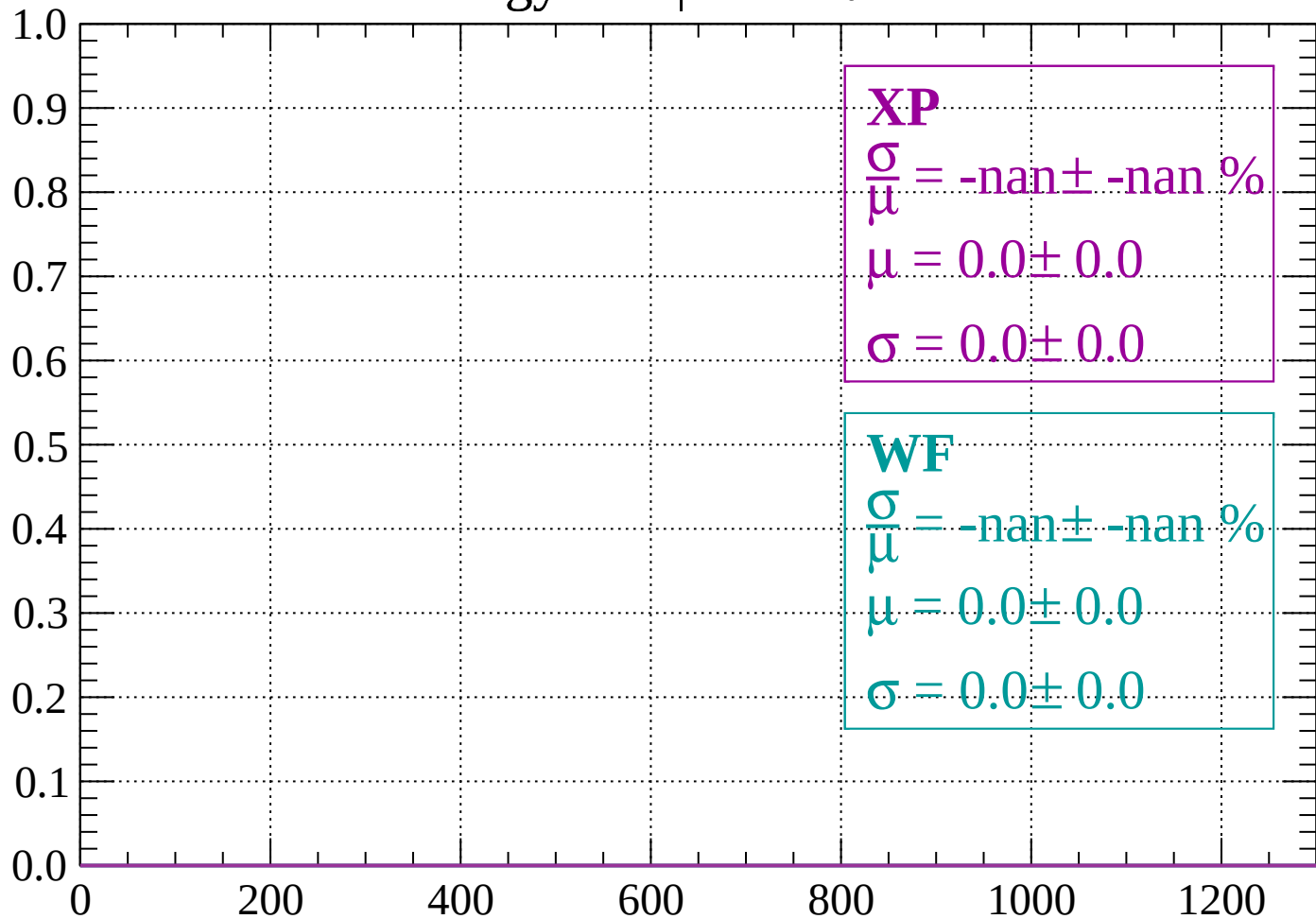
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

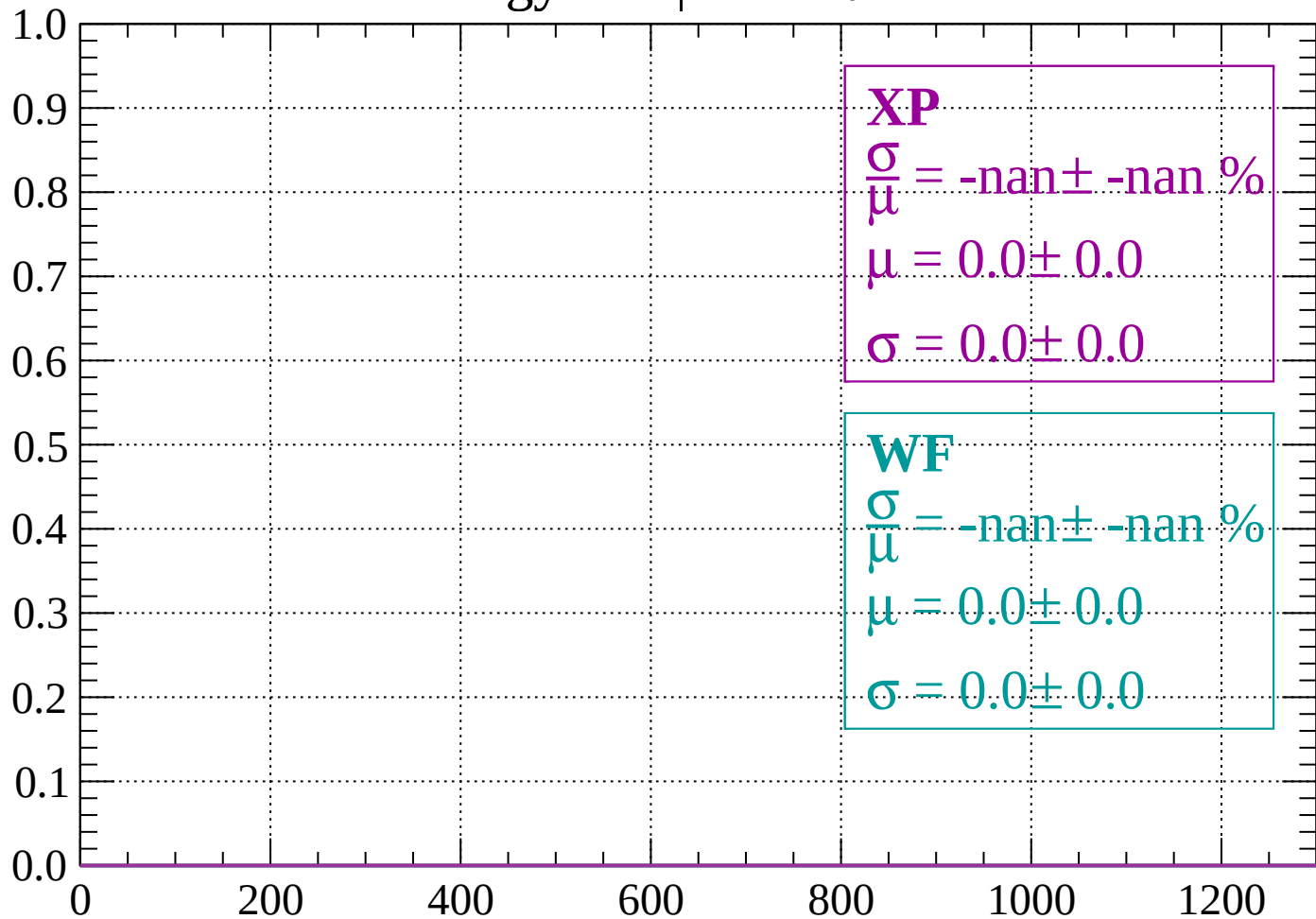
Count



dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

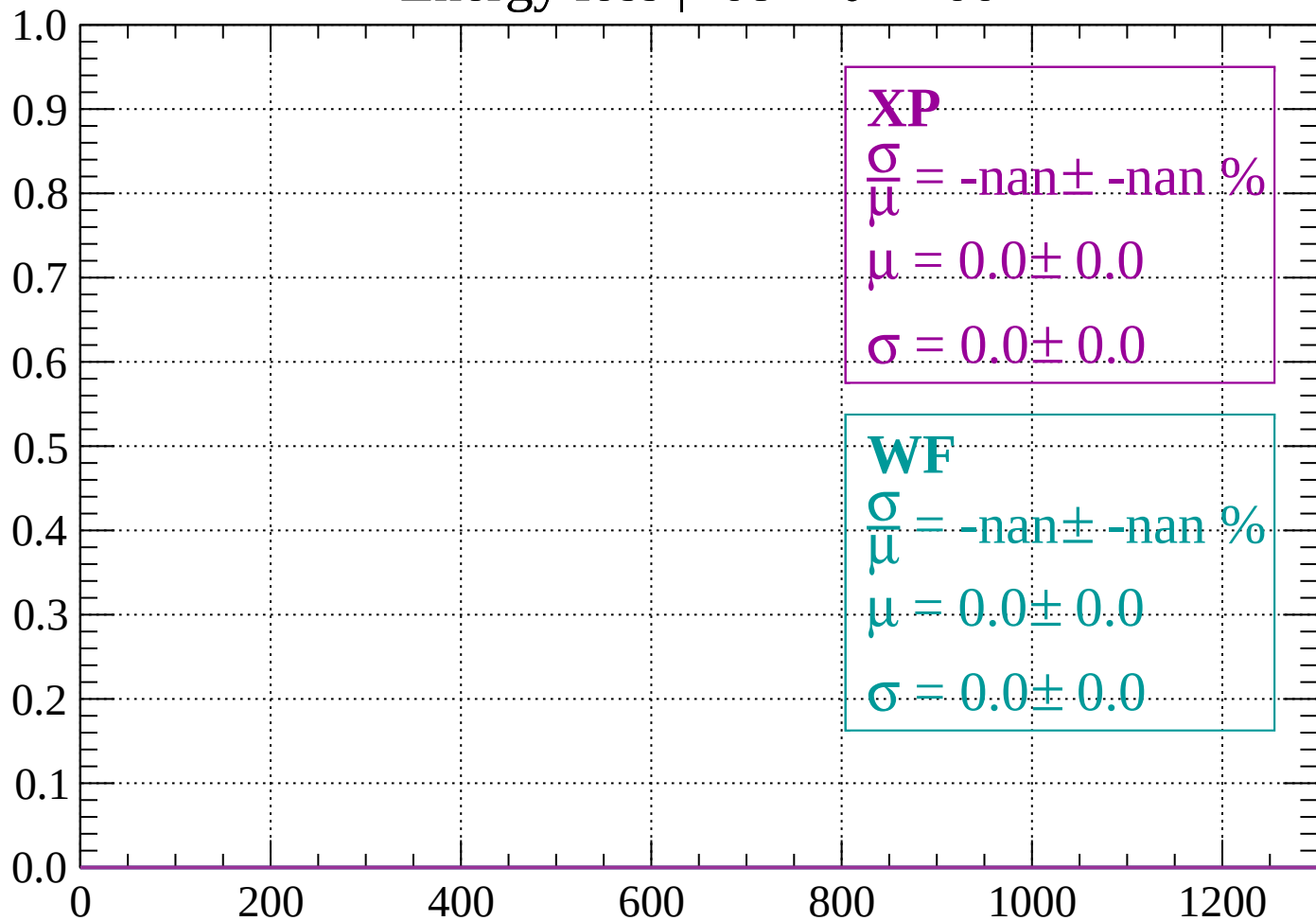
Count



dE/dx (ADC counts/cm)

Energy loss | $-68 < \theta < -66$

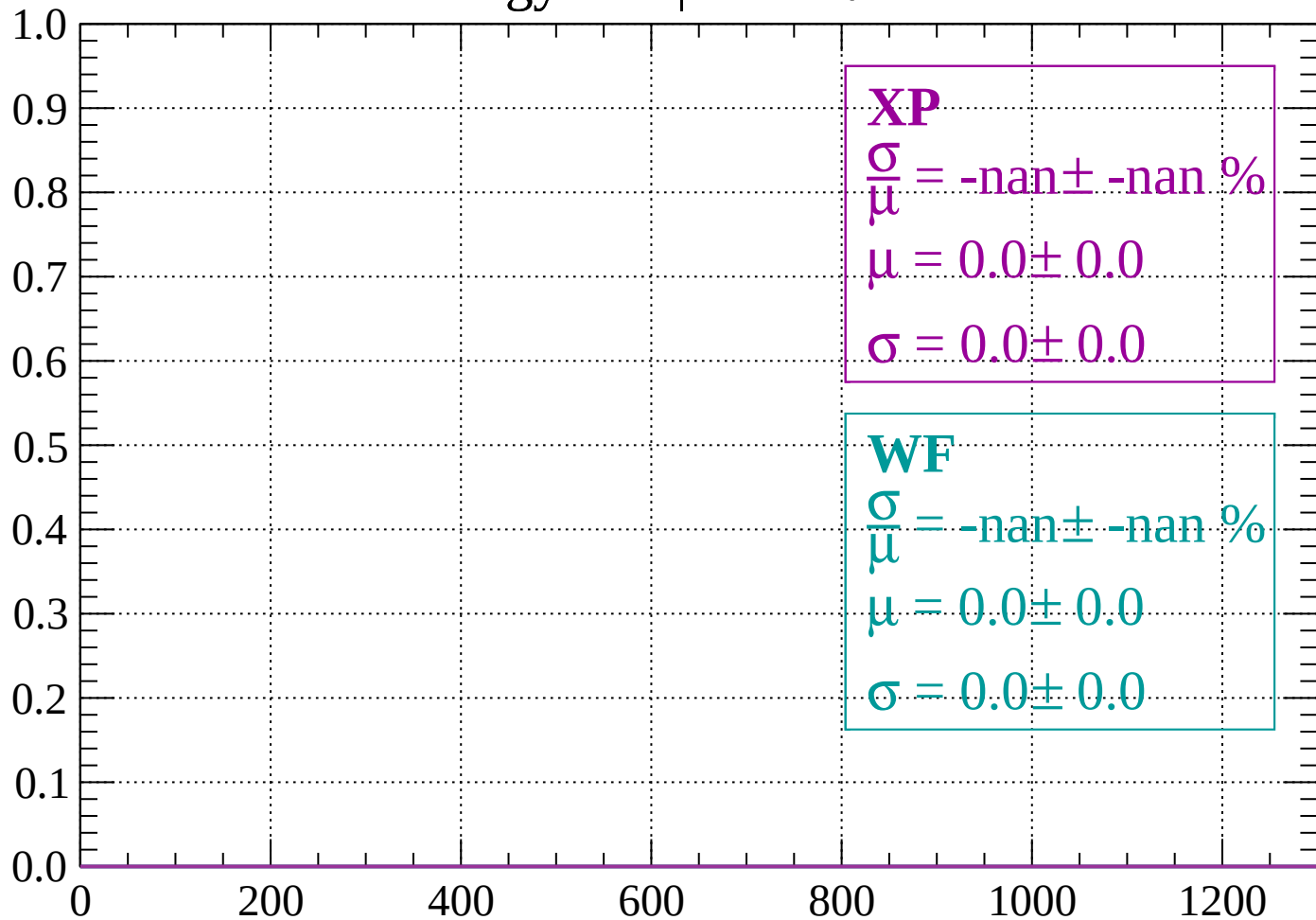
Count



dE/dx (ADC counts/cm)

Energy loss | $-66 < \theta < -64$

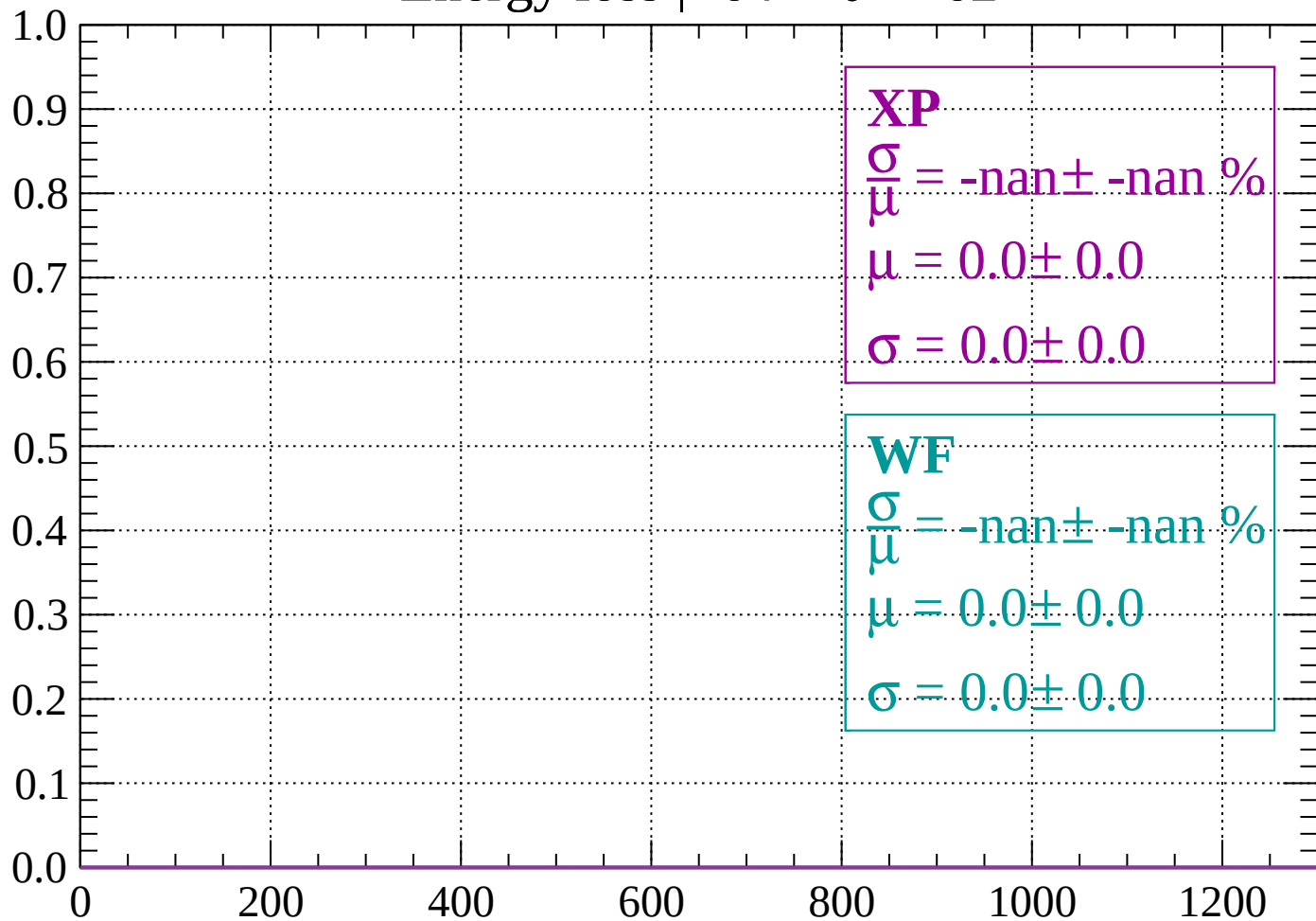
Count



dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

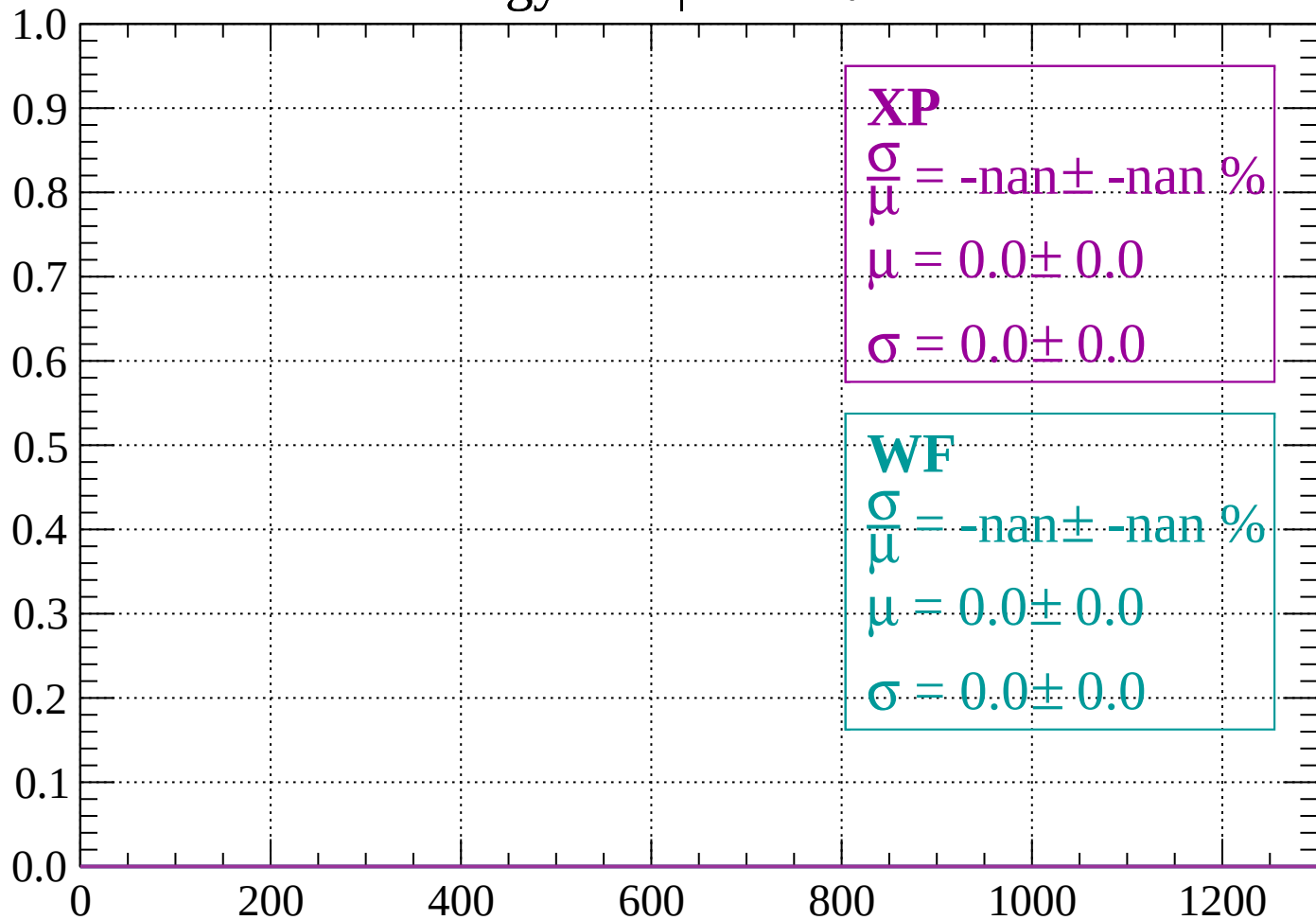
Count



dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

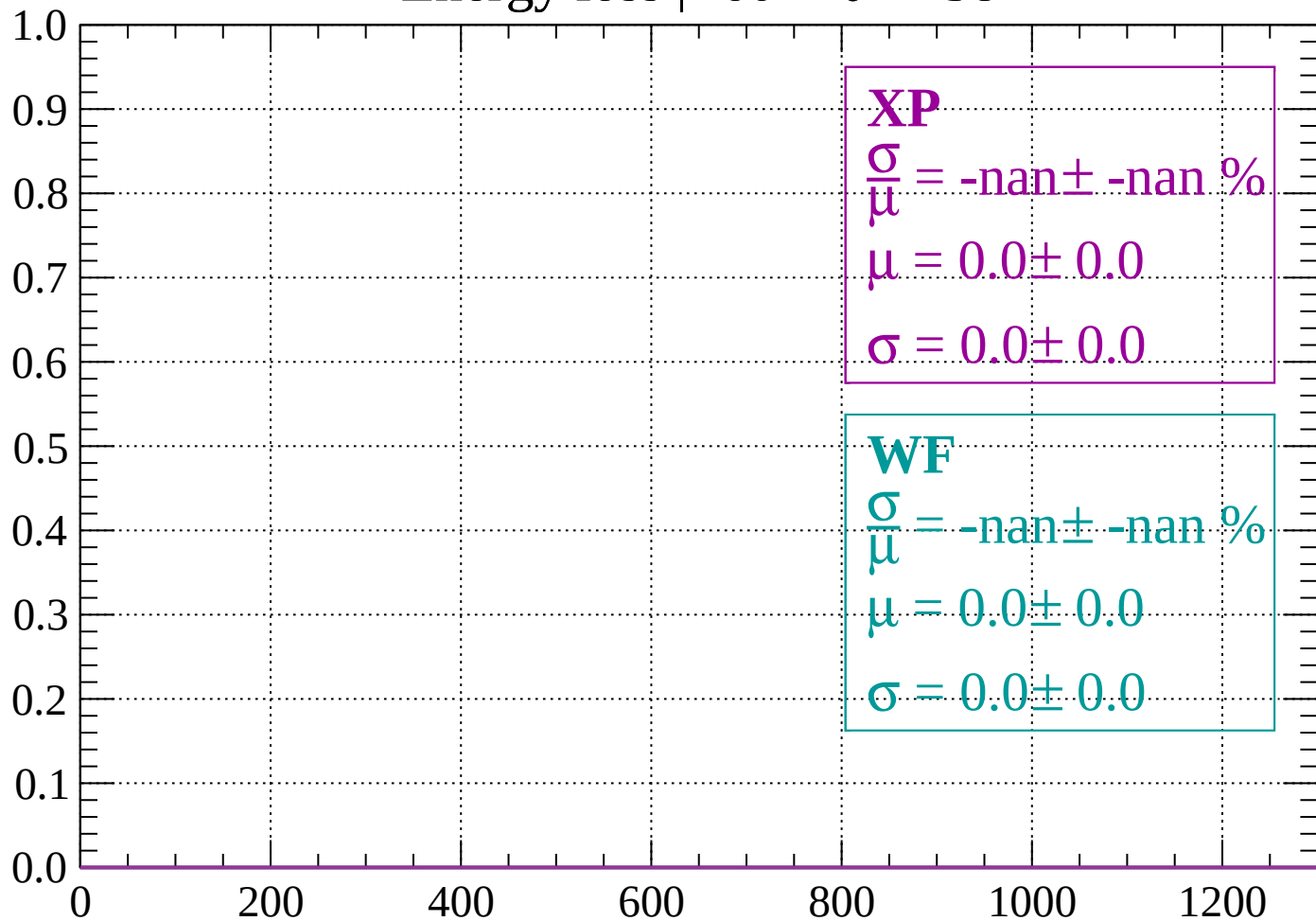
Count



dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

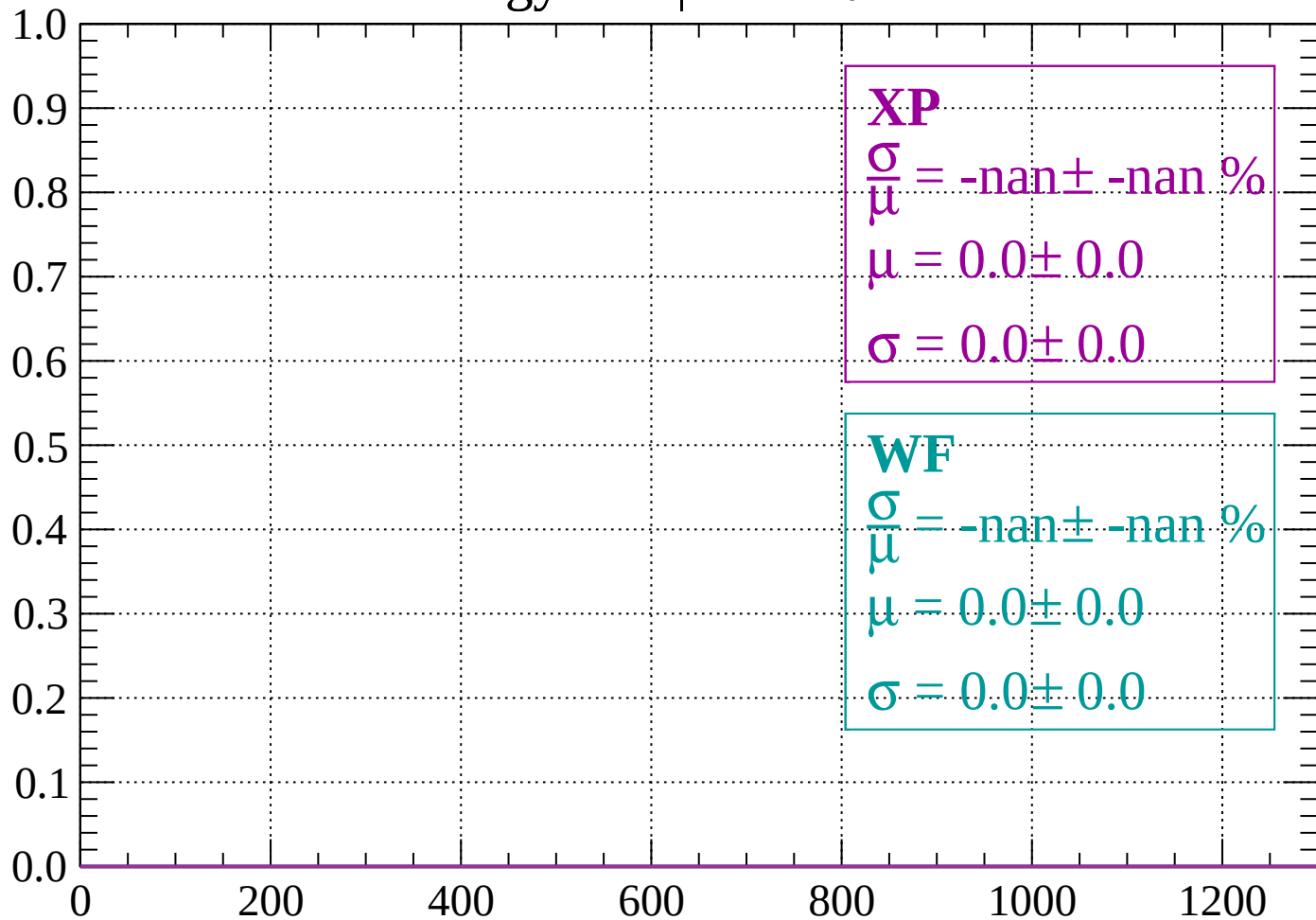
Count



dE/dx (ADC counts/cm)

Energy loss | $-58 < \theta < -56$

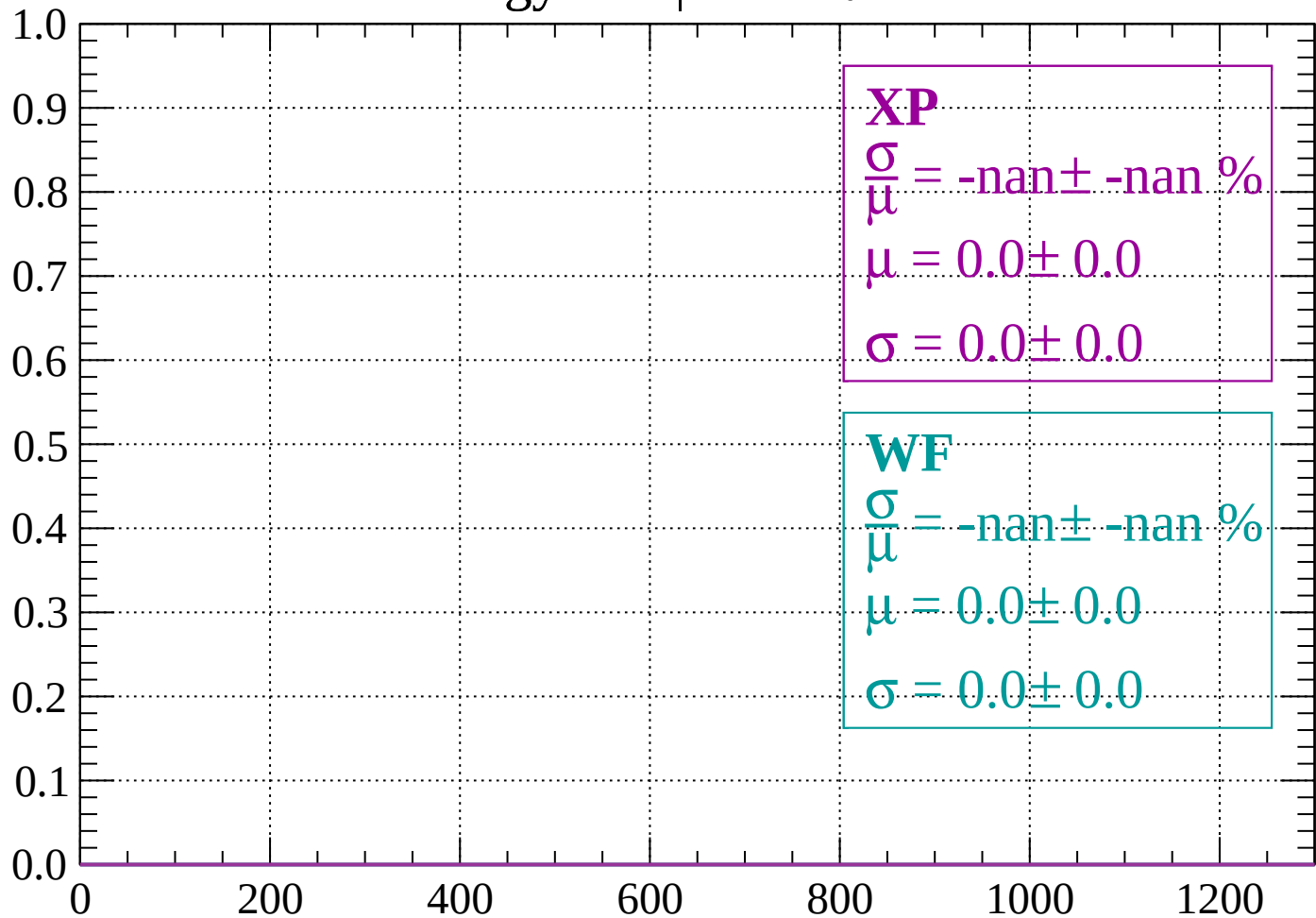
Count



dE/dx (ADC counts/cm)

Energy loss | $-56 < \theta < -54$

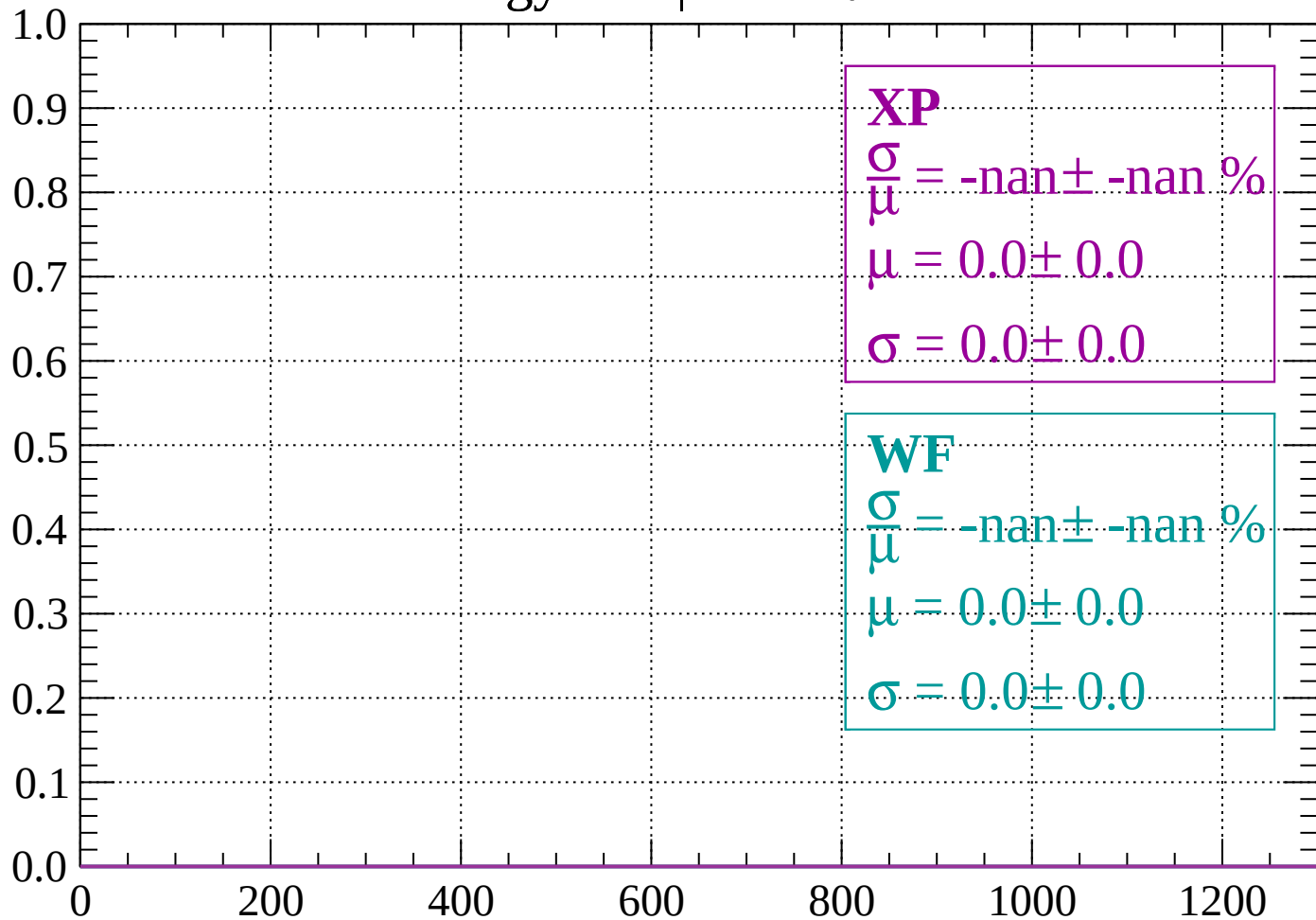
Count



dE/dx (ADC counts/cm)

Energy loss | $-54 < \theta < -52$

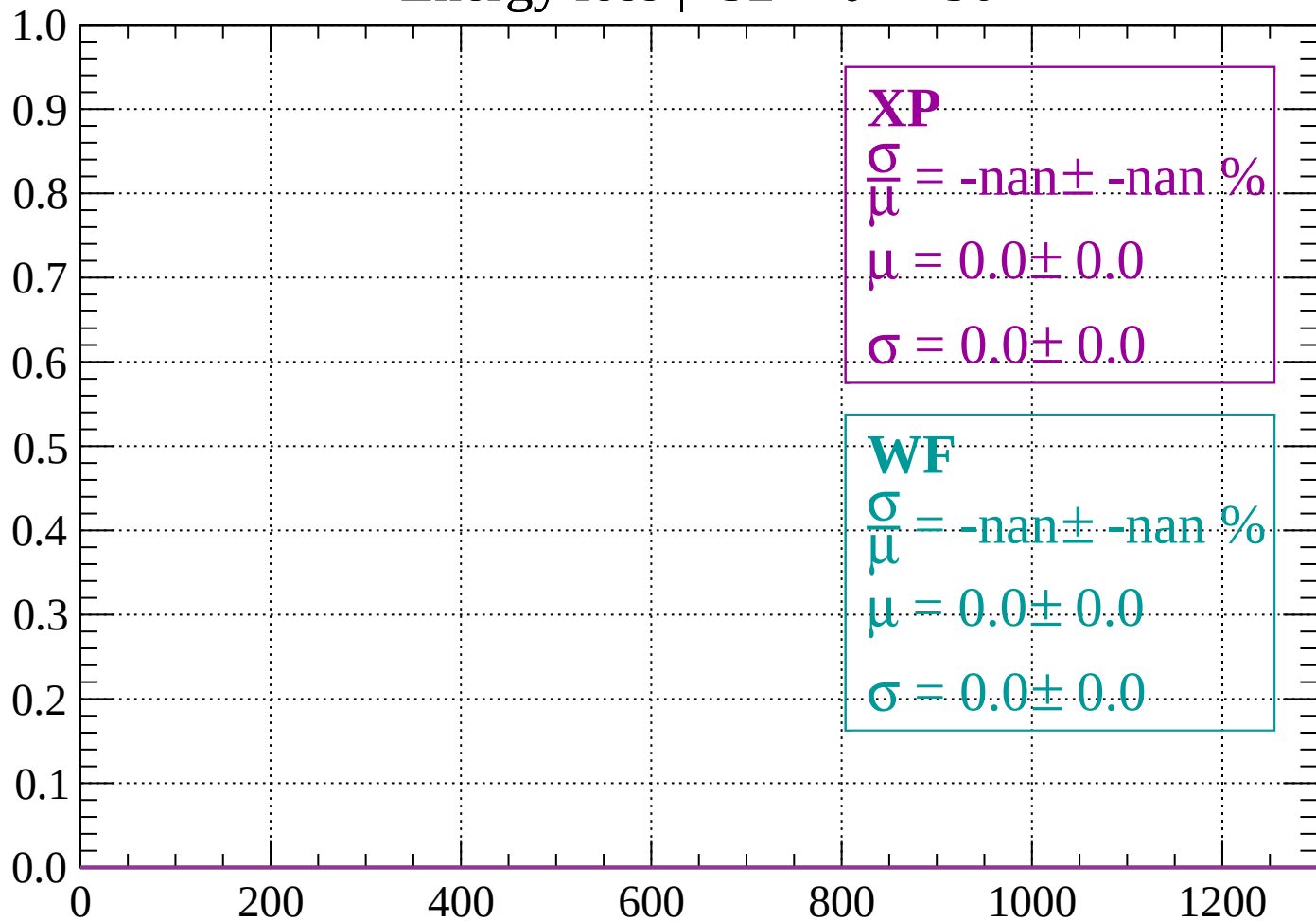
Count



dE/dx (ADC counts/cm)

Energy loss | $-52 < \theta < -50$

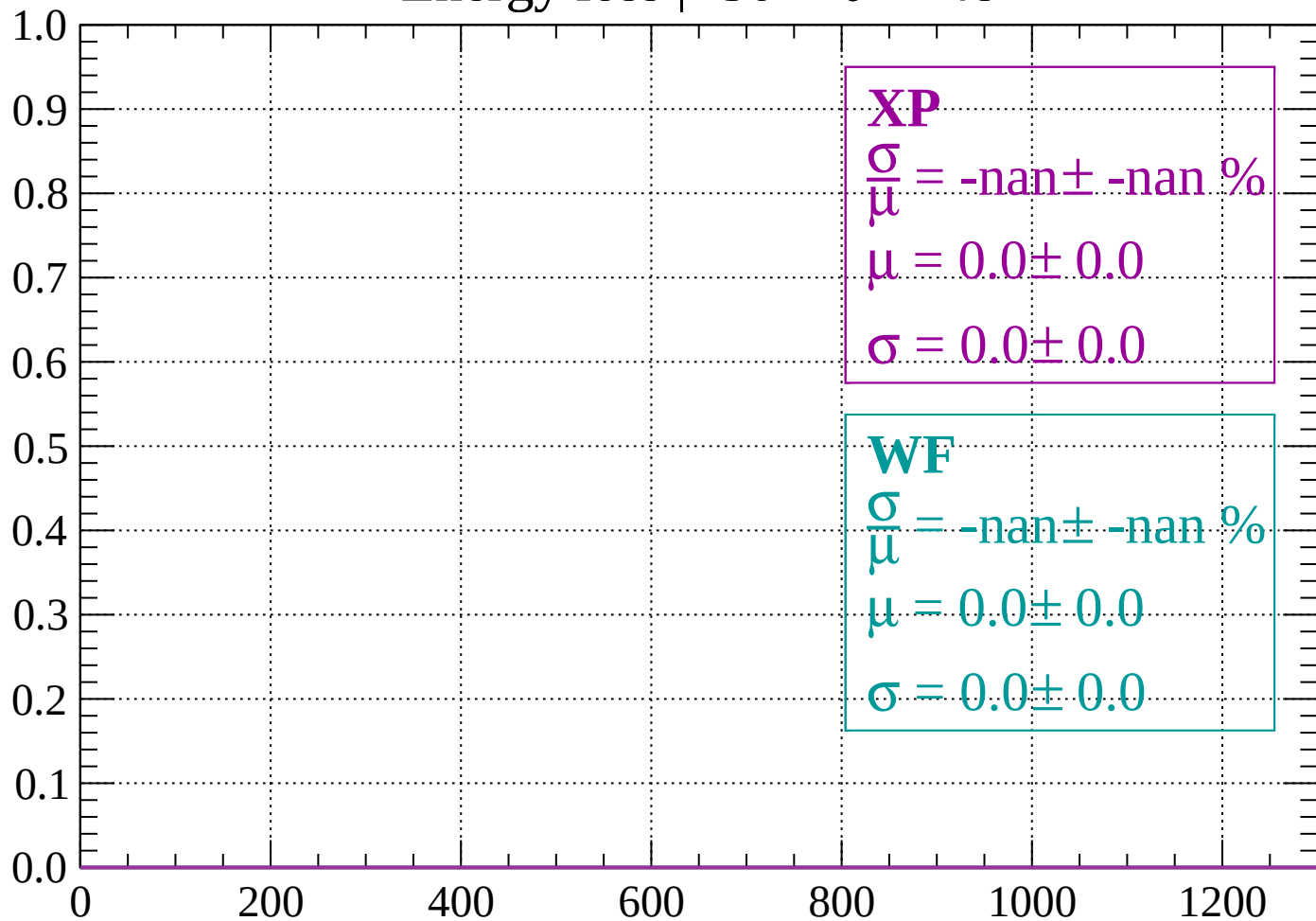
Count



dE/dx (ADC counts/cm)

Energy loss | $-50 < \theta < -48$

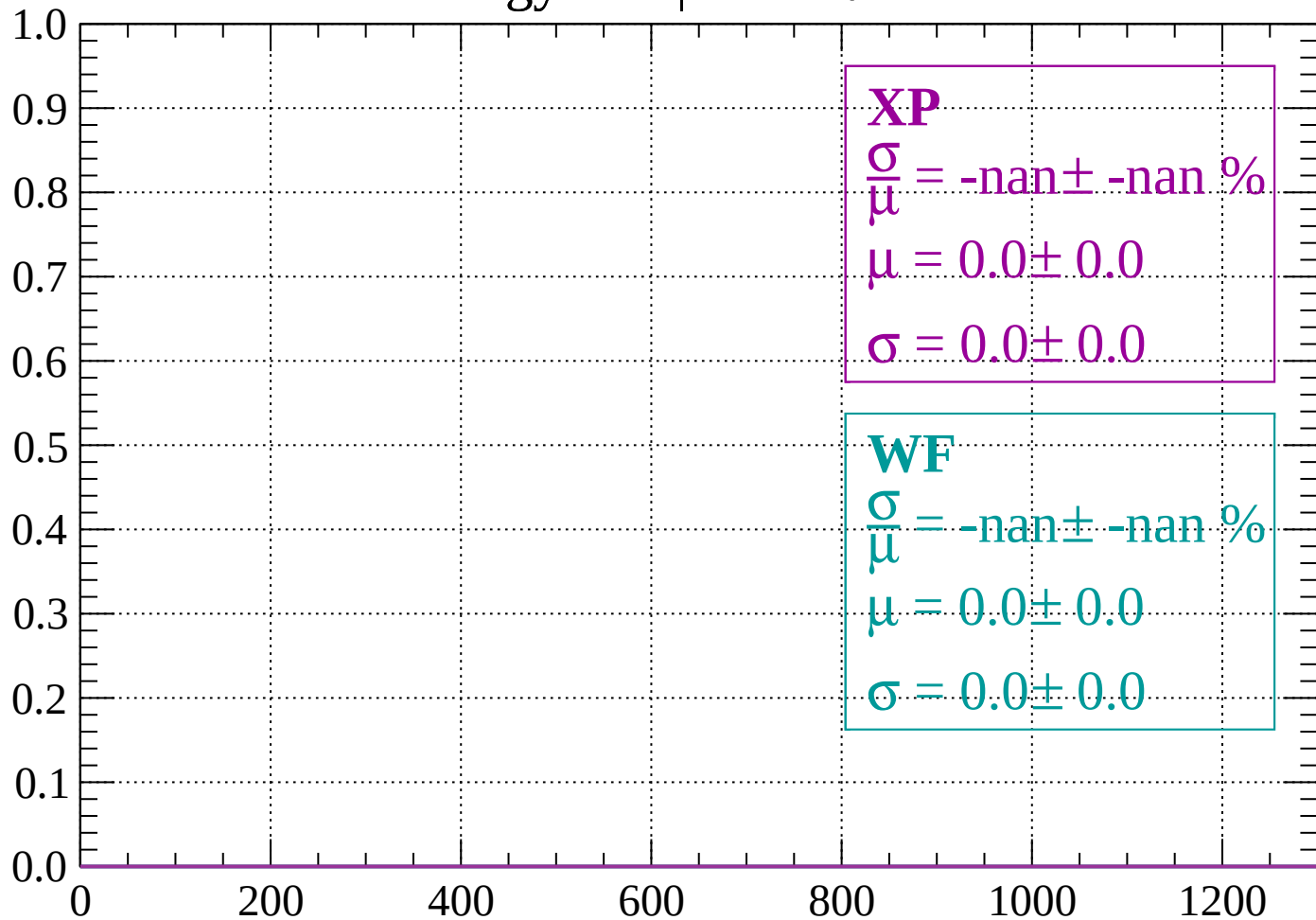
Count



dE/dx (ADC counts/cm)

Energy loss | $-48 < \theta < -46$

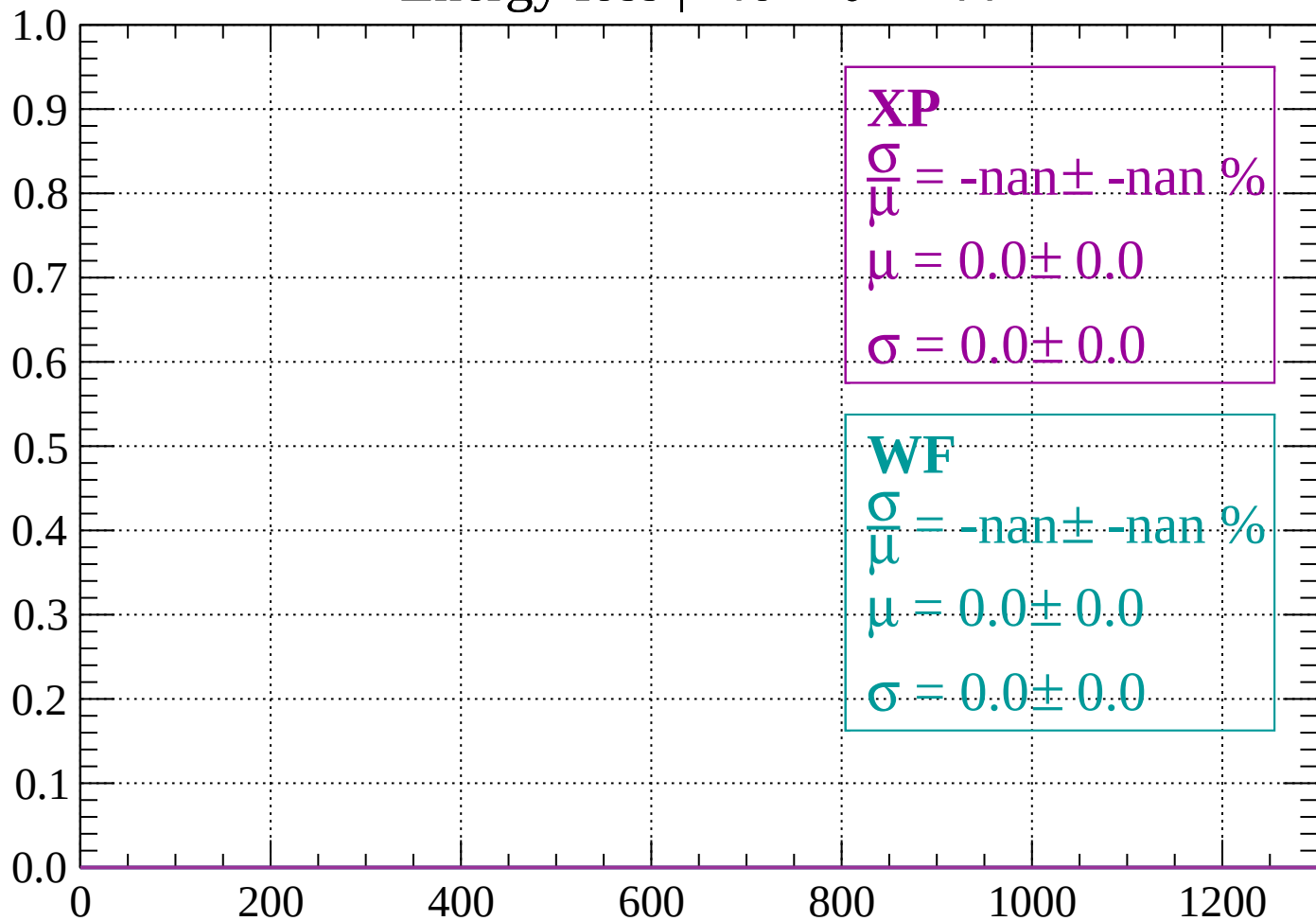
Count



dE/dx (ADC counts/cm)

Energy loss | $-46 < \theta < -44$

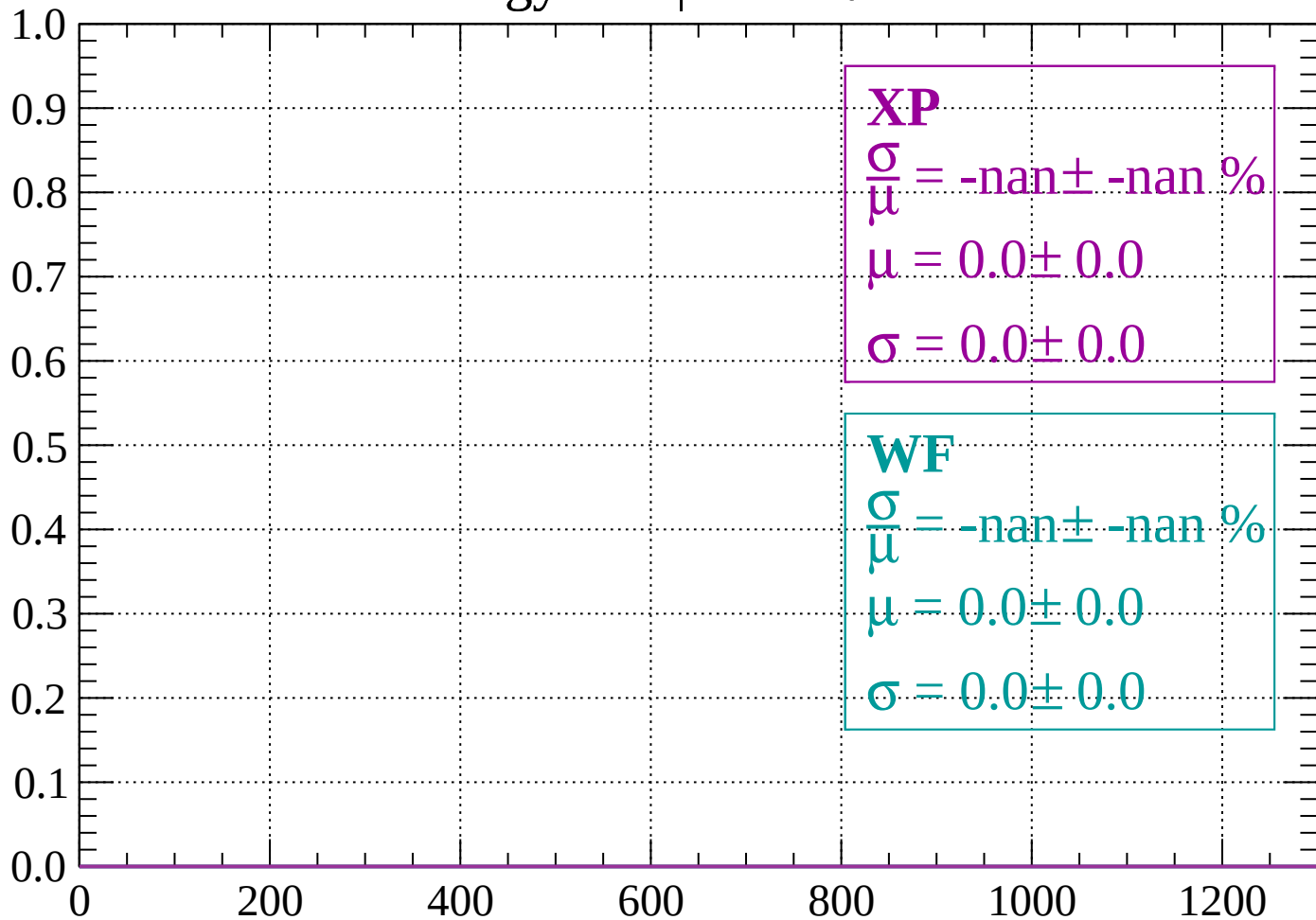
Count



dE/dx (ADC counts/cm)

Energy loss | $-44 < \theta < -42$

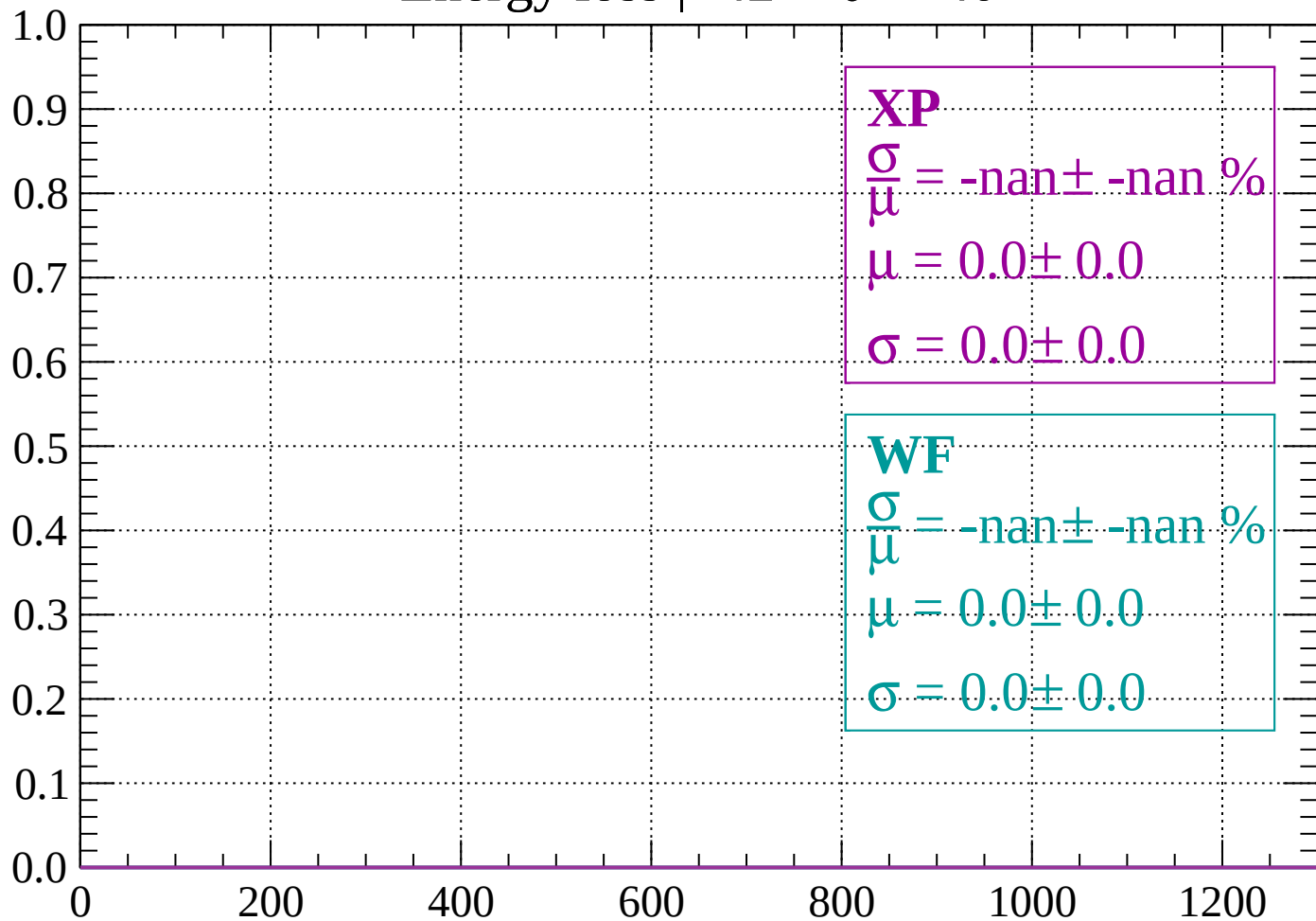
Count



dE/dx (ADC counts/cm)

Energy loss | $-42 < \theta < -40$

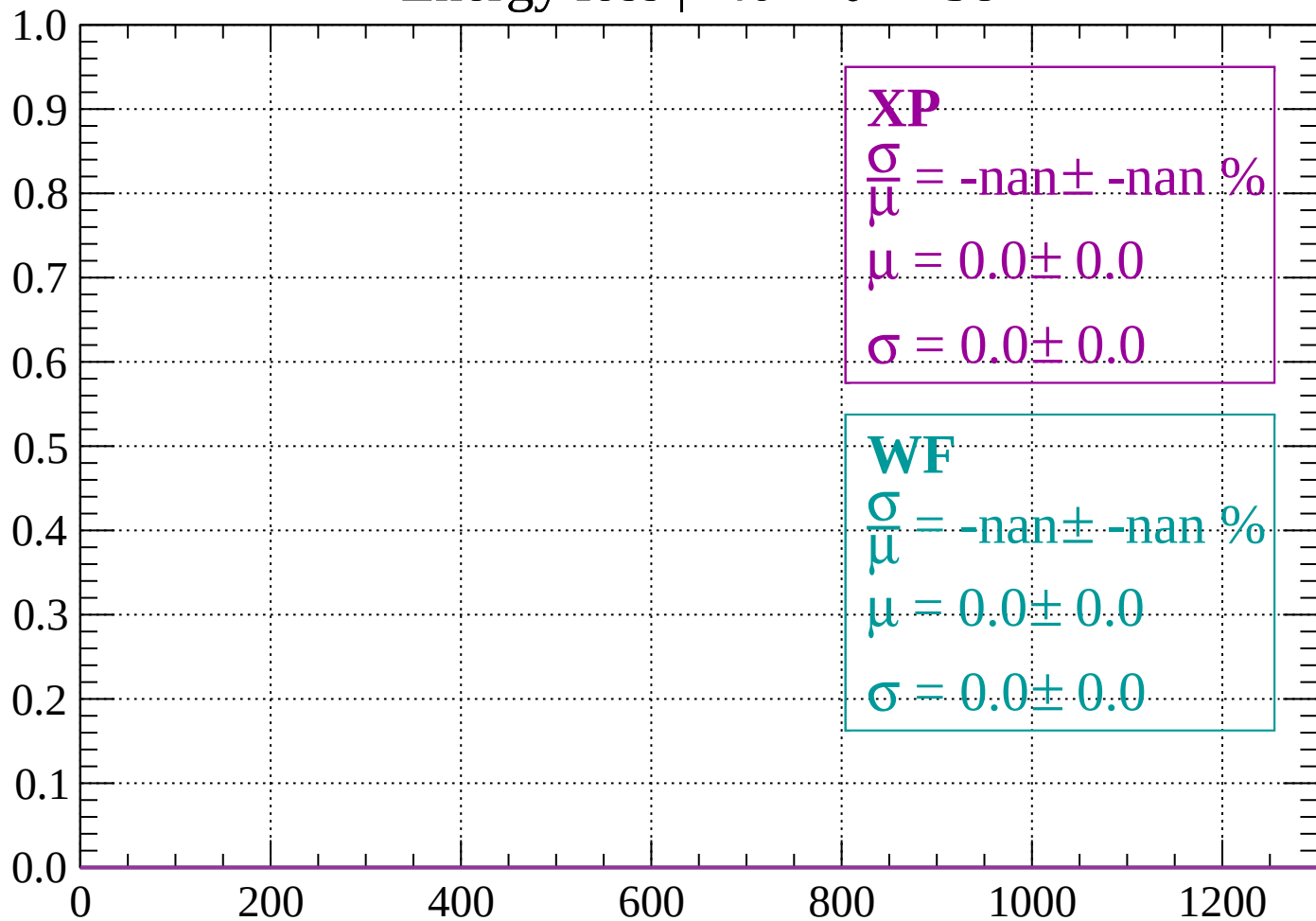
Count



dE/dx (ADC counts/cm)

Energy loss | $-40 < \theta < -38$

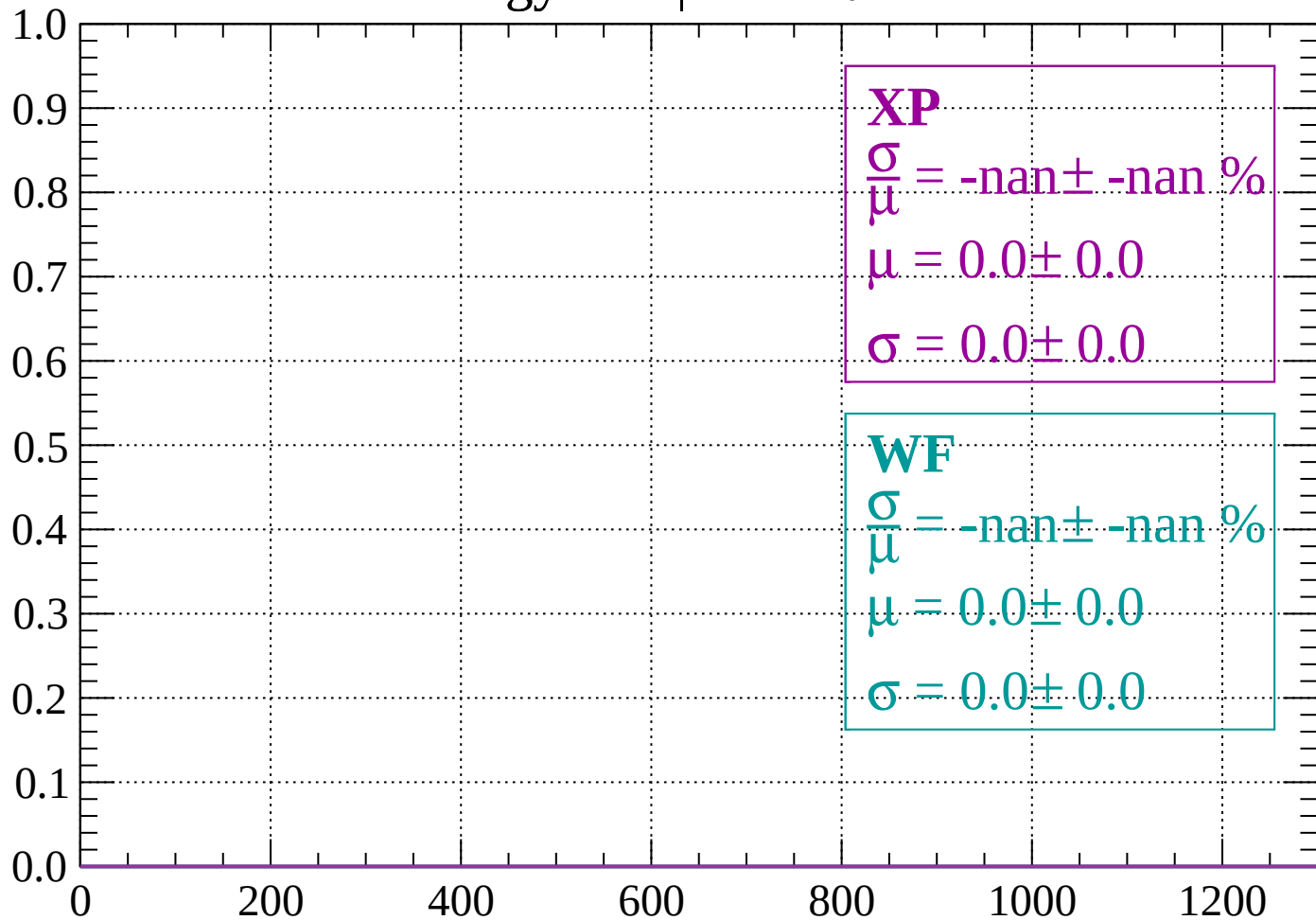
Count



dE/dx (ADC counts/cm)

Energy loss | $-38 < \theta < -36$

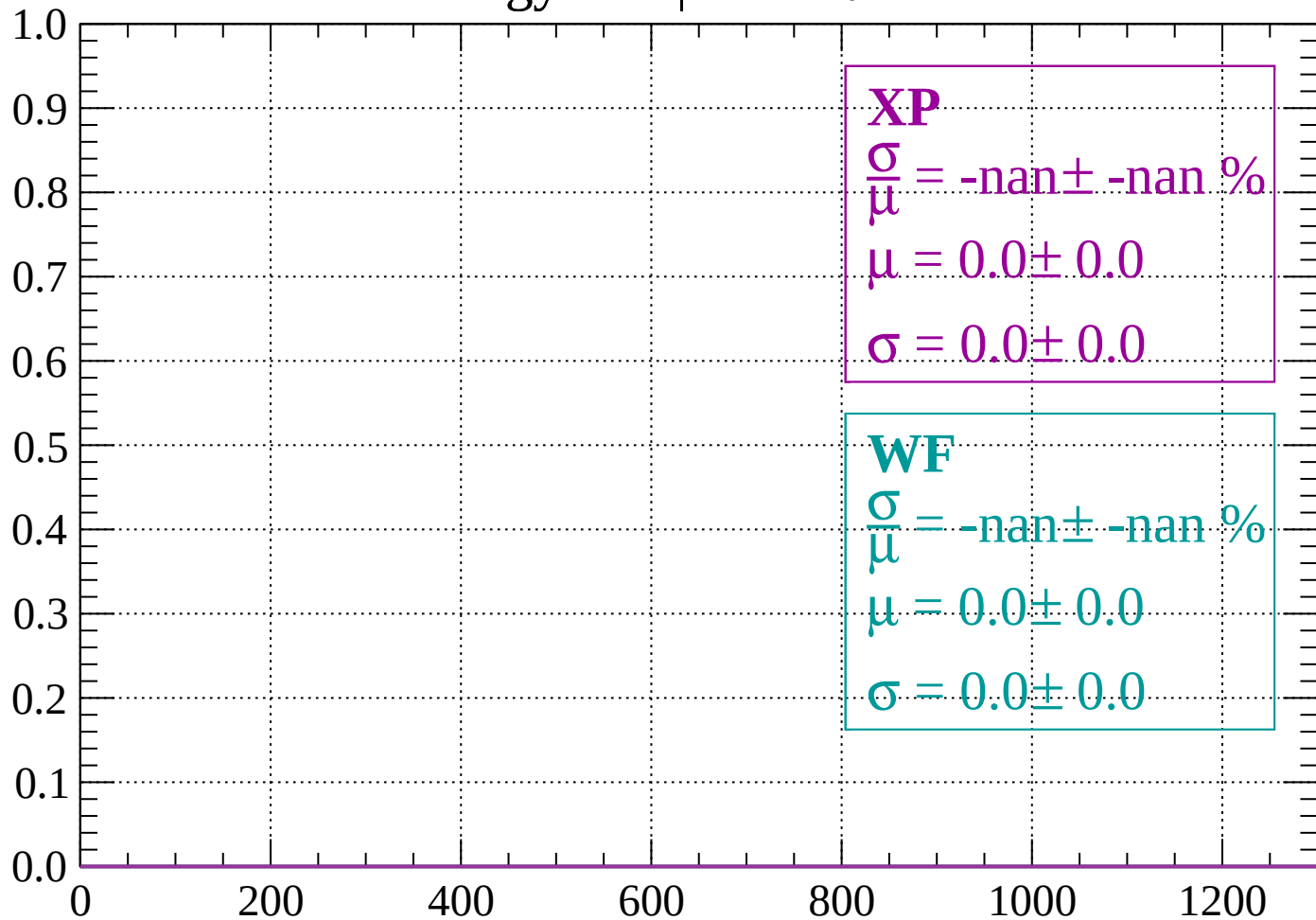
Count



dE/dx (ADC counts/cm)

Energy loss | $-36 < \theta < -34$

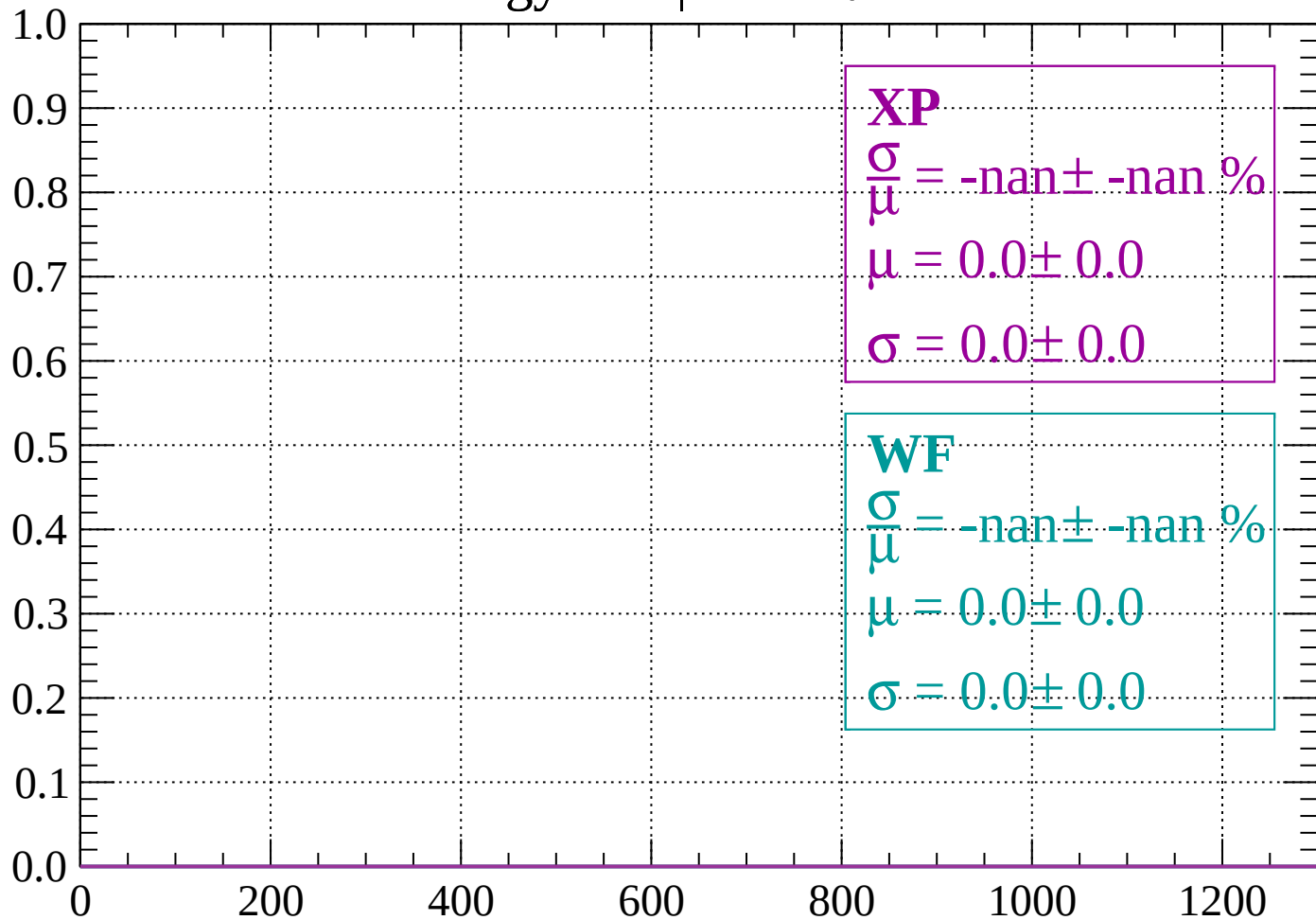
Count



dE/dx (ADC counts/cm)

Energy loss | $-34 < \theta < -32$

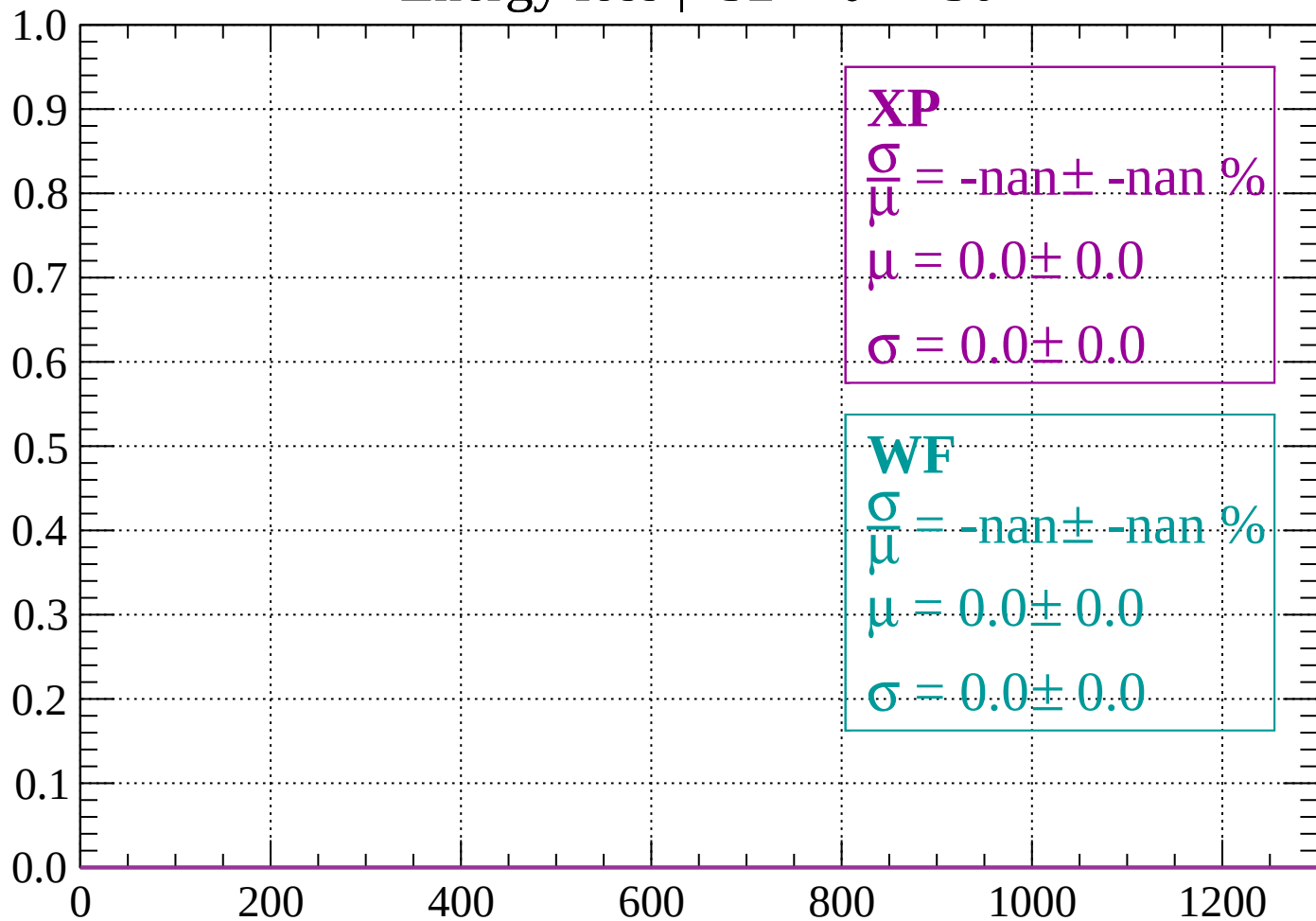
Count



dE/dx (ADC counts/cm)

Energy loss | $-32 < \theta < -30$

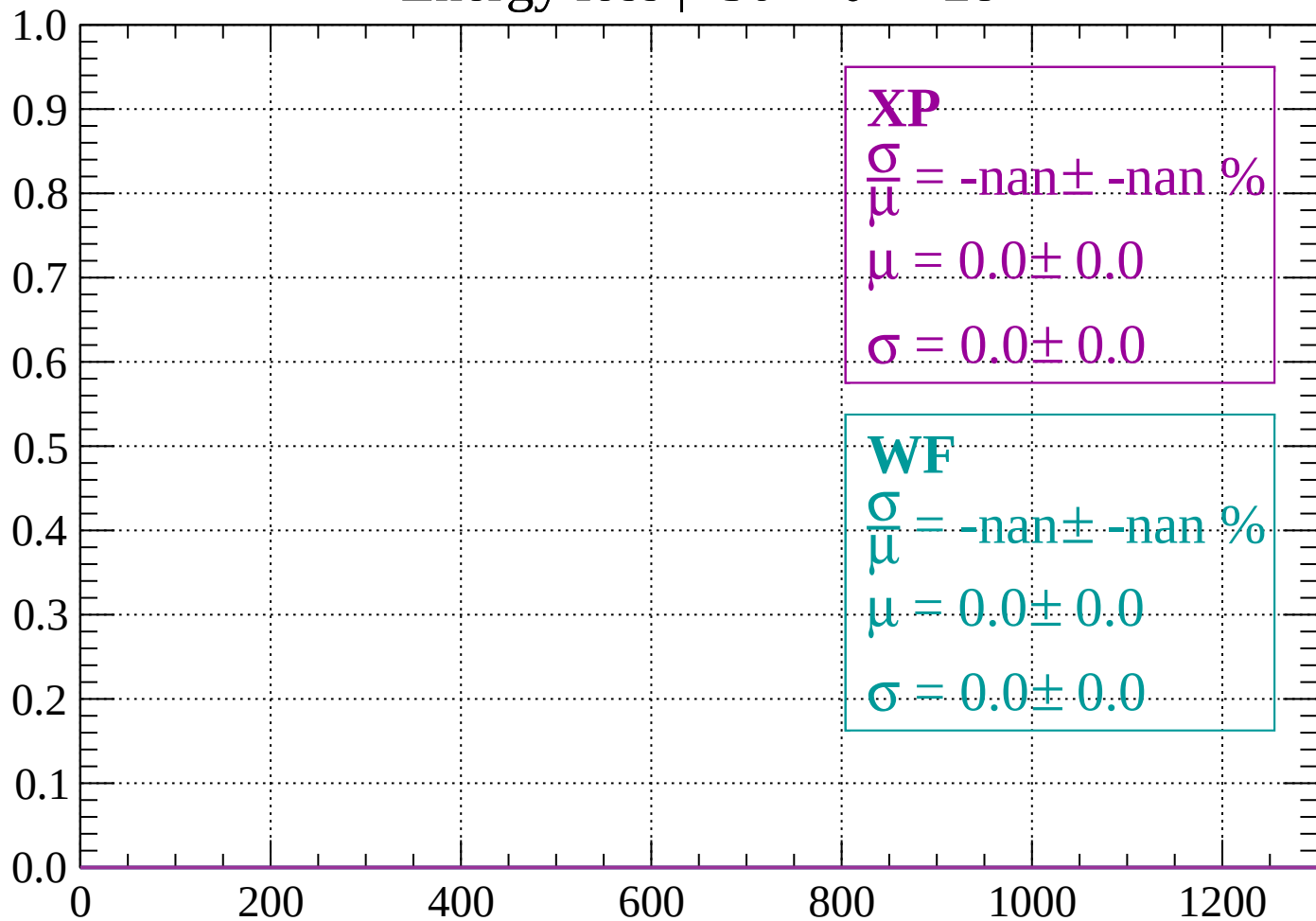
Count



dE/dx (ADC counts/cm)

Energy loss | $-30 < \theta < -28$

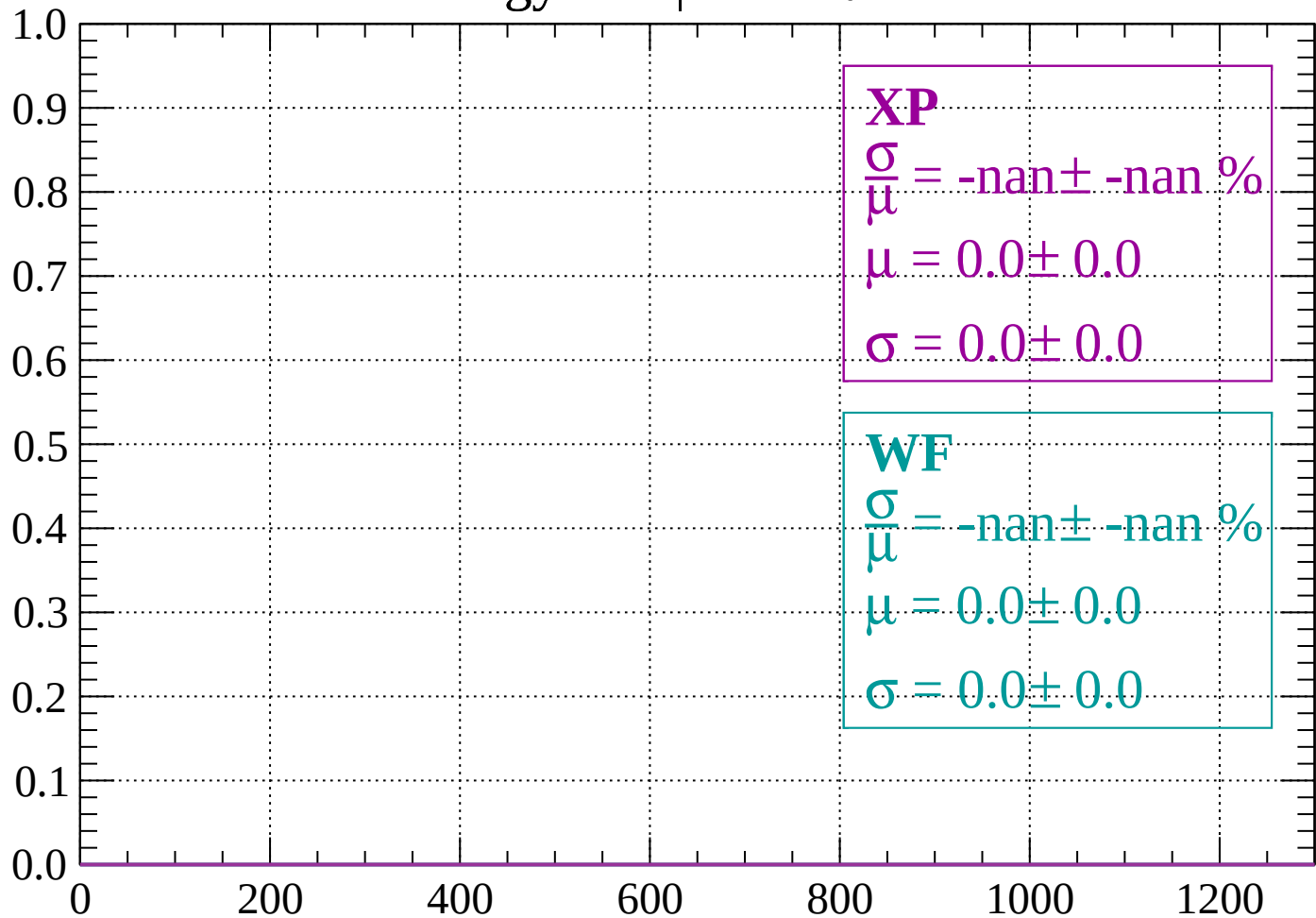
Count



dE/dx (ADC counts/cm)

Energy loss | $-28 < \theta < -26$

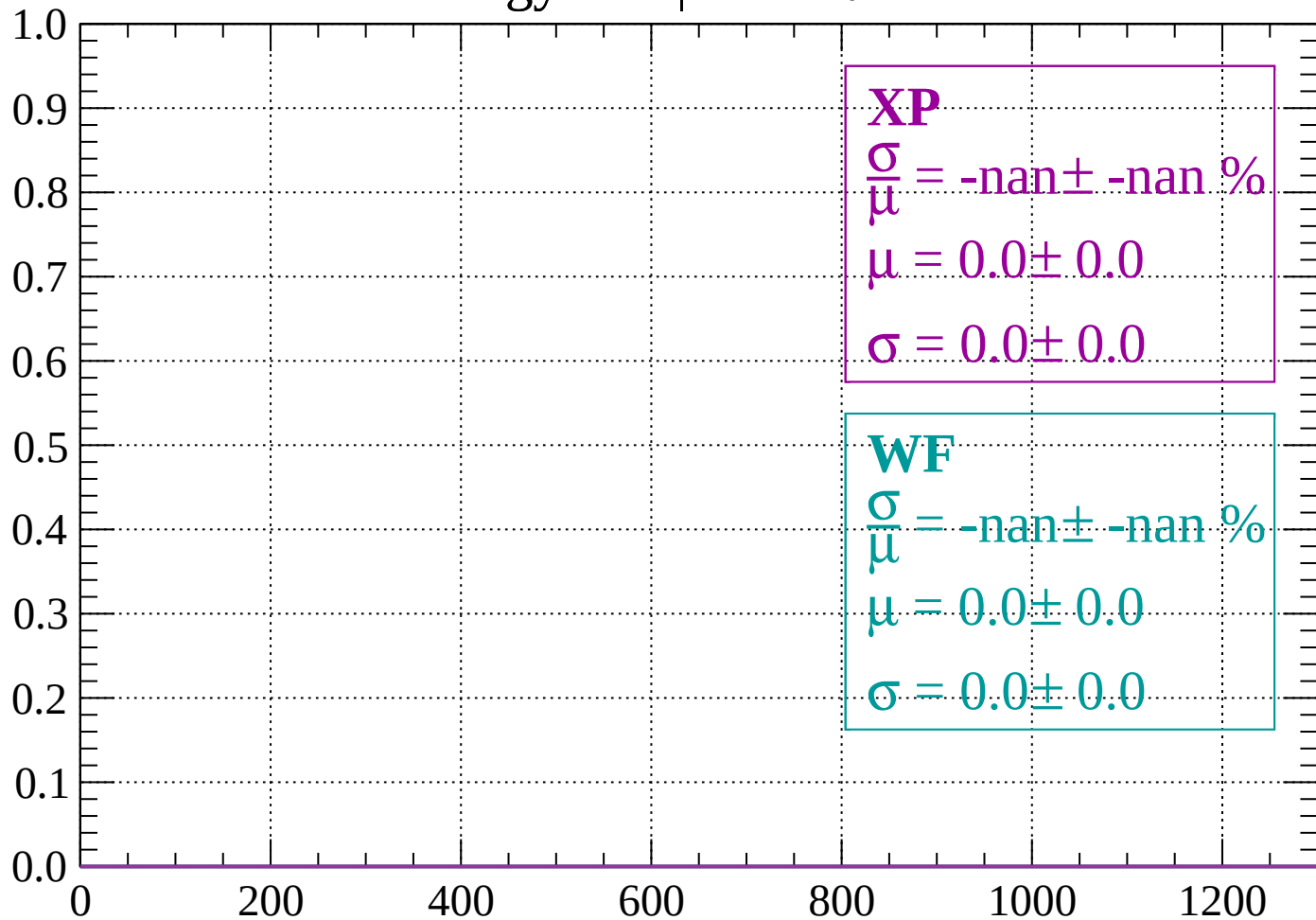
Count



dE/dx (ADC counts/cm)

Energy loss | $-26 < \theta < -24$

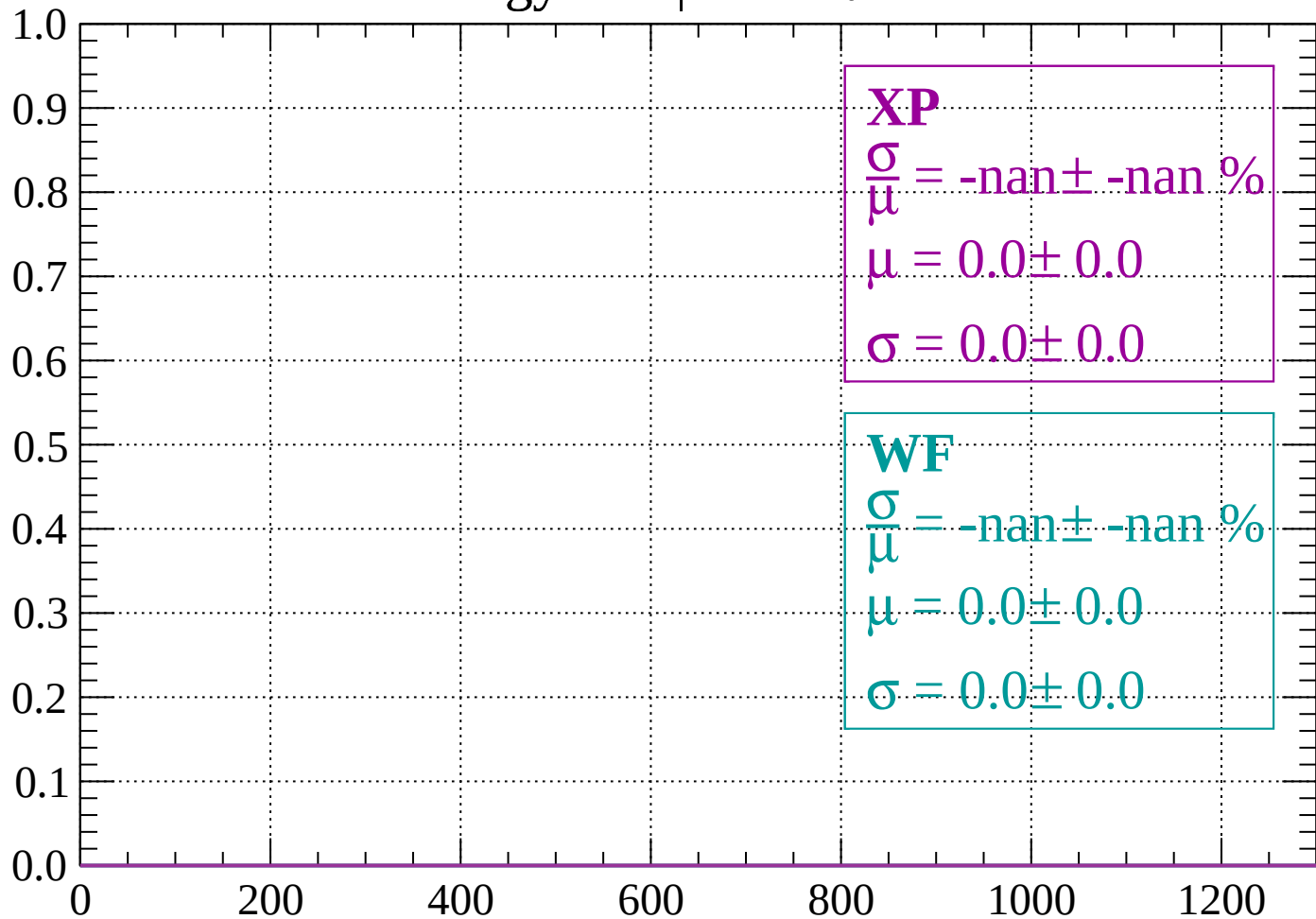
Count



dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

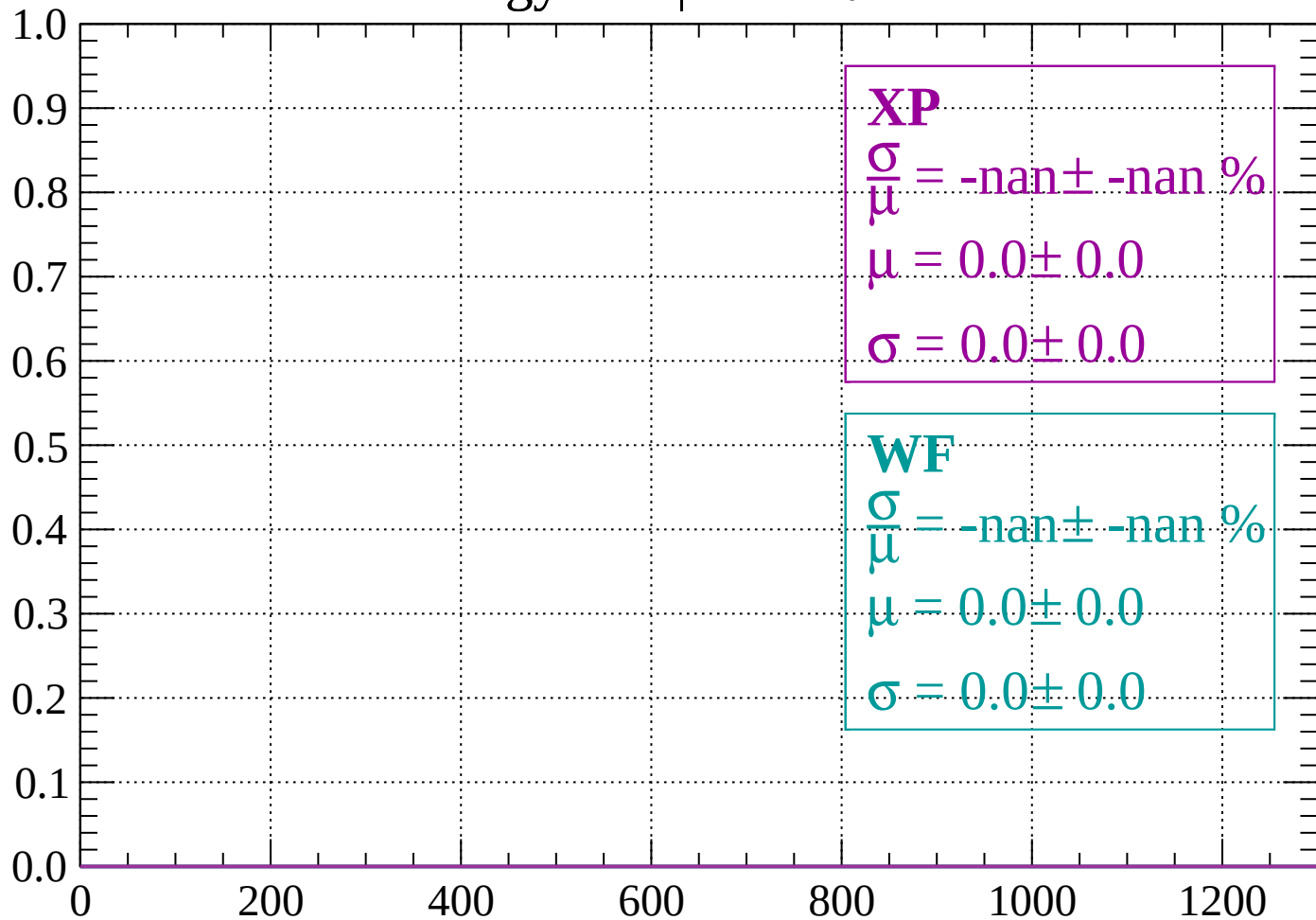
Count



dE/dx (ADC counts/cm)

Energy loss | $-22 < \theta < -20$

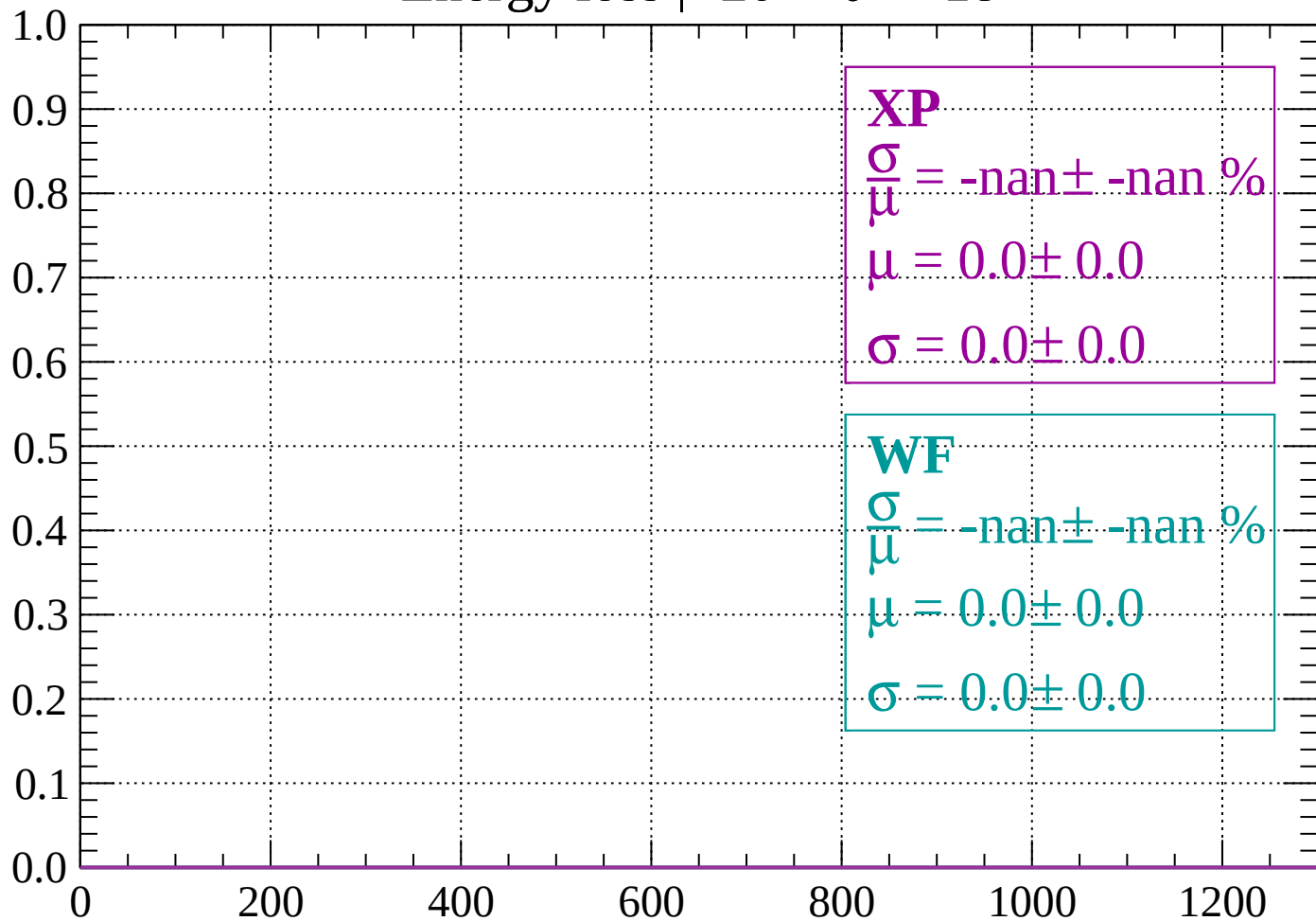
Count



dE/dx (ADC counts/cm)

Energy loss | $-20 < \theta < -18$

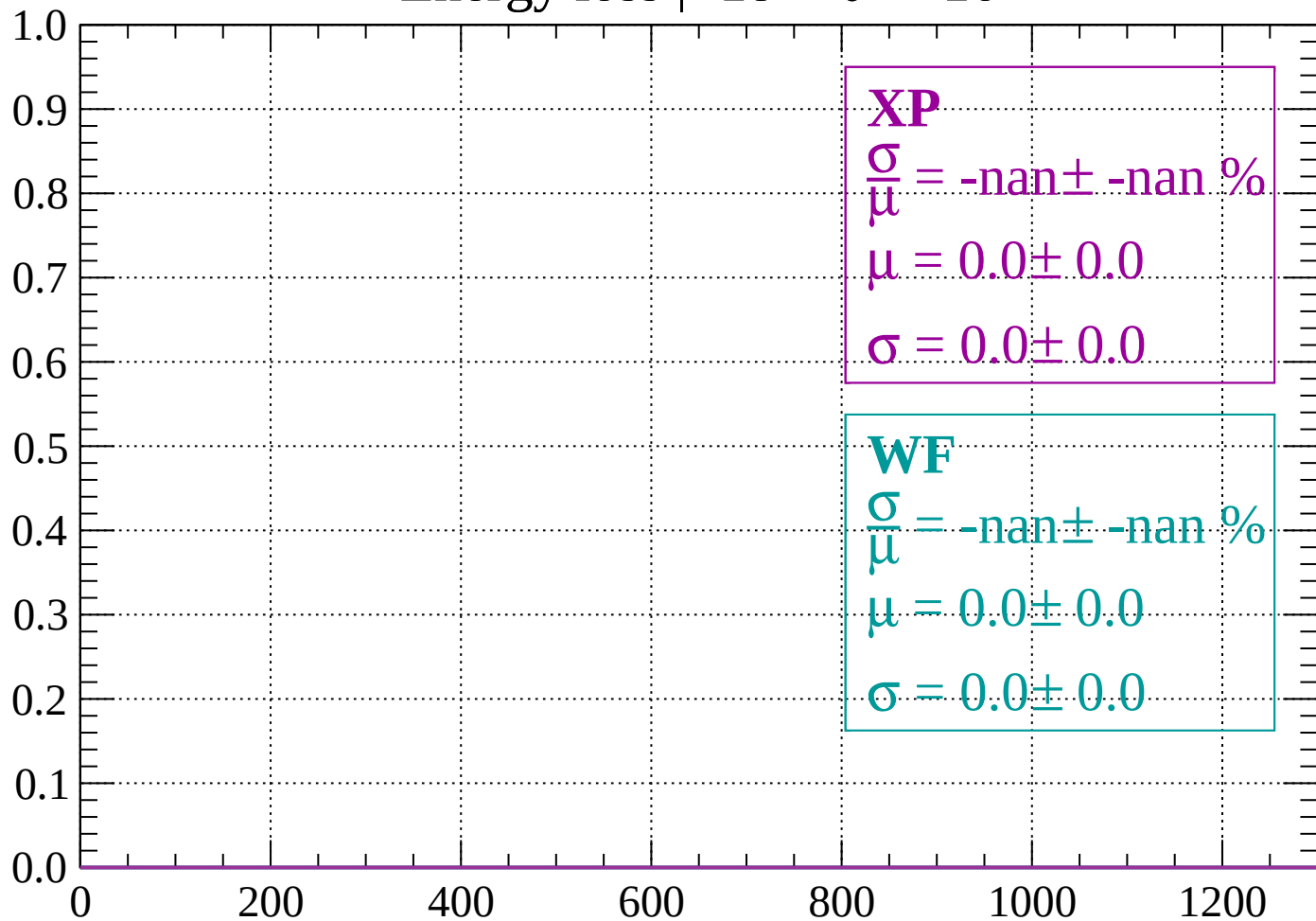
Count



dE/dx (ADC counts/cm)

Energy loss | $-18 < \theta < -16$

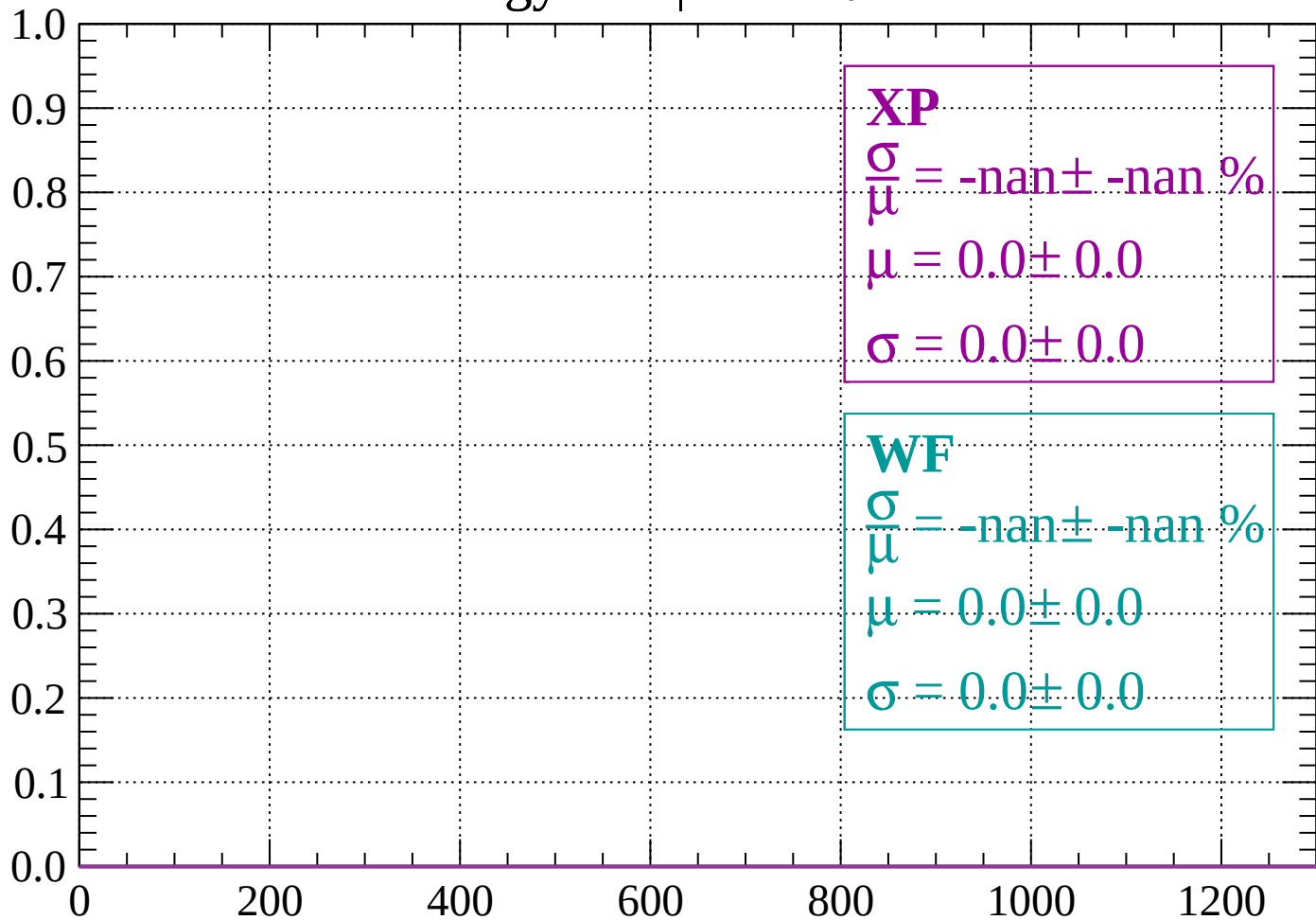
Count



dE/dx (ADC counts/cm)

Energy loss | $-16 < \theta < -14$

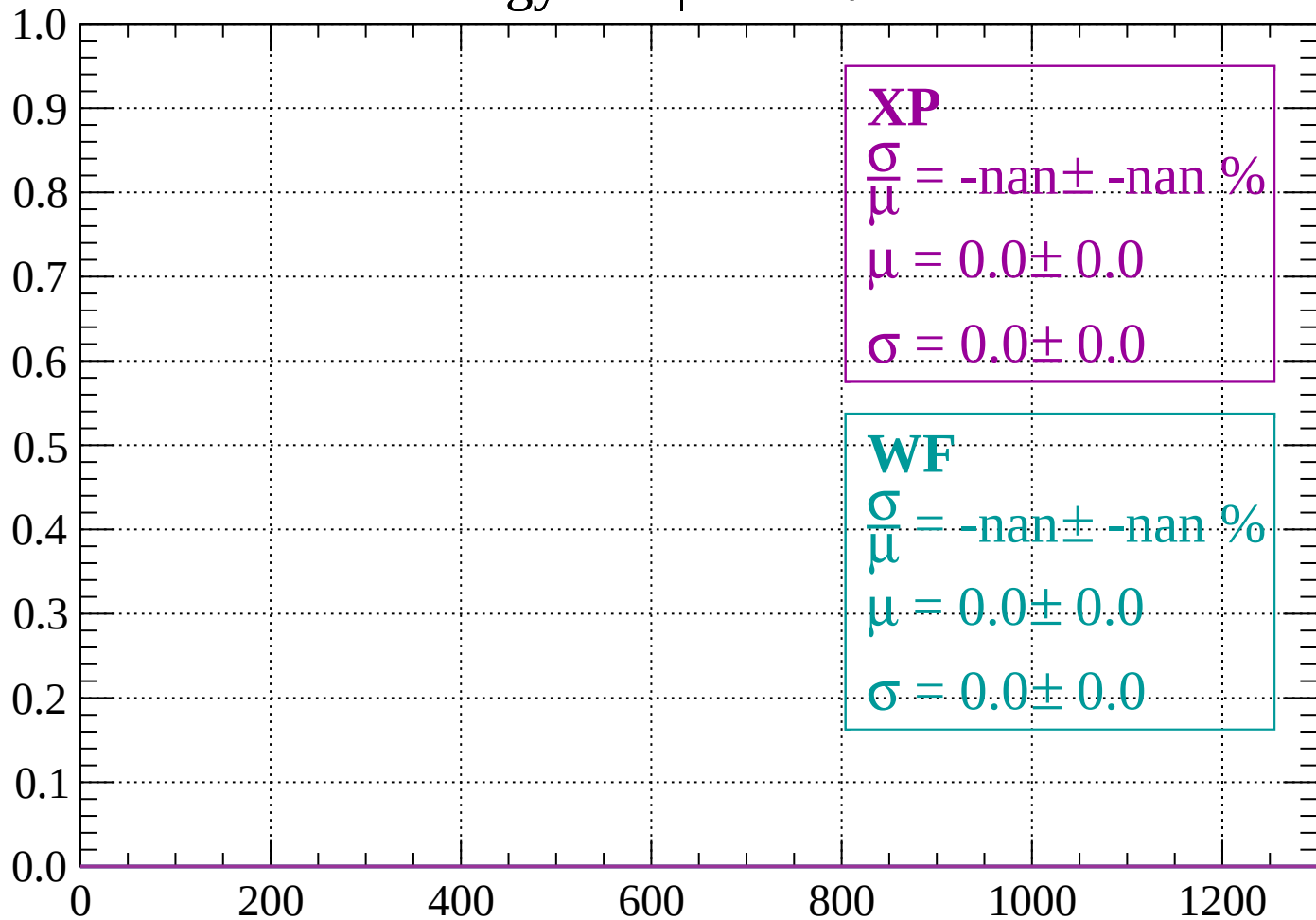
Count



dE/dx (ADC counts/cm)

Energy loss | $-14 < \theta < -12$

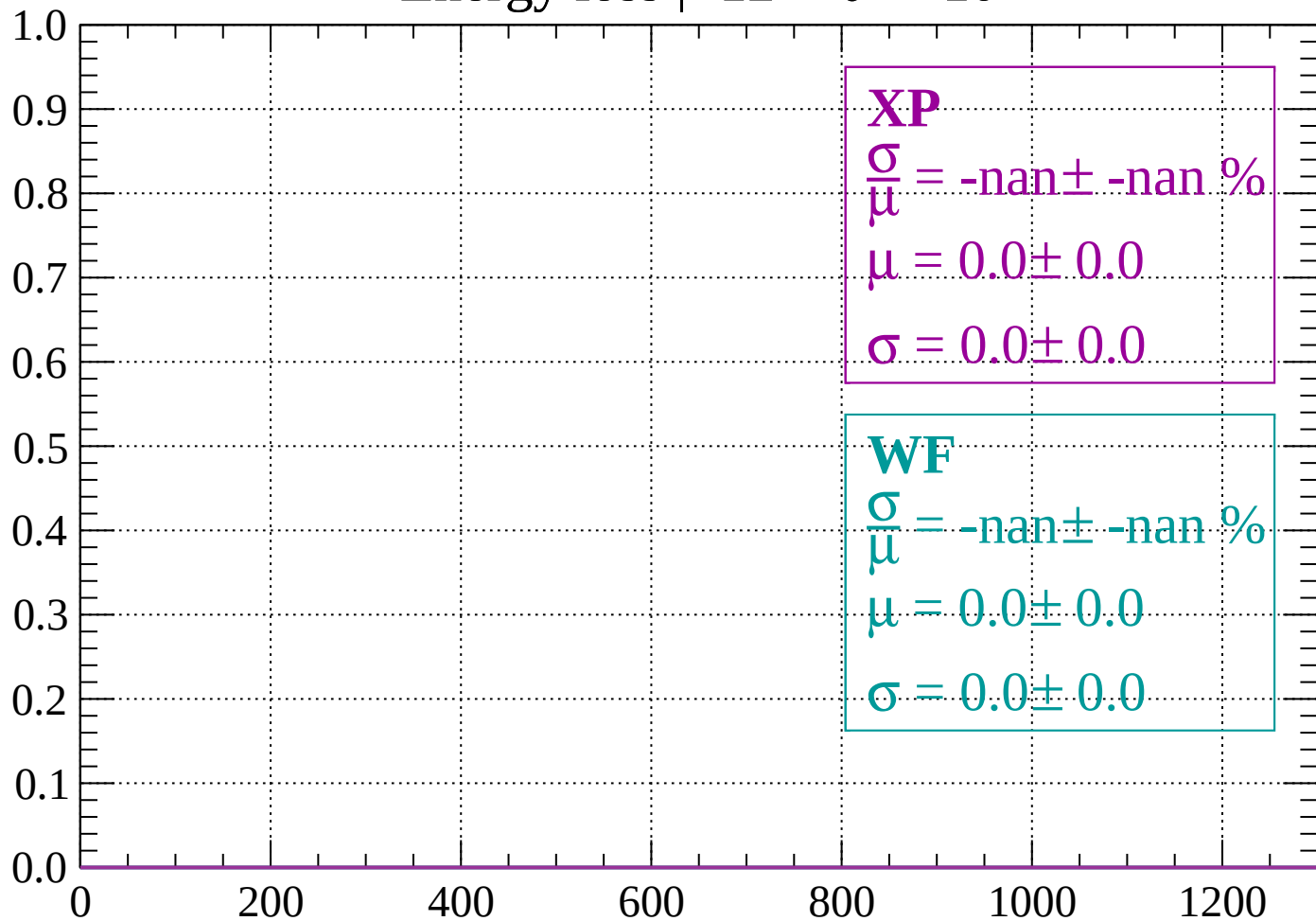
Count



dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -10$

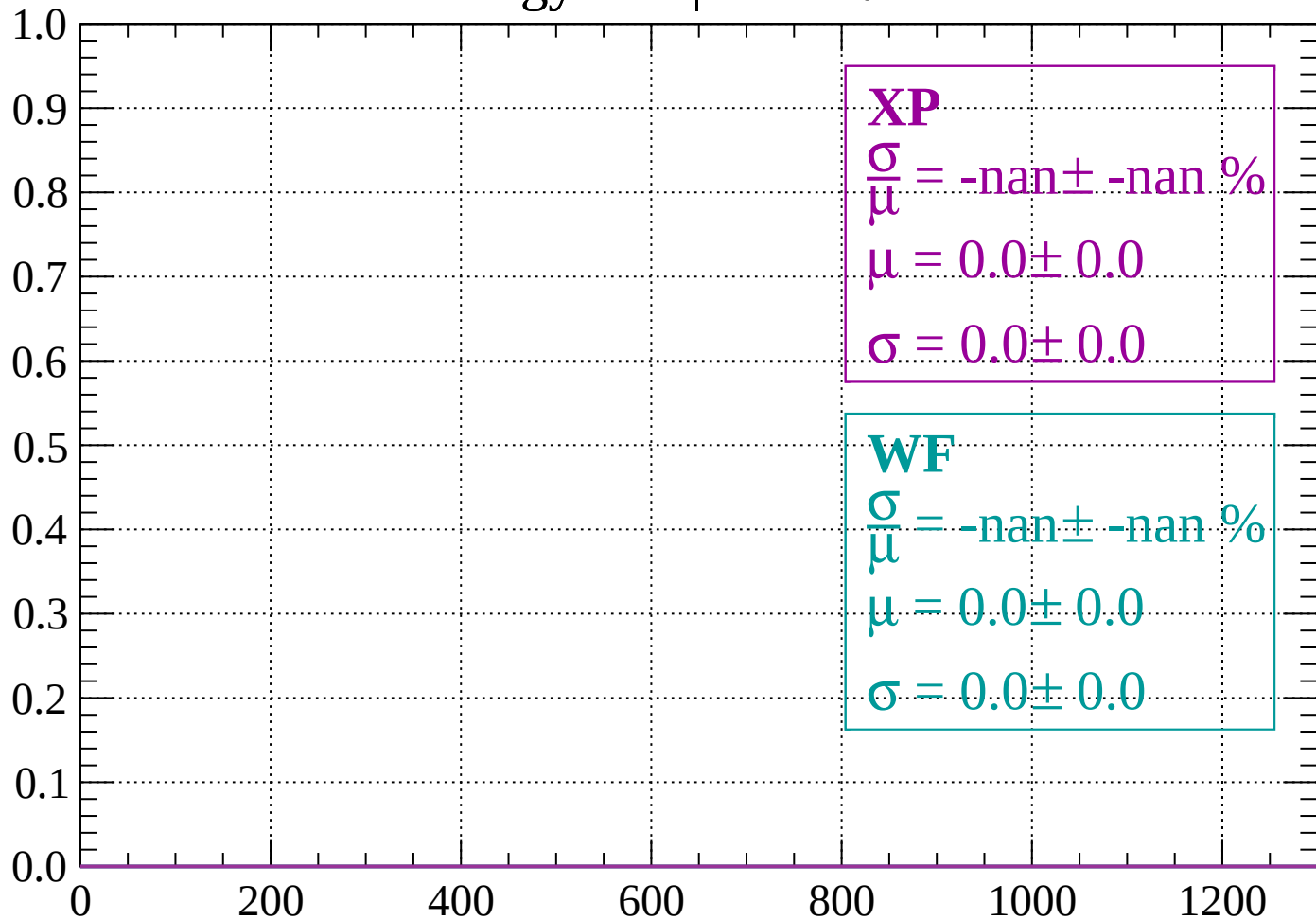
Count



dE/dx (ADC counts/cm)

Energy loss | $-10 < \theta < -8$

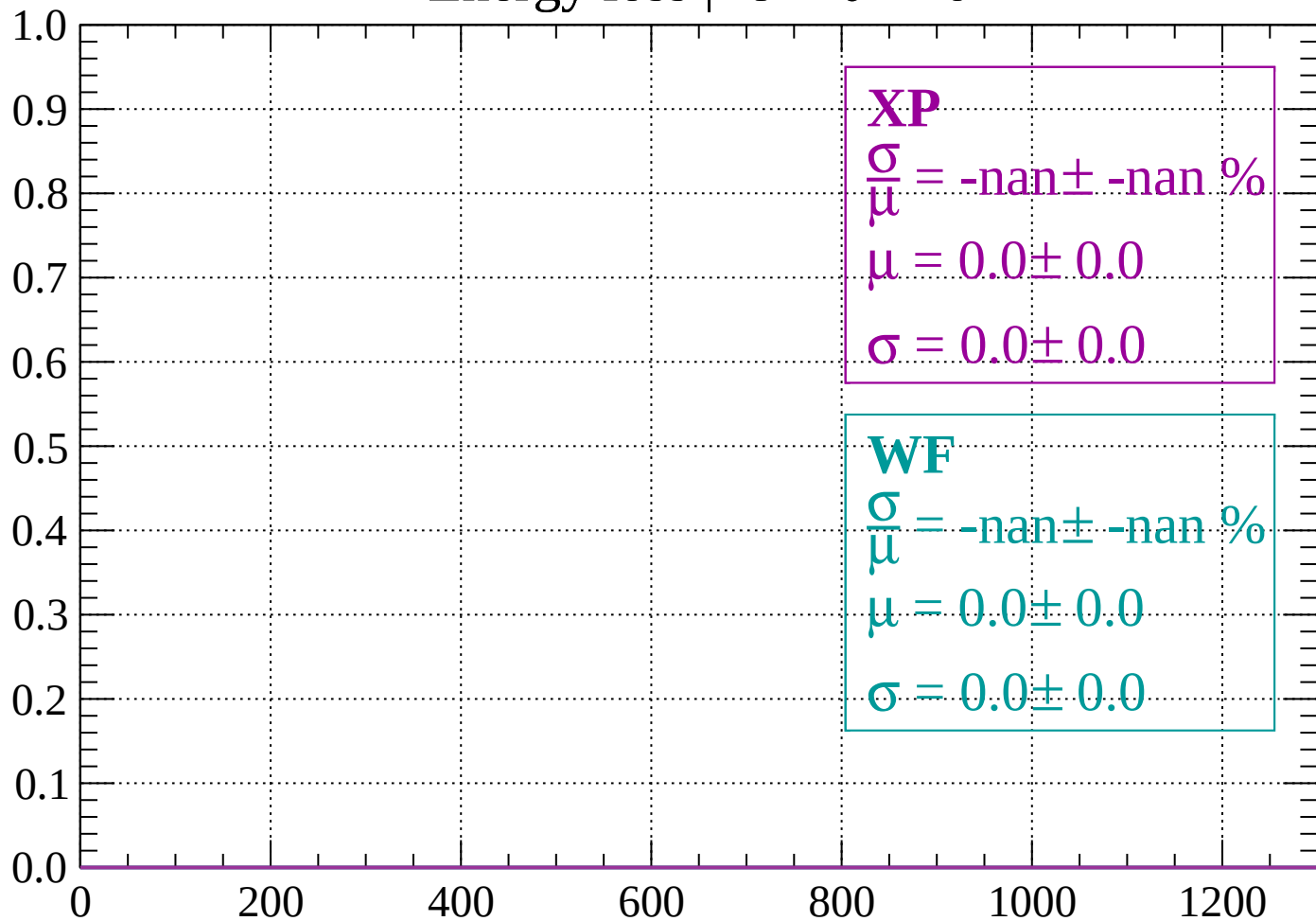
Count



dE/dx (ADC counts/cm)

Energy loss | $-8 < \theta < -6$

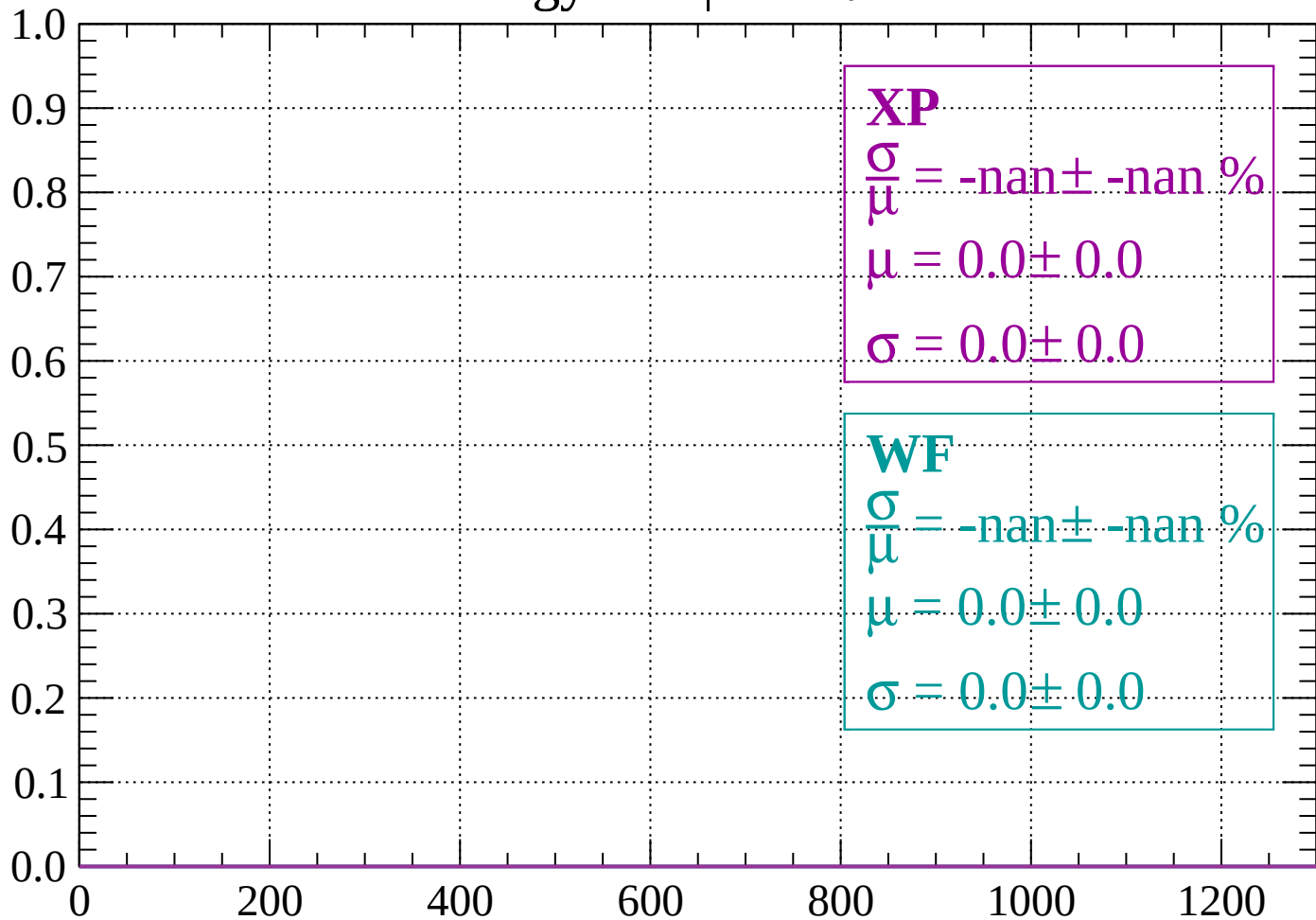
Count



dE/dx (ADC counts/cm)

Energy loss | $-6 < \theta < -4$

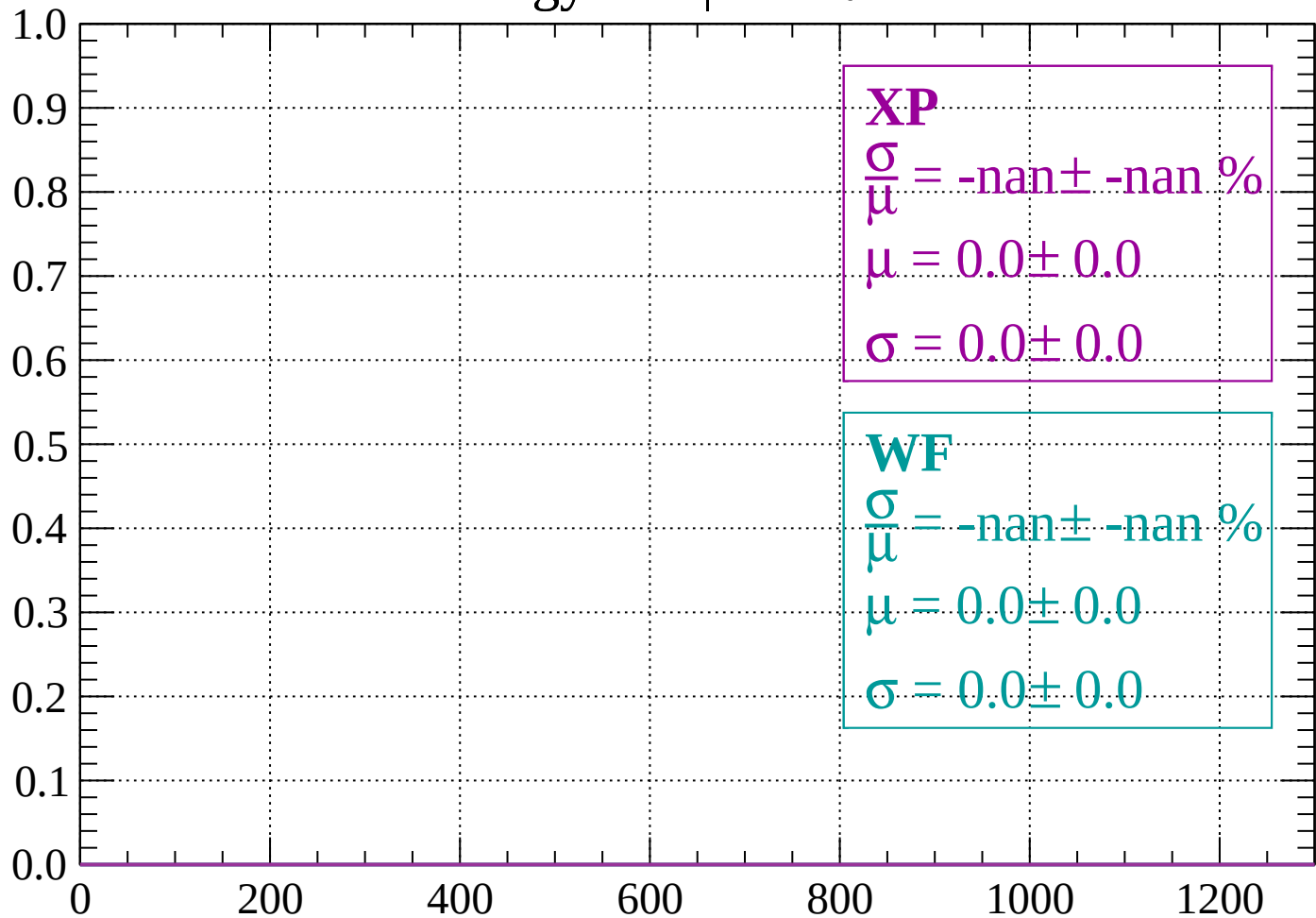
Count



dE/dx (ADC counts/cm)

Energy loss | $-4 < \theta < -2$

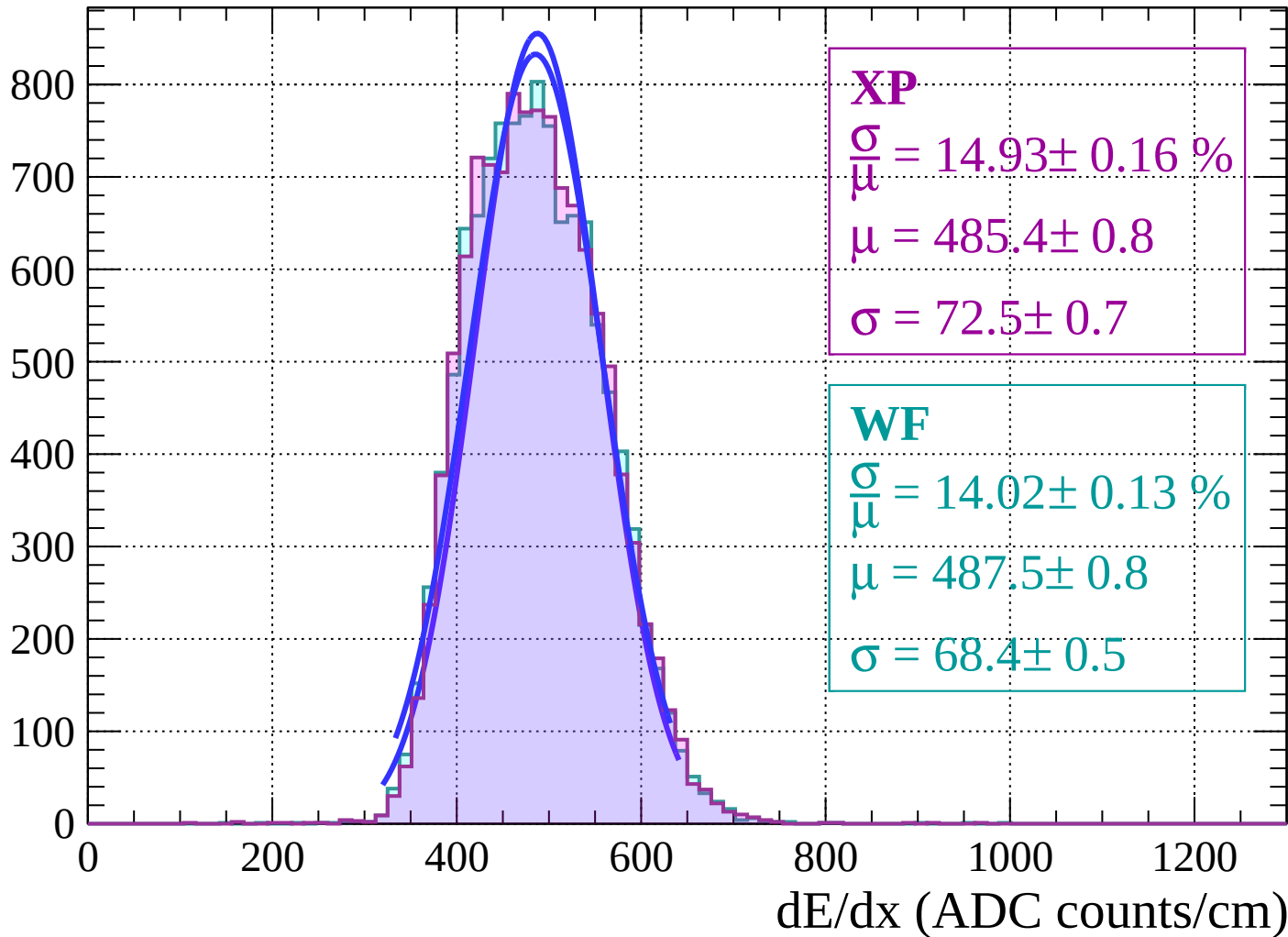
Count



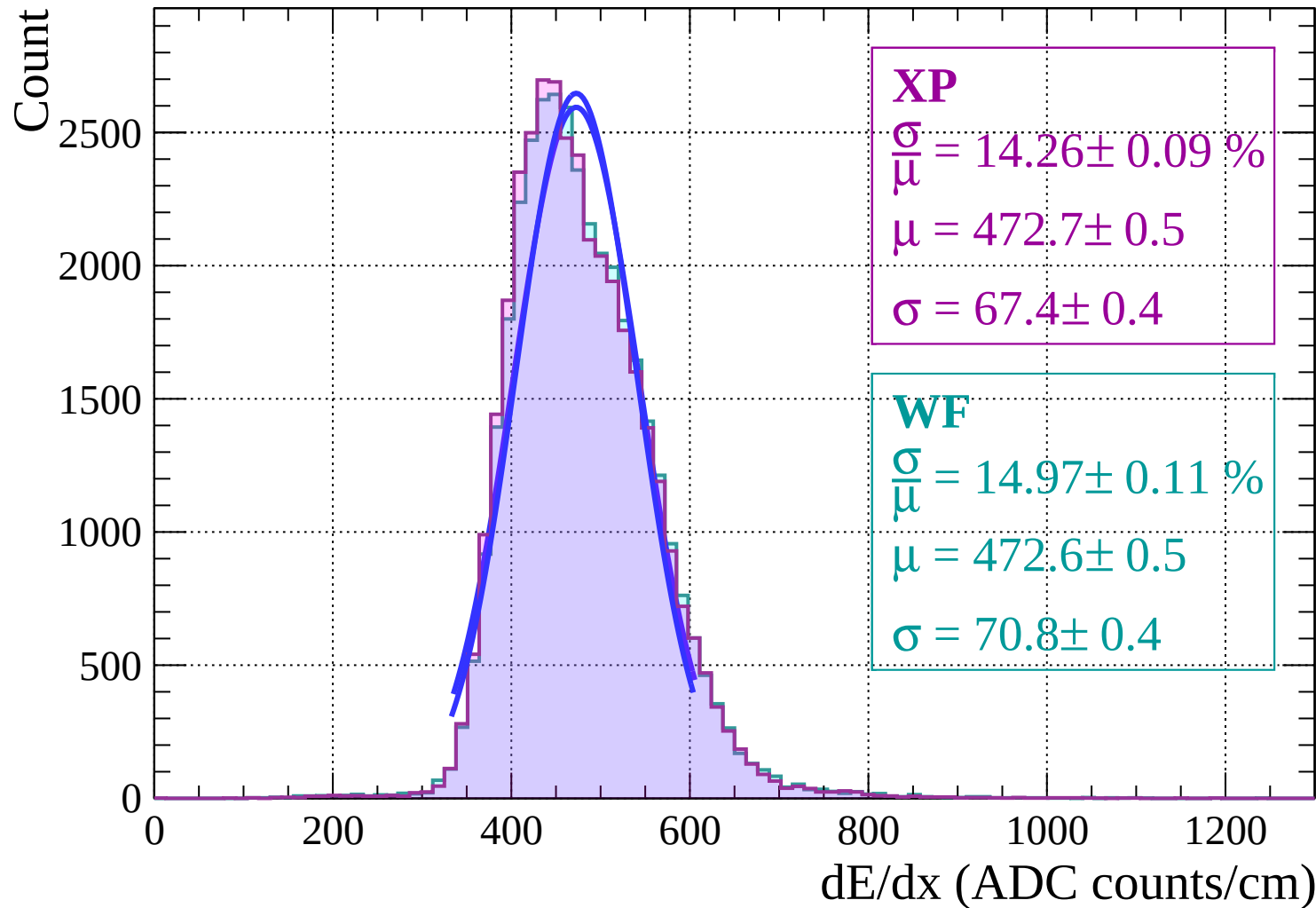
dE/dx (ADC counts/cm)

Energy loss | $-2 < \theta < 0$

Count

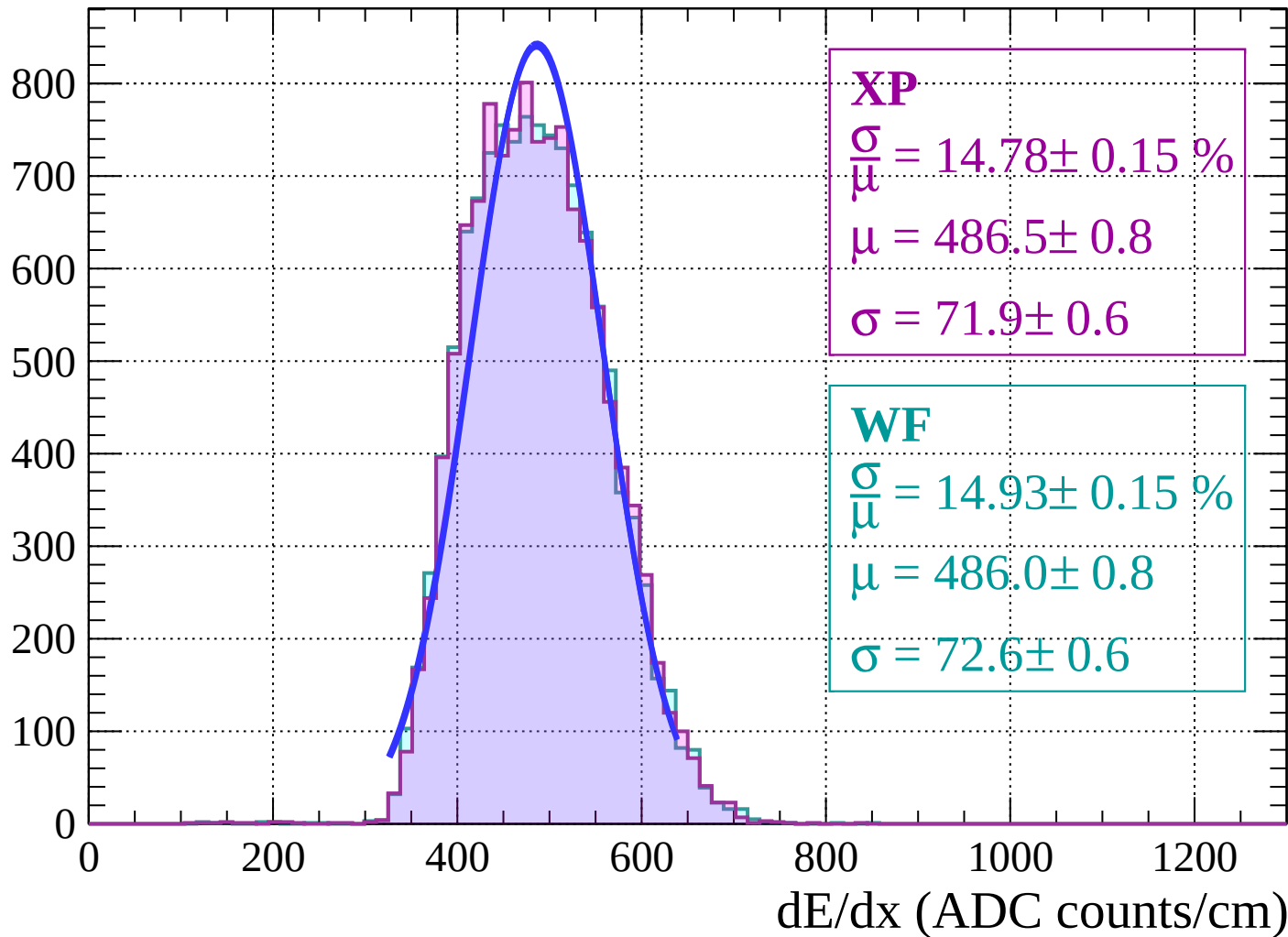


Energy loss | $0 < \theta < 2$



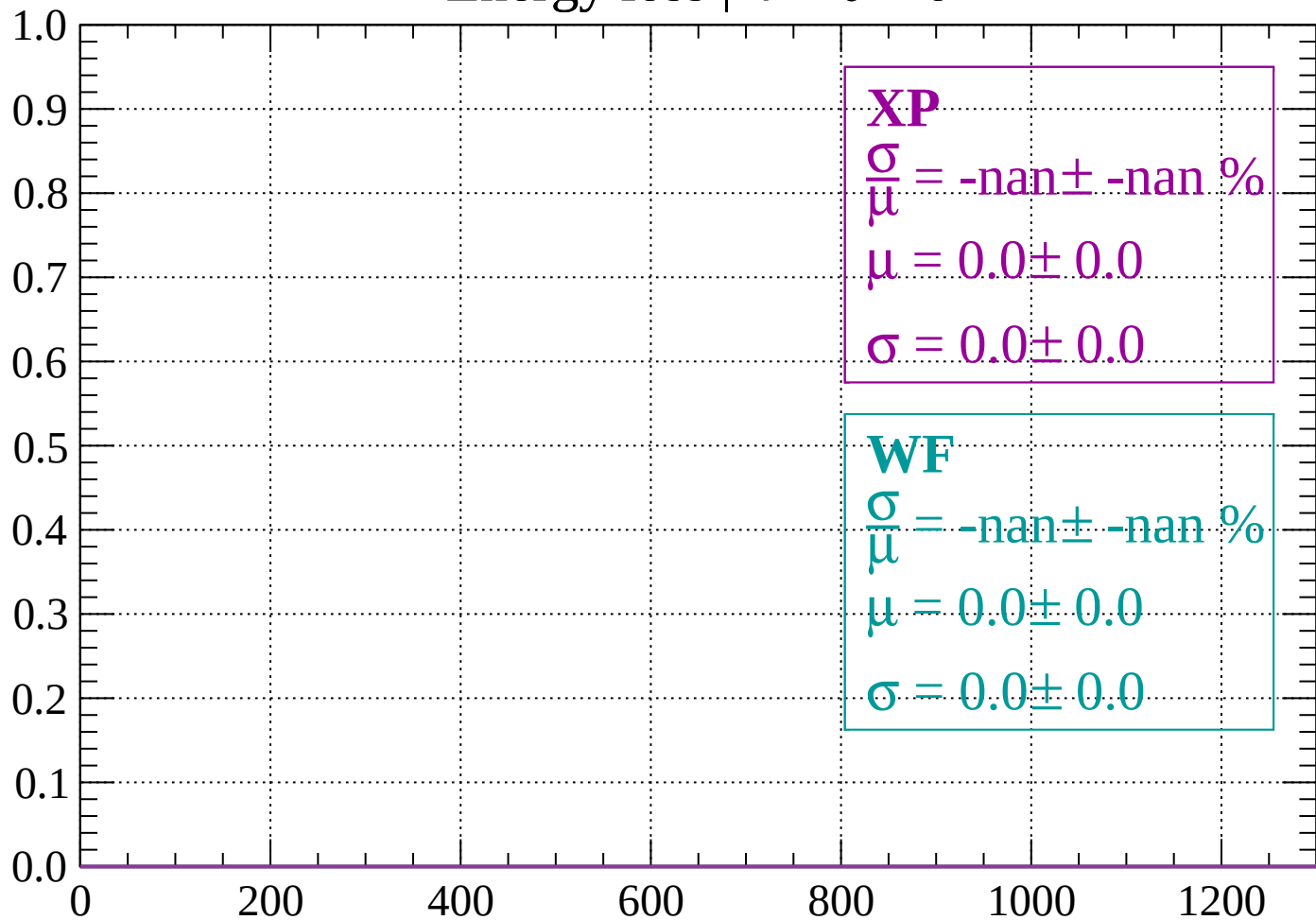
Energy loss | $2 < \theta < 4$

Count



Energy loss | $4 < \theta < 6$

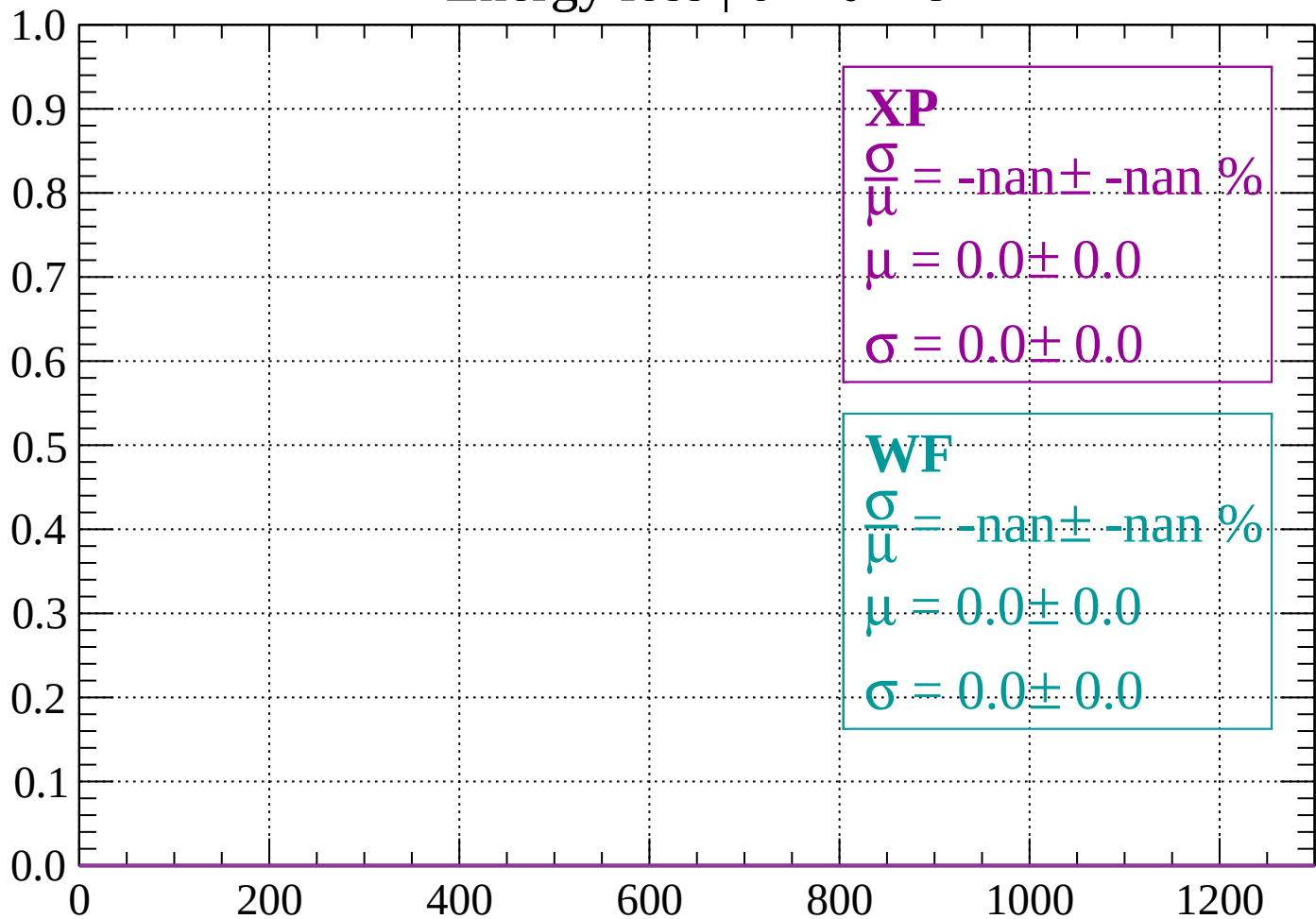
Count



dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$

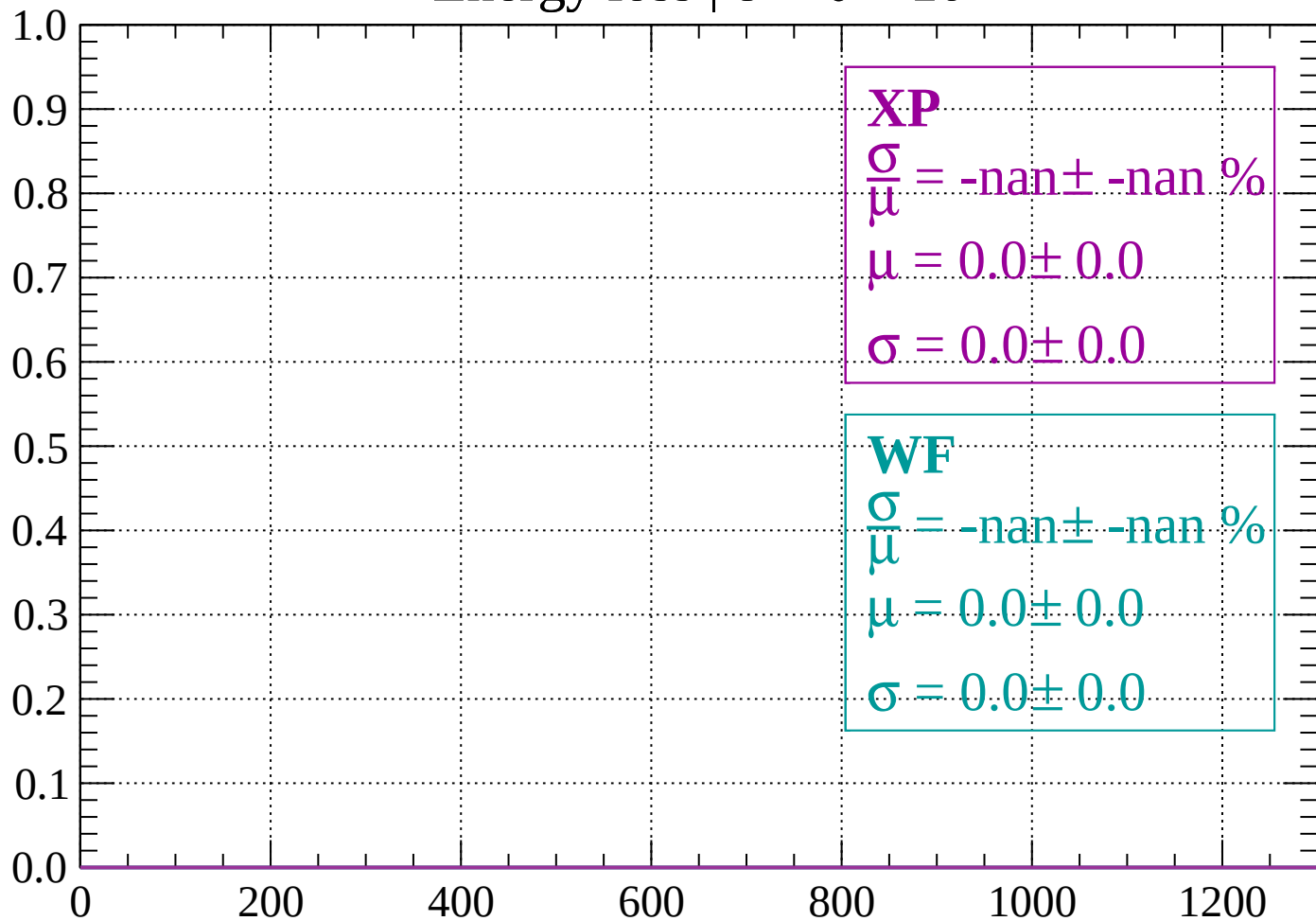
Count



dE/dx (ADC counts/cm)

Energy loss | $8 < \theta < 10$

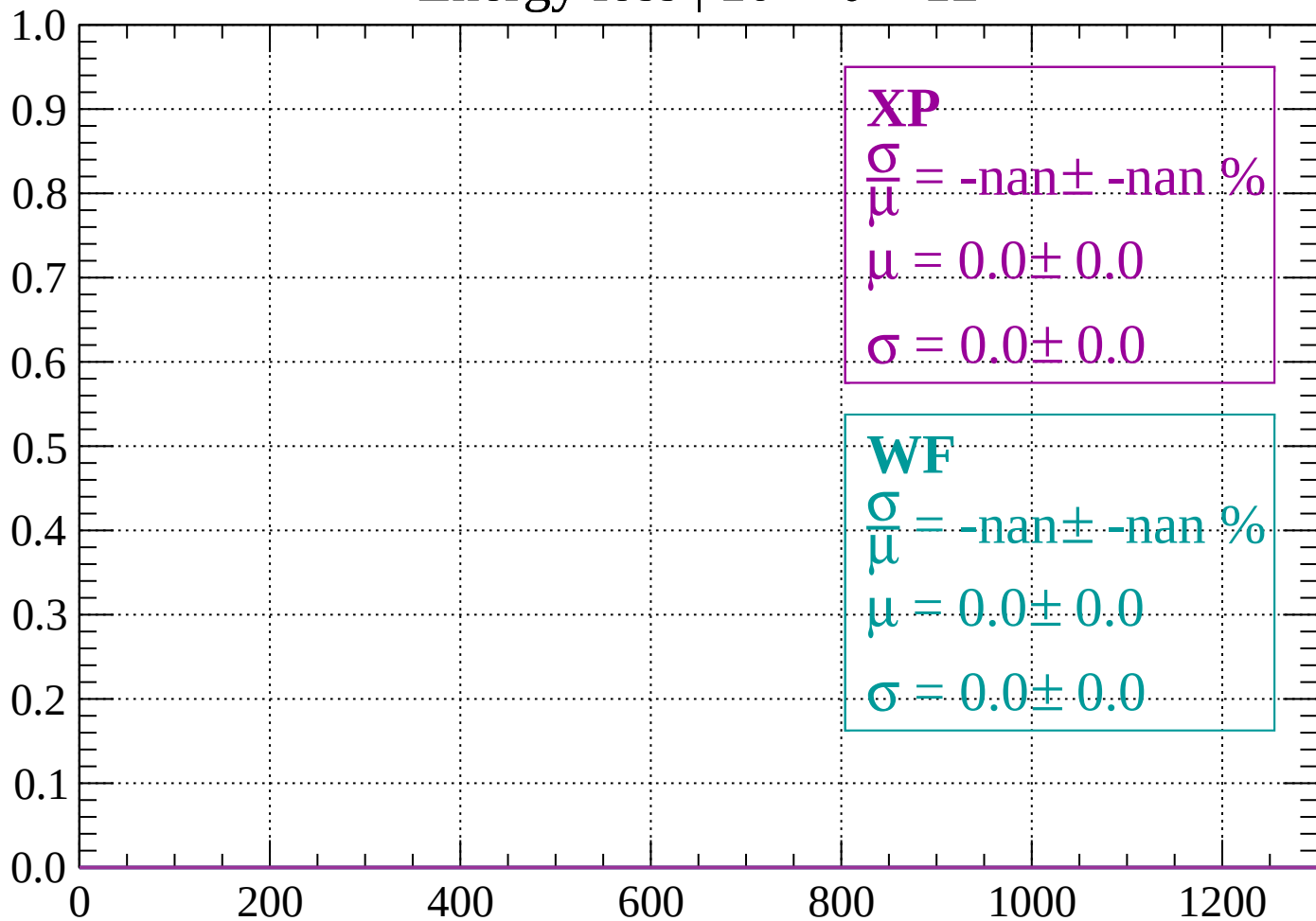
Count



dE/dx (ADC counts/cm)

Energy loss | $10 < \theta < 12$

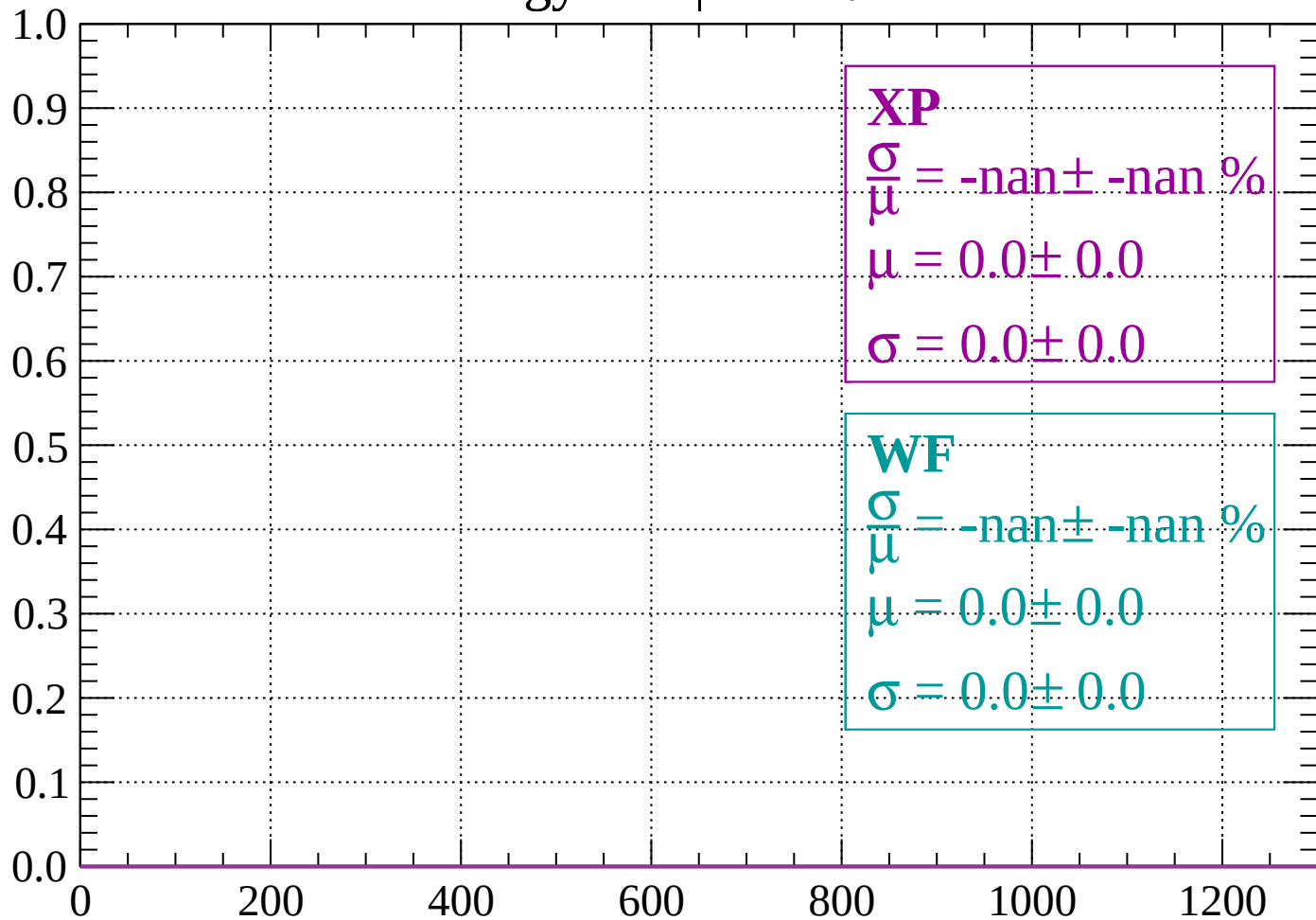
Count



dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$

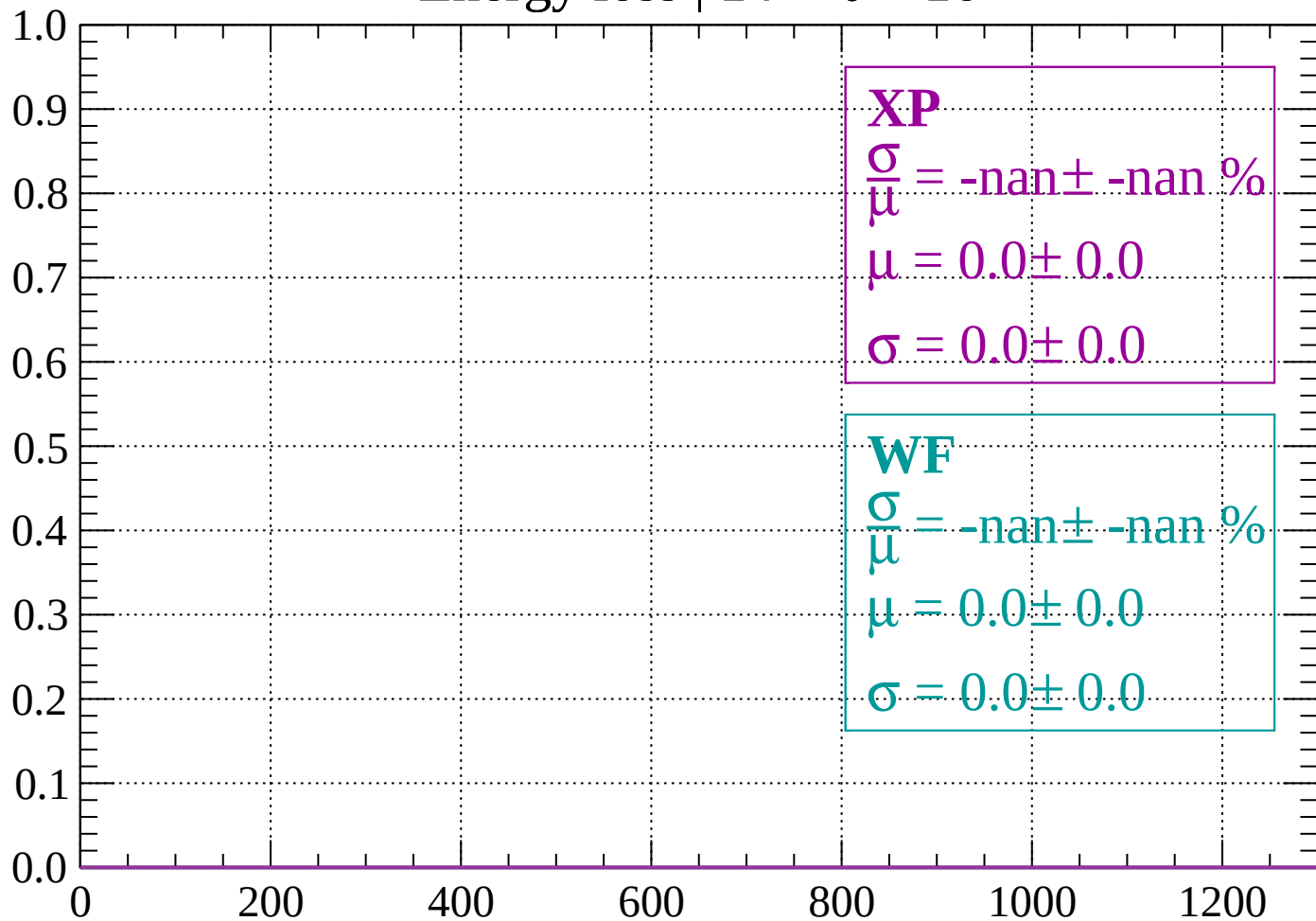
Count



dE/dx (ADC counts/cm)

Energy loss | $14 < \theta < 16$

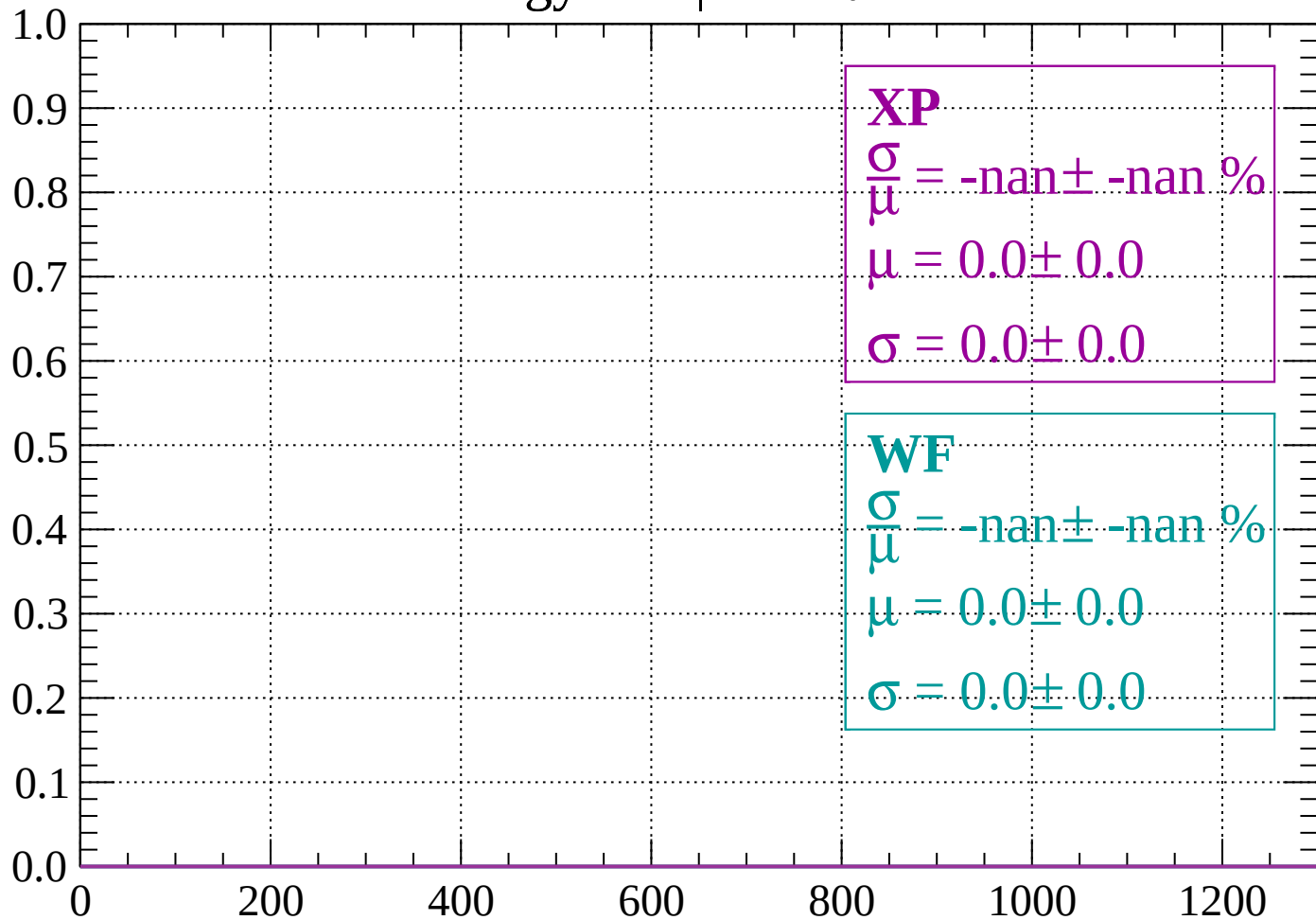
Count



dE/dx (ADC counts/cm)

Energy loss | $16 < \theta < 18$

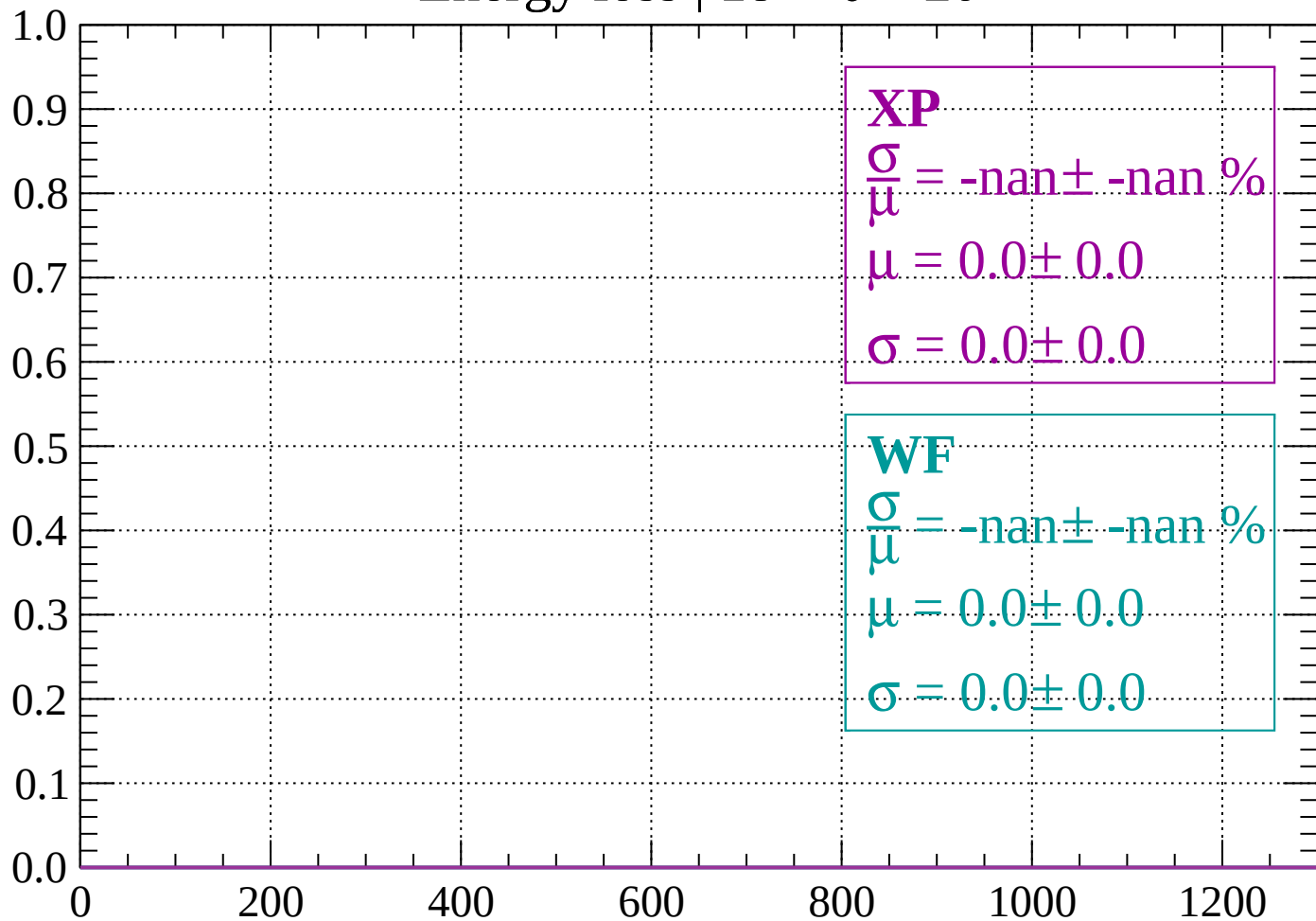
Count



dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$

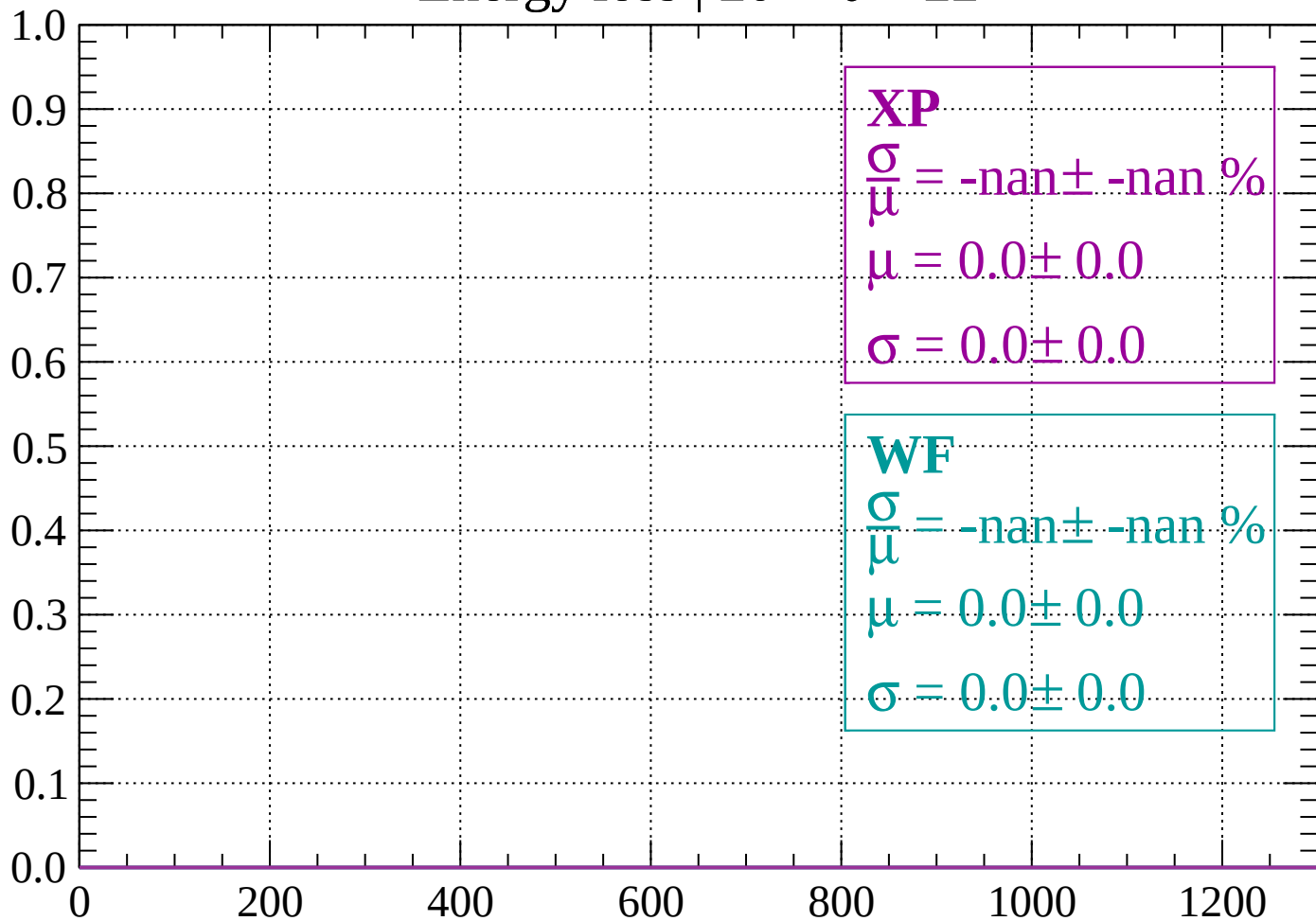
Count



dE/dx (ADC counts/cm)

Energy loss | $20 < \theta < 22$

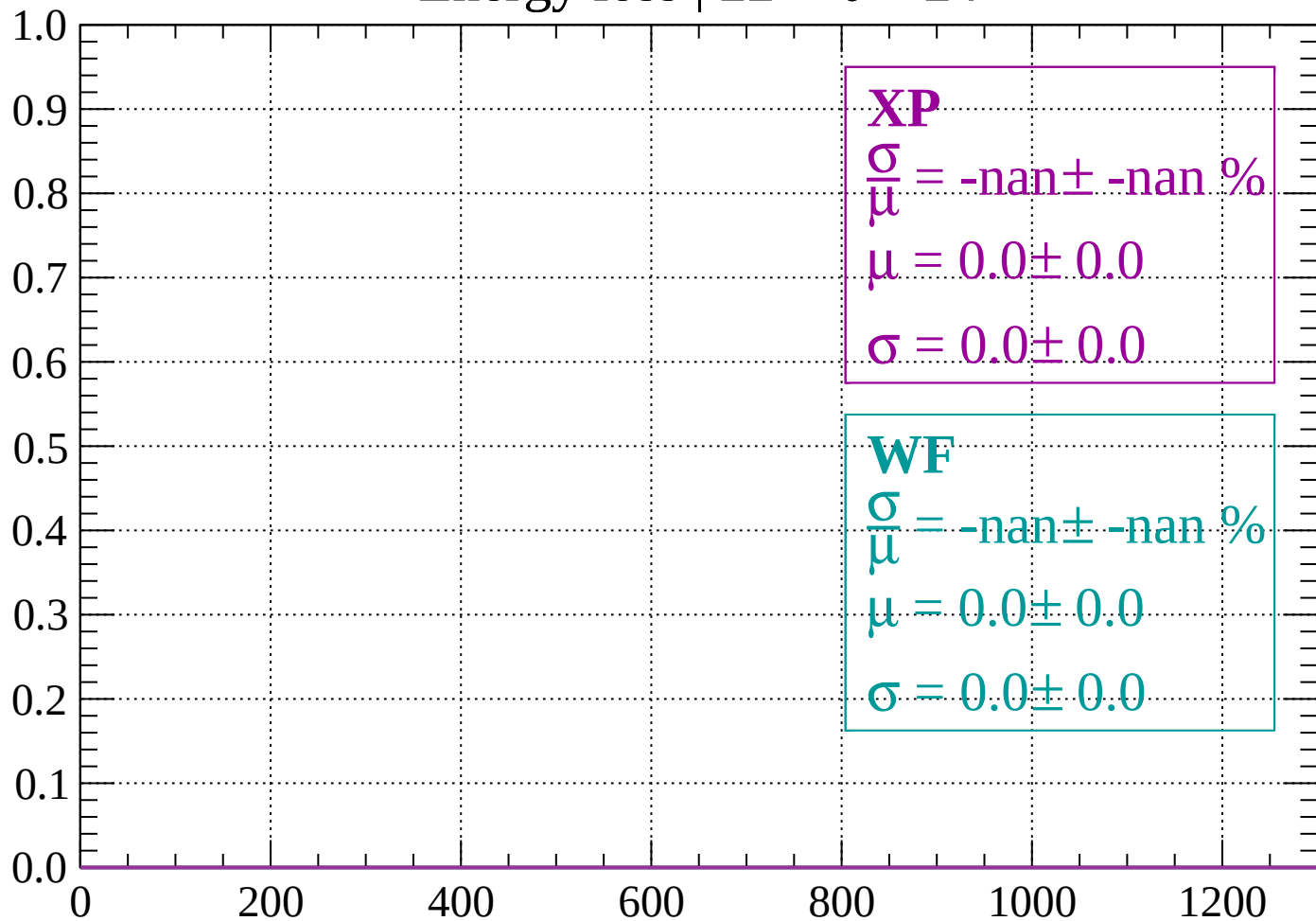
Count



dE/dx (ADC counts/cm)

Energy loss | $22 < \theta < 24$

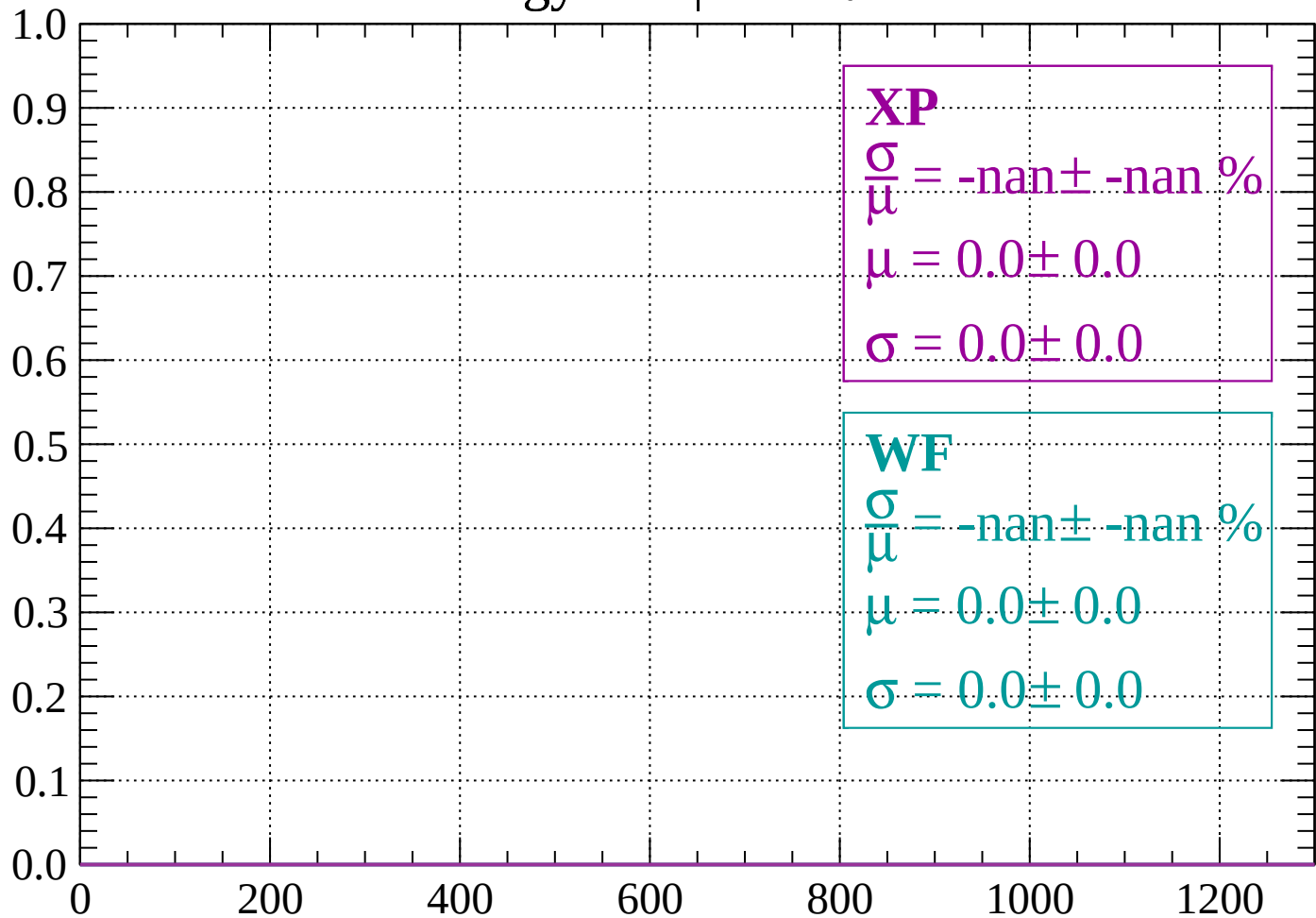
Count



dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 26$

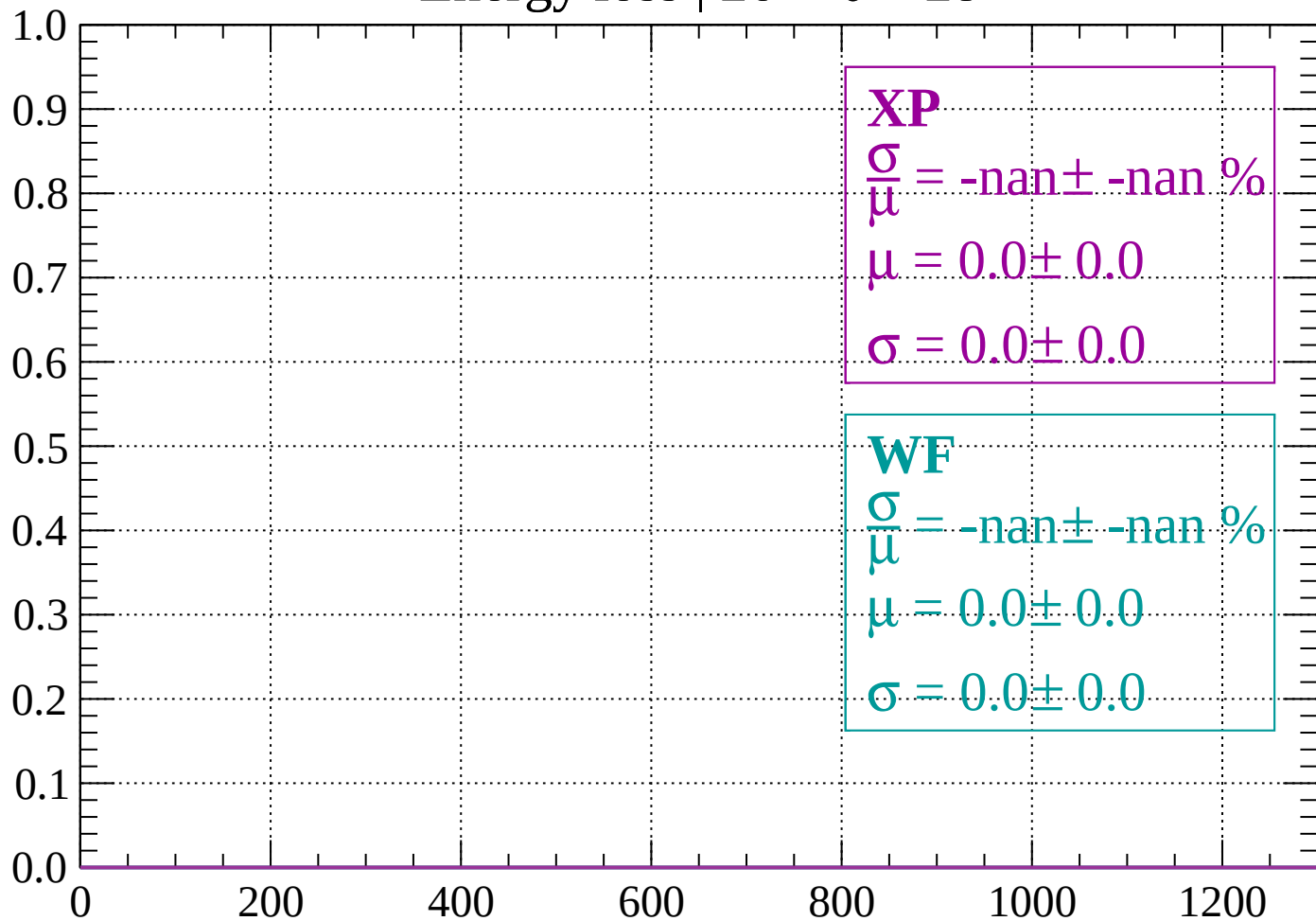
Count



dE/dx (ADC counts/cm)

Energy loss | $26 < \theta < 28$

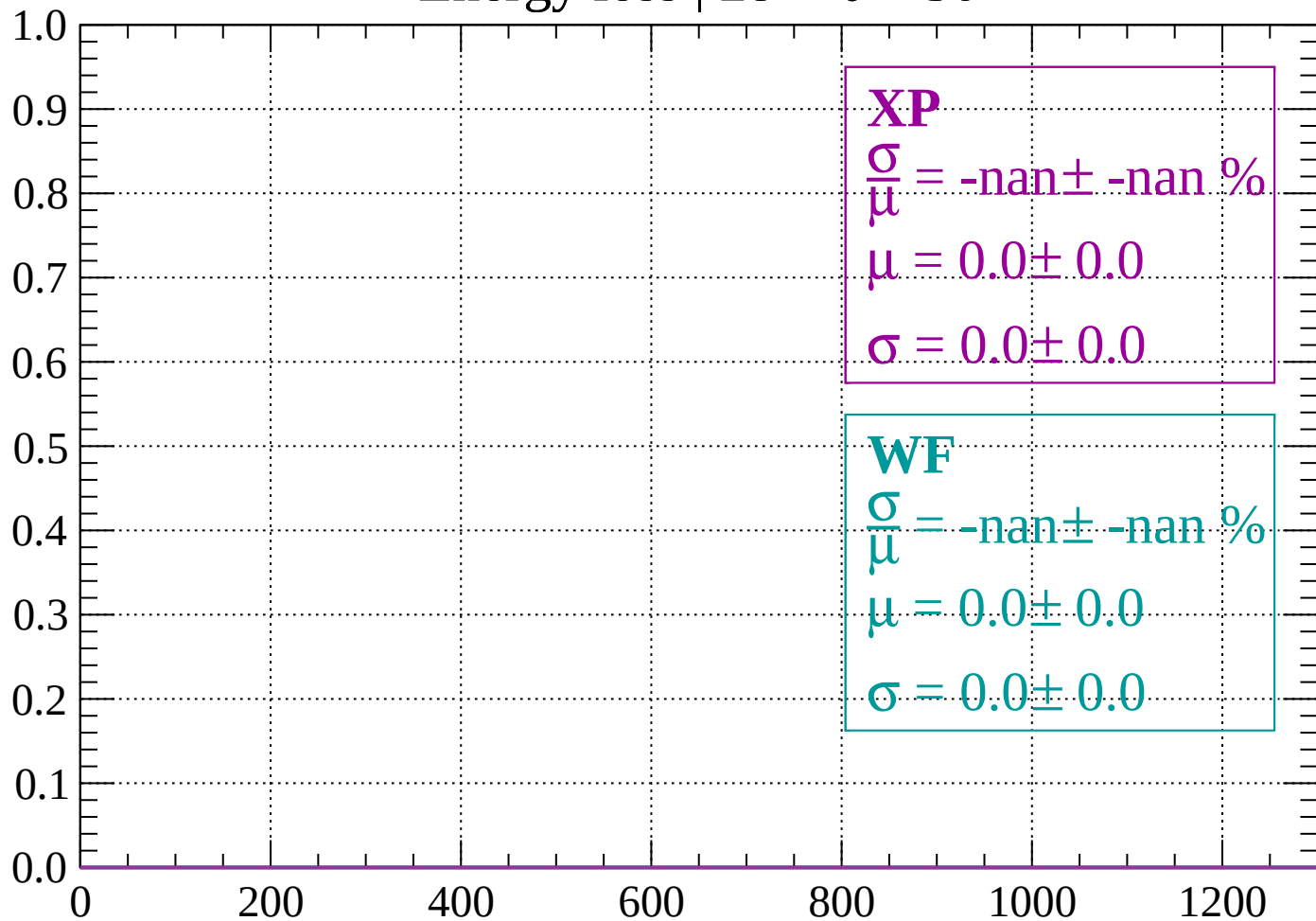
Count



dE/dx (ADC counts/cm)

Energy loss | $28 < \theta < 30$

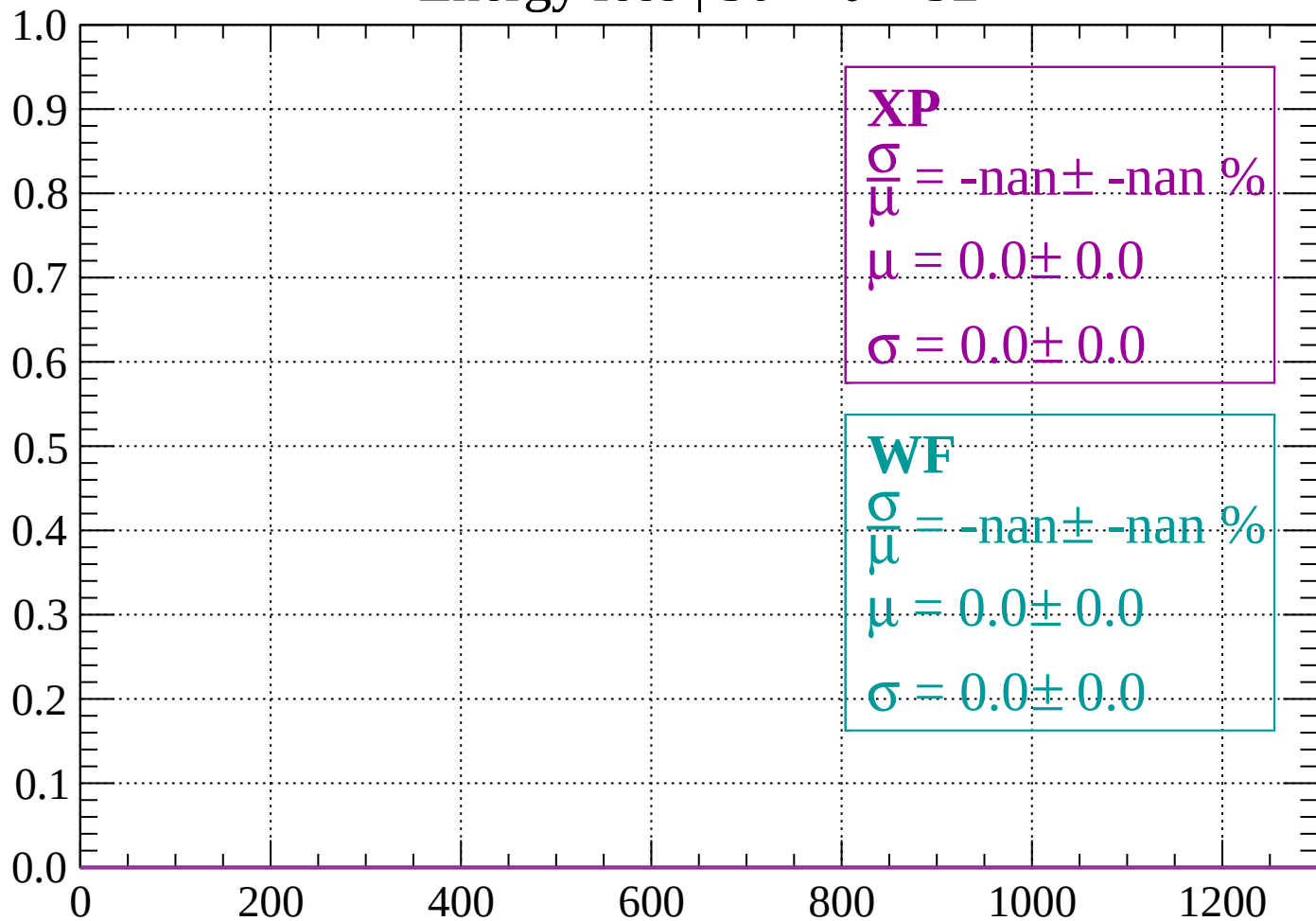
Count



dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 32$

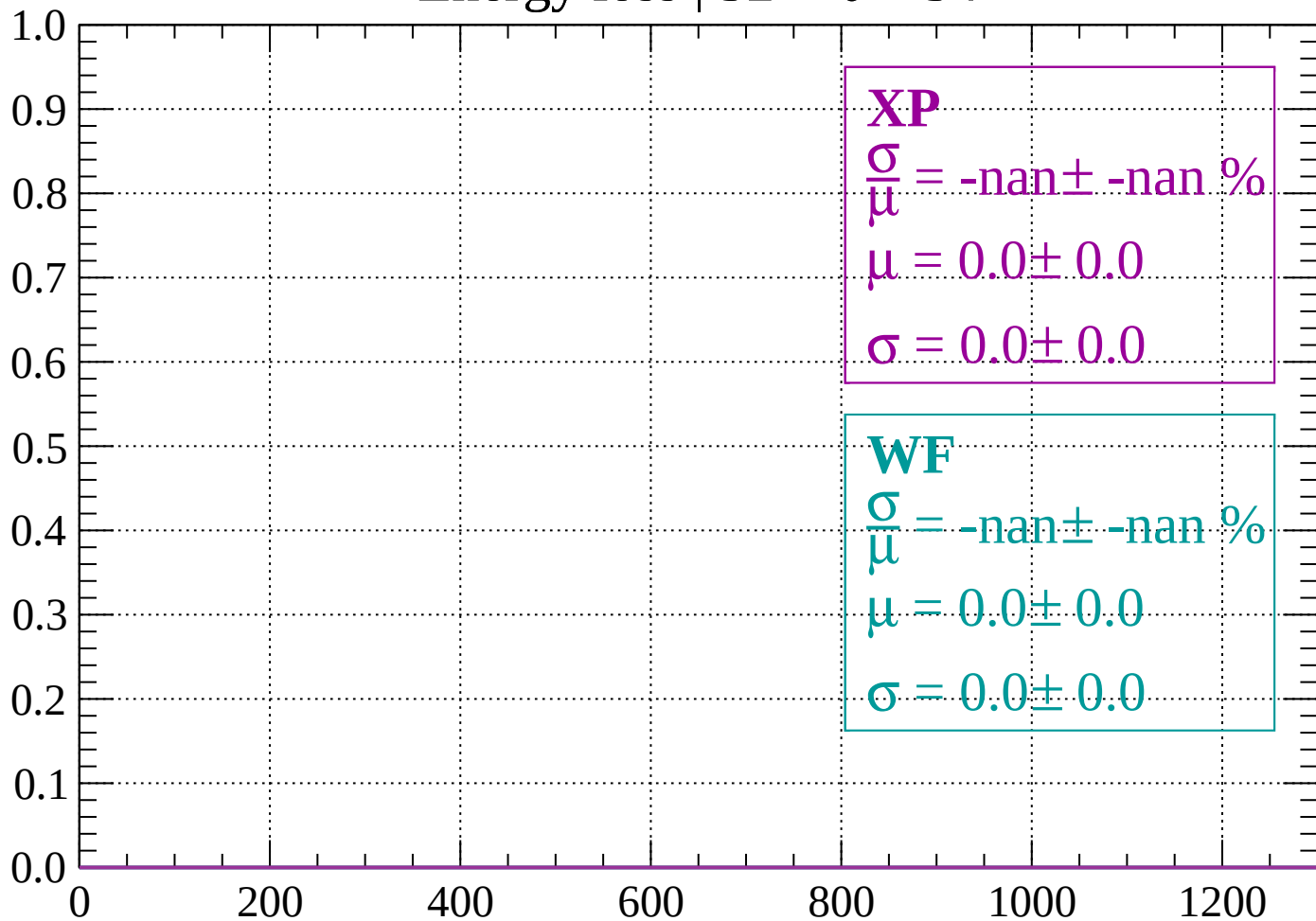
Count



dE/dx (ADC counts/cm)

Energy loss | $32 < \theta < 34$

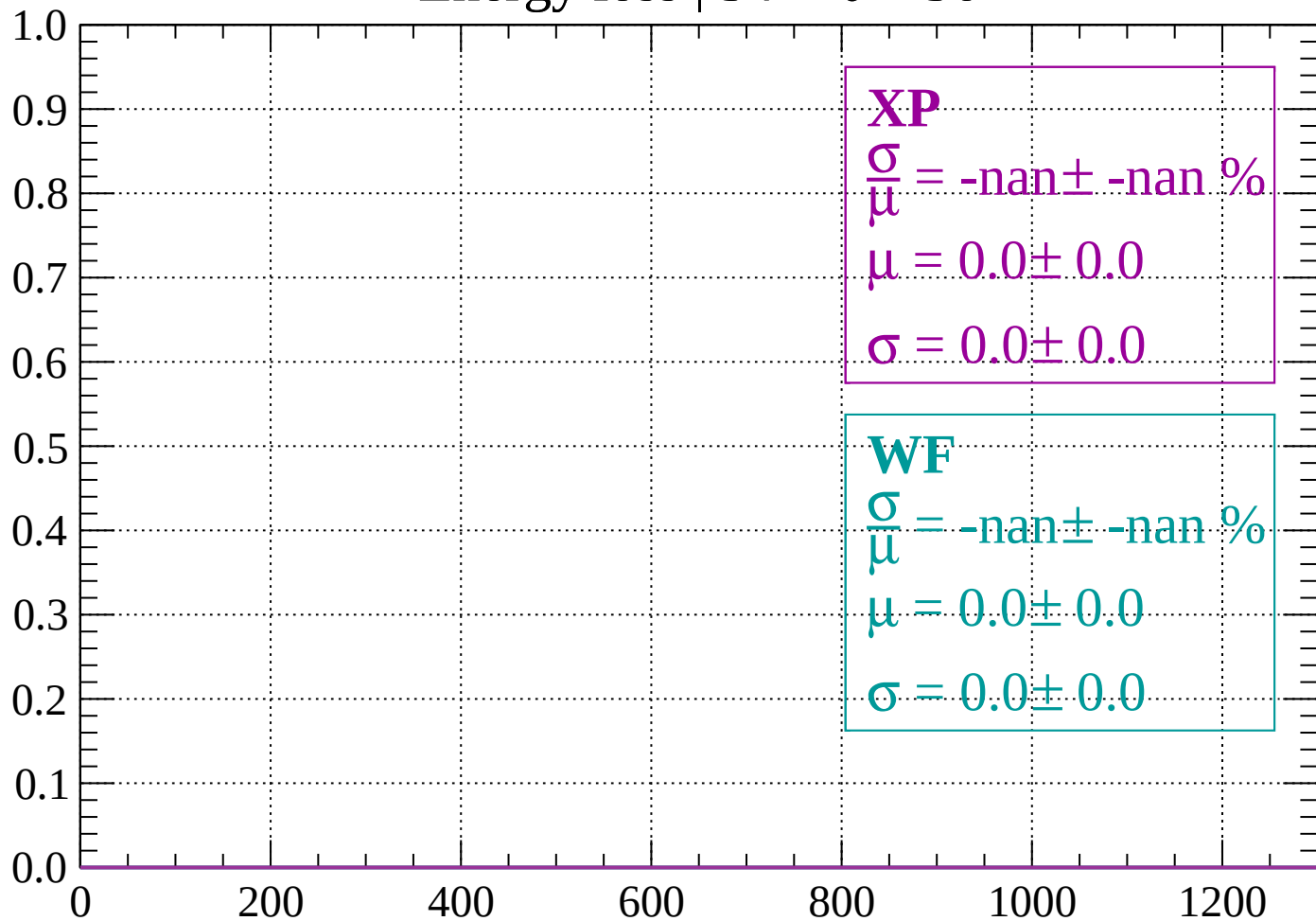
Count



dE/dx (ADC counts/cm)

Energy loss | $34 < \theta < 36$

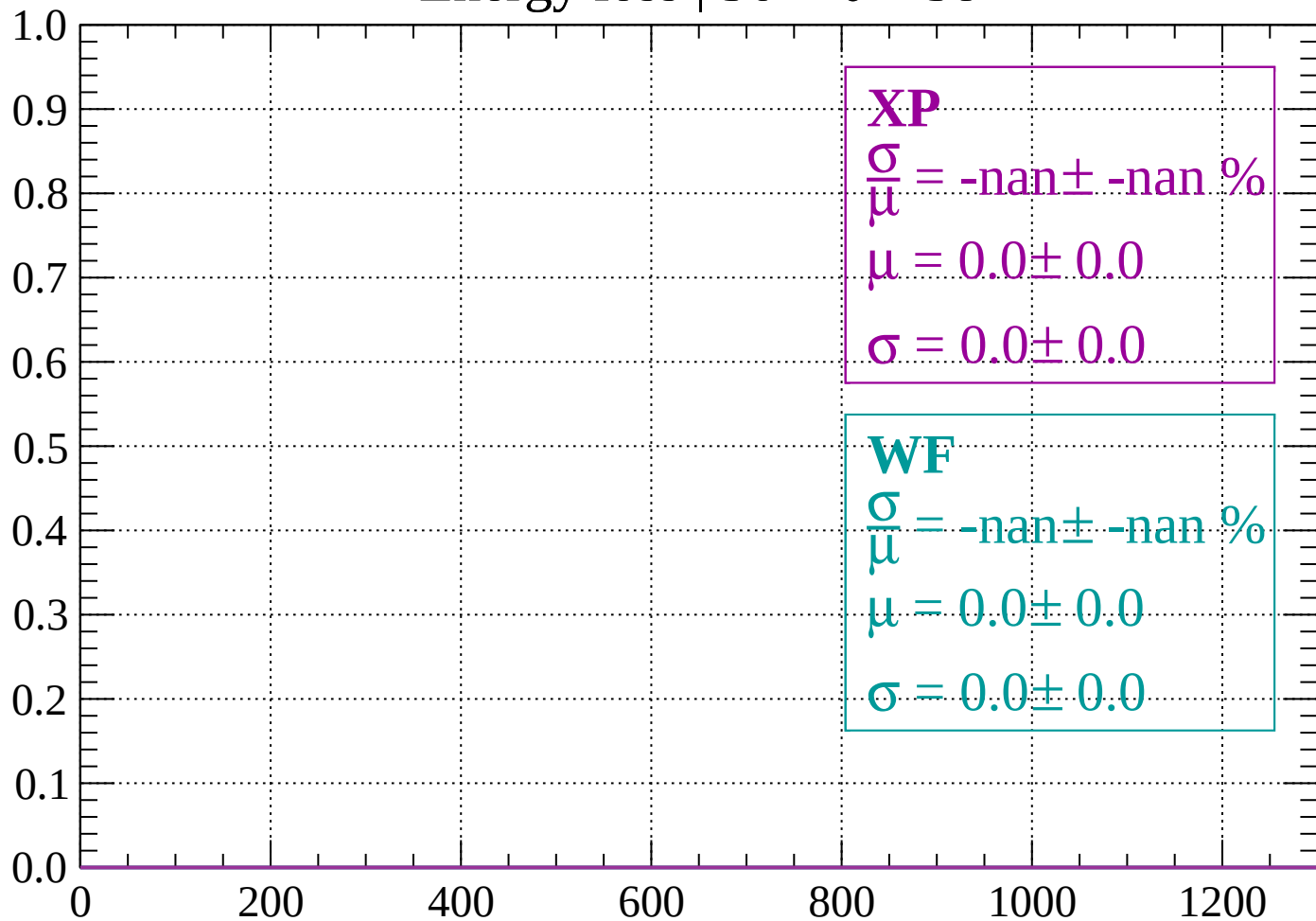
Count



dE/dx (ADC counts/cm)

Energy loss | $36 < \theta < 38$

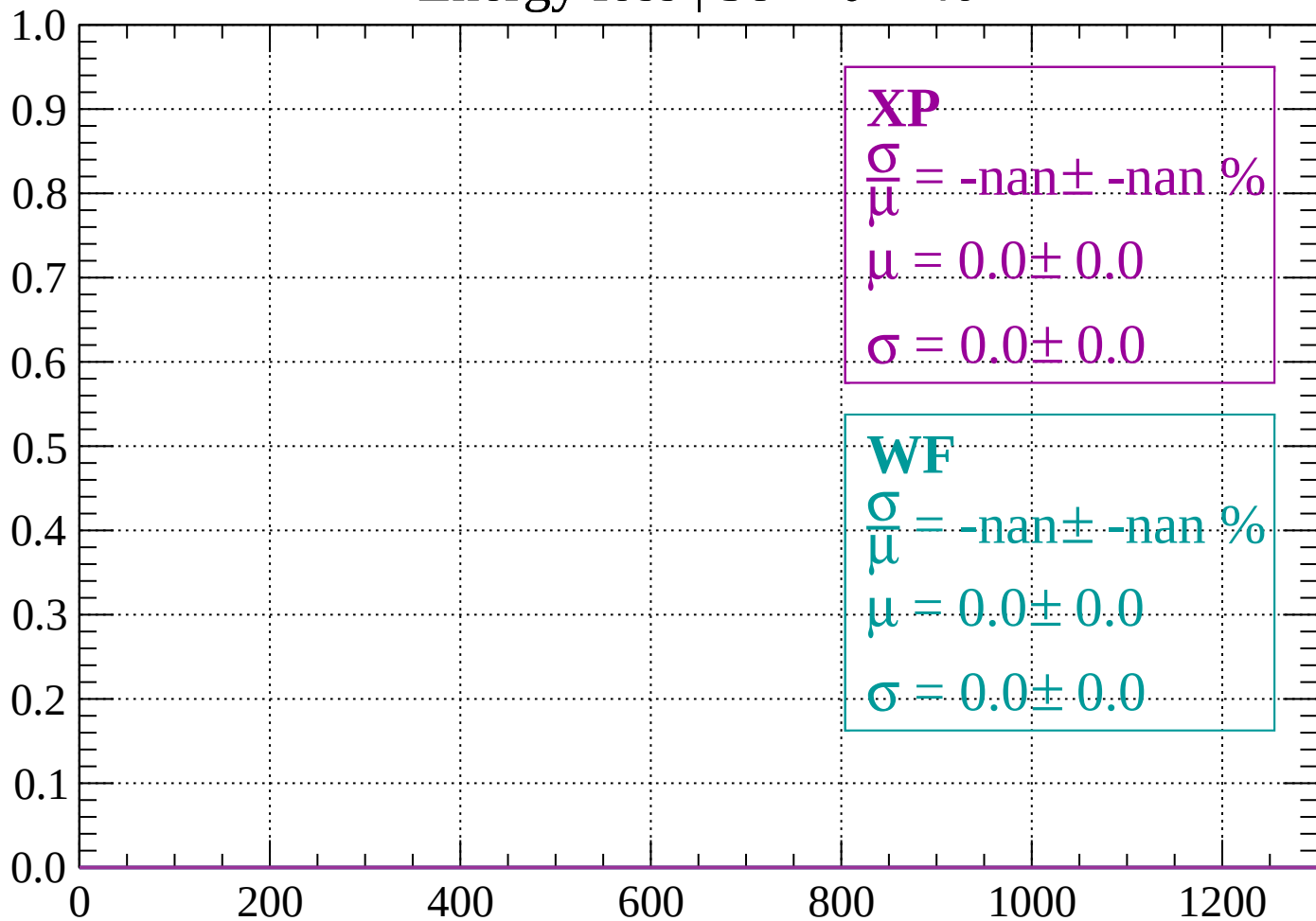
Count



dE/dx (ADC counts/cm)

Energy loss | $38 < \theta < 40$

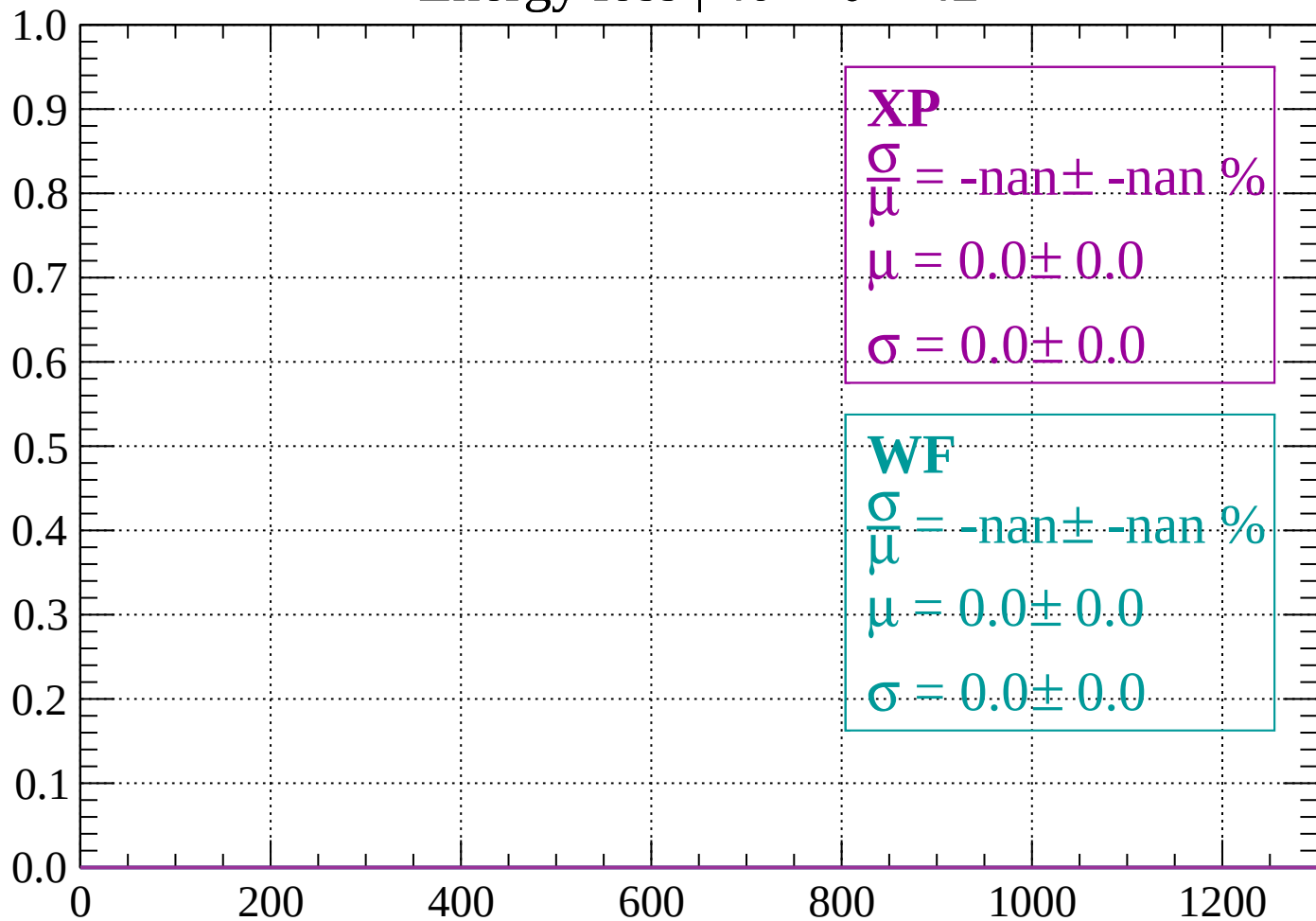
Count



dE/dx (ADC counts/cm)

Energy loss | $40 < \theta < 42$

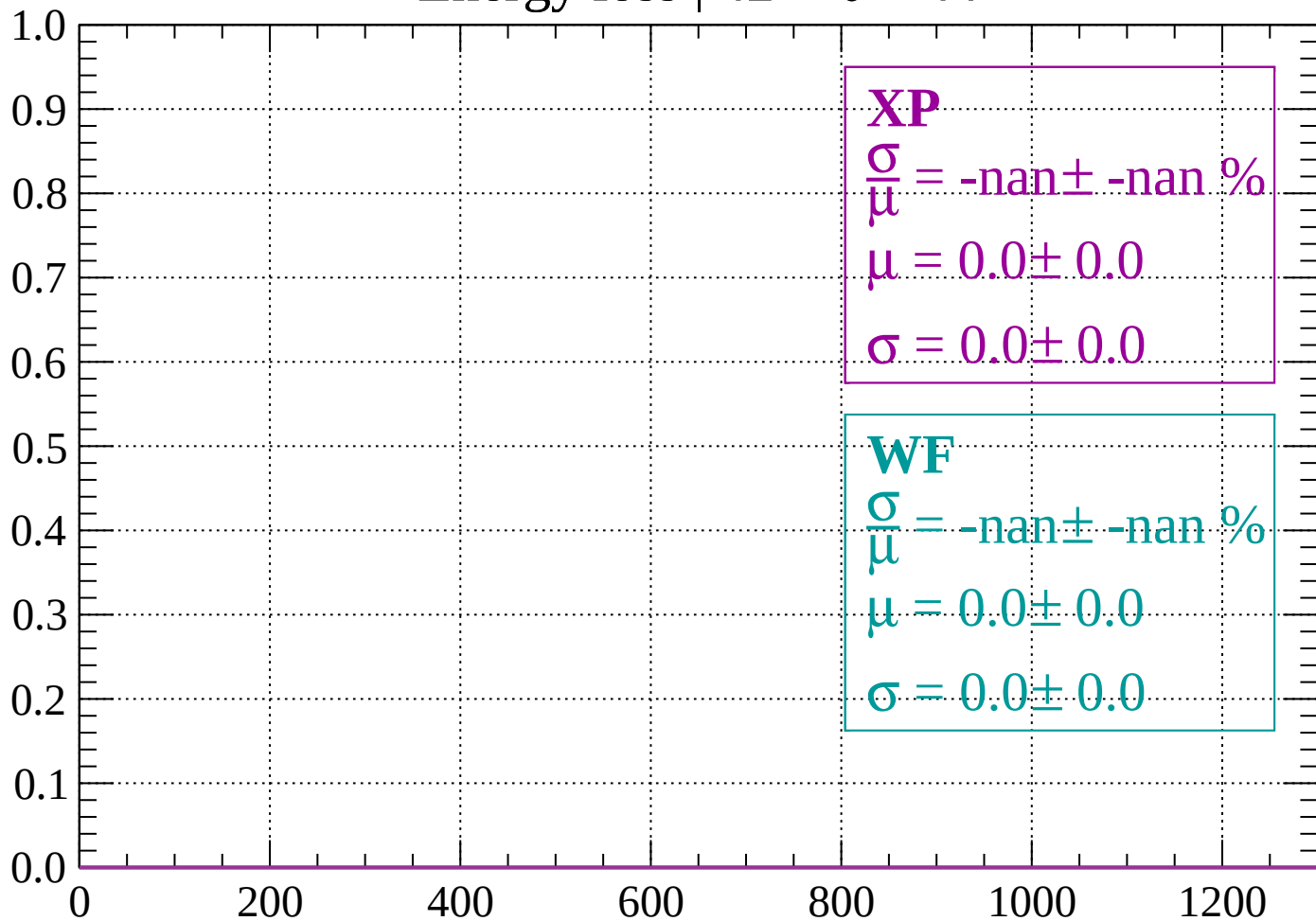
Count



dE/dx (ADC counts/cm)

Energy loss | $42 < \theta < 44$

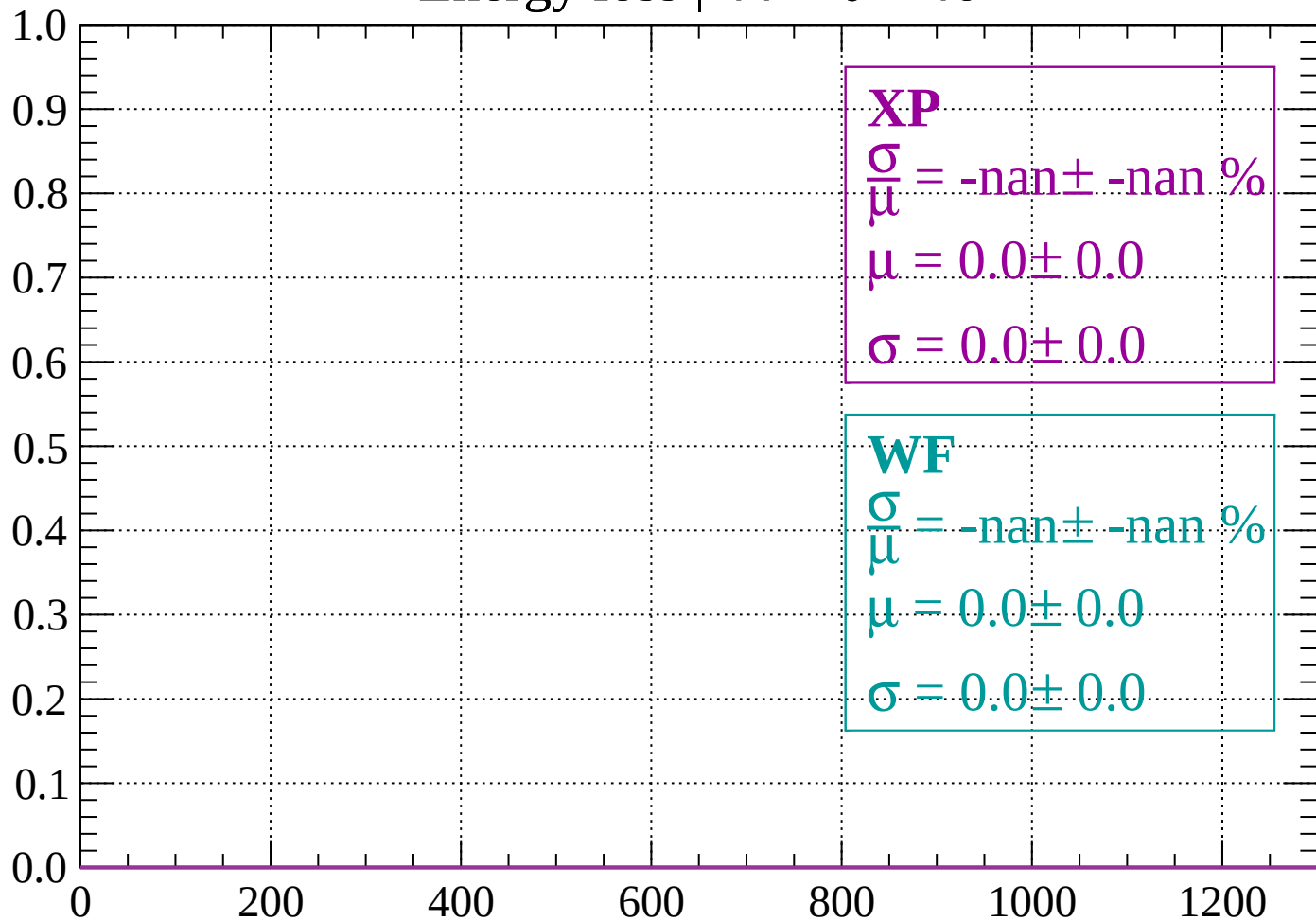
Count



dE/dx (ADC counts/cm)

Energy loss | $44 < \theta < 46$

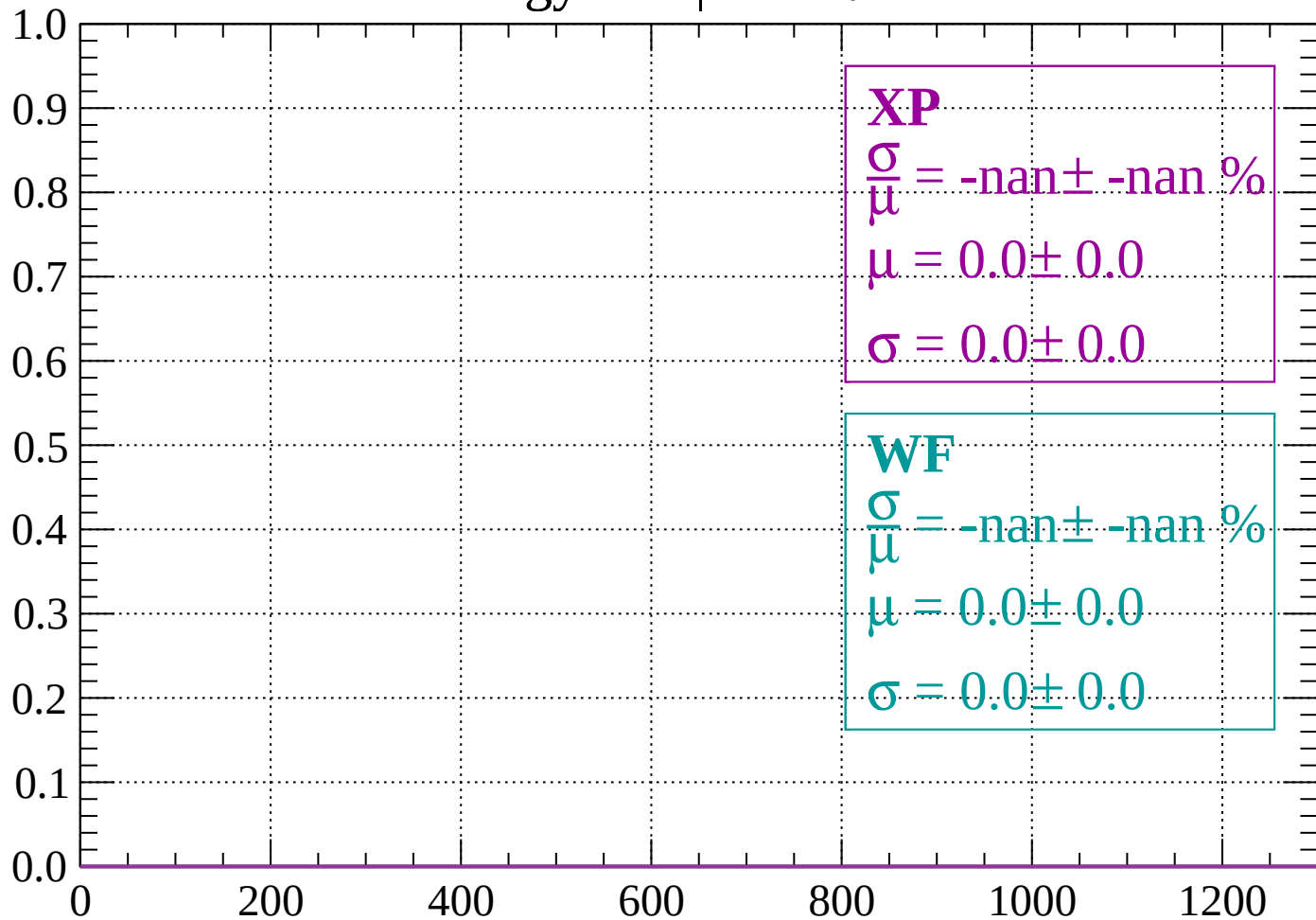
Count



dE/dx (ADC counts/cm)

Energy loss | $46 < \theta < 48$

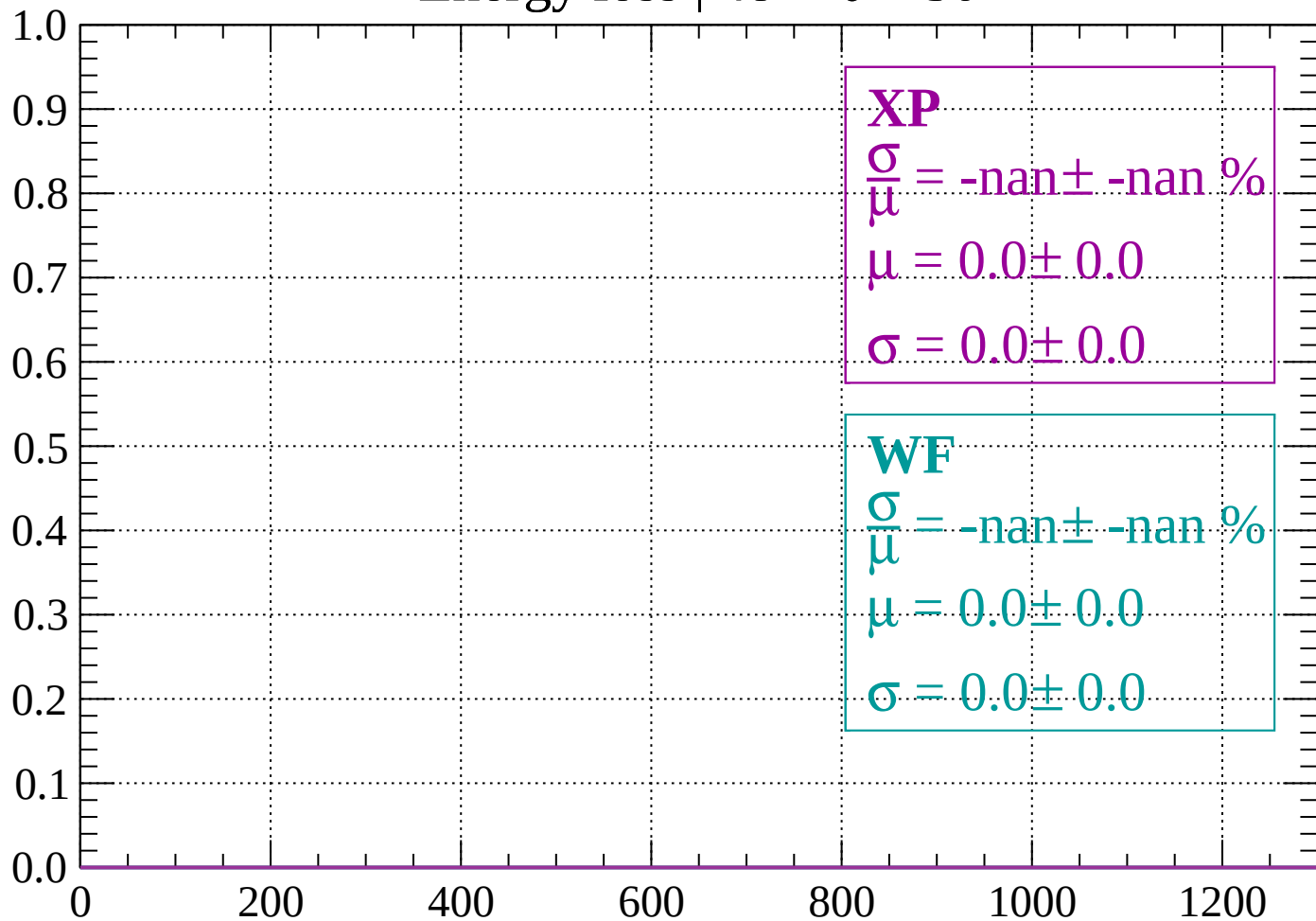
Count



dE/dx (ADC counts/cm)

Energy loss | $48 < \theta < 50$

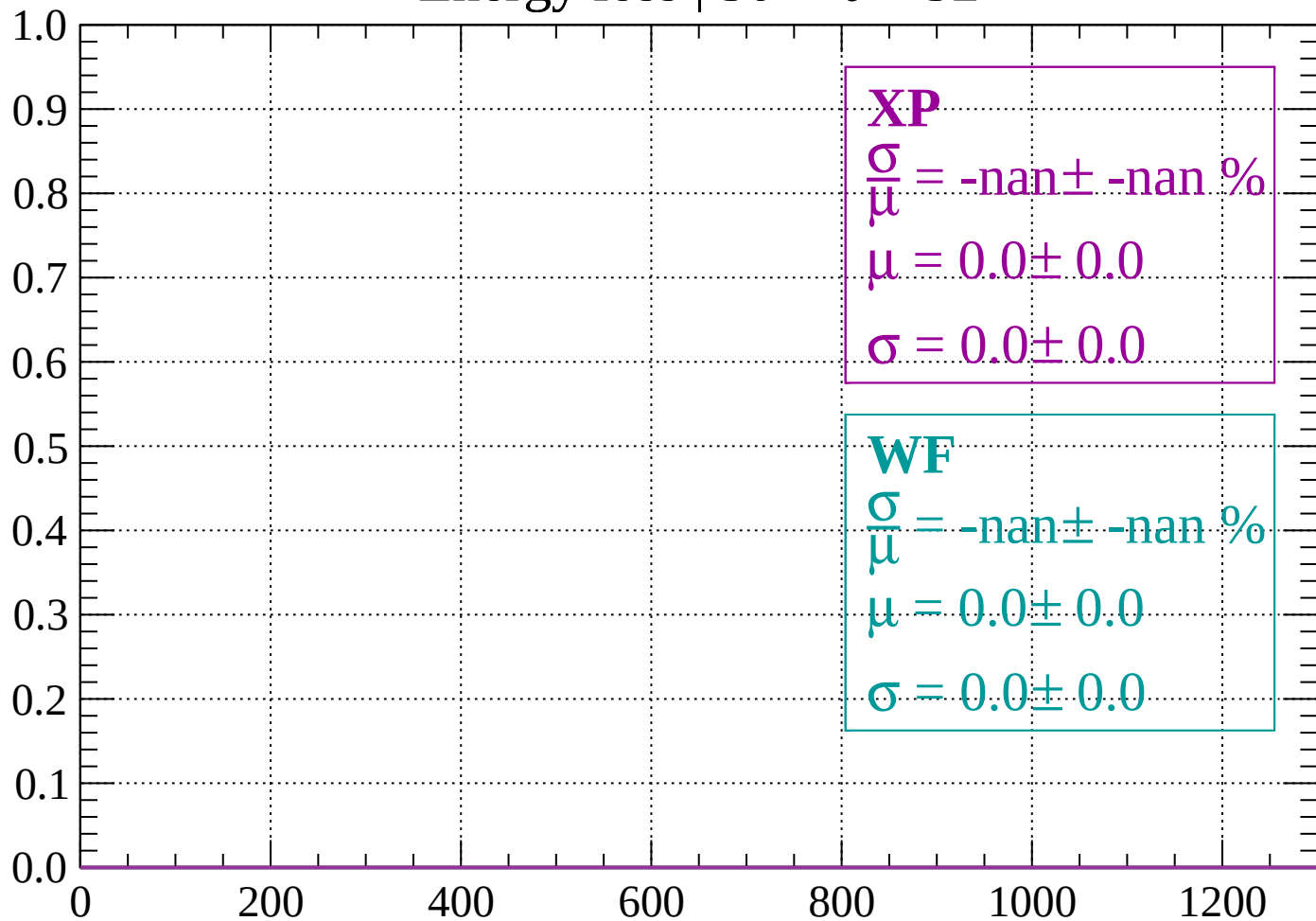
Count



dE/dx (ADC counts/cm)

Energy loss | $50 < \theta < 52$

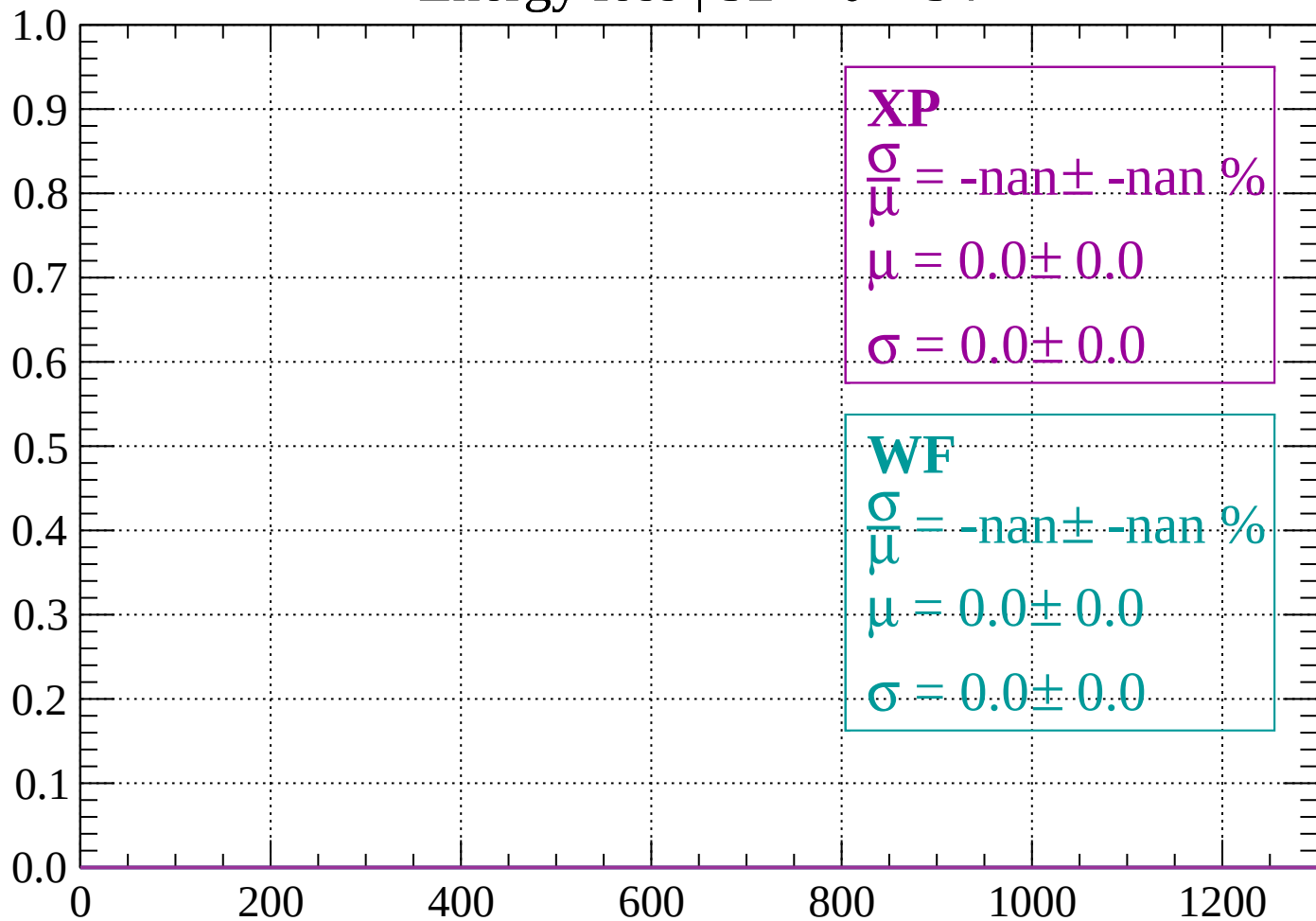
Count



dE/dx (ADC counts/cm)

Energy loss | $52 < \theta < 54$

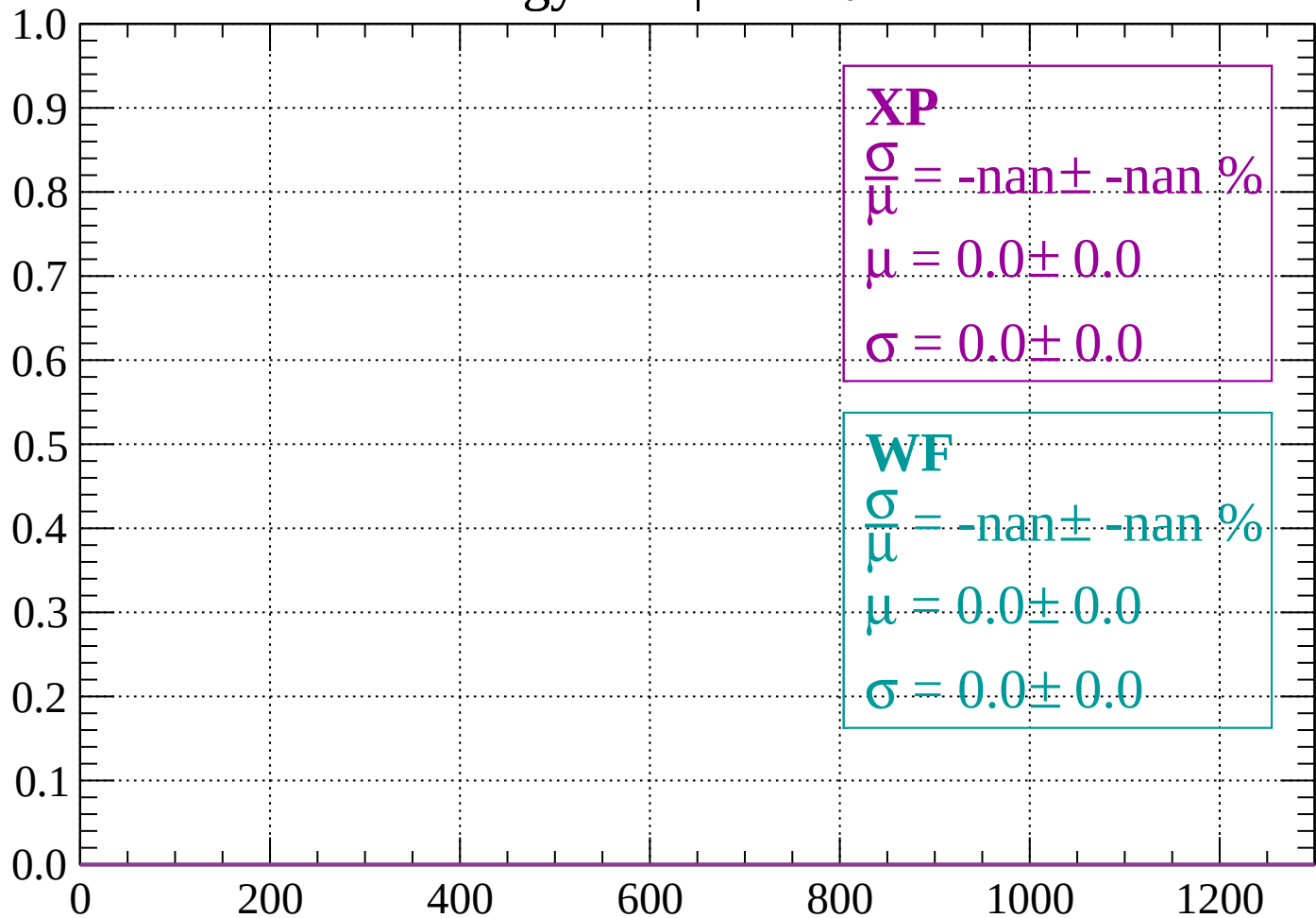
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 56$

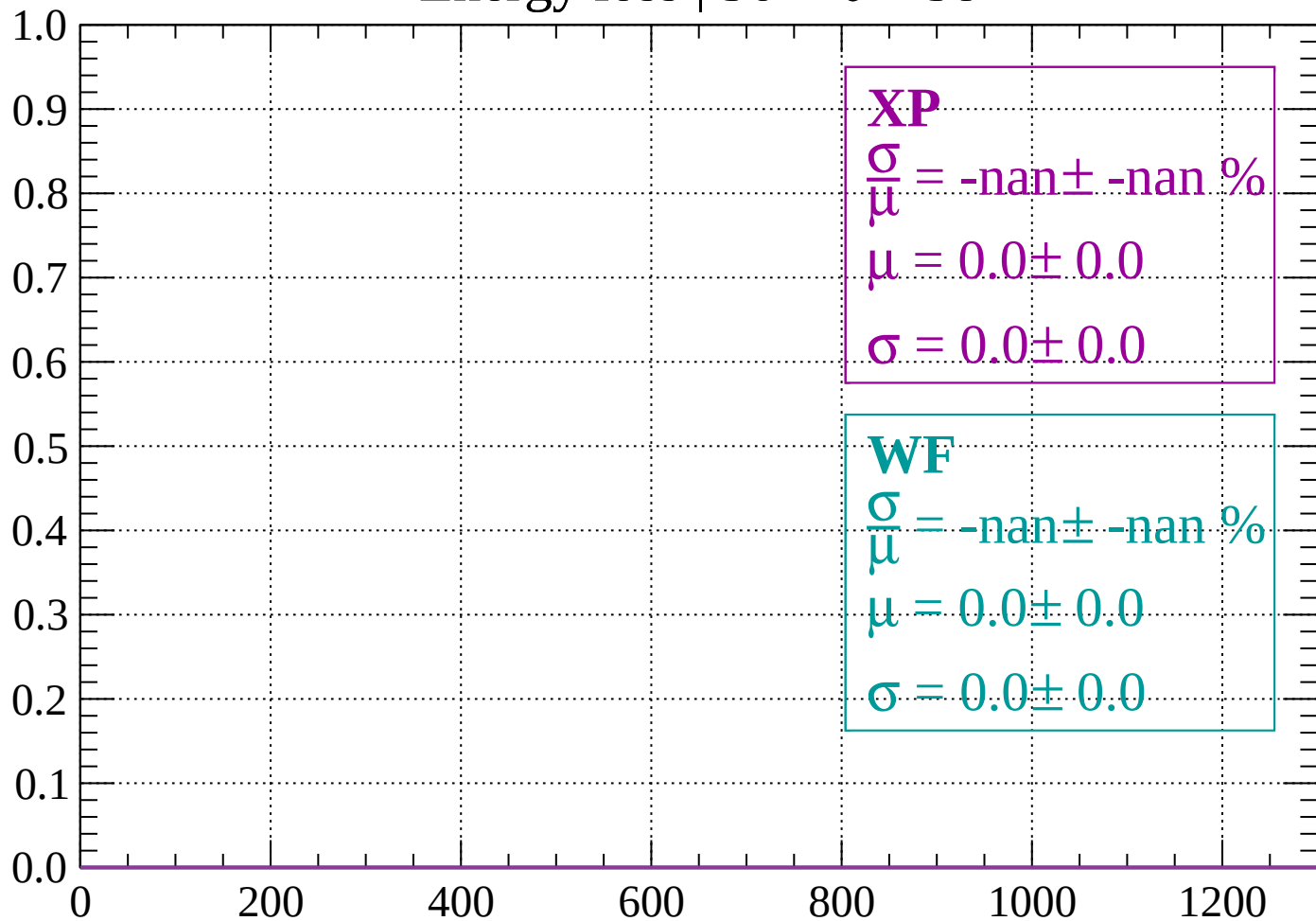
Count



dE/dx (ADC counts/cm)

Energy loss | $56 < \theta < 58$

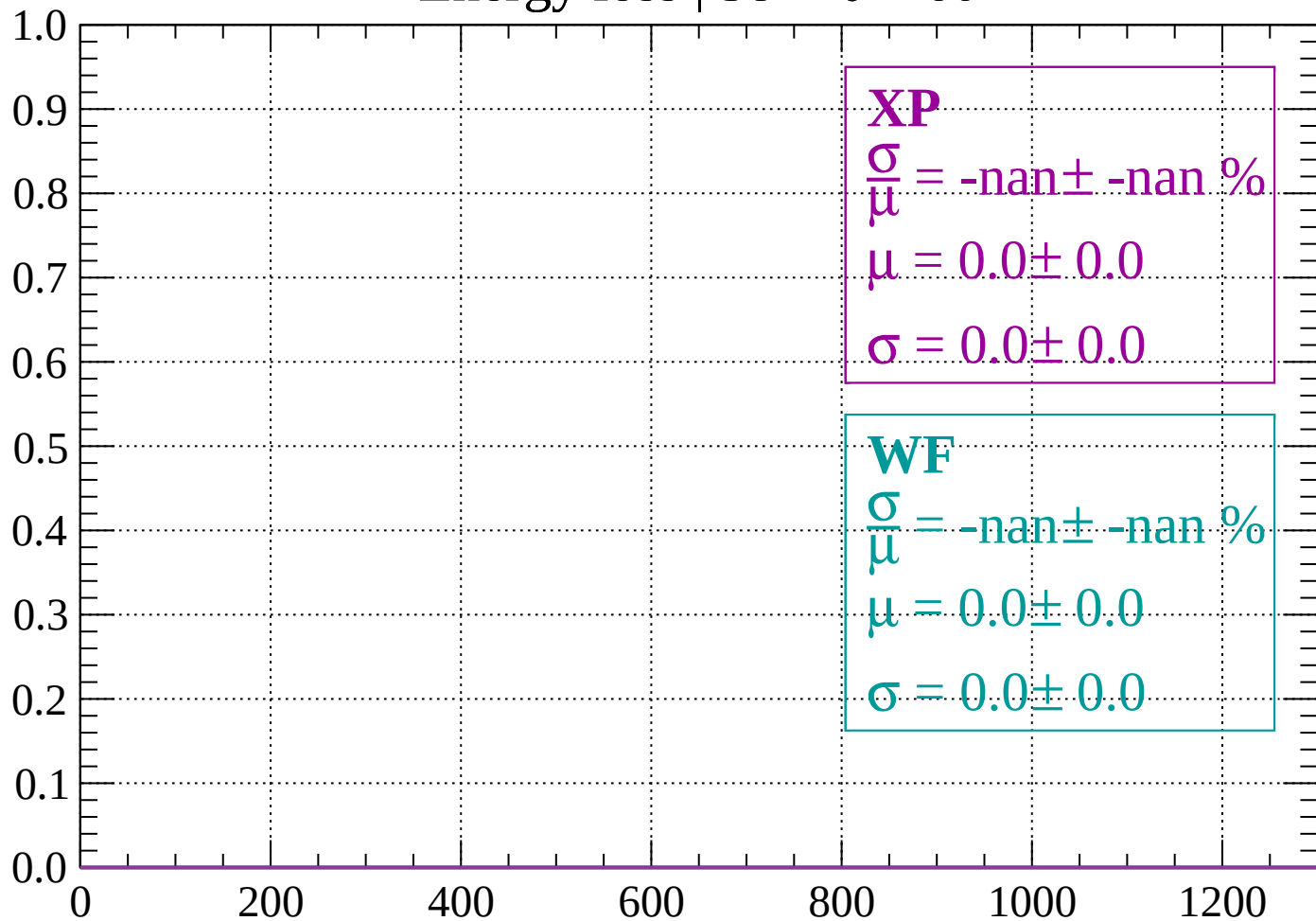
Count



dE/dx (ADC counts/cm)

Energy loss | $58 < \theta < 60$

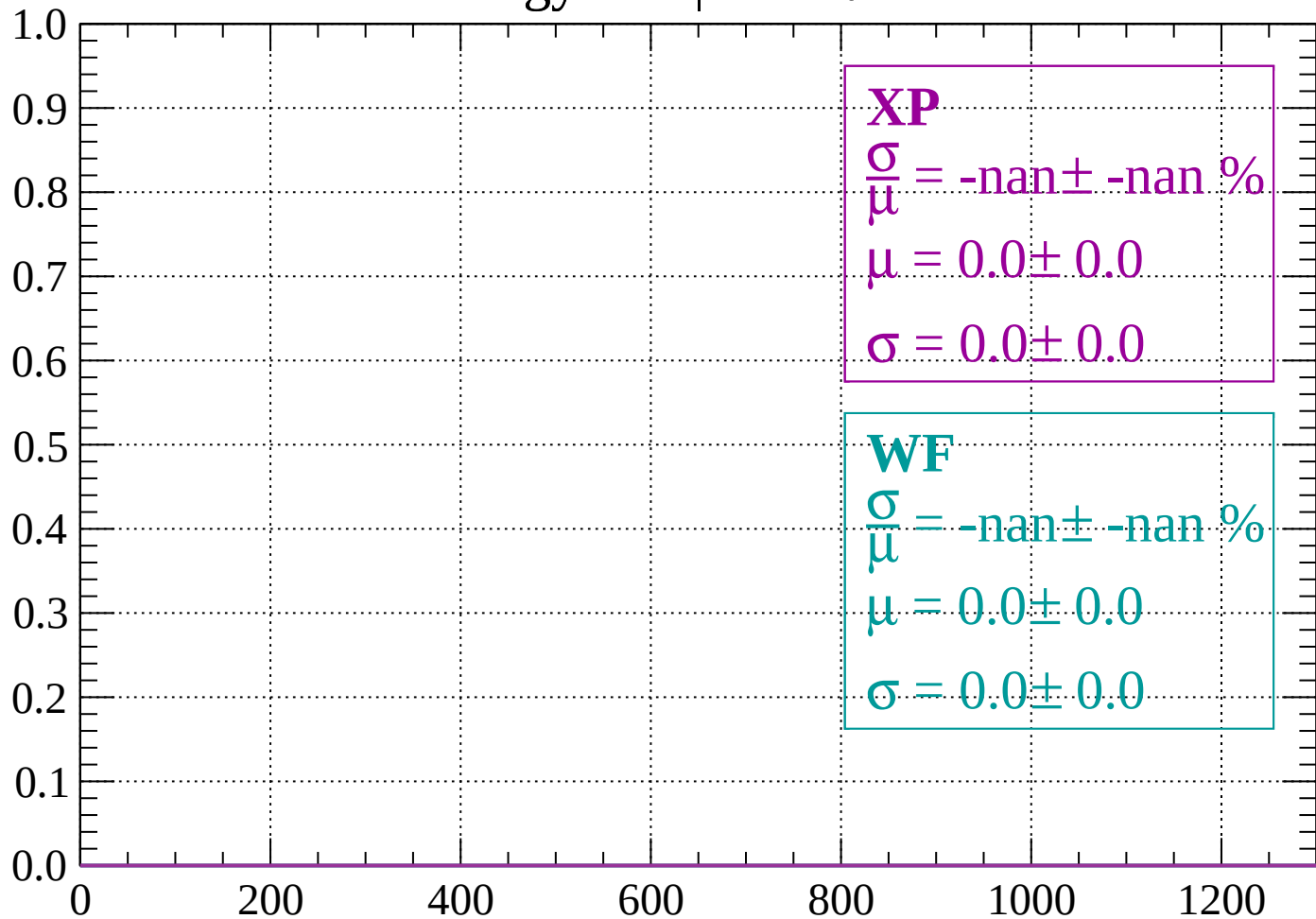
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 62$

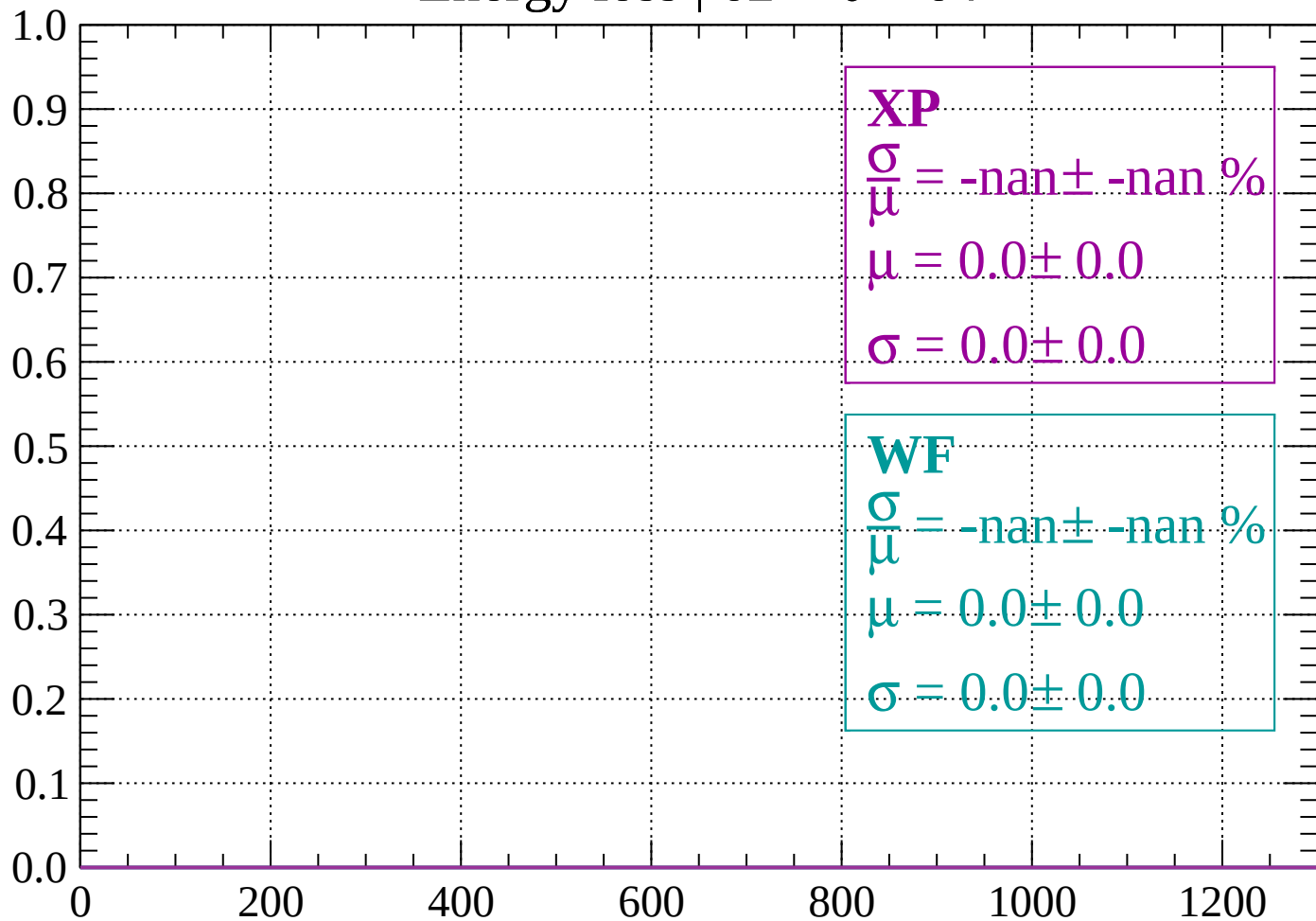
Count



dE/dx (ADC counts/cm)

Energy loss | $62 < \theta < 64$

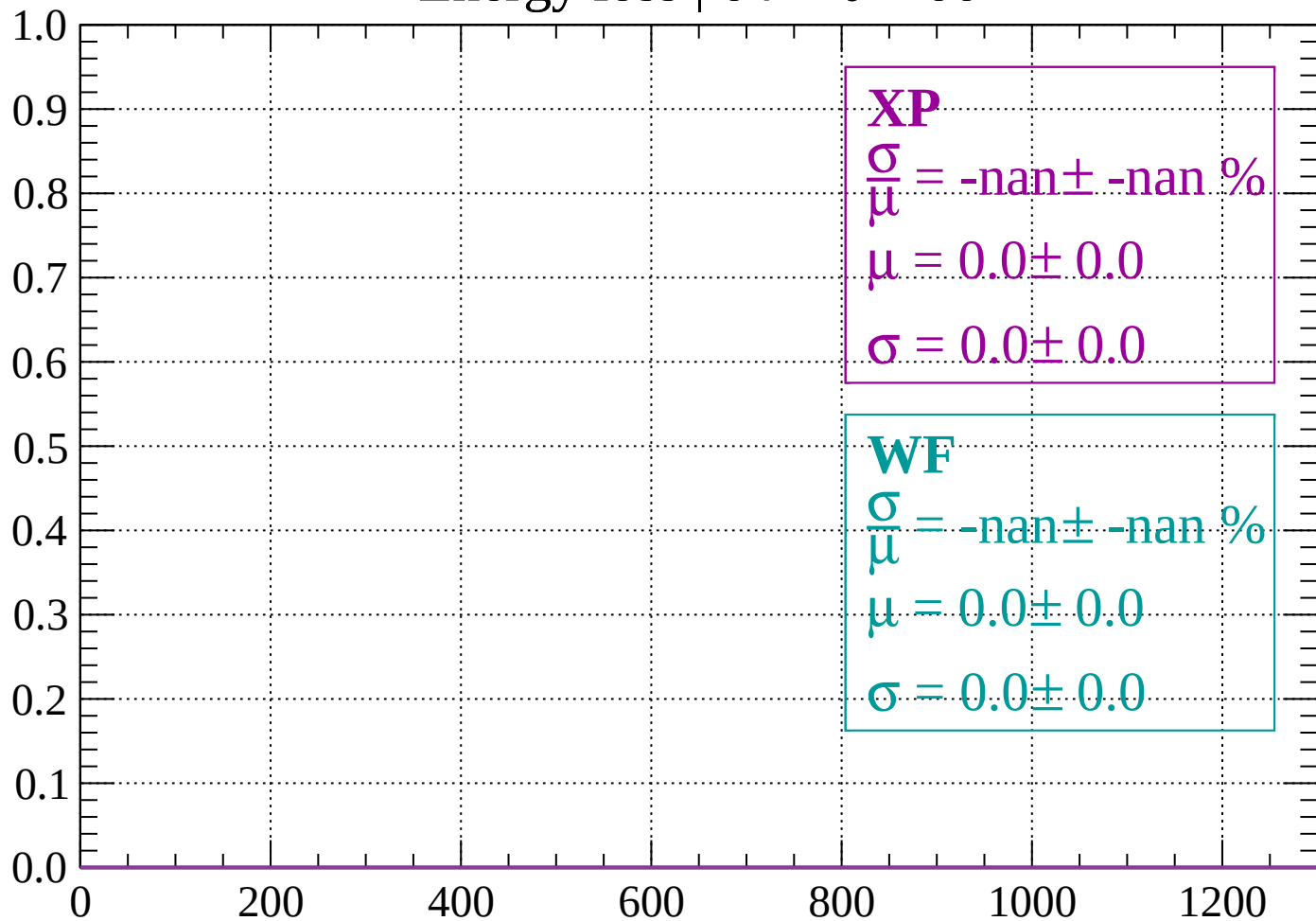
Count



dE/dx (ADC counts/cm)

Energy loss | $64 < \theta < 66$

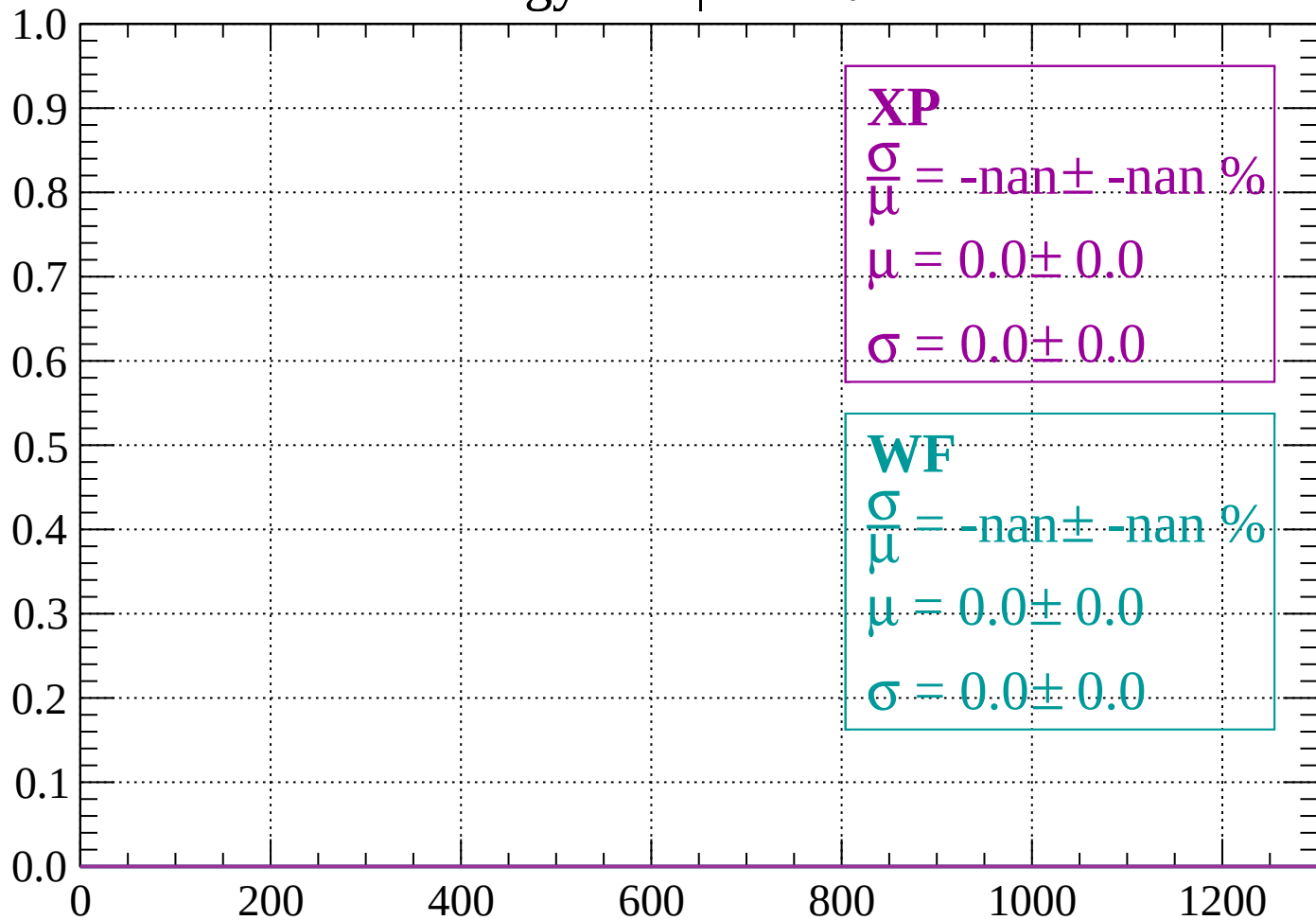
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 68$

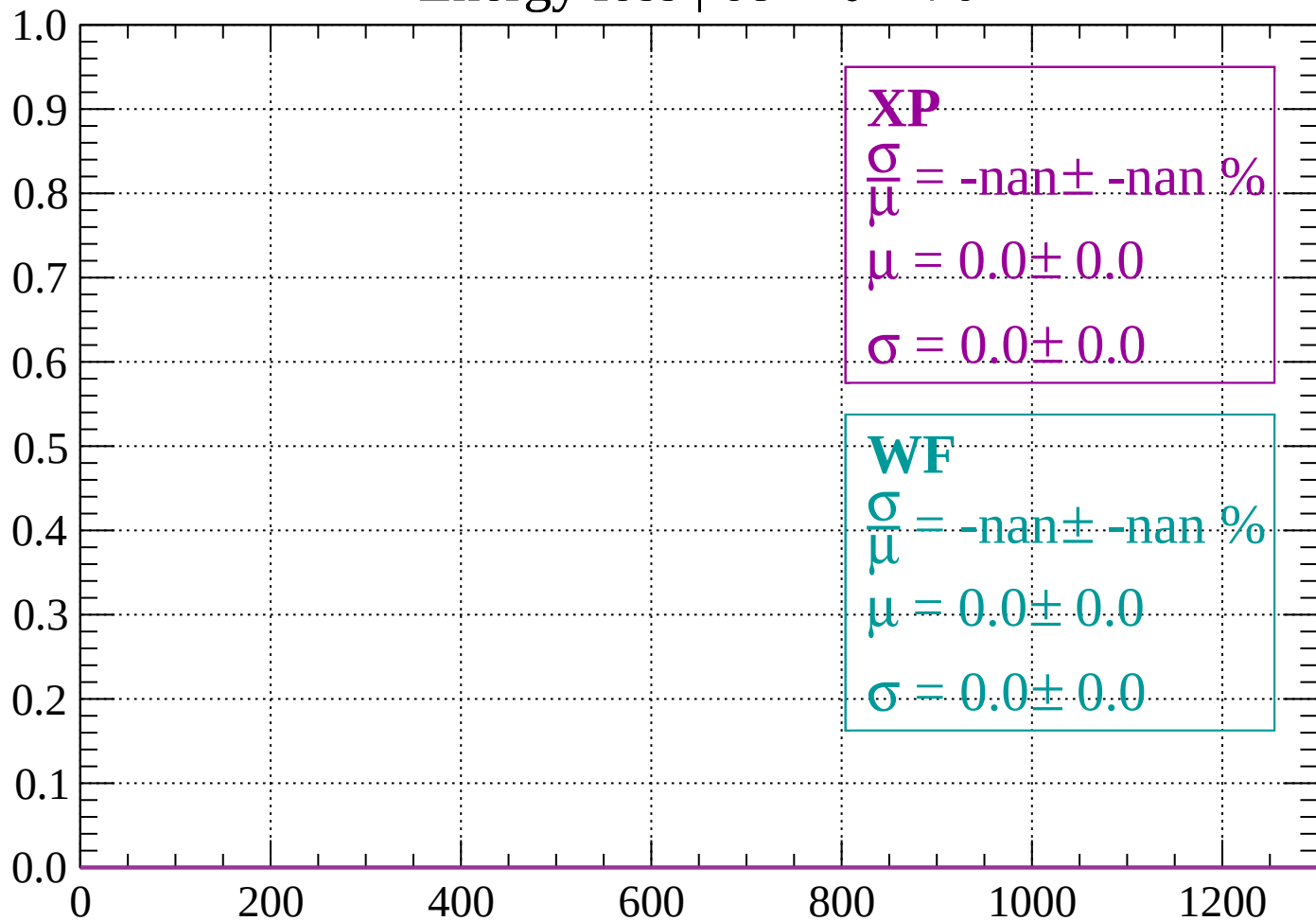
Count



dE/dx (ADC counts/cm)

Energy loss | $68 < \theta < 70$

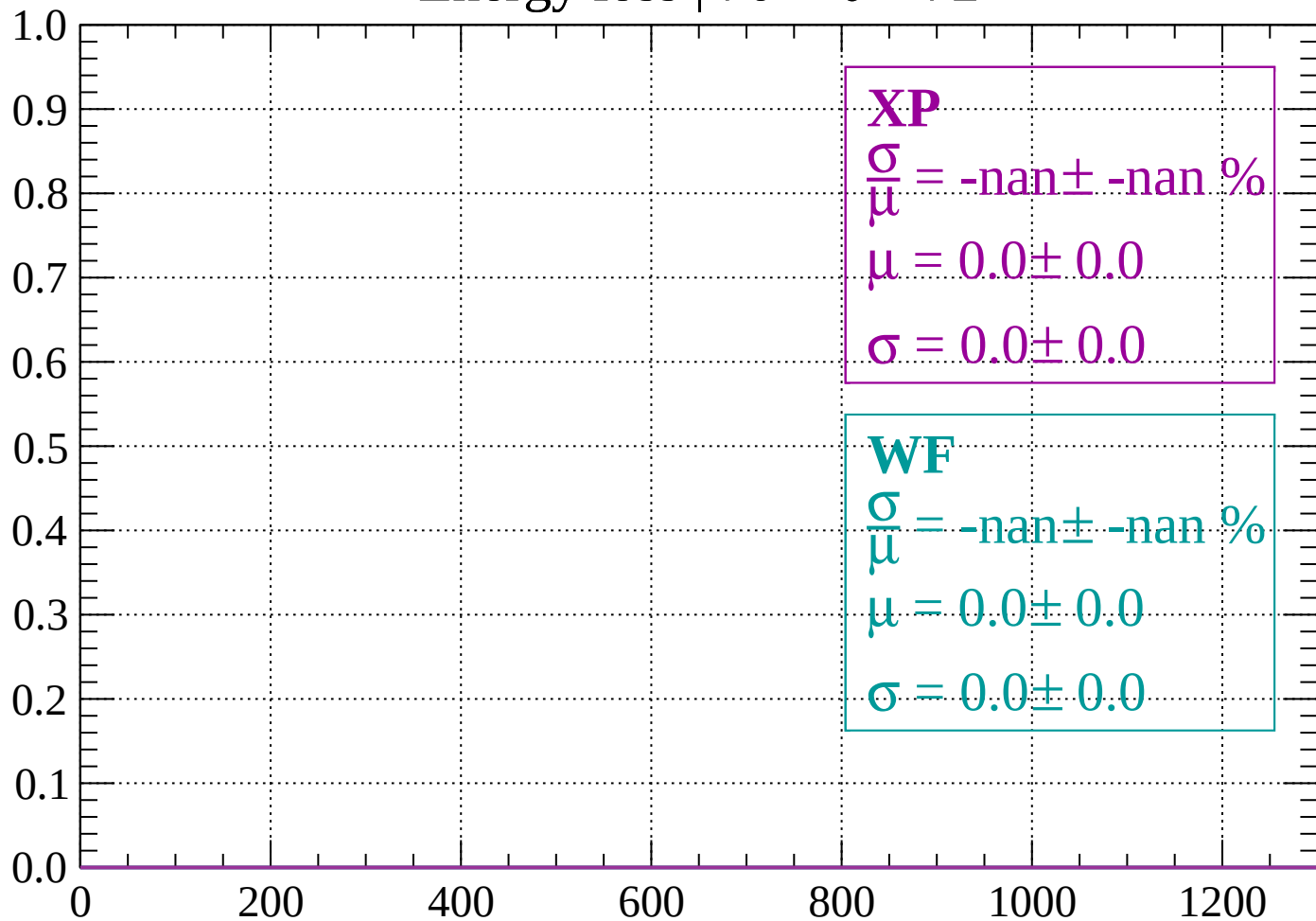
Count



dE/dx (ADC counts/cm)

Energy loss | $70 < \theta < 72$

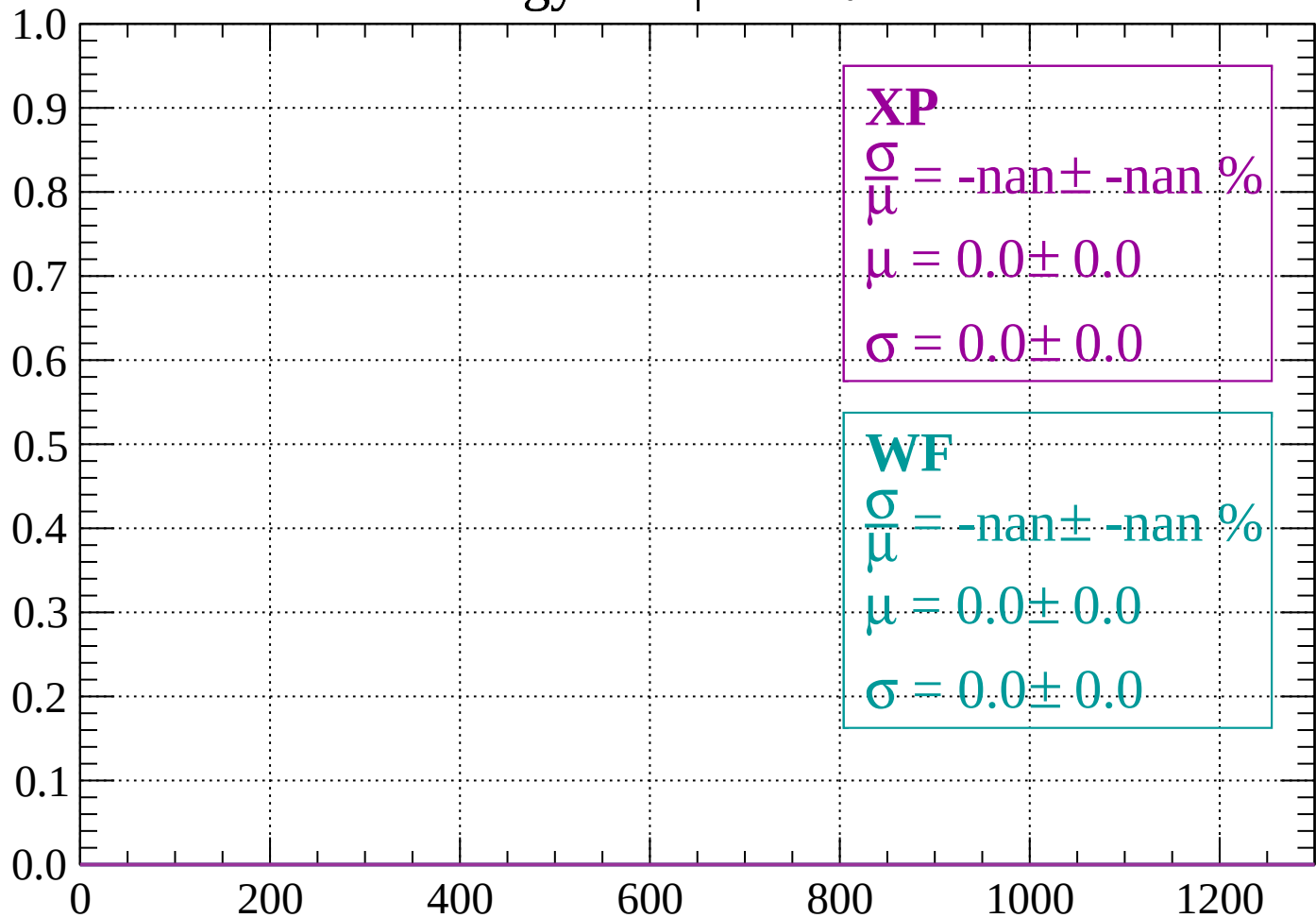
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 74$

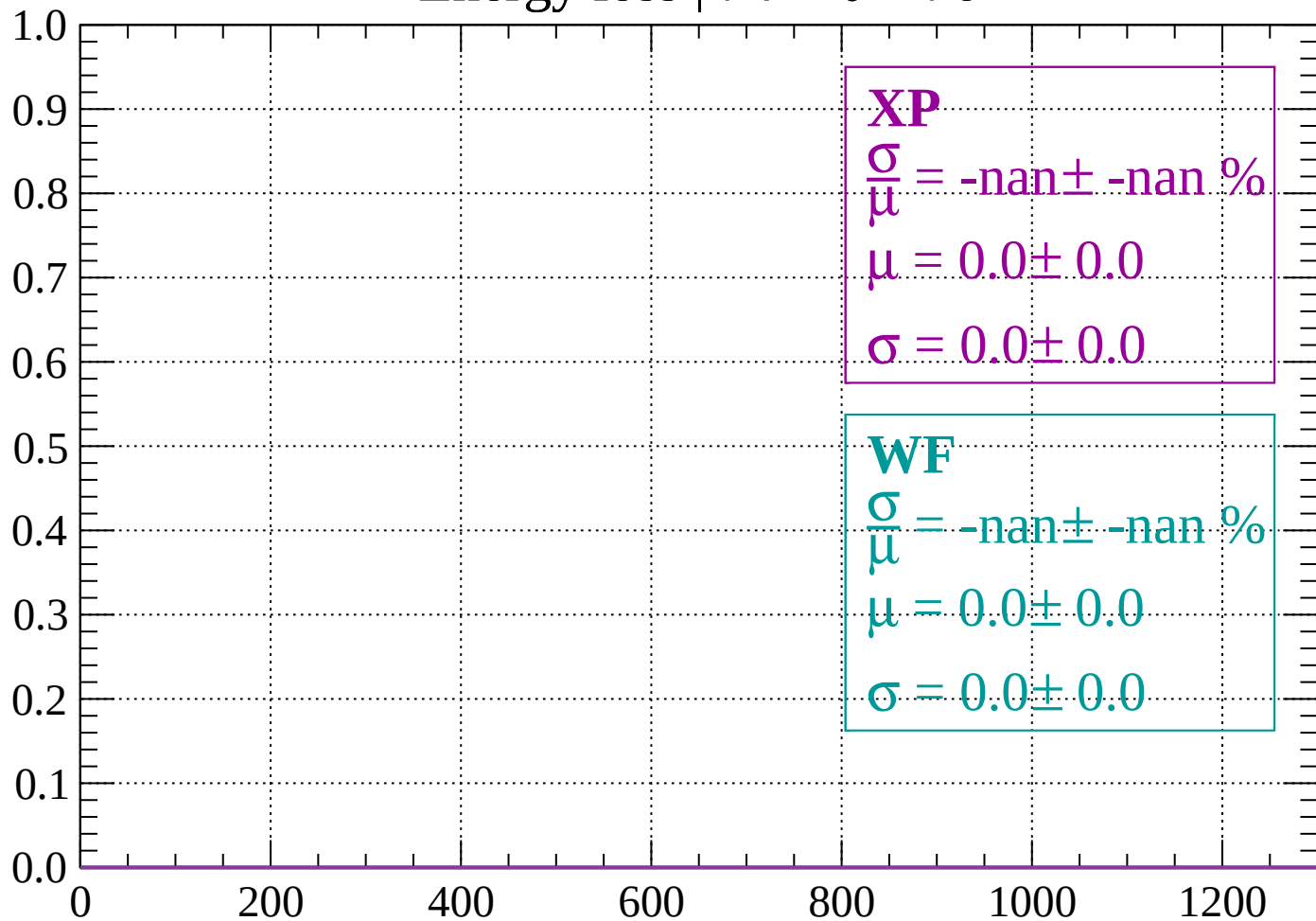
Count



dE/dx (ADC counts/cm)

Energy loss | $74 < \theta < 76$

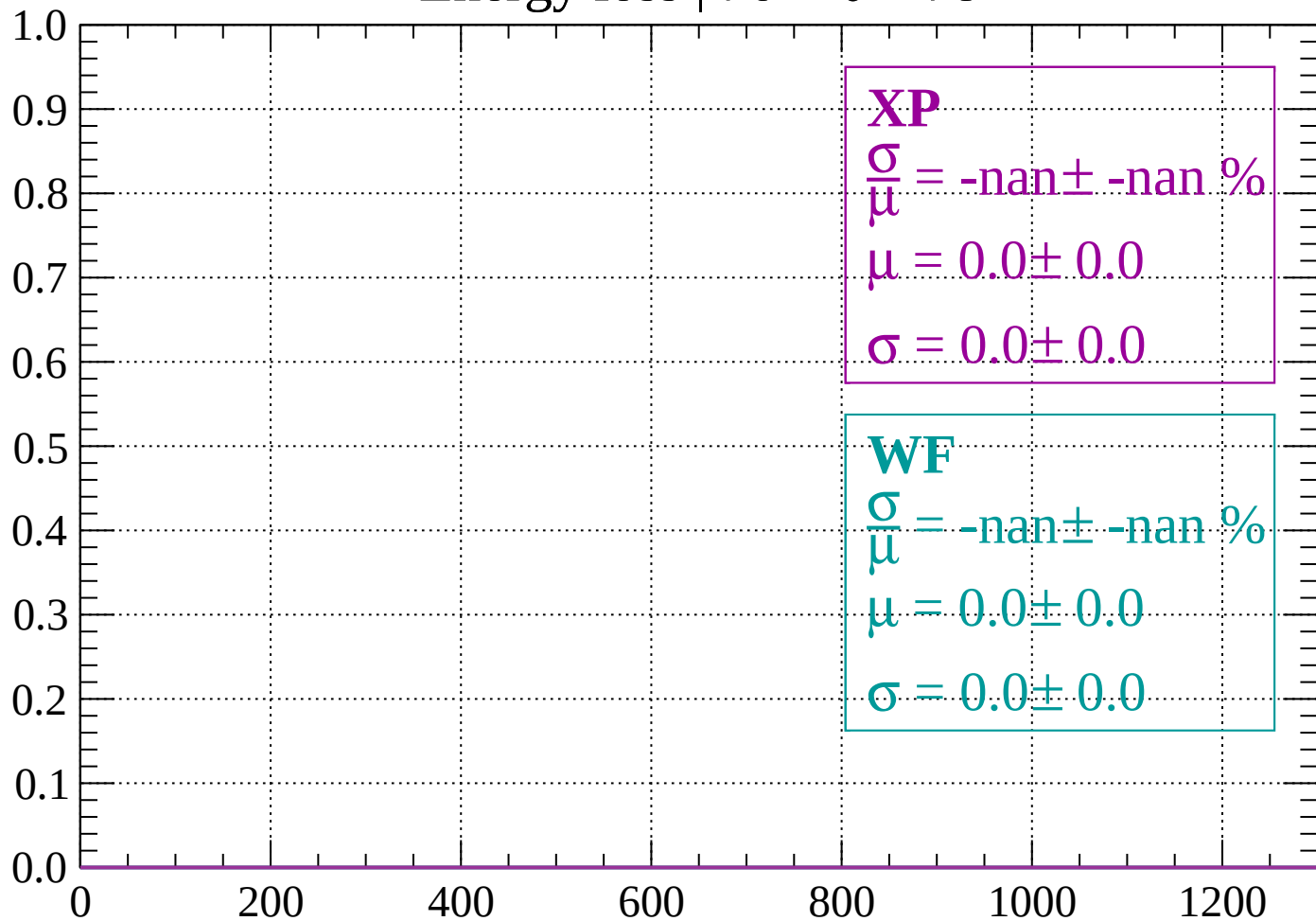
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

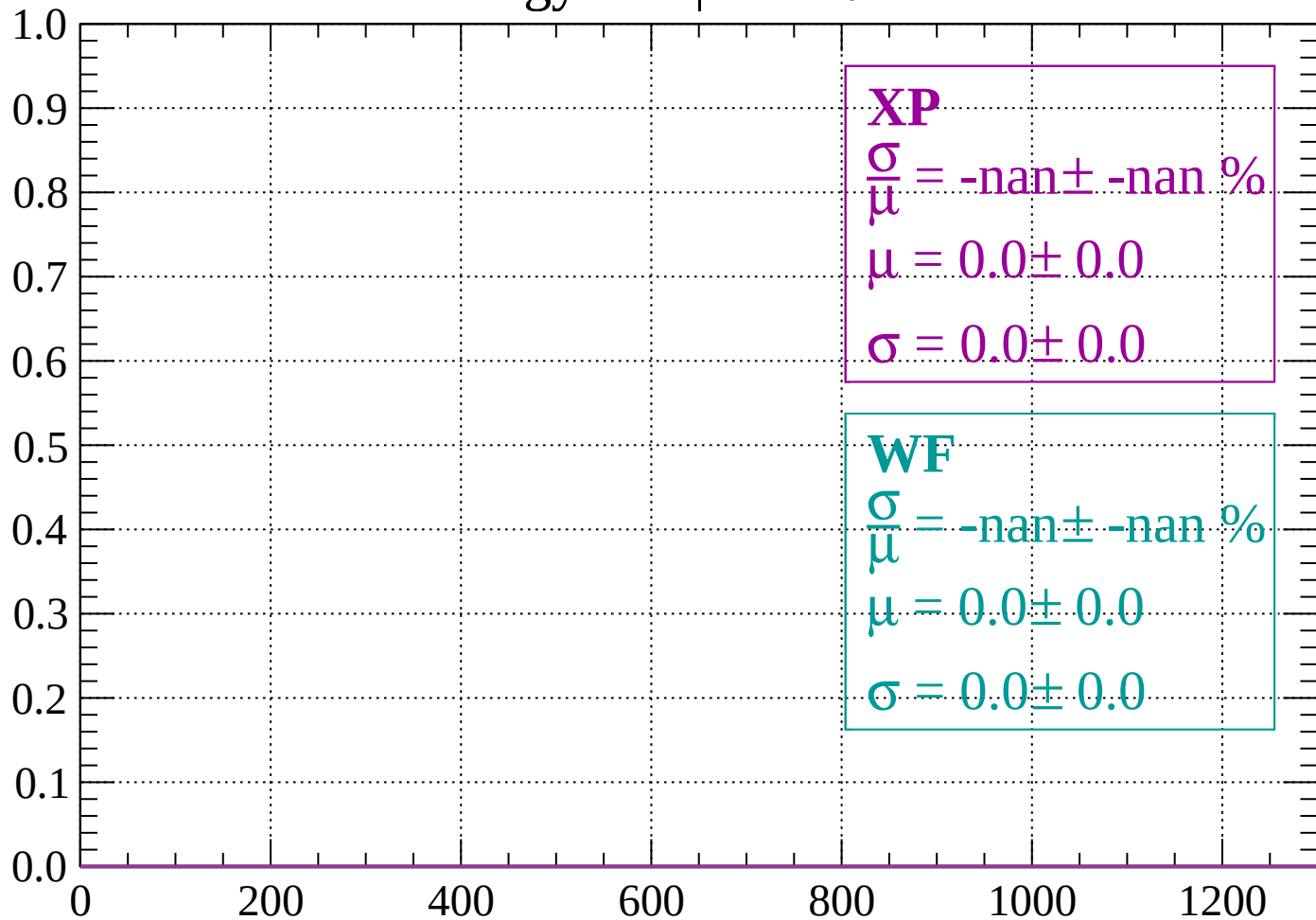
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

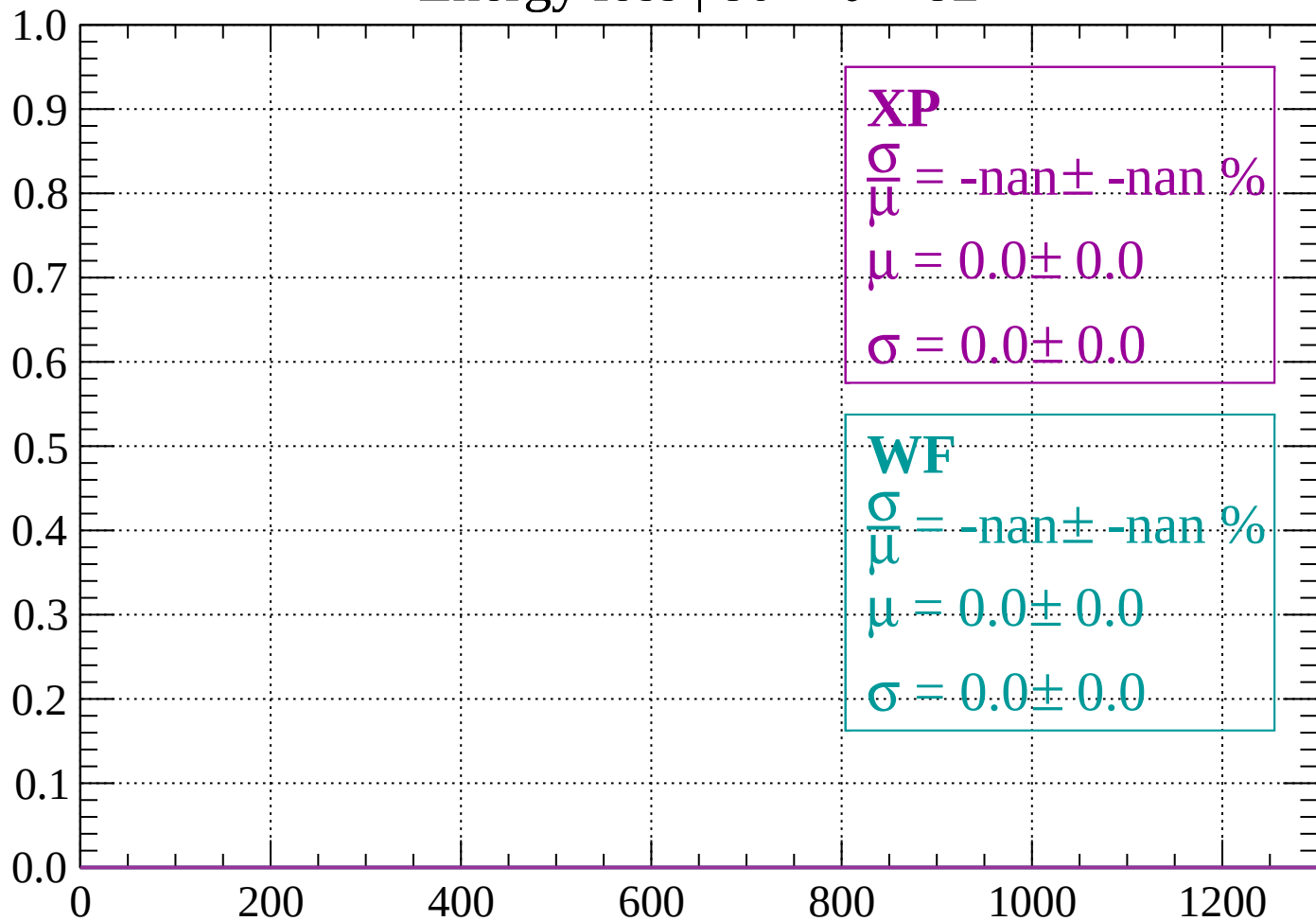
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

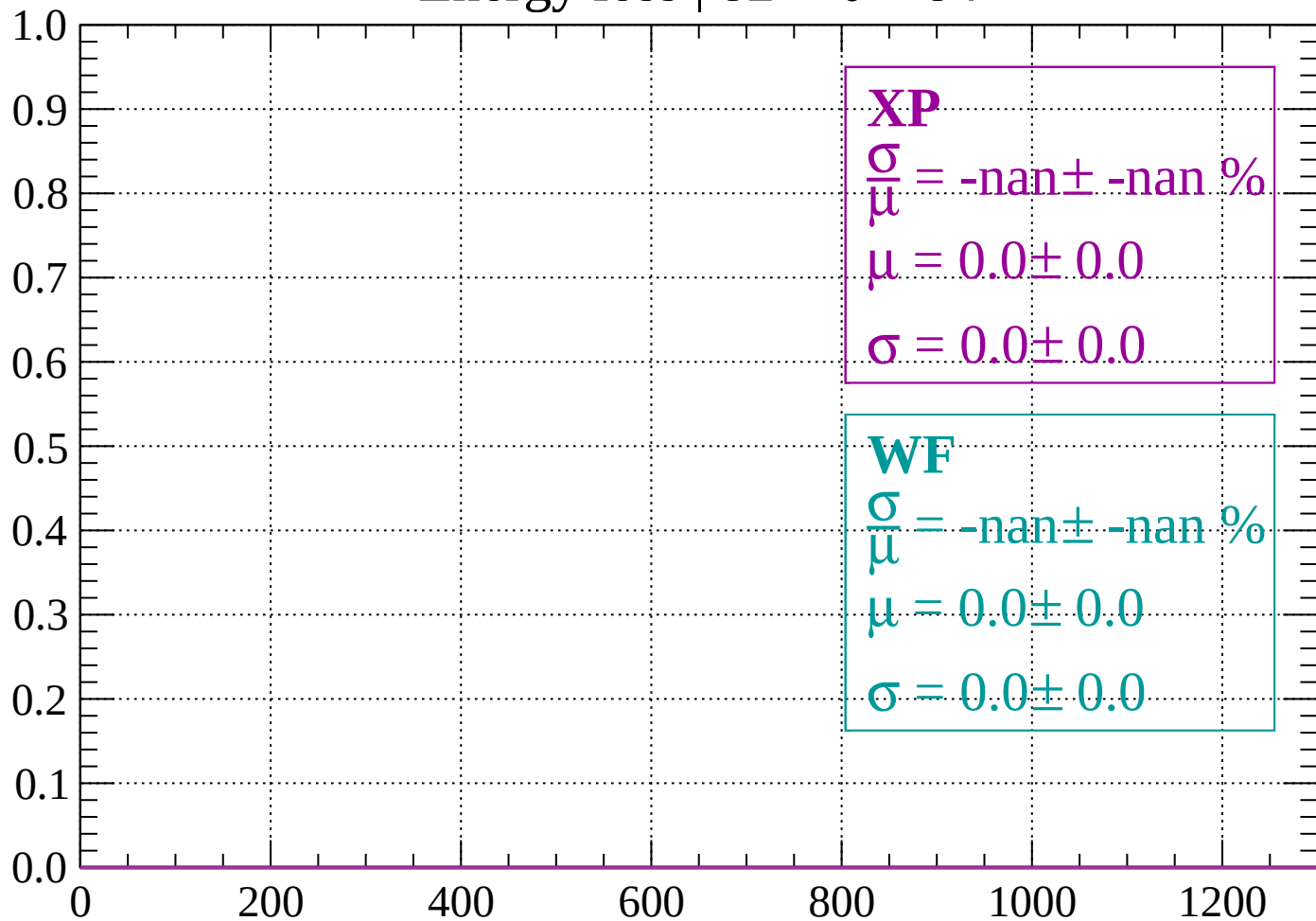
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

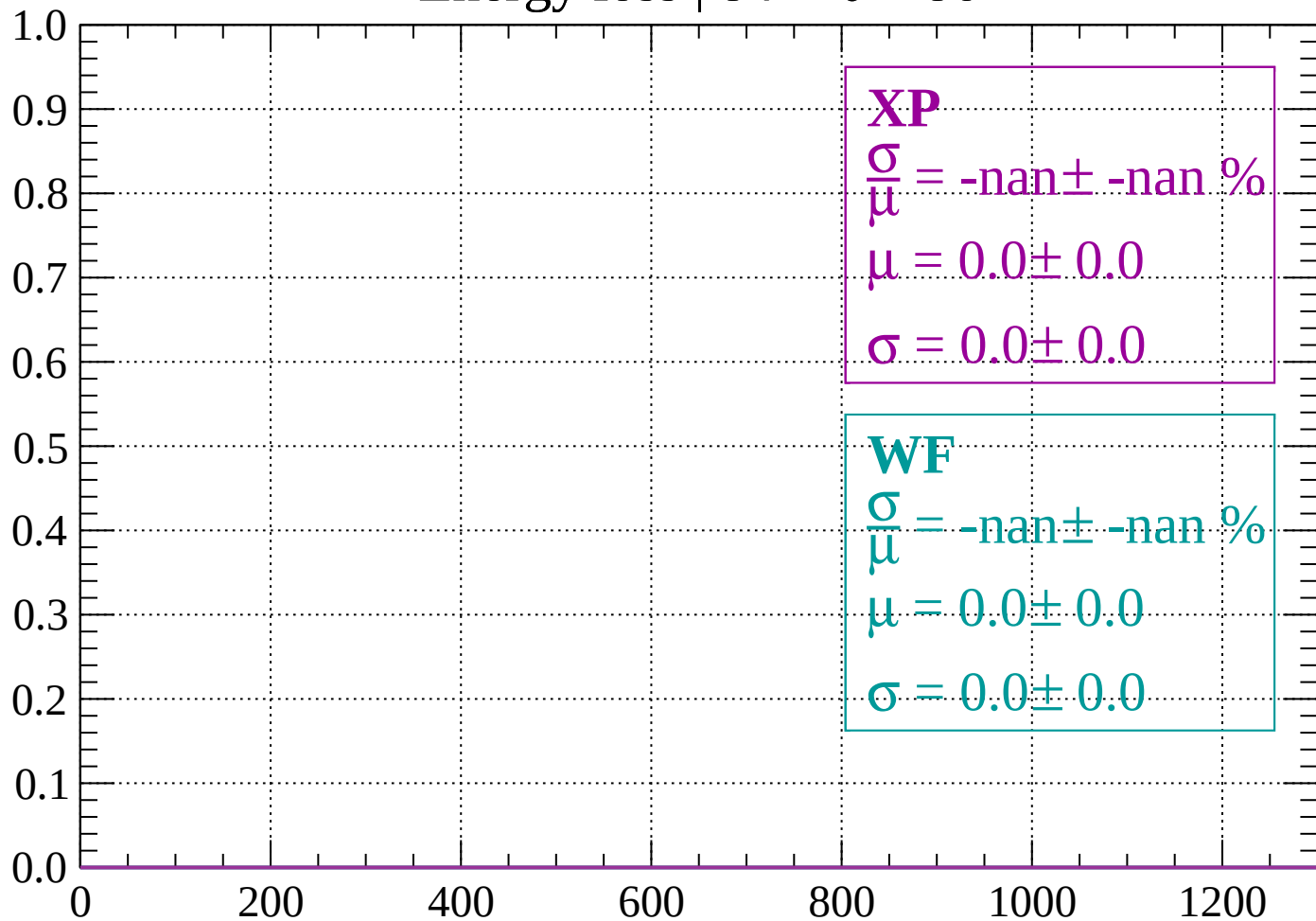
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

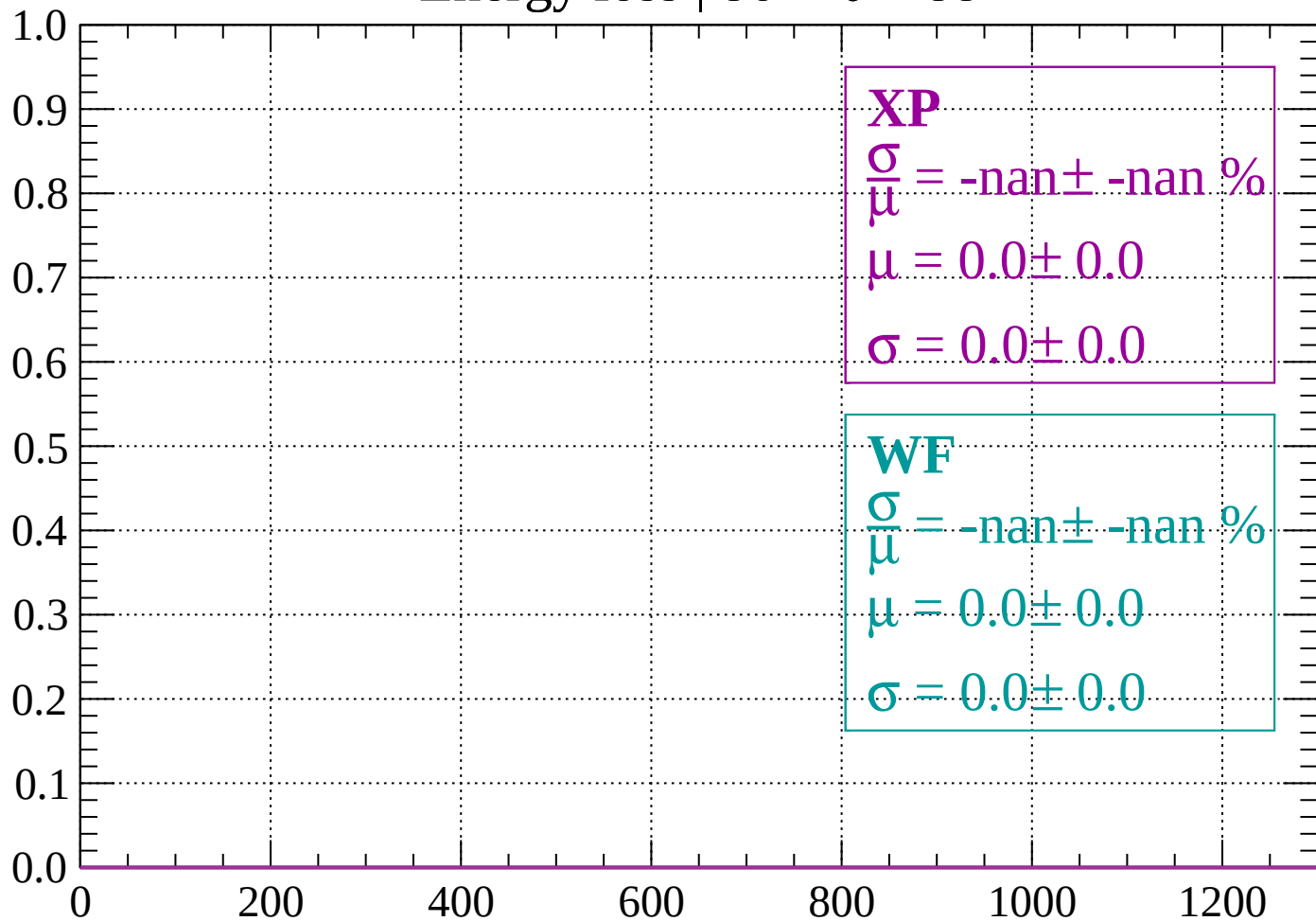
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

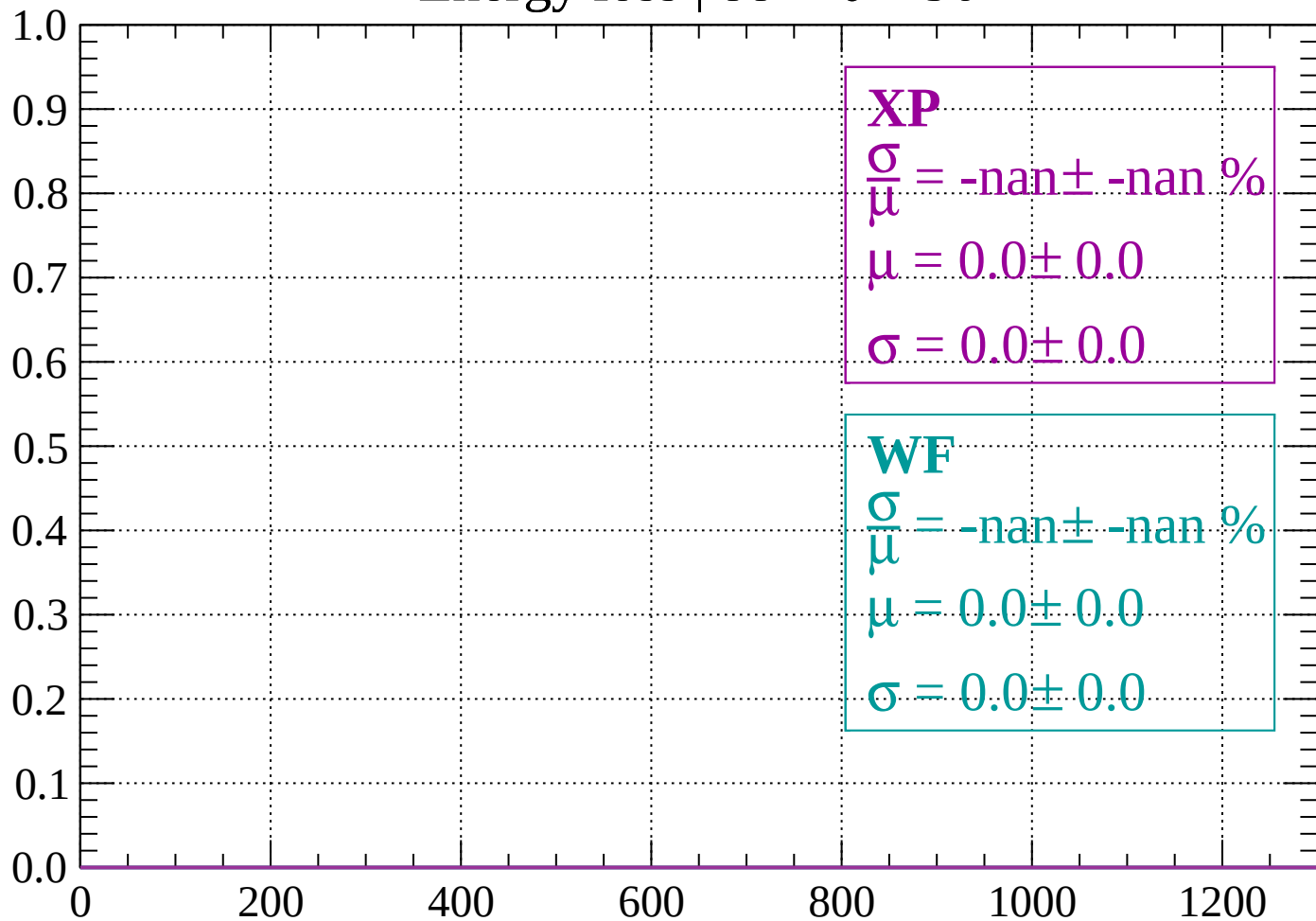
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

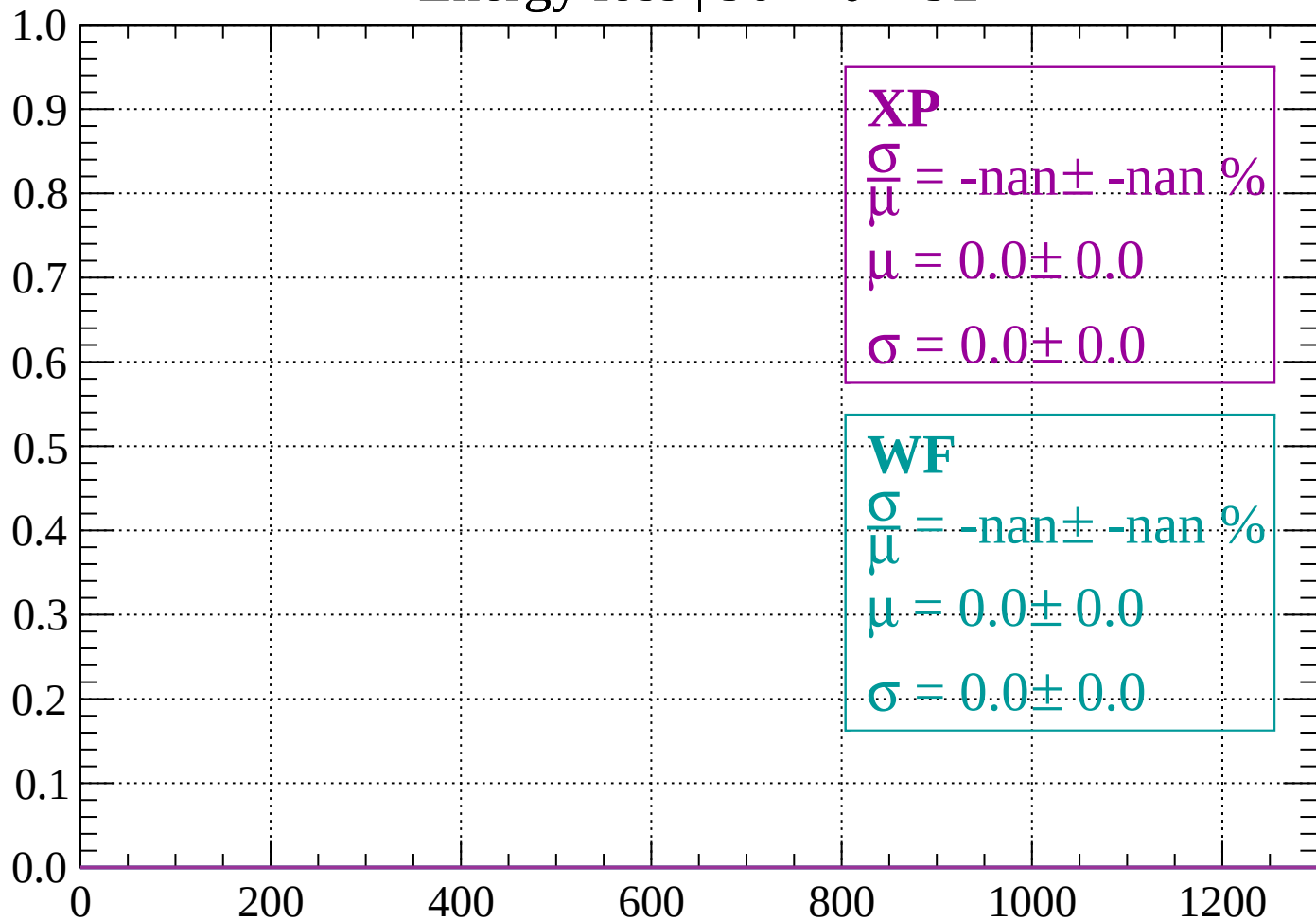
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

Count

